

Spectral Theory, Chapters 1–2

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Introduction

Spectral theory aims to study the properties of certain normed algebras, the fundamental examples being the algebra of complex-valued continuous functions on a topological space, the algebra of endomorphisms of a Banach space, and the convolution algebra of functions integrable with respect to a Haar measure on a locally compact group. The scope of this study is not necessarily apparent from this simple indication, but to grasp its importance one need only observe that two contemporary mathematicians, unless they belong to the same school, will often have as common knowledge, besides linear algebra and the elementary part of differential and integral calculus, only certain aspects of Fourier analysis.

In the first two chapters of this Book, we develop the most fundamental ideas of spectral theories, and consider in greater detail the third example mentioned above, namely, the Fourier transform, in the setting of a locally compact abelian group.

The following chapters will first continue the general theory of compact linear maps, then that of operators on Hilbert spaces, this case including linear maps defined only on a subspace (generally dense) of such a space. As is well known, this theory is the basis of the mathematical formalism of Quantum Mechanics. Finally, we will present the elementary theory of unitary representations of topological groups, particularly compact ones, which we have already had occasion to use in Chapter IX of the book *Lie Groups and Lie Algebras*.

The titles of the first two chapters are the same as those of the first edition of this Book, published in 1967. They have been revised, corrected, and expanded.

For ease of reading, we indicate below the most important changes found in this edition, and in particular the additions.

For Chapter I:

- In no. 3 of §3, we introduce the notion of proper partial map, which allows the Gelfand correspondence between locally compact spaces and commutative C^* -algebras to be stated in a functorial manner.

- In §4, we added nos. 13 and 14, which deal with holomorphic functional calculus in real or complex normable algebras, and in algebras without a unit element.

- In §6, the presentation of the continuous functional calculus has been reworked (no. 6). Nos. 9 to 11 are new; they consider the notion of a positive element in a general C^* -algebra, approximate units of such algebras, and the quotient of a C^* -algebra by a closed two-sided ideal.

- §7, concerning the most elementary spectral properties of endomorphisms of Banach spaces, is essentially new; numbers 7 and 8 partially take up certain numbers of the old paragraph 6. It is this new paragraph that will serve as the basis for the more in-depth study of endomorphisms of Hilbert spaces in Chapter IV.

For Chapter II:

- §1 on the Fourier transform has been revised in detail, for example no. 6, which deals with the functoriality of Pontryagin duality. In the new no. 9, we make more explicit the statements most important for the groups $\mathbf{R}^n$ and $(\mathbf{R}/\mathbf{Z})^n$. Fourier theory on $\mathbf{R}^n$, whose most natural framework lies in the theory of tempered distributions, is stated here through its essential properties, without giving the proofs, which will appear in Chapter IV.

- We have finally added a Fourier theory summary sheet that brings together the definitions and properties of the main function spaces involved in this theory, along with the essential formulas.

Many of the exercises are new. In particular, we note the considerable increase in the number of exercises in §1 of Chapter II, reflecting the mathematical importance of Fourier theory.

Chapter 1

Normed algebras

Unless stated otherwise, the algebras considered in this chapter are assumed to be associative.

1.1 SPECTRA AND CHARACTERS

In this paragraph, the letter K denotes a commutative field. If E and F are K -vector spaces, we denote by $E \otimes F = E \otimes_K F$.

1.1.1 Unital algebras

A unital algebra over K is a pair (A, e) where A is an algebra over K with identity element, and e is the identity element of A . Since e is uniquely determined by A , we will sometimes say, by abuse of language, that A is a unital algebra. If (A, e) and (A', e') are unital algebras, a unital morphism of (A, e) into (A', e') is a morphism φ from A to A' such that $\varphi(e) = e'$. A unital subalgebra of (A, e) is a pair (A', e) , where A' is a subalgebra of A containing e .

We will often denote the identity element by 1 .

Lemma 1. — Let A be an algebra. For every idempotent j of A , the subspace jAj of A is the set of $x \in A$ such that $xj = jx = x$. This is a subalgebra of A which admits the unit element j .

The proof is elementary.

1.1.2 Spectrum of an element in a unital algebra

DEFINITION 1. — Let A be a unital algebra over K and e its identity element. For every $x \in A$ we call the spectrum of x relative to A the set of $\lambda \in K$ such that $\lambda e - x$ is not invertible.

The spectrum of x will be denoted $\text{Sp}_A(x)$, or $\text{Sp}(x)$ if no confusion results. The complement of $\text{Sp}_A(x)$ in K is called the resolvent set of x .

Remarks. — 1) If $A = \{0\}$, then we have $\text{Sp}(0) = \emptyset$.

2) If $A \neq \{0\}$, we have $\text{Sp}(\lambda e) = \{\lambda\}$ for all $\lambda \in K$.

3) For $x \in A$ to be invertible, it is necessary and sufficient that $0 \notin \text{Sp}(x)$.

4) Let $R \in K(X)$ be a rational fraction and let $x \in A$ be an element that is substitutable in R , that is (A, IV, p. 20) there exist P and $Q \in K[X]$ such that $R = P/Q$ and $Q(x)$ is invertible; one can then form the element $R(x) = P(x) \cdot Q(x)^{-1} = Q(x)^{-1} \cdot P(x)$ of A ; it does not depend on the choices of P and Q . We have $0 \notin Q(\text{Sp}(x))$, so that every element of $\text{Sp}(x)$ is substitutable in R .

We have $R(\text{Sp}(x)) \subset \text{Sp}(R(x))$. Indeed, let $\lambda \in \text{Sp}(x)$; there exists a polynomial P_1 such that $R(\lambda) - R(x) = (\lambda - X)P_1(X)/Q(X)$; then, $R(\lambda)e - R(x) = (\lambda e - x)(P_1(x)/Q(x))$, so that $R(\lambda) - R(x)$ is not invertible, hence $R(\lambda) \in \text{Sp}(R(x))$.

Conversely, suppose that the field K is algebraically closed. Suppose first that R is not constant and let us prove that one has $\text{Sp}(R(x)) = R(\text{Sp}(x))$. Let $\mu \in \text{Sp}(R(x))$. Since R is not constant, $P - \mu Q$ is not the zero polynomial; let $\mu Q - P = \alpha \prod (\lambda_i - X)$ be a decomposition into factors of degree 1, so that $\mu e - R(x) = \alpha \prod (\lambda_i e - x) Q(x)^{-1}$. Since $\mu e - R(x)$ is not invertible, there exists i such that $\lambda_i e - x$ is not invertible, hence $\lambda_i \in \text{Sp}(x)$, then $R(\lambda_i) = \mu \in R(\text{Sp}(x))$.

When R is constant, the equality $\text{Sp}(R(x)) = R(\text{Sp}(x))$ also holds, provided that $\text{Sp}(x)$ is nonempty.

5) Suppose that the algebra A is nonzero. Let $x \in A$ be a nilpotent element. Let n be an integer such that $x^n = 0$. The spectrum of x^n is reduced to 0, hence so is the spectrum of x by remark 4.

- 6) Let A and B be unital algebras over K and $\varphi : A \rightarrow B$ a unital morphism. For every $x \in A$, one has $\text{Sp}_B(\varphi(x)) \subset \text{Sp}_A(x)$.
- 7) Let A be a unital algebra, R its radical (A, VIII, p. 150, def. 2) and let φ be the canonical morphism from A onto $B = A/R$. If $x \in A$, one has $\text{Sp}_B(\varphi(x)) = \text{Sp}_A(x)$. Indeed, it suffices to prove that if $\varphi(x)$ is invertible in B , then x is invertible in A . Now, if $y \in A$ is such that $\varphi(x)\varphi(y) = \varphi(y)\varphi(x) = \varphi(e)$, one has $xy \in e + R$, $yx \in e + R$, hence xy and yx are invertible (A, VIII, p. 151, th. 1) and consequently x is invertible. In particular, if $x \in R$, one has $\text{Sp}_A(x) = \{0\}$ if $A \neq \{0\}$.
- 8) Let $(B_i)_{i \in I}$ be a family of unital algebras, with $B_i = (A_i, e_i)$ for $i \in I$. Set $A = \prod_i A_i$, $e = (e_i)_{i \in I}$. Then (A, e) is a unital algebra called the product of the B_i . If $x = (x_i)_{i \in I} \in A$, one has $\text{Sp}_A(x) = \bigcup_i \text{Sp}_{A_i}(x_i)$.

Examples. — 1) Let X be a set and let $A = K^X$ be the algebra of functions with values in K defined on X . The spectrum of an element f of A is the set of values of f .

- 2) Let A be a unital algebra of finite rank over K . For $x \in A$ to be invertible, it is necessary and sufficient that the linear map $y \mapsto xy$ from A to A have nonzero determinant. It follows that the spectrum of x is the set of roots of the characteristic polynomial of x (A, III, p. 110). If A is the algebra of endomorphisms of a vector space V of finite dimension over K , the spectrum of x is therefore the set of eigenvalues of x . This is not always the case when V is of infinite dimension (cf. I, p. 153, exercise 2).

1.1.3 Resolvent

DEFINITION 2. — Let A be a unital algebra over K and $x \in A$. For every $\lambda \in K - \text{Sp}(x)$, set

$$R(x, \lambda) = (\lambda e - x)^{-1}.$$

The map from $K - \text{Sp}(x)$ to A given by $\lambda \mapsto R(x, \lambda)$ is called the resolvent of x .

For x fixed, the values of $R(x, \lambda)$ are pairwise permutable. If $\lambda, \mu \in K$, we have:

$$(\lambda - \mu)e = (\lambda e - x) - (\mu e - x)$$

hence, if $\lambda, \mu \in K - \text{Sp}(x)$, one has the relation

$$(\lambda - \mu)R(x, \lambda)R(x, \mu) = R(x, \mu) - R(x, \lambda). \quad (1)$$

If $x, y \in A$ and $\lambda \in K$, one has:

$$y - x = (\lambda e - x) - (\lambda e - y)$$

hence, if $\lambda \in K - (\text{Sp}(x) \cup \text{Sp}(y))$, one has the relation

$$R(y, \lambda)(y - x)R(x, \lambda) = R(y, \lambda) - R(x, \lambda). \quad (2)$$

1.1.4 Spectrum of an element in an algebra

Let A be an algebra over K . Recall (A, III, p. 4) that one defines on the vector space $\tilde{A} = K \times A$ an algebra structure such that:

$$(\lambda, a)(\mu, b) = (\lambda\mu, \lambda b + \mu a + ab).$$

Let $e = (1, 0)$. Then $(\tilde{A}, e)$ is the unital algebra obtained from A by adjoining a unit element. The algebra A is identified with the two-sided ideal $\{0\} \times A$ of $\tilde{A}$; the algebra A is commutative if and only if $\tilde{A}$ is.

If A' is a second algebra over K , let $(\tilde{A}', e')$ be the unital algebra obtained from A' by adjoining an identity element, and let φ be a morphism from A to A' ; there exists one and only one unital morphism from $(\tilde{A}, e)$ into $(\tilde{A}', e')$ which extends φ .

Let A be an algebra over K and $x \in A$. We call the spectrum of x relative to A the spectrum of x relative to the unital algebra $\tilde{A}$ deduced from A by adjoining an identity element. This set will be denoted $\text{Sp}'_A(x)$, or $\text{Sp}'(x)$ if no confusion arises. We have $0 \in \text{Sp}'_A(x)$ for all $x \in A$.

If φ is a morphism from A into an algebra B , we have $\text{Sp}'_B(\varphi(x)) \subset \text{Sp}'_A(x)$.

Remarks. — 1) Let $(A, 1)$ be a unital algebra. If $x \in A$, one has

$$\text{Sp}'_A(x) = \text{Sp}_A(x) \cup \{0\}.$$

We verify in fact that $(e - 1) \cdot A = A \cdot (e - 1) = 0$, so that $\tilde{A}$ is the unital algebra product of A and $K(e - 1)$. Our assertion therefore follows from Remark 8 of I, p. 3.

2) It follows from Remark 1 that, if B is an algebra over K and if $x \in B$, we have:

$$\text{Sp}'_B(x) = \text{Sp}_{\tilde{B}}(x) = \text{Sp}_{\tilde{B}}(x) \cup \{0\} = \text{Sp}'_{\tilde{B}}(x).$$

3) If x belongs to the radical of A (A, VIII, p. 430, def. 3), then $\text{Sp}'_A(x) = \{0\}$. This follows from Remark 7 of I, p. 3.

PROPOSITION 1. — Let A be an algebra and $x, y \in A$. We have $\text{Sp}'(xy) = \text{Sp}'(yx)$.

Passing to $\tilde{A}$ we reduce to the case where A has a unit element e . It then suffices to prove that, if $\lambda \neq 0$ is such that $xy - \lambda e$ admits an inverse u , then $yx - \lambda e$ is invertible. Set $z = yux - e$. Since $xyu = \lambda u + e$, one has

$$\begin{aligned} (yx - \lambda e)z &= y(xy u)x - yx - \lambda yux + \lambda e \\ &= y(\lambda u + e)x - yx - \lambda yux + \lambda e = \lambda e \end{aligned}$$

and likewise $z(yx - \lambda e) = \lambda e$. Since $\lambda \neq 0$, we see that $yx - \lambda e$ is invertible.

If A is a unital algebra and if $x, y \in A$, the preceding proposition implies that $\text{Sp}(xy) \cup \{0\} = \text{Sp}(yx) \cup \{0\}$, but one may have $\text{Sp}(xy) \neq \text{Sp}(yx)$ (cf. I, p. 153, exerc. 3).

1.1.5 Full subalgebras

Let A be a unital algebra over K .

DEFINITION 3. — A full subalgebra of A is a unital subalgebra B such that every element of B which is invertible in A is invertible in B .

In other words, B is a full subalgebra of A if and only if $\text{Sp}_B(x) = \text{Sp}_A(x)$ for all $x \in B$.

The intersection of a family of full subalgebras of A is a full subalgebra of A .

Let M be a subset of A . The intersection of the full subalgebras of A containing M is the smallest full subalgebra of A containing M .

It is called the full subalgebra of A generated by M . The commutant M' of M in A is a full subalgebra of A (for, if x is invertible in A and commutes with M , then x^{-1} commutes with M). Therefore, the bicommutant M'' of M is a full subalgebra of A that contains the full subalgebra of A generated by M .

If the elements of M are pairwise permutable, we have $M \subset M'$ and $M'' \subset M'''$; the algebra M'' is therefore commutative, and the same is then true of the full subalgebra generated by M .

A maximal commutative subalgebra of A is a full subalgebra, because it is equal to its commutant.

Lemma 2. — Let $(x_\lambda)_{\lambda \in \Lambda}$ be a family of pairwise commuting elements of A . The full subalgebra generated by (x_λ) is the set of elements of the form $R((x_\lambda))$, where $R \in K((X_\lambda))$ ranges over the set of rational fractions in which the family (x_λ) is substitutable.

Let B be the full subalgebra of A generated by the family (x_λ) , and denote by B_1 the set of elements of the form $R((x_\lambda))$, where $R((X_\lambda))$ is a rational fraction in which (x_λ) is substitutable. Explicitly, B_1 is the set of elements of A of the form $P((x_\lambda))Q((x_\lambda))^{-1}$, where $P, Q \in K[(X_\lambda)]$ and $Q((x_\lambda))$ is invertible in A . The set B_1 is a unital subalgebra of A containing the family (x_λ) . It is a full subalgebra: if $P((x_\lambda))Q((x_\lambda))^{-1}$ is invertible in A , then $P((x_\lambda))$ is invertible in A and the inverse $Q((x_\lambda))P((x_\lambda))^{-1}$ of $P((x_\lambda))Q((x_\lambda))^{-1}$ belongs to B_1 . We therefore have $B \subset B_1$. On the other hand, if $P, Q \in K[(X_\lambda)]$, and if $Q((x_\lambda))$ is invertible in A , then $P((x_\lambda)) \in B$ and $Q((x_\lambda)) \in B$, hence $Q((x_\lambda))^{-1} \in B$ and $P((x_\lambda))Q((x_\lambda))^{-1} \in B$, hence $B_1 \subset B$.

1.1.6 Characters of a commutative unital algebra

DEFINITION 4. — Let A be a commutative unital algebra over K . A unital character is a unital morphism from A to K .

When no confusion can arise, one simply says character instead of unital character. The set of unital characters of A is denoted $X(A)$. If A is the zero algebra, then $X(A)$ is empty.

Let A and B be commutative unital algebras over K and h a unital morphism from A to B .

The map $\chi \mapsto \chi \circ h$ of $X(B)$ into $X(A)$ is denoted $X(h)$. If k is a morphism from B to a commutative unital algebra, one has $X(k \circ h) = X(h) \circ X(k)$. The map $X(\text{Id}_A)$ is the identity map of $X(A)$.

If h is surjective, $X(h)$ is a bijection from $X(B)$ onto the set of characters of A that vanish on the kernel of h .

Let $(A_1, e_1), \dots, (A_n, e_n)$ be commutative unital algebras over K and let A be the unital algebra $A_1 \times \dots \times A_n$, with identity element $(e_1, \dots, e_n)$. For each i , identify A_i with an ideal of A and let π_i be the canonical map from A onto A_i . Then $X(\pi_i)$ is a bijection from $X(A_i)$ onto the set X_i of characters of A vanishing on $\prod_{j \neq i} A_j$. The sets X_i are pairwise disjoint. On the other hand, let $\chi \in X(A)$. Since $1 = \sum \chi(e_i)$, there exists i such that $\chi(e_i) \neq 0$. For every $j \neq i$ and every $y \in A_j$, one has $\chi(e_i)\chi(y) = \chi(e_i y) = \chi(0) = 0$, hence $\chi(A_j) = 0$. Thus, χ vanishes on $\prod_{j \neq i} A_j$, so that $X(A)$ is the union of the X_i .

Let B be the unital algebra $A_1 \otimes \dots \otimes A_n$. Denote by h_i the canonical morphism $A_i \rightarrow B$. Then

$$\chi \mapsto (\chi \circ h_1, \dots, \chi \circ h_n)$$

is a map from $X(B)$ into $X(A_1) \times \dots \times X(A_n)$, and

$$(\chi_1, \dots, \chi_n) \mapsto \chi_1 \otimes \dots \otimes \chi_n$$

is a map from $X(A_1) \times \dots \times X(A_n)$ into $X(B)$. One verifies that these maps are inverse bijections of one another, by which one identifies $X(B)$ with $X(A_1) \times \dots \times X(A_n)$.

Let A be a commutative unital algebra over K . Let Y be the set of ideals of codimension 1 of A . For every $\chi \in X(A)$, one has $\text{Ker}(\chi) \in Y$. The map $\chi \mapsto \text{Ker}(\chi)$ is a bijection from $X(A)$ onto Y . Indeed, if $I \in Y$, there exists a unique isomorphism of the unital K -algebra A/I onto K and the composite morphism

$$A \rightarrow A/I \rightarrow K$$

is the unique character of A with kernel I .

DEFINITION 5. — Let A be a commutative unital algebra over K . For every $x \in A$, we denote by $\mathcal{G}_A(x)$, or simply $\mathcal{G}(x)$, the map $\chi \mapsto \chi(x)$ from $X(A)$ to K . It is called the Gelfand transform of x .

The map $\mathcal{G}$ is a unital morphism from A into the unital algebra $K^{X(A)}$ of maps from $X(A)$ to K . It is called the Gelfand transform of A .

If $x \in A$, the image of the Gelfand transform $\mathcal{G}_A(x)$ of x is contained in $\text{Sp}_A(x)$. Indeed, let $\chi \in X(A)$; since $\chi(x - \chi(x)e) = 0$, the element $x - \chi(x)e$ is not invertible.

Let B be a unital commutative algebra over K and h a unital morphism from A to B ; then $X(h): X(B) \rightarrow X(A)$ defines a unital morphism $h_*: K^{X(A)} \rightarrow K^{X(B)}$, and the diagram:

$$\begin{array}{ccc} A & \xrightarrow{\mathcal{G}_A} & K^{X(A)} \\ \downarrow h & & \downarrow h_* \\ B & \xrightarrow{\mathcal{G}_B} & K^{X(B)} \end{array} \quad (3)$$

is commutative. Indeed, for every $x \in A$ and every $\chi \in X(B)$, we have:

$$\begin{aligned} \mathcal{G}_B(h(x))(\chi) &= \chi(h(x)) = (\chi \circ h)(x) \\ &= (X(h)(\chi))(x) \\ &= \mathcal{G}_A(x)(X(h)(\chi)) \\ &= h_*(\mathcal{G}_A(x))(\chi). \end{aligned} \quad (4)$$

Suppose now that K is a topological field. We then equip $X(A)$ with the topology of point-wise convergence on A (cf. EVT, III, p. 14, example 1), and the topological space $X(A)$ is called the character space of A . The topology of $X(A)$ is therefore the coarsest for which the functions $\mathcal{G}_A(x)$ for $x \in A$ are continuous, and the map $\chi \mapsto (\chi(a))_{a \in A}$ identifies the space $X(A)$ with a subset of K^A .

When $K = \mathbb{R}$ or $\mathbb{C}$, this topology is the topology induced on $X(A) \subset A^*$ by the weak topology $\sigma(A^*, A)$ on A^* (EVT, II, p. 45, def. 2); as such, we will also say that it is the weak topology on $X(A)$.

If h is a unital morphism from A to B , the map $X(h): X(B) \rightarrow X(A)$ is continuous. If h is surjective, the image of $X(h)$, namely, the set of characters of A vanishing on the kernel of h , is closed in $X(A)$; on the other hand, the topology on $X(h)(X(B))$ deduced from that of $X(B)$ by

the bijection $X(h)$ is the topology of pointwise convergence in A , that is, the topology induced by that of $X(A)$. In other words, $X(h)$ is a homeomorphism from $X(B)$ onto a closed subset of $X(A)$.

If $A_1, \dots, A_n$ are unital commutative algebras over K , the space $X(A_1 \times \dots \times A_n)$ is thus identified with the topological sum space of $X(A_1), \dots, X(A_n)$. Likewise, $X(A_1 \otimes \dots \otimes A_n)$ is identified with the product topological space $X(A_1) \times \dots \times X(A_n)$.

1.1.7 Case of algebras without a unit element

DEFINITION 6. — Let A be a commutative algebra over K . A character of A is an algebra homomorphism from A to K .

The set of characters of A will be denoted by $X'(A)$.

The zero map is an algebra homomorphism. If A has a unit element, e , a nonzero algebra morphism from A into K is unital, that is, it is a unital character of A in the sense of Definition 4: indeed, for $\chi \in X'(A)$, χ is nonzero if and only if $\chi(e) = 1$.

We set $X(A) = X'(A) - \{0\}$; by the preceding, the notation is compatible with that introduced when A is unital.

If $h: A \rightarrow B$ is a morphism of commutative algebras, the map $\chi \mapsto \chi \circ h$ is a map $X'(h): X'(B) \rightarrow X'(A)$. It maps 0 to 0. If $k: B \rightarrow C$ is a morphism of commutative algebras, then one has $X'(k \circ h) = X'(h) \circ X'(k)$. If h is surjective, $X'(h)$ is a bijection from $X'(B)$ onto the set of characters of A vanishing on the kernel of h . Let $A_1, \dots, A_n$ be commutative algebras, $A = A_1 \times \dots \times A_n$ and $\pi_i: A \rightarrow A_i$ the canonical morphism; then $X'(\pi_i)$ is a bijection from $X'(A_i)$ onto a subset X'_i of $X'(A)$, namely the set of characters of A vanishing on $\prod_{j \neq i} A_j$; we see as in no. 6 that $X'(A)$ is the union

of X'_i ; on the other hand, $X'_i \cap X'_j = \{0\}$ for $i \neq j$; in particular the $X'_i - \{0\}$ form a partition of $X'(A) - \{0\} = X(A)$.

For every $x \in A$, let $\mathcal{G}'_A(x)$, or simply $\mathcal{G}'(x)$, be the map $\chi \mapsto \chi(x)$ from $X'(A)$ to K . The map $\mathcal{G}'$ is a morphism from A into the algebra A_1 of the maps $X'(A) \rightarrow K$ vanishing at 0. Let B be a commutative algebra, let B_1 be the algebra of maps $X'(B) \rightarrow K$ vanishing at 0, and let h be a morphism from A to B ; then $X'(h)$ defines a morphism $h_1: A_1 \rightarrow B_1$, and we have

$h_1 \circ \mathcal{G}'_A = \mathcal{G}'_B \circ h$. One denotes by $\mathcal{G}_A(x)$, or simply $\mathcal{G}(x)$ the restriction of $\mathcal{G}'_A(x)$ onto $X(A)$ and it is called the Gelfand transform of x .

Let $\tilde{A}$ be the unital algebra deduced from A by adjoining an identity element. By restriction, every character of $\tilde{A}$ defines a character of A ; conversely, every character of A extends uniquely to a character of $\tilde{A}$. This defines a canonical bijection from $X'(A)$ onto $X(\tilde{A})$, by which these two sets are identified. The character 0 of A is identified with the unique character of $\tilde{A}$ with kernel A .

If $x \in A$ and $\chi \in X'(A)$, one has $\chi(x) \in \text{Sp}_{\tilde{A}}(x)$, hence $\chi(x) \in \text{Sp}'_A(x)$.

Lemma 3. — The map $\chi \mapsto \text{Ker}(\chi)$ is a bijection from $X(A)$ onto the set of regular ideals of codimension 1 of A .

Recall (A, VIII, p. 426, def. 1) that an ideal I of A is said to be regular if the quotient algebra A/I admits a unit element.

Let us prove the lemma. On the one hand, $X(A)$ is identified with the set of characters of $\tilde{A}$ nonzero on A . On the other hand, by A, VIII, p. 428, prop. 4, the map $I \mapsto A \cap I$ is a bijection from the set of maximal ideals of $\tilde{A}$ distinct from A onto the set of regular maximal ideals of A . The lemma then follows from the results of no. 6.

Suppose now that K is a topological field. We then equip $X'(A)$ with the topology of pointwise convergence on A ; the notation $X'(A)$ will henceforth denote the topological space thus obtained. When $K = \mathbb{R}$ or $\mathbb{C}$, we shall also call it the weak topology. For every $x \in A$, the function $\mathcal{G}'_A(x)$ on $X'(A)$ is continuous.

If h is a morphism from A to B , the map $X'(h): X'(B) \rightarrow X'(A)$ is continuous. If h is surjective, $X'(h)$ is a homeomorphism from $X'(B)$ onto its image and this image is closed in $X'(A)$.

Let $A = A_1 \times \cdots \times A_n$; with the same notations as above, $X'(\pi_i)$ is a homeomorphism from $X'(A_i)$ onto X'_i and X'_i is closed in $X'(A)$. Therefore $X'_i - \{0\}$ is open in $X'(A)$; the $X'(\pi_i)$ define a continuous map from the sum space S of the $X'(A_i)$ onto $X'(A)$, and one verifies immediately that the image of a union of neighborhoods of the points $0 \in X'(A_1), \dots, 0 \in X'(A_n)$ is a neighborhood of $0 \in X'(A)$; from all this it follows that $X'(A)$ is canonically identified with a quotient space of S . In particular, the space $X(A)$ is identified with the sum

space of the $X(A_i)$.

The canonical bijection from $X'(A)$ onto $X(\tilde{A})$ is a homeomorphism. Let B be a unital algebra over K and B' the underlying algebra; then the space $X(B)$ is identified with the subspace $X(B')$ of $X'(B')$.

1.1.8 Primitive ideals

Let A be an algebra over K and E a vector space over K . One calls a representation of A in E a morphism from A to the algebra $\mathcal{L}(E)$ of endomorphisms of E . An injective representation is said to be faithful. Let π_1 and π_2 be representations of A in spaces E_1, E_2 . A morphism of π_1 into π_2 is a K -linear map $u : E_1 \rightarrow E_2$ such that $u(\pi_1(a)x) = \pi_2(a)u(x)$ for all $a \in A$ and $x \in E_1$. Representations are said to be equivalent if there exists a morphism of π_1 into π_2 which is an isomorphism of vector spaces. Its inverse is then a morphism of π_2 into π_1 . A representation π of A in E is said to be irreducible if $E \neq \{0\}$ and if the only vector subspaces of E stable under $\pi(A)$ are $\{0\}$ and E .

Example. — The zero map from A to $\mathcal{L}(E)$ is a representation, called the trivial representation, of A . It is irreducible if and only if E has dimension 1.

Lemma 4. — Let π be an irreducible, non-trivial representation of A in E . For every non-zero element ξ of E , one has $\pi(A)\xi = E$.

The subspace $\pi(A)\xi$ of E is stable under $\pi(A)$. Suppose it is zero. The non-zero subspace $K\xi$ of E would then be stable under $\pi(A)$, and hence equal to E ; but this would imply that π is the zero representation. We therefore have $\pi(A)\xi = E$.

Let π be an irreducible, non-trivial representation of A in E . By this lemma, the annihilator R of ξ in A is a regular left ideal (A, VIII, p. 425, n° 1) of A , and the representation π is equivalent to the representation defined by the A -pseudomodule A/R . Since π is irreducible, the ideal R is a maximal regular left ideal.

DEFINITION 7. — Let A be an algebra over K . One calls a primitive ideal of A the kernel of a non-trivial irreducible representation of A .

If A is commutative, the primitive ideals of A are the maximal regular ideals of A . Indeed, the non-trivial irreducible representations of A are, up to equivalence, the representations π_R

defined by the A -pseudomodules A/R , where R is a maximal regular ideal of A . The kernel of π_R contains R . It is even equal to it since, by A, VIII, p. 426, prop. 2, the commutativity of A implies that A/R is a field. Hence $\text{Ker}(\pi_R)$ is maximal regular.

Lemma 5. — Let π be an irreducible representation of A in a vector space E over K .

- a) Let I be a two-sided ideal of A . If $\pi(I) \neq \{0\}$, then $\pi|_I$ is irreducible;
- b) Let I_1 and I_2 be two-sided ideals of A such that $\pi(I_1) \neq 0$ and $\pi(I_2) \neq 0$. Then $\pi(I_1 I_2) \neq 0$.

The set of elements of E annihilated by $\pi(I)$ is stable under $\pi(A)$ and distinct from E , hence equal to 0. Therefore, if ξ is a non-zero element of E , one has $\pi(I)\xi \neq 0$; since $\pi(I)\xi$ is stable under $\pi(A)$, one has $\pi(I)\xi = E$, which proves *a*). On the other hand, what precedes proves that $\pi(I_2)E = E$, $\pi(I_1)\pi(I_2)E = E$, hence $\pi(I_1 I_2) \neq 0$, whence *b*).

Lemma 6. — Let I_1 and I_2 be two-sided ideals of A , and I a primitive ideal of A . If I contains $I_1 I_2$ (in particular, if I contains $I_1 \cap I_2$), then I contains I_1 or I_2 .

Let π be an irreducible representation with kernel I . If $I \not\supseteq I_1$ and $I \not\supseteq I_2$, Lemma 5, *b*) proves that $\pi(I_1 I_2) \neq 0$, whence $I \not\supseteq I_1 I_2$.

Lemma 7. — Suppose that A has a unit element. Let I be a maximal two-sided ideal of A . Then I is a primitive ideal.

There exists a maximal left ideal R of A containing I (A, I, p. 99, th. 1). Let π be the canonical representation of A in A/R , which is irreducible and nonzero. Since $IA \subset R$, the kernel I' of π contains I , hence $I' = I$ and I is primitive.

Let $J(A)$ be the set of primitive ideals of A . For any subset M of A , we denote by $V(M)$ the set of primitive ideals of A containing M ; if I is the two-sided ideal of A generated by M , then $V(M) = V(I)$. If M is reduced to a single element x , we write $V(x)$ instead of $V(\{x\})$.

The mapping $M \mapsto V(M)$ is decreasing with respect to inclusion relations. We have:

$$(5) \quad V(\emptyset) = J(A), \quad V(A) = \emptyset$$

$$(6) \quad V\left(\bigcup_{i \in I} M_i\right) = V\left(\sum_{i \in I} M_i\right) = \bigcap_{i \in I} V(M_i)$$

for any family $(M_i)_{i \in I}$ of subsets of A . On the other hand, by Lemma 6,

$$(7) \quad V(I_1 \cap I_2) = V(I_1 I_2) = V(I_1) \cup V(I_2)$$

for all two-sided ideals I_1, I_2 of A . Formulas (5) through (7) show that the subsets $V(M)$ of $J(A)$ are the closed subsets of a topology called the Jacobson topology on $J(A)$.

Let T be a subset of $J(A)$, and let $\Upsilon(T)$ be the intersection of the elements of T , so that $\Upsilon(T)$ is a two-sided ideal of A . Then the closure of T in $J(A)$ is the smallest closed subset of $J(A)$ containing T , that is, $V(\Upsilon(T))$. In particular, T is closed if and only if $T = V(\Upsilon(T))$.

PROPOSITION 2. — Let I_1 and I_2 be distinct points of $J(A)$. Then one of these two points is not adherent to the other.

Indeed, we have, for example, $I_1 \not\subset I_2$. The set $V(I_1)$ of $I \in J(A)$ such that $I_1 \subset I$ is closed in $J(A)$ and it contains I_1 but not I_2 .

PROPOSITION 3. — Let $I \in J(A)$. For $\{I\}$ to be closed in $J(A)$, it is necessary and sufficient that I be a primitive maximal ideal.

Indeed, the closure of $\{I\}$ consists of the primitive ideals of A containing I .

The relation

« π_1, π_2 are representations of A , which are isomorphic.

is an equivalence relation with respect to π_1 and π_2 . For any representation π of A , we denote by $\text{cl}(\pi)$ the equivalence class of π , which is therefore a representation of A isomorphic to π , such that two representations π_1 and π_2 are isomorphic if and only if $\text{cl}(\pi_1) = \text{cl}(\pi_2)$. We say that $\text{cl}(\pi)$ is the class of π .

Let c be the cardinal of A . Let π be a nonzero irreducible representation of A in a K -vector space E . Let ξ be a nonzero element of E . Since $\pi(A)\xi = E$ (lemma 4), the dimension of E is $\leq c$ (A , II, p. 97, corollary). The relation

“ λ is a class of irreducible representations of A

in a K -vector space of dimension $\leq c$ »

is collectivizing in λ (E , II, p. 3). Indeed, every vector space of dimension $\leq c$ is isomorphic to a space K^B where B is a subset of A (A , II, p. 25, def. 10), and the assertion then follows from E , II, p. 47.

We denote by $\widehat{A}$ the set of classes of irreducible, nontrivial representations of A . According to what precedes, for every nontrivial irreducible representation π of A , there exists a unique representation $\widehat{\pi} \in \widehat{A}$ which is isomorphic to π .

The map from $\widehat{A}$ into $J(A)$ which associates to π its kernel is surjective. If A is commutative, it follows from the fact that primitive ideals are regular maximal ideals that this map is a bijection.

We endow $\widehat{A}$ with the inverse-image topology under the map $\widehat{A} \rightarrow J(A)$.

If A possesses a unit element, the spaces $J(A)$ and $\widehat{A}$ are quasi-compact.

It suffices to give the proof for $J(A)$. Let (T_j) be a family of closed subsets of $J(A)$ whose intersection is empty. If the sum $\sum_j \Upsilon(T_j)$ was different from A , then this sum would be contained in a maximal two-sided ideal I . The ideal I would be primitive (Lemma 7); since each T_j is closed, hence equal to $V(\Upsilon(T_j))$, we would have $I \in T_j$ for all j , which contradicts the hypothesis. Thus we have $\sum_j \Upsilon(T_j) = A$, and therefore we can write $1 = x_1 + \dots + x_n$ with $n \geq 1$ and $x_i \in \Upsilon(T_{j_i})$ for all i . This implies that $\Upsilon(T_{j_1}) + \dots + \Upsilon(T_{j_n}) = A$, whence $T_{j_1} \cap \dots \cap T_{j_n} = \emptyset$.

Suppose the algebra A is commutative and unital. The Jacobson topology on $J(A)$ is the topology induced on $J(A)$ by the Zariski topology of the prime spectrum of A (AC, II, def. 4, p. 125).

Suppose that A is commutative and that K is a topological field. The canonical isomorphism from K onto $\mathcal{L}(K)$ allows one to identify an element of $X(A)$ with a representation of A in the vector space K , which defines an injective map from $X(A)$ into $\widehat{A}$. We can therefore identify $X(A)$ with a subset of $\widehat{A}$.

Proposition 5. — The topology induced on $X(A)$ by that of $\widehat{A}$ is coarser than the topology of $X(A)$.

Indeed, let T be a closed subset of $\widehat{A}$. Then T is the set of $\pi \in \widehat{A}$ whose kernel contains a subset M of A . Therefore $T \cap X(A)$ is the set of $\chi \in X(A)$ which vanish on M , that is, a closed subset of $X(A)$. Whence the proposition.

In general, the topology of $X(A)$ does not coincide with the topology induced by the topology of $\widehat{A}$ (cf. I, p. 193, exercise 6, c)).

1.2 NORMED ALGEBRAS

In this section, K denotes either of the fields $\mathbb{R}$ or $\mathbb{C}$.

1.2.1 Generalities

Recall (cf. TG, IX, p. 37, def. 9) that a normed algebra is an algebra A over K equipped with a norm $x \mapsto \|x\|$ such that:

$$\|xy\| \leq \|x\|\|y\| \quad (1)$$

for all $x, y \in A$. If A is complete, A is said to be a Banach algebra.

Let A be a complete normable algebra over K . The topology of A can be defined by a norm satisfying (1) (cf. TG, IX, p. 38). When A is equipped with the structure of a normed algebra defined by such a norm, we shall also say that the algebra A is a Banach algebra.

We also recall the following facts (cf. TG, IX, p. 38–39):

- (1) If A is a nonzero normed algebra and has a unit element e , then $\|e\| \geq 1$.
- (2) Let A be a normed K -algebra. The opposite algebra of A , equipped with the same norm, is a normed algebra. The completion of A , equipped with the norm obtained by continuous extension of the norm of A , is a K -Banach algebra. Every subalgebra of A , equipped with the induced norm, is a normed algebra. If I is a closed two-sided ideal of A , the algebra A/I , equipped with the norm defined by $\|\dot{x}\| = \inf_{x \in \dot{x}} \|x\|$ for all $\dot{x} \in A/I$ is a normed algebra.
- (3) Let $(A_i)_{i \in I}$ be a family of normed algebras and B the product algebra of the algebras A_i . The subalgebra of elements $(x_i)_{i \in I}$ of B such that $\|(x_i)\| = \sup_{i \in I} \|x_i\| < +\infty$ is a normed algebra, called the product normed algebra of the algebras $(A_i)_{i \in I}$. If A_i is a Banach algebra for every i , then B is a Banach algebra.

Let A be a normed algebra. Let $(y_i)_{i \in I}$ be a family of elements of A ; the smallest closed subalgebra B of A containing the elements y_i is called the closed subalgebra of A generated by the y_i ; if $B = A$, one says that the y_i topologically generate the normed algebra A , or that the

family $(y_i)_{i \in I}$ is a topological generating system of the normed algebra A .

Similarly, if A is a unital normed algebra, the smallest closed unital subalgebra containing the elements y_i is called the closed unital subalgebra of A generated by the y_i . If it is equal to A , one says that the y_i topologically generate the unital normed algebra A .

Let A be a normed K -algebra. Denote by $\tilde{A}$ the algebra obtained from A by adjoining an identity element. On $\tilde{A}$, one defines a norm by setting $\|(\lambda, x)\| = |\lambda| + \|x\|$ for all $\lambda \in K$ and every $x \in A$. We have

$$\begin{aligned} \|(\lambda, x)(\mu, y)\| &= |\lambda\mu| + \|xy + \mu x + \lambda y\| \\ &\leq |\lambda|\|\mu\| + \|x\|\|y\| + |\mu|\|x\| + |\lambda|\|y\| \\ &= \|(\lambda, x)\|\|(\mu, y)\|. \end{aligned}$$

Thus $\tilde{A}$ becomes a normed algebra, called the unital normed algebra deduced from A by adjoining an identity element. The algebra A is identified with the closed two-sided ideal $\{0\} \times A$ of $\tilde{A}$.

DEFINITION 1. — Let A be a normed algebra. For $a \in A$, let γ_a and δ_a be the maps $x \mapsto ax$, and $x \mapsto xa$ from A to A . The map $\gamma : a \mapsto \gamma_a$ is a representation of A in A , called the left regular representation of A . The map $\delta : a \mapsto \delta_a$ is a representation of the opposite algebra of A in A , called the right regular representation of A .

Lemma 1. — Let A be a normed algebra. The left regular representation of A and the right regular representation of A are continuous with norm ≤ 1 .

If the algebra A is unital, the maps $x \mapsto \|\gamma_x\|$ and $x \mapsto \|\delta_x\|$ are norms on A , defining the topology of A . They satisfy the inequality (1).

If the unital normed algebra A is nonzero, that is, if $e \neq 0$, we also have $\|\gamma_e\| = \|\delta_e\| = 1$.

It is immediate that $\|\gamma_x\| \leq \|x\|$ and that $\|\delta_x\| \leq \|x\|$.

If A has an identity element e , then we have $x = \gamma_x e = \delta_x e$, hence:

$$\|x\| \leq \|\gamma_x\| \cdot \|e\| \quad \|x\| \leq \|\delta_x\| \cdot \|e\|. \quad (2)$$

In this case, $x \mapsto \|\gamma_x\|$ and $x \mapsto \|\delta_x\|$ are therefore norms equivalent to the norm of A . They satisfy (1) because γ and δ are representations.

The last assertion follows from the fact that $\gamma_e = \delta_e = Id_A$.

1.2.2 Examples

- 1) Let E be a normed space. Equip E with the product defined by $ab = 0$ for all a and b in E ; this defines on E a normed algebra structure.
- 2) Let X be a set. We denote by $\mathcal{B}(X; K)$ the normed algebra of bounded functions on X with values in K , equipped with the norm

$$\|f\| = \sup_{x \in X} |f(x)|$$

(TG, X, p. 22). It is a unital Banach algebra over K (TG, X, p. 21, cor. 1). It is commutative. Let f be an element of $\mathcal{B}(X; K)$. Then f is invertible in $\mathcal{B}(X; K)$ if and only if one has

$$\inf_{x \in X} |f(x)| > 0.$$

The spectrum of f is therefore the set of $\lambda \in K$ such that

$$\inf_{x \in X} |f(x) - \lambda| = 0,$$

that is, the closure in K of the set $f(X)$ of values of f .

- 3) Let X be a topological space. We denote by $\mathcal{C}_b(X; K)$ the unital subalgebra of $\mathcal{B}(X; K)$ of continuous and bounded functions on X with values in K , $\mathcal{C}_0(X; K)$ the subalgebra of $\mathcal{C}_b(X; K)$ consisting of functions that tend to 0 at infinity (cf. INT, III, §1, n°2 and TG, X, p. 40, cor. 2). We recall that $\mathcal{H}(X; K)$ denotes the subalgebra of $\mathcal{C}_b(X; K)$ of continuous functions with compact support. The algebras $\mathcal{C}_b(X; K)$ and $\mathcal{C}_0(X; K)$ are commutative Banach algebras over K ; indeed, they are closed subspaces of $\mathcal{B}(X; K)$ (TG, X, p. 21, cor. 2 and INT, III, §1, n°2).

Since the inverse of a continuous function that is nowhere zero is continuous, the subalgebra $\mathcal{C}_b(X; K)$ is a full subalgebra of $\mathcal{B}(X; K)$. In particular, the spectrum of an element

f of $\mathcal{C}_b(X; K)$ is equal to $\overline{f(X)}$.

If X is discrete, one has $\mathcal{C}_b(X; K) = \mathcal{B}(X; K)$.

If X is compact, one has $\mathcal{C}_b(X; K) = \mathcal{H}(X; K) = \mathcal{C}(X; K)$, the unital normed algebra $\mathcal{C}(X; K)$ of continuous functions on X with values in K . In this case, an element $f \in \mathcal{C}(X; K)$ is invertible if and only if f does not take the value 0, and the spectrum of f is equal to the set $f(X)$ of values of f .

Suppose henceforth that X is not compact. The algebra $\mathcal{C}_0(X; K)$ is then not unital; the algebra obtained from it by adjoining a unit element is identified with the subalgebra $K \cdot 1 \oplus \mathcal{C}_0(X; K)$ of $\mathcal{C}(X; K)$ formed by functions that have a limit at infinity. For an element of this subalgebra to be invertible, it is necessary and sufficient that it does not vanish and that its limit at infinity is not zero. It follows that for every $f \in \mathcal{C}_0(X; K)$, the spectrum of f is equal to $f(X) \cup \{0\}$.

The spectrum of $f \in \mathcal{H}(X; K)$ is equal to $f(X)$ (indeed, if X is not compact, 0 belongs to $f(X)$).

- 4) Let $n \geq 0$ be an integer. Let A_n be the algebra of functions $f : [0, 1] \rightarrow K$ admitting continuous derivatives in $[0, 1]$ up to order n , equipped with the norm

$$\|f\| = \sum_{k=0}^n \frac{1}{k!} \sup_{0 \leq t \leq 1} |f^{(k)}(t)|.$$

If $f, g \in A_n$, one has

$$\begin{aligned} \|fg\| &= \sum_{k=0}^n \frac{1}{k!} \sup |(fg)^{(k)}(t)| = \sum_{k=0}^n \frac{1}{k!} \sup \left| \sum_{s=0}^k \binom{k}{s} f^{(s)}(t) g^{(k-s)}(t) \right| \\ &\leq \sum_{k=0}^n \sum_{s=0}^k \frac{1}{s!(k-s)!} \sup_{0 \leq t \leq 1} |f^{(s)}(t)| \sup_{0 \leq t \leq 1} |g^{(k-s)}(t)| = \|f\| \|g\|, \end{aligned}$$

by Leibniz's formula (FVR, I, p. 28, prop. 2), hence A_n is a unital commutative Banach algebra.

- 5) Let E be a Banach space whose norm we denote by p . The algebra $\mathcal{L}(E)$ of continuous

endomorphisms of E , equipped with the norm

$$\|u\| = \sup_{p(x) \leq 1} p(u(x))$$

is a unital Banach algebra (EVT, III, p. 14 and p. 24, cor. 2; TG, X, p. 23, formula (3)).

- 6) Let G be a locally compact group. Denote by e its identity element. Let $\mathcal{M}^1(G)$ be the Banach space of bounded complex measures on G (INT, III, p. 57). The convolution product (INT, VIII, p. 120, def. 1) equips $\mathcal{M}^1(G)$ with a complex Banach algebra structure (INT, VIII, §3, n° 1, prop. 2) admitting as identity element the measure ε_e defined by the unit mass placed at the point e . If G is commutative, this Banach algebra is commutative. The space $\mathcal{C}'(G)$ of measures with compact support is a subalgebra of $\mathcal{M}^1(G)$ (loc. cit.).
- 7) Let G be a locally compact group equipped with a Haar measure μ . Then $L_K^1(G, \mu)$ is a Banach algebra for the convolution product (INT, VIII, prop. 12, p. 166). If $K = \mathbb{C}$, the map defined by $f \mapsto f\mu$ allows one to identify $L_{\mathbb{C}}^1(G, \mu)$ with a subalgebra of the Banach algebra $\mathcal{M}^1(G)$. If G is commutative, the Banach algebra $L_K^1(G, \mu)$ is commutative.
- 8) Take $G = \mathbb{Z}$ and $K = \mathbb{C}$ in Example 7. Then $L_{\mathbb{C}}^1(\mathbb{Z})$ is the complex commutative Banach algebra of sequences $(x_n)_{n \in \mathbb{Z}}$ such that $\sum_n |x_n| < +\infty$, the product of elements (x_n) and (y_n) being (z_n) , where

$$z_n = \sum_{k \in \mathbb{Z}} x_k y_{n-k}$$

and the norm $\|(x_n)\| = \sum_n |x_n|$. This algebra admits as its identity element the sequence $\varepsilon = (\varepsilon_n)$ such that $\varepsilon_0 = 1$ and $\varepsilon_n = 0$ for $n \neq 0$.

Let U denote the unit circle in $\mathbb{C}$. If $x = (c_n)$ is an element of A , let $\varphi(x)$ be the continuous function on U whose value at e^{it} is

$$\varphi(x)(e^{it}) = \sum_{n \in \mathbb{Z}} c_n e^{int}.$$

One checks that φ is a morphism of $L_{\mathbb{C}}^1(\mathbb{Z})$ onto an algebra A of continuous functions on U , multiplication in A being the usual multiplication. Integrating the equality term by

term

$$\left(\sum_{m \in \mathbf{Z}} c_m e^{imt} \right) \cdot e^{-int} = \varphi(x)(e^{it}) \cdot e^{-int},$$

it comes

$$c_n = \frac{1}{2\pi} \int_0^1 \varphi(x)(e^{it}) e^{-int} dt.$$

In particular, it follows that the morphism φ is injective. The algebra A , endowed with the norm induced by that of $L^1_{\mathbf{C}}(\mathbf{Z})$ via φ , is called the Banach algebra of absolutely convergent Fourier series. It admits as identity element the function $1 = \varphi(\varepsilon)$.

- 9) Let Δ be the disk of complex numbers z satisfying $|z| \leq 1$. The algebra A of continuous functions on Δ that are analytic in the interior of Δ (VAR, R1, p. 26, 3.2.1) is equipped with the norm $\|f\| = \sup_{z \in \Delta} |f(z)|$. Then A is a commutative unital Banach algebra.

1.2.3 Spectral radius

Lemma 2 (Fekete's Lemma). — Let $(a_n)_{n \geq 1}$ be a sequence of real numbers. Suppose that

$$a_{n+m} \leq a_n + a_m$$

for all $n \geq 1$ and every $m \geq 1$. Then the sequence $(a_n/n)_{n \geq 1}$ converges and satisfies

$$\lim_{n \rightarrow +\infty} \frac{a_n}{n} = \inf_{n \geq 1} \frac{a_n}{n}.$$

Let us set $a_0 = 0$; the inequality $a_{n+m} \leq a_n + a_m$ remains valid for every $n \geq 0$ and every $m \geq 0$. Fix an integer $m \geq 1$. For every integer $n \geq 1$, let $q(n)$ and $r(n)$ be the integers such that $n = q(n)m + r(n)$ and $0 \leq r(n) < m$ (E, III, p. 39, th. 1). The hypothesis then implies

$$\frac{a_n}{n} \leq \frac{a_{q(n)m}}{n} + \frac{a_{r(n)}}{n} \leq \frac{q(n)a_m}{n} + \frac{a_{r(n)}}{n} \leq \frac{q(n)}{n} a_m + \frac{m}{n}.$$

Letting n tend to $+\infty$, one deduces that $\limsup_n (a_n/n) \leq a_m/m$ since $q(n)/n \rightarrow 1/m$.

Since this holds for every $m \geq 1$, we therefore have

$$\limsup_{n \rightarrow +\infty} \frac{a_n}{n} \leq \inf_{m \geq 1} \frac{a_m}{m} \leq \liminf_{n \rightarrow +\infty} \frac{a_n}{n}.$$

These inequalities prove the convergence of the sequence $(a_n/n)_{n \geq 1}$ as well as the formula $\lim a_n/n = \inf_{n \geq 1} a_n/n$.

PROPOSITION 1. — Let A be a normed algebra. For every $x \in A$ the sequence $(\|x^n\|^{1/n})_{n \geq 1}$ is convergent and its limit $\varrho(x)$ is equal to $\inf_{n \geq 1} \|x^n\|^{1/n}$. Moreover, for every norm $x \mapsto \|x\|_1$ defining the topology of A , one also has

$$\varrho(x) = \lim_{n \rightarrow +\infty} \|x^n\|_1^{1/n} = \inf_{n \geq 1} \|x^n\|_1^{1/n}.$$

If x is nilpotent, one has $\|x^n\|_1^{1/n} = 0$ for every integer n sufficiently large and every norm $x \mapsto \|x\|_1$ defining the topology of A .

Suppose now that x is not nilpotent, and let us set $\alpha_n = \|x^n\|$. We have $\alpha_n > 0$ for every integer $n \geq 1$, and $\alpha_{n+m} \leq \alpha_n \alpha_m$ for all $n, m \in \mathbb{N}$ by (1). Lemma 2, applied to the sequence $a_n = \log(\alpha_n)$ shows the existence of the limit $\varrho(x)$ and the formula $\varrho(x) = \inf_{n > 0} \alpha_n^{1/n}$.

Let $x \mapsto \|x\|_1$ be a norm defining the topology of A . There exist real numbers $a > 0$ and $b > 0$ such that

$$a\|x\| \leq \|x\|_1 \leq b\|x\|$$

for all $x \in A$ (EVT, II, p. 7, cor. 2). Consequently,

$$a^{1/n} \|x^n\|^{1/n} \leq \|x^n\|_1^{1/n} \leq b^{1/n} \|x^n\|^{1/n},$$

for all $n \geq 1$; whence, passing to the limit, or taking the infimum, the equality

$$\varrho(x) = \lim_{n \rightarrow +\infty} \|x^n\|_1^{1/n} = \inf_{n > 0} \|x^n\|_1^{1/n}.$$

DEFINITION 2. — For every element x of a normed algebra A , the real number

$$\varrho(x) = \lim_{n \rightarrow \infty} \|x^n\|^{1/n} = \inf_{n > 0} \|x^n\|^{1/n}$$

is called the spectral radius of x .

For every element x of A , we have

$$(3) \quad \varrho(x) \leq \|x\|$$

$$(4) \quad \varrho(x^n) = \varrho(x)^n, \text{ for every integer } n \geq 1.$$

DEFINITION 3. — An element x of A is quasi-nilpotent if $\varrho(x) = 0$.

This amounts to saying that, for any $\lambda \in \mathbf{K}$, the numbers $\|(\lambda x)^n\|$ are bounded for $n \geq 1$, or equivalently that, for any $\lambda \in \mathbf{K}$, the sequence $(\lambda x)^n$ tends to 0 as $n \rightarrow +\infty$.

Remark 1. — Let A be a normed algebra. If an element $x \in A$ satisfies $\varrho(x) = \|x\|$, one has $\|x^n\| = \|x\|^n$ for all $n \in \mathbf{N}$, by (3) and (4).

Conversely, suppose that $\|x^2\| = \|x\|^2$ for all $x \in A$. Then for every integer $n \geq 0$, we have the equality $\|x^{2^n}\| = \|x\|^{2^n}$, hence $\|x\| = \|x^{2^n}\|^{2^{-n}}$; as n tends to $+\infty$, we obtain $\|x\| = \varrho(x)$.

Remark 2. — The function $x \mapsto \varrho(x)$ on A , being the lower envelope of continuous functions $x \mapsto \|x^n\|^{1/n}$ for $n \geq 1$, is upper semicontinuous (TG, IV, p. 31, cor.), but in general it is not continuous. It can even happen (cf. exerc. 12 of I, p. 157) that a sequence of nilpotent elements of A tends to an element which is not quasi-nilpotent.

1.2.4 Inverses

Let A be a unital Banach algebra, whose identity element is denoted by 1. Recall (TG, IX, prop. 14, p. 40) that the group G of invertible elements of A is an open subset of A , that the topology induced on G by that of A is compatible with the group structure, and that the topological group G is complete.

PROPOSITION 2. — Let A be a Banach algebra and x an element of A . The series $\sum_{n=1}^{\infty} \lambda^n x^n$ considered as a power series in λ , has radius of convergence $\varrho(x)^{-1}$. If A is unital and if $\varrho(x) < 1$, then $1-x$ is invertible and has inverse $\sum_{n=0}^{\infty} x^n$.

The series $\sum_{n=1}^{\infty} \lambda^n x^n$ has radius of convergence

$$(\limsup_{n \rightarrow +\infty} \|x^n\|^{1/n})^{-1} = \varrho(x)^{-1}$$

(cf. VAR, R1, p. 23, 3.1.4). Suppose that A has a unit element. If $\varrho(x) < 1$, the series $\sum_{n=0}^{\infty} x^n$ is therefore absolutely convergent. Since

$$(1 - x) \left(\sum_{n=0}^k x^n \right) = \left(\sum_{n=0}^k x^n \right) (1 - x) = 1 - x^{k+1}$$

for every integer $k \geq 0$, the element $1 - x$ is invertible and its inverse is equal to $\sum_{n=0}^{\infty} x^n$.

COROLLARY 1. — If A is a unital Banach algebra, then the group of invertible elements of A contains the open ball with center 1 and radius 1.

This is immediate since $\|x\| < 1$ implies $\varrho(x) < 1$.

COROLLARY 2. — Let A be a Banach algebra and I a maximal regular left (resp. right) ideal of A . Then I is closed.

Let $(\tilde{A}, e)$ be the unital Banach algebra obtained from A by adjoining an identity element. There exists a maximal left (resp. right) ideal J of $\tilde{A}$ such that $J \cap A = I$ (A, VIII, p. 428, prop. 4). Then J is disjoint from the open ball with center e and radius 1 (cor. 1), and hence $\bar{J} \neq \tilde{A}$. Since J is a maximal ideal, this implies that $\bar{J} = J$, and consequently $I = J \cap A = \bar{J} \cap A$ is closed in A .

COROLLARY 3. — The radical of a Banach algebra is closed.

Indeed, the radical is the intersection of the regular maximal left ideals (A, VIII, p. 430, def. 3).

PROPOSITION 3. — Let A be a unital Banach algebra.

- a) If $x \in A$ admits a left inverse (resp. right inverse) y , every element $x' \in A$ such that $\|x' - x\| < \|y\|^{-1}$ admits a left inverse (resp. right inverse).
- b) The set of elements of A that are invertible (resp. left-invertible, resp. right-invertible) is open in A .
- c) Let (x_n) be a sequence of elements of A admitting left inverses (resp. right inverses) y_n , such that (x_n) converges to an element $x \in A$ and the sequence (y_n) is bounded. Then x

is left-invertible (resp. right-invertible).

It suffices to treat the case of left inverses; the case of right inverses follows by considering the opposite algebra.

Let $x, y, x' \in A$ such that $yx = 1$ and $\|x' - x\| < \|y\|^{-1}$. We have

$$\|1 - yx'\| = \|yx - yx'\| \leq \|y\| \cdot \|x - x'\| < 1,$$

hence yx' is invertible: there exists $z \in A$ such that $z(yx') = 1$. Thus the element x' is left-invertible with inverse zy . This proves assertion a), and assertion b) follows immediately.

Let (x_n) be a sequence of elements of A admitting left inverses y_n such that (x_n) converges to an element $x \in A$ and the sequence (y_n) is bounded. If $M \geq 1$ is a real number such that $\|y_n\| \leq M$ for all $n \geq 1$, then we have $\|x_n - x\| < M^{-1} \leq \|y_n\|^{-1}$ for n large enough, and hence x admits a left inverse by a).

DEFINITION 4. — Let A be a normed algebra, and let x be an element of A . Denote by γ_x and δ_x the maps $y \mapsto xy$ and $y \mapsto yx$ from A to A . We say that x is a left (resp. right) topological zero divisor if γ_x (resp. δ_x) is not a homeomorphism from A onto $\gamma_x(A)$ (resp. onto $\delta_x(A)$).

Remark. — By TG, IX, p. 36, cor. 2, x is a left (resp. right) topological zero divisor if and only if there exists a sequence (z_n) in A such that $\|z_n\| = 1$ and such that xz_n tends to 0 (resp. that z_nx tends to 0) when $n \rightarrow +\infty$.

A left (resp. right) topological zero divisor is a left (resp. right) zero divisor. Suppose that A is nonzero and unital. A left (resp. right) topological zero divisor x is not left (resp. right) invertible. Indeed, if for example $yx = 1$ and if xz_n tends toward 0, then $z_n = y(xz_n)$ tends to 0 and one cannot have $\|z_n\| = 1$ for all n .

PROPOSITION 4. — Let A be a unital Banach algebra. Let x be an element of A that is not left-invertible. If there exists a sequence (x_n) of left-invertible elements of A that converges to x , then x is a right topological zero divisor.

Let y_n be a left inverse of x_n . By prop. 3 (ii), $\|y_n\|$ tends to $+\infty$. Let $z_n = \|y_n\|^{-1}y_n$. We have $\|z_n\| = 1$, and $z_nx_n = \|y_n\|^{-1}$ tends to 0, hence $z_nx = z_nx_n + z_n(x - x_n)$ tends to 0. One then concludes with the aid of the remark following definition 4.

PROPOSITION 5. — Let A be a unital Banach algebra and let B be a full subalgebra of A .

Then $\overline{B}$ is a full subalgebra of A .

Indeed, let x be an element of $\overline{B}$ that is invertible in A , and let (x_n) be a sequence of elements of B tending to x . Then, for n sufficiently large, x_n is invertible in A and x_n^{-1} tends to x^{-1} . Since the subalgebra B is full, we have $x_n^{-1} \in B$, whence $x^{-1} \in \overline{B}$.

Let A be a unital Banach algebra and let $(y_i)_{i \in I}$ be a family of elements of A . Let B be the full subalgebra of A generated by the elements y_i . Then $\overline{B}$ is the smallest closed full subalgebra of A containing the y_i . It is called the closed full subalgebra generated by the elements y_i .

1.2.5 Spectrum of an element in a normed algebra

In this section, we assume that $K = \mathbb{C}$.

THEOREM 1. — Let A be a unital Banach algebra and let $x \in A$.

- a) The set $\text{Sp}_A(x)$ is a compact subset of $\mathbb{C}$;
- b) The spectral radius $\varrho(x)$ is the radius of the smallest closed disk with center 0 in $\mathbb{C}$ which contains $\text{Sp}_A(x)$;
- c) The resolvent $\lambda \mapsto R(x, \lambda) = (\lambda - x)^{-1}$ of x is holomorphic in $\mathbb{C} - \text{Sp}_A(x)$ and zero at infinity. Moreover, for every integer $k \geq 0$, we have the following formula:

$$\frac{\partial^k}{\partial \lambda^k} R(x, \lambda) = (-1)^k k! R(x, \lambda)^{k+1};$$

- d) For every complex number λ such that $|\lambda| > 1/\varrho(x)$, one has

$$R(x, \lambda) = \sum_{n=0}^{+\infty} \lambda^{-n-1} x^n.$$

The complement of the spectrum $\text{Sp}_A(x)$ is the inverse image of the group G of invertible elements of A under the continuous map $\lambda \mapsto x - \lambda$ from $\mathbb{C}$ to A ; by Proposition 3, b) of I, p. 23, the spectrum $\text{Sp}_A(x)$ is a closed subset of $\mathbb{C}$. Furthermore, let $\lambda \in \mathbb{C}$ such that $|\lambda| > \varrho(x)$. We have $\lambda - x = \lambda(1 - \lambda^{-1}x)$. Since $\varrho(\lambda^{-1}x) = |\lambda|^{-1}\varrho(x) < 1$, the element $1 - \lambda^{-1}x$, hence also the element $\lambda - x$, is invertible and

$$R(x, \lambda) = (\lambda - x)^{-1} = \sum_{n=0}^{\infty} \lambda^{-n-1} x^n \quad (5)$$

(I, p. 22, Prop. 2). In particular, $\lambda \notin \text{Sp}_A(x)$. This proves that $\text{Sp}_A(x)$ is contained in the disk with center 0 and radius $\varrho(x)$. Consequently, $\text{Sp}_A(x)$ is compact. This formula (5) also shows that the resolvent of x is defined and holomorphic in the complement of the closed disk $\Delta_{\varrho(x)}$ of center 0 and radius $\varrho(x)$ and tends to 0 at infinity.

Let $\lambda_0 \in \mathbb{C} - \text{Sp}_A(x)$. Set $y = \lambda_0 - x$. Let $\mu \in \mathbb{C}$ such that $|\lambda_0 - \mu| < \|y^{-1}\|^{-1}$. We have

$$\mu - x = y - (\lambda_0 - \mu) = y(1 - (\lambda_0 - \mu)y^{-1}),$$

hence $\mu - x$ is invertible and has as its inverse

$$(\mu - x)^{-1} = y^{-1} \sum_{n=0}^{\infty} (\lambda_0 - \mu)^n y^{-n} \quad (6)$$

By prop. 2 of I, p. 22. Hence the resolvent of x is defined and holomorphic in the open disk centered at λ_0 and of radius $\|y^{-1}\|^{-1}$. Consequently, the resolvent of x is a holomorphic map from $\mathbb{C} - \text{Sp}_A(x)$ into A .

Formula (1) of I, p. 4 implies $\frac{\partial}{\partial \lambda} R(x, \lambda) = -R(x, \lambda)^2$, whence, by induction on k ,

$$\frac{\partial^k}{\partial \lambda^k} R(x, \lambda) = (-1)^k k! R(x, \lambda)^{k+1}.$$

Let $a > 0$ be a real number such that $\text{Sp}_A(x)$ is contained in the closed disk Δ_a of center 0 and radius a . The function $\lambda \mapsto (\lambda^{-1} - x)^{-1}$ is then defined and holomorphic for $0 < |\lambda| < a^{-1}$ and tends to 0 when λ tends to 0. The unique continuous function on the open disk centered at 0 with radius a^{-1} which extends this holomorphic function is then holomorphic (VAR, R1, 3.3.9), hence the radius of convergence of the series (5) defining it is $\geq a^{-1}$ (VAR, R1, 3.2.9). By prop. 2 of I, p. 22, we therefore have $a \geq \varrho(x)$.

Remarks. — 1) The spectrum of an element in a unital Banach algebra can be any arbitrary nonempty compact subset F of $\mathbb{C}$ (cf. Example 3 of I, p. 17; for $A = C(F; \mathbb{C})$ and $f \in A$ the canonical inclusion of F into $\mathbb{C}$, one has $\text{Sp}_A(f) = F$).

2) Let A be a unital Banach algebra, and let $x \in A$. By Theorem 1 of I, p. 24, $\mathbf{C} - \text{Sp}_A(x)$ is an open subset of $\mathbf{C}$, hence is locally connected. Therefore the connected components of $\mathbf{C} - \text{Sp}_A(x)$ are open. By Thm. 1, one of these connected components contains the set of $\lambda \in \mathbf{C}$ such that $|\lambda| > \varrho(x)$; all the other connected components are therefore bounded.

COROLLARY 1. — Let A be a nonzero unital normed algebra. For every $x \in A$ the spectrum $\text{Sp}_A(x)$ is nonempty.

Suppose first that A is complete. If one had $\text{Sp}(x) = \emptyset$, the resolvent of x would be holomorphic in $\mathbf{C}$ and vanish at infinity, hence identically zero (VAR, R., 3.3.6, p. 29). Since $R(x, \lambda) = (\lambda - x)^{-1}$ is invertible, it would follow that $1=0$ and hence that $A = \{0\}$.

In the general case, let $\widehat{A}$ be the completed algebra of A ; the relation $\text{Sp}_A(x) = \emptyset$ would entail $\text{Sp}_{\widehat{A}}(x) = \emptyset$, whence $\widehat{A} = \{0\}$ and $A = \{0\}$.

COROLLARY 2 (Gelfand–Mazur theorem). — Let A be a normed algebra over $\mathbf{C}$. If A is a field, then $A = \mathbf{C} \cdot 1$.

If $x \in A$, there exists $\lambda \in \mathbf{C}$ such that $x - \lambda$ is non-invertible (cor. 1), hence $x - \lambda = 0$ and $x \in \mathbf{C} \cdot 1$.

COROLLARY 3. — Let A be a unital Banach algebra and let x be an invertible element of A such that $\|x\| = \|x^{-1}\| = 1$. Then $\text{Sp}(x) \subset \mathbf{U}$.

Let Δ be the disk with center 0 and radius 1 in $\mathbf{C}$. By th. 1 b) and the fact that $\varrho(x) \leq \|x\|$, one has $\text{Sp}(x) \subset \Delta$. Likewise, $\text{Sp}(x)^{-1} = \text{Sp}(x^{-1}) \subset \Delta$, whence the corollary (cf. I, p. 2, remark 4).

COROLLARY 4. — Let E be a complex Banach space, $\mathcal{L}(E)$ the Banach algebra of continuous endomorphisms of E and A a nonzero subalgebra of $\mathcal{L}(E)$ such that E is a simple A -pseudomodule (A, II, p. 176, Appendix).

- a) Let u be an endomorphism of E , not necessarily continuous, that commutes with A . Then u is a homothety.
- b) Let u be an endomorphism of E , not necessarily continuous. For every integer $n \geq 1$ and for every $(\xi_1, \dots, \xi_n) \in E^n$, there exists $v \in A$ such that

$$(v(\xi_1), \dots, v(\xi_n)) = (u(\xi_1), \dots, u(\xi_n)).$$

Let us show *a*). Let $\tilde{A}$ be the algebra obtained from A by adjoining an identity element. Since E is a simple A -pseudomodule, it is a simple $\tilde{A}$ -module.

Let B be the commutant of $\tilde{A}$ in the ring of endomorphisms of the C -vector space E . The algebra B contains 1 and is the algebra of endomorphisms of the $\tilde{A}$ -module E . Since E is a simple $\tilde{A}$ -module, Schur's lemma (*A*, VIII, p. 43, corollary) shows that B is a field.

Let $\xi_0 \in E$ be such that $A\xi_0 \neq \{0\}$. We therefore have $A\xi_0 = E$. For every $u \in B$, let A_u be the set of $v \in A$ such that $v(\xi_0) = u(\xi_0)$. This set is nonempty, since $A\xi_0 = E$. We then set

$$\|u\|_B = \inf_{v \in A_u} \|v\|.$$

The map $u \mapsto \|u\|_B$ is a seminorm on B .

Let us show that this map is a norm. Let u be a nonzero element of B . For every $v \in A_u$, one has $\|v\| \geq \|v(\xi_0)\|/\|\xi_0\| = \|u(\xi_0)\|/\|\xi_0\|$, so that $\|u\|_B \geq \|u(\xi_0)\|/\|\xi_0\|$. It therefore suffices to show that $u(\xi_0) \neq 0$. Let $\xi_1 \in E$ be such that $u(\xi_1) \neq 0$. Since $A\xi_0 = E$, there exists $w \in A$ such that $\xi_1 = w(\xi_0)$. Then, $wu(\xi_0) = uw(\xi_0) = u(\xi_1) \neq 0$, hence $u(\xi_0) \neq 0$.

On the other hand, let u and u' be elements of B . For every $\varepsilon > 0$, there exists $v, v' \in A$ such that $v(\xi_0) = u(\xi_0)$, $v'(\xi_0) = u'(\xi_0)$ and $\|v\| \leq \|u\|_B + \varepsilon$, $\|v'\| \leq \|u'\|_B + \varepsilon$. Then we have $vv'(\xi_0) = vu'(\xi_0) = u'v(\xi_0) = u'u(\xi_0)$, whence

$$\|u'u\|_B \leq \|v'v\| \leq \|v'\|\|v\| \leq (\|u\|_B + \varepsilon)(\|u'\|_B + \varepsilon),$$

and finally $\|u'u\|_B \leq \|u\|_B\|u'\|_B$. This shows that B , equipped with the norm $u \mapsto \|u\|_B$, is a normed algebra. Since it is a field, Corollary 2 implies that $B = C \cdot 1$, which is the desired conclusion.

We prove *b*). By *a*), the commutant of A in $\text{End}_C(E)$ is reduced to the homotheties of E . Its bicommutant is therefore $\text{End}_C(E)$. Assertion *b*) therefore follows from the Jacobson density theorem (Theorem 1 of *A*, VIII, p. 434).

COROLLARY 5. — Let A be a Banach algebra and $x \in A$.

- a) $\text{Sp}'(x)$ is a compact subset of $\mathbb{C}$;
- b) The spectral radius $\varrho(x)$ is the radius of the smallest closed disk centered at 0 in $\mathbb{C}$ that contains $\text{Sp}'(x)$;
- c) For x to be quasi-nilpotent, it is necessary and sufficient that one has $\text{Sp}'(x) = \{0\}$.

The assertions *a)* and *b)* follow from th. 1 by considering the Banach algebra deduced from A by adjoining an identity element. The assertion *c)* follows from *b)* .

1.2.6 Spectrum relative to a subalgebra

In this section, we assume that $K = \mathbb{C}$.

Lemma 3. — Let X_1 and X_2 be compact subsets of $\mathbb{C}$. If X_2 is contained in X_1 and if the boundary of X_1 in $\mathbb{C}$ is contained in X_2 , then X_1 is the union of X_2 and of certain bounded connected components of the complement of X_2 in $\mathbb{C}$.

Let U be a connected component of $\mathbb{C} - X_2$. Every boundary point of $X_1 \cap U$ in the open set U is also a boundary point of X_1 in $\mathbb{C}$, hence belongs to X_2 by hypothesis; since $U \cap X_2 = \emptyset$, we see that $X_1 \cap U$ has no boundary point in the space U . Since U is connected, the intersection $X_1 \cap U$ is either empty or equal to U (TG, I, p. 82, cor.), and the lemma follows.

PROPOSITION 6. — Let A be a unital Banach algebra and B a closed unital subalgebra of A . For every $x \in B$, one has $\text{Sp}_B(x) \supset \text{Sp}_A(x)$, and the boundary of $\text{Sp}_A(x)$ in $\mathbb{C}$ contains the boundary of $\text{Sp}_B(x)$ in $\mathbb{C}$. In particular, if $\text{Sp}_B(x) \subset \mathbb{R}$, then we have $\text{Sp}_B(x) = \text{Sp}_A(x)$.

We have $\text{Sp}_B(x) \supset \text{Sp}_A(x)$ (remark 6 of I, p. 3). If λ is a point on the boundary of $\text{Sp}_B(x)$ in $\mathbb{C}$, there exists a sequence (λ_n) of points exterior to $\text{Sp}_B(x)$ tending to λ . Then $x - \lambda_n$ is invertible in B and tends to $x - \lambda$, which is not invertible in B ; hence $x - \lambda$ is a left or right topological zero divisor in B (prop. 4 of I, p. 24), hence in A . Thus, $\lambda \in \text{Sp}_A(x)$. But since $\text{Sp}_A(x) \subset \text{Sp}_B(x)$, the complex number $\lambda \in \text{Fr}_{\mathbb{C}}(\text{Sp}_B(x))$ cannot be interior to $\text{Sp}_A(x)$, hence belongs to its boundary.

COROLLARY. — The set $\text{Sp}_B(x)$ is the union of $\text{Sp}_A(x)$ and of certain bounded connected components of $\mathbb{C} - \text{Sp}_A(x)$.

This follows from prop. 6 and lemma 3.

This corollary will be completed by propositions 13 of I, p. 46 and 14 of I, p. 46.

1.3 COMMUTATIVE BANACH ALGEBRAS

In this section, the base field is $\mathbb{C}$.

1.3.1 Characters of a commutative Banach algebra

THEOREM 1. — Let A be a Banach algebra and let $\chi : A \rightarrow \mathbb{C}$ be an algebra homomorphism (cf. I, p. 9). Then χ is continuous, with norm at most 1. If A is unital and if χ is a unital homomorphism, then χ has norm 1.

Let us prove that $|\chi(x)| \leq \|x\|$ for all $x \in A$. Replacing A , if necessary, by the Banach algebra generated by x , we may assume that A is commutative; then $\chi \in X'(A)$. For every $x \in A$, one has $\chi(x) \in \text{Sp}'_A(x)$ (I, p. 9, no. 7), hence $|\chi(x)| \leq \varrho(x) \leq \|x\|$ (I, p. 28, cor. 5), whence the first assertion.

If, moreover, A is unital, the equality $\chi(1) = 1$ implies that $\|\chi\| \geq 1$, whence the desired equality.

Remark. — Let A be a commutative Banach algebra. It follows from this theorem that the topology of pointwise convergence on $X'(A)$ coincides with the restriction to $X'(A)$ of the weak topology $\sigma(A', A)$ of the dual A' of A .

COROLLARY. — Let A be a commutative Banach algebra. The space $X'(A)$ is compact. The space $X(A)$ is locally compact, and is compact if A has a unit element.

Let A' be the dual of the Banach space A . Its unit ball A'_1 is weakly compact (EVT, III, p. 17, cor. 2). By th. 1, we have $X'(A) \subset A'_1$. Moreover, $X'(A)$ is closed in A' for the weak topology, since it is the intersection of the weakly closed sets

$$X_{x,y} = \{f \in A' \mid f(xy) - f(x)f(y) = 0\} \quad (\text{for } x, y \in A).$$

It follows that $X'(A)$ is compact and that $X(A) = X'(A) - \{0\}$ is locally compact.

If A has a unit element 1, then $X(A)$ is the set of $\chi \in X'(A)$ such that $\chi(1) = 1$, and is

therefore a closed subset, and hence compact, of $X'(A)$.

Every compact space is homeomorphic to $X(A)$ for a suitable commutative unital Banach algebra A (cf. I, p. 32, cor. 2).

THEOREM 2. — Let A be a commutative Banach algebra. The map $\chi \mapsto \text{Ker}(\chi)$ is a bijection from $X(A)$ onto the set $J(A)$ of the regular maximal ideals of A .

Let I be a regular maximal ideal of A . It is closed (I, p. 22, cor. 2), hence A/I is a Banach algebra. Since it is a field (A, VIII, p. 426, prop. 2), the Gelfand–Mazur theorem (I, p. 26, cor. 2) implies that A/I has dimension 1 over $\mathbb{C}$. Hence I has codimension 1 in A . The theorem then follows from lemma 3 of I, p. 10.

It may happen that a nonzero commutative Banach algebra A , without unit element, has no maximal ideal; then $X'(A)$ is reduced to $\{0\}$ (cf. I, p. 186, exercise 31).

This theorem shows that the sets $J(A)$ and $\hat{A}$ from I, p. 11, which are identified since A is commutative, can also be identified with the set $X(A)$. There is therefore reason to consider on $X(A)$ the weak topology and the Jacobson topology, which is coarser (I, p. 15, prop. 5). The closed sets $V(M)$ of the Jacobson topology, for $M \subset A$, are the sets

$$\{\chi \in X(A) \mid \chi(M) = 0\}$$

of $X(A)$, which will sometimes still be denoted $V(M)$. Similarly, for any subset M of $X(A)$, we shall denote by $\Upsilon(M)$ the ideal intersection of the kernels of the $\chi \in M$; it is equal to the set $\Upsilon(M')$ defined in I, p. 13 for the subset $M' \subset J(A)$ corresponding to M when one identifies $J(A)$ and $X(A)$.

When a topological notion is used in $X(A)$ without specifying which topology is meant, it will always be the weak topology.

1.3.2 Continuous functions vanishing at infinity on a locally compact space

In this section, X is a locally compact space. We denote by $\mathcal{C}_0(X)$ the commutative Banach algebra of continuous complex-valued functions vanishing at infinity on X , equipped with the

norm $\|f\| = \sup_{x \in X} |f(x)|$ (example 3 of I, p. 17).

PROPOSITION 1. — For every closed subset Φ of X , let I_Φ be the set of $f \in \mathcal{C}_0(X)$ vanishing on Φ . Then $\Phi \mapsto I_\Phi$ is a bijection from the set of closed subsets of X onto the set of closed ideals of $\mathcal{C}_0(X)$.

The set I_Φ is a closed ideal of $\mathcal{C}_0(X)$.

Let $\Phi \neq \Phi'$ be closed subsets of X . If necessary, by exchanging Φ and Φ' one may assume that there exists $x \in \Phi'$ such that $x \notin \Phi$; there then exists a function $f \in \mathcal{C}_0(X)$ vanishing on Φ and nonzero at x (TG, IX, p. 43, prop. 1). We have $f \in I_\Phi$ and $f \notin I_{\Phi'}$, so that the map $\Phi \mapsto I_\Phi$ is injective.

Let I be a closed ideal of $\mathcal{C}_0(X)$. Let Φ be the set of $x \in X$ such that $f(x) = 0$ for all $f \in I$; it is a closed subset of X , and we have $I \subset I_\Phi$. Let us prove that $I_\Phi \subset I$, which will imply that $I = I_\Phi$ and will finish the proof of the proposition.

Let $f \in I_\Phi$. For every real number $\varepsilon > 0$, let C_ε be the set of $x \in X$ such that $|f(x)| \geq \varepsilon$. Since f tends to 0 at infinity, the set C_ε is compact. Let $x \in C_\varepsilon$; since $f(x) \neq 0$ and $f \in I_\Phi$, one has $x \notin \Phi$; by definition of Φ , there then exists a function $\varphi_x \in I$ such that $|\varphi_x(x)| > 1$, hence such that $|\varphi_x(y)| > 1$ for all y belonging to a neighborhood V_x of x . The open sets $V_x \cap C_\varepsilon$ cover C_ε . Since the set C_ε is compact, there exists a finite subset $T_\varepsilon \subset X$ such that

$$C_\varepsilon \subset \bigcup_{x \in T_\varepsilon} V_x.$$

Then the element

$$g_\varepsilon = \frac{1}{\varepsilon} \sum_{x \in T_\varepsilon} \varphi_x \overline{\varphi_x} \geq 0$$

of $\mathcal{C}_0(X)$ belongs to I , and we have $g_\varepsilon \geq \varepsilon^{-1}$ on C_ε . The function

$$f_\varepsilon = \frac{f g_\varepsilon}{1 + g_\varepsilon}$$

belongs to I . For $x \notin C_\varepsilon$, one has

$$|f(x) - f_\varepsilon(x)| \leq 2\varepsilon,$$

and for $x \in C_\varepsilon$, one has

$$|f(x) - f_\varepsilon(x)| = \frac{|f(x)|}{1 + g_\varepsilon(x)} \leq \varepsilon |f(x)|.$$

Thus f_ε converges uniformly to f on X as ε tends to 0. We therefore have $f \in \bar{I}$, whence $f \in I$ since I is closed.

COROLLARY 1. — For every $x \in X$, let I_x be the set of $f \in \mathcal{C}_0(X)$ vanishing at x . Then $x \mapsto I_x$ is a bijection from X onto the set of maximal closed ideals of $\mathcal{C}_0(X)$. These ideals are regular.

This follows immediately from prop. 1.

Let X' denote the Alexandroff compactification of X , that is, the compact space obtained from X by adjoining a point at infinity ω_X (TG, I, pp. 67–68). The algebra $\mathcal{C}_0(X)$ is identified with the Banach algebra of continuous complex-valued functions on X' vanishing at ω_X .

For every $x \in X'$, we denote by ev_x the character of $\mathcal{C}_0(X)$ defined by $ev_x(f) = f(x)$ for all $f \in \mathcal{C}_0(X)$.

COROLLARY 2. — The map $x \mapsto ev_x$ is a homeomorphism from X' onto $X'(\mathcal{C}_0(X))$ and its restriction to X is a homeomorphism from X onto $X(\mathcal{C}_0(X))$. Moreover, the weak topology and the Jacobson topology coincide on $X(\mathcal{C}_0(X))$.

The map $ev: x \mapsto ev_x$ of X' into $X'(\mathcal{C}_0(X))$ is injective. It is surjective by Cor. 1 and Th. 2 of I, p. 30. It is continuous, because for every function $f \in \mathcal{C}_0(X)$ and every open set U of $\mathbb{R}$, we have

$$ev^{-1}(\{\chi \in X'(\mathcal{C}_0(X)) \mid \chi(f) \in U\}) = f^{-1}(U)$$

which is open in X . The map ev is therefore a homeomorphism since X' is compact. The restriction of ev to X is then a homeomorphism onto $X(\mathcal{C}_0(X))$.

If F is a weakly closed subset of $X(\mathcal{C}_0(X))$ it corresponds, via the homeomorphism ev , to a closed subset Φ of X ; precisely, by Prop. 1, we have $F = \{\chi \in X(\mathcal{C}_0(X)) \mid I_\Phi \subset \text{Ker } \chi\}$ which is closed for the Jacobson topology.

COROLLARY 3. — Suppose X is compact. Then the map $x \mapsto ev_x$ is a homeomorphism

from X onto $X(\mathcal{C}(\mathbb{X}))$. The weak topology and the Jacobson topology coincide on $X(\mathcal{C}(\mathbb{X}))$.

1.3.3 Proper partial maps

In this section, X and Y are locally compact topological spaces. We denote by X' (resp. Y') the compact space obtained from X (resp. Y) by adjoining a point at infinity ω_X (resp. ω_Y) (TG, I, p. 67–68). We identify X' and Y' with $X'(\mathcal{C}_0(X))$ and $X'(\mathcal{C}_0(Y))$ respectively (Corollary 2 of I, p. 32).

DEFINITION 1. — A proper partial map from X to Y is a correspondence $f = (\Gamma, X, Y)$ (E, II, p. 10, def. 2) between X and Y such that

- (i) The graph Γ is functional;
- (ii) The domain of definition of f is an open set U of X ;
- (iii) The map $x \mapsto f(x)$ from U to Y is proper.

The identity map on X is a proper partial map from X into X . Let Z be a locally compact topological space and f (resp. g) a proper partial map from X to Y (resp. from Y to Z). Then the composite correspondence $g \circ f$ (E, II, p. 11, def. 6) is a proper partial map from X into Z (TG, I, p. 72, prop. 3, and p. 73, prop. 5).

Lemma 1. — For every proper partial map f from X to Y , with domain of definition U , denote by $\tilde{f}$ the mapping from X' to Y' defined by $\tilde{f}(x) = f(x)$ if $x \in U$ and $\tilde{f}(x) = \omega_Y$ if $x \notin U$; it is continuous.

The map $f \mapsto \tilde{f}$ is a bijection between the set of proper partial maps f from X to Y and the set of continuous maps g from X' to Y' such that $g(\omega_X) = \omega_Y$.

Let f a proper partial map from X to Y , and let U be its domain. Let us prove that the map $\tilde{f}$ is continuous. It is continuous at every point of U , since U is open in X' . Let us prove that it is also continuous at every point x of $X' - U$; we then have $\tilde{f}(x) = \omega_Y$. Let V be an open neighborhood of ω_Y into Y' ; let us prove that $\tilde{f}^{-1}(V)$ is a neighborhood of x . By definition of the topological space Y' , one may assume that V is of the form $Y' - K$ where K is a compact subset of Y . Since f defines a proper map from U to Y , the set $f^{-1}(K)$ is compact in U (TG, I,

p. 77, prop. 6), hence in X' . This is in particular a closed subset of X' and $\tilde{f}^{-1}(V) = X' - f^{-1}(K)$ is an open subset of X' and is therefore a neighborhood of x .

Conversely, let $g: X' \rightarrow Y'$ be a continuous map such that $g(\omega_X) = \omega_Y$ and $\Gamma_g \subset X' \times Y'$ its graph. The set $U = X - \bar{g}^{-1}(\omega_Y)$ is open in X . The correspondence $f = (\Gamma_g \cap (U \times Y), X, Y)$ is a proper partial map from X into Y such that $\tilde{f} = g$, and it is the only one.

We will identify the proper partial maps from X to Y with the continuous maps from X' into Y' that map ω_X to ω_Y . In particular, the proper maps from X to Y are the proper partial maps with domain X ; they are identified with the continuous maps f from X' into Y' such that $f^{-1}(\omega_Y) = \{\omega_X\}$. If X is compact, these are simply the continuous maps from X to Y .

Let A be a commutative complex Banach algebra. Recall that $X'(A)$ is identified with the compact space obtained from $X(A)$ by adjoining a point at infinity (I, p. 29, corollary).

PROPOSITION 2. — Let A and B be commutative complex Banach algebras. For every algebra homomorphism $\pi: A \rightarrow B$, the map $X'(\pi)$ is a proper partial map from $X(B)$ into $X(A)$.

Indeed, $X'(\pi)$ is a continuous map from $X'(B)$ into $X'(A)$ (I, p. 10). The point at infinity of $X'(B)$ (resp. of $X'(A)$) is the zero character, and we have $X'(\pi)(0) = 0$.

PROPOSITION 3. — a) For every proper partial map φ from X to Y , the map $f \mapsto f \circ \varphi$ of $\mathcal{C}(Y')$ into $\mathcal{C}(X')$ induces an algebra morphism φ^* of $\mathcal{C}_0(Y)$ into $\mathcal{C}_0(X)$;

b) The map $\varphi \mapsto \varphi^*$ is a bijection from the set of proper partial maps from X to Y onto the set of algebra morphisms from $\mathcal{C}_0(Y)$ into $\mathcal{C}_0(X)$. Its inverse bijection is the map $\pi \mapsto X'(\pi)$.

Let us prove a). Let φ be a proper partial map from X to Y , identified with a continuous map from X' to Y' such that $\varphi(\omega_X) = \omega_Y$. For $f \in \mathcal{C}_0(Y)$, one has $(f \circ \varphi)(\omega_X) = f(\omega_Y) = 0$, hence the map φ^* is well defined. It is an algebra morphism.

Let us prove that $X'(\varphi^*)$ is identified with φ . Let $x \in X$. For every function $f \in \mathcal{C}_0(Y)$, the character $X'(\varphi^*)(\text{ev}_x)$ associates to f the complex number

$$(\text{ev}_x \circ \varphi^*)(f) = \text{ev}_x(f \circ \varphi) = f(\varphi(x)),$$

hence $X'(\varphi^*)(\text{ev}_x) = \text{ev}_{\varphi(x)}$. This proves the assertion.

Conversely, let $\pi: \mathcal{C}_0(Y) \rightarrow \mathcal{C}_0(X)$ be an algebra morphism. Let us prove that $X'(\pi)^* = \pi$. Let $f \in \mathcal{C}_0(Y)$, and denote $g = X'(\pi)^*(f) \in \mathcal{C}_0(X)$. For every $x \in X$, one has

$$g(x) = (f \circ X'(\pi))(x) = \text{ev}_{X'(\pi)(x)}(f) = (\text{ev}_x \circ \pi)(f) = \pi(f)(x),$$

since $X'(\pi)$ satisfies $\text{ev}_{X'(\pi)(x)} = \text{ev}_x \circ \pi$. We therefore have $g = \pi(f)$, which allows us to conclude that $X'(\pi)^* = \pi$.

In a completely similar way, we have:

PROPOSITION 4. — Suppose that X and Y are compact. Identify the space X (resp. the space Y) with $X(\mathcal{C}(X))$ (resp. $X(\mathcal{C}(Y))$) (corollary 3 of I, p. 33).

- a) For every continuous map $\varphi: X \rightarrow Y$, the map $\varphi^*: f \mapsto f \circ \varphi$ is an algebra morphism from $\mathcal{C}(Y)$ into $\mathcal{C}(X)$;
- b) The maps $\varphi \mapsto \varphi^*$ and $\pi \mapsto X(\pi)$ are reciprocal bijections between the set of continuous maps from X into Y and the set of algebra morphisms from $\mathcal{C}(Y)$ into $\mathcal{C}(X)$.

Remark. — *In the language of category theory, the preceding results are interpreted as follows. Let G be the category whose objects are locally compact topological spaces and whose morphisms are proper partial maps. The functor $X \mapsto \mathcal{C}_0(X)$ is a contravariant, fully faithful functor from the category G to the category of commutative complex Banach algebras. Moreover, $A \mapsto X(A)$ is a contravariant functor from the category of commutative complex Banach algebras to the category G . If to a locally compact topological space X one associates the homeomorphism

$$\text{ev}: X \rightarrow X(\mathcal{C}_0(X)),$$

one obtains an isomorphism from the identity functor of the category G to the composite functor $X \mapsto X(\mathcal{C}_0(X))$.

It is not true that the composite functor $A \mapsto \mathcal{C}_0(X(A))$ is isomorphic to the identity functor of the category of commutative complex Banach algebras (cf. example 2 of I, p. 36 and exercise

2 of I, p. 155). One will nevertheless see a statement of this type for commutative stellar algebras (number 5 of I, p. 107).*

1.3.4 Gelfand transformation

Let A be a commutative Banach algebra. Recall that, for every $x \in A$, we denote by $\mathcal{G}_A(x)$, or $\mathcal{G}(x)$, the function $\chi \mapsto \chi(x)$ on $X(A)$, that $\mathcal{G}(x)$ is called the Gelfand transform of x , and that the map $x \mapsto \mathcal{G}(x)$ is called the Gelfand transformation (cf. def. 5 of I, p. 7). One therefore has by definition:

$$\mathcal{G}(x)(\chi) = \chi(x).$$

Examples. — 1) Let X be a locally compact topological space and consider the commutative Banach algebra $\mathcal{C}_0(X)$ (example 3 of I, p. 17 and number 2 of I, p. 31). By cor. 1 of I, p. 32, the space of characters $X(\mathcal{C}_0(X))$ is identified with X by means of the map associating to an element $x \in X$ the character $f \mapsto f(x)$ of $\mathcal{C}_0(X)$, and the Gelfand transformation of $\mathcal{C}_0(X)$ is then identified with the identity map.

2) Let $n \geq 0$ be an integer. Let A_n be the algebra of functions $f : [0, 1] \rightarrow K$ admitting continuous derivatives in $[0, 1]$ up to order n . Equipped with the norm

$$\|f\| = \sum_{k=0}^n \frac{1}{k!} \sup_{0 \leq t \leq 1} |f^{(k)}(t)|,$$

it is a Banach algebra (example 4 of I, p. 18). For every n , the space of characters $X(A_n)$ is identified with $[0, 1]$ and G under inclusion A_n into $\mathcal{C}([0, 1])$ (cf. example 1 of I, p. 144).

3) Let Δ be the disk of complex numbers z satisfying $|z| \leq 1$ and let A be the complex Banach algebra of continuous functions on Δ that are analytic in the interior of Δ , equipped with the norm $\|f\| = \sup_{z \in \Delta} |f(z)|$ (example 9 of I, p. 20). Then $X(A)$ is identified with Δ and $\mathcal{G}$ with the inclusion of A into $\mathcal{C}(\Delta)$ (cf. exerc. 6 of I, p. 193).

4) Consider the complex Banach algebra A of absolutely convergent Fourier series (example 8 of I, p. 19). For every element u of the unit circle U , the map $f \mapsto f(u)$ is a character ev_u

of A . If $f_0 \in A$ is the identity map of U , one has $ev_u(f_0) = u$, hence the map $ev: u \mapsto ev_u$ from U into $X(A)$ is injective; it is continuous. Let $\chi \in X(A)$. We have $\|f_0\| = \|f_0^{-1}\| = 1$, hence $|\chi(f_0)| \leq 1$ and $|\chi(f_0)^{-1}| \leq 1$. This shows that $\chi(f_0) \in U$ and there exists $u \in U$ such that $\chi(f_0) = ev_u(f_0)$. Since $\{f_0, f_0^{-1}\}$ topologically generates the unital algebra A , one has $\chi = ev_u$. Thus, the map ev is a homeomorphism from U onto $X(A)$, by which one identifies these spaces. The Gelfand transformation of A is then identified with the inclusion of A into $\mathcal{C}(U)$.

Since A is isomorphic to the Banach algebra $L^1(\mathbf{Z})$ (example 8 of I, p. 19), the space $X(L^1(\mathbf{Z}))$ is identified with U and, for every element $(c_n) \in L^1(\mathbf{Z})$, the Gelfand transform $\mathcal{G}_{L^1(\mathbf{Z})}((c_n))$ is identified with the function $u \mapsto \sum_{n \in \mathbf{Z}} c_n u^n$ on U .

- 5) Let Δ the unit disk of complex numbers z such that $|z| \leq 1$. Its boundary in $\mathbf{C}$ is U . Let A be the Banach algebra of complex functions f on U such that there exists a continuous function $\tilde{f} \in \mathcal{C}(\Delta)$ extending f that is analytic in $\mathring{\Delta}$, equipped with the norm $\|f\| = \sup_{z \in U} |f(z)|$. By virtue of the maximum principle (VAR, R1, p. 30, 3.3.7), one then has $\|f\| = \sup_{z \in \Delta} |\tilde{f}(z)|$, and hence A coincides with the algebra of example 9 of I, p. 20. The set $X(A)$ is identified with Δ and, if $f \in A$, the map $\mathcal{G}(f)$ is identified with the continuous extension of f into Δ that is analytic in $\mathring{\Delta}$ (cf. exerc. 6 of I, p. 193).

PROPOSITION 5. — Let A be a commutative Banach algebra. For every $x \in A$, the function $\mathcal{G}(x)$ belongs to the commutative Banach algebra $\mathcal{C}_0(X(A))$ of continuous functions on $X(A)$ vanishing at infinity.

By definition (cf. no. 7 of I, p. 9), the function $\mathcal{G}'_A(x): \chi \mapsto \chi(x)$ is continuous on $X'(A)$ and vanishes at 0. Since $X'(A)$ is identified with the Alexandroff compactification of $X(A)$ according to cor. 1 of I, p. 29, the proposition follows.

PROPOSITION 6. — Let A be a commutative Banach algebra and let $x \in A$.

- a) The union of the set of values of $\mathcal{G}(x)$ and of $\{0\}$ is equal to $\text{Sp}'_A(x)$;
- b) If A has a unit element, the set of values of $\mathcal{G}(x)$ is $\text{Sp}_A(x)$. In particular, for x to be invertible, it is necessary and sufficient that $\mathcal{G}(x)$ does not vanish.

Suppose that A has a unity element. We know that, for every $\chi \in X(A)$, one has $\chi(x) \in \text{Sp}_A(x)$. Conversely, let $\lambda \in \text{Sp}_A(x)$. Then $x - \lambda$ is not invertible, hence belongs to a maximal ideal of A . There then exists $\chi \in X(A)$ such that $\chi(x - \lambda) = 0$ (th. 2 of I, p. 30), whence b).

Moving to the general case. Let $\tilde{A}$ the Banach algebra obtained from A by adjoining an identity element; it is commutative. The set $\text{Sp}'_{\tilde{A}}(x)$ is equal to $\text{Sp}_{\tilde{A}}(x)$, that is, the set of values of $\mathcal{G}_{\tilde{A}}(x)$ on $X(\tilde{A}) = X'(A)$. Hence a).

Example. — Consider the Banach algebra A of absolutely convergent Fourier series (example 4). Prop. 6, b) implies that if φ is a function on the unit circle U admitting an absolutely convergent Fourier series, and if φ does not vanish, the function $1/\varphi$ also admits an absolutely convergent Fourier series (“Wiener’s theorem”).

PROPOSITION 7. — Let A be a commutative Banach algebra.

a) The Gelfand transformation $\mathcal{G}$ defines a morphism from A to $\mathcal{C}_0(X(A))$ such that

$$\|\mathcal{G}(x)\| = \varrho(x) \leq \|x\| \text{ for all } x \in A ;$$

b) For the Gelfand transform to $\mathcal{G}$ it is isometric, it is necessary and sufficient that $\|x^2\| = \|x\|^2$ for all $x \in A$.

The map $\mathcal{G}$ is a morphism from A to $\mathcal{C}_0(X(A))$ by no. 7 of I, p. 9 and prop. 5 of I, p. 37, and verifies $\|\mathcal{G}(x)\| = \varrho(x)$ by prop. 6 and cor. 5 of I, p. 28. The assertion b) follows from a) and of Remark 1 of I, p. 21.

COROLLARY. — Let A be a Banach algebra, and let x and y be commuting elements of A .

a) We have $\varrho(xy) \leq \varrho(x)\varrho(y)$ and $\varrho(x + y) \leq \varrho(x) + \varrho(y)$;

b) If y is quasi-nilpotent, then $\text{Sp}'_A(x) = \text{Sp}'_A(x + y)$; if moreover A is unital, then $\text{Sp}_A(x) = \text{Sp}_A(x + y)$.

By considering the Banach algebra obtained from A by adjoining an identity element, one first reduces to the case where the algebra A is unital. Then, by considering the closed full subalgebra of A generated by x and y , one reduces to the case where A is commutative and unital. Assertion (a) is then a consequence of Prop. 7(a), and assertion (b) follows from Prop. 6 of I, p. 37 and Cor. 5 of I, p. 28.

PROPOSITION 8. — Let A be a commutative Banach algebra. The following four sets are equal:

- (i) The kernel of the Gelfand transformation;
- (ii) The set of elements x of A such that $\text{Sp}'_A(x) = \{0\}$;
- (iii) The set of quasi-nilpotent elements of A ;
- (iv) The radical of A .

We denote N_1, N_2, N_3, N_4 , respectively, these sets. We have $N_1 = N_2$ (prop. 6, a)), and $N_2 = N_3$ (I, p. 28, cor. 5). By definition, the set N_4 is the intersection of the regular maximal ideals of A ; it is therefore the intersection of the kernels of the characters of A (th. 2 of I, p. 30), which is equal to N_1 .

Remarks. — 1) In general, the image of the Gelfand transformation is neither closed in $\mathcal{C}_0(X(A))$, nor dense in $\mathcal{C}_0(X(A))$ (exerc. 7 of I, p. 193).

- 2) The image of the Gelfand transformation separates the points of $X(A)$, since if $\chi_1 \neq \chi_2$ are elements of $X(A)$, there exists $x \in A$ such that $\chi_1(x) \neq \chi_2(x)$.
- 3) If $\chi \in X(A)$, there exists an element of the image of the Gelfand transformation which does not vanish at χ .
- 4) If A has an identity element, the image of the Gelfand transformation is a full subalgebra of the algebra of continuous functions on $X(A)$ (prop. 6, b)).

Lemma 2. — Let A be a commutative Banach algebra. Let M be a subset of $X(A)$. Then M is closed for the Jacobson topology if and only if, for every character $\chi \in X(A) \setminus M$, there exists an element x of A such that $\mathcal{G}(x)$ vanishes on M and is nonzero at χ .

Let $\Upsilon(M)$ the intersection of the kernels of the elements of M . The set M is closed for the Jacobson topology if and only if $M = V(\Upsilon(M))$ (cf. I, p. 13 and I, p. 30). This condition is equivalent to saying that the elements χ of M are precisely the characters which vanish on $\Upsilon(M)$. Consequently M is closed if and only if for every character $\chi \notin M$, there exists $x \in \Upsilon(M)$ such that $\chi(x) \neq 0$. This translates into $\mathcal{G}(x)(\chi) \neq 0$ and $\mathcal{G}(x)|_M = 0$.

1.3.5 Morphisms of commutative Banach algebras

PROPOSITION 9. — Let A be a Banach algebra, and let B be a commutative semisimple Banach algebra. Every morphism from the underlying algebra of A into the underlying algebra of B is continuous.

Let $h: A \rightarrow B$ an algebra homomorphism and let $(a, b) \in A \times B$ a point adherent to the graph Γ of h . Let $\chi \in X'(B)$. The function $x \mapsto \chi(h(x))$ of A into $\mathbb{C}$ is an algebra homomorphism, hence is continuous (I, p. 29, th. 1). The map from $A \times B$ into $\mathbb{C}$ given by $(x, y) \mapsto \chi(h(x)) - \chi(y)$ is then continuous, it vanishes on Γ therefore zero at (a, b) . We thus have $\chi(h(a)) = \chi(b)$ for all $\chi \in X'(B)$. Since B is semisimple, we have $h(a) = b$. Thus the graph of h is closed and therefore h is continuous (EVT, I, p. 19, cor. 5).

COROLLARY. — On a commutative complex algebra without radical, two norms defining Banach algebra structures are equivalent.

It suffices to apply Prop. 9 to the identity map of the algebra.

Let A and B be commutative Banach algebras. By No. 7 of I, p. 9, if $h: A \rightarrow B$ is a surjective morphism, $X'(h)$ is a homeomorphism from $X'(B)$ on a closed subspace of $X'(A)$ which maps 0 to 0. (In the case where h is the injection of a subalgebra into $\mathcal{C}(X)$, the map $X'(h)$ is injective under much weaker hypotheses, cf. I, p. 142, prop. 1, d).)

Now let $h: A \rightarrow B$ an injective morphism. In general, $X'(h)$ is not surjective, but the following proposition provides a necessary condition for this to be the case.

PROPOSITION 10. — Let A and B be unital commutative Banach algebras, $h: A \rightarrow B$ a unital algebra homomorphism, not necessarily continuous. If $X(h)$ is surjective, then $h(A)$ is a full subalgebra of B .

Let $x \in A$ such that $h(x)$ be invertible in B . For every $\chi \in X(A)$, there exists $\xi \in X(B)$ such that $\chi = X(h)(\xi)$, hence $\chi(x) = \xi(h(x)) \neq 0$. Prop. 6 of I, p. 37 then shows that x is invertible in A , and therefore that $h(x)$ is invertible in $h(A)$.

The necessary condition of the proposition is not sufficient, even if h is isometric (I, p. 168, exerc. 14). We nevertheless have the following result:

PROPOSITION 11. — Let A and B be commutative unital Banach algebras, let a be an element of A and let $h: A \rightarrow B$ be an injective unital morphism (not necessarily continuous).

Suppose that the closed full subalgebra of A generated by a is equal to A . The following conditions are equivalent:

- (i) $X(h)$ is surjective;
- (ii) $h(A)$ is a full subalgebra of B ;
- (iii) $\text{Sp}_A(a) = \text{Sp}_B(h(a))$.
- (iv) $\Rightarrow$ (ii) follows from prop. 10.
- (v) $\Rightarrow$ (iii) follows from the formula $\text{Sp}_A(a) = \text{Sp}_{h(A)}(h(a)) = \text{Sp}_B(h(a))$, valid since $h(A)$ is a full subalgebra of B .
- (vi) $\Rightarrow$ (i) by formula (4) of no. 6 of I, p. 6, we have the commutative diagram

$$\begin{array}{ccc} X(B) & \xrightarrow{X(h)} & X(A) \\ \downarrow \mathcal{G}_B(h(a)) & & \downarrow \mathcal{G}_A(a) \\ \text{Sp}_B(h(a)) & \xrightarrow{i} & \text{Sp}_A(a) \end{array}$$

where the vertical arrows denote surjective maps (I, p. 37, prop. 6) and i is the canonical inclusion. Hypothesis (iii) means that i is bijective. Moreover, the surjective map $\mathcal{G}_A(a): X(A) \rightarrow \text{Sp}_A(a)$ is bijective: indeed, for every characters χ_1 and χ_2 of A , the set $A_{\chi_1, \chi_2} = \{x \in A \mid \chi_1(x) = \chi_2(x)\}$ is a closed full subalgebra of A . By hypothesis, we therefore have $A_{\chi_1, \chi_2} = A$ if $a \in A_{\chi_1, \chi_2}$, that is, $\chi_1 = \chi_2$ if $\mathcal{G}_A(a)\chi_1 = \mathcal{G}_A(a)\chi_2$. The diagram then implies that the map $X(h)$ is surjective.

1.3.6 Simultaneous spectrum

Let Λ a set. Let $C_\Lambda = \mathbf{C}[(X_\lambda)_{\lambda \in \Lambda}]$ be the unital algebra of complex polynomials in a family of indeterminates $(X_\lambda)_{\lambda \in \Lambda}$. For every $\chi \in X(C_\Lambda)$, one has $(\chi(X_\lambda))_{\lambda \in \Lambda} \in C^\Lambda$; the map $\chi \mapsto (\chi(X_\lambda))_{\lambda \in \Lambda}$ is a homeomorphism from $X(C_\Lambda)$ onto the product space C^Λ , by which these spaces are identified.

On the other hand, let A be a commutative unital Banach algebra and let $x = (x_\lambda)_{\lambda \in \Lambda}$ be

a family of elements of A . There exists a unique unital morphism h from C_Λ to A such that $h(X_\lambda) = x_\lambda$ for all λ . The continuous map $X(h)$ of $X(A)$ into C^Λ is the map which to χ associates the family $(\chi(x_\lambda))_{\lambda \in \Lambda}$. It is called the map of $X(A)$ into C^Λ defined by x .

DEFINITION 2. — The image of the map $X(h)$ is called the simultaneous spectrum of x , and is denoted $\text{Sp}_A^\Lambda(x)$ or $\text{Sp}^\Lambda(x)$.

The simultaneous spectrum of x is a compact subset of C^Λ . An element $c = (c_\lambda) \in C^\Lambda$ belongs to $\text{Sp}_A^\Lambda(x)$ if and only if the elements $x_\lambda - c_\lambda$ belong to a same maximal ideal of A , in other words if the family $(x_\lambda - c_\lambda)_{\lambda \in \Lambda}$ does not generate the algebra A .

If Λ contains a single element, so that the family x reduces to a single element $x \in A$, one has $\text{Sp}_A^\Lambda(x) = \text{Sp}_A(x)$ (I, p. 37, prop. 6, b)). If $\Lambda' \subset \Lambda$, then $\text{Sp}_A^{\Lambda'}((x_\lambda)_{\lambda \in \Lambda'})$ is the image of $\text{Sp}_A^\Lambda((x_\lambda)_{\lambda \in \Lambda})$ by the canonical projection map of C^Λ on $C^{\Lambda'}$. In particular, we have

$$\text{Sp}_A^\Lambda(x) \subset \prod_{\lambda \in \Lambda} \text{Sp}_A(x_\lambda).$$

We denote z_λ , for $\lambda \in \Lambda$, the coordinate functions on C^Λ . If $\chi \in X(A)$, the value at χ of $z_\lambda \circ X(h)$ is $\chi(x_\lambda)$, hence $z_\lambda \circ X(h) = \mathcal{G}(x_\lambda)$.

Let A and B be commutative unital Banach algebras, φ a unital morphism from A to B , and $x = (x_\lambda)_{\lambda \in \Lambda}$ a family of elements of A . Let $\varphi(x)$ the family $(\varphi(x_\lambda))_{\lambda \in \Lambda}$ of elements of B . We have, for every $\chi \in X(B)$, and every $\lambda \in \Lambda$

$$\chi(\varphi(x_\lambda)) = (X(\varphi)(\chi))(x_\lambda),$$

hence $\text{Sp}_B^\Lambda(\varphi(x)) \subset \text{Sp}_A^\Lambda(x)$. The diagram

$$\begin{array}{ccc} X(B) & \xrightarrow{X(\varphi)} & X(A) \\ \downarrow & & \downarrow \\ \text{Sp}_B^\Lambda(\varphi(x)) & \xrightarrow{i} & \text{Sp}_A^\Lambda(x) \end{array} \quad (1)$$

where i denotes the inclusion, and the vertical arrows denote the maps defined by the families $\varphi(x)$ and x , is therefore commutative.

Example. — Let $K \subset C^\Lambda$ a compact subset. Let $z = (z_\lambda)_{\lambda \in \Lambda}$ the family in $\mathcal{C}(K)$ of the restrictions to K of the coordinate functions of C^Λ . Then the simultaneous spectrum $\text{Sp}_{\mathcal{C}(K)}^\Lambda(z)$

is equal to K . Indeed, by cor. 2 of I, p. 32, every character χ of $\mathcal{C}(K)$ is of the form $f \mapsto f(x)$ for an element $x \in K$, and one then has $(\chi(z_\lambda))_{\lambda \in \Lambda} = x$.

PROPOSITION 12. — Let Λ a set and A a commutative unital Banach algebra. Let $x = (x_\lambda)_{\lambda \in \Lambda}$ be a family of elements of A .

- a) We assume that the full subalgebra of A generated by the family x is dense in A . The map from $X(A)$ to $\mathbf{C}^\Lambda$ defined by x is a homeomorphism from $X(A)$ onto the simultaneous spectrum $\text{Sp}_A^\Lambda(x)$;
- b) We assume that the unital subalgebra of A generated by the family x is dense in A . For every $c \in \mathbf{C}^\Lambda$, the following conditions are equivalent:
 - (i) $c \in \text{Sp}_A^\Lambda(x)$;
 - (ii) $|P(c)| \leq \varrho(P(x))$ for every polynomial $P \in \mathbf{C}[(X_\lambda)_{\lambda \in \Lambda}]$;
 - (iii) $|P(c)| \leq \|P(x)\|$ for every polynomial $P \in \mathbf{C}[(X_\lambda)_{\lambda \in \Lambda}]$.
- a) The map from $X(A)$ into $\text{Sp}_A^\Lambda(x)$ defined by the family x is continuous and surjective. Let $\chi, \chi' \in X(A)$ be characters having the same image, that is, such that $\chi(x_\lambda) = \chi'(x_\lambda)$ for all $\lambda \in \Lambda$. The characters χ and χ' coincide on elements of the form $P(x)Q(x)^{-1}$, where $P \in \mathbf{C}[(X_\lambda)]$, $Q \in \mathbf{C}[(X_\lambda)]$ and $Q(x)$ is invertible in A , that is, on the full subalgebra of A generated by the elements x_λ (lemma 2 of I, p. 6). Since χ and χ' are continuous (th. 1 of I, p. 29), they are therefore equal on A . This proves that $X(h)$ is a continuous bijection from $X(A)$ on $\text{Sp}_A^\Lambda(x)$, and consequently it is a homeomorphism since $X(A)$ is compact.
- b) Let us show that (i) implies (ii): if $c = (c_\lambda)_{\lambda \in \Lambda} \in \text{Sp}_A^\Lambda(x)$, there exists $\chi \in X(A)$ such that $c_\lambda = \chi(x_\lambda)$ for all λ . For every $P \in \mathbf{C}[(X_\lambda)]$, we therefore have $|P(c)| = |P((\chi(x_\lambda))_{\lambda \in \Lambda})| = |\chi(P(x))| \leq \varrho(P(x))$.

Assertion (ii) implies (iii) by virtue of the inequality $\varrho(x) \leq \|x\|$, valid for all $x \in A$ by definition of the spectral radius.

Let us finally show that (iii) implies (i). Let $c = (c_\lambda)_{\lambda \in \Lambda} \in \mathbf{C}^\Lambda$ such that

$$|P(c)| \leq \|P(x)\| \quad (2)$$

for all $P \in \mathbf{C}[(X_\lambda)]$. Let A' be the unital subalgebra of A generated by the family x ; its elements are of the form $P(x)$ for $P \in \mathbf{C}[(X_\lambda)]$. The estimate (2) implies that the condition $P(x) = 0$ implies $P(c) = 0$. There therefore exists a unital algebra morphism ξ of A' into $\mathbf{C}$ such that $\xi(x_\lambda) = c_\lambda$ for all $\lambda \in \Lambda$. By (2), the morphism ξ is continuous. It therefore extends by continuity to a character χ of $\overline{A'} = A$, which satisfies $c = (\chi(x_\lambda))_{\lambda \in \Lambda} \in \text{Sp}_A^\Lambda(x)$. This completes the proof.

1.3.7 Polynomially convex sets

DEFINITION 3. — Let Λ a set and V a subset of $\mathbf{C}^\Lambda$. We say that V is polynomially convex if V is the set of points $(c_\lambda)_{\lambda \in \Lambda}$ of $\mathbf{C}^\Lambda$ such that

$$|P((c_\lambda))| \leq \sup_{c \in V} |P(c)|$$

for all $P \in \mathbf{C}[(X_\lambda)]$

Lemma 3. — Let Λ a set. A subset V of $\mathbf{C}^\Lambda$ is polynomially convex if and only if there exist a family $(P_i)_{i \in I}$ of elements of $\mathbf{C}[(X_\lambda)_{\lambda \in \Lambda}]$ and a family $(M_i)_{i \in I}$ of elements of $[0, +\infty]$ such that V is the set of $c \in \mathbf{C}^\Lambda$ satisfying

$$|P_i(c)| \leq M_i$$

for all $i \in I$.

If the subset V of $\mathbf{C}^\Lambda$ is polynomially convex, it satisfies the above condition for the family consisting of the elements P of $\mathbf{C}[(X_\lambda)_{\lambda \in \Lambda}]$ by setting $M_P = \sup_{c \in V} |P(c)|$.

Conversely, let $(P_i)_{i \in I}$ be a family of elements of $\mathbf{C}[(X_\lambda)_{\lambda \in \Lambda}]$ and $(M_i)_{i \in I}$ be a family of elements of $[0, +\infty]$. Let V be the set of c in $\mathbf{C}^\Lambda$ such that $|P_i(c)| \leq M_i$ for all $i \in I$. We then have $\sup_{c \in V} |P_i(c)| \leq M_i$ for $i \in I$. Suppose that $x \in \mathbf{C}^\Lambda$ satisfies

$$|P(x)| \leq \sup_{c \in V} |P(c)|$$

for all $P \in \mathbf{C}[(X_\lambda)]$. For $i \in I$, we have in particular $|P_i(x)| \leq M_i$, hence $x \in V$. Conversely, for every element $x \in V$ and every polynomial $P \in \mathbf{C}[(X_\lambda)]$, one has $|P(x)| \leq \sup_{c \in V} |P(c)|$. Consequently, the set V is polynomially convex.

Lemma 4. — Let A be a unital commutative Banach algebra. Let Λ a set and $x = (x_\lambda)_{\lambda \in \Lambda}$ a family of elements of A . If the unital subalgebra generated by the family x is dense in A , then the simultaneous spectrum $\text{Sp}_A^\Lambda(x)$ is polynomially convex.

This follows from the assertion. b) of prop. 12 of I, p. 43 and of definition 3.

Any intersection of polynomially convex subsets of C^Λ is polynomially convex (Lemma 3). This justifies the following definition:

DEFINITION 4. — Let Λ a set and V a subset of C^Λ . The polynomially convex hull of V is the smallest polynomially convex subset of C^Λ containing V .

The polynomially convex hull of V is the set of c belonging to C^Λ such that $|P(c)| \leq \sup_W |P|$ for all $P \in \mathbf{C}[(X_\lambda)]$. Indeed, this set is polynomially convex by Lemma 3, and is contained in every polynomially convex set containing V by definition.

Example. — Let Λ a finite set and $V \subset C^\Lambda$ a compact convex subset. Then V is polynomially convex. Indeed, let W be the polynomially convex hull of V . Let us prove that $W \subset V$, which will imply the assertion. Let $x \in C^\Lambda - V$. There exists a real hyperplane H into C^Λ which strictly separates x and V (EVT, II, p. 41, prop. 4). Let $f_{\mathbf{R}}$ a form $\mathbf{R}$ -linear on C^Λ and $\alpha \in \mathbf{R}$ such that H is the set of $y \in C^\Lambda$ satisfying $f_{\mathbf{R}}(y) = \alpha$. Let f a linear form on C^Λ such that $f_{\mathbf{R}} = \Re(f)$. We therefore have

$$\Re(f(x)) > \sup_{y \in V} \Re(f(y)).$$

For every $t \in \mathbf{R}$ and $y \in V$, set $f_t(y) = t + f(y)$. We have $|f_t| - \Re(f_t) \rightarrow 0$ into $\mathcal{C}(C^\Lambda, \mathbf{R})$ endowed with the topology of compact convergence when $t \rightarrow +\infty$. For t sufficiently large, it follows that $|f_t(x)| > \sup_{y \in V} |f_t(y)|$ since V is compact. Thus $x \in C^\Lambda - W$ since f_t is a polynomial function.

Lemma 5. — Let K be a compact subset of C . We denote by $\widehat{K}$ the union of K and the connected components of $C - K$ which are relatively compact. Then the set $\widehat{K}$ is compact.

Since K is compact, there exists a real number $r > 0$ such that K is contained in the open disk D centered at 0 with radius r . Then $C - K$ contains $C - D$. Since the space $C - D$ is connected (being homeomorphic to $[r, +\infty[\times S^1$), it is contained in a connected component U of $C - K$. Every other connected component of $C - K$ is contained in D , hence is bounded. The connected component U is then the unique unbounded connected component of $C - K$, that is, $U = C - \widehat{K}$. Since U is open and contains the complement of the disk D , the set $\widehat{K}$ is compact.

PROPOSITION 13. — Let $n \geq 1$ be an integer. Let $K \subset C^n$ be a closed subset and V its polynomially convex hull.

- a) Every bounded connected component of $C^n - K$ is contained in V ;
- b) If $n = 1$, and if K is compact, then V is the union $\widehat{K}$ of K and the bounded connected components of $C - K$.

Since $K \subset C^n$ is closed, its complement $C^n - K$ is open, hence locally connected, so that each connected component of $C^n - K$ is open. The maximum principle (VAR, R1, p. 29, 3.3.7) then shows that every bounded connected component of $C^n - K$ is contained in V , which proves assertion a).

Now suppose $n = 1$ and K compact. Let $\widehat{K}$ denote the union of K and the bounded connected components of $C - K$, so that $\widehat{K} \subset V$; by the preceding, the set $\widehat{K}$ is compact (lemma 5). Let A be the commutative unital Banach algebra $\mathcal{C}(\widehat{K})$. Let $x \in A$ the identity function on $\widehat{K}$ and B the closed unital subalgebra of A generated by x . We have $\text{Sp}_A(x) = \widehat{K}$ (example 3 of I, p. 17), hence $C - \text{Sp}_A(x)$ is connected. Consequently we have $\text{Sp}_B(x) = \text{Sp}_A(x)$ (cor. of prop. 6 of I, p. 28). Since $\text{Sp}_B(x)$ is polynomially convex by lemma 4, this shows that $\widehat{K}$ is polynomially convex and therefore that $V \subset \widehat{K}$.

The second part of the proposition does not extend to the case $n \geq 2$ (cf. exerc. 23 of I, p. 170).

PROPOSITION 14. — Let Λ be a set. Let A be a commutative unital Banach algebra, $x = (x_\lambda)_{\lambda \in \Lambda}$ a family of elements of A and A' the unital Banach subalgebra generated by x .

Then the simultaneous spectrum $\text{Sp}_A^\Lambda(x)$ is the polynomially convex hull of $\text{Sp}_A^\Lambda(x)$.

Indeed, prop. 12 of I, p. 43, b) proves that $\text{Sp}_A^\Lambda(x)$ is the set of $c \in \mathbf{C}^\Lambda$ such that $|P(c)| \leq \varrho(P(x))$ for all $P \in \mathbf{C}[(X_\lambda)]$.

Now we have

$$\varrho(P(x)) = \sup_{\chi \in X(A)} |\chi(P(x))| = \sup_{\chi \in X(A)} |P((\chi(x_\lambda))_{\lambda \in \Lambda})| = \sup_{c \in \text{Sp}_A^\Lambda(x)} |P(c)|,$$

by prop. 7 of I, p. 38, a) and prop. 12 of I, p. 43, a). The result then follows from lemma 3.

PROPOSITION 15. — Let Λ be a set. Let K be a compact polynomially convex subset of $\mathbf{C}^\Lambda$. Let A_0 the set of restrictions to K of polynomial functions on $\mathbf{C}^\Lambda$, and let A be the closure of A_0 in algebra $\mathcal{C}(K)$.

Let $\text{ev}: K \rightarrow X(A)$ be the map defined by $x \rightarrow \text{ev}_x$, where ev_x is the character $f \rightarrow f(x)$ of A . Let $z = (z_\lambda)_{\lambda \in \Lambda}$ be the family in A of restrictions to K of the coordinate functions on $\mathbf{C}^\Lambda$, and let φ be the surjective map from $X(A)$ to $\text{Sp}_A^\Lambda(z)$ defined by the family z .

Then we have $K = \text{Sp}_A^\Lambda(z)$, and the maps $\text{ev}: K \rightarrow X(A)$ and $\varphi: X(A) \rightarrow K$ are reciprocal homeomorphisms.

The map $\varphi \circ \text{ev}$ is the identity map of K . In particular, K is contained in the image $\text{Sp}_A^\Lambda(z)$ of φ .

Prop. 14 implies that $\text{Sp}_A^\Lambda(z)$ is the polynomially convex hull of $\text{Sp}_{\mathcal{C}(K)}^\Lambda(z)$. Since, on the other hand, we have $\text{Sp}_{\mathcal{C}(K)}^\Lambda(z) = K$ (I, p. 42, example), which is polynomially convex by hypothesis, we deduce that $\text{Sp}_A^\Lambda(z) = K$.

Since the unital subalgebra generated by the family of elements z_λ is dense in A , the map φ is a homeomorphism from $X(A)$ on $\text{Sp}_A^\Lambda(z) = K$ Proposition 12 of Book I, p. 43(a). The identity $\varphi \circ \text{ev} = \text{Id}_K$ then shows that ev is the inverse homeomorphism of φ .

For every subset Λ' of Λ , we denote by $\text{pr}_{\Lambda'}$ the canonical projection $\mathbf{C}^\Lambda \rightarrow \mathbf{C}^{\Lambda'}$. Let W be a subset of $\mathbf{C}^\Lambda$ and let V be its polynomially convex hull. Set $W' = \text{pr}_{\Lambda'} W$. Since every element of $\mathbf{C}[(X_\lambda)_{\lambda \in \Lambda'}]$ is identified with an element of $\mathbf{C}[(X_\lambda)_{\lambda \in \Lambda}]$ the polynomially convex hull of W' is contained in $\text{pr}_{\Lambda'} V$.

Lemma 6. — Let $K \subset \mathbf{C}^\Lambda$ a compact polynomially convex subset and U a neighborhood of K . There exists a finite subset Λ_0 of Λ such that, for every subset Λ' of Λ containing Λ_0 , the

set $\text{pr}_{\Lambda'}(U)$ contains the polynomially convex hull of $\text{pr}_{\Lambda'}(K)$.

Since K is compact, there exists a family of compact disks D_λ in $\mathbb{C}$ with center 0 and radii R_λ such that K is contained in the product $D = \prod_\lambda D_\lambda$. For every $P \in \mathbb{C}[(X_\lambda)]$, let K_P the set of $x \in \mathbb{C}^\Lambda$ such that

$$|P(x)| \leq \sup_{c \in K} |P(c)|.$$

We have

$$D \cap \bigcap_P K_P = K \quad (3)$$

which is contained in U . Thus, the space D is the union of the open set $D \cap U$ and of the family of open sets $D - K_P$ for $P \in \mathbb{C}[(X_\lambda)]$. Since D is compact, there exists an integer $q \geq 1$ and polynomials $P_1, \dots, P_q \in \mathbb{C}[(X_\lambda)]$ such that:

$$D \cap K_{P_1} \cap \dots \cap K_{P_q} \subset U. \quad (4)$$

There exists a finite set $\Lambda_0 \subset \Lambda$ such that $P_i \in \mathbb{C}[(X_\lambda)_{\lambda \in \Lambda_0}]$ for $1 \leq i \leq q$. Let us show that Λ_0 has the required property.

Let Λ' a subset of Λ containing Λ_0 . Let E be the subset of $\mathbb{C}^{\Lambda'}$ consisting of the elements $c = (c_\lambda)_{\lambda \in \Lambda'}$ defined by the inequalities $|c_\lambda| \leq R_\lambda$ for $\lambda \in \Lambda'$, and

$$|P_i(c)| \leq \sup_{c \in K} |P_i(c)|$$

for $i = 1, \dots, q$. The set E is polynomially convex (lemma 3), and formula (3) shows that $\text{pr}_{\Lambda'}(K) \subset E$.

On the other hand, let $c = (c_\lambda)_{\lambda \in \Lambda'} \in E$; let $d = (d_\lambda)_{\lambda \in \Lambda}$ be the element of $\mathbb{C}^\Lambda$ defined by $d_\lambda = c_\lambda$ for $\lambda \in \Lambda'$ and $d_\lambda = 0$ for $\lambda \in \Lambda - \Lambda'$. Then (4) implies that $d \in U$ and therefore that $c \in \text{pr}_{\Lambda'}(U)$. Thus, $E \subset \text{pr}_{\Lambda'}(U)$, which completes the proof.

Lemma 7. — Let $n \geq 1$ an integer and K a polynomially convex compact subset of $\mathbb{C}^n$. Then K admits a fundamental system of compact neighborhoods that are polynomially convex.

There exists a compact polydisk (cf. VAR, R1, p. 24) Δ of $\mathbb{C}^n$ which is a neighborhood of K . Since K is polynomially convex, there exists a family $(P_i)_{i \in I}$ of elements of $\mathbb{C}[X_1, \dots, X_n]$, and a family $(M_i)_{i \in I}$ of positive real numbers such that K is the set of $z \in \Delta$ satisfying $|P_i(z)| \leq M_i$ for all i (Lemma 3). For every finite subset J of I and every $\varepsilon > 0$, let $K_{J,\varepsilon}$ the set of $z \in \Delta$ such that $|P_i(z)| \leq M_i + \varepsilon$ for $i \in J$. Then each set $K_{J,\varepsilon}$ is a compact polynomially convex neighborhood of K (loc. cit.), and the intersection of the sets $K_{J,\varepsilon}$ is K . The sets $K_{J,\varepsilon}$ therefore form a fundamental system of polynomially convex neighborhoods of K (TG, I, p. 60, th. 1).

1.4 HOLOMORPHIC FUNCTIONAL CALCULUS

1.4.1 Germs of holomorphic functions

Let E and F be complex Banach spaces. We recall (cf. VAR, R1, p. 26, 3.2.1, p. 22, 3.1 and p. 88, App.) that a holomorphic mapping defined on an open subset U of E and taking values in F is a mapping $f: U \rightarrow F$ such that, for all $x \in U$ there exists a convergent series

$$f_x = \sum_{k \geq 0} f_{x,k}$$

satisfying $f(x + y) = f_x(y)$ for all $y \in E$ sufficiently close to 0, where $f_{x,k}: E \rightarrow F$ is a continuous homogeneous polynomial of degree k on E with values in F , that is, a mapping of the form

$$f_{x,k}(y) = \tilde{f}_{x,k}(y, \dots, y)$$

where $\tilde{f}_{x,k}: E^k \rightarrow F$ is a map k continuous multilinear. We denote $\mathcal{O}(U; F)$ the complex vector space of holomorphic functions on U with values in F , equipped with the topology of compact convergence. It is a locally convex topological vector space, whose topology is defined by the seminorms $f \mapsto \sup_{z \in K} \|f(z)\|$ where K ranges over the set of compact subsets of U .

Let G be a complex Banach space and V an open subset of G . For every holomorphic map $\varphi: V \rightarrow U$, the map $\varphi^*: f \mapsto f \circ \varphi$ is a continuous linear map from $\mathcal{O}(U; F)$ into $\mathcal{O}(V; F)$.

If H is a complex Banach space and $\varphi: F \rightarrow H$ a continuous linear map, then the map

$f \mapsto \varphi \circ f$ is a continuous linear map from $\mathcal{O}(U; F)$ into $\mathcal{O}(U; H)$, denoted φ_* .

Let n be a natural number and set $E = C^n$. Let K be a compact subset of C^n and let U be the decreasing filtered set of open neighborhoods of K . If $U, U' \in U$ and $U' \subset U$, the restriction map of functions from $\mathcal{O}(U; F)$ into $\mathcal{O}(U'; F)$ is continuous. The inductive limit of the spaces $\mathcal{O}(U; F)$ for these maps is denoted $\mathcal{O}(K; F)$. The elements of $\mathcal{O}(K; F)$ are called the germs of holomorphic functions in a neighborhood of K with values in F .

The space $\mathcal{O}(K; F)$ is endowed with the inductive limit topology of the locally convex topologies of the $\mathcal{O}(U; F)$. Let X be a locally convex topological vector space and $\varphi: \mathcal{O}(K; F) \rightarrow X$ a map. For every open neighborhood U of K , the map $\mathcal{O}(U; F) \rightarrow X$ derived from φ by composition with the canonical map $\mathcal{O}(U; F) \rightarrow \mathcal{O}(K; F)$ is denoted φ^U . The map φ is continuous if and only if φ^U is continuous for every U (EVT, II, p. 29, prop. 5, (iii)).

Let m be a natural number. Let L be a compact subset of C^m and let V be an open neighborhood of L . Let $\varphi: V \rightarrow C^n$ a holomorphic map such that $\varphi(L) \subset K$. The continuous linear maps

$$\mathcal{O}(U; F) \xrightarrow{\varphi^*} \mathcal{O}(\varphi^{-1}(U); F) \xrightarrow{\varphi^{-1}\varphi^U} \mathcal{O}(L; F)$$

for U an open neighborhood of K , induce a continuous linear map $\varphi^*: \mathcal{O}(K; F) \rightarrow \mathcal{O}(L; F)$ (loc. cit.).

Let H be a complex Banach space and $\varphi: F \rightarrow H$ a continuous linear map. The continuous linear maps

$$\mathcal{O}(U; F) \xrightarrow{\varphi_*} \mathcal{O}(U; H) \xrightarrow{\varphi^U} \mathcal{O}(K; F)$$

where U ranges over the set of open neighborhoods of K in C^n induce a continuous linear map φ_* of $\mathcal{O}(K; F)$ into $\mathcal{O}(K; H)$ (loc. cit.). We shall sometimes denote $\varphi \circ f = \varphi_*(f)$.

For every open neighborhood U of K , the restriction to K is a continuous linear map $\mathcal{O}(U; F) \rightarrow \mathcal{C}(K; F)$; these maps induce a continuous linear map $\mathcal{O}(K; F) \rightarrow \mathcal{C}(K; F)$ called evaluation of germs of holomorphic functions on K .

Let A be a unital complex Banach algebra. The spaces $\mathcal{O}(U; A)$ and $\mathcal{O}(K; A)$ are unital

algebras. If $A \neq \{0\}$ one can canonically identify $\mathcal{O}(U; \mathbb{C})$ (resp. $\mathcal{O}(K; \mathbb{C})$) with the subalgebra $\mathcal{O}(U; \mathbb{C}) \cdot 1$ of $\mathcal{O}(U; A)$ (resp. to the subalgebra $\mathcal{O}(K; \mathbb{C}) \cdot 1$ of $\mathcal{O}(K; A)$). We shall set $\mathcal{O}(U) = \mathcal{O}(U; \mathbb{C})$ and $\mathcal{O}(K) = \mathcal{O}(K; \mathbb{C})$.

1.4.2 Statement of the main theorem

Let X be a set. If $m \leq n$, we shall denote by $\pi_{m,n}$ the map from X^n into X^m such that $\pi_{m,n}(x) = (x_1, \dots, x_m)$ for all $x = (x_1, \dots, x_n) \in X^n$.

Let A be a unital Banach algebra over $\mathbb{C}$. For every integer $n \geq 1$ and every $a \in A^n$, we denote by $\text{Sp}^n(a)$ the simultaneous spectrum $\text{Sp}_A^{\{1, \dots, n\}}(a)$ (def. 2 of I, p. 42). It is a compact subset of C^n . For every integer m such that $1 \leq m \leq n$, one has $\pi_{m,n}(\text{Sp}^n(a)) = \text{Sp}^m(\pi_{m,n}(a))$ (I, p. 41, no. 6). The continuous linear map

$$\pi_{m,n}^*: \mathcal{O}(\text{Sp}^m(\pi_{m,n}(a)); A) \rightarrow \mathcal{O}(\text{Sp}^n(a); A)$$

is a morphism of unital algebras.

Let A be a commutative unital Banach algebra over $\mathbb{C}$. Let $n \geq 1$ be an integer. One calls holomorphic functional calculus in n variables on A the data, for every $a \in A^n$, of a map

$$\Theta_a: \mathcal{O}(\text{Sp}^n(a); A) \rightarrow A$$

satisfying the conditions:

(CF1) For every $a \in A^n$, the map Θ_a is a continuous morphism of unital algebras.

(CF2) If $a = (a_1, \dots, a_n)$, and if $z_1, \dots, z_n$ denote the germs in a neighborhood of $\text{Sp}^n(a)$ of the coordinate functions on C^n , one has

$$\Theta_a(z_1) = a_1, \dots, \Theta_a(z_n) = a_n.$$

Remark. — If the radical of the algebra A is zero, one may omit the continuity condition in (CF1) (cf. prop. 9 of I, p. 40).

One calls holomorphic functional calculus on A the data, for every integer $n \geq 1$, of a holomorphic functional calculus in n variables on A , satisfying:

(CF3) Whatever the integers m and n such that $1 \leq m \leq n$, and whatever $a \in A^n$ and $f \in \mathcal{O}(\mathrm{Sp}^m(\pi_{m,n}(a); A))$, one has

$$\Theta_a(\pi_{m,n}^*(f)) = \Theta_{\pi_{m,n}(a)}(f).$$

The object of this paragraph is to prove the following theorem:

THEOREM 1. — Let A be a commutative complex unital Banach algebra. There exists a unique holomorphic functional calculus on A .

The proof of this theorem will occupy Nos. 3 to 7.

1.4.3 Adapted sequences and associated differential forms

In this number, and up to number 5, we denote by A a commutative complex unital Banach algebra and by n an integer ≥ 1 .

When we speak of infinitely differentiable functions on an open subset of $\mathbb{C}^n$, it will be a question of functions infinitely differentiable for the underlying structure of real manifold. The notions of differential calculus used will be relative to this structure.

DEFINITION 1. — Let $a = (a_1, \dots, a_n) \in A^n$, let $h: \mathbb{C}^n \rightarrow \mathbb{C}$ be a map and let $u_1, \dots, u_n$ be maps from $\mathbb{C}^n$ into A . One says that the sequence $(h, u_1, \dots, u_n)$ is adapted to a if

- (i) The map h is infinitely differentiable, with compact support, and equal to 1 in a neighborhood of $\mathrm{Sp}^n(a)$;
- (ii) The maps $u_1, \dots, u_n$ are infinitely differentiable;
- (iii) For every $z = (z_1, \dots, z_n) \in \mathbb{C}^n$, we have

$$h(z) + (z_1 - a_1)u_1(z) + \dots + (z_n - a_n)u_n(z) = 1. \quad (1)$$

The differential form of degree $2n$ on $\mathbb{C}^n$, with coefficients in A , defined by

$$\omega = \bigwedge_{i=1}^n (du_i \wedge dz_i)$$

is called the differential form associated with $(h, u_1, \dots, u_n)$.

If $(h, u_1, \dots, u_n)$ is adapted to a , then by differentiating (1) we obtain the equality

$$dh = - \sum_{i=1}^n u_i dz_i - \sum_{i=1}^n (z_i - a_i) du_i \quad (2)$$

whence, for every i such that $1 \leq i \leq n$, the relation

$$dh \wedge dz_i \wedge \bigwedge_{j \neq i} (du_j \wedge dz_j) = -(z_i - a_i) \omega. \quad (3)$$

Lemma 1. — Let U be an open subset of $\mathbb{C}^n$. Let K be a compact subset of U . There exists an infinitely differentiable function h from $\mathbb{C}^n$ to $\mathbb{C}$, equal to 1 on K and with compact support contained in U .

Let V be a relatively compact open neighborhood of K such that $\overline{V}$ is contained in U (TG, I, p. 65, prop. 10). There exists an infinitely differentiable function h of $\mathbb{C}^n$ in $\mathbb{C}$ whose support is contained in V and which equals 1 on K (VAR, R1, p. 40, 5.3.6). This function has the required properties.

Example. — Suppose that $n = 1$. Let $a \in A$. For every open neighborhood U of $\text{Sp}(a)$, there exists an infinitely differentiable function h from $\mathbb{C}$ to $\mathbb{C}$ with compact support contained in U , equal to 1 in a neighborhood of $\text{Sp}(a)$ (VAR, R1, p. 40, 5.3.6). Set

$$u(z) = (1 - h(z))(z - a)^{-1}$$

for $z \in \mathbb{C} - \text{Sp}(a)$ and $u(z) = 0$ if $z \in \text{Sp}(a)$. The pair (h, u) is adapted to a and the associated differential form is $\omega = du \wedge dz$.

Lemma 2. — Let $a = (a_1, \dots, a_n) \in A^n$. There exist infinitely differentiable maps $v_1, \dots, v_n$ of $\mathbb{C}^n - \text{Sp}^n(a)$ in A such that

$$(z_1 - a_1)v_1(z) + \dots + (z_n - a_n)v_n(z) = 1$$

for all $z = (z_1, \dots, z_n) \in \mathbb{C}^n - \text{Sp}^n(a)$.

Let $w = (w_1, \dots, w_n) \in \mathbb{C}^n - \text{Sp}^n(a)$. By definition of the simultaneous spectrum, there

exist $b_1, \dots, b_n$ in A such that

$$(w_1 - a_1)b_1 + \dots + (w_n - a_n)b_n = 1$$

(I, p. 41, no. 6). There exists an open neighborhood W_w of w such that the element $(z_1 - a_1)b_1 + \dots + (z_n - a_n)b_n$ of A is invertible if $z = (z_1, \dots, z_n)$ belongs to W_w . For every integer j such that $1 \leq j \leq n$ and every z in W_w , then let

$$u_j(z) = b_j \left(\sum_{i=1}^n (z_i - a_i)b_i \right)^{-1}.$$

The functions $u_1, u_2, \dots, u_n$ of W_w in A thus defined are infinitely differentiable in W_w , and we have

$$(z_1 - a_1)u_1(z) + \dots + (z_n - a_n)u_n(z) = 1$$

for all z in W_w .

Since the family $(W_w)_{w \in \mathbf{C}^n - \text{Sp}^n(a)}$ is an open covering of $\mathbf{C}^n - \text{Sp}^n(a)$, there exists a locally finite open refinement $\mathcal{W} = (W_\lambda)_{\lambda \in L}$ (TG, I, p. 70, th. 5) and, for every $\lambda \in L$, functions $u_{1\lambda}, \dots, u_{n\lambda}$, with values in A , defined and infinitely differentiable in W_λ , such that $(z_1 - a_1)u_{1\lambda}(z) + \dots + (z_n - a_n)u_{n\lambda}(z) = 1$ for all z into W_λ . Let $(f_\lambda)_{\lambda \in L}$ a partition of unity subordinate to the covering $\mathcal{W}$ consisting of infinitely differentiable functions (VAR, R1, p. 40, 5.3.6). Let i be an integer such that $1 \leq i \leq n$. For every $\lambda \in L$, let $u'_{i\lambda}$ the map from $\mathbf{C}^n - \text{Sp}^n(a)$ in A obtained by extending by 0 the function $f_\lambda u_{i\lambda}$ into $(\mathbf{C}^n - \text{Sp}^n(a)) - W_\lambda$. The functions $u'_{i\lambda}$ are infinitely differentiable. Since the family $(\text{Supp}(u'_{i\lambda}))_{\lambda \in L}$ is locally finite, the function $v_i = \sum_{\lambda \in L} u'_{i\lambda}$ is defined and infinitely differentiable in $\mathbf{C}^n - \text{Sp}^n(a)$.

Let $z \in \mathbf{C}^n - \text{Sp}^n(a)$. Let L' denote the finite set of $\lambda \in L$ such that $z \in W_\lambda$. Then

$$\begin{aligned} \sum_{i=1}^n (z_i - a_i)v_i(z) &= \sum_{\lambda \in L'} \sum_{i=1}^n (z_i - a_i)u'_{i\lambda}(z) \\ &= \sum_{\lambda \in L'} f_\lambda(z) \sum_{i=1}^n (z_i - a_i)u_{i\lambda}(z) = \left(\sum_{\lambda \in L'} f_\lambda(z) \right) \cdot 1 = 1. \end{aligned}$$

Lemma 3. — Let $a = (a_1, \dots, a_n) \in A^n$. Let h a map of $\mathbf{C}^n$ into $\mathbf{C}$ infinitely differentiable, equal to 1 in a neighborhood of $\text{Sp}^n(a)$ and with compact support. There exist infinitely

differentiable maps $u_1, \dots, u_n$ of $\mathbf{C}^n$ in A such that the sequence $(h, u_1, \dots, u_n)$ is adapted to a .

Let $v_1, \dots, v_n$ be infinitely differentiable maps from $\mathbf{C}^n - \text{Sp}^n(a)$ into A such that

$$\sum_{j=1}^n (z_j - a_j) v_j(z) = 1$$

for z into $\mathbf{C}^n - \text{Sp}^n(a)$ (lemma 2). Let i be an integer such that $1 \leq i \leq n$. Set $u_i(z) = (1 - h(z))v_i(z)$ if $z \in \mathbf{C}^n - \text{Sp}^n(a)$ and $u_i(z) = 0$ if $z \in \text{Sp}^n(a)$. The mappings u_i are infinitely differentiable in $\mathbf{C}^n - \text{Sp}^n(a)$ and vanish in a neighborhood of $\text{Sp}^n(a)$, hence infinitely differentiable in $\mathbf{C}^n$. Equality (1) holds in $\text{Sp}^n(a)$ because the functions u_i are zero on $\text{Sp}^n(a)$ and h is equal to 1 in a neighborhood of $\text{Sp}^n(a)$. It is also true on $\mathbf{C}^n - \text{Sp}^n(a)$ by construction.

Lemma 4. — Let $a \in A^n$. Let $(h, u_1, \dots, u_n)$ be a sequence adapted to a and ω the associated differential form.

- a) For $i = 1, 2, \dots, n$, there exists a differential form β_i on $\mathbf{C}^n$, of degree $n - 1$, with coefficients in A , such that

$$(z_i - a_i) \omega = d(h\beta_i \wedge dz_1 \wedge \dots \wedge dz_n);$$

- b) The differential form ω has compact support contained in the support of h ;
c) There exists a differential form β on $\mathbf{C}^n$, of degree $n - 1$, with coefficients in A , such that

$$(n + 1)h\omega - \omega = d(h\beta \wedge dz_1 \wedge \dots \wedge dz_n).$$

Let i be an integer such that $1 \leq i \leq n$. There exists $\varepsilon_i \in \{-1, 1\}$ such that

$$\varepsilon_i \bigwedge_{j \neq i} du_j \wedge dz_1 \wedge \dots \wedge dz_n = dz_i \wedge \bigwedge_{j \neq i} (du_j \wedge dz_j).$$

Let us set $\beta_i = \varepsilon_i \bigwedge_{j \neq i} du_j$, so that the left-hand term in this formula is $\beta_i \wedge dz_1 \wedge \dots \wedge dz_n$ and that $d\beta_i = 0$. Thus

$$d(h\beta_i \wedge dz_1 \wedge \cdots \wedge dz_n) = dh \wedge \beta_i \wedge dz_1 \wedge \cdots \wedge dz_n =$$

$$dh \wedge dz_i \wedge \bigwedge_{j \neq i} (du_j \wedge dz_j) = (z_i - a_i)\omega,$$

by formula (3), whence the assertion a).

From assertion a) and formula (1) we deduce the relation

$$\begin{aligned} \omega &= h\omega + (1 - h)\omega = h\omega + \sum_{i=1}^n (z_i - a_i)u_i\omega \\ &= h\omega + \sum_{i=1}^n u_i d(h\beta_i \wedge dz_1 \wedge \cdots \wedge dz_n), \end{aligned}$$

whence $\text{Supp}(\omega) \subset \text{Supp}(h)$ which proves b).

Finally, set

$$\beta = \sum_{i=1}^n \varepsilon_i u_i \bigwedge_{j \neq i} du_j = \sum_{i=1}^n u_i \beta_i, \text{ and } \tau = h\beta dz_1 \wedge \cdots \wedge dz_n.$$

We have $d\beta = \sum_i du_i \wedge \beta_i$ and therefore

$$d\beta \wedge dz_1 \wedge \cdots \wedge dz_n = \sum_i du_i \wedge dz_i \wedge \bigwedge_{j \neq i} (du_j \wedge dz_j) = n\omega.$$

Thus

$$\begin{aligned} d\tau &= dh \wedge \beta \wedge dz_1 \wedge \cdots \wedge dz_n + h d\beta \wedge dz_1 \wedge \cdots \wedge dz_n \\ &= \sum_{i=1}^n u_i dh \wedge dz_i \wedge \bigwedge_{j \neq i} (du_j \wedge dz_j) + nh\omega \\ &= - \sum_{i=1}^n u_i (z_i - a_i)\omega + nh\omega = (h - 1)\omega + nh\omega = (n + 1)h\omega - \omega, \end{aligned}$$

Given formulas (3) and (1), whence c).

We now propose to study how the differential form ω associated with a sequence adapted to a varies as a function of this sequence. We will say that sequences $(h, u_1, \dots, u_n)$ and $(h', u'_1, \dots, u'_n)$ adapted to a are linked if there exists a differential form ψ of degree $n - 1$ on $\mathbb{C}^n$, with coefficients in A and with support contained in the union of the supports of h and of h' , such that the associated differential forms ω and ω' satisfy

$$\omega - \omega' = d(\psi \wedge dz_1 \wedge dz_2 \wedge \cdots \wedge dz_n).$$

Let us begin with an elementary modification:

Lemma 5. — Let $a \in A^n$, let $(h, u_1, \dots, u_n)$ be a sequence adapted to a , and let ω the associated differential form.

Let w an infinitely differentiable map from $\mathbb{C}^n$ into A and let i and j be distinct integers between 1 and n . Define $u'_1, \dots, u'_n$ by

$$\begin{aligned} u'_i &= u_i + (z_j - a_j)w, & u'_j &= u_j - (z_i - a_i)w, \\ u'_k &= u_k \text{ for } k \neq i, j. \end{aligned}$$

Then the sequence $(h, u'_1, \dots, u'_n)$ is adapted to a and is linked to the sequence $(h, u_1, \dots, u_n)$.

We denote $dz = dz_1 \wedge \cdots \wedge dz_n$. Since

$$\begin{aligned} \sum_{k=1}^n (z_k - a_k)u'_k(z) &= \sum_{k=1}^n (z_k - a_k)u_k(z) + w(z)(z_j - a_j)(z_i - a_i) \\ &\quad - w(z)(z_i - a_i)(z_j - a_j) = 1 - h(z) \end{aligned}$$

for all $z \in \mathbb{C}^n$ the sequence $(h, u'_1, \dots, u'_n)$ is adapted to a . Moreover, we have

$$\begin{aligned} du'_i \wedge du'_j \wedge dz_1 \wedge \cdots \wedge dz_n &= \\ (du_i + wdz_j + (z_j - a_j)dw) \wedge (du_j - wdz_i - (z_i - a_i)dw) \wedge dz &= \\ = (du_i \wedge du_j - (z_i - a_i)du_i \wedge dw - (z_j - a_j)du_j \wedge dw) \wedge dz & \end{aligned}$$

There therefore exists $\varepsilon \in \{-1, 1\}$ such that $\varepsilon(\omega - \omega')$ is equal to

$$\begin{aligned} du'_i \wedge du'_j \wedge \bigwedge_{k \neq i, j} du'_k \wedge dz &- du_i \wedge du_j \wedge \bigwedge_{k \neq i, j} du_k \wedge dz \\ &= -\left((z_i - a_i)du_i \wedge dw + (z_j - a_j)du_j \wedge dw\right) \wedge \bigwedge_{k \neq i, j} du_k \wedge dz \\ &= -\left(\sum_{k=1}^n (z_k - a_k)du_k\right) \wedge dw \wedge \bigwedge_{k \neq i, j} du_k \wedge dz \end{aligned}$$

and, taking (2) into account, this is equal to

$$dh \wedge dw \wedge \bigwedge_{k \neq i, j} du_k \wedge dz = d\left(hdw \wedge \bigwedge_{k \neq i, j} du_k \wedge dz\right),$$

whence the result.

Lemma 6. — Let $\mathbf{a} \in A^n$. All sequences adapted to $\mathbf{a}$ are linked.

Let $(h, u_1, \dots, u_n)$ and $(h', u'_1, \dots, u'_n)$ of sequences adapted to $\mathbf{a}$, and denote ω and ω' the associated differential forms.

Let us define the infinitely differentiable maps

$$w_{ij} = u'_i u_j - u_i u'_j,$$

$$s_i = u'_i h - u_i h',$$

$$1 \leq i \leq n, 1 \leq j \leq n,$$

$$1 \leq i \leq n,$$

so that $w_{ji} = -w_{ij}$, and $\text{Supp}(s_i) \subset \text{Supp}(h) \cup \text{Supp}(h')$.

Let us set $u''_i = u'_i - s_i$, $u = (u_1, \dots, u_n)$ and $u'' = (u''_1, \dots, u''_n)$. Let us also note v_{ij} the map from $\mathbf{C}^n$ into A^n whose i -th component is $(z_j - a_j)w_{ij}$, whose j -th component is $(z_i - a_i)w_{ji} = -(z_i - a_i)w_{ij}$ and whose other components are zero. Then we have

$$u'' = u + \sum_{i < j} v_{ij}.$$

Indeed, for every integer k such that $1 \leq k \leq n$, the k -th component of the right-hand side is

$$\begin{aligned} u_k + \sum_{i=1}^{k-1} (z_i - a_i)w_{ki} + \sum_{j=k+1}^n (z_j - a_j)w_{kj} &= u_k + \sum_{i=1}^n (z_i - a_i)w_{ki} \\ &= u_k + u'_k \sum_{i=1}^n (z_i - a_i)u_i - u_k \sum_{i=1}^n (z_i - a_i)u'_i \\ &= u_k + (1 - h)u'_k - (1 - h')u_k = u'_k - s_k. \end{aligned}$$

By induction, we deduce from Lemma 5, applied to the integers i and j and to the maps w_{ij} , that the sequence $(h, u''_1, \dots, u''_n)$ is adapted to $\mathbf{a}$ and is linked to $(h, u_1, \dots, u_n)$. Let ω'' the differential form associated with $(h, u''_1, \dots, u''_n)$. Since $u''_i = u'_i - s_i$, one has

$$\begin{aligned} \omega'' - \omega' &= d(u'_1 - s_1) \wedge dz_1 \wedge \dots \wedge d(u'_n - s_n) \wedge dz_n - \\ &\quad du'_1 \wedge dz_1 \wedge \dots \wedge du'_n \wedge dz_n, \end{aligned}$$

which is expressed as a linear combination, with coefficients 1 or -1, of differential forms of

the form

$$\xi_{I_1, I_2} = \bigwedge_{i \in I_1} ds_i \wedge \bigwedge_{i \in I_2} du'_i \wedge dz_1 \wedge \cdots \wedge dz_n,$$

where I_1 (resp. I_2) is a nonempty subset (resp. a subset) of $\{1, \dots, n\}$. Each differential form ξ_{I_1, I_2} can also be written in the form

$$d(\tilde{\psi} \wedge dz_1 \wedge \cdots \wedge dz_n),$$

where the support of the differential form $\tilde{\psi}$ is contained in the support of s_i for all $i \in I_1$. Since I_1 is nonempty, this support is contained in $\text{Supp}(h) \cup \text{Supp}(h')$. Consequently, $(h, u''_1, \dots, u''_n)$ is linked to $(h', u'_1, \dots, u'_n)$, and the lemma follows by writing

$$\omega - \omega' = (\omega - \omega'') + (\omega'' - \omega').$$

1.4.4 Construction of the maps Θ_a

Let $a = (a_1, \dots, a_n) \in A^n$ and let U be an open neighborhood of $\text{Sp}^n(a)$. Let h be an infinitely differentiable function, equal to 1 in a neighborhood of $\text{Sp}^n(a)$ and such that the support of h is compact and contained in U (Lemma 1 of I, p. 52). By Lemma 3 of I, p. 54, there exist infinitely differentiable functions $(u_1, \dots, u_n)$ of C^n in A such that the sequence $(h, u_1, \dots, u_n)$ adapted to a . Let ω be the associated differential form; it has compact support contained in U (Lemma 4 of I, p. 54). There exists an infinitely differentiable function ψ with compact support in U and with values in A such that

$$\omega = \psi dx_1 \wedge dy_1 \wedge \cdots \wedge dx_n \wedge dy_n$$

where $x_j + iy_j$ are the coordinate functions on C^n identified with R^{2n} . Let μ the Lebesgue measure on R^{2n} .

Let $f \in \mathcal{C}(U; A)$. The differential form $f\omega|_U$ on U is continuous and compactly supported. The vector measure associated with this differential form (VAR, R2, 10.4.3 and 10.4.4) is the vector measure $f\psi \cdot \mu$; it is a basic measure μ (INT, VI, §2, no. 4, def. 4), its support is compact

and contained in the support of ω . This measure is bounded relative to the norm of A (INT, VI, §2, no. 4, prop. 8); we denote by $\|f\omega\|$ the positive measure on C^n associated with it (INT, VI, §2, no. 3, def. 3).

By INT, VI, §2, no. 4, prop. 8, b), we have

$$\|f\omega\| = \|f\psi \cdot \mu\| = \|f\psi\| \cdot \mu = \|f\|\|\omega\|.$$

In particular, the integral of the differential form $f\omega$ over U satisfies

$$\left\| \int_U f\omega \right\| \leq \int_U \|f\|\|\omega\|$$

(INT, VI, §2, no. 3, prop. 5).

Lemma 7. — For every function $f \in \mathcal{O}(U; A)$, the integral $\int_U f\omega$ is an element of A which depends only on a and on the germ of f in a neighborhood of $\text{Sp}^n(a)$. It satisfies the inequality

$$\left\| \int_U f\omega \right\| \leq \left(\int_U \|\omega\| \right) \sup_{z \in \text{Supp}(h)} \|f(z)\|, \quad (4)$$

and

$$\int_U (af)\omega = a \int_U f\omega$$

for all $a \in A$.

We saw above that the integral is defined for $f \in \mathcal{C}(U; A)$ and satisfies

$$\left\| \int_U f\omega \right\| \leq \int_U \|f\|\|\omega\| \leq \left(\int_U \|\omega\| \right) \sup_{z \in \text{Supp}(h)} \|f(z)\|.$$

Moreover, for every $a \in A$ and every $f \in \mathcal{C}(U; A)$, one has

$$\int_U (af)\omega = a \int_U f\omega$$

(INT, VI, §2, no. 2, prop. 2 applied to multiplication by a).

Let $(h', u'_1, \dots, u'_n)$ a sequence adapted to a such that $\text{Supp}(h') \subset U$. Let ω' the associated differential form. By lemma 6 of I, p. 57, there exists a differential form ψ on C^n of degree $n-1$,

with coefficients in A and with support contained in $\text{Supp}(h) \cup \text{Supp}(h') \subset U$, such that

$$\omega - \omega' = d(\psi \wedge dz_1 \wedge \cdots \wedge dz_n).$$

Let $f \in \mathcal{O}(U; A)$. The map f being holomorphic, we have

$$df = \sum_{i=1}^n \frac{\partial f}{\partial z_i} dz_i,$$

(VAR, R2, p. 24, 8.8.9) and hence

$$f(\omega - \omega') = f d(\psi \wedge dz_1 \wedge \cdots \wedge dz_n) = d(f\psi \wedge dz_1 \wedge \cdots \wedge dz_n).$$

By Stokes' formula (VAR, R2, p. 48, 11.2.3), we then have

$$\int_U f(\omega - \omega') = 0.$$

Thus the element $\int_U f \omega$ does not depend on the choice of the sequence $(h, u_1, \dots, u_n)$. To conclude, let us show that $\int_U f \omega$ depends only on the germ of f in a neighborhood of $\text{Sp}^n(a)$. Let U and U' be open neighborhoods of $\text{Sp}^n(a)$. Let $f \in \mathcal{O}(U; A)$ and $f' \in \mathcal{O}(U'; A)$ such that f and f' coincide on an open neighborhood U'' of $\text{Sp}^n(a)$. There exists a map h of $\mathbb{C}^n$ into $\mathbb{C}$ infinitely differentiable, equal to 1 in a neighborhood of $\text{Sp}^n(a)$, with compact support contained in U'' (lemma 1 of I, p. 52), and there exists $(u_1, \dots, u_n)$ such that the sequence $(h, u_1, \dots, u_n)$ is adapted to a (lemma 3 of I, p. 54). Let ω be the associated differential form. Since $\text{Supp}(\omega) \subset \text{Supp}(h) \subset U''$ (lemma 4, b) of I, p. 54), we have

$$\int_U f \omega = \int_{U''} f \omega = \int_{U''} f' \omega = \int_{U'} f' \omega,$$

which completes the proof.

This lemma shows that there exists a unique A -linear map Θ_a of $\mathcal{O}(\text{Sp}^n(a); A)$ to A such that

$$\Theta_a(f) = \frac{n!}{(2i\pi)^n} \int_U \tilde{f} \omega \quad (5)$$

for every open set U and every representative $\tilde{f} \in \mathcal{O}(U; A)$ of a germ $f \in \mathcal{O}(\mathrm{Sp}^n(a); A)$. The linear map Θ_a is continuous by inequality (4) and EVT, II, p. 29, prop. 5.

1.4.5 Properties of mappings Θ_a

Recall (VAR, R2, pp. 46 and 47, 11.1.3, d)) that if K is a compact subset of C , there exists a fundamental system of compact neighborhoods V of K which are pieces of C (that is, for every $x \in V$ there exists a chart (U, φ, C) from C at x such that $\varphi(U \cap K)$ is an open subset of a closed half-space of C). We then denote by ∂V the boundary of the piece V equipped with the orientation induced by the orientation of C (VAR, R2, p. 47) and let dz denote the differential of the injection $\partial V \rightarrow C$.

PROPOSITION 1. — Let a be an element of A , let U be an open neighborhood of $\mathrm{Sp}(a)$, and let $f \in \mathcal{O}(U; A)$. Let V be a compact neighborhood of $\mathrm{Sp}(a)$ contained in U and such that V is a piece of C .

Then $z \mapsto f(z)(z - a)^{-1}$ is continuous on ∂V ; the differential form $f(z)(z - a)^{-1}dz$ is integrable on ∂V , and one has

$$\Theta_a(f) = \frac{1}{2i\pi} \int_{\partial V} f(z)(z - a)^{-1} dz.$$

Let h be a map of C into C infinitely differentiable, equal to 1 in a neighborhood of $\mathrm{Sp}(a)$ and with compact support contained in the interior of V (lemma 1 of I, p. 52). Let u be a map of C to A such that (h, u) is adapted to a (cf. example 3 of I, p. 53). The associated differential form is $\omega = du \wedge dz$. It follows that $f\omega = fdu \wedge dz = d(fudz)$ since f is holomorphic. Moreover, $u(z) = (z - a)^{-1}$ on the boundary of V . Furthermore, the differential form $fudz$ is of class C^1 on U . Therefore,

$$2i\pi \Theta_a(f) = \int_V d(fudz) = \int_{\partial V} fudz = \int_{\partial V} f(z)(z - a)^{-1} dz$$

by formula (5) and Stokes' formula for the piece V (VAR, R2, p. 47, 11.2.3).

COROLLARY. — Let $a \in A$. One has $\Theta_a(1) = 1$.

Let $R > \varrho(a)$ be a real number. Let V be the closed disk with center 0 and radius R , so that $\text{Sp}(a) \subset \overset{\circ}{V}$. This is a piece of $\mathbb{C}$ whose boundary ∂V is the circle with center 0 and radius R . For $z \in \mathbb{C} - \overset{\circ}{V}$, we have the formula $(z - a)^{-1} = z^{-1}(1 - z^{-1}a)^{-1} = \sum_{j=1}^{+\infty} z^{-j}a^{j-1}$. The series converges uniformly for $z \in \partial V$. We therefore have

$$\Theta_a(1) = \frac{1}{2i\pi} \sum_{j=1}^{+\infty} a^{j-1} \int_{\partial V} z^{-j} dz = 1$$

since

$$\int_{\partial V} z^j dz = 0$$

for every integer $j \neq -1$ and

$$\int_{\partial V} z^{-1} dz = 2i\pi$$

(VAR, R2, p. 44, 10.4.5, and p. 47, 11.2.1, example).

Lemma 8. — Let U be an open subset of C^n . Let ω_1 be a continuous differential form of degree n in U with compact support and values in A (resp. let ω_2 be a continuous differential form of degree 2 in C with compact support and values in C). Let us denote by π_1 and π_2 the canonical projections of $U \times C$ onto U and C . The differential form $\pi_1^* \omega_1 \wedge \pi_2^* \omega_2$ on $U \times C$ is continuous with compact support and takes values in A . We have

$$\int_{U \times C} \pi_1^* \omega_1 \wedge \pi_2^* \omega_2 = \left(\int_C \omega_2 \right) \left(\int_U \omega_1 \right).$$

Let μ_n be the Lebesgue measure on C^n and μ_1 the Lebesgue measure on C . There exists $\psi_1 \in \mathcal{H}(U; A)$ and $\psi_2 \in \mathcal{H}(C)$ such that the vector measure associated with ω_1 is equal to $\psi_1 \cdot \mu_n$, and the vector measure associated with ω_2 is equal to $\psi_2 \cdot \mu_1$. The vector measure associated with the differential form $\pi_1^* \omega_1 \wedge \pi_2^* \omega_2$ is $(\psi_1 \otimes \psi_2) \cdot \mu_n \otimes \mu_1$.

Let ℓ be a continuous linear form on A . By INT, VI, §2, no. 2, def. 2 and the definition of the product measure (INT, III, §4, no. 1, def. 1), it follows

$$\begin{aligned}
\ell\left(\int_{U \times C} \pi_1^* \omega_1 \wedge \pi_2^* \omega_2\right) &= \int_{U \times C} \ell \circ (\psi_1 \otimes \psi_2) \mu_n \otimes \mu_1 \\
&= \int_{U \times C} \psi_2(z) \ell(\psi_1(x)) d\mu_n(x) d\mu_1(z) \\
&= \left(\int_C \psi_2(z) d\mu_1(z)\right) \left(\int_U \ell(\psi_1(x)) d\mu_n(x)\right) \\
&= \left(\int_C \psi_2 \mu_1\right) \ell\left(\int_U \psi_1 \mu_n\right),
\end{aligned}$$

whence the result (INT VI, loc. cit.).

Lemma 9. — Let $a = (a_1, \dots, a_n) \in \mathbb{A}^n$. Let $p \in \mathbb{N}$ and $a' = (a_{n+1}, \dots, a_{n+p}) \in \mathbb{A}^p$. Then we have $\Theta_{(a, a')} \circ \pi_{n, n+p}^* = \Theta_a$. In particular, we have $\Theta_a(1) = 1$.

As we have $\pi_{n, n+p} = \pi_{n, n+1} \circ \dots \circ \pi_{n+p-1, n+p}$, it suffices to prove the first assertion when $p = 1$, which we now assume. We simply denote $\pi = \pi_{n, n+1}$. It then suffices to show that, for every open neighborhood U of $\text{Sp}^n(a)$, and every function $f \in \mathcal{O}(U; A)$, one has $\Theta_{(a, a_{n+1})}(f \circ \pi) = \Theta_a(f)$. Denote by $g = f \circ \pi$. Let h (resp. h') be a mapping from $\mathbb{C}^n$ into $\mathbb{C}$ (resp. from $\mathbb{C}$ into $\mathbb{C}$), infinitely differentiable, equal to 1 in a neighborhood of $\text{Sp}^n(a)$ (resp. of $\text{Sp}(a')$), with compact support contained in U (resp. in $\mathbb{C}$). There exist mappings $(u_1, \dots, u_n)$ from $\mathbb{C}^n$ into A , infinitely differentiable, such that the sequence $(h, u_1, \dots, u_n)$ is adapted to a (lemma 3 of I, p. 54), and an infinitely differentiable mapping u_{n+1} of $\mathbb{C}$ into A such that the pair (h', u_{n+1}) is adapted to a_{n+1} (loc. cit.)

For $z \in \mathbb{C}^n$ and $z_{n+1} \in \mathbb{C}$, let $h''(z, z_{n+1}) = h(z)h'(z_{n+1})$ and $u_{n+1}''(z, z_{n+1}) = h(z)u_{n+1}(z_{n+1})$. The functions h'' and u_{n+1}'' are infinitely differentiable in $\mathbb{C}^{n+1}$. The function h'' is equal to 1 in a neighborhood of $\text{Sp}^{n+1}(a, a_{n+1})$, and has compact support contained in $U \times \mathbb{C}$. For every $w = (z, z_{n+1}) \in \mathbb{C}^{n+1}$, one has

$$\begin{aligned}
&(z_1 - a_1)(u_1 \circ \pi)(w) + \dots + (z_n - a_n)(u_n \circ \pi)(w) \\
&+ (z_{n+1} - a_{n+1})u_{n+1}''(w) = 1 - h(z) + h(z)(1 - h'(z_{n+1})) \\
&= 1 - h''(w),
\end{aligned}$$

which shows that the sequence $(h'', u_1 \circ \pi, \dots, u_n \circ \pi, u_{n+1}'')$ is adapted to (a, a_{n+1}) . Let ω be the associated differential form.

The differential form $du_1 \wedge dz_1 \wedge \dots \wedge du_n \wedge dz_n \wedge dh$ on $\mathbb{C}^n$ has degree $2n + 1$, hence is zero. Consequently,

$$\begin{aligned}
\omega &= d(u_1 \circ \pi) \wedge dz_1 \wedge \cdots \wedge d(u_n \circ \pi) \wedge dz_n \wedge du_{n+1}' \wedge dz_{n+1} \\
&= (h \circ \pi) d(u_1 \circ \pi) \wedge dz_1 \wedge \cdots \wedge d(u_n \circ \pi) \wedge dz_n \wedge du_{n+1} \wedge dz_{n+1}.
\end{aligned}$$

Since $g = f \circ \pi$, formula (5) and lemma 8 imply

$$\begin{aligned}
\Theta_{a,a'}(g) &= \frac{(n+1)!}{(2i\pi)^{n+1}} \int_{U \times \mathbb{C}} g \omega = \frac{(n+1)!}{(2i\pi)^{n+1}} \left(\int_{\mathbb{C}} du_{n+1} \wedge dz_{n+1} \right) \\
&\quad \times \left(\int_U f h du_1 \wedge dz_1 \wedge \cdots \wedge du_n \wedge dz_n \right).
\end{aligned}$$

On the one hand, we have

$$\int_{\mathbb{C}} du_{n+1} \wedge dz_{n+1} = 2i\pi \Theta_{a_{n+1}}^{\mathbb{C}}(1) = 2i\pi \cdot 1$$

by corollary 5. On the other hand, part c) of lemma 4 of I, p. 54 and the fact that the integral of a closed form is zero (VAR, R2, p. 48, 11.2.4) imply

$$\begin{aligned}
(n+1) \int_U f h du_1 \wedge dz_1 \wedge \cdots \wedge du_n \wedge dz_n &= \\
\int_U f du_1 \wedge dz_1 \wedge \cdots \wedge du_n \wedge dz_n &= \frac{(2i\pi)^n}{n!} \Theta_a(f).
\end{aligned}$$

Thus one obtains

$$\Theta_{a,a'}(g) = \frac{(n+1)!}{(2i\pi)^{n+1}} \times \frac{(2i\pi)^n}{(n+1)!} \Theta_a(f) \times 2i\pi = \Theta_a(f).$$

Finally, the formula $\Theta_a(1) = 1$ follows from the foregoing and from the corollary of Prop. 1.

Lemma 10. — Let $a \in A^n$. Let g be a polynomial function on $\mathbb{C}^n$ with coefficients in A and $f \in \mathcal{O}(\text{Sp}^n(a); A)$. We have $\Theta_a(gf) = g(a)\Theta_a(f)$. In particular, we have $\Theta_a(g) = g(a)$.

By Lemma 9, it suffices to prove the first assertion.

We denote by $z_1, \dots, z_n$ the coordinate functions on $\mathbb{C}^n$. Since the map Θ_a is A -linear, it suffices to prove the assertion of the lemma when $g = z_1^{e_1} \cdots z_n^{e_n}$, where $(e_1, \dots, e_n) \in \mathbb{N}^n$. Proceeding by induction on $e_1 + \cdots + e_n$, we reduce to the case where there exists an integer i such that $1 \leq i \leq n$ and $g = z_i$.

Let U be an open neighborhood of $\text{Sp}^n(a)$. Let $(h, u_1, \dots, u_n)$ be a sequence adapted to a such that the support of h is contained in U (Lemma 1 of I, p. 52 and Lemma 3 of I, p. 54), and

let ω be the associated differential form. By lemma 4, a) of I, p. 54, there exists a differential form β such that

$$(z_i - a_i)\omega = d(h\beta \wedge dz_1 \wedge \cdots \wedge dz_n).$$

Consequently, for every function $f \in \mathcal{O}(U; A)$, one has

$$(z_i - a_i)f\omega = fd(h\beta \wedge dz_1 \wedge \cdots \wedge dz_n) = d(fh\beta \wedge dz_1 \wedge \cdots \wedge dz_n)$$

since f is holomorphic, so that $df \wedge dz_1 \wedge \cdots \wedge dz_n = 0$. Applying Stokes' formula (VAR, R2, p. 48, 11.2.4), we obtain $\int_U (z_i - a_i)f\omega = 0$, whence

$$\Theta_a(z_i f) = \frac{n!}{(2i\pi)^n} \int_U z_i f \omega = \frac{n!}{(2i\pi)^n} \int_U a_i f \omega = a_i \Theta_a(f)$$

by formula (5). The result follows.

PROPOSITION 2. — Let $\varrho_1, \dots, \varrho_n$ be real numbers > 0 and let $U \subset \mathbb{C}^n$ be the polydisc product of the open disks with center 0 and radius ϱ_i . Let

$$\sum_{(k_1, \dots, k_n) \in \mathbb{N}^n} c(k_1, \dots, k_n) X_1^{k_1} \cdots X_n^{k_n} \in A[[X_1, \dots, X_n]]$$

a formal series with coefficients in A . Suppose that this series converges in U , and denote by f the holomorphic function in U which is its sum.

Let $a = (a_1, \dots, a_n) \in A^n$ such that $\varrho(a_i) < \varrho_i$ for $1 \leq i \leq n$. Then $\text{Sp}^n(a) \subset U$, the family $(c(k_1, \dots, k_n) a_1^{k_1} \cdots a_n^{k_n})$ of elements of A is absolutely summable, and

$$\Theta_a(f) = \sum_{(k_1, \dots, k_n) \in \mathbb{N}^n} c(k_1, \dots, k_n) a_1^{k_1} \cdots a_n^{k_n}.$$

For every character χ of A and every integer i such that $1 \leq i \leq n$, one has $|\chi(a_i)| \leq \varrho(a_i) < \varrho_i$, hence $\text{Sp}^n(a) \subset U$ by definition of the simultaneous spectrum. Let $z_1, \dots, z_n$ be the restrictions to U of the coordinate functions on $\mathbb{C}^n$. Then the family $(c(k_1, \dots, k_n) z_1^{k_1} \cdots z_n^{k_n})$ is summable in $\mathcal{O}(U; A)$ with sum f . Taking into account Lemma 10 and the continuity of the map Θ_a^U , the family $(c(k_1, \dots, k_n) a_1^{k_1} \cdots a_n^{k_n})$ is therefore summable in A with sum $\Theta_a(f)$. For

$1 \leq i \leq n$, let λ_i be a real number such that $\varrho(a_i) < \lambda_i < \varrho_i$. There exists $M_i < +\infty$ such that $\|a_i^k\| \leq M_i \lambda_i^k$ for every integer $k \geq 0$. We then have

$$\sum_{(k_1, \dots, k_n) \in \mathbb{N}^n} \|c(k_1, \dots, k_n)\| \|a_1^{k_1} \dots a_n^{k_n}\| \leq$$

$$M_1 \dots M_n \sum_{(k_1, \dots, k_n) \in \mathbb{N}^n} \|c(k_1, \dots, k_n)\| \lambda_1^{k_1} \dots \lambda_n^{k_n}$$

which is finite by hypothesis, hence the family $(c(k_1, \dots, k_n) a_1^{k_1} \dots a_n^{k_n})$ is absolutely summable.

COROLLARY. — Suppose A is nonzero. Let $a \in C^n \subset A^n$. We have $\text{Sp}_A^n(a) = \{a\}$. For every germ $f \in \mathcal{O}(\{a\}; A)$, one has $\Theta_a(f) = f(a)$.

PROPOSITION 3. — Let B be a commutative unital Banach algebra and φ a continuous unital morphism from A to B . Let $a = (a_1, \dots, a_n) \in A^n$. Set $b = (\varphi(a_1), \dots, \varphi(a_n))$, so that $\text{Sp}_B^n(b) \subset \text{Sp}_A^n(a)$. For every $f \in \mathcal{O}(\text{Sp}_A^n(a); A)$, one has

$$\varphi(\Theta_a(f)) = \Theta_b(\varphi_*(f)),$$

where $\varphi_*(f)$ denotes the germ of $\varphi \circ f$ in a neighborhood of $\text{Sp}_B^n(b)$.

It suffices to prove that for every open neighborhood U of $\text{Sp}_A^n(a)$ and every $f \in \mathcal{O}(U; A)$, one has $\varphi(\Theta_a(f)) = \Theta_b(\varphi \circ f)$, where $\varphi \circ f \in \mathcal{O}(U; B)$. Let $(h, u_1, \dots, u_n)$ be a sequence adapted to a , where the support of h is contained in U (Lemma 1 of I, p. 52 and Lemma 3 of I, p. 54). Denote by ω the associated differential form. For every $z \in C^n$, one has

$$\sum_{j=1}^n (z_j - b_j) \varphi(u_j(z)) = \varphi \left(\sum_{j=1}^n (z_j - a_j) u_j(z) \right) = 1 - h(z),$$

so that the sequence $(h, \varphi \circ u_1, \dots, \varphi \circ u_n)$ is adapted to b . Let ω' be the associated differential form. Denote by μ the Lebesgue measure on C^n . Let $f \in \mathcal{O}(U; A)$. Write $\psi \cdot \mu$ for the vector measure associated with the differential form $f\omega$. The vector measure associated with the differential form

$$(\varphi \circ f)\omega' = (\varphi \circ f)d(\varphi \circ u_1) \wedge dz_1 \wedge \cdots \wedge d(\varphi \circ u_n) \wedge dz_n$$

is equal to $(\varphi \circ \psi) \cdot \mu$. Therefore, by formula (5), and INT, VI, §2, no. 2, prop. 2, we have

$$\Theta_b(\varphi \circ f) = \frac{n!}{(2i\pi)^n} \int_U (\varphi \circ f) \mu = \frac{n!}{(2i\pi)^n} \varphi \left(\int_U \psi \mu \right) = \varphi(\Theta_a(f)),$$

as was required.

COROLLARY 1. — Let $\chi \in X(A)$ and $a \in A^n$. For every germ $f \in \mathcal{O}(\mathrm{Sp}^n(a))$, one has $\chi(\Theta_a(f)) = f(\chi(a_1), \dots, \chi(a_n))$.

This is a consequence of Proposition 3, applied to the continuous unital morphism $\chi: A \rightarrow C$ (th. 1 of I, p. 29), and of the corollary of prop. 2, applied to the Banach algebra C .

Remark. — Suppose that the algebra A is semisimple. Let $a = (a_1, \dots, a_n) \in A^n$. By prop. 8 of I, p. 38, the map Θ_a^U is the unique map φ of $\mathcal{O}(U)$ to A such that $\chi(\varphi(f)) = f(\chi(a_1), \dots, \chi(a_n))$ for all $\chi \in X(A)$ and every function $f \in \mathcal{O}(U)$.

COROLLARY 2. — Let p be an integer ≥ 1 . For every family $(f_1, \dots, f_p)$ of elements of $\mathcal{O}(\mathrm{Sp}^n(a))$, one has

$$\mathrm{Sp}^p((\Theta_a(f_1), \dots, \Theta_a(f_p))) = (f_1, \dots, f_p)(\mathrm{Sp}^n(a)).$$

In particular, for every $f \in \mathcal{O}(\mathrm{Sp}^n(a))$, one has $\mathrm{Sp}(\Theta_a(f)) = f(\mathrm{Sp}^n(a))$.

This follows from Cor. 1 and the definition of the simultaneous spectrum.

Example. — Let A be the complex Banach algebra of functions on the unit circle with absolutely convergent Fourier series (I, p. 19, example 8). Let $\varphi \in A$. Let f be a germ of a holomorphic function in a neighborhood of the set of values of φ . Then $\psi = \Theta_\varphi(f)$ is an absolutely convergent Fourier series which for every $u \in U$ satisfies $\psi(u) = f(\varphi(u))$ (cor. 1, applied to the characters $\varphi \mapsto \varphi(u)$). In other words, the function $f \circ \varphi$ on the unit circle also has an absolutely convergent Fourier series (“P. Lévy’s theorem”). This result generalizes Wiener’s theorem (I, p. 38, example 4), which concerns the case of the function $f(z) = 1/z$ on $C - \{0\}$ when φ does not vanish.

1.4.6 Approximation theorems

In this issue, A is a commutative unital complex Banach algebra.

PROPOSITION 4. — Let L be a compact polynomially convex subset of $\mathbf{C}^n$ and let U be an open neighborhood of L . For every function $f \in \mathcal{O}(U; A)$, there exists a sequence of polynomial functions on $\mathbf{C}^n$ with coefficients in A that converges to $f|_L$ into $\mathcal{C}(L; A)$.

We may assume that L is nonempty and that A is nonzero. Let P (resp. P_0) be the set of restrictions to L of polynomial functions on $\mathbf{C}^n$ with coefficients in A (resp. with coefficients in $\mathbf{C}$). Let B (resp. B_0) be the Banach algebra closure of P (resp. of P_0)

into $\mathcal{C}(L; A)$. Denote by ι the injection of A onto the normed subalgebra of B consisting of constant functions.

Let $z_1, \dots, z_n$ be the restrictions to L of the coordinate functions on $\mathbf{C}^n$; these are elements of B_0 , and by setting $z = (z_1, \dots, z_n)$ it follows $\text{Sp}_{B_0}^n(z) = L$ according to prop. 15 of I, p. 47.

Let $f \in \mathcal{O}(U; A)$. By composition with ι , the function f defines an element $f_B = \iota \circ f$ of $\mathcal{O}(U; B)$. Since $\text{Sp}_B^n(z) \subset \text{Sp}_{B_0}^n(z) \subset U$, one can form the element $b = \Theta_z(f_B)$ of B . Let $w = (w_1, \dots, w_n) \in L$, and let φ be the continuous unital morphism $g \mapsto g(w)$ from B to A . We have $\varphi \circ \iota = \text{Id}_A$, so that $\varphi \circ f_B = f$. Since $\varphi(z_i) = w_i$, prop. 3 of I, p. 66 implies $\varphi(\Theta_z(f_B)) = \Theta_w(\varphi \circ f_B)$. We therefore have

$$b(w) = \varphi(b) = \varphi(\Theta_z(f_B)) = \Theta_w(\varphi \circ f_B) = \Theta_w(f) = f(w),$$

by the corollary of prop. 2 of I, p. 65. Thus, we have $f|_L = b$; in particular, $f|_L$ belongs to B . This proves the proposition.

THEOREM 2 (Oka–Weil). — Let K be a polynomially convex compact subset of $\mathbf{C}^n$ and P the set of germs in a neighborhood of K of polynomial functions on $\mathbf{C}^n$ with coefficients in A . Then P is dense in $\mathcal{O}(K; A)$. More precisely, every element of $\mathcal{O}(K; A)$ is the limit of a sequence of elements of P .

Consider an element of $\mathcal{O}(K; A)$, the germ of a function $f \in \mathcal{O}(U; A)$, where U is an open neighborhood of K . By Lemma 7 of I, p. 48, there exists a compact neighborhood L of K contained in U which is polynomially convex. Let V be the interior of L ; it is a neighborhood of K .

By the preceding proposition, there exists a sequence (P_k) of polynomial functions on $\mathbb{C}^n$ with coefficients in A that converges to $f|_L$ into $\mathcal{C}(L; A)$. In particular, the sequence (P_k) converges to $f|_V$ into $\mathcal{O}(V; A)$.

By definition of the topology on $\mathcal{O}(K; A)$ (cf. EVT, II, p. 29, prop. 5), the canonical mapping from $\mathcal{O}(V; A)$ into $\mathcal{O}(K; A)$ is continuous. Consequently, the sequence of germs in a neighborhood of K of the functions P_k converges to the germ of f in the neighborhood of K in the space $\mathcal{O}(K; A)$, which is what had to be proved.

COROLLARY 1. — Let U be an open neighborhood of K . Let u_1 and u_2 be continuous maps from $\mathcal{O}(U; A)$ into a topological space X factoring through $\mathcal{O}(K; A)$. Then $u_1 = u_2$ if and only if u_1 and u_2 coincide on the set of restrictions to K of polynomial functions on $\mathbb{C}^n$ with coefficients in A .

COROLLARY 2. — Let E be a Banach space. Let K be a compact polynomially convex subset of $\mathbb{C}^n$. Let P be the set of germs in a neighborhood of K of polynomial functions on $\mathbb{C}^n$ with values in E . Then every element of $\mathcal{O}(K; E)$ is the limit of a sequence of elements of P .

Equip E with the multiplication defined by $ab = 0$ for all a and b in E (example 1 of I, p. 17). This is a commutative Banach algebra. Let A be the unital commutative Banach algebra obtained from E by adjoining a unit element. Since the canonical map $\mathcal{O}(K; A) \rightarrow \mathcal{O}(K; E)$ is continuous, the assertion follows from the Oka-Weil theorem applied to the algebra A .

For $n = 1$ and $A = \mathbb{C}$, we also have the following result, which will be made more precise by Corollary 2 of I, p. 150.

THEOREM 3 (Runge). — Let K be a compact subset of $\mathbb{C}$, and let Q be the set of germs of rational functions holomorphic in a neighborhood of K . Then Q is dense in $\mathcal{O}(K)$.

According to the definition of the topology on $\mathcal{O}(K)$, it suffices to show that for every open neighborhood U of K , and every compact subset L of U , every function $f \in \mathcal{O}(U)$ is a limit of rational functions continuous on L . One may assume that L is a compact neighborhood of K .

Let Q' be the set of restrictions to L of rational functions on $\mathbb{C}$ that are continuous on L , and let C be the closure of Q' in $\mathcal{C}(L)$. This is an algebra without radical.

Let $z \in C$ be the identity map of L . Then C is the full closed subalgebra of $\mathcal{C}(L)$ generated by z (lemma 2 of I, p. 6). We therefore have $\text{Sp}_C(z) = \text{Sp}_{\mathcal{C}(L)}(z) = L$. We can then form the

element $c = \Theta_z(f)$ in C . Since C is radical-free, applying cor. 1 of I, p. 66 to the characters $g \mapsto g(w)$ of C for all $w \in L$ shows that c coincides with the restriction of f to L . By definition of C , this proves that $f|_L$ is a uniform limit on L of elements of Q' , and this completes the proof of the theorem.

1.4.7 Existence and uniqueness of the holomorphic functional calculus

Suppose that A is a commutative unital complex Banach algebra.

DEFINITION 2. — Let $n \geq 1$ be an integer and $a \in A^n$. Let U be an open neighborhood of $\text{Sp}^n(a)$. We say that a family a' is an envelope of (a, U) if $a' \in \mathbb{C}^{n+p}$ extends a and if $U \times \mathbb{C}^p$ contains the polynomially convex envelope of $\text{Sp}^{n+p}(a')$.

Lemma 11. — Let $n \geq 1$ be an integer. Let $a \in A^n$. For every open neighborhood U of $\text{Sp}^n(a)$, there exists an envelope of (a, U) .

Let $(a_\lambda)_{\lambda \in \Lambda}$ be a family of elements of A extending the family a and topologically generating the unital Banach algebra A . Let π be the canonical projection from C^Λ onto C^n and let $U' = \pi^{-1}(U)$. Then U' is a neighborhood of $\text{Sp}^\Lambda((a_\lambda))$, and $\text{Sp}^\Lambda((a_\lambda))$ is polynomially convex (I, p. 44, lemma 4). By lemma 6 of I, p. 47, there exists a finite subset Λ_0 of Λ containing $\{1, 2, \dots, n\}$ such that $\text{pr}_{\Lambda_0}(U')$ contains the polynomially convex hull S of $\text{pr}_{\Lambda_0}(\text{Sp}^\Lambda((a_\lambda)_{\lambda \in \Lambda})) = \text{Sp}^{\Lambda_0}((a_\lambda)_{\lambda \in \Lambda_0})$. Let $p \geq 0$ be the integer such that Λ_0 has cardinality $n + p$ and let j be a bijection from $\{1, \dots, n + p\}$ into Λ_0 which coincides with the identity map on $\{1, \dots, n\}$. Since the projection of S is contained in U , the family $(a_{j(k)})_{1 \leq k \leq n+p}$ is an envelope of (a, U) .

PROPOSITION 5. — The data of the mappings Θ_a , for $n \geq 1$ and $a \in A^n$, is a holomorphic functional calculus on A , that is, conditions (CF1), (CF2), and (CF3) of I, p. 51 are satisfied.

Let $n \geq 1$ be an integer and $a = (a_1, \dots, a_n) \in A^n$. The map Θ_a satisfies $\Theta_a(z_i) = a_i$ for every i such that $1 \leq i \leq n$ by lemma 10 of I, p. 64, which proves property (CF2). Lemma 9 of I, p. 63 implies property (CF3) for the maps Θ_a .

The map Θ_a is A -linear and continuous (I, p. 61, no. 5). It satisfies $\Theta_a(1) = 1$ (lemma 9 of I, p. 63). To verify condition (CF1), it remains to establish that Θ_a is an algebra morphism. To do this, we will prove that Θ_a^U is an algebra morphism for every open neighborhood U of $\text{Sp}^n(a)$

Suppose first that U contains the polynomially convex hull K of $\text{Sp}^n(a)$. Let f_1 and f_2 be elements of $\mathcal{O}(U; A)$. There exists a sequence $(f_{1,k})$ (resp. $(f_{2,k})$) of polynomial functions which converges to f_1 (resp. to f_2) in $\mathcal{O}(K; A)$ (Theorem 2 of I, p. 68), hence in $\mathcal{O}(\text{Sp}^n(a); A)$. For every integer k , one has

$$\Theta_a^U(f_{1,k})\Theta_a^U(f_{2,k}) = \Theta_a^U(f_{1,k}f_{2,k})$$

by Lemma 10 of I, p. 64, whence $\Theta_a^U(f_1)\Theta_a^U(f_2) = \Theta_a^U(f_1f_2)$ by passing to the limit.

Consider the general case. Let $a' \in C^{n+p}$ be an envelope of (a, U) (lemma 11) and $\pi : U \times C^p \rightarrow U$ the canonical projection. Since $U \times C^p$ contains the polynomially convex envelope of $\text{Sp}^{n+p}(a')$, one has

$$\Theta_{a'}^{U \times C^p}(f_1 \circ \pi)\Theta_{a'}^{U \times C^p}(f_2 \circ \pi) = \Theta_{a'}^{U \times C^p}(f_1f_2 \circ \pi)$$

for f_1 and f_2 in $\mathcal{O}(U; A)$ by the first case. As, for every function $f \in \mathcal{O}(U; A)$, one has $\Theta_{a'}^{U \times C^p}(f \circ \pi) = \Theta_a^U(f)$ (condition (CF3) previously proved), the conclusion follows, and hence condition (CF1).

We can now prove Theorem 1 of I, p. 51. Prop. 5 shows that the family of maps $(\Theta_a)_a$ is a holomorphic functional calculus on A . It therefore remains only to establish the uniqueness of the holomorphic functional calculus on A .

Let $(\Psi_a)_a$ be a family of maps defined for every integer $n \geq 1$ and every $a \in A^n$ and satisfying conditions (CF1), (CF2), (CF3) of the holomorphic functional calculus on A (I, p. 51). It suffices to prove that for every integer $n \geq 1$, every $a \in A^n$, and every open neighborhood U of a , we have $\Theta_a^U = \Psi_a^U$.

Let $n \geq 1$ and $a = (a_1, \dots, a_n) \in A^n$. Let U be an open neighborhood of $\text{Sp}^n(a)$. Suppose first that U contains the polynomially convex hull K of $\text{Sp}^n(a)$. The morphisms Θ_a^U and Ψ_a^U coincide on polynomial functions by properties (CF1) and (CF2). By the corollary of Theorem 2 of I, p. 68 and the continuity property (CF1), these morphisms are therefore equal.

Let us prove the general case. Let $a' \in C^{n+p}$ be an envelope of (a, U) and $\pi : U \times C^p \rightarrow U$

the canonical projection. We have

$$\Theta_a^U = \Theta_{a'}^{U \times C^p} \circ \pi^* = \Psi_{a'}^{U \times C^p} \circ \pi^* = \Psi_a^U$$

by property (CF3) and the preceding case. This concludes the proof of Theorem 1 of I, p. 51.

Note that Th. 2 of I, p. 68 also entails the following uniqueness result:

PROPOSITION 6. — Let $a \in A^n$. We suppose $\text{Sp}^n(a)$ polynomially convex. Let $z_1, \dots, z_n$ be the germs in a neighbourhood of $\text{Sp}^n(a)$ of the coordinate functions on $\mathbb{C}^n$. Then the map Θ_a is the unique continuous morphism of unital algebras φ of $\mathcal{O}(\text{Sp}^n(a); A)$ to A such that $\varphi(z_1) = a_1, \dots, \varphi(z_n) = a_n$.

Lemma 10 of I, p. 64 and the corollary of Proposition 2 of I, p. 65 justify the following notations for the holomorphic functional calculus. Let $n \geq 1$ and $a \in A^n$. For every germ $f \in \mathcal{O}(K; A)$ (resp. for every holomorphic function $f \in \mathcal{O}(U; A)$ on an open neighbourhood U of $\text{Sp}^n(a)$), we set

$$f(a) = \Theta_a(f). \quad (6)$$

This notation is consistent with the notation introduced in A, IV, p. 4, n°3, if f is a polynomial, by properties (CF1) and (CF2). Properties (CF2) and (CF3) of I, p. 51 can then be written

$$z_i(a) = a_i, \quad 1 \leq i \leq n, \quad (f \circ \pi_{m,n})(a) = f(\pi_{m,n}(a)).$$

1.4.8 Substitution in the functional calculus

With the notations introduced above, the statements of Cor. 1 of I, p. 66 and Cor. 2 of I, p. 67 become respectively

$$\chi(g(a)) = g(\chi(a_1), \dots, \chi(a_n)), \quad \text{Sp}(g(a)) = g(\text{Sp}^n(a))$$

for $f \in \mathcal{O}(\text{Sp}^n(a); A)$, $\chi \in X(A)$ and $g \in \mathcal{O}(\text{Sp}^n(a))$.

We shall now prove a more general substitution property.

THEOREM 4. — Let A be a commutative unital complex Banach algebra, let $n \geq 1$ be

an integer and $a = (a_1, \dots, a_n) \in A^n$. Let $f = (f_1, \dots, f_p)$ where $f_1, \dots, f_p$ are elements of $\mathcal{O}(\text{Sp}^n(a))$. The image of $\text{Sp}^n(a)$ by the map $z \mapsto f(z) = (f_1(z), \dots, f_p(z))$ is equal to $\text{Sp}^p(f(a))$.

For every $g \in \mathcal{O}(\text{Sp}^p(f(a)); A)$, the composed germ $g \circ f$ is an element of $\mathcal{O}(\text{Sp}^n(a); A)$ and we have $g(f(a)) = (g \circ f)(a)$.

The first assertion concerning the image of $\text{Sp}^n(a)$ follows from Cor. 2 of I, p. 67. To prove the second, we shall use the following lemma.

Lemma 12. — Let K be the polynomially convex hull of $\text{Sp}^p(f(a))$. We have $g(f(a)) = (g \circ f)(a)$ for every germ $g \in \mathcal{O}(K; A)$.

Let Ψ be the map from $\mathcal{O}(K; A)$ to A such that $\Psi(g) = (g \circ f)(a)$. It is a continuous unital morphism, such that $\Psi(z_j) = f_j(a)$, where z_j is the germ of the j -th coordinate function on $\mathbb{C}^p$. When g is the germ of a polynomial function, we therefore have $\Psi(g) = g(f(a))$. By th. 2 of I, p. 68, this formula remains valid for all $g \in \mathcal{O}(K; A)$.

Now let us prove the theorem. Let V be an open neighborhood of $\text{Sp}^p(f(a))$ and $\tilde{g} \in \mathcal{O}(V; A)$ be a holomorphic function whose germ in a neighborhood of $\text{Sp}^p(f(a))$ is equal to g . Let $b \in \mathbb{C}^{p+q}$ be an envelope of $(f(a), V)$ (Lemma 11 of I, p. 70) and $\pi: V \times \mathbb{C}^q \rightarrow V$ the canonical projection.

Let $\tilde{f}_1, \dots, \tilde{f}_p$ be holomorphic functions whose germs are $f_1, \dots, f_p$ and let U be an open neighborhood of $\text{Sp}^n(a)$ such that $(\tilde{f}_1, \dots, \tilde{f}_p)(U) \subset V$. Let π' be the canonical projection from $U \times \mathbb{C}^q$ onto U . Denote by $z_{n+1}, \dots, z_{n+q}$ the q last coordinate functions on $\mathbb{C}^{n+q}$. Denote by $h = \tilde{g} \circ (\tilde{f}_1, \dots, \tilde{f}_p)$ and

$$c = (a_1, \dots, a_n, b_{p+1}, \dots, b_{p+q}) \in A^{n+q}.$$

The map $\tilde{g} \circ \pi$ is holomorphic in the open neighborhood $V \times \mathbb{C}^q$ of the polynomially convex hull L of $\text{Sp}^{p+q}(b)$. By Lemma 12, applied to c and to the germs in the neighborhood of L of the functions

$$(\tilde{f}_1 \circ \pi', \dots, \tilde{f}_p \circ \pi', z_{n+1}, \dots, z_{n+q}),$$

and to the germ of $\tilde{g} \circ \pi$, one has

$$(g \circ \pi) \left((f_1 \circ \pi')(c), \dots, (f_p \circ \pi')(c), z_{n+1}(c), \dots, z_{n+q}(c) \right) = (h \circ \pi')(c).$$

Since $\pi'(c) = a$, one has $(h \circ \pi')(c) = h(a)$ and $(f_i \circ \pi')(c) = f_i(a)$ for $1 \leq i \leq p$ (property (CF3) of the holomorphic functional calculus). Moreover, since $z_{n+j}(c) = b_{p+j}$ for $1 \leq j \leq q$. By property (CF2), we have

$$(g \circ \pi)(f_1(a), \dots, f_p(a), b_{p+1}, \dots, b_{p+q}) = h(a),$$

from which we deduce $g(f_1(a), \dots, f_p(a)) = h(a)$ by applying property (CF3) again.

1.4.9 Holomorphic functional calculus in one variable

THEOREM 5. — Let A be a unital Banach algebra, not necessarily commutative. Let a be an element of A and z the germ of the identity function of C in a neighborhood of $\text{Sp}_A(a)$. There exists a unique unital continuous morphism φ_a of $\mathcal{O}(\text{Sp}_A(a))$ to A such that $\varphi_a(z) = a$.

The image of φ_a is contained in the full closed subalgebra of A generated by a . In particular, it is contained in the bicommutant of a .

Let us prove the existence of the morphism φ_a . Let B be the closed full subalgebra of A generated by a . It is commutative, and one has $\text{Sp}_B(a) = \text{Sp}_A(a)$ (I, p. 5, no. 5). The mapping Θ_a of holomorphic functional calculus on B is a continuous unital morphism from $\mathcal{O}(\text{Sp}_B(a))$ to B such that $\Theta_a(z) = a$ (Theorem 1 of I, p. 51). The composed morphism of Θ_a and of the canonical injection of B into A is a continuous unital morphism φ_a of $\mathcal{O}(\text{Sp}_A(a))$ into A such that the image of z is a .

Let us prove uniqueness. Let φ'_a be a continuous unital morphism from $\mathcal{O}(\text{Sp}_A(a))$ to A such that $\varphi'_a(z) = a$. Then φ_a and φ'_a coincide on the set of germs of polynomials in a neighborhood of $\text{Sp}_A(a)$, hence on the set of germs of holomorphic rational fractions in a neighborhood of $\text{Sp}_A(a)$. Now these germs are dense in $\mathcal{O}(\text{Sp}_A(a))$ (I, p. 69, Th. 3). This implies that $\varphi_a = \varphi'_a$.

The construction of φ_a shows that its image is contained in the commutative subalgebra B ,

which is contained in the bicommutant of a (I, p. 6).

If the radical of the algebra A is zero, the uniqueness of the morphism φ_a is valid without requiring it to be continuous (cf. prop. 9 of I, p. 40). This is not the case in general, cf. G. R. ALLAN, Embedding the algebra of formal power series in a Banach algebra, Proc. London Math. Soc. (3) 25 (1972), 329–340.

For every Banach algebra A , every element a of A and every germ $f \in \mathcal{O}(\text{Sp}_A(a))$, we denote by $f(a)$ the element $\varphi_a(f)$ of Theorem 5. If A is a commutative Banach algebra, this element $f(a)$ coincides with the element $f(a)$ provided by the holomorphic functional calculus on a commutative Banach algebra (Theorem 1 of I, p. 51).

Let B be the closed full subalgebra of A generated by a , so that $\text{Sp}_A(a) = \text{Sp}_B(a)$. The element $f(a)$ of A belongs to B , and coincides with the element $f(a)$ computed relative to the algebra B .

PROPOSITION 7. — Let A and B be unital Banach algebras and φ a continuous unital morphism from A to B . Let $a \in A$. Then $\text{Sp}_B(\varphi(a)) \subset \text{Sp}_A(a)$ and we have $\varphi(f(a)) = f(\varphi(a))$ for all $f \in \mathcal{O}(\text{Sp}_A(a))$. In particular, for every $\chi \in X(A)$, one has $\chi(f(a)) = f(\chi(a))$.

This follows from prop. 3 of I, p. 66.

PROPOSITION 8. — Let A be a unital Banach algebra and $a \in A$. Let $f \in \mathcal{O}(\text{Sp}(a))$. We have $f(\text{Sp}_A(a)) = \text{Sp}_A(f(a))$. Moreover, for every $g \in \mathcal{O}(\text{Sp}_A(f(a)))$, one has $g \circ f \in \mathcal{O}(\text{Sp}_A(a))$ and $g(f(a)) = (g \circ f)(a)$.

This follows from th. 4.

PROPOSITION 9. — Let A be a unital Banach algebra and $a \in A$. Let U be an open neighborhood of $\text{Sp}_A(a)$ and $f \in \mathcal{O}(U)$. Let moreover V be a compact neighborhood of $\text{Sp}_A(a)$ contained in U such that V is a piece of U with oriented boundary ∂V .

For every integer $n \geq 0$, the map $z \mapsto f(z)(z - a)^{-n-1}$ is continuous on ∂V ; the differential form $z \mapsto f(z)(z - a)^{-n-1}dz$ is integrable on ∂V , and we have

$$f^{(n)}(a) = \frac{n!}{2i\pi} \int_{\partial V} f(z)(z - a)^{-n-1}dz \quad (7)$$

where $f^{(n)} \in \mathcal{O}(U)$ is the n -th derivative of f .

Proceed by induction on n . When $n = 0$, the result follows from prop. 1 of I, p. 61. Now

suppose that the assertion of the proposition is true for the integer $n \geq 0$. Let $g \in \mathcal{O}(\mathbb{C} - \text{Sp}_A(a); A)$ be the holomorphic function defined by $g(z) = (z - a)^{-n-1}f(z)$. The differential form $g'(z)dz = dg$ is of class C^1 ; since the piece V is compact, Stokes' formula (VAR, R2, p. 47, 11.2.3) implies

$$\int_{\partial V} g'(z)dz = \int_{\partial V} dg = 0.$$

Since $g'(z) = (z - a)^{-n-1}f'(z) - (n+1)(z - a)^{-n-2}f(z)$, from which one deduces

$$\int_{\partial V} f'(z)(z - a)^{-n-1}dz = (n+1) \int_{\partial V} f(z)(z - a)^{-n-2}dz.$$

Applying the induction hypothesis to f' , one therefore obtains

$$\frac{2i\pi}{n!} f^{(n+1)}(a) = (n+1) \int_{\partial V} f(z)(z - a)^{-n-2}dz,$$

which is the assertion of the proposition for the integer $n+1$. This concludes the proof.

PROPOSITION 10. — Let A be a unital Banach algebra and U an open subset of $\mathbb{C}$.

- a) The set Ω of $a \in A$ such that $\text{Sp}_A(a) \subset U$ is open in A ;
- b) Let $f \in \mathcal{O}(U)$. The map $a \mapsto f(a)$ of Ω in A is holomorphic, and in particular continuous.

Let $a \in \Omega$. There exists a compact neighborhood V of $\text{Sp}_A(a)$ contained in U which is a piece of U (VAR, R2, p. 46 and p. 47, 11.1.3, d)).

Since the resolvent of a tends to 0 at infinity (th. 1, c) of I, p. 24), the map $z \mapsto \|(z - a)^{-1}\|$ is bounded on $\mathbb{C} - \mathring{V}$. Let M denote its upper bound. If $h \in A$ is such that $\|h\| \leq (2M)^{-1}$ and if $z \in \mathbb{C} - \mathring{V}$, one has

$$z - (a + h) = (1 - h(z - a)^{-1})(z - a)$$

and $\|h(z - a)^{-1}\| \leq \frac{1}{2}$, hence $z - (a + h)$ is invertible and its inverse satisfies

$$(z - (a + h))^{-1} = (z - a)^{-1} \sum_{n=0}^{\infty} (h(z - a)^{-1})^n \quad (8)$$

with $\|(h(z-a)^{-1})^n\| \leq 2^{-n}$ (prop. 2 of I, p. 22). Thus, $\text{Sp}_A(a+h)$ is contained in V , hence in U , which proves that Ω is open in A .

Let $f \in \mathcal{O}(U)$. Denote by m the supremum of $|f(z)|$ for $z \in \partial V$. Let $a \in A$. For every $h \in A$ such that $\|h\| \leq (2M)^{-1}$, one has

$$f(a+h) = \frac{1}{2i\pi} \int_{\partial V} f(z)(z-(a+h))^{-1} dz$$

(prop. 9). The series (8) converges uniformly on the boundary of V , hence

$$f(a+h) = \sum_{n=0}^{+\infty} f_{a,n}(h)$$

where the map $f_{a,n}$ from A to A is defined by

$$f_{a,n}(h) = \frac{1}{2i\pi} \int_{\partial V} f(z)(z-a)^{-1} (h(z-a)^{-1})^n dz.$$

For every $n \in \mathbb{N}$, the function $f_{a,n}$ is a continuous homogeneous polynomial function of degree n . Moreover, it follows that

$$\|f_{a,n}(h)\| \leq \frac{mM}{\pi} \left(\int_{\partial V} \|dz\| \right) 2^{-(n+1)}$$

(INT, VI, §2, no. 3, prop. 5). The series $\sum_n f_{a,n}(h)$ is therefore absolutely convergent for $\|h\| \leq (2M)^{-1}$. This proves that the map which to a , associates $f(a)$ is holomorphic on Ω (VAR, R1, p. 26, 3.2.1).

PROPOSITION 11. — Let A be a unital Banach algebra, $a \in A$ and U an open neighborhood of $\text{Sp}_A(a)$. Denote by δ the distance from $\text{Sp}_A(a)$ to $\mathbb{C} - U$. Let $f \in \mathcal{O}(U)$.

a) For every real number η such that $0 < \eta < \delta$, there exists a real number $C \geq 0$ such that

$$\|f^{(n)}(a)\| \leq Cn!\eta^{-n} \text{ for every integer } n \in \mathbb{N};$$

b) If $b \in A$ commutes with a and if $\varrho(b) < \delta$, one has $\text{Sp}_A(a+b) \subset U$, and

$$f(a+b) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} b^n,$$

where the series converges absolutely.

Let η be a real number such that $0 < \eta < \delta$. Denote by $\varepsilon = \delta - \eta > 0$. Let K be the compact neighborhood of $\text{Sp}_A(a)$ consisting of the points of C whose distance to $\text{Sp}_A(a)$ is $\leq \varepsilon/2$. Since f is holomorphic in every open disk of radius $\eta + \varepsilon/2$ whose center belongs to K , there exists, by the Cauchy inequalities (VAR, R1, p. 29, 3.3.4), a real number $C \geq 0$ such that

$$\sup_{z \in K} \frac{|f^{(n)}(z)|}{n!} \leq \frac{C}{\eta^n}$$

for every integer $n \geq 0$. Then assertion a) follows from prop. 1 of I, p. 61 applied to $f^{(n)}$ and to a piece V contained in K .

Let b be an element of A commuting with a such that $\varrho(b) < \delta$. By replacing A by the closed full subalgebra B generated by a and b , which satisfies $\text{Sp}_A(a) = \text{Sp}_B(a)$ and $\text{Sp}_A(a+b) = \text{Sp}_B(a+b)$, one reduces, in order to prove b), to the case where A is commutative.

Since $\varrho(b) < \delta$, one can choose η such that $\varrho(b) < \eta < \delta$. Let V_1 be the set of points of C whose distance to $\text{Sp}_A(a)$ is $< \delta - \eta$, and V_2 be the open disk with center 0 and radius η in C . Let g be the map $(z_1, z_2) \mapsto z_1 + z_2$ from $V_1 \times V_2$ into U . Then $h = f \circ g$ is the map $(z_1, z_2) \mapsto f(z_1 + z_2)$ from $V_1 \times V_2$ into C . We have $\text{Sp}_A^2(a, b) \subset V_1 \times V_2$, hence $\text{Sp}_A(a+b) \subset U$ (cf. Cor. 2 of I, p. 67), and moreover $f(a+b) = h(a, b)$. By th. 4 of I, p. 72. Now, in the space $\mathcal{O}(V_1 \times V_2)$, one has

$$h(z_1, z_2) = \sum_{n \geq 0} \frac{f^{(n)}(z_1)}{n!} z_2^n,$$

therefore the series

$$\sum_{n \geq 0} \frac{f^{(n)}(a)}{n!} b^n$$

converges in A and its sum is $h(a, b) = f(a+b)$. Moreover, this series is absolutely convergent by assertion (a).

1.4.10 Exponential and logarithm

We denote by $\exp$ the complex exponential function from $\mathbb{C}$ to $\mathbb{C}$ (FVR, III, p. 8, def. 2). It is differentiable and satisfies $\exp' = \exp$ (FVR, III, p. 9, (26)); therefore it is holomorphic in $\mathbb{C}$. Let A be a unital Banach algebra and a an element of A . By prop. 2 of I, p. 65 and formula (9) of FVR, III, p. 16, we have

$$\exp(a) = \sum_{n=0}^{\infty} \frac{a^n}{n!}. \quad (9)$$

Since $\|a^n\| \leq \|a\|^n$ we see that $\|\exp(a)\| \leq \exp(\|a\|)$ and that the series (9) converges uniformly in every ball of A . The map $a \mapsto \exp(a)$ from A to A is holomorphic (prop. 10 of I, p. 76). One also sometimes denotes e^a the exponential of $a \in A$.

When a is an endomorphism of a Banach space E , the exponential $\exp(a)$ thus defined in the Banach algebra $\mathcal{L}(E)$ coincides with that defined in FVR, IV, p. 27, def. 1, according to loc. cit. prop. 7 (3).

For every element b of A that is permutable with a , we also have

$$\exp(a + b) = \sum_{n=0}^{\infty} \frac{b^n}{n!} \exp(a),$$

(prop. 11 of I, p. 77), whence

$$\exp(a + b) = \exp(a) \cdot \exp(b). \quad (10)$$

In particular, $\exp(a)$ is invertible and

$$\exp(a)^{-1} = \exp(-a). \quad (11)$$

Let B be the set of $z \in \mathbb{C}$ such that $-\pi < \operatorname{Im} z < \pi$. Let F be the complement in $\mathbb{C}$ of the interval R_- . The restriction of the exponential to B induces, by passing to subspaces, a bijection from B onto F (FVR, III, p. 10, n°7), whose inverse bijection will be denoted by $\log$.

If $a \in A$ is such that $\operatorname{Sp}_A(a) \subset F$, one can form the element $\log(a)$ of A . We have $\operatorname{Sp}_A(\log(a)) \subset B$, and

$$\exp(\log(a)) = a \quad (12)$$

by prop. 8 of I, p. 75. Conversely, let b be an element of A such that $\text{Sp}_A(b) \subset B$. We have $\text{Sp}_A(\exp(b)) \subset F$ and

$$\log(\exp(b)) = b \quad (13)$$

(loc. cit.).

In particular, if $a \in A$ is such that $\varrho(a) < 1$, one has $\text{Sp}_A(1 - a) \subset F$ and we can form $\log(1 - a)$. For $n \geq 1$, the n th derivative of $z \mapsto \log(1 - z)$ is $z \mapsto -(n - 1)!(1 - z)^{-n}$. The power series expansion of $z \mapsto \log(1 - z)$ at the point 0 is therefore

$$\log(1 - z) = - \sum_{n=1}^{\infty} \frac{z^n}{n},$$

valid for $|z| < 1$ (VAR, R1, p. 30, 3.3.9). By prop. 2 of I, p. 65, we obtain

$$\log(1 - a) = - \sum_{n=1}^{\infty} \frac{a^n}{n}. \quad (14)$$

PROPOSITION 12. — Let A be a commutative unital Banach algebra. The image of the exponential map is the connected component of the identity in the group G of invertible elements of A .

Formulas (10) and (11) prove that $\exp(A)$ is a subgroup of G . By the preceding (see formula (12)), this subgroup contains the open ball of center 1 and radius 1. It is therefore an open subgroup, and consequently closed, of G . Moreover, A is connected and the map $a \mapsto \exp(a)$ is continuous, so that $\exp(A)$ is connected. Therefore $\exp(A)$ is the identity component of G .

1.4.11 Partitions of the character space

PROPOSITION 13. — Let A be a commutative unital Banach algebra. Let U_1 and U_2 be open subsets of $X(A)$ forming a partition of $X(A)$. Then there exists a unique idempotent j in A such that the Gelfand transform $G(j)$ equals 1 on U_1 and 0 on U_2 .

Let us identify the space $X(A)$ with a compact subset of $\mathbb{C}^A$ by the map $\chi \mapsto (\chi(a))_{a \in A}$ (cf. no. 6 of I, p. 6 and cor. of th. 1 of I, p. 29). The subsets U_1 and U_2 of the uniform space $\mathbb{C}^A$ are compact and disjoint.

By TG, II, p. 31, prop. 4, there exists a finite subset M of A and disjoint open sets V_1 and V_2 of $\mathbb{C}^M$ such that

$$p(U_1) \subset V_1, \quad p(U_2) \subset V_2,$$

where p is the canonical projection of $\mathbb{C}^A$ on $\mathbb{C}^M$.

Let $a_1, \dots, a_n$ be the distinct elements of M , and identify $\mathbb{C}^M$ with $\mathbb{C}^n$. We have $\text{Sp}_A^n(a_1, \dots, a_n) \subset p(X(A)) \subset V_1 \cup V_2$ since $U_1 \cup U_2 = X(A)$. Let f be the function on $V_1 \cup V_2$ equal to 1 on V_1 and to 0 on V_2 . We have $f \in \mathcal{O}(V_1 \cup V_2)$. Set $j = f(a_1, \dots, a_n)$. Since $f^2 = f$, one has $j^2 = j$. By cor. 1 of I, p. 66, we have $\chi(j) = 1$ if $\chi \in U_1$ and $\chi(j) = 0$ if $\chi \in U_2$, which proves the existence of the required idempotent.

On the other hand, if j_1 is an idempotent of A , the relations $j^2 = j$ and $j_1^2 = j_1$ imply $(j - j_1)(j + j_1 - 1) = 0$. If $\mathcal{G}(j_1) = \mathcal{G}(j)$, the Gelfand transform of $j + j_1 - 1$ takes values in $\{-1, 1\}$, hence $j + j_1 - 1$ is invertible (prop. 6 of I, p. 37), whence $j = j_1$.

COROLLARY. — Let A be a commutative unital Banach algebra. The following assertions are equivalent:

- a) The space of characters $X(A)$ is not connected;
- b) There exists an idempotent element of A different from 0 and 1.
- c) The algebra A is isomorphic to the product of two nonzero Banach algebras.

The proposition demonstrates that *a*) implies *b*). If j is an idempotent of A , let $I_1 = jA$ and $I_2 = (1 - j)A$. Then I_1 and I_2 are closed ideals of A , and $I_1 + I_2 = A$. If $j \notin \{0, 1\}$, the ideals I_1 and I_2 are distinct from A . On the other hand, the ideal I_1 (resp. I_2) is the set of elements x of A such that $jx = x$ (resp. $(1 - j)x = x$), hence $I_1 \cap I_2 = \{0\}$. The algebra A is then identified with the product $A/I_1 \times A/I_2$. Thus, the assertion *b*) implies *c*). Finally, if A is isomorphic to $A_1 \times A_2$, the space $X(A)$ is identified with the sum space of $X(A_1)$ and of $X(A_2)$ (I, p. 6, no.

6), hence c) implies a).

PROPOSITION 14. — Let A be a commutative Banach algebra without radical. For A to admit an identity element, it is necessary and sufficient that $X(A)$ be compact.

The condition is necessary (I, p. 29, corollary). Suppose $X(A)$ compact. Let $\tilde{A}$ be the Banach algebra obtained from A by adjoining an identity element, and identify $X'(A)$ with $X(\tilde{A})$. The complement of $X(A)$ in $X(\tilde{A})$ is reduced to the character χ_0 of $\tilde{A}$ whose kernel is A . The sets $X(A)$ and $\{\chi_0\}$ are open in $X(\tilde{A})$. By prop. 13, there exists an element $j \in A$ such that $\chi(j) = 1$ for $\chi \in X(A)$, and $\chi_0(j) = 0$. We therefore have $j \in A$.

Let then x in A . We have $\chi(jx) = \chi(x)$ for all $\chi \in X(A)$, hence $jx = x$ since A is without radical. Thus, j is an identity element of A .

PROPOSITION 15. — Let A be a commutative Banach algebra, let I_1 be an ideal of A and F_1 be the set of $\chi \in X(A)$ which vanish on I_1 . Let F_2 be a subset of $X(A)$ disjoint from F_1 , closed for the Jacobson topology, and compact for the weak topology. Then there exists $u \in I_1$ such that $\mathcal{G}(u) = 1$ on F_2 .

Let I_2 be the intersection of the kernels of the characters belonging to F_2 . The Banach algebra A/I_2 is without radical (prop. 8 of I, p. 38). Since F_2 is closed for the Jacobson topology, the only elements of $X(A)$ vanishing on I_2 are those of F_2 (cf. I, p. 13). Hence F_2 , equipped with the topology induced by the weak topology of $X(A)$, is identified with $X(A/I_2)$ equipped with the weak topology (I, p. 9, no. 7). Since F_2 is weakly compact, the algebra A/I_2 possesses an identity element (prop. 14).

We then have $I_1 + I_2 = A$. Indeed, in the contrary case, $(I_1 + I_2)/I_2$ would be a proper ideal, hence contained in the kernel of a nonzero character of A/I_2 (I, p. 30, th. 2). This character would define, by composition with the canonical projection $A \rightarrow A/I_2$, a nonzero character χ of A vanishing on I_1 and I_2 , and would therefore belong to $F_1 \cap F_2$, contrary to the hypothesis.

Since $I_1 + I_2 = A$, there exists $u \in I_1$ whose class in A/I_2 is an identity element of A/I_2 . Then $\chi(u) = 1$ for all $\chi \in F_2$, which concludes the proof.

COROLLARY. — Let A be a commutative Banach algebra. Let F_1 and F_2 two disjoint subsets of $X(A)$, closed for the Jacobson topology. We assume F_2 weakly compact. Then there exists $u \in A$ such that $\mathcal{G}(u) = 1$ on F_2 and $\mathcal{G}(u) = 0$ on F_1 .

1.4.12 Partitions of the spectrum of an element

Let A be a unital Banach algebra, $x \in A$, and $K = \text{Sp}_A(x)$. Let Π denote the set of subsets of K that are both open and closed in K . Let B be the closed full subalgebra of A generated by x ; it is commutative.

For every $H \in \Pi$, there exists a unique element f_H of $\mathcal{O}(K)$ equal to 1 in a neighborhood of H and to 0 in a neighborhood of $K \setminus H$. We set $j_H = f_H(x)$. The element j_H is an idempotent of A , called associated with x and H , and we have the following formulas:

$$j_{H \cap H'} = j_H j_{H'} = j_{H'} j_H \quad (15)$$

$$(H, H' \in \Pi)$$

$$j_{H \cup H'} = j_H + j_{H'} - j_{H'} j_H \quad (16)$$

$$(H, H' \in \Pi)$$

$$j_\emptyset = 0, \quad j_K = 1.$$

Let $H \in \Pi$. We define $A_H = j_H A j_H$. This is a closed subalgebra of A , admitting the identity element j_H (cf. Lemma 1 of I, p. 2). One also sets $B_H = j_H B j_H$ and $x_H = x j_H = j_H x = j_H x j_H \in B_H$.

Let g_H be the element of $\mathcal{O}(K)$ defined by $g_H(z) = z$ in the neighborhood of H and $g_H(z) = 0$ in the neighborhood of $K \setminus H$. We have $g_H(z) = f_H(z)z$ on K , and therefore $x_H = g_H(x)$. It follows that, if $H \neq K$, one has

$$\text{Sp}_A(x_H) = g_H(K) = H \cup \{0\}.$$

Let $\lambda \in \mathbb{C} \setminus H$. Let $h_{H,\lambda}$ be the element of $\mathcal{O}(K)$ equal to $(\lambda - z)^{-1}$ in a neighborhood of H and to 0 in a neighborhood of $K \setminus H$. We have $h_{H,\lambda} = f_H h_{H,\lambda}$ and $(\lambda f_H - g_H) h_{H,\lambda} = f_H$. If we denote $R_H(x, \lambda) = h_{H,\lambda}(x)$, we therefore have $R_H(x, \lambda) \in B_H$ and

$$\begin{aligned} R_H(x, \lambda)(\lambda j_H - x_H) &= (\lambda j_H - x_H)R_H(x, \lambda) = j_H, \\ R_H(x, \lambda)j_{K-H} &= j_{K-H}R_H(x, \lambda) = 0. \end{aligned} \tag{17}$$

In particular, $\lambda \in \mathbf{C} - \text{Sp}_{A_H}(x_H)$.

Now let $\lambda \in H$. Suppose that $\lambda j_H - x_H$ admits an inverse y in A_H . Using the formulas $j_H y = y$ (since $y \in A_H$) and $j_{K-H} R_{K-H}(x, \lambda) = R_{K-H}(x, \lambda)$ (since $R_{K-H}(x, \lambda) \in A_{K-H}$), we find

$$(\lambda - x)(y + R_{K-H}(x, \lambda)) = (\lambda - x)(j_H y + j_{K-H} R_{K-H}(x, \lambda)) =$$

$$(\lambda j_H - x j_H)y + (\lambda j_{K-H} - x j_{K-H})R_{K-H}(x, \lambda) = j_H + j_{K-H} = 1,$$

(thanks to formula (17) applied to $K-H$). One checks similarly that

$$(y + R_{K-H}(x, \lambda))(\lambda - x) = 1.$$

This proves that $\lambda - x$ admits in A the inverse $y + R_{K-H}(x, \lambda)$, which is absurd. Thus we have $\lambda \in \text{Sp}_{A_H}(x_H)$. We therefore conclude that

$$\text{Sp}_{A_H}(x_H) = H. \tag{18}$$

In particular, if H is nonempty, the idempotent j_H is nonzero.

Formulas (17) and (18) prove that the function $\lambda \mapsto R_H(x, \lambda)$, defined in $\mathbf{C} - H$, is the resolvent of x_H with respect to A_H .

PROPOSITION 16. — We keep the previous notation. Let $(H_i)_{1 \leq i \leq n}$ be a partition of $\text{Sp}_A(x)$ into elements of Π .

- a) The algebra B is canonically identified with the algebra $B_{H_1} \times \cdots \times B_{H_n}$
- b) We have $x_{H_i} x_{H_j} = 0$ for $i \neq j$, and

$$x = x_{H_1} + x_{H_2} + \cdots + x_{H_n};$$

c) We have

$$R(x, \lambda) = R_{H_1}(x, \lambda) + \cdots + R_{H_n}(x, \lambda) \quad (19)$$

for all $\lambda \in \mathbb{C} - \text{Sp}_A(x)$. In particular, if $H \in \Pi$, the resolvent $\lambda \mapsto R(x, \lambda)$ is equal in the neighborhood of H to the sum of $R_H(x, \lambda)$ and of a holomorphic function.

The relation $1 = j_{H_1} + \cdots + j_{H_n}$ is a decomposition of 1 into pairwise orthogonal idempotents of B , so the algebra B is canonically identified with the product algebra $B_{H_1} \times \cdots \times B_{H_n}$ (A, I, p. 105, prop. 10).

Assertion b) follows from the corresponding relations for the functions g_{H_i} ; assertion c) is a consequence of a) and of the equality $R(x_H, \lambda) = R_H(x, \lambda)$.

PROPOSITION 17. — Let μ be an isolated point of $\text{Sp}_A(x)$. Then

a) For every $\lambda \in \mathbb{C} - \text{Sp}_A(x)$, we have

$$R(x, \lambda) = R_{\{\mu\}}(x, \lambda) + R_{\text{Sp}_A(x) - \{\mu\}}(x, \lambda);$$

b) The function $\lambda \mapsto R_{\text{Sp}_A(x) - \{\mu\}}(x, \lambda)$ is holomorphic in $\mathbb{C} - \text{Sp}_A(x)$ and in a neighborhood of μ ; moreover, the function $\lambda \mapsto R_{\{\mu\}}(x, \lambda)$ is holomorphic in $\mathbb{C} - \{\mu\}$;

c) We have

$$\lim_{n \rightarrow +\infty} \|(x - \mu)^n j_{\{\mu\}}\|^{1/n} = 0$$

and, for $\lambda \in \mathbb{C} - \{\mu\}$, the formula

$$R_{\{\mu\}}(x, \lambda) = \sum_{n=0}^{\infty} (\lambda - \mu)^{-n-1} (x - \mu)^n j_{\{\mu\}}. \quad (20)$$

The preceding implies the assertions a) and b). Let us prove c). By replacing x by $x - \mu$, one reduces to the case where $\mu = 0$. Set $H = \{0\}$; this is an open and closed subset of $\text{Sp}_A(x)$. By formula (18), the spectrum of x_H in A_H is $\{0\}$, hence x_H is quasi-nilpotent, that is, $\|x^n j_H\|^{1/n} = \|(x j_H)^n\|^{1/n}$ tends to 0 as n tends to $+\infty$. Moreover, for $\lambda \neq 0$, in A one has

$$(\lambda j_H - x_H)^{-1} = \sum_{n=0}^{\infty} \lambda^{-n-1} x_H^n$$

(Theorem 1 of I, p. 24, d)), whence (20).

COROLLARY 1. — Let μ be an isolated point of $\text{Sp}_A(x)$ and p be a strictly positive integer. For μ to be a pole of order p of the resolvent of x (cf. VAR, R1, p. 30, 3.3.9), it is necessary and sufficient that $(x - \mu)^{p-1} j_{\{\mu\}} \neq 0$ and $(x - \mu)^p j_{\{\mu\}} = 0$.

COROLLARY 2. — Let μ be an isolated point of $\text{Sp}_A(x)$. Let Γ be the oriented boundary of an open disk Δ with center μ such that

$$\text{Sp}_A(x) \cap (\Gamma \cup \Delta) = \{\mu\}.$$

Then the idempotent $j_{\{\mu\}}$ associated with x and $\{\mu\}$ is given by

$$j_{\{\mu\}} = \frac{1}{2i\pi} \int_{\Gamma} (z - x)^{-1} dz.$$

In other words, the idempotent $j_{\{\mu\}}$ is the residue at μ of the resolvent of x .

For $z \in \mathbb{C} - \text{Sp}_A(x)$, one has

$$(z - x)^{-1} = R(x, z) = R_{\{\mu\}}(x, z) + R_H(x, z),$$

where $H = \text{Sp}_A(x) - \{\mu\}$ (formula (19)). The function $z \mapsto R_H(x, z)$ is holomorphic in $\mathbb{C} - H$ and in a neighborhood of $\{\mu\}$ (prop. 17, b)), hence

$$\frac{1}{2i\pi} \int_{\Gamma} R_H(x, z) dz = 0$$

(VAR, R2, p. 48, 11.2.5). The function $z \mapsto R_{\{\mu\}}(x, z)$ is the resolvent of the element $j_{\{\mu\}} x j_{\{\mu\}}$ of the unital algebra $A_{\{\mu\}}$. We then have

$$j_{\{\mu\}} = \frac{1}{2i\pi} \int_{\Gamma} R_{\{\mu\}}(x, z) dz$$

by prop. 9 of I, p. 75 applied to $A_{\{\mu\}}$ and to the constant function 1 in a neighborhood of $\Delta \cup \Gamma$. The corollary follows.

1.4.13 Holomorphic functional calculus in a complete normable real or complex algebra

Let E be a real topological vector space. The topological vector space $C \otimes E$, the complexification of E (EVT, II, p. 65), is denoted $E_{(C)}$ and E is identified with a real topological vector subspace of $E_{(C)}$ by the map $x \mapsto 1 \otimes x$.

PROPOSITION 18. — The complex topological vector space $E_{(C)}$ is normable (resp. complete) if and only if E is normable (resp. complete).

The underlying real topological vector space of $E_{(C)}$ is isomorphic to $E \times E$. Thus $E_{(C)}$ is complete if and only if E is complete, and E is normable if $E_{(C)}$ is.

Suppose conversely that E is normable. Let p be a norm defining the topology of E and B the unit ball of p . There exists a closed balanced neighborhood V of 0 in $E_{(C)}$ contained in $B + iB$ (EVT, II, p. 66). The sets λV , where λ describes R_+^* , therefore form a fundamental system of neighborhoods of 0 in $E_{(C)}$. The gauge of V is a norm on $E_{(C)}$ that defines the topology of $E_{(C)}$, hence $E_{(C)}$ is normable.

Remark. — Let E and F be normable topological vector spaces over K . The vector space $\mathcal{L}(E; F)$ of continuous linear maps from E into F , equipped with the topology of bounded convergence, is a normable topological vector space (EVT, III, p. 14).

Let E and F be normable topological vector spaces over R . The C -linear map $\varphi: \mathcal{L}(E; F)_{(C)} \rightarrow \mathcal{L}(E_{(C)}; F_{(C)})$ defined by $\varphi(\lambda \otimes u) = \lambda u_{(C)}$ is an isomorphism of complex topological vector spaces. In particular, the dual of $E_{(C)}$ is identified with the complexification of the dual of E , and the normed algebra $\mathcal{L}(E_{(C)})$ with the complexification of the normed algebra $\mathcal{L}(E)$.

Let S be a compact subset of C stable under complex conjugation. Consider the C -algebra $\mathcal{O}(S)$ of germs of complex-valued holomorphic functions in a neighborhood of S , endowed with the structure of a complex locally convex space defined in No. 1 of I, p. 49. If U is an open neighborhood of S in C , and $h: U \rightarrow C$ be a holomorphic function, the image V of U under complex conjugation is an open neighborhood of S in C and $h^*: w \mapsto \overline{h(\overline{w})}$ is a holomorphic function on V . Passing to the inductive limit yields a continuous involution $f \mapsto f^*$ on the algebra $\mathcal{O}(S)$. In particular, we have:

$$(f + g)^* = f^* + g^* \quad (fg)^* = f^* g^* \quad (\lambda f)^* = \bar{\lambda} f^*$$

for f, g in $\mathcal{O}(S)$ and λ in $\mathbf{C}$.

We denote by $\mathcal{O}_{\mathbf{R}}(S)$ the set of germs $f \in \mathcal{O}(S)$ such that $f = f^*$. It is a closed full sub- $\mathbf{R}$ -algebra of $\mathcal{O}(S)$.

PROPOSITION 19. — Let z denote the germ in $\mathcal{O}(S)$ of the identity mapping of $\mathbf{C}$. Then $\mathcal{O}_{\mathbf{R}}(S)$ is the smallest closed full sub- $\mathbf{R}$ -algebra of $\mathcal{O}(S)$ containing z .

We have $z^* = z$, hence z belongs to $\mathcal{O}_{\mathbf{R}}(S)$. Let B be a closed full sub- $\mathbf{R}$ -algebra of $\mathcal{O}(S)$ containing z . The map $f \mapsto f + f^*$ of $\mathcal{O}(S)$ into $\mathcal{O}_{\mathbf{R}}(S)$ is continuous and surjective, and the set of germs of rational holomorphic functions in a neighborhood of S is dense in $\mathcal{O}(S)$ (Theorem 3 of I, p. 69). To show that B contains $\mathcal{O}_{\mathbf{R}}(S)$ it therefore suffices to show that if f is the germ of such a rational function, then we have $f + f^* \in B$.

There exist polynomials P and Q in $\mathbf{C}[X]$ such that Q does not vanish at any point of S and such that we have $f = \frac{P(z)}{Q(z)}$. Denote by P^* and Q^* the polynomials obtained by replacing the coefficients of P and Q by their conjugates. We then have $P(z)^* = P^*(z)$ and $Q(z)^* = Q^*(z)$. Since S is stable under complex conjugation, the polynomial Q^* vanishes at no point of S . The germs $Q^*(z)$ and $(QQ^*)(z)$ are therefore invertible in $\mathcal{O}(S)$, and

$$f + f^* = \frac{P(z)}{Q(z)} + \frac{P^*(z)}{Q^*(z)} = \frac{(PQ^* + P^*Q)(z)}{(QQ^*)(z)}.$$

Since the polynomials $PQ^* + P^*Q$ and QQ^* have real coefficients and B is a full $\mathbf{R}$ -subalgebra of $\mathcal{O}(S)$ containing z , the element $f + f^*$ belongs to B . This concludes the proof of the proposition.

Let A be a complete normable unital algebra over $\mathbf{R}$. Let x be an element of A . The spectrum of the element $1 \otimes x$ of the algebra $A_{(\mathbf{C})}$ is called the complex spectrum of x , and is denoted $\text{Sp}_{A_{(\mathbf{C})}}(x)$. Its intersection with the set $\mathbf{R}$ is none other than the spectrum $\text{Sp}_A(x)$ of x relative to A , which is sometimes called the real spectrum of x . The complex spectrum $\text{Sp}_{A_{(\mathbf{C})}}(x)$ is a compact subset of $\mathbf{C}$, stable under complex conjugation. It is nonempty when the algebra A is not reduced to 0.

Let x be an element of A . The spectral radius of $1 \otimes x \in A_{(C)}$ is equal to the spectral radius $\varrho(x)$ of x . It is the smallest real number $r \geq 0$ such that $|\lambda| \leq r$ for all $\lambda \in \text{Sp}_{A_{(C)}}(x)$. We have

$$\varrho(x) = \lim_{n \rightarrow +\infty} \|x^n\|^{1/n} = \inf_{n > 0} \|x^n\|^{1/n}$$

for every norm on A that defines the topology of A . Indeed, one may assume that the norm on A is the restriction of a norm on $A_{(C)}$ that defines the topology of $A_{(C)}$ and apply prop. 1 of I, p. 20.

We denote by $u \mapsto \bar{u}$ the endomorphism of the $\mathbf{R}$ -algebra $A_{(C)}$ which maps $\lambda \otimes a$ to $\bar{\lambda} \otimes a$. It is continuous.

Lemma. — For every $f \in \mathcal{O}(\text{Sp}_{A_{(C)}}(x))$, one has $f^*(1 \otimes x) = \overline{f(1 \otimes x)}$.

The maps $f \mapsto f(1 \otimes x)$ and $f \mapsto \overline{f^*(1 \otimes x)}$ are continuous unital homomorphisms of $\mathbf{C}$ -algebras from $\mathcal{O}(\text{Sp}(x))$ into $A_{(C)}$ which map z to $1 \otimes x$; therefore they are equal (I, p. 74, th. 5).

PROPOSITION 20. — For every $f \in \mathcal{O}_{\mathbf{R}}(\text{Sp}_{A_{(C)}}(x))$, there exists a unique element $f(x)$ of A such that $f(1 \otimes x) = 1 \otimes f(x)$ in $A_{(C)}$. The map $f \mapsto f(x)$ of $\mathcal{O}_{\mathbf{R}}(\text{Sp}_{A_{(C)}}(x))$ in A is the unique continuous unital homomorphism of $\mathbf{R}$ -algebras that maps the germ in $\mathcal{O}_{\mathbf{R}}(\text{Sp}_{A_{(C)}}(x))$ of the identity map of $\mathbf{C}$ to x .

We denote $S = \text{Sp}_{A_{(C)}}(x)$. By the above lemma, for every germ $f \in \mathcal{O}_{\mathbf{R}}(\text{Sp}(x))$, one has $f(1 \otimes x) = \overline{f^*(1 \otimes x)}$. The first assertion follows from this. Let us denote by z the germ in $\mathcal{O}_{\mathbf{R}}(S)$ of the identity mapping of $\mathbf{C}$. The map $f \mapsto f(x)$ is a continuous unital homomorphism of the $\mathbf{R}$ -algebra $\mathcal{O}_{\mathbf{R}}(\text{Sp}(x))$ into A , which maps z to x . This is the unique one by Prop. 19, since any morphism with these properties is uniquely determined on every sub- $\mathbf{R}$ -algebra of $\mathcal{O}(S)$ containing z .

Let $f \in \mathcal{O}_{\mathbf{R}}(\text{Sp}_{A_{(C)}}(x))$. The element $f(x)$ belongs to every closed full subalgebra of A containing x (prop. 19), hence belongs to the bicommutant of x in A . The complex spectrum of $f(x)$ is equal to $f(\text{Sp}(x))$ (I, p. 75, prop. 8). For all $g \in \mathcal{O}_{\mathbf{R}}(f(\text{Sp}_{A_{(C)}}(x)))$, one has $g \circ f \in \mathcal{O}_{\mathbf{R}}(\text{Sp}_{A_{(C)}}(x))$ and (loc. cit.) $g \circ f(x) = g(f(x))$.

Let U be an open subset of $\mathbf{C}$, stable under complex conjugation. The set Ω of elements x of A whose complex spectrum is contained in U is open in A (I, p. 76, prop. 10). Let f be a

holomorphic function on U such that $f^* = f$. The map $x \mapsto f(x)$ of Ω in A is analytic (loc. cit.).

Let A, B be complete normable unital associative algebras over R and $\varphi : A \rightarrow B$ a continuous unital algebra morphism. Let $x \in A$. The complex spectrum of $\varphi(x)$ is contained in that of x and, for every $f \in \mathcal{O}_{\mathbf{R}}(\text{Sp}_{A(C)}(x))$, one has $f(\varphi(x)) = \varphi(f(x))$. This follows immediately from the analogous statement in the complex case (I, p. 75, prop. 8).

1.4.14 Case of an algebra without unit element

Let A be a complete normable algebra, not necessarily unital, over $K = R$ or C . Denote by $(\tilde{A}, e)$ the unital algebra obtained from A by adjoining an identity element. It is normable and complete.

Let x be an element of A . If $K = C$, denote by $\text{Sp}'(x) = \text{Sp}_{\tilde{A}}(x)$ the spectrum of x with respect to $\tilde{A}$, and consider a germ $f \in \mathcal{O}(\text{Sp}'(x))$. If $K = R$, denote by $\text{Sp}'(x)$ the complex spectrum of the element x of $\tilde{A}$, and consider a germ $f \in \mathcal{O}_{\mathbf{R}}(\text{Sp}'(x))$. In these two cases, 0 belongs to $\text{Sp}'(x)$, and the element $f(x)$ of $\tilde{A}$ belongs to A if and only if f satisfies $f(0) = 0$. Indeed, the projection $\pi : \tilde{A} \rightarrow Ke$ is a continuous morphism whose kernel is A , and we have $\pi(f(x)) = f(\pi(x)) = f(0)$.

1.5 REGULAR COMMUTATIVE BANACH ALGEBRAS

In this section, the base field is C .

1.5.1 Definition

PROPOSITION 1. — Let A be a commutative Banach algebra. The following conditions are equivalent:

- (i) The weak topology and the Jacobson topology on $X(A)$ coincide;
- (ii) For every $\chi \in X(A)$ and every weakly closed subset F of $X(A)$ such that $\chi \notin F$, there exists an $x \in A$ such that $\mathcal{G}(x)$ is equal to 1 at χ and to 0 on F ;
- (iii) For every weakly compact subset K and every weakly closed subset F of $X(A)$ such that

$K \cap F = \emptyset$, there exists an element $x \in A$ such that $\mathcal{G}(x)$ is equal to 1 on K and to 0 on F .

Let $M \subset X(A)$. Saying that M is closed for the Jacobson topology means that, for every $\chi \in X(A) - M$, there exists an $x \in A$ such that $\mathcal{G}(x)$ vanishes on M but not at χ (lemma 2 of I, p. 39). Condition (ii) therefore means that every subset of $X(A)$ weakly closed is closed for the Jacobson topology, which shows that (ii) $\Rightarrow$ (i). Moreover, (iii) $\Rightarrow$ (ii) since the set $\{\chi\}$ is weakly compact in $X(A)$. Finally, (i) $\Rightarrow$ (iii) by the corollary of prop. 15 of I, p. 81.

DEFINITION 1. — Let A be a commutative Banach algebra. It is said to be regular if it satisfies the equivalent conditions of Proposition 1.

Note. — Let $\tilde{A}$ be the Banach algebra obtained from A by adjoining an identity element e . Condition (ii) of Prop. 1 shows that if $\tilde{A}$ is regular, then A is regular. Suppose A is regular and let us show that $\tilde{A}$ is regular. Consider subsets F and F' of $X(\tilde{A})$ which are disjoint and weakly closed (hence weakly compact), and let us construct an $x \in \tilde{A}$ such that $\mathcal{G}(x)$ vanishes on F , and is equal to 1 on F' . Let $\chi_0 \in X(\tilde{A})$ be the zero character on A . If $\chi_0 \notin F'$, by condition (iii) of prop. 1 and the hypothesis on A , there exists an element $x \in A$ such that $\mathcal{G}(x)$ vanishes on F and equals 1 on F' . If $\chi_0 \in F'$, one has $\chi_0 \notin F$; there therefore exists an element $y \in A$ such that $\mathcal{G}(y)$ vanishes on F' and equals 1 on F . The element $x = e - y$ of $\tilde{A}$ then has the required property.

Examples. — Let us return to the examples of no. 2 of I, p. 17.

The algebra of continuous complex-valued functions tending to 0 at infinity on a locally compact space X (example 3 of I, p. 17) is regular (cf. I, p. 36, example 1).

The algebra of functions n times differentiable on $[0, 1]$ (example 4 of I, p. 18) is regular (cf. I, p. 36, example 2).

If G is a commutative locally compact group and μ be a Haar measure on G , then the algebra $L^1(G, \mu)$ (example 7 of I, p. 19) is regular (cf. II, p. 219, cor. 2).

The algebra of functions that are continuous in the disk $|z| \leq 1$ and analytic in the interior (example 9 of I, p. 20) is not regular (cf. I, p. 193, exerc. 6).

PROPOSITION 2. — Let A be a commutative unital regular Banach algebra. Let $n \geq 1$ be an integer and $(U_1, \dots, U_n)$ be an open covering of $X(A)$. There exist elements $x_1, \dots, x_n$ of A with sum 1 such that $\text{Supp}(\mathcal{G}(x_i)) \subset U_i$ for $i = 1, \dots, n$.

We prove the proposition by induction on n . The assertion is valid for $n = 1$. Suppose that $n \geq 2$ and that the assertion is established for $n - 1$.

There exists an open covering $(V_1, \dots, V_n)$ of $X(A)$ such that $\overline{V_i} \subset U_i$ for every i . By the induction hypothesis, there exist elements $x, x_3, \dots, x_n \in A$ such that $x + x_3 + \dots + x_n = 1$ and $\text{Supp}(\mathcal{G}(x)) \subset V_1 \cup V_2$, $\text{Supp}(\mathcal{G}(x_i)) \subset V_i$ for $i \geq 3$. Let $K = \text{Supp}(\mathcal{G}(x)) \subset V_1 \cup V_2$. Let K_1 (resp. K_2) be the set of elements of K that do not belong to V_1 (resp. V_2). Then K_1 and K_2 are disjoint compact subsets of K . Since the Banach algebra A is regular, there therefore exists $y \in A$ such that $\mathcal{G}(y) = 1$ on K_1 and $\mathcal{G}(y) = 0$ on K_2 . Then $\mathcal{G}(xy)$ is zero on $X(A) - K$ and on K_2 , hence $\text{Supp } \mathcal{G}(xy) \subset \overline{V_2} \subset U_2$. Likewise, $\mathcal{G}(x(1 - y))$ is zero on $X(A) - K$ and on K_1 , hence $\text{Supp } \mathcal{G}(x(1 - y)) \subset \overline{V_1} \subset U_1$. The elements $x_1 = x(1 - y)$, $x_2 = xy$, and $x_3, \dots, x_n$ then satisfy the properties of the proposition.

COROLLARY 1. — Let A be a commutative unital regular Banach algebra, let I be an ideal of A , and let $f : X(A) \rightarrow \mathbb{C}$ be a continuous function. Suppose that, for every $\chi \in X(A)$, there exists an element $y_\chi \in I$ such that $f = \mathcal{G}(y_\chi)$ in a neighborhood of χ . Then there exists an element $y \in I$ such that $f = \mathcal{G}(y)$.

Since $X(A)$ is compact, there exists a finite open covering $(U_1, \dots, U_n)$ of $X(A)$, and elements $y_1, \dots, y_n$ of I such that $f = \mathcal{G}(y_i)$ on U_i . By prop. 2, there exist elements $x_1, \dots, x_n$ of A with sum 1 such that $\text{Supp}(\mathcal{G}(x_i)) \subset U_i$ for every i . Let $y = x_1 y_1 + \dots + x_n y_n$. This is an element of I that has the required property. Indeed, let $\chi \in X(A)$. For $1 \leq i \leq n$, we have $\mathcal{G}(x_i)(\chi) \mathcal{G}(y_i)(\chi) = \mathcal{G}(x_i)(\chi) f(\chi)$ since $\mathcal{G}(y_i)(\chi) = f(\chi)$ if $\chi \in U_i$, and $\mathcal{G}(x_i)(\chi) = 0$ if $\chi \notin U_i$. It follows that

$$\mathcal{G}(y)(\chi) = \sum_{i=1}^n \mathcal{G}(x_i)(\chi) \mathcal{G}(y_i)(\chi) = f(\chi) \sum_{i=1}^n \mathcal{G}(x_i)(\chi) = f(\chi).$$

COROLLARY 2. — Let A be a commutative regular Banach algebra, I an ideal of A , and $f : X'(A) \rightarrow \mathbb{C}$ be a continuous function. Suppose that, for every $\chi \in X'(A)$, there exists an element $y_\chi \in I$ such that $f = \mathcal{G}'(y_\chi)$ in a neighborhood of χ . Then there exists an element $y \in I$ such that $f = \mathcal{G}'(y)$.

Let $\tilde{A}$ be the Banach algebra obtained from A by adjoining an identity element. Then $\tilde{A}$ is regular (remark 1), and $X'(A) = X(\tilde{A})$; it therefore suffices to apply cor. 1 to $\tilde{A}$ and to the ideal

I.

If I is an ideal of a commutative Banach algebra, recall (cf. I, p. 30) that we denote by $V(I)$ the set of $\chi \in X(A)$ whose kernel contains I , in other words the set of $\chi \in X(A)$ where all the functions $\mathcal{G}(x)$ for $x \in I$ vanish. This is a subset of $X(A)$ closed for the Jacobson topology.

PROPOSITION 3. — Let A be a commutative regular Banach algebra, I an ideal of A , and K a compact subset of $X(A)$ disjoint from $V(I)$. There exists an element $x \in I$ such that $\mathcal{G}(x) = 1$ for every x in K .

This is a special case of prop. 15 of I, p. 81, taking into account the fact that the Jacobson topology coincides with the weak topology on $X(A)$.

1.5.2 Harmonic synthesis

Let A be a commutative Banach algebra. Recall that if M is a subset of $X(A)$, we denote by $\Upsilon(M)$ the intersection of the kernels of the elements of M (cf. I, p. 30); this is an ideal of A .

PROPOSITION 4. — Let A be a regular commutative Banach algebra without radical. Let F be a closed subset of $X(A)$. The set of ideals I of A such that $V(I) = F$, ordered by inclusion, admits a greatest element, namely $\Upsilon(F)$, and a least element, namely the set J of $x \in A$ such that $\mathcal{G}(x)$ has compact support disjoint from F .

By construction, $\Upsilon(F)$ is an ideal of A such that $V(\Upsilon(F))$ contains F , and it is the largest ideal of A having this property. Since F is closed, there exists an ideal I of A such that $V(I) = F$; we therefore have $I \subset \Upsilon(F)$, whence $V(\Upsilon(F)) \subset V(I) = F$ so that $V(\Upsilon(F)) = F$. This proves the first assertion.

The set J is an ideal of A and $V(J)$ contains F . Let us show that $V(J) = F$. Let $\chi \in X(A)$ not belong to F . Let U be a compact neighborhood of χ not meeting F (TG, I, p. 65, cor. of prop. 9). By assertion (ii) of prop. 1 of I, p. 88, there exists $x \in A$ such that $\mathcal{G}(x)$ is equal to 1 at χ and to 0 outside of U . We then have $x \in J$ and therefore $\chi \notin V(J)$. This shows that $V(J) \subset F$ and hence $V(J) = F$.

Finally, let I be an ideal of A such that $V(I) = F$. We show that $J \subset I$. Let $x \in J$ and let C be the support of $\mathcal{G}(x)$; the set C is a compact subset of $X(A)$ disjoint from F . By Prop. 3, there exists an element $u \in I$ such that $\mathcal{G}(u) = 1$ on C . We then have $\mathcal{G}(x) = \mathcal{G}(ux)$ and therefore

$x = ux$ since A is radical-free (prop. 8 of I, p. 38). Consequently, we have $x \in I$, which shows that $J \subset I$.

COROLLARY 1. — Let A be a regular commutative Banach algebra without radical. Let J be the set of $x \in A$ such that $\mathcal{G}(x)$ has compact support. We assume that $\bar{J} = A$. Then every closed ideal of A distinct from A is contained in a regular maximal ideal.

If I is a closed ideal of A which is contained in no regular maximal ideal, then $V(I) = \emptyset$, hence $I \supset J$ (prop. 4 applied to $F = \emptyset$), whence $I \supset \bar{J} = A$.

COROLLARY 2. — Let A be a commutative regular Banach algebra without radical. Let $x, y \in A$. If the support of $\mathcal{G}(x)$ is compact and contained in the set of characters χ such that $\mathcal{G}(y)(\chi) \neq 0$, then x is a multiple of y in A .

Let I be the ideal Ay of A . Then $V(I)$ is the set of zeros of $\mathcal{G}(y)$. Since the support F of $\mathcal{G}(x)$ is compact and disjoint from $V(I)$, we have $x \in I$ (prop. 4 applied to F).

DEFINITION 2. — Let A be a commutative Banach algebra.

Let I be an ideal of A , $x \in A$, and $\chi \in X'(A)$. We say that x belongs to I in a neighborhood of χ if there exists an element $y \in I$ such that $\mathcal{G}'(y)$ and $\mathcal{G}'(x)$ coincide in a neighborhood of χ .

We say that A satisfies the Ditkin condition if, for every $\chi \in X'(A)$ and every $x \in A$ such that $\mathcal{G}'(x)$ vanishes at χ there exists a sequence (x_n) in A such that $x = \lim_{n \rightarrow \infty} x_n x$ and such that each $\mathcal{G}'(x_n)$ vanishes in a neighborhood V_n of χ .

Remarks. — Let A be a commutative Banach algebra.

- 1) If χ is such that $\mathcal{G}'(x)$ vanishes in a neighborhood of χ , then x belongs to I in a neighborhood of χ .
- 2) If x belongs to I in a neighborhood of χ and $y \in A$ is an arbitrary element, then xy belongs to I in a neighborhood of χ .
- 3) The set of χ such that x belongs to I in a neighborhood of χ is open in $X'(A)$.
- 4) Suppose that A is regular and radical-free. If x belongs to I in a neighborhood of χ for all $\chi \in X'(A)$, then x belongs to I (cor. 2 of I, p. 91 applied to the function $f = \mathcal{G}'(x)$ and prop. 8 of I, p. 38).

5) Suppose that A is regular. Let I be an ideal of A and let χ be an element of $X(A)$ such that $\chi \notin V(I)$. Then every element x of A belongs to I in the neighborhood of χ . Indeed, by def. 1 of I, p. 89, there exists a $z \in A$ such that $\mathcal{G}'(z)$ is equal to 1 in a neighborhood of χ and equal to 0 in a neighborhood of $V(I)$. The support of $\mathcal{G}(z)$ is compact and hence we have $z \in I$ (prop. 4 applied to $V(I)$); therefore $xz \in I$, and $\mathcal{G}'(xz) = \mathcal{G}'(x)$ in a neighborhood of χ .

Recall that a subspace K of a topological space X is said to be perfect if it is closed and has no isolated point (TG, I, p. 8).

Lemma 1. — Let A be a commutative regular Banach algebra without radical, satisfying Ditkin's condition. Let I be a closed ideal of A and let x be an element of $\Upsilon(V(I))$. Let K be the set of $\chi \in X'(A)$ such that x does not belong to I in a neighborhood of χ . Then the set K is a perfect subset of $X'(A)$.

Let G denote the complement of K in $X'(A)$. The set G is open in $X'(A)$ (remark 3), hence K is closed.

Proceed by contradiction, and suppose that K has an isolated point χ_0 . Let U be a neighborhood of χ_0 such that $U - \{\chi_0\} \subset G$. Since x does not belong to I in a neighborhood of χ_0 , Remark 5 shows that $\chi_0 \in V(I)$. In particular, we have $\chi_0(x) = 0$ since $x \in \Upsilon(V(I))$.

We are going to show that there exists an element y of A which belongs to I in a neighborhood of every point of $X'(A) - \{\chi_0\}$, which does not belong to I in a neighborhood of χ_0 , and such that $\chi_0(y) = 0$.

Suppose first that the existence of such an element y has been proved. Since A satisfies Ditkin's condition, there then exists a sequence (x_n) in A such that $x_n y$ tends to y and such that each $\mathcal{G}'(x_n)$ vanishes in a neighborhood of χ_0 . For every n , the element $x_n y$ then belongs to I in a neighborhood of every point of $X'(A)$ (Remarks 1 and 2) and therefore $x_n y \in I$ (remark 4). Since I is closed, we deduce that $y \in I$, which contradicts the fact that y does not belong to I in a neighborhood of χ_0 .

It remains to prove the existence of y . If $\chi_0 \neq 0$, by assertion (iii) of prop. 1 of I, p. 88, there exists a $u \in A$ such that $\mathcal{G}'(u)$ is equal to 1 in a neighborhood of χ_0 and equal to 0 in a neighborhood of $X'(A) - U$. Let $y = ux$. Since x belongs to I in a neighborhood of χ for all

$\chi \in U - \{\chi_0\}$, the same is true of y . Moreover, if $\chi \in X'(A) - U$, then $\mathcal{G}'(y)$ vanishes in a neighborhood of χ . Therefore (Remark 5) the element $y = ux$ belongs to I in a neighborhood of every $\chi \neq \chi_0$. Since $\mathcal{G}'(y)$ coincides with $\mathcal{G}'(x)$ in a neighborhood of χ_0 , the fact that χ_0 belongs to K implies that y does not belong to I in a neighborhood of χ_0 . Finally, we have $\chi_0(y) = \chi_0(u)\chi_0(x) = 0$.

If $\chi_0 = 0$, there similarly exists an element $v \in A$ such that $\mathcal{G}'(v)$ is zero in a neighborhood of χ_0 and equal to 1 in a neighborhood of $X'(A) - U$; as before, we deduce from this that the element $y = x - vx$ belongs to I in a neighborhood of every $\chi \neq \chi_0$, that it does not belong to I in a neighborhood of χ_0 , and that $\chi_0(y) = 0$.

Lemma 2. — Let X be a topological space. Let F and D be disjoint subspaces of X such that F is closed and D is discrete. If F contains no nonempty perfect subspace, then neither does $F \cup D$.

Indeed, suppose that K is a nonempty perfect subspace of $F \cup D$. Let x be a point of K . If x belongs to D , it is isolated in D , hence also in $F \cup D$ since F is closed. Therefore x is isolated in K , which contradicts the hypotheses.

PROPOSITION 5. — Let A be a regular commutative Banach algebra without radical, satisfying Ditkin's condition. Let I be a closed ideal of A such that the boundary F of $V(I)$ contains no nonempty perfect set. Then $I = Y(V(I))$, that is, I is the set of $x \in A$ such that $G(x)$ vanishes on $V(I)$. In particular, if $V(I)$ reduces to a single point χ , we have $I = \text{Ker}(\chi)$.

We have $I \subset \Upsilon(V(I))$. Now let $x \in \Upsilon(V(I))$. Let G be the set of characters $\chi \in X'(A)$ such that x belongs to I in a neighborhood of χ . It is open and its complement K is perfect (lemma 1). Since $\mathcal{G}'(x)$ is zero on $V(I)$, the set G contains the interior of $V(I) \cup \{0\}$ (remark 1). It also contains $X(A) - V(I)$ by remark 5. Therefore $K = X'(A) - G$ is contained in the boundary F_0 of $V(I) \cup \{0\}$.

We have $F_0 \subset F \cup \{0\}$. The hypothesis therefore implies that F_0 contains no nonempty perfect set (lemma 2). It follows therefore from lemma 1 that the perfect set K is empty. Hence x belongs to I in a neighborhood of every $\chi \in X'(A)$, which means that $x \in I$ (remark 4). Thus $\Upsilon(V(I)) \subset I$, which concludes the proof.

1.6 STELLAR ALGEBRAS

In this paragraph, the base field is $\mathbb{C}$.

1.6.1 Semilinear involutions

Let E be a complex vector space. A semilinear involution on E is an $\mathbb{R}$ -linear map from E to E such that $u \circ u = Id_E$ and $u(\lambda x) = \overline{\lambda}u(x)$ for all $\lambda \in \mathbb{C}$ and every $x \in E$. We then denote by E^u the real vector subspace of E formed by the elements $x \in E$ such that $u(x) = x$.

Lemma 1. — Let E be a complex vector space and let u be a semilinear involution on E . Let $x \in E$; set

$$x_1 = \frac{1}{2}(x + u(x)), \quad x_2 = \frac{1}{2i}(x - u(x)).$$

The pair (x_1, x_2) is the unique element of $E^u \times E^u$ such that $x = x_1 + ix_2$. The elements x_1 and x_2 satisfy $x_1 + ix_2 = x$ and belong to E^u since $u(u(x)) = x$. Conversely, if y_1 and y_2 are in E^u and satisfy $x = y_1 + iy_2$, it follows that $u(x) = u(y_1) + u(iy_2) = y_1 - iy_2$, hence

$$y_1 = \frac{1}{2}(x + u(x)) = x_1, \quad iy_2 = \frac{1}{2}(x - u(x)) = ix_2.$$

PROPOSITION 1. — Let E_1 and E_2 be complex vector spaces, and let u_1 and u_2 be semilinear involutions on E_1 and E_2 respectively. Let f be a linear map from E_1 into E_2 . Then $f \circ u_1 = u_2 \circ f$ if and only if $f(E_1^{u_1}) \subset E_2^{u_2}$.

If $f \circ u_1 = u_2 \circ f$, we immediately obtain $f(E_1^{u_1}) \subset E_2^{u_2}$. Conversely, suppose this condition is satisfied. Let $x \in E_1$. Write $x = x_1 + ix_2$ with $(x_1, x_2) \in E_1^{u_1} \times E_1^{u_1}$ (lemma above). We then have $f(u_1(x)) = f(x_1) - if(x_2)$ and $u_2(f(x)) = u_2(f(x_1)) - iu_2(f(x_2)) = f(x_1) - if(x_2) = f(u_1(x))$.

1.6.2 Involutive algebras

DEFINITION 1. — Let A be an algebra over $\mathbb{C}$. An involution on A is a map $x \mapsto x^*$ from A to A such that:

$$(x^*)^* = x, \quad (x + y)^* = x^* + y^*, \quad (\lambda x)^* = \overline{\lambda} x^* \\ (xy)^* = y^* x^*$$

for all $x, y \in A$ and $\lambda \in C$. An algebra over C equipped with an involution is called an involutive algebra.

An involution on A is in particular an isomorphism of the ring A onto the opposite ring A° .

Let A be an involutive algebra. One says that x^* is the adjoint of x . A subset of A stable under the involution is said to be self-adjoint. If A has a unit element e , one has $e^* = e$; one says that (A, e) is a unital involutive algebra. An element u of a unital involutive algebra is said to be unitary if $uu^* = u^*u = e$, in other words if u is invertible and its inverse is u^* .

An element $x \in A$ is said to be hermitian if $x = x^*$ and normal if $xx^* = x^*x$. This terminology generalizes that of A , IX, § 7, no. 3. Every hermitian element is normal, and every unitary element is normal. The set A_h of hermitian elements of A is a real vector subspace of A . If x and y are hermitian and commute, one has $(xy)^* = y^*x^* = yx = xy$, hence xy is hermitian. For every $x \in A$, the elements xx^* and x^*x of A are hermitian.

If $A = C$ equipped with the involution $z \mapsto \bar{z}$, one has $A_h = R$.

Lemma 2. — Let A be an involutive algebra and $x \in A$. The elements

$$x_1 = \frac{1}{2}(x + x^*), \quad x_2 = \frac{1}{2i}(x - x^*)$$

are hermitian and satisfy $x = x_1 + ix_2$. If $x = y_1 + iy_2$, where y_1 and y_2 are hermitian, then $x_1 = y_1$ and $x_2 = y_2$. Moreover, the element x is normal if and only if x_1 and x_2 commute.

The first two assertions follow from lemma 1 of I, p. 95. One computes that $xx^* - x^*x = 2i(x_2x_1 - x_1x_2)$, hence x is normal if and only if x_1 and x_2 commute.

Let A be a unital involutive algebra. For $x \in A$ to be invertible, it is necessary and sufficient that x^* be so, and one then has $(x^*)^{-1} = (x^{-1})^*$. Since $(x - \lambda e)^* = x^* - \overline{\lambda}e$ for all $\lambda \in C$, one deduces that $\text{Sp}_A(x^*) = \overline{\text{Sp}_A(x)}$.

Let A be an involutive algebra and $\widetilde{A}$ the algebra obtained from A by adjoining a unit element. There exists on $\widetilde{A}$ a unique involution extending that of A , given by $(\lambda, x)^* = (\overline{\lambda}, x^*)$ for $\lambda \in C$ and $x \in A$. If $x \in A$, one has $\text{Sp}'_A(x^*) = \overline{\text{Sp}'_A(x)}$.

Let A and B be involutive algebras. A morphism from A to B is an algebra morphism φ from A to B such that $\varphi(x^*) = \varphi(x)^*$ for all x in A . The identity map of A is a morphism of involutive algebras. If C is an involutive algebra and $\pi : B \rightarrow C$ a morphism of involutive algebras, then $\pi \circ \varphi$ is a morphism of involutive algebras. If φ is an isomorphism of algebras, then φ^{-1} is a morphism of involutive algebras, and one says that φ is an isomorphism of involutive algebras.

By prop. 1 of I, p. 95, if A and B are involutive algebras, an algebra morphism φ from A to B is a morphism of involutive algebras if and only if $\varphi(A_h) \subset B_h$. One calls an involutive subalgebra of A a self-adjoint subalgebra. The center of A is an involutive subalgebra. If A_1 is a self-adjoint two-sided ideal of A , the involution of A defines by passage to the quotient an involution on the algebra A/A_1 , and the canonical map from A onto A/A_1 is a morphism of involutive algebras.

Let A be an involutive algebra. The radical of A is equal to the radical of the opposite algebra (A , VIII, p. 431, prop. 7), and is therefore self-adjoint.

Let A be an involutive algebra. If $M \subset A$ is self-adjoint, its commutant M' is an involutive subalgebra of A . If $x \in A$, the bicommutant of $\{x, x^*\}$ is an involutive subalgebra containing x and x^* and this subalgebra is commutative if and only if x is normal (no. 5 of I, p. 5).

Remark. — Let A be an involutive algebra and let B be a maximal commutative involutive subalgebra of A . Then B is a maximal commutative subalgebra. In particular, if A is unital, then the algebra B is full.

Indeed, let $x \in A$ be an element commuting with B . Then x^* commutes with B . Write $x = x_1 + ix_2$ with x_1 and x_2 hermitian; the elements x_1 and x_2 commute with B (lemma 2). The subalgebra of A generated by B and x_1 is therefore commutative and involutive. Consequently, it is equal to B , so that $x_1 \in B$. Similarly, we have $x_2 \in B$, and hence finally $x \in B$.

Let A be an involutive algebra. If f is a linear form on A , the map $x \mapsto \overline{f(x^*)}$ on A is a linear form on A , which is denoted by f^* . The map $f \mapsto f^*$ is a semilinear involution on A' . We say that f is Hermitian if $f = f^*$. By lemma 1 of I, p. 95, every linear form f on A has a unique representation $f = f_1 + if_2$ where f_1 and f_2 are Hermitian, namely $f_1 = \frac{1}{2}(f + f^*)$ and $f_2 = \frac{1}{2i}(f - f^*)$.

For a linear form f to be Hermitian, it is necessary and sufficient that the restriction of f to

A_h be real-valued (proposition 1 of I, p. 95). The map $f \mapsto f|_{A_h}$ is an isomorphism of the real vector space of Hermitian forms on the dual vector space of the real vector space A_h .

In particular, we denote by $X'(A)_h$ (resp. $X(A)_h$) the set of Hermitian characters of A (resp. the set of nonzero Hermitian characters of A). A character is therefore Hermitian if its restriction to A_h takes real values.

If A is commutative and if χ is a character of A , then χ^* is a character of A , and the map $\chi \mapsto \chi^*$ is a homeomorphism from $X'(A)$ onto $X'(A)$.

Lemma 3. — Let A be a commutative involutive algebra. The Gelfand transform of A to $\mathcal{C}_0(X(A))$ is a morphism of involutive algebras if and only if every character of A is Hermitian.

Indeed, saying that G_A is a morphism of involutive algebras amounts to saying that, for all $x \in A$ and $\chi \in X(A)$, one has

$$\chi(x^*) = \mathcal{G}_A(x^*)(\chi) = \overline{\mathcal{G}_A(x)(\chi)} = \overline{\chi(x)},$$

that is, every χ is Hermitian.

Examples. — 1) Let A be the algebra of complex-valued functions on a set X . The map $f \mapsto \bar{f}$ is an involution in A . The subalgebra of bounded functions in X is an involutive subalgebra of A . If X is a locally compact topological space, the subalgebras $\mathcal{C}(X)$, $\mathcal{C}_b(X)$, $\mathcal{C}_0(X)$ and $\mathcal{K}(X)$ are involutive subalgebras of A .

2) Let X be a locally compact topological space and μ a positive measure on X . The map $f \mapsto \bar{f}$ is an involution on the algebra $\mathcal{L}^\infty(X, \mu)$; it induces, by passing to the quotient, an involution on the unital algebra $L^\infty(X, \mu)$.

3) Let E be a complex Hilbert space. On the Banach algebra $\mathcal{L}(E)$, the map $x \mapsto x^*$ (EVT, V, p. 37, prop. 1) is an involution.

4) Let G be a locally compact group. Let $\mathcal{M}^1(G)$ be the Banach algebra of bounded complex measures on G (example 6 of I, p. 19).

The map $x \mapsto x^{-1}$ from G to G maps every measure $\mu \in \mathcal{M}^1(G)$ into a measure $\check{\mu} \in \mathcal{M}^1(G)$ (INT, VII, p. 12, formula (13)). We denote by μ^* the complex conjugate measure of $\check{\mu}$. The map

$\mu \mapsto \check{\mu}$ is an isometric isomorphism from the Banach algebra $\mathcal{M}^1(G)$ onto the Banach algebra $\mathcal{M}^1(G^\circ)$ (INT, VIII, §3, no. 1, cor. of prop. 7); hence $\mu \mapsto \mu^*$ is an isometric involution of the Banach algebra $\mathcal{M}^1(G)$.

The set A of bounded measures admitting a density with respect to a Haar measure is a closed subalgebra of $\mathcal{M}^1(G)$ stable under the involution (cf. INT, VIII, §4, no. 5); it does not depend on the choice of a Haar measure.

Let ν be a left Haar measure on G and denote by Δ the module of G . We equip $L^1(G, \nu)$ with the product $(f, g) \mapsto f * \nu g$ and with the involution $f \mapsto f^* = \tilde{f} \cdot \Delta^{-1}$, where $\tilde{f}(x) = \overline{f(x^{-1})}$ for all $x \in G$. Then the map $f \mapsto f \cdot \nu$ is an isomorphism of the involutive algebra $L^1(G, \nu)$ onto A . This isomorphism is isometric. In particular, $L^1(G, \nu)$ is identified with an involutive subalgebra of $\mathcal{M}^1(G)$.

5) Let U be an open subset of $\mathbb{C}$ stable under complex conjugation. Consider the algebra $\mathcal{O}(U)$ of complex-valued holomorphic functions on U . For every function $f \in \mathcal{O}(U)$, the map $f^* : z \mapsto \overline{f(\bar{z})}$ is a holomorphic function on U . The map $f \mapsto f^*$ is an involution on $\mathcal{O}(U)$.

Similarly, let S be a compact subset of $\mathbb{C}$ stable under complex conjugation. Consider the algebra $\mathcal{O}(S)$ of germs of complex-valued holomorphic functions in a neighborhood of S . The map $f \mapsto f^*$ is an involution on $\mathcal{O}(S)$.

The subalgebra $\mathcal{O}_{\mathbf{R}}(U)$ (resp. the subalgebra $\mathcal{O}_{\mathbf{R}}(S)$) defined in no. 13 of I, p. 85 is the set of Hermitian elements of $\mathcal{O}(U)$ (resp. of $\mathcal{O}(S)$).

1.6.3 Normed involutive algebras

DEFINITION 2. — A normed involutive algebra is a normed algebra A equipped with an involution $x \mapsto x^*$ such that $\|x^*\| = \|x\|$ for all x . If A is a Banach algebra, then A is said to be an involutive Banach algebra.

Examples. — 1) Let X be a locally compact topological space. The Banach algebra $\mathcal{C}_b(X)$ of continuous and bounded complex functions on X , equipped with the norm $\|f\| = \sup_{x \in X} |f(x)|$ and with the involution $f \mapsto \bar{f}$, is an involutive Banach algebra. The subalgebra $\mathcal{C}_0(X)$ of

continuous functions tending to 0 at infinity is a closed involutive subalgebra of $\mathcal{C}_b(X)$.

- 2) Let X be a locally compact topological space and μ a positive measure on X . The involutive algebra $L^\infty(X, \mu)$ (example 2 of I, p. 99) is an involutive Banach algebra, since $|f| = |\bar{f}|$ for every element $f \in L^\infty(X, \mu)$.
- 3) The involutive algebra $\mathcal{L}(E)$ of continuous endomorphisms of a complex Hilbert space E (I, p. 99, example 3), equipped with the norm

$$\|u\| = \sup_{\substack{x \in E \\ \|x\| \leq 1}} \|u(x)\|$$

(EVT, III, p. 14) is an involutive Banach algebra (EVT, V, p. 37, prop. 1).

- 4) The involutive algebra $\mathcal{M}^1(G)$ of bounded measures on a locally compact group (I, p. 99, example 4), equipped with the usual norm (example 6 of I, p. 19), is an involutive Banach algebra. Let ν be a left Haar measure on G . The involutive Banach algebra $L^1(G, \nu)$ identifies with a closed subalgebra of $\mathcal{M}^1(G)$.
- 5) Let (A_i) be a family of involutive normed algebras. Let A be the product normed algebra of the A_i (No. 1 of I, p. 15). The algebra A , equipped with the involution $(x_i)^* = (x_i^*)$, is an involutive normed algebra. If each of the algebras A_i is an involutive Banach algebra, then A is an involutive Banach algebra. We say that A is the product involutive normed algebra (resp. the product involutive Banach algebra) of A_i .
- 6) Let A be a normed involutive algebra, and let $\tilde{A}$ be the normed algebra deduced from A by adjoining a unit element. Equipped with the involution defined in no. 2, the algebra $\tilde{A}$ is a normed involutive algebra. If A is an involutive Banach algebra, then $\tilde{A}$ is also an involutive Banach algebra.

If A is a normed involutive algebra, the closure of an involutive subalgebra is an involutive subalgebra. If $M \subset A$, the smallest closed involutive subalgebra containing M is called the closed involutive subalgebra generated by M ; it is the closure of the subalgebra generated by $M \cup M^*$. If M consists of a single normal element, the closed involutive subalgebra generated by

M is commutative, and all its elements are normal.

Similarly, if A is a unital involutive normed algebra and M a subset of A , the smallest closed unital involutive subalgebra containing M is called the closed unital involutive subalgebra generated by M ; it is the closure of the unital subalgebra generated by $M \cup M^*$. If M consists of a single normal element, the closed unital involutive subalgebra generated by M is commutative, and all its elements are normal.

The quotient of an involutive normed algebra by a closed two-sided self-adjoint ideal, the completion, and the opposite of an involutive normed algebra are naturally involutive normed algebras.

If A is a normed involutive algebra, the set A_h of the Hermitian elements of A is a real normed vector space.

Lemma 4. — Let A be a normed involutive algebra. For every continuous linear form f on A , we have $\|f^*\| = \|f\|$. If moreover f is Hermitian, then $\|f\| = \|f|_{A_h}\|$.

The first assertion follows from the definitions. For the second, let g be the restriction of f to A_h . We have $\|f\| \geq \|g\|$. Let us prove the reverse inequality. For every $\varepsilon > 0$, there exists $x \in A$ such that $\|x\| \leq 1$ and $|f(x)| \geq \|f\| - \varepsilon$. By multiplying x by a complex number of modulus 1, we may assume $f(x) \geq 0$. Then the element $\frac{1}{2}(x + x^*)$ belongs to A_h and has norm ≤ 1 . We have

$$\left| g\left(\frac{1}{2}(x + x^*)\right) \right| = \frac{1}{2}|f(x) + f(x^*)| = f(x) \geq \|f\| - \varepsilon$$

hence $\|g\| \geq \|f\| - \varepsilon$. From this we deduce that $\|g\| \geq \|f\|$.

In what follows, we shall identify continuous Hermitian linear forms on A with continuous real linear forms on A_h .

Lemma 5. — Let A be an involutive Banach algebra.

- a) For every $x \in A$, one has $\exp(x)^* = \exp(x^*)$;
- b) Let $x \in A_h$ be a Hermitian element. Then $\exp(ix)$ is unitary.

Indeed, since the involution on A is continuous, we have

$$\exp(x)^* = \left(\sum_{n=0}^{\infty} \frac{x^n}{n!} \right)^* = \sum_{n=0}^{\infty} \frac{(x^*)^n}{n!} = \exp(x^*)$$

for all $x \in A$ (formula (9) of I, p. 78). If $x \in A_h$, it then follows that

$$\exp(ix)^* = \exp((ix)^*) = \exp(-ix) = \exp(ix)^{-1}$$

(formula (11) of I, p. 78).

1.6.4 Stellar algebras

DEFINITION 3. — A stellar algebra is an involutive Banach algebra A such that $\|x\|^2 = \|x^*x\|$ for all $x \in A$.

If A and B are stellar algebras, a morphism, or morphism of stellar algebras, from A to B is a morphism of involutive algebras from A to B . An isomorphism from A to B is an isomorphism of involutive algebras from A to B .

Some authors speak of "C*-algebra".

Let A be a stellar algebra. A stellar subalgebra of A is a closed involutive subalgebra of A .

Examples. — 1) The involutive Banach algebra of continuous endomorphisms of a complex Hilbert space (example 3 of I, p. 100) is a stellar algebra (EVT, V, p. 39, prop. 2).

2) Let X be a locally compact topological space. The involutive Banach algebra $\mathcal{C}_b(X)$ of continuous bounded complex-valued functions on X (example 1 of I, p. 100) is a stellar algebra. Indeed, for every function $f \in \mathcal{C}_b(X)$, one has $f^*f = |f|^2$, and hence $\|f^*f\| = \||f|^2\| = \|f\|^2$.

Let $A = \mathcal{C}_0(X)$ be the involutive Banach subalgebra of functions tending to 0 at infinity. It is a stellar subalgebra of $\mathcal{C}_b(X)$. For every function $f \in A$, one has $\|f\| = \varrho(f)$ since $\text{Sp}'_A(f) = f(X) \cup \{0\}$.

Let X and Y be locally compact topological spaces. For every proper partial map φ from X into Y (def. 1 of I, p. 33), the algebra morphism φ^* of $\mathcal{C}_0(Y)$ into $\mathcal{C}_0(X)$ (prop. 3 of I, p. 34) is a morphism of involutive algebras. Conversely, every morphism of stellar algebras $\pi: \mathcal{C}_0(Y) \rightarrow \mathcal{C}_0(X)$ is of this form (loc. cit.).

- 3) Let X be a compact topological space and $x_0 \in X$ a fixed element of X . The subalgebra $\mathcal{C}'(X)$ of $\mathcal{C}(X)$ of continuous functions $f : X \rightarrow \mathbb{C}$ such that $f(x_0) = 0$ is a stellar algebra.
- 4) Let X be a separated topological space and μ a positive measure on X . The involutive Banach algebra $L^\infty(X, \mu)$ is a commutative unital stellar algebra.
- 5) Let (A_i) be a family of stellar algebras. The product involutive Banach algebra A of the A_i (example 5 of I, p. 101) is a stellar algebra, called the product stellar algebra of the A_i .
- 6) Let A be a stellar algebra. If B is a closed involutive subalgebra of A , then B is a stellar algebra. It will be seen (V, to appear) that every stellar algebra is isomorphic to a closed involutive subalgebra of the stellar algebra of endomorphisms of a Hilbert space (example 1).
- 7) Let A be a stellar algebra. If $M \subset A$ is any subset, then the closed involutive subalgebra of A generated by M is a stellar algebra, called the stellar subalgebra of A generated by M . If A is moreover unital, then the closed unital involutive subalgebra generated by M is a unital stellar algebra, called the unital stellar subalgebra of A generated by M .
- 8) In general, the involutive Banach algebra $\mathcal{M}^1(G)$ (example 4 of I, p. 100) is not a stellar algebra (I, p. 181, exerc. 8).

Lemma 6. — Let A be a Banach algebra equipped with an involution satisfying

$$\|x\|^2 \leq \|x^*x\| \quad (1)$$

for all $x \in A$. Then A is a stellar algebra.

Let $x \in A$. We then have $\|x\|^2 \leq \|x^*\| \cdot \|x\|$, whence $\|x\| \leq \|x^*\|$. By exchanging the roles of x and x^* we see that $\|x\| = \|x^*\|$. Thus $\|x^*x\| \leq \|x^*\| \|x\| \leq \|x\|^2$ and the hypothesis implies the equality $\|x\|^2 = \|x^*x\|$.

Lemma 7. — Let A be a stellar algebra.

- a) The regular representation γ of A (def. 1 of I, p. 16) is isometric, that is,

$$\|x\| = \sup_{\|y\| \leq 1} \|xy\|,$$

for all $x \in A$.

b) For every $x \in A_h$, one has

$$\varrho(x) = \|x\|. \quad (2)$$

Let $x \in A$. We have $\sup_{\|y\| \leq 1} \|xy\| \leq \|x\|$. To prove that $\|x\| \leq \sup_{\|y\| \leq 1} \|xy\|$, we may assume $\|x\| = 1$. Then the element $y = x^*$ satisfies $\|y\| = \|x^*\| = 1$, and $\|xy\| = \|x\|^2 = 1$, whence the assertion *a*).

Suppose that x is Hermitian. It follows $\|x^2\| = \|x^*x\| = \|x\|^2$, whence by induction $\|x^{2^n}\|^{2^{-n}} = \|x\|$ for every integer $n \geq 1$, whence the assertion *b*) by prop. 1 of I, p. 20.

Remarks. — 1) Let A be a unital stellar algebra. Then we have

$$\|1\|^2 = \|1^*1\| = \|1\|,$$

therefore the norm $\|1\|$ is zero or equal to 1. If $A \neq \{0\}$, we deduce $\|1\| = 1$. Consequently, for every unitary element u of A , we have $\|u\| = \|u^*u\|^{1/2} = 1$.

2) Let A be a normed involutive algebra. If $\|x\|^2 = \|x^*x\|$ for all $x \in A$, the completion $\widehat{A}$ of A is a stellar algebra.

PROPOSITION 2. — Let A be an involutive Banach algebra, let B be a stellar algebra, and let π be a morphism of involutive algebras from A to B . Then one has $\|\pi(x)\| \leq \|x\|$ for all $x \in A$, and in particular π is continuous.

For every $x \in A$, one has $\text{Sp}'_B(\pi(x)) \subset \text{Sp}'_A(x)$, hence $\varrho(\pi(x)) \leq \varrho(x) \leq \|x\|$. Since $\pi(x^*x) \in B_h$, one has $\|\pi(x^*x)\| = \varrho(\pi(x^*x))$ (formula (2)), hence

$$\|\pi(x)\|^2 = \|\pi(x^*x)\| = \varrho(\pi(x^*x)) \leq \|x^*x\| = \|x\|^2.$$

PROPOSITION 3. — Let A be a stellar algebra and let $\widetilde{A}$ be the involutive algebra obtained

from A by adjoining a unit element. There exists a unique norm on $\tilde{A}$ extending that of A and making $\tilde{A}$ a stellar algebra.

The uniqueness of such a norm follows from Prop. 2. We now show its existence. Let us denote by $\tilde{e}$ the identity element of $\tilde{A}$.

First, suppose that A has a unit element e . The product of the involutive normed algebras A and $\mathbb{C}(\tilde{e} - e)$ is identified with $\tilde{A}$ and is a stellar algebra (example 5). The norm on $\tilde{A}$ extends that of A , whence the assertion.

Now suppose that A has no identity element. For every $x \in \tilde{A}$, let γ_x be the multiplication operator $y \mapsto xy$ from A to A , and set $\|x\|_{\tilde{A}} = \|\gamma_x\|$. The map $x \mapsto \|x\|_{\tilde{A}}$ is a semi-norm on $\tilde{A}$. For all x and x' in $\tilde{A}$, one has $\|xx'\|_{\tilde{A}} \leq \|x\|_{\tilde{A}}\|x'\|_{\tilde{A}}$. Moreover, by Lemma 7, we have $\|x\|_{\tilde{A}} = \|x\|$ for all $x \in A$.

Let us show that the map $x \mapsto \|x\|_{\tilde{A}}$ is a norm on $\tilde{A}$. Let $\lambda \in \mathbb{C}$ and $x \in A$ such that $\|\lambda\tilde{e} + x\|_{\tilde{A}} = 0$. If $\lambda \neq 0$ the condition $(\lambda\tilde{e} + x)y = 0$ for all $y \in A$ implies that $-\lambda^{-1}x$ is a left identity element in A . Likewise, the element $-\overline{\lambda}^{-1}x^*$ is a right identity element. Thus, the algebra A would then have an identity element, contrary to the hypothesis. Hence, $\lambda = 0$. But then $0 = \|x\|_{\tilde{A}} = \|x\|$, and hence $x = 0$.

Since A is complete and of codimension 1 in $\tilde{A}$, the space $\tilde{A}$ equipped with the norm $x \mapsto \|x\|_{\tilde{A}}$ is complete. To conclude, it is therefore sufficient to show that one has $\|x\|_{\tilde{A}}^2 \leq \|x^*x\|_{\tilde{A}}$ for all $x \in \tilde{A}$ (lemma 6). One may assume that $\|x\|_{\tilde{A}} = 1$. For every real number $r < 1$, there therefore exists $y \in A$ such that $\|y\| = \|y^*\| \leq 1$ and $\|xy\|^2 \geq r$. Since $xy \in A$, one has

$$\|x^*x\|_{\tilde{A}} \geq \|x^*xy\| \geq \|y^*(x^*x)y\| = \|(xy)^*(xy)\| = \|xy\|^2 \geq r.$$

From this one deduces that $\|x^*x\|_{\tilde{A}} \geq 1$, and hence the involutive algebra $\tilde{A}$ endowed with the norm $x \mapsto \|x\|_{\tilde{A}}$ is a stellar algebra.

DEFINITION 4. — One says that $\tilde{A}$, endowed with the norm of prop. 3, is the stellar algebra deduced from A by adjoining an identity element.

When $A \neq \{0\}$, the stellar-algebra norm on the normed involutive algebra $\tilde{A}$ is not the one considered in example 6 of I, p. 101 (cf. exercise 10 of I, p. 181).

PROPOSITION 4. — Let A be a stellar algebra.

a) If A has a unit element and u is a unitary element of A , then $\text{Sp}(u) \subset \mathbf{U}$;

b) If h is a Hermitian element of A , then $\text{Sp}'(h) \subset \mathbf{R}$.

We may assume that A is nonzero. Let us prove the assertion a). Let u be a unitary element of A . We have $\|u\| = \|u^{-1}\| = 1$ (remark 1), hence $\text{Sp}(u) \subset \mathbf{U}$ (I, p. 26, cor. 3). To prove b), we may assume that A is unital (prop. 3). Let h be a Hermitian element of A . Then $\exp(ih)$ is unitary (Lemma 5 of I, p. 102). Thus, by Cor. 2 of I, p. 67 and a), we have $\exp(i\text{Sp}(h)) = \text{Sp}(\exp(ih)) \subset \mathbf{U}$, which means that $\text{Sp}(h) \subset \mathbf{R}$.

PROPOSITION 5. — Let A be a unital stellar algebra and B a stellar subalgebra of A containing the identity element of A . Then B is a full subalgebra of A . In particular, one has $\text{Sp}_B(x) = \text{Sp}_A(x)$ for all x in B .

Let x be a Hermitian element of B . Since $\text{Sp}_B(x) \subset \mathbf{R}$ (prop. 4), prop. 6 of I, p. 28 shows that $\text{Sp}_B(x) = \text{Sp}_A(x)$. In particular, x is invertible in B if and only if it is invertible in A .

Now let x be any element of B that is invertible in A . Then x^* is invertible in A , and xx^* is invertible in A . Since xx^* is Hermitian, the preceding shows that xx^* is invertible in B . This implies that x is right-invertible in B . Similarly, one verifies that x is left-invertible in B , and consequently that x is invertible in B . Thus, B is a full subalgebra of A .

COROLLARY. — Let A be a stellar algebra and let B be a stellar subalgebra of A . Then we have $\text{Sp}'_B(x) = \text{Sp}'_A(x)$ for all x in B .

By adjoining an identity element (prop. 3), this follows from prop. 5.

PROPOSITION 6 (Fuglede–Putnam Theorem)

Let A be a unital stellar algebra. Let a and b be normal elements of A . If $c \in A$ satisfies $ac = cb$, then $a^*c = cb^*$.

The hypothesis implies $(wa)^k c = c(wb)^k$ for every integer $k \geq 0$ and every $w \in \mathbf{C}$, hence $e^{wa}c = ce^{wb}$ (formula (9) of I, p. 78). Consider the function f from $\mathbf{C}$ to A defined by $z \mapsto e^{-za^*}ce^{zb^*}$. This is a holomorphic function on $\mathbf{C}$, whose derivative satisfies

$$f'(z) = -a^*e^{-za^*}ce^{zb^*} + e^{-za^*}cb^*e^{zb^*}$$

for all $z \in \mathbf{C}$. Since $c = e^{-\bar{z}a}ce^{\bar{z}b}$ one can write

$$f(z) = e^{-za^*} e^{\bar{z}a} c e^{-\bar{z}b} e^{zb^*}.$$

Since a and b are normal, the elements $e^{-za^*} e^{\bar{z}a} = e^{-za^* + \bar{z}a}$ and $e^{-\bar{z}b} e^{zb^*} = e^{-\bar{z}b + zb^*}$ of A are unitary for all $z \in \mathbf{C}$ (Lemma 5 of I, p. 102), hence of norm 1. Consequently, one has $\|f(z)\| \leq \|c\|$ for all $z \in \mathbf{C}$. The function f is therefore constant (VAR, R1, 3.3.6, p. 29), that is, $f(z) = f(0) = c$ for all $z \in \mathbf{C}$. But then $-a^*c + cb^* = f'(0) = 0$.

COROLLARY. — Let A be a stellar algebra and let a be a normal element of A . The commutant (resp. bicommutant) of $\{a, a^*\}$ coincides with the commutant (resp. bicommutant) of a .

It suffices to take $b = a$ in the proposition.

1.6.5 Commutative stellar algebras

Lemma 8. — Let X and Y be metric spaces, with X complete. Let f be a mapping from X into Y such that

$$d(f(x), f(y)) \geq d(x, y)$$

for all x and y in X and such that the graph of f is closed in $X \times Y$. Then f is a closed map.

Let F be a closed subset of X and let $(y_n)_{n \in \mathbf{N}}$ a sequence in $f(F)$ which converges to $y \in Y$. For every $n \in \mathbf{N}$, let $x_n \in F$ such that $f(x_n) = y_n$. The hypothesis implies that the sequence $(x_n)_{n \in \mathbf{N}}$ is a Cauchy sequence; let $x \in X$ be its limit; it is an element of F , because F is closed. Moreover, we have $(x_n, f(x_n)) \rightarrow (x, y)$ into $X \times Y$. Since the graph of f is closed, it follows that $y = f(x)$ belongs to $f(F)$.

Lemma 9. — Let A be a commutative stellar algebra. Every character of A is Hermitian, and the Gelfand transformation is a morphism of stellar algebras from A into $\mathcal{C}_0(X(A))$.

It suffices to prove the first assertion (proposition 7 of I, p. 38 and lemma 3 of I, p. 98). Let χ be a character of A . For every Hermitian element y of A , we have $\chi(y) = \mathcal{G}_A(y)(\chi) \in \text{Sp}'(y) \subset \mathbf{R}$ (prop. 4 of I, p. 106). Consequently, the character χ is Hermitian (prop. 1 of I, p. 95).

THEOREM 1. — Let A be a commutative stellar algebra. The Gelfand transformation is an isometric isomorphism of the stellar algebra A onto the stellar algebra $\mathcal{C}_0(X(A))$ of continuous functions on $X(A)$ vanishing at infinity.

The Gelfand transform is an involutive algebra morphism from A into $\mathcal{C}_0(X(A))$ (lemma 9). Let B be its image. It is an involutive subalgebra of $\mathcal{C}_0(X(A))$. The elements of B separate the points of $X(A)$ by definition. Let $\chi \in X(A)$. There exists $x \in A$ such that $\chi(x) \neq 0$, hence $f = \mathcal{G}(x)$ is an element of B such that $f(\chi) \neq 0$. By TG, X, p. 40, cor. 2, the subalgebra B is therefore dense in $\mathcal{C}_0(X(A))$.

For every Hermitian element y of A , we have $\|\mathcal{G}(y)\| = \varrho(y) = \|y\|$ (prop. 7 of I, p. 38 and formula (2) of I, p. 104), whence, for every $x \in A$, the equalities

$$\|x\|^2 = \|x^*x\| = \|\mathcal{G}(x^*x)\| = \|\overline{\mathcal{G}(x)} \cdot \mathcal{G}(x)\| = \|\mathcal{G}(x)\|^2.$$

Thus G is isometric, and its image B is therefore closed (Lemma 8). We conclude that $B = \mathcal{C}_0(X(A))$.

COROLLARY 1. — Let A be a stellar algebra and let x be a normal element of A . Then $\|x\| = \varrho(x)$.

Since the stellar subalgebra of A generated by x and x^* is commutative, we may assume that A is commutative. In this case, the corollary follows from Th. 1, Example 2 of I, p. 102, and Th. 1 of I, p. 24.

COROLLARY 2. — Let A be a commutative stellar algebra.

- a) There exists a locally compact topological space X such that A is isomorphic to the stellar algebra $\mathcal{C}_0(X)$;
- b) Let B be a commutative stellar algebra. The map $\pi \mapsto X'(\pi)$ is a bijection from the set of stellar algebra morphisms from A to B onto the set of proper partial maps from $X(B)$ into $X(A)$ (def. 1 of I, p. 33).

Theorem 1 establishes the first assertion, and the second assertion follows from Prop. 3 of I, p. 34 and from Example 2 of I, p. 102.

COROLLARY 3. — Let A be a unital commutative stellar algebra.

- a) There exists a compact topological space X such that A is isomorphic to the stellar algebra $\mathcal{C}(X)$;
- b) Let B be a commutative unital stellar algebra. The map $\pi \mapsto X(\pi)$ is a bijection from the set of unital morphisms of stellar algebras from A to B to the set of continuous maps from $X(B)$ into $X(A)$.

This follows from the foregoing and from prop. 4 of I, p. 35.

Remark. — *Let G be the category whose objects are locally compact spaces and whose morphisms are proper partial maps (def. 1 of I, p. 33), and let S be the category of commutative stellar algebras, whose morphisms are morphisms of involutive algebras. Consider the functor from S to the opposite category G° which associates to a commutative stellar algebra A the locally compact space $X(A)$ of nonzero characters of A , and to a morphism $\varphi: A \rightarrow B$ of commutative stellar algebras the continuous map $X'(\varphi)$. Th. 1 and Cor. 2 mean that this functor is an equivalence of categories, and that a quasi-inverse of this functor is the functor which associates to a locally compact topological space X the commutative stellar algebra $C_0(X)$.

Similarly, Corollary 3 means that the opposite category of the category of compact spaces is equivalent to the category of unital commutative stellar algebras.*

1.6.6 Functional calculus in unital stellar algebras

In this number, A is a unital stellar algebra and x is a normal element of A .

Let B be the unital stellar subalgebra of A generated by x ; it is commutative and contained in the bicommutant of $\{x, x^*\}$, hence in the bicommutant of x (corollary of Proposition 6 in I, p. 106). The Gelfand transform $\mathcal{G}_B : B \rightarrow \mathcal{C}(X(B))$ is an isomorphism of stellar algebras (th. 1 of I, p. 108). We have $\text{Sp}_B(x) = \text{Sp}_A(x)$ (prop. 5 of I, p. 106).

Lemma 10. — The map $\text{ev}_x : \chi \mapsto \chi(x)$ induces a homeomorphism of $X(B)$ onto $\text{Sp}_A(x)$.

The map $x \mapsto \chi(x)$ of $X(B)$ into $\mathbb{C}$ is continuous, and its image is equal to $\text{Sp}_B(x)$ by prop. 6 of I, p. 37, hence to $\text{Sp}_A(x)$. Since the characters of B are Hermitian (lemma 9 of I, p. 107), characters of B that agree on x are also equal on x^* and therefore are equal on the unital stellar subalgebra B generated by x . This proves that the map ev_x is injective. Since $X(B)$ is compact

and $\mathbf{C}$ is separated, it induces a homeomorphism from $X(B)$ onto its image, hence the lemma.

From the lemma we deduce an isomorphism of stellar algebras $\varphi_x: \mathcal{C}(\text{Sp}_A(x)) \rightarrow \mathcal{C}(X(B))$. It maps a function $f \in \mathcal{C}(\text{Sp}_A(x))$ to the function $\chi \mapsto f(\chi(x))$. The map $\mathcal{G}_B^{-1} \circ \varphi_x$ is an isomorphism of the stellar algebra $\mathcal{C}(\text{Sp}_A(x))$ onto B .

DEFINITION 5. — The involutive morphism $f \mapsto (\mathcal{G}_B^{-1} \circ \varphi_x)(f)$ of $\mathcal{C}(\text{Sp}_A(x))$ to A is called the continuous functional calculus map of x in A . We denote it $f \mapsto f(x)$.

Remark. — The map $f \mapsto f(x)$ is isometric; its image is the unital stellar subalgebra B generated by x , which is contained in the bicommutant of x .

If f is the restriction to $\text{Sp}_A(x)$ of a function of the form $z \mapsto P(z, \bar{z})$, where $P \in \mathbf{C}[X, Y]$ is a polynomial, then we have $f(x) = P(x, x^*)$ in the usual algebraic sense.

Examples. — 1) Suppose that there exists $\lambda \in \mathbf{C}$ such that $\text{Sp}_A(x)$ is reduced to λ . We then have $x = \lambda \cdot 1$. Indeed, the identity function of $\text{Sp}_A(x)$ is equal to λ , hence its image under the functional calculus map, that is, x is equal to $\lambda \cdot 1$.

2) For x to be Hermitian, it is necessary and sufficient that $\text{Sp}_A(x)$ be contained in $\mathbf{R}$. Indeed, let f be the continuous map on $\text{Sp}_A(x)$ given by $f(z) = z - \bar{z}$. Then x is Hermitian if and only if $f(x) = 0$, that is, if f is zero, that is, if $\text{Sp}_A(x)$ is contained in $\mathbf{R}$.

3) For x to be unitary, it is necessary and sufficient that its spectrum be contained in the unit circle of $\mathbf{C}$. Indeed, let $f \in \mathcal{C}(\text{Sp}_A(x))$ be the function defined by $f(z) = z\bar{z} - 1$; the element x is unitary if and only if $f(x) = 0$, that is, if f is zero.

Proposition 7. — The map $f \mapsto f(x)$ is the unique unital morphism of involutive algebras from $\mathcal{C}(\text{Sp}_A(x))$ into A such that the identity map z of $\text{Sp}_A(x)$ has image x .

Indeed, the unital subalgebra of $\mathcal{C}(\text{Sp}_A(x))$ generated by the elements z and $\bar{z}$ of $\mathcal{C}(\text{Sp}_A(x))$ is dense in $\mathcal{C}(\text{Sp}_A(x))$ (TG, X, p. 40, cor. 1). Since every morphism of involutive algebras from $\mathcal{C}(\text{Sp}_A(x))$ to A is continuous (I, p. 104, prop. 2), there exists at most one morphism of involutive algebras from $\mathcal{C}(\text{Sp}_A(x))$ to A which maps z to x .

The following corollary shows that when f is the restriction of a holomorphic function in a neighborhood of $\text{Sp}_A(x)$ the definition of $f(x)$ coincides with that of the holomorphic functional calculus in one variable from no. 9 of I, p. 74.

COROLLARY 1. — Let $f \in \mathcal{O}(\mathrm{Sp}_A(x))$ be a germ of a holomorphic function in a neighborhood of $\mathrm{Sp}_A(x)$ and let $\tilde{f} \in \mathcal{C}(\mathrm{Sp}_A(x))$ be the continuous function on $\mathrm{Sp}_A(x)$ associated with f . We have $\tilde{f}(x) = f(x)$, where $f(x)$ is the element of A given by the holomorphic functional calculus.

Indeed, the map $f \mapsto \tilde{f}(x)$ is a unital continuous morphism from $\mathcal{O}(\mathrm{Sp}_A(x))$ to A which maps the germ of the identity function in a neighborhood of $\mathrm{Sp}_A(x)$ to x . The result then follows from th. 5 of I, p. 74.

COROLLARY 2. — Let $f \in \mathcal{C}(\mathrm{Sp}_A(x))$.

a) We have

$$\mathrm{Sp}_A(f(x)) = f(\mathrm{Sp}_A(x));$$

b) For every $g \in \mathcal{C}(\mathrm{Sp}_A(f(x)))$, one has $(g \circ f)(x) = g(f(x))$.

Since $f(x)$ belongs to the full subalgebra B of A , we have $\mathrm{Sp}_A(f(x)) = \mathrm{Sp}_B(f(x))$ (prop. 5 of I, p. 106). The isomorphism $f \mapsto f(x)$ of $\mathcal{C}(\mathrm{Sp}_A(x))$ to B preserves the spectrum; we therefore have $\mathrm{Sp}_B(f(x)) = \mathrm{Sp}_{\mathcal{C}(\mathrm{Sp}_A(x))}(f) = f(\mathrm{Sp}_A(x))$ (example 3 of I, p. 17). This proves assertion a).

The map $g \mapsto (g \circ f)(x)$ is a unital morphism of involutive algebras from $\mathcal{C}(\mathrm{Sp}_A(f(x)))$ to A which maps the identity function of $\mathrm{Sp}_A(f(x))$ to $f(x)$. By prop. 7, we therefore have $(g \circ f)(x) = g(f(x))$ for all $g \in \mathcal{C}(\mathrm{Sp}_A(f(x)))$.

Example 4. — Let X be a locally compact space and let $A = \mathcal{C}_b(X)$ be the commutative unital stellar algebra of continuous bounded functions on X (example 2 of I, p. 102). Let $g \in A$; its spectrum S is the closure in $\mathbb{C}$ of the set $g(X)$ of the values of g (example 3 of I, p. 17). The functional calculus mapping of g is then the mapping $f \mapsto f \circ g$, for $f \in \mathcal{C}(S)$. Indeed, this map is a unital morphism of stellar algebras such that the identity map of S has image g .

In the case where X is compact, we have $A = \mathcal{C}(X)$ and $S = g(X)$.

Let $\pi: A \rightarrow A'$ be a unital morphism of unital stellar algebras. The element $\pi(x)$ of A' is normal and its spectrum relative to A' is contained in $\mathrm{Sp}_A(x)$. We then have:

PROPOSITION 8. — Let $f \in \mathcal{C}(\mathrm{Sp}_A(x))$. Still denoting f the restriction of f to $\mathrm{Sp}_{A'}(\pi(x))$

, we have the equality $\pi(f(x)) = f(\pi(x))$. In particular, for every $\chi \in X(A)$, one has $\chi(f(x)) = f(\chi(x))$.

Let z be the identity map of $\text{Sp}_A(x)$. The maps defined by $f \mapsto \pi(f(x))$ and $f \mapsto f(\pi(x))$ are continuous unital morphisms of involutive algebras from $\mathcal{C}(\text{Sp}_A(x))$ to B that map z to $\pi(x)$. These morphisms therefore coincide on the unital involutive subalgebra of $\mathcal{C}(\text{Sp}_A(x))$ generated by z . Since this one is dense in $\mathcal{C}(\text{Sp}_A(x))$ (TG, X, p. 40, cor. 1), these morphisms are equal.

COROLLARY. — Suppose that A is commutative. For every $f \in \mathcal{C}(\text{Sp}_A(x))$, one has $\mathcal{G}_A(f(x)) = f \circ \mathcal{G}_A(x)$.

It suffices to apply Prop. 8 to the Gelfand transform $\mathcal{G}_A$ from A to $\mathcal{C}(X(A))$ and to note (example above) that $f(\mathcal{G}_A(x)) = f \circ \mathcal{G}_A(x)$.

1.6.7 Applications of functional calculus

PROPOSITION 9. — Every injective morphism of stellar algebras is isometric and, in particular, has closed image.

Let A and A' be stellar algebras, and let $\pi: A \rightarrow A'$ be a morphism of involutive algebras from A to A' .

Suppose first that A and A' are unital and that π is unital. We have $\|\pi\| \leq 1$ (prop. 2). Suppose there exists x in A such that $\|\pi(x)\| < \|x\|$. Let $y = x^*x$; it is a Hermitian element of A . Since A and A' are stellar algebras, we have $\|\pi(y)\| = \|\pi(x)\|^2 < \|x\|^2 = \|y\|$, that is, $\varrho(\pi(y)) < \varrho(y)$ (lemma 7 of I, p. 104). In particular, $\text{Sp}_{A'}(\pi(y))$ is a closed subset of $\text{Sp}_A(y)$, distinct from $\text{Sp}_A(y)$.

(Remark 6 of I, p. 3 and Th. 1 of I, p. 24). There then exists a nonzero function $f \in \mathcal{C}(\text{Sp}_A(y))$ such that $f|_{\text{Sp}_{A'}(\pi(y))} = 0$ (TG, IX, p. 13, Prop. 3). Let $w = f(y) \in A$. We have $w \neq 0$ since $f \neq 0$ but $\pi(w) = \pi(f(y)) = f(\pi(y)) = 0$ since f is zero on $\text{Sp}_{A'}(\pi(y))$ (Prop. 8). Hence π is not injective.

We now treat the general case. Let $\tilde{A}$ and $\tilde{A}'$ be the stellar algebras deduced from A and from A' respectively by adjoining a unit element (def. 4 of I, p. 106). There exists a unique unital morphism of involutive algebras $\tilde{\pi}: \tilde{A} \rightarrow \tilde{A}'$ extending π . This morphism is injective,

hence is isometric by what precedes. For every $x \in A$, we then have $\|\pi(x)\| = \|\widetilde{\pi}(x)\| = \|x\|$.

Lemma 11. — Let X be a completely regular topological space, that is, uniformizable and separated (TG, IX, p. 8, def. 4), containing at least two points. There exist nonzero continuous functions f and g in $\mathcal{C}(X)$ such that $fg = 0$.

Let $x \neq y$ be distinct points of X . Let U and V be open neighborhoods of x and y , respectively, such that U and V are disjoint. Since X is uniformizable, by TG, IX, p. 7, th. 2, there exists a function $f \in \mathcal{C}(X)$ such that $f(x) = 1$ and $f|_{X-U} = 0$. Similarly, there exists $g \in \mathcal{C}(X)$ such that $g(y) = 1$ and $g|_{X-V} = 0$. We then have $fg = 0$.

PROPOSITION 10. — Let A be a unital stellar algebra. Suppose that for every pair (x, y) of permutable elements of A , the condition $xy = 0$ implies that $x = 0$ or $y = 0$. Then $A = \mathbb{C} \cdot 1$.

If A is not equal to $\mathbb{C} \cdot 1$, there exists a hermitian element x in A which does not belong to $\mathbb{C} \cdot 1$ (lemma 2 of I, p. 96). Let B be the unital stellar subalgebra of A generated by x . It is commutative, and isomorphic to $\mathcal{C}(\text{Sp}_A(x))$ (I, p. 110, remark). Since x is not scalar, its spectrum in B is not reduced to a single element (example 1 of I, p. 110). There therefore exist nonzero continuous functions f and g on $\text{Sp}_A(x)$ such that $fg = 0$ (lemma 11). The elements $f(x)$ and $g(x)$ of A are nonzero, permutable, and satisfy $f(x)g(x) = 0$ in A .

PROPOSITION 11. — Let A be a unital stellar algebra, and let a, x and y be elements of A . Suppose that x and y are normal. If $xa = ay$, then we have $f(x)a = af(y)$ for every continuous function f on the union of the spectrum of x and the spectrum of y . In particular, we have $f(aa^*)a = af(a^*a)$ for every function $f \in \mathcal{C}(\text{Sp}'(a^*a))$.

Let $S = \text{Sp}(x) \cup \text{Sp}(y)$. Proposition 6 of I, p. 106 implies that $x^*a = ay^*$. Consequently, we obtain $f(x)a = af(y)$ for every function f which is of the form $z \mapsto P(z, \bar{z})$, where $P \in \mathbb{C}[X, Y]$ is a polynomial. Since the set of functions $f \in \mathcal{C}(S)$ satisfying $f(x)a = af(y)$ is a closed subalgebra of $\mathcal{C}(S)$, it coincides with $\mathcal{C}(S)$ by TG, X, p. 40, cor. 1.

The second assertion is a consequence of the first, applied to the hermitian elements $x = aa^*$ and $y = a^*a$, taking into account the fact that $\text{Sp}'(a^*a) = \text{Sp}'(aa^*)$ (prop. 1 of I, p. 5).

1.6.8 Functional calculus in a non-unital algebra

Let A be a stellar algebra and let $\tilde{A}$ be the unital stellar algebra deduced from A by adjoining a unit element e . Denote by π the hermitian character $x + \lambda e \mapsto \lambda$ of $\tilde{A}$ into $\mathbb{C}$; we have $\text{Ker}(\pi) = A$.

Let $x \in A$ be a normal element. It is normal in $\tilde{A}$ and $\text{Sp}'_{\tilde{A}}(x) = \text{Sp}'_A(x)$. Denote by $\mathcal{C}'(\text{Sp}'_A(x))$ the stellar algebra of continuous functions f on $\text{Sp}'_A(x)$ such that $f(0) = 0$.

Let $f \in \mathcal{C}'(\text{Sp}'_A(x))$. Since $\pi(f(x)) = f(\pi(x))$ (prop. 8 of I, p. 112), we have $f(x) \in A$ if and only if $f(0) = 0$. The map $f \mapsto f(x)$ defines a morphism of involutive algebras from the stellar algebra $\mathcal{C}'(\text{Sp}'_A(x))$ to A for which the image of the identity map z of $\text{Sp}'_A(x)$ is x . This morphism is isometric and its image is the stellar subalgebra of A generated by x .

PROPOSITION 12. — The map $f \mapsto f(x)$ is the unique morphism of involutive algebras from the stellar algebra $\mathcal{C}'(\text{Sp}'_A(x))$ to A such that the identity map z of $\text{Sp}'_A(x)$ has image x .

The elements z and $\bar{z}$ of $\mathcal{C}'(\text{Sp}'_A(x))$ generate a dense subalgebra of $\mathcal{C}'(\text{Sp}'_A(x))$ (cf. TG, X, p. 40, cor. 2). Since every morphism of involutive algebras from the stellar algebra $\mathcal{C}'(\text{Sp}'_A(x))$ to the stellar algebra A is continuous (I, p. 104, prop. 2), the result follows.

The results of the preceding number concerning functional calculus extend to the general case. We shall simply state them and leave it to the readers to complete the proofs, *mutatis mutandis*.

PROPOSITION 13. — One has the following properties:

- a) For every $f \in \mathcal{C}'(\text{Sp}'_A(x))$, one has $\text{Sp}'_A(f(x)) = f(\text{Sp}'_A(x))$;
- b) For every $f \in \mathcal{C}'(\text{Sp}'_A(x))$ and for every $g \in \mathcal{C}'(\text{Sp}'_A(f(x)))$, one has $(g \circ f)(x) = g(f(x))$;
;
- c) Let A' be a stellar algebra and let π be a morphism from A to A' ; then $\pi(x)$ is normal in A' , one has $\text{Sp}'_{A'}(\pi(x)) \subset \text{Sp}'_A(x)$ and $\pi(f(x)) = f(\pi(x))$ for every $f \in \mathcal{C}'(\text{Sp}'_A(x))$;
- d) If A is commutative, and if $f \in \mathcal{C}'(\text{Sp}'_A(x))$, then $\mathcal{G}'_A(f(x)) = f \circ \mathcal{G}'_A(x)$.

Remark. — Let A be a unital stellar algebra and let $\tilde{A}$ be the unital stellar algebra deduced from A by adjoining a unit element e . For every $x \in A$, one has $\text{Sp}'_{\tilde{A}}(x) = \text{Sp}_A(x) \cup \{0\}$.

Let x be a normal element of A . It is then a normal element of $\widetilde{A}$ and one therefore has two functional calculus maps in A , the first defined on $\mathcal{C}(\mathrm{Sp}_A(x))$ and the second on $\mathcal{C}'(\mathrm{Sp}'_A(x))$. Let $f' \in \mathcal{C}'(\mathrm{Sp}'_A(x))$; if one denotes by f its restriction to $\mathrm{Sp}_A(x)$, we then have $f'(x) = f(x)$.

1.6.9 Positive elements in stellar algebras

DEFINITION 6. — Let A be a stellar algebra. An element x of A is said to be positive if it is Hermitian and if $\mathrm{Sp}'_A(x) \subset \mathbf{R}_+$. One denotes by A_+ the set of positive elements of A . It is a subset of A_h .

We denote $x \geq y$ if $x - y \in A_+$.

If the stellar algebra A is unital, its identity element is positive.

If B is a stellar subalgebra of A , one has $B_+ = B \cap A_+$ (cor. of prop. 5 of I, p. 106).

If $\pi : A \rightarrow B$ is a morphism of stellar algebras, then $\pi(A_+) \subset B_+$.

Examples. — 1) Let X be a locally compact space. In the stellar algebra $\mathcal{C}_0(X)$ of continuous functions on X tending to 0 at infinity, resp. in the stellar algebra $\mathcal{C}_b(X)$ of bounded continuous functions on X , a function f is a positive element if and only if it is real-valued and if $f(x) \geq 0$ for all $x \in X$ (cf. example 3 of I, p. 17).

2) Let A be a commutative stellar algebra. Let a be an element of A . Since $\mathrm{Sp}'_A(x)$ is the union of $\{0\}$ and of the image of the Gelfand transform $\mathcal{G}(a)$ (prop. 6 of I, p. 37), the element a is positive if, and only if, the Gelfand transform $\mathcal{G}(a)$ is a positive function.

3) Let E be a complex Hilbert space. An element x of the stellar algebra $\mathcal{L}(E)$ (example 1 of I, p. 102) is positive if and only if it is a positive endomorphism of E in the sense of EVT, V, p. 45, def. 6 (prop. 8 of I, p. 138).

Lemma 12. — Let A be a unital stellar algebra and let $x \in A$ be a Hermitian element.

- a) The element x is positive if and only if $\|x\| \cdot 1 - x \leq \|x\|$;
- b) If $\|x\| \leq 1$, then x is positive if and only if $\|1 - x\| \leq 1$.
- c) If x is positive, then $1 - x$ is positive if and only if $\|x\| \leq 1$;

d) If x is positive and if $y \in A_+$ commutes with x , then xy is positive.

The element x is Hermitian, hence normal. By considering the stellar subalgebra generated by x , which is commutative, one reduces to the case where the algebra A is commutative, that is, to the case where $A = \mathcal{C}_0(X)$ for a locally compact topological space X (th. 1 of I, p. 108). The first three assertions then follow immediately from example 1 above. Likewise, to prove assertion d), one may consider the stellar subalgebra generated by x and y , which is commutative.

PROPOSITION 14. — Let A be a stellar algebra. The set A_+ is a closed pointed salient convex cone in the real Banach space A_h (EVT, II, p. 11).

Let $\tilde{A}$ be the stellar algebra deduced from A by adjoining an identity element. Since $A_+ = A \cap \tilde{A}_+$, it suffices to prove the proposition for $\tilde{A}$. One may therefore suppose that A has a unit element.

We have $0 \in A_+$. For every $\lambda \in \mathbf{R}_+^*$ and every $x \in A$, one has $\text{Sp}'_A(\lambda x) = \lambda \text{Sp}'_A(x)$ which implies that A_+ is a cone in the real Banach space A_h .

To show that A_+ is convex, it suffices to show that if x and y are positive, then $x + y \geq 0$ (EVT, II, p. 11, prop. 10). By homothety, it suffices to prove that if $x \geq 0$ and $y \geq 0$ additionally satisfy $\|x\| \leq 1$, $\|y\| \leq 1$, then the element $\frac{1}{2}(x + y)$ is positive. But we have

$$\left\| 1 - \frac{1}{2}(x + y) \right\| \leq \frac{1}{2}\|1 - x\| + \frac{1}{2}\|1 - y\| \leq 1$$

according to the assertion b) of Lemma 12, and this same assertion then shows that $\frac{1}{2}(x + y)$ is positive.

Finally, the assertion a) of Lemma 12 also implies that A_+ is closed.

Since A_+ is a pointed cone in A_h , it is salient if and only if $A_+ \cap (-A_+)$ is reduced to 0. But if $x \in A_+ \cap (-A_+)$, one has $\text{Sp}'_A(x) = \{0\}$, hence $\varrho(x) = 0$, and $\|x\| = 0$. Since x is Hermitian (Lemma 7, (2) of I, p. 104), it follows that $x=0$.

Proposition 14 means that the relation $x \geq y$ is an order relation on A_h (EVT, II, p. 13, prop. 13).

PROPOSITION 15. — Let A be a stellar algebra. Let x be a normal element of A .

- a) Suppose that A is unital and let f be a continuous function from $\text{Sp}_A(x)$ to $\mathbb{C}$. For $f(x)$ to be positive, it is necessary and sufficient that the image of f be contained in $\mathbb{R}_+$;
- b) Let f be a continuous function from $\text{Sp}'_A(x)$ to $\mathbb{C}$ such that $f(0) = 0$. For $f(x)$ to be a positive element of A , it is necessary and sufficient that the image of f be contained in $\mathbb{R}_+$.

Assertion a) follows from assertion a) of Cor. 2 of I, p. 111, and assertion b) follows from Prop. 13 of I, p. 114.

Let x be a Hermitian element of the stellar algebra A . Its spectrum is contained in $\mathbb{R}$ (Prop. 4 of I, p. 106). Consider the continuous functions from $\text{Sp}'_A(x)$ into $\mathbb{R}$ defined by

$$f_1 : t \mapsto \sup(t, 0), \quad f_2 : t \mapsto \sup(-t, 0), \quad f_3 : t \mapsto |t|.$$

We denote

$$x^+ = f_1(x), \quad x^- = f_2(x), \quad |x| = f_3(x). \quad (3)$$

Since the functions f_1, f_2, f_3 are real-valued, positive, and vanish at 0, the elements x^+, x^- and $|x|$ are positive elements of A (Prop. 15, a)) which belong to the stellar subalgebra of A generated by x .

We have $f_1(t) - f_2(t) = t$ for all $t \in \mathbb{R}$, as well as the relations $f_1 + f_2 = f_3$ and $f_1 f_2 = 0$. It follows that

$$x = x^+ - x^-, \quad |x| = x^+ + x^-, \quad x^+ x^- = x^- x^+ = 0. \quad (4)$$

Since the functional calculus map is isometric, we have

$$\| |x| \| = \|x\|, \quad \|x^+\| \leq \|x\|, \quad \|x^-\| \leq \|x\|.$$

Let x be a positive element of A . It is Hermitian, hence normal. Let $\alpha \in \mathbb{R}_+^*$, and let g be the restriction to $\text{Sp}'_A(x)$ of the function $t \mapsto t^\alpha$; we denote $x^\alpha = g(x)$. This is a positive element of the stellar subalgebra of A generated by x . Let α and β be in $\mathbb{R}_+^*$. Since the functional calculus

map is an algebra homomorphism, and by Prop. 13 of I, p. 114, we have

$$x^\alpha x^\beta = x^{\alpha+\beta} \quad (x^\alpha)^\beta = x^{\alpha\beta}. \quad (5)$$

We shall also write $\sqrt{x} = x^{1/2}$.

PROPOSITION 16. — Let x be a positive element of A . Let $\alpha \in \mathbf{R}_+^*$. Then $x^{1/\alpha}$ is the unique positive element y of A such that $y^\alpha = x$.

We saw above that $x^{1/\alpha}$ satisfies the required properties. Conversely, let y be a positive element of A such that $y^\alpha = x$. By formula (5), we have $y = (y^\alpha)^{1/\alpha} = x^{1/\alpha}$, which is what had to be proved.

Lemma 13. — Let A be a unital stellar algebra. Every element of A is a sum of unitary elements.

Let x be a Hermitian element of A . Suppose first that $\|x\| \leq 2$. By Lemma 12, c), we have $1 - \frac{1}{4}x^2 \in A_+$. Let $y = \frac{1}{2}x + i\sqrt{1 - \frac{1}{4}x^2}$. We have $y^* = \frac{1}{2}x - i\sqrt{1 - \frac{1}{4}x^2}$, hence $yy^* = 1$ and $x = y + y^*$ is a sum of two unitary elements. In the general case, let k be an integer such that $\|\frac{1}{k}x\| \leq 2$; the element x is then a sum of $2k$ unitary elements. By Lemma 2 of I, p. 96, the lemma follows.

THEOREM 2. — Let A be a stellar algebra. An element x of A is positive if and only if there exists $y \in A$ such that $x = y^*y$.

Suppose that x is positive. Let $y = x^{1/2}$; this is a Hermitian element of A and we have $y^*y = y^2 = x$.

Conversely, let y be an element of A and set $x = y^*y$. This is a Hermitian element of A . Let us show that x is positive. For this, denote $z = x^-$ and set $w = yz$. We then have

$$w^*w = z^*y^*yz = zxz = z(x^+ - z)z = -z^3.$$

Since $z \geq 0$, one has $z^3 \geq 0$ and we deduce that $\text{Sp}'_A(w^*w) \subset \mathbf{R}_-$. Let us also write $w = w_1 + iw_2$ where w_1 and w_2 are Hermitian (lemma 2 of I, p. 96). The elements w_1^2 and w_2^2 are positive. We have $ww^* + w^*w = 2w_1^2 + 2w_2^2$, and prop. 14 therefore shows that $ww^* = 2w_1^2 + 2w_2^2 + (-w^*w)$ is positive. Since $\text{Sp}'_A(ww^*) = \text{Sp}'_A(w^*w)$ (I, p. 5, prop. 1), which is

contained in $\mathbf{R}_+$, we conclude that $\text{Sp}'_A(w^*w) = \{0\}$. Since w^*w is Hermitian, this implies (cor. 1 of I, p. 108) that $\|w^*w\| = \varrho(w^*w) = 0$, so that $z^3 = 0$. Since z is Hermitian, we have $z = 0$. Thus, $x = x^+$ is positive.

Remark. — Let A be a stellar algebra and let $x \in A$. The element x^*x is positive; we therefore set $|x| = (x^*x)^{1/2}$. We have $\|x\|^2 = \|x^*x\| = \||x|^2\| = \||x|\|^2$, hence $\|x\| = \||x|\|$.

When x is normal, we also have $|x| = f(x)$, where f is the map from C to C , vanishing at 0, given by $f(z) = |z|$. In particular, when x is Hermitian, $|x|$ coincides with the element defined by formula (3).

Suppose further that A is unital and that x is invertible. Then $|x|$ is also invertible, the element $u = x|x|^{-1}$ is unitary and we have $x = u|x|$ (polar decomposition; see also I, p. 139, no. 8 for the case of endomorphisms of Hilbert spaces).

Lemma 14. — Let A be a stellar algebra, and let x and y be Hermitian elements of A .

- a) If $x \leq y$, then for every element w of A , we have $w^*xw \leq w^*yw$. In particular, if $y \geq 0$, one has $w^*yw \geq 0$;
- b) Suppose that A is unital. If $0 \leq x \leq y$ and if x is invertible, then y is invertible and $y^{-1} \leq x^{-1}$;
- c) If $0 \leq x \leq y$ then $\|x\| \leq \|y\|$.

Let $u = (y - x)^{1/2}$. We have $w^*yw - w^*xw = w^*u^2w = (uw)^*(uw)$ and the assertion a) follows from th. 2.

Let us prove assertion b). Suppose first that $x = 1$. Let B be the unital stellar subalgebra generated by y . By the Gelfand isomorphism, y corresponds to a continuous function ≥ 1 on the compact space $X(B)$. This function is therefore invertible, and its inverse is ≤ 1 . This implies that y is invertible and $y^{-1} \leq 1 = x^{-1}$. In the general case, we observe that $0 \leq 1 \leq x^{-1/2}yx^{-1/2}$ according to a), so the preceding case implies that $z = x^{-1/2}yx^{-1/2}$ is invertible and $z^{-1} \leq 1$. Thus, y is invertible and $y^{-1} \leq x^{-1}$ according to a) again.

To prove the assertion c), we may assume that A is unital (prop. 3 of I, p. 105). Suppose first that y is invertible. Let $b = \sqrt{y}$. According to a), the conditions $0 \leq x \leq y$ imply $0 \leq b^{-1}xb^{-1} \leq b^{-1}yb^{-1} = 1$, hence $\|b^{-1}xb^{-1}\| \leq 1$ by Lemma 12, c) of I, p. 116. We then

have

$$\|x\| = \|b(b^{-1}xb^{-1})b\| \leq \|b\|\|b^{-1}xb^{-1}\|\|b\| \leq \|b\|^2 = \|b^2\| = \|y\|.$$

In the general case, for every real $\varepsilon > 0$, the element $y + \varepsilon$ is invertible and $0 \leq x \leq y + \varepsilon$. From what precedes, we therefore have $\|x\| \leq \|y + \varepsilon\|$ for every real number $\varepsilon > 0$, whence the result.

Note. — In general, if x and y are positive elements of a stellar algebra A , the condition $0 \leq x \leq y$ does not imply $x^2 \leq y^2$ (cf. exercise 15 of I, p. 182).

1.6.10 Approximate units in stellar algebras

DEFINITION 7. — Let A be a normed algebra. An approximate unit of A is a filter base F on the unit ball of A such that, for every x in A , the filter bases xF and Fx on A converge to x , in other words:

$$\lim_{f, \mathfrak{F}} fx = \lim_{f, \mathfrak{F}} xf = x.$$

If A is a stellar algebra, an approximate unit F is said to be increasing if F is a filter base on A_+ .

Let A be a stellar algebra. We denote by $A_+^{\leq 1}$ (resp. $A_+^{< 1}$) the set of positive elements of A of norm ≤ 1 (resp. of norm < 1); these are the Hermitian elements of A whose spectrum is contained in $[0, 1]$ (respectively in $[0, 1)$).

PROPOSITION 17. — Let A be a stellar algebra, let $\mathfrak{F}$ be a filter base on $A_+^{\leq 1}$. For $\mathfrak{F}$ to be an increasing approximate unit of A , it is necessary and sufficient that one have

$$\lim_{f, \mathfrak{F}} fx = x$$

for every positive element x of A .

The condition is obviously necessary; let us prove that it is sufficient. Let $\tilde{A}$ be the stellar algebra deduced from A by adjoining a unit element. Let x be an element of A and let $f \in A_+^{\leq 1}$. We have

$$\begin{aligned}\|fx - x\|^2 &= \|(fx - x)(fx - x)^*\| = \|(f - 1)xx^*(f - 1)\| \\ &\leq \|(f - 1)xx^*\|\end{aligned}$$

because $\|f - 1\| \leq 1$ (Lemma 12, c) of I, p. 116). We therefore have

$$\limsup_{f, \mathfrak{F}} \|fx - x\|^2 \leq \limsup_{f, \mathfrak{F}} \|fxx^* - xx^*\|.$$

Since xx^* is positive (th. 2 of I, p. 118), the hypothesis implies that

$$\limsup_{f, \mathfrak{F}} \|fx - x\|^2 \leq \|xx^* - xx^*\| = 0,$$

so that

$$\lim_{f, \mathfrak{F}} fx = x.$$

Since the involution of A is continuous and since $\mathfrak{F}$ is a filter base on A_h , it follows that

$$\lim_{f, \mathfrak{F}} xf = \lim_{f, \mathfrak{F}} (f^*x^*)^* = \lim_{f, \mathfrak{F}} (fx^*)^* = (x^*)^* = x.$$

The proposition follows from this.

PROPOSITION 18. — Let A be a stellar algebra. The ordered set $A_+^{\leq 1}$ is directed to the right (E, III, p. 12, def. 7) and the filter of its sections (TG, I, p. 38, example 2) is an increasing approximate unit of A.

Let $\widetilde{A}$ be the stellar algebra deduced from A by adjoining a unit element denoted 1 (def. 4 of I, p. 106).

Let g be the function on $[0, 1)$ into $\mathbf{R}_+$ defined by $g(t) = t(1 - t)^{-1}$. This is a continuous increasing bijection, whose inverse bijection is given by $t \mapsto 1 - (1 + t)^{-1}$.

Let us prove that the ordered set $A_+^{\leq 1}$ is directed to the right. Let x and y be elements of $A_+^{\leq 1}$. Since $g(0) = 0$ and since $\text{Sp}'_A(x)$ and $\text{Sp}'_A(y)$ are contained in $[0, 1)$, the elements $g(x)$ and $g(y)$ are defined; they are positive, so that $g(x) + g(y) \geq 0$. We can therefore form the element $z = g^{-1}(g(x) + g(y))$ of A. We have $\text{Sp}'_A(z) \subset [0, 1)$, and hence $z \in A_+^{\leq 1}$.

We have $0 \leq g(x) \leq g(z)$, whence $1 \leq 1 + g(x) \leq 1 + g(z)$. Assertion b) of Lemma 14 of

I, p. 119 implies that $1 + g(x)$ and $1 + g(z)$ are invertible and that $(1 + g(z))^{-1} \leq (1 + g(x))^{-1}$. Consequently, $z = 1 - (1 + g(z))^{-1} \geq 1 - (1 + g(x))^{-1} = x$. Similarly, we have $z \geq y$. Therefore, z majorizes x and y in $A_+^{\leq 1}$. The ordered set $A_+^{\leq 1}$ is thus directed to the right. Let $\mathfrak{F}$ denote the filter of its sections

Let x be a positive element of A . For every integer $n \geq 1$, let $e_n = g^{-1}(nx)$; we have $e_n \in A_+^{\leq 1}$. Let h_n be the continuous function on $\mathbf{R}_+$ defined for all $t \in \mathbf{R}_+$ by $h_n(t) = t^2(1 - g^{-1}(nt)) = t^2/(1 + nt)$. We have $|h_n(t)| \leq t/n$ for all $t \geq 0$, and hence $\|x(1 - e_n)x\| = \|h_n(x)\| \leq \|x\|/n$. In particular, $x(1 - e_n)x$ tends to 0 as n tends to infinity.

Let $\varepsilon > 0$ be a real number. Let n be an integer such that $\|x(1 - e_n)x\| < \varepsilon$. For every $f \in A_+^{\leq 1}$ such that $f \geq e_n$, we then have

$$\begin{aligned} \|x - fx\|^2 &= \|(1 - f)x\|^2 = \|((1 - f)x)^*(1 - f)x\| = \|x^*(1 - f)^2x\| \\ &= \|x(1 - f)^2x\|. \end{aligned}$$

Additionally, since $0 \leq f \leq 1$, it follows that $(1 - f) - (1 - f)^2 = (1 - f)f \geq 0$ (Lemma 12, d) of I, p. 116), hence $(1 - f)^2 \leq 1 - f$. Since $1 - f \leq 1 - e_n$, it follows that $0 \leq (1 - f)^2 \leq 1 - e_n$. By lemma 14, a) and c) of I, p. 119, it follows

$$\|x - fx\|^2 = \|x(1 - f)^2x\| \leq \|x(1 - e_n)x\| < \varepsilon.$$

We therefore have $\lim_{f, \mathfrak{F}} fx = x$ for all $x \in A_+$. The filter $\mathfrak{F}$ is therefore an approximate unit of A by proposition 17.

1.6.11 Quotient by a closed two-sided ideal

Lemma 15. — Let A be a stellar algebra and let I be a closed two-sided ideal of A . Then I is self-adjoint.

Let $J = I \cap I^*$. The set J is a self-adjoint two-sided ideal of A , which contains I^*I . In particular, J is a stellar algebra. Let $\mathfrak{F}$ be an increasing approximate unit of J (prop. 18 of I, p. 121). For all $x \in I$ and every $f \in J_+^{\leq 1}$, one has

$$\begin{aligned}\|xf - x\|^2 &= \|(xf - x)^*(xf - x)\| = \|f(x^*xf - x^*x) - (x^*xf - x^*x)\| \\ &\leq 2\|x^*xf - x^*x\|.\end{aligned}$$

Since $x^*x \in J$, we therefore have

$$\lim_{f, \mathfrak{F}} \|xf - x\|^2 = 0.$$

Since $xf \in J$ for all $f \in J_+^{\leq 1}$ and since J is closed, this implies that $x \in J$. Therefore $I = J$, and the ideal I is self-adjoint.

PROPOSITION 19. — Let A be a stellar algebra and let I be a closed two-sided ideal of A . Then the quotient involutive Banach algebra A/I is a stellar algebra.

The ideal I is self-adjoint (Lemma 15). We consider the stellar algebra $\widetilde{A}$ deduced from A by adjoining an identity element (def. 4 of I, p. 106). In this algebra, the set I is a closed self-adjoint two-sided ideal, and A/I is identified with a closed involutive subalgebra of $\widetilde{A}/I$. We may therefore assume that A is unital.

The Banach algebra A/I is involutive. Let $\pi: A \rightarrow A/I$ be the canonical projection. The self-adjoint two-sided ideal I is a stellar subalgebra of A . Let $\mathfrak{F}$ be an increasing approximate unit of I (prop. 18 of I, p. 121). Let us first show that for every $x \in A$, one has

$$\|\pi(x)\|_{A/I} = \lim_{f, \mathfrak{F}} \|x - xf\|. \quad (6)$$

On the one hand, since $xf \in I$ for all $f \in I_+^{\leq 1}$, one has

$$\|\pi(x)\|_{A/I} = \inf_{a \in I} \|x - a\| \leq \liminf_{f, \mathfrak{F}} \|x - xf\|.$$

On the other hand, for every $a \in I$, one has

$$\begin{aligned}\|x - xf\| &\leq \|(x - a) - (x - a)f\| + \|a - af\| \\ &= \|(x - a)(1 - f)\| + \|a - af\|\end{aligned}$$

and therefore, since $\|1 - f\| \leq 1$ (Lemma 12 of I, p. 116) and $a \in I$, from which we deduce

$$\limsup_{f, \mathfrak{F}} \|x - xf\| \leq \|x - a\|.$$

Thus, it follows $\limsup_{f, \mathfrak{F}} \|x - xf\| \leq \|\pi(x)\|_{A/I}$ since a is arbitrary in I . Formula (6) is therefore proved.

Now let x be an element of A . By formula (6), we have

$$\begin{aligned} \|\pi(x)\|_{A/I}^2 &= \lim_{f, \mathfrak{F}} \|x - xf\|^2 = \lim_{f, \mathfrak{F}} \|x(1 - f)\|^2 \\ &= \lim_{f, \mathfrak{F}} \|(1 - f)x^*x(1 - f)\| \leq \lim_{f, \mathfrak{F}} \|x^*x(1 - f)\| = \|\pi(x^*x)\|_{A/I}. \end{aligned}$$

Lemma 6 of I, p. 103 then implies that the involutive Banach algebra A/I is a stellar algebra.

1.6.12 Enveloping stellar algebra of an involutive Banach algebra

Lemma 16. — Let A be an involutive algebra and let p be a semi-norm on A . The following conditions are equivalent:

- (i) We have $p(xy) \leq p(x)p(y)$, $p(x^*) = p(x)$ and $p(x)^2 = p(x^*x)$ for all $x, y \in A$;
- (ii) The set R of elements x of A such that $p(x) = 0$ is a self-adjoint two-sided ideal of A , and the seminorm on A/R induced by p makes A/R an involutive normed algebra whose completion is a C^* -algebra;
- (iii) There exists a stellar algebra B and a morphism of involutive algebras φ from A to B such that $p(x) = \|\varphi(x)\|$ for all $x \in A$.

The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i) are all elementary.

A semi-norm satisfying the conditions of Lemma 16 will be called a stellar semi-norm on the involutive algebra A .

Let A be a normed involutive algebra and let S be the set of stellar seminorms on A . We have $p(x) \leq \|x\|$ for all $x \in A$ and every $p \in S$ (I, p. 104, prop. 2). The map $x \mapsto \|x\|_* = \sup_{p \in S} p(x)$ is a stellar seminorm on A . It is the largest stellar seminorm on A .

Let R be the set of $x \in A$ such that $\|x\|_* = 0$. It is a closed two-sided ideal of A . We denote by $\text{Stell}(A)$ the completed stellar algebra of A/R for the norm induced by $x \mapsto \|x\|_*$ (Lemma 16,

(ii)). The canonical map from A into $\text{Stell}(A)$ is continuous, its image is dense in $\text{Stell}(A)$, and its kernel is equal to R .

DEFINITION 8. — The stellar algebra $\text{Stell}(A)$ is called the enveloping stellar algebra of the normed involutive algebra A .

If A is commutative, then $\text{Stell}(A)$ is commutative; if A is unital, then $\text{Stell}(A)$ is unital.

PROPOSITION 20. — Let A be a normed involutive algebra and let j be the canonical morphism from A into $\text{Stell}(A)$. For every stellar algebra B and for every morphism φ of involutive algebras from A to B , there exists a unique morphism φ' of stellar algebras from $\text{Stell}(A)$ into B such that $\varphi = \varphi' \circ j$.

We denote by $x \mapsto \|x\|_*$ the norm on $\text{Stell}(A)$. Let R be the kernel of j . The map $x \mapsto \|\varphi(x)\|$ is a stellar seminorm on A . We therefore have $\|\varphi(x)\| \leq \|x\|_*$ for all $x \in A$. The morphism φ therefore defines, by passing to the quotient, a continuous morphism from A/R to B , which extends by continuity to a morphism φ' from $\text{Stell}(A)$ to B satisfying $\varphi = \varphi' \circ j$. The uniqueness of φ' results from the fact that the image of j is dense in $\text{Stell}(A)$.

COROLLARY. — Let A be a commutative involutive Banach algebra and let j be the canonical morphism from A into $\text{Stell}(A)$. The map $X(j)$ is a homeomorphism from $X(\text{Stell}(A))$ onto the subspace $X(A)_h$ of $X(A)$ consisting of the Hermitian characters of A .

The Hermitian characters of A are the morphisms of involutive algebras from A to the stellar algebra C . Proposition 20 therefore implies that $X(j)$ is a bijection from $X(\text{Stell}(A))$ onto $X(A)_h$. Since $X(j)$ is a homeomorphism onto its image (cf. I, p. 10), the corollary follows.

We identify $X(\text{Stell}(A))$ with $X(A)_h$ by the map $X(j)$. For every $x \in A$, the map $\mathcal{G}_{\text{Stell}(A)}(j(x))$ is none other than the restriction to $X(A)_h$ of $\mathcal{G}_A(x)$.

PROPOSITION 21. — Let A be an involutive Banach algebra and let j be the canonical morphism from A into $\text{Stell}(A)$. The radical of A is contained in the kernel of j .

Let x be an element of the radical of A . Then x^*x belongs to the radical of A , and therefore $\text{Sp}'_A(x^*x) = \{0\}$ (I, p. 5, remark 3). Since $\text{Sp}'_{\text{Stell}(A)}(j(x^*x)) \subset \text{Sp}'_A(x^*x)$, we therefore have $\text{Sp}'_{\text{Stell}(A)}(j(x)^*j(x)) = \{0\}$, whence $\|j(x)\|^2 = \|j(x)^*j(x)\| = \varrho(j(x)^*j(x)) = 0$ (formula (2) of I, p. 104), and therefore $j(x) = 0$.

1.6.13 Stellar algebra of a locally compact group

DEFINITION 9. — Let G be a locally compact group and let A be the involutive Banach algebra of bounded measures on G admitting a density with respect to a Haar measure on G (Example 4 of I, p. 99). The enveloping stellar algebra of the involutive Banach algebra A is called the stellar algebra of G . It is denoted by $\text{Stell}(G)$.

Note. — Let ν be a left Haar measure on G and let Δ be its modulus. The map $f \mapsto f \cdot \nu$ is an isometric isomorphism of the algebra $L^1(G, \nu)$ onto A (loc. cit.). One can therefore also define $\text{Stell}(G)$ as the enveloping stellar algebra of the normed involutive algebra $L^1(G, \nu)$.

Let G be a locally compact group and let ν be a left Haar measure on G . For $\mu \in \mathcal{M}^1(G)$ and $f \in L^2(G, \nu)$, one has $\mu * f \in L^2(G, \nu)$ (INT, VIII, §4, prop. 6). Let us then denote by $\gamma(\mu)$ the endomorphism $f \mapsto \mu * f$ of $L^2(G, \nu)$. The map $\mu \mapsto \gamma(\mu)$ is a representation of the algebra $\mathcal{M}^1(G)$ into the Banach algebra $\mathcal{L}(L^2(G, \nu))$ of continuous endomorphisms of $L^2(G, \nu)$ (INT, VIII, §4, cor. of prop. 6). On the other hand, $\gamma(\check{\mu})$ is the transpose of the endomorphism $\gamma(\mu)$ (INT, VIII, §4, no. 3, prop. 8). It follows that $\gamma(\mu^*)$ is the adjoint endomorphism of $\gamma(\mu)$, and therefore that the map $\gamma: \mu \mapsto \gamma(\mu)$ is a morphism of involutive algebras from $\mathcal{M}^1(G)$ into the stellar algebra $\mathcal{L}(L^2(G, \nu))$, called the left regular representation of $\mathcal{M}^1(G)$ in $L^2(G, \nu)$. According to INT, VIII, §4, no. 7, prop. 19, this representation is faithful.

Let j be the canonical map from $L^1(G, \nu)$ into $\text{Stell}(G)$. By restriction to $L^1(G, \nu)$, the regular representation γ defines an injective morphism of involutive algebras from $L^1(G, \nu)$ into $\mathcal{L}(L^2(G, \nu))$, called the left regular representation of $L^1(G)$ in $L^2(G)$. According to prop. 20, there exists a unique morphism $\gamma': \text{Stell}(G) \rightarrow \mathcal{L}(L^2(G, \nu))$ such that $\gamma = \gamma' \circ j$. We say that γ' is the left regular representation of $\text{Stell}(G)$ in $L^2(G, \nu)$. By abuse of notation, we will still denote it by

$$\gamma'(\varphi)(f) = \varphi * f \quad (7)$$

for $f \in L^2(G, \nu)$ and $\varphi \in \text{Stell}(G)$. We have

$$\|\varphi * f\|_2 \leq \|\varphi\|_* \|f\|_2. \quad (8)$$

Remark. — In general, the left regular representation γ' of $\text{Stell}(G)$ in $L^2(G, \nu)$ is not faithful. One can show that it is faithful if and only if there exists a positive linear form f on $L^\infty_\mathbf{R}(G)$ such that $f(1) = 1$ and $f(\gamma(g)x) = f(x)$ for all $(g, x) \in G \times G$ (we then say that the group G is amenable, cf. EVT, IV, p. 73, exercise 4).

PROPOSITION 22. — The canonical map j from $L^1(G, \nu)$ into $\text{Stell}(G)$ is injective and has dense image.

The image of j is dense by definition of the enveloping stellar algebra of a normed involutive algebra. Since the left regular representation γ is faithful, the injectivity of j follows from the equality $\gamma = \gamma' \circ j$.

COROLLARY. — The algebra $L^1(G, \nu)$ is radical-free.

This follows from prop. 21 of I, p. 125 and from prop. 22.

Thus one can identify $L^1(G, \nu)$ with a dense involutive subalgebra of $\text{Stell}(G)$, and the canonical injection of $L^1(G, \nu)$ into $\text{Stell}(G)$ is then continuous.

Suppose now that the group G is unimodular (INT, VII, §1, no. 3, def. 3). One can then repeat the same arguments starting from the right regular representation $(f, \mu) \mapsto \delta(\mu)(f) = f * \check{\mu}$ of $L^2(G, \nu) \times \mathcal{M}^1(G)$ into $L^2(G, \nu)$. We then define a morphism δ' of $\text{Stell}(G)$ into $\mathcal{L}(L^2(G, \nu))$ such that $\delta = \delta' \circ j$, and define $\delta'(\varphi)(f) = f * \varphi$ for $f \in L^2(G, \nu)$ and $\varphi \in \text{Stell}(G)$.

For $\varphi, \psi \in \text{Stell}(G)$, one has $\delta'(\psi) \circ \gamma'(\varphi) = \gamma'(\psi) \circ \delta'(\varphi)$, that is,

$$(\varphi * f) * \psi = \varphi * (f * \psi) \quad (9)$$

for all $f \in L^2(G, \nu)$. Indeed, this formula is true for $\varphi, \psi \in L^1(G, \nu)$, and the maps $(\varphi, \psi) \mapsto (\varphi * f) * \psi$ and $(\varphi, \psi) \mapsto \varphi * (f * \psi)$ are continuous bilinear maps from $\text{Stell}(G) \times \text{Stell}(G)$ into $L^2(G, \nu)$.

1.7 SPECTRUM OF ENDOMORPHISMS OF BANACH SPACES

Unless stated otherwise, the vector spaces considered in this paragraph are vector spaces over $\mathbb{C}$. We denote by 1_E the identity map of a vector space E . An endomorphism of a topological vector space E is a continuous linear map from E to E .

Let E be a topological vector space and u an endomorphism of E . If F is a subspace of E stable under u , we shall say that the endomorphism of F deduced from u by restriction is the endomorphism of F deduced from u . We denote it by $u|_F$.

1.7.1 Spectrum of an endomorphism

DEFINITION 1. — Let E be a topological vector space and let u be an endomorphism of E . We call the spectrum of u , and denote by $\text{Sp}(u)$, the spectrum of u relative to the unital algebra $\mathcal{L}(E)$.

Let E be a topological vector space and let $u \in \mathcal{L}(E)$. The spectrum of u is the set of complex numbers λ such that $u - \lambda 1_E$ is not an automorphism of E . If E is metrizable and complete, it is also the set of complex numbers λ such that $u - \lambda 1_E$ is not bijective (EVT, I, p. 19, cor. 1).

Every eigenvalue of u belongs to the spectrum of u , but the converse is false in general.

In the rest of this paragraph, we shall restrict ourselves to studying the notion of spectrum in the case where E is a Banach space.

Lemma 1. — Let E be a Banach space and let u be an endomorphism of E . Let $(E_i)_{i \in I}$ be a finite family of closed subspaces of E , stable under u , such that $E = \bigoplus_{i \in I} E_i$. For every $i \in I$, let u_i be the endomorphism of E_i induced by u . We have $\text{Sp}(u) = \bigcup_{i \in I} \text{Sp}(u_i)$, and for every $f \in \mathcal{O}(\text{Sp}(u))$, the endomorphism $f(u)$ stabilizes the spaces E_i , and $f(u)$ coincides with $f(u_i)$ on E_i .

The endomorphism u is an isomorphism if and only if u_i is an isomorphism for every $i \in I$. Applying this property to $u - \lambda 1_E$, one deduces that $\text{Sp}(u)$ is the union of the sets $\text{Sp}(u_i)$ for $i \in I$.

Let $f \in \mathcal{O}(\text{Sp}(u))$. The endomorphism $f(u)$ of E belongs to the bicommutant of u in $\mathcal{L}(E)$ (theorem 5 of I, p. 74), hence it commutes with the projections p_i . It then stabilizes the spaces E_i . Consider the continuous unital morphism ϖ of the product algebra $\prod_{i \in I} \mathcal{L}(E_i)$ into $\mathcal{L}(E)$ defined by $(v_i)_{i \in I} \mapsto \bigoplus_i v_i$. It maps the family (u_i) to u . Proposition 7 of I, p. 75 then implies that $\text{Sp}(u) \subset \bigcup_{i \in I} \text{Sp}(u_i)$ and

$$f(u) = f(\varpi((u_i)_{i \in I})) = \varpi((f(u_i))_{i \in I}) = \bigoplus_{i \in I} f(u_i),$$

which concludes the proof of the lemma.

PROPOSITION 1. — Let E be a complex Banach space and let u be an endomorphism of E . Let $\lambda \in \mathbb{C}$ and $f \in \mathcal{O}(\text{Sp}(u))$. We have

$$\text{Ker}(u - \lambda 1_E) \subset \text{Ker}(f(u) - f(\lambda) 1_E).$$

Let $x \in E$ be nonzero. The set A of $v \in \mathcal{L}(E)$ such that x is an eigenvector of v is a unital subalgebra of $\mathcal{L}(E)$.

The algebra A is full: if $v \in A$ is invertible in $\mathcal{L}(E)$ and if x is an eigenvector of v for the eigenvalue λ , then $\lambda \neq 0$ and $v^{-1}(x) = \lambda^{-1}x$ which proves that $v^{-1} \in A$.

The algebra A is closed in $\mathcal{L}(E)$. Indeed, if $(v_n)_{n \in \mathbb{N}}$ is a sequence in A such that v_n converges to $v \in \mathcal{L}(E)$, the sequence $(\lambda_n)_{n \in \mathbb{N}}$ such that $v_n(x) = \lambda_n x$ is bounded, hence admits a subsequence converging to a complex number μ , so that $v(x) = \mu x$.

Suppose that x is an eigenvector of u and that $u(x) = \lambda x$. The algebra A contains u , hence contains the closed unital full subalgebra B generated by u , which is commutative. The map $\chi: B \rightarrow \mathbb{C}$ which associates with v the eigenvalue of v relative to x is a character of B such that $\chi(u) = \lambda$. For every $f \in \mathcal{O}(\text{Sp}(u))$, one has $f(u) \in B$ and $\chi(f(u)) = f(\chi(u)) = f(\lambda)$ by prop. 7 of I, p. 75, whence the proposition.

1.7.2 Spectral projectors

Let E be a Banach space. We denote by A the unital Banach algebra $\mathcal{L}(E)$ of endomorphisms of E . Let $u \in A$.

Let H be a subset of $\text{Sp}(u)$ which is open and closed in $\text{Sp}(u)$; denote by K its complement in $\text{Sp}(u)$.

The idempotent element associated with u and H (no. 12 of I, p. 81) is a continuous projector of E , called the spectral projector associated with u and H ; it is denoted $e_H(u)$, or simply e_H . Its image is called the spectral subspace of E associated with u and H , and is denoted $E_H(u)$, or simply E_H . Its kernel is $E_K(u)$, and is also denoted $\tilde{E}_H(u)$, or simply $\tilde{E}_H$. The space E is the topological direct sum of the closed subspaces E_H and E_K . The condition $E_H = 0$ is equivalent to $e_H = 0$, that is, to the indicator function f_H of H on $\text{Sp}(u)$ being the zero function, and hence to $H = \emptyset$.

Every endomorphism v of E that commutes with u also commutes with $e_H(u)$ (th. 5 of I, p. 74), hence stabilizes E_H and E_K . In particular, the endomorphism u leaves the subspaces E_H and E_K stable. The space E_K is the only topological supplement of E_H in E that is stable under u .

The unital algebra $A_H = e_H A e_H$ (loc. cit.) is the subalgebra of A consisting of the endomorphisms of E that leave E_H stable and that vanish on E_K . For every $v \in A_H$, we denote by $v|_{E_H}$ the endomorphism of E_H deduced from v . The map $v \mapsto v|_{E_H}$ is an isomorphism from A_H onto $\mathcal{L}(E_H)$. In particular, we have $\text{Sp}(u|_{E_H}) = \text{Sp}_{A_H}(u|_{E_H}) = H$ and $\text{Sp}_{A_K}(u|_{E_K}) = K$ by formula (18) of I, p. 82.

PROPOSITION 2. — Let E be a Banach space and let u be an endomorphism of E . Let E_1 and E_2 be closed subspaces of E stable under u such that $E = E_1 \oplus E_2$. Suppose that the endomorphisms u_1 and u_2 of E_1 and E_2 deduced from u have disjoint spectra $H_1 = \text{Sp}(u_1)$

and $H_2 = \text{Sp}(u_2)$. Then $\text{Sp}(u) = H_1 \cup H_2$ and $e_{H_1}(u)$ is the projector with image E_1 and kernel E_2 . In particular, we have $E_{H_1} = E_1$ and $E_{H_2} = E_2$.

We have $\text{Sp}(u) = H_1 \cup H_2$ (I, p. 128, lemma 1). Since the sets H_1 and H_2 are compact, they are open and closed in $\text{Sp}(u)$. For every holomorphic function f in a neighborhood of $\text{Sp}(u)$, the endomorphism $f(u)$ stabilizes E_1 and E_2 and coincides on E_1 with $f(u_1)$ and on E_2 with $f(u_2)$ (loc. cit.). Take in particular for f the germ of a holomorphic function f_{H_1} which equals 1 in a neighborhood of H_1 and 0 in a neighborhood of H_2 (cf. no. 12 of I, p. 81); then $f_{H_1}(u_1)$ is the identity map of E_1 since $f_{H_1} = 1$ in a neighborhood of $H_1 = \text{Sp}(u_1)$, and $f_{H_1}(u_2)$ is zero

since $f_{H_1} = 0$ in a neighborhood of H_2 . Therefore $e_{H_1}(u) = f_{H_1}(u)$ is the projector with image E_1 and kernel E_2 , and we therefore have $E_1 = E_{H_1}$ and $E_2 = E_{H_2}$.

Let E be a Banach space and u an endomorphism of E . Let $(H_i)_{i \in I}$ be a finite family of open and closed subsets of $\text{Sp}(u)$, pairwise disjoint, and let H be their union. Relations (15) and (16) of I, p. 81 imply the following assertions:

- a) The family of projectors $(e_{H_i}(u))_{i \in I}$ is orthogonal (that is, $e_{H_i}(u)e_{H_j}(u) = 0$ for all (i, j) in I^2 such that $i \neq j$, cf. A, II, p. 18, def. 7) and its sum is $e_H(u)$;
- b) The vector space E_H is the topological direct sum of the family $(E_{H_i})_{i \in I}$;
- c) For every $j \in I$, one has $E_{\text{Sp}(u) - H_j} = E_{\text{Sp}(u) - H} \oplus \bigoplus_{i \neq j} E_{H_i}$;
- d) The vector space $E_{\text{Sp}(u) - H}$ is the intersection of the family $(E_{\text{Sp}(u) - H_i})_{i \in I}$.

When $H = \text{Sp}(u)$, the topological direct sum decomposition $\bigoplus_{i \in I} E_{H_i}$ of E is called the spectral decomposition of E associated with u and the finite partition $(H_i)_{i \in I}$ of $\text{Sp}(u)$.

Let f be an element of $\mathcal{O}(\text{Sp}(u))$. The spectrum of the endomorphism $f(u)$ of E is the image under f of the spectrum of u (I, p. 75, prop. 8). Let L be an open and closed subset of $\text{Sp}(f(u))$. The set H of elements $\lambda \in \text{Sp}(u)$ such that $f(\lambda)$ belongs to L is open and closed in $\text{Sp}(u)$, and we have $e_L(f(u)) = e_H(u)$ since $f_L \circ f = f_H$ in $\mathcal{O}(\text{Sp}(u))$ (loc. cit.).

1.7.3 Isolated points of the spectrum

Let E be a Banach space and let u be an endomorphism of E . Let $\lambda \in C$ be an isolated point of $\text{Sp}(u)$. We then denote by $E_\lambda(u) = E_{\{\lambda\}}(u)$ and $e_\lambda(u)$ the spectral projector with image $E_\lambda(u)$ associated with u and $\{\lambda\}$.

We also denote by $\tilde{E}_\lambda(u) = E_{\text{Sp}(u) - \{\lambda\}}(u)$. The spectrum of the endomorphism of $\tilde{E}_\lambda(u)$ induced by u is $\text{Sp}(u) - \{\lambda\}$; in particular, $u - \lambda 1_E$ induces an automorphism of $\tilde{E}_\lambda(u)$.

The space $E_\lambda(u)$ is not zero. The spectrum of the endomorphism of $E_\lambda(u)$ induced by u is reduced to λ , hence $u - \lambda 1_E$ induces a quasi-nilpotent endomorphism of $E_\lambda(u)$. The endomorphism $u - \lambda 1_E$ induces an automorphism of $\tilde{E}_\lambda(u)$, hence $\text{Ker}(u - \lambda 1_E)^n \subset E_\lambda(u)$ for all $n \in \mathbb{N}$. In particular, we have $\text{Ker}(u - \lambda 1_E) \subset E_\lambda(u)$.

For λ to be a pole of order $p > 0$ of the resolvent of u , it is necessary and sufficient that $(u - \lambda 1_E)^{p-1} e_\lambda(u) \neq 0$ and $(u - \lambda 1_E)^p e_\lambda(u) = 0$ (corollary of proposition 17 of I, p. 83). In this case, we have $E_\lambda(u) = \text{Ker}((u - \lambda 1_E)^p)$ and $\tilde{E}_\lambda(u) = \text{Im}((u - \lambda 1_E)^p)$ since $(u - \lambda 1_E)^p$ induces an automorphism of $\tilde{E}_\lambda(u)$. We also have

$$(u - \lambda 1_E)^{p-1} e_\lambda(u) = \lim_{z \rightarrow \lambda} (z - \lambda 1_E)^p R(u, z)$$

according to proposition 17 of I, p. 83.



Note that in general $E_\lambda(u)$ is not the union of the family $(\text{Ker}((u - \lambda 1_E)^n))_{n \in \mathbb{N}}$, nor even the closure of this union; in particular, an isolated point of $\text{Sp}(u)$ is not necessarily an eigenvalue of u (I, p. 187, exerc. 1). Likewise, there may exist eigenvalues of u that are not isolated points of $\text{Sp}(u)$ (I, p. 188, exerc. 2).

1.7.4 Spectrum of the transpose of an endomorphism

PROPOSITION 3. — Let E be a Banach space and let E' be its dual. Let u be an endomorphism of E .

- a) We have $\text{Sp}(u) = \text{Sp}({}^t u)$;
- b) For every $f \in \mathcal{O}(\text{Sp}(u))$, one has $f({}^t u) = {}^t f(u)$;
- c) We have $e_H({}^t u) = {}^t e_H(u)$ for every subset H of $\text{Sp}(u)$ that is open and closed.

For an endomorphism of E to be an automorphism, it is necessary and sufficient that its transpose be an automorphism of E' (EVT, IV, p. 30, cor. 5), whence assertion a).

The map $v \mapsto {}^t v$ is a unital continuous homomorphism of the Banach algebra $\mathcal{L}(E)$ into the opposite Banach algebra of $\mathcal{L}(E')$ (EVT, IV, p. 7, prop. 8). Since the algebra $\mathcal{O}(\text{Sp}({}^t u))$ is commutative, the map $f \mapsto {}^t f(u)$ is a unital continuous homomorphism of the algebra $\mathcal{O}(\text{Sp}({}^t u))$ into the algebra $\mathcal{L}(E')$. This homomorphism maps the germ of the identity map of $\mathbb{C}$ to ${}^t u$,

hence coincides with the homomorphism $f \mapsto f({}^t u)$ (th. 5 of I, p. 74). This proves b).

Assertion c) follows from b), applied to the function f_H equal to 1 in a neighborhood of H and to 0 in a neighborhood of its complement in $\text{Sp}(u)$.

Remark. — Let H be an open and closed subset of $\text{Sp}(u)$, whose associated spectral decomposition is $E = E_H(u) \oplus \tilde{E}_H(u)$. It follows from assertion c) that if one identifies E' with $E_H(u)' \oplus \tilde{E}_H(u)'$, then we have $E'_H({}^t u) = E_H(u)'$ and $\tilde{E}'_H({}^t u) = \tilde{E}_H(u)'$.

1.7.5 The case of Hilbert spaces

In this section, we consider Hilbert spaces over $K = \mathbb{R}$ or $\mathbb{C}$. We denote by $\langle x_1 | x_2 \rangle$ the scalar product of two vectors x_1 and x_2 in a Hilbert space E .

If E is a complex Hilbert space, the Banach algebra $\mathcal{L}(E)$ equipped with the involution $u \mapsto u^*$ is a stellar algebra (example 1 of I, p. 102). In particular, if $u \in \mathcal{L}(E)$, one has $\varrho(u^*) = \varrho(u)$ and $\text{Sp}(u^*) = \overline{\text{Sp}(u)}$.

Let $u \in \mathcal{L}(E)$ be a normal endomorphism and let $\lambda \in \mathbb{C}$. The eigenspace of u corresponding to λ coincides with the eigenspace of the adjoint u^* corresponding to $\overline{\lambda}$ (EVT, V, p. 43, cor. of prop. 8). However, these spaces do not coincide in general if u is not normal (exercise 3 of I, p. 188).

Let λ and μ be complex numbers such that $\lambda \neq \mu$. Let x be an eigenvector of u corresponding to λ and let y be an eigenvector of u corresponding to μ . Then, $u^*(x) = \overline{\lambda}x$, hence

$$\mu \langle x | y \rangle = \langle x | u(y) \rangle = \langle u^*(x) | y \rangle = \langle \overline{\lambda}x | y \rangle = \lambda \langle x | y \rangle.$$

Consequently, $\langle x | y \rangle = 0$: the eigenspaces of u are pairwise orthogonal. Moreover, for every $\lambda \in \mathbb{C}$, the eigenspace of u relative to λ coincides with the primary subspace of u relative to λ , that is (LIE, VII, §1, no. 1) the union over $k \in \mathbb{N}$ of the kernels of $(u - \lambda 1_E)^k$ (EVT, V, p. 43, cor. of prop. 8).

Lemma 2. — Let E be a complex Hilbert space and let u be a normal endomorphism of E . Let $(E_i)_{i \in I}$ be a finite family of closed subspaces of E , stable under u and pairwise orthogonal, such that $E = \bigoplus_{i \in I} E_i$. For every $i \in I$, let u_i be the endomorphism of E_i induced by u . We have $\text{Sp}(u) = \bigcup_{i \in I} \text{Sp}(u_i)$, and for every $f \in \mathcal{C}(\text{Sp}(u))$, the endomorphism $f(u)$ stabilizes the

spaces E_i , and $f(u)$ coincides with $f(u_i)$ on E_i .

The proof follows that of lemma 1 of I, p. 128, using remark 6 of I, p. 110 and prop. 8 of I, p. 112.

PROPOSITION 4. — Let E be a complex Hilbert space and let u be a normal endomorphism of E . For every function $f \in \mathcal{C}(\text{Sp}(u))$ and every $\lambda \in \mathbf{C}$, one has

$$\text{Ker}(u - \lambda 1_E) \subset \text{Ker}(f(u) - f(\lambda)1_E).$$

The proof is analogous to that of Proposition 1 of I, p. 128; let us take up its arguments again. The algebra A introduced in loc. cit. is here a unital stellar subalgebra of $\mathcal{L}(E)$ (EVT, V, p. 43, cor.) . It therefore contains the unital stellar subalgebra B generated by u , which is commutative. The map $\chi : B \rightarrow \mathbf{C}$ which associates with v the eigenvalue of v relative to x is a character of B such that $\chi(u) = \lambda$. For every $f \in \mathcal{C}(\text{Sp}(u))$, one has $f(u) \in B$ and $\chi(f(u)) = f(\chi(u)) = f(\lambda)$ by prop. 8 of I, p. 112, whence the assertion.

Lemma 3. — Let E be a Hilbert space and let $p \in \mathcal{L}(E)$ be a projector. The following assertions are equivalent:

- (i) The projector p is an orthogonal projector, that is, $\text{Ker}(p) = \text{Im}(p)^\circ$ (EVT, V, p. 13);
- (ii) The projector p is Hermitian;
- (iii) The projector p is normal;
- (iv) We have $\text{Ker}(p) \subset \text{Ker}(p^*)$;
- (v) We have $\text{Im}(p) \subset \text{Im}(p^*)$;
- (vi) The projector p is positive;
- (vii) We have $\|p\| \leq 1$.

Let us first recall that $\text{Ker}(p^*) = \text{Im}(p)^\circ$ and $\text{Ker}(p) = \text{Im}(p^*)^\circ$ (EVT, V, p. 41, prop. 4). Moreover, the image of p (resp. p^*) is closed, since it coincides with the kernel of $1 - p$ (resp. $1 - p^*$). Hence we have

$$\text{Im}(p) = \text{Ker}(p^*)^\circ, \quad \text{Im}(p^*) = \text{Ker}(p)^\circ. \quad (1)$$

(i) $\Rightarrow$ (ii): p^* is a projector with kernel $\text{Im}(p)^\circ = \text{Ker}(p)$ and whose image is $\text{Im}(p^*)^\circ = \text{Ker}(p) = \text{Im}(p)^\circ$; therefore $p^* = p$.

(ii) $\Rightarrow$ (iii): since every Hermitian endomorphism is normal.

(iii) $\Rightarrow$ (iv): since p is normal, we have $\|p(x)\|^2 = \|p^*(x)\|^2$ for all x in E (EVT, V, p. 43, prop. 7), whence the required inclusion.

(iv) $\Rightarrow$ (v) follows from equalities (1) above.

(v) $\Rightarrow$ (vi): for all $x \in E$, one has $p(x) \in \text{Im}(p^*)$, and consequently $\langle p(x)|x \rangle = \langle p^*(p(x))|x \rangle = \|p(x)\|^2 \geq 0$.

(vi) $\Rightarrow$ (vii): let $x \in E$ and $y = x - p(x) \in \text{Ker}(p)$. For every $t \in R$, by hypothesis

$$\langle x + ty|p(x) \rangle = \langle x + ty|p(x + ty) \rangle \geq 0,$$

which is possible only if $\langle y|p(x) \rangle = 0$. But then

$$\|p(x)\|^2 = \langle x|p(x) \rangle \leq \|x\|\|p(x)\|,$$

and therefore $\|p\| \leq 1$.

(vii) $\Rightarrow$ (i): let $y \in \text{Im}(p)$; let z be the orthogonal projection of y on $\text{Ker}(p)^\circ$ and set $x = y - z \in \text{Ker}(p)$. We have $p(z) = p(y) = y$, hence $\|y\| \leq \|z\|$ by hypothesis. But, since x and z are orthogonal, we have $\|y\|^2 = \|x\|^2 + \|z\|^2$, whence $\|x\| = 0$, that is, $y = z$. Thus, $\text{Im}(p) \subset \text{Ker}(p)^\circ$. Since moreover $\|p^*\| = \|p\| \leq 1$, we likewise have $\text{Im}(p^*) \subset \text{Ker}(p^*)^\circ$ which provides the reverse inclusion by (1).

PROPOSITION 5. — Let E be a complex Hilbert space and let u be a normal endomorphism of E .

- a) For every open and closed subset H of the spectrum of u , the spectral projector $e_H(u)$ is an orthogonal projector whose kernel is the image of the spectral projector $e_{\text{Sp}(u)-H}(u)$;
- b) If H_1 and H_2 are disjoint, open, and closed subsets of the spectrum of u , then the spectral subspaces E_{H_1} and E_{H_2} are orthogonal;
- c) If $\lambda \in \mathbb{C}$ is an isolated point of the spectrum of u , then λ is an eigenvalue of u and the image of the spectral projector $e_\lambda(u)$ is the eigenspace of u corresponding to λ .

We prove *a*). Since the holomorphic functional calculus is compatible with the continuous functional calculus (I, p. 111, cor. 1), we have $e_H(u) = \varphi_H(u)$, where $\varphi_H \in \mathcal{C}(\text{Sp}(u))$ is the characteristic function of H . This implies $e_H(u)^* = \overline{\varphi_H}(u) = \varphi_H(u) = e_H(u)$, hence $e_H(u)$ is an orthogonal projector (Lemma 3, (ii)). Its kernel is the image of the projector $1 - e_H(u) = e_{\text{Sp}(u)-H}(u)$.

We prove *b*). The characteristic functions φ_{H_1} and φ_{H_2} of H_1 and H_2 on $\text{Sp}(u)$ are continuous and their product is zero, which implies $e_{H_1}(u) \circ e_{H_2}(u) = e_{H_2}(u) \circ e_{H_1}(u) = 0$. The inclusions $E_{H_2}(u) \subset E_{H_1}(u)^\circ$ and $E_{H_1}(u) \subset E_{H_2}(u)^\circ$ result from them.

Let us finally prove the assertion *c*). The characteristic function φ_λ of $\{\lambda\}$ is continuous and nonzero on $\text{Sp}(u)$; it satisfies $(z - \lambda)\varphi_\lambda(z) = 0$ for all $z \in \text{Sp}(u)$. We therefore have $(u - \lambda 1_E)\varphi_\lambda(u) = 0$. The nonzero image of $\varphi_\lambda(u)$ is therefore contained in the eigenspace of u corresponding to λ . Since we have $e_\lambda(u) = \varphi_\lambda(u)$ and since the image of $e_\lambda(u)$ contains the eigenspace of u corresponding to λ , the assertion follows from this.

Lemma 4. — Let E be a Hilbert space and let u be a normal endomorphism of E . Let F be a closed subspace of E containing a total set of eigenvectors of u . Then F° is stable under u , and the endomorphism $\tilde{u}$ of F° induced by u is normal.

Since u is normal, every eigenvector of u is also an eigenvector of u^* (EVT, V, p. 43, cor.). The hypothesis therefore implies that F is stable under u and u^* . By EVT, V, p. 41, prop. 4 (ii), we therefore have $u(F^\circ) \subset F^\circ$ and $u^*(F^\circ) \subset F^\circ$. It follows that the adjoint of $\tilde{u}$ is the endomorphism of F° induced by u^* . Since u is normal, the endomorphism $\tilde{u}$ is normal.

1.7.6 Numerical range

DEFINITION 2. — Let E be a complex Hilbert space and let u be an endomorphism of E . We call the numerical range of u the set of complex numbers of the form $\langle x|u(x) \rangle$, where x ranges over the unit sphere of E . We denote it by $\iota(u)$.

The numerical range of u^* is the image of $\iota(u)$ by complex conjugation. For all complex numbers λ and μ , the numerical range of $\lambda u + \mu 1_E$ is equal to $\lambda \iota(u) + \mu$.

PROPOSITION 6. — Let E be a complex Hilbert space and let u be an endomorphism of E .

- a) The set of eigenvalues of u is contained in $\iota(u)$;
- b) The spectrum of u is contained in the closure of $\iota(u)$ in $\mathbb{C}$.

Let λ be an eigenvalue of u and let $x \in E$ be a nonzero vector such that $u(x) = \lambda x$. Replacing x by $x/\|x\|$, we may assume that $\|x\| = 1$. Then, $\langle x|u(x) \rangle = \lambda$, hence $\lambda \in \iota(u)$.

We prove *b*). By considering $u - \lambda 1_E$, we reduce to proving that if 0 belongs to the spectrum of u , then 0 is adherent to $\iota(u)$.

Suppose first that there exists a real number $c > 0$ such that $\|u(x)\| \geq c$ for all x of norm 1 in E . The endomorphism u is then injective and has closed range (Lemma 8 of I, p. 107). Since it is not invertible by hypothesis, it is not surjective. Consequently, the orthogonal complement of the kernel of u^* is not equal to E (EVT, V, p. 41, prop. 4), which proves that the kernel of u^* is nonzero. Thus 0 belongs to $\iota(u^*)$, hence to $\iota(u)$.

If the preceding hypothesis is not valid, then for every integer $n \geq 1$, there exists a vector x_n of norm 1 in E such that $\|u(x_n)\| \leq 1/n$. We then have $|\langle x_n|u(x_n) \rangle| \leq 1/n$, which implies that 0 belongs to the closure of $\iota(u)$.

PROPOSITION 7 (Hausdorff-Toeplitz Theorem)

Let E be a complex Hilbert space and let $u \in \mathcal{L}(E)$. The numerical range $\iota(u)$ is a convex subset of $\mathbb{C}$.

We will need two lemmas to prove this proposition.

Lemma 5. — Let E be a complex Hilbert space of dimension 2. Equip the real vector space $\mathcal{L}(E)_h$ of Hermitian endomorphisms of E with the pre-Hilbert norm $u \mapsto \text{Tr}(u^*u)^{1/2}$. The set of rank-1 orthogonal projectors of E is the sphere S of radius $1/\sqrt{2}$ centered at $\frac{1}{2}1_E$ in the

3-dimensional affine subspace of endomorphisms of trace 1 in $\mathcal{L}(E)_h$.

Let F be the real affine subspace of $\mathcal{L}(E)_h$ consisting of elements of trace 1. The rank-1 orthogonal projectors of E belong to F (lemma 3, (ii)).

Let $u \in F$. We have $\|u - \frac{1}{2}1_E\|^2 = \text{Tr}(u^2 - u + \frac{1}{4}) = \text{Tr}(u^2) - \frac{1}{2}$. Consequently, $u \in S$ if and only if $\text{Tr}(u^2) = 1$. Since $2 \det(u) = \text{Tr}(u)^2 - \text{Tr}(u^2) = 1 - \text{Tr}(u^2)$, this condition is equivalent to $\det(u) = 0$. By the Hamilton–Cayley theorem (A, III, p. 107, prop. 20), we therefore have $u \in S$ if and only if $u^2 - u = 0$, which means that u is a rank-1 Hermitian projector (loc. cit.), whence the result.

Lemma 6. — Let E be a real normed vector space, let F be a real vector space, and let $u: E \rightarrow F$ be a non-injective affine map. Let B be a ball in E and S the corresponding sphere. We have $u(S) = u(B)$, and in particular, $u(S)$ is convex.

We reduce to the case where u is linear and where B is the unit ball of E . We have $u(S) \subset u(B)$. Conversely, let $x \in B$ and let y be a nonzero element of $\text{Ker}(u)$. The image of the continuous map $t \mapsto \|x + ty\|$ of $\mathbf{R}$ into $\mathbf{R}_+$ is an unbounded interval containing the real number $\|x\| \leq 1$. Hence there exists $t \in \mathbf{R}$ such that $\|x + ty\| = 1$. We then have $x + ty \in S$ and $u(x + ty) = u(x)$, hence $u(x) \in u(S)$.

Let us prove Prop. 7. Let x and y be elements of the unit sphere of E ; we prove that the segment with endpoints $\langle x|u(x) \rangle$ and $\langle y|u(y) \rangle$ is contained in the numerical range of u .

Let F be the subspace of E spanned by x and y . If $\dim(F) = 1$, one has $\langle x|u(x) \rangle = \langle y|u(y) \rangle$, whence the assertion. Otherwise, we have $\dim(F) = 2$; let p be the orthogonal projector of E with image F , and denote by u_F the endomorphism of F given by $x \mapsto p(u(x))$. Since p is Hermitian (Lemma 3 of I, p. 133), we have $\langle z|u_F(z) \rangle = \langle z|u(z) \rangle$ for all $z \in F$, so that $\iota(u_F) \subset \iota(u)$. We may therefore assume that $E = F$.

For every element z of E , let v_z be the Hermitian endomorphism of E defined by $t \mapsto \langle z|t \rangle z$; we have $\langle z|u(z) \rangle = \text{Tr}(u \circ v_z)$. When z traverses the unit sphere of E , v_z describes the set of rank-1 orthogonal projectors of E , which is a sphere S in the real affine subspace V of $\mathcal{L}(E)$ formed by the Hermitian endomorphisms of E with trace 1 (Lemma 5). The numerical range of E is therefore the set of $\text{Tr}(u \circ v)$, for $v \in S$. The map $v \mapsto \text{Tr}(u \circ v)$ of V in $\mathbf{C}$ is linear. Since $\dim_{\mathbf{R}}(V) = 3 > \dim_{\mathbf{R}}(\mathbf{C})$, it is not injective; it then follows from Lemma 6 that $\iota(u)$ is convex.

1.7.7 Positive elements

Let E be a complex Hilbert space. Let u be an endomorphism of E . Recall (EVT, V, p. 45, def. 6) that u is said to be positive if one has $\langle x|u(x) \rangle \geq 0$ for all $x \in E$. The endomorphism u is then Hermitian (loc. cit.). Moreover, if F is a complex Hilbert space and if $v \in \mathcal{L}(F; E)$, then the endomorphism v^*uv of F is positive (EVT, V, p. 45, prop. 12).

PROPOSITION 8. — Let E be a complex Hilbert space. Let u be an endomorphism of E . The following conditions are equivalent:

- (i) The endomorphism u is positive;
- (ii) The image of u under the numerical map is contained in $\mathbf{R}_+$;
- (iii) The endomorphism u is a positive element of the stellar algebra $\mathcal{L}(E)$;
- (iv) There exists a Hermitian element v of $\mathcal{L}(E)$ such that $u = v^2$;
- (v) There exists a continuous linear map v from E into a complex Hilbert space F such that $u = v^*v$.

By EVT, V, p. 45, def. 6, u is positive if and only if it is Hermitian and if $\langle x|u(x) \rangle \geq 0$ for all $x \in E$. (i) The implication (i) $\Rightarrow$ (ii) therefore follows from the definition of the numerical image.

(ii) $\Rightarrow$ (iii): the hypothesis implies that u is Hermitian (EVT, V, p. 45, and remark, p. 2) and its spectrum is contained in $\mathbf{R}_+$ (prop. 6); consequently, u is a positive element of the stellar algebra $\mathcal{L}(E)$.

(iii) $\Rightarrow$ (iv): this is a special case of prop. 16 of I, p. 118.

(iv) $\Rightarrow$ (v) is immediate.

(v) $\Rightarrow$ (i): let F be a complex Hilbert space and let $v \in \mathcal{L}(E; F)$ such that $u = v^*v$. Let $x \in E$. We have $\langle x|u(x) \rangle = \langle x|(v^*v)(x) \rangle = \|v(x)\|^2$, which proves that u is positive.

Recall (EVT, V, p. 45, remark 1) that, for every hermitian element u of $\mathcal{L}(E)$, we set

$$m(u) = \inf_{\substack{x \in E \\ \|x\|=1}} \langle x|u(x) \rangle = \inf \imath(u) = \inf_{x \in E - \{0\}} \frac{\langle x|u(x) \rangle}{\|x\|^2},$$

$$M(u) = \sup_{\substack{x \in E \\ \|x\|=1}} \langle x|u(x) \rangle = \sup \imath(u) = \sup_{x \in E - \{0\}} \frac{\langle x|u(x) \rangle}{\|x\|^2}.$$

If $E = \{0\}$, then $M(u) = -\infty$, $m(u) = +\infty$, and $\imath(u) = \emptyset$.

Suppose E is nonzero; we then have $m(u) \leq M(u)$ and the numerical range of u is an interval with endpoints $m(u)$ and $M(u)$. By prop. 6, $\text{Sp}(u)$ is contained in the interval $[m(u), M(u)]$. More precisely:

PROPOSITION 9. — Let E be a complex Hilbert space and let u be a hermitian element of $\mathcal{L}(E)$.

a) We have $m(u) = \inf \text{Sp}(u)$ and $M(u) = \sup \text{Sp}(u)$;

b) If E is not zero, we have $\|u\| = \sup(|m(u)|, |M(u)|)$.

Let $\lambda \in \mathbf{R}$. For λ to be a lower bound of the spectrum of u , it is necessary and sufficient that $u - \lambda \geq 0$. This is equivalent (prop. 8, (ii)) to the condition $\langle x|u(x) \rangle \geq \lambda \|x\|^2$ for all $x \in E$, that is, to $m(u) \geq \lambda$. This proves that $m(u)$ is the lower bound of $\text{Sp}(u)$. Similarly, one verifies that $M(u)$ is the upper bound of $\text{Sp}(u)$.

Since u is normal, we have $\varrho(u) = \|u\|$ (cor. 1 of I, p. 108). Since $E \neq \{0\}$, the spectrum of u is not empty (cor. 1 of I, p. 26) and $\varrho(u)$ is the radius of the smallest disk with center 0 that contains $\text{Sp}(u)$ (th. 1 of I, p. 24), hence b) follows from a).

1.7.8 Polar decomposition

In this section, we consider complex Hilbert spaces.

Let E_1 and E_2 be Hilbert spaces and $u \in \mathcal{L}(E_1; E_2)$. The endomorphism u^*u of E_1 is positive (prop. 8), hence one can form the positive element $(u^*u)^{1/2}$ of $\mathcal{L}(E_1)$.

DEFINITION 3. — One says that $(u^*u)^{1/2}$ is the absolute value of u , and one denotes it by $|u|$.

In the case where $E_1 = E_2$, this definition coincides with that given in remark 9 of I, p. 119.

For an element u of $\mathcal{L}(E_1; E_2)$, recall (EVT, V, p. 41, def. 2) that the initial subspace of u is the closed subspace $\text{Ker}(u)^\circ$ of E_1 and the final subspace of u is the closed subspace $\overline{\text{Im}(u)}$ of E_2 .

PROPOSITION 10. — Let E_1 and E_2 be complex Hilbert spaces and $u \in \mathcal{L}(E_1; E_2)$.

- a) The initial subspace and the final subspace of $|u|$ are both equal to the initial subspace of u and we have $\| |u| \| = \|u\|$;
- b) There exists a unique partial isometry j of E_1 into E_2 such that $\text{Ker}(j) = \text{Ker}(u)$ and $u = j|u|$;
- c) The initial (resp. final) subspace of j is equal to that of u ;
- d) Let u_1 be a positive element of $\mathcal{L}(E_1)$ and j_1 be a partial isometry of $\mathcal{L}(E_1; E_2)$ such that $u = j_1 u_1$ and $\text{Ker}(j_1) = \text{Ker}(u_1)$. Then $u_1 = |u|$ and $j_1 = j$.

For every $x \in E_1$, one has

$$\|u(x)\|^2 = \langle x | (u^* u) (x) \rangle = \langle x | |u|^2 (x) \rangle = \| |u|(x) \|^2. \quad (2)$$

This proves that $\text{Ker}(u) = \text{Ker}(|u|)$ and $\| |u| \| = \|u\|$. Since $|u|$ is Hermitian, the closure of the image of $|u|$ is the orthogonal complement of its kernel (EVT, V, p. 41, prop. 4), that is, the initial space of u , whence a).

Formula (2) implies that there exists an isometric map v of $\text{Im}(|u|)$ on $\text{Im}(u)$ such that $v(|u|(x)) = u(x)$ for all $x \in E_1$. Let j be the unique element of $\mathcal{L}(E_1; E_2)$ which extends v and vanishes on $\text{Im}(|u|)^\circ = \text{Ker}(|u|) = \text{Ker}(u)$. Then j has the properties of b). The uniqueness of j follows from the decomposition $E = \text{Ker}(u) \oplus \overline{\text{Im}(|u|)}$.

The initial subspace of j is $\text{Ker}(j)^\circ = \text{Ker}(u)^\circ$, the initial space of u . Its final subspace is $\overline{j(\text{Ker}(u)^\circ)} = \overline{j(\text{Im}(|u|))} = \overline{\text{Im}(u)}$, the final subspace of u . This proves c).

Now let u_1 and j_1 be as in d). We have $u^* u = u_1 j_1^* j_1 u_1$. The map $j_1^* j_1$ is the orthogonal projector with kernel $\text{Ker}(j_1) = \text{Ker}(u_1)$ (EVT, V, p. 41, prop. 5 (ii)) and hence with image $\overline{\text{Im}(u_1)}$. Therefore $u^* u = u_1^2$ and consequently $u_1 = (u^* u)^{1/2} = |u|$ (I, p. 118, prop. 16). The

uniqueness assertion of b) finally implies that $j_1 = j$.

DEFINITION 4. — Let E_1 and E_2 be complex Hilbert spaces and $u \in \mathcal{L}(E_1; E_2)$. The pair $(j, |u|)$, where j is the unique partial isometric map from E_1 into E_2 such that $u = j|u|$ and $\text{Ker}(j) = \text{Ker}(u)$, is called the polar decomposition of u .

PROPOSITION 11. — Let E_1 and E_2 be complex Hilbert spaces and $u \in \mathcal{L}(E_1; E_2)$. Let $(j, |u|)$ be the polar decomposition of u .

- a) We have $|u| = j^*u = u^*j$;
- b) We have $|u^*| = ju^* = uj^*$;
- c) The polar decomposition of u^* is $(j^*, |u^*|)$.

Let $I = \text{Ker}(u)^\circ$ and $F = \overline{\text{Im}(u)}$ be the initial subspace and the final subspace of u ; moreover, we have $I = \text{Ker}(|u|)^\circ = \overline{\text{Im}(|u|)}$ (prop. 10, a)). The map j^*j is the orthogonal projector of E_1 onto I (loc. cit. and

EVT, V, p. 41, prop. 5 (ii)). We therefore have $j^*u = j^*j|u| = |u|$, and then $u^*j = (j^*u)^* = |u|^* = |u|$, whence a).

Similarly, we compute $u^* = |u|j^* = (j^*j|u|)j^* = j^*(j|u|j^*)$. The endomorphism $j|u|j^*$ of E_2 is positive. The linear map j^* is partially isometric with initial space F and final space I (EVT, V, p. 41, prop. 5), and the linear maps from I to F (resp. from F to I) deduced from j and j^* by passing to subspaces are mutually inverse isomorphisms (loc. cit.). We therefore have $\text{Ker}(j|u|j^*) = \text{Ker}(|u|j^*) = \text{Ker}(j^*)$, since the image of j^* is contained in $\text{Ker}(u)^\circ = \text{Ker}(|u|)^\circ$. By prop. 10, d), the pair $(j^*, j|u|j^*)$ is the polar decomposition of u^* . This proves c), and assertion b) then follows from assertion a) applied to u^* .

COROLLARY. — Let E_1 and E_2 be complex Hilbert spaces and let $u \in \mathcal{L}(E_1; E_2)$. We have $\text{Im}(u) = \text{Im}(|u^*|)$.

Let $(j, |u|)$ be the polar decomposition of u . We have $|u^*| = j|u|j^* = uj^*$ by the preceding proposition. By EVT, V, p. 41, prop. 5, the map j^* is partially isometric. Its final space is $\text{Ker}(j)^\circ = \text{Ker}(u)^\circ$ (prop. 10, c)). The assertion follows.

PROPOSITION 12. — Let E_1 and E_2 be complex Hilbert spaces and $u \in \mathcal{L}(E_1; E_2)$. Let $(j, |u|)$ be the polar decomposition of u . For u to be bijective, it is necessary and sufficient that

$|u|$ be invertible in $\mathcal{L}(E_1)$ and that j be an isomorphism from E_1 onto E_2 .

The condition is sufficient. Conversely, if u is bijective, then u^*u is invertible in $\mathcal{L}(E_1)$, and $|u| = (u^*u)^{1/2}$ is also bijective since its spectrum is contained in R_+^* . Moreover, $\text{Ker}(j) = \text{Ker}(u) = \{0\}$ and $\text{Im}(j) = \text{Im}(u) = F$, so j maps E_1 isometrically onto E_2 .

PROPOSITION 13. — Let E be a complex Hilbert space and let u be an endomorphism of E . The following conditions are equivalent:

- (i) The endomorphism u is normal;
- (ii) There exists a unitary element v of $\mathcal{L}(E)$, commuting with $|u|$, such that $u = v|u|$.

Let $(j, |u|)$ be the polar decomposition of u . Suppose that u is normal. Then we have $|u^*| = (uu^*)^{1/2} = (u^*u)^{1/2} = |u|$. Prop. 11 then implies $|u|j = |u^*|j = ju^*j = j|u|$. Moreover, j stabilizes the supplementary orthogonal subspaces $\text{Ker}(|u|) = \text{Ker}(j)$ and $\overline{\text{Im}(|u|)} = \overline{\text{Im}(j)}$ (prop. 10). Let v be the element of $\mathcal{L}(E)$ which coincides with j on $\text{Ker}(u)^\circ$ and with the identity map on $\text{Ker}(u)$. Since j induces an isometry of $\text{Ker}(u)^\circ$ onto $\overline{\text{Im}(u)} = \text{Ker}(u^*)^\circ = \text{Ker}(u)^\circ$ (since u is normal), the endomorphism v is unitary; moreover, it commutes with $|u|$ since $j|u| = |u|j$, and we have $u = v|u|$.

Conversely, let v be a unitary element of $\mathcal{L}(E)$, commuting with $|u|$, such that $u = v|u|$. We have $uu^* = v|u|^2v^* = |u|^2vv^* = |u|^2 = u^*u$, hence u is normal.



Let E be a complex Hilbert space and $u \in \mathcal{L}(E)$. Let $(j, |u|)$ be the polar decomposition of u . It is possible that j commutes with $|u|$ without u being normal (exercise 11 of I, p. 189).

1.8 ALGEBRAS OF CONTINUOUS FUNCTIONS ON A COMPACT SPACE

In this paragraph, the base field is $\mathbb{C}$.

1.8.1 Subalgebras of the algebra of continuous functions on a compact space

Let X be a compact topological space and B a unital subalgebra of $\mathcal{C}(X)$. We denote by ev the map $x \mapsto \text{ev}_x$ from X into $X(B)$ such that $\text{ev}_x(f) = f(x)$ for all $f \in B$. We denote by j the injection of B into $\mathcal{C}(X)$, so $X(j) = \text{ev}$.

PROPOSITION 1. — Let $f \mapsto \|f\|_B$ be a norm endowing B with the structure of a Banach algebra.

- a) The map j has norm ≤ 1 in $\mathcal{L}(B; \mathcal{C}(X))$;
- b) The radical of the algebra B is zero;
- c) The map ev is continuous from X into $X(B)$;
- d) If B separates the points of X , then the map ev is a homeomorphism from X onto a closed subset of $X(B)$.

One can identify X with $X(\mathcal{C}(X))$ and $\mathcal{G}_{\mathcal{C}(X)}$ with the identity map (example 1 of I, p. 36). The map ev is then identified with $X(j)$, which proves c).

For every function $f \in B$ and every $x \in X$, one has $\mathcal{G}_B(f)(\text{ev}_x) = f(x)$, whence

$$\|f\|_{\mathcal{C}(X)} = \sup_{x \in X} |\mathcal{G}_B(f)(\text{ev}_x)| \leq \varrho_B(f) \leq \|f\|_B,$$

(cf. prop. 7 of I, p. 38, a)) which entails a). Moreover, this shows that the Gelfand transform of B is injective, whence b) (prop. 8 of I, p. 38). If B separates the points of X , then the map ev is injective, whence d) since X is compact.

We now turn to the surjectivity of the map ev .

PROPOSITION 2. — Let $f \mapsto \|f\|_B$ be a norm such that $(B, \|\cdot\|_B)$ is a Banach algebra.

- a) If the map ev is surjective, then the algebra B is a full subalgebra of $\mathcal{C}(X)$;
- b) Suppose that B is a full subalgebra of $\mathcal{C}(X)$, and that there exists an element $a \in B$ such that the set of elements $f(a)$, where f ranges over the set of rational functions on $\mathbb{C}$ having no pole on $\text{Sp}_B(a)$, is dense in B . Then the map ev is surjective;

c) If B is a full involutive subalgebra of $\mathcal{C}(X)$ then the map ev is surjective.

Assertion a) follows from prop. 10 of I, p. 40, and assertion b) of prop. 11 of I, p. 41, since the full subalgebra of B generated by a is the set of elements $f(a)$, where f ranges over the set of rational functions on $\mathbb{C}$ having no pole on $Sp_B(a)$ (lemma I, p. 6 and proposition 6 of I, p. 37, b)).

We prove c). Suppose then that B is a full involutive subalgebra of $\mathcal{C}(X)$. To prove that ev is surjective, it suffices to prove that for every $\chi \in X(B)$, there exists $y \in X$ such that $Ker(\chi) = Ker(ev_y)$ (th. 2 of I, p. 30). Let $I = Ker(\chi)$. This is a maximal ideal of B . Let Φ be the set of $x \in X$ such that $f(x) = 0$ for all $f \in I$. We show that Φ is not empty. In the contrary case, since X is compact, there would exist an integer $n \geq 1$, an open covering $(V_1, \dots, V_n)$ of X and, for every integer i such that $1 \leq i \leq n$, a function $f_i \in I$ such that $f_i(x) \neq 0$ for all $x \in V_i$. Since the algebra B is an involutive subalgebra of $\mathcal{C}(X)$, the function

$$f = \sum_{i=1}^n f_i \bar{f}_i$$

would belong to I . But $f(x) > 0$ for all $x \in X$, and therefore f would be invertible in $\mathcal{C}(X)$. Since B is assumed to be a full subalgebra of $\mathcal{C}(X)$, the function $f \in I$ would be invertible in B , which is impossible. Consequently, the set Φ is not empty. Let y be an element of Φ ; the kernel of the character ev_y contains I , and is therefore equal to I .

Example. — Let $X = [0, 1]$ and let $n \geq 0$ be an integer. Let B be the algebra of functions $f : X \rightarrow \mathbb{C}$ admitting continuous derivatives on $[0, 1]$ up to order n , equipped with the norm considered in Example 4 of I, p. 18. Then B is a full involutive subalgebra of $\mathcal{C}(X)$ separating the points of X , hence $X(B)$ is identified with X .

We consider the logarithm function as defined on R_+ and taking values in $R \cup \{-\infty\}$ by setting $\log(0) = -\infty$.

PROPOSITION 3. — Let X be a compact space. Let B be a unital Banach subalgebra of $\mathcal{C}(X)$, equipped with the induced norm, separating the points of X . One identifies X with a closed subset of $X(B)$ (prop. 1, d)).

a) For every $f \in B$, the Gelfand transform $\mathcal{G}_B(f)$ is a continuous function in $X(B)$ which

extends f and satisfies $\|f\| = \sup |\mathcal{G}_B(f)|$. In particular, $\mathcal{G}_B$ is an isometric isomorphism from B onto a Banach subalgebra of $\mathcal{C}(X(\mathbf{B}))$;

- b) Let B^* be the set of invertible elements of B . For every character $\chi \in X(B)$ there exists a positive measure μ of mass 1 on X such that, for every $f \in B^*$, one has

$$\log(|\chi(f)|) = \int_X \log(|f|) d\mu.$$

Moreover, for every $f \in B$, one has

$$\chi(f) = \int_X f d\mu;$$

- c) We assume that every element of $\mathcal{C}_R(X)$ is a uniform limit of real parts of functions belonging to B . Then, for every $\chi \in X(B)$, there exists a unique measure $\mu_\chi \geq 0$ on X such that, for every function $f \in B$, one has

$$\chi(f) = \int_X f d\mu_\chi.$$

Moreover, for every function $f \in B$, one has

$$\log(|\chi(f)|) \leq \int_X \log(|f|) d\mu_\chi;$$

the function $\log |f|$ being bounded above, the integral exists in $R \cup \{-\infty\}$.

Assertion *a*) follows from the identification of X with a closed subset of $X(B)$ and the inequalities

$$\|f\| = \sup_{x \in X} |f(x)| \leq \sup_{\chi \in X(B)} |\chi(f)| = \sup |\mathcal{G}_B(f)| = \varrho_B(f) \leq \|f\|$$

for all $f \in B$.

Let $\chi \in X(B)$ and let n be a positive integer. Let $\lambda_1, \dots, \lambda_n \in \mathbf{R}$ and $f_1, \dots, f_n \in B^*$. We then have

$$\sum_{i=1}^n \lambda_i \log(|\chi(f_i)|) \leq \sup_{x \in X} \left(\sum_{i=1}^n \lambda_i \log(|f_i(x)|) \right). \quad (1)$$

Indeed, by continuity, it suffices to prove this inequality when the real numbers λ_i are rational. By reducing to a common denominator, we reduce to the case where $\lambda_i \in \mathbb{Z}$ for all i . The inequality then becomes

$$\log \left(\left| \chi \left(f_1^{\lambda_1} \dots f_n^{\lambda_n} \right) \right| \right) \leq \sup_{x \in X} \log \left(\left| \left(f_1^{\lambda_1} \dots f_n^{\lambda_n} \right) (x) \right| \right),$$

and follows from the fact that $\|\chi\| = 1$ (th. 1 of I, p. 29).

Let B' be the vector subspace of $\mathcal{C}_{\mathbf{R}}(X)$ generated by the functions $\log(|f|)$ for $f \in B^*$. The estimate (1) proves that there exists a linear form h of norm ≤ 1 on B' such that $\log(|\chi(f)|) = h(\log(|f|))$ for all $f \in B^*$. By the Hahn–Banach theorem (EVT, II, p. 24, cor. 2), the linear form h extends to a linear form μ of norm ≤ 1 on $\mathcal{C}_{\mathbf{R}}(X)$, that is, to a real measure μ on X such that $\|\mu\| \leq 1$ (INT, III, §1, no. 5). Taking as the element f of B^* the constant $e = \exp(1)$ we see that $1 = \mu(1)$. Hence, writing $\mu = \mu^+ - \mu^-$ as the difference of two mutually singular positive measures (INT, III, §1, no. 6, th. 3), it follows

$$1 = \mu^+(1) - \mu^-(1) \leq \mu^+(1) + \mu^-(1) = \|\mu\| \leq 1,$$

whence $\mu = \mu^+ \geq 0$ and $\|\mu\| = 1$.

For every $f \in B$, one has $\exp(f) \in B^*$, hence

$$\begin{aligned} \int_X \mathcal{R}(f) d\mu &= \int_X \log(|\exp(f)|) d\mu = \log(|\chi(\exp(f))|) \\ &= \log(|\exp(\chi(f))|) = \mathcal{R}(\chi(f)), \end{aligned}$$

where we used Cor. 1 of I, p. 66. Applying this equality to if , we conclude that $\int_X f d\mu = \chi(f)$. This establishes b).

Let us place ourselves under the assumptions of c). The existence of μ_χ follows from b . On the other hand, we have $\mu_\chi(\mathcal{R}(f)) = \mathcal{R}(\chi(f))$ for all $f \in B$. Since the real parts of functions $f \in B$ are dense in $\mathcal{C}_{\mathbf{R}}(X)$ by hypothesis, the measure μ_χ is uniquely determined by

χ .

Let $f \in B$. Let $\varepsilon > 0$ be a real number. By hypothesis, there exists a function $g \in B$ such that

$$\mathcal{R}(g) - \varepsilon \leq \log(|f| + \varepsilon) \leq \mathcal{R}(g) + \varepsilon. \quad (2)$$

Let $h = \exp(g) \in B^*$. By (2), we have

$$|h|e^{-\varepsilon} \leq |f| + \varepsilon \leq |h|e^{\varepsilon}. \quad (3)$$

The bound implies $|fh^{-1}| \leq e^{\varepsilon}$, whence $|\chi(fh^{-1})| \leq e^{\varepsilon}$ and consequently

$$\log(|\chi(f)|) \leq \log(|\chi(h)|) + \varepsilon = \int_X \log(|h|) d\mu_{\chi} + \varepsilon. \quad (4)$$

The lower bound in (3) then implies

$$\log(|\chi(f)|) \leq \int_X \log(|f| + \varepsilon) d\mu_{\chi} + 2\varepsilon.$$

Letting ε tend to 0, we deduce that

$$\log(|\chi(f)|) \leq \int_X \log(|f|) d\mu_{\chi}.$$

This completes the proof.

1.8.2 Continuous functions on a compact subset of C^{Λ}

Let Λ be a set and X a compact subset of C^{Λ} . We denote by $P(X)$ the unital Banach subalgebra of $\mathcal{C}(X)$ consisting of the functions on X that are uniform limits on X of polynomial functions on C^{Λ} . The coordinate functions $z_{\lambda}|_X$ topologically generate $P(X)$, and $P(X)$ separates the points of X . Let Y be the polynomially convex hull of X (def. 4 of I, p. 45). Since

$$\sup_{z \in Y} |p(z)| = \sup_{z \in X} |p(z)|$$

(cf. no. 7 of I, p. 44) for all $p \in \mathbf{C}[(X_\lambda)_{\lambda \in \Lambda}]$, uniformly convergent sequences of polynomials on X extend uniquely to uniformly convergent sequences of polynomials on Y . There therefore exists a unique isometric isomorphism from $P(X)$ onto $P(Y)$ which, for every coordinate function z_λ on $\mathbf{C}^\Lambda$, maps $z_\lambda|_X$ to $z_\lambda|_Y$. This isomorphism will be said to be canonical.

PROPOSITION 4. — Let X be a compact space. Let B be a unital Banach subalgebra of $\mathcal{C}(X)$, equipped with the induced norm, separating the points of X . One identifies X with a closed subset of $X(B)$ (prop. 1 of I, p. 142, d)). Let $(x_\lambda)_{\lambda \in \Lambda}$ be a family of elements of B topologically generating the unital algebra B . We consider the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{i} & X(B) \\ \downarrow \varphi & & \downarrow \varphi' \\ \mathrm{Sp}_{\mathcal{C}(X)}^\Lambda((x_\lambda)) & \xrightarrow{j} & \mathrm{Sp}_B^\Lambda((x_\lambda)) \end{array}$$

where φ and φ' are the mappings defined by the family $(x_\lambda)_{\lambda \in \Lambda}$ (cf. no. 6 of I, p. 41), and i and j are the canonical injections. Then:

- a) The maps φ and φ' are homeomorphisms;
- b) The simultaneous spectrum $\mathrm{Sp}_B^\Lambda((x_\lambda))$ is the polynomially convex hull of $\mathrm{Sp}_{\mathcal{C}(X)}^\Lambda((x_\lambda))$;
- c) The map φ maps $\mathcal{C}(X)$ to $\mathcal{C}(\mathrm{Sp}_{\mathcal{C}(X)}^\Lambda((x_\lambda)))$ and B to $P(\mathrm{Sp}_{\mathcal{C}(X)}^\Lambda((x_\lambda)))$;
- d) The map φ' maps $\mathcal{G}_B(B)$ to $P(\mathrm{Sp}_B^\Lambda((x_\lambda)))$;
- e) The maps φ and φ' map $\mathcal{G}_B$ to the canonical isomorphism from $P(\mathrm{Sp}_{\mathcal{C}(X)}^\Lambda((x_\lambda)))$ onto $P(\mathrm{Sp}_B^\Lambda((x_\lambda)))$.

The maps φ and φ' are continuous and surjective (I, no. 6, p. 41). The map φ' is a homeomorphism by assertion a) of prop. 12 of I, p. 43, and i is injective, therefore φ is injective. Therefore φ and φ' are homeomorphisms.

Let $X_1 = \text{Sp}_{\mathcal{C}(X)}^\Lambda((x_\lambda))$. The polynomially convex hull of X_1 is $Y_1 = \text{Sp}_B^\Lambda((x_\lambda))$ by prop. 14 of I, p. 46.

For every $\lambda \in \Lambda$, let z_λ be the corresponding coordinate function on $\mathbf{C}^\Lambda$. The map φ maps x_λ to $z_\lambda|_{X_1}$, and φ' maps $\mathcal{G}_B(x_\lambda)$ to $z_\lambda|_{Y_1}$. Therefore φ maps B onto $P(X_1)$, and φ' maps $\mathcal{G}_B(B)$ onto $P(Y_1)$. Finally, φ and φ' map $\mathcal{G}_B$ to the canonical isomorphism from $P(X_1)$ onto $P(Y_1)$.

COROLLARY 1. — Let Λ be a set and X a compact subset of $\mathbf{C}^\Lambda$. We identify X with a subset of $X(P(X))$ (prop. 1 of I, p. 142, d)). Let $(z_\lambda)_{\lambda \in \Lambda}$ be the family of coordinate functions on $\mathbf{C}^\Lambda$.

- a) The map θ from $X(P(X))$ onto $\text{Sp}_{P(X)}^\Lambda((z_\lambda))$ defined by the family (z_λ) is a homeomorphism from $X(P(X))$ onto the polynomially convex hull Y of X . Its restriction to X is the identity map of X ;
- b) For every function $f \in P(X)$, the homeomorphism θ transforms the extension $\mathcal{G}_{P(X)}(f)$ of f to $X(P(X))$ into an extension $\tilde{f}$ of f to Y . The map $f \mapsto \tilde{f}$ is the canonical isomorphism from $P(X)$ onto $P(Y)$.

In prop. 4, take $B = P(X)$ and $x_\lambda = z_\lambda$. Then φ becomes the identity map and φ' becomes the map θ . The assertions of the corollary then reduce to those of loc. cit.

COROLLARY 2. — Let Λ be a set and let $X \subset \mathbf{C}^\Lambda$ be a compact set. If X is connected, then its polynomially convex hull is connected.

If X is connected, the only idempotents of $\mathcal{C}(X)$, hence of $P(X)$, are 0 and 1 (cor. of prop. I, p. 79). Consequently, the space $X(P(X))$ is connected (loc. cit.); but this set is homeomorphic to the polynomially convex hull of X (cor. 1, a)).

1.8.3 Continuous functions on a compact subset of $\mathbf{C}$

Lemma 1. — Let X be a compact subset of the plane and let O be a bounded connected component of $\mathbf{C} - X$. The boundary of O is contained in X .

The closure of the set O in $\mathbf{C} - X$ is equal to $\overline{O} \cap (\mathbf{C} - X)$ where $\overline{O}$ is its closure in $\mathbf{C}$. Since O is a connected component of $\mathbf{C} - X$, we therefore have $\overline{O} \cap (\mathbf{C} - X) = O$, which indeed proves

that $\overline{O} - O \subset X$.

Let X be a compact subset of C . Let O_∞ be the unbounded connected component of $C - X$, and let $(O_i)_{i \in I}$ be the family of bounded connected components of $C - X$, the sets O_i being pairwise distinct.

Let E be a subset of $C - X$. We denote by $R_E(X)$ the closure in $\mathcal{C}(X)$ of the set of functions $f|X$, where f is a rational function on C all of whose poles belong to E . The algebra $R_E(X)$ is a unital Banach subalgebra of $\mathcal{C}(X)$ which separates the points of X . Let z be the identity function on X . The full closed subalgebra of $R_E(X)$ generated by z is equal to $R_E(X)$ (lemma 2 of I, p. 6). The elements of $R_E(X)$ are holomorphic in the interior of X .

In particular, we have $R_\emptyset(X) = P(X)$. We denote $R(X) = R_{C-X}(X)$. We denote by $I(E)$ the set of elements $i \in I$ such that $E \cap O_i \neq \emptyset$, and

$$X_E = X \cup \left(\bigcup_{i \in I(E)} O_i \right).$$

The set X_E is compact, since it is bounded and closed, its complement in C being the union of the open O_∞ and open sets O_i that meet E .

PROPOSITION 5. — With the above notation:

- a) The restriction map h of $R_E(X_E)$ into $R_E(X)$ is an isometric isomorphism;
- b) The algebra $R_E(X_E)$ is a full subalgebra of $\mathcal{C}(X_E)$;
- c) Every character of $R_E(X_E)$ is of the form $f \mapsto f(w)$, where w is an element of X_E ;
- d) The map $\chi \mapsto \chi(z)$ is a homeomorphism from $X(R_E(X))$ onto X_E ;
- e) If E' is a subset of $C - X$, then we have $R_E(X) = R_{E'}(X)$ if and only if $X_E = X_{E'}$, which is also equivalent to $I(E) = I(E')$.

The restriction map h of $R_E(X_E)$ into $R_E(X)$ is a Banach algebra morphism such that $\|h(f)\| \leq \|f\|$ for every function $f \in R_E(X_E)$. Let $f \in R_E(X_E)$. Let $i \in I(E)$. Since f is holomorphic in a neighborhood of the bounded open set O_i , the maximum principle (VAR, I, p. 29) implies that there exists an element z_0 in the boundary of O_i such that $|f(z_0)| = \sup_{z \in O_i} |f(z)|$. Since the boundary of O_i is contained in X (lemma 1), it follows that $\sup_{z \in O_i} |f(z)| \leq \|h(f)\|$. Since this

inequality holds for every $i \in I(E)$, it results that $\|f\| \leq \|h(f)\|$. The morphism h is therefore isometric, and in particular injective. Let us now prove that it is surjective. Let $g \in R_E(X)$. There exists a sequence (f_n) of rational functions whose poles belong to E which converges uniformly to g on X . The $f_n|_{X_E}$ are elements of $R_E(X_E)$. Since h is isometric, the sequence $(f_n|_{X_E})$ converges in $\mathcal{C}(X_E)$. If f is its limit, we have $f \in R_E(X_E)$ and $g = f|_X = h(f)$. Hence $a)$.

Let us prove assertion d). Apply prop. 3 of I, p. 144 to the algebra $B = R_E(X)$. Assertion b) of loc. cit. implies that the map which associates $\chi(z)$ to χ is a homeomorphism from $X(R_E(X))$ onto $\text{Sp}_{R_E(X)}(z)$. Let z_E be the identity map of X_E . By loc. cit., a), we have $\text{Sp}_{R_E(X)}(z) = \text{Sp}_{R_E(X_E)}(z_E)$. It therefore suffices to show that $\text{Sp}_{R_E(X_E)}(z_E) = X_E$. The corollary of prop. 6 of I, p. 28 shows that $\text{Sp}_{R_E(X_E)}(z_E)$ is the union of $\text{Sp}_{\mathcal{C}(X_E)}(z_E) = X_E$ and of certain bounded connected components of the complement of X_E . Let O_i be one of the bounded connected components of the complement of X_E . The intersection $E \cap O_i$ is therefore nonempty. Let $\lambda \in E \cap O_i$; since $(\lambda - z_E)^{-1} \in R_E(X_E)$, we have $\lambda \notin \text{Sp}_{R_E(X_E)}(z_E)$. Thus, O_i is not contained in $\text{Sp}_{R_E(X_E)}(z_E)$. This shows that $\text{Sp}_{R_E(X_E)}(z_E) = X_E$.

This equality also implies that $\text{Sp}_{R_E(X_E)}(z_E) = \text{Sp}_{\mathcal{C}(X_E)}(z_E)$. This establishes condition (iii) of prop. 11 of I, p. 41, applied to the canonical injection of $R_E(X_E)$ into $\mathcal{C}(X_E)$ and to the element z_E . Assertions b) and c) are the equivalent conditions (i) and (ii) of loc. cit.

Assertion d) shows that $X_E = X_{E'}$ if $R_E(X) = R_{E'}(X)$. Conversely, suppose that $X_E = X_{E'}$. By replacing E' by $E \cup E'$, we may assume that $E \subset E'$. According to b), the algebra $R_E(X_E)$ is a full closed subalgebra of $\mathcal{C}(X_E)$, and therefore also of $R_{E'}(X_E)$. It contains z_E , and hence $R_E(X_E) = R_{E'}(X_E)$. Applying a), we deduce that $R_E(X) = R_{E'}(X)$. Finally, the equivalence of $X_E = X_{E'}$ and $I(E) = I(E')$ is a consequence of the definitions.

COROLLARY 1. — The following conditions are equivalent:

- (i) The set E meets every bounded connected component of $C - X$;
- (ii) The map $\chi \mapsto \chi(z)$ is a homeomorphism from $X(R_E(X))$ onto X ;
- (iii) We have $R_E(X) = R(X)$.

Let $E' = C - X$. Conditions (i), (ii), and (iii) are respectively equivalent to $I(E) = I(E')$,

to $X_E = X_{E'}$ (by prop. 5, d)) and to $R_E(X) = R_{E'}(X)$. They are therefore equivalent to each other by prop. 5, e).

The following corollary makes the Runge theorem more precise (th. 3 of I, p. 69).

COROLLARY 2 (Runge's theorem). — For every $i \in I$, let λ_i be a point of O_i . Let f be a complex function holomorphic in an open neighborhood of X . Then $f|_X$ is the uniform limit on X of restrictions of rational functions whose poles are among the λ_i .

With $E = \{\lambda_i\}_{i \in I}$, the hypothesis is condition (i) of Cor. 1. We therefore have $R_E(X) = R(X)$, and Th. 3 of I, p. 69 shows that $R(X) = \mathcal{O}(X)$.

Exercises for Section 1: Spectra and Characters

- 1) Let (e_1, e_2) be the canonical basis of R^2 and let u be the element of $A = \mathcal{L}(R^2)$ defined by $u(e_1) = e_2$, $u(e_2) = -e_1$. Prove that $\text{Sp}_A(u) = \emptyset$ and $\text{Sp}_A(u^2) = \{-1\}$.
- 2) Let μ be the Lebesgue measure on $[0, 1]$, H the Hilbert space $L^2_{\mathbb{C}}([0, 1], \mu)$, $A = \mathcal{L}(H)$, and $x \in A$ the operator which transforms the function $f(t)$ into the function $tf(t)$. Prove that $\text{Sp}_A(x) = [0, 1]$, but that x has no eigenvalue.
- 3) a) Let H be a Hilbert space admitting a countable orthonormal basis $(e_0, e_1, e_2, \dots)$, and $A = L(H)$. Let $u \in \mathcal{L}(H)$ be such that $u(e_i) = e_{i+1}$ for every i . Let $u' \in \mathcal{L}(H)$ be such that $u'(e_i) = e_{i-1}$ for $i \geq 1$ and $u'(e_0) = 0$. We have $u'u = 1$ but $uu'(e_0) = 0$, so that $\text{Sp}_A(u'u) \neq \text{Sp}_A(uu')$.
b) Let A be a unital Noetherian algebra over a commutative field. If $x, y \in A$, one has $\text{Sp}_A(xy) = \text{Sp}_A(yx)$ (use exerc. 6 of A, VIII, p. 37).
- 4) Let A be a commutative algebra over $\mathbb{C}$, I an ideal of A , h the canonical injection of I into A , and S the set of $\chi \in X'(A)$ which vanish on I .
a) $X'(h)$ is surjective.
b) $X'(h)|(X'(A) - S)$ is a homeomorphism from $X'(A) - S$ onto $X(I)$. (Let $\chi_0 \in X'(A) - S$. Let $x_1, \dots, x_n \in A$, $\varepsilon > 0$, and let V be the neighborhood of χ_0 in $X'(A)$ defined by

$|(\chi - \chi_0)(x_i)| \leq \varepsilon, (i = 1, \dots, n)$. Let $u_0 \in I$ be such that $\chi_0(u_0) = 1$. We have $u_i = u_0 x_i \in I$. Then if $|(\chi - \chi_0)(u_i)| \leq \delta$ ($i = 0, 1, \dots, n$) with δ small enough, we have $\chi \in V$.)

5) In exerc. 4, take $A = \mathbb{C}[X, Y]$ and $I = AX$. The space $X(A)$ is identified with $\mathbb{C}^2$, $S = \{0\}$ is identified with $\{0\} \times \mathbb{C}$. Let U be the set of $(\xi, \eta) \in \mathbb{C}^2$ such that $|\xi| < e^{-|\eta|}$. Prove that $X'(h)(U)$ is not a neighborhood of 0 in $X'(I)$. Deduce that $X'(I)$ does not identify with the quotient space of $X'(A)$ by the equivalence relation defined by $X'(h)$. (On this subject, cf. I, p. 169, exerc. 17.)

6) Let A be a commutative algebra over a field and let I be a maximal ideal of A . Prove that we have $A^2 \subset I$ with $\dim(A/I) = 1$ or else that A/I is a field. (Apply to A/I exercise 3 of A, I, p. 153.) In particular, if $A^2 = A$, every maximal ideal of A is regular.

7) Let A be a commutative algebra over a commutative field K , $x \in A$ and $\alpha \in K$. The set of $\chi \in X(A)$ such that $(\mathcal{G}x)(\chi) = \alpha$ is closed for the Jacobson topology. (Reduce to the case where A has a unit element, then to the case where $\alpha = 0$.)

¶ 8) Let A be an algebra, I a two-sided ideal of A , $J^I(A)$ the set of primitive ideals of A that do not contain I , and $\widehat{A}^I$ the set of irreducible representations $\pi \in \widehat{A}$ such that $\pi(I) \neq 0$.

a) If $\pi \in \widehat{A}^I$, one has $\pi|I \in \widehat{I}$ (lemma 5, a) of I, p. 12). If representations π and $\pi' \in \widehat{A}$ are such that $\pi|I$ and $\pi'|I$ are equivalent, then π and π' are equivalent. (One may assume that $\pi(x) = \pi'(x)$ for all $x \in I$. Then, for every $y \in A$, $\pi(y)$ and $\pi'(y)$ coincide on $\sum_{x \in I} \text{Im } \pi(x)$, and this subspace is equal to the space of π .)

b) If $\pi \in \widehat{I}$, π extends to an element of $\widehat{A}^I$. (Let R be a maximal regular left ideal of I . Let g be a right unit of I modulo R . Assuming that A admits a unit element, show that the relation $AR + A(1 - g) = A$ would entail $g^2 \in R$. Therefore $AR + A(1 - g)$ is contained in a maximal left ideal M of A . Prove that $M \cap I = R$ and $M + I = A$.)

c) Deduce from a) and b) that the map $I' \mapsto I' \cap I$ is a homeomorphism from $J^I(A)$ onto $J(I)$, and that the map $\pi \mapsto \pi|I$ is a homeomorphism from $\widehat{A}^I$ onto $\widehat{I}$.

Exercises for Section 2: Normed Algebras

1) Let (A, e) be a unital complex algebra.

a) For a linear form $\lambda \in A^*$ such that $\lambda(e) = 1$, the following conditions are equivalent:

- (i) For every $x \in \text{Ker}(\lambda)$, one has $x^2 \in \text{Ker}(\lambda)$;
- (ii) For every $x \in A$, one has $\lambda(x^2) = \lambda(x)^2$;
- (iii) The kernel of λ is a right ideal of A ;
- (iv) The linear form λ is a unital morphism from A to $\mathbb{C}$.

(Observe that $\lambda(x - \lambda(x)e) = 0$ for all $x \in A$.)

b) Let $\lambda \in A^*$ a linear form such that $\lambda(1) = 1$ and such that for every $x \in A$, one has

$\lambda(x) \in \text{Sp}_A(x)$. Let $x \in \text{Ker}(\lambda)$ such that $\lambda(x^2) \neq 0$. The spectrum of x is not bounded in $\mathbb{C}$. (For $n \geq 2$ integer, consider the polynomial $p \in \mathbb{C}[X]$ such that $p(z) = \lambda((ze - x)^n)$ for all $z \in \mathbb{C}$; observe that its roots belong to the spectrum of x and bound their Euclidean norm below in terms of $|\lambda(x^2)|$.)

c) Suppose that A is a Banach algebra. For a linear form λ on A to be a unital morphism from A into $\mathbb{C}$, it is necessary and sufficient that $\lambda(e) = 1$ and that $\lambda(x) \in \text{Sp}_A(x)$ for all $x \in A$.

2) Let $n \geq 0$ be an integer. Let A_n the commutative complex Banach algebra of functions $f : [0, 1] \rightarrow \mathbb{C}$ admitting continuous derivatives in $[0, 1]$ up to order n , equipped with the norm

$$\|f\| = \sum_{k=0}^n \frac{1}{k!} \sup_{0 \leq t \leq 1} |f^{(k)}(t)|$$

(example 4 of I, p. 18). Show that A_n is isomorphic to A_m if, and only if, $n = m$. (However, according to Example 1 of I, p. 144, these Banach algebras all have the same character space.)

3) Let G be a group equal to its derived subgroup (A, I, p. 67). For every $g \in G$, we denote

by $\ell(g)$ the smallest integer $k \geq 0$ such that there exist pairs $(a_1, b_1), \dots, (a_k, b_k)$ into $G \times G$ satisfying $g = [a_1, b_1] \cdots [a_k, b_k]$.

a) The map ℓ is a metric on G .

b) For every $g \in G$, the limit

$$\text{lsc}(g) = \lim_{n \rightarrow +\infty} \frac{\ell(g^n)}{n}$$

exists. It is called the stable commutator length of g .

c) Let S be a set and $G = D(F(S))$ the derived subgroup of the free group generated by S .

The group G is equal to its derived subgroup, and for every $g \in G$, $g \neq e$, one has $\text{lsc}(g) \geq 1/2$. There exists $g \in G$ such that $\text{lsc}(g) = 1/2$.

4) Let $n \geq 1$ an integer. We denote $S(n)$ the largest integer $k \geq 0$ such that, for every set X of n real numbers, there exists a subset $Y \subset X$ of cardinality k such that the equation $x + y = z$ has no solution in Y . We therefore have $S(n) \leq n$.

a) Show that the limit

$$\sigma = \lim_{n \rightarrow +\infty} \frac{S(n)}{n}$$

exists.

b) Show that $\sigma \leq 2/5$.

c) Show that $\sigma \geq 1/3$.

(One can show that $\sigma = 1/3$, cf. S. EBERHARD, B. GREEN, F. MANNERS, Sets of integers with no large sum-free subset, *Annals of Mathematics* 180 (2014), 621–652.)

5) Let A be a commutative ring and $n \geq 1$ an integer. For a matrix $M = (m_{i,j})$ of type (n, n) with coefficients in A , we call the permanent of M the element

$$\text{per}(M) = \sum_{\sigma \in S_n} m_{1,\sigma(1)} \dots m_{n,\sigma(n)}$$

of A.

Let $k \geq 1$ an integer. We denote $X_{n,k}$ the set of matrices M of type (n, n) with real coefficients such that each column of M contains k coefficients equal to 1, all the others being zero.

Show that the limit

$$\lim_{n \rightarrow +\infty} \left(\inf_{M \in X_{n,k}} \text{per}(M) \right)^{1/n}$$

exists.

- 6) Let $k \geq 1$ an integer. For every integer $N \geq 1$, we denote by $S_k(N)$ the largest cardinality of a set $A \subset \{1, \dots, N\}$ that does not contain k elements in arithmetic progression, that is, for any $n_0 \geq 0$, $h \geq 1$, $\ell \geq 1$, if $\{n_0 + h, n_0 + 2h, \dots, n_0 + \ell h\} \subset A$, then $\ell < k$.

a) Show that the limit

$$s_k = \lim_{N \rightarrow +\infty} \frac{S_k(N)}{N}$$

exists.

- b) Show that $s_1 = s_2 = 0$. One can show ("Szemerédi's theorem", cf. E. SZEMERÉDI, On sets of integers containing no k elements in arithmetic progression, Acta Arithmetica XXVII, 1975, 199–245) that $s_k = 0$ for all $k \geq 1$; for the case $k=3$, cf. exercise 23 of II, p. 270.

7) Let A be a normed algebra.

a) Let S be a nonempty bounded subset of A . Show that the limit

$$\varrho(S) = \lim_{n \rightarrow +\infty} \sup_{x \in S^n} \|x\|^{1/n}$$

exists and that $\varrho(S) = \varrho(x)$ if S is reduced to a single element x . We say that $\varrho(S)$ is the

spectral radius of the part S.

- b) Let N be the set of norms p on A , equivalent to the norm of A , and such that A , equipped with p , is a normed algebra. Let X be a subset of A . For there to exist $p \in N$ such that $p(x) \leq 1$ for all $x \in X$, it is necessary and sufficient that there exist a real number $c \geq 0$ such that

$$\sup_{n \geq 1} \sup_{x \in X^n} \|x\| \leq c.$$

- c) Show that

$$\varrho(S) = \inf_{p \in N} \sup_{x \in S} p(x).$$

- 8) Let A be the algebra of continuous complex-valued functions on $\mathbf{R}$ tending to 0 at infinity, equipped with the norm $\|f\| = \sup_{t \in \mathbf{R}} |f(t)|$. Then $\widetilde{A}$ is identified with the algebra of continuous complex-valued functions on $\mathbf{R}$ tending toward a finite limit at infinity. On $\widetilde{A}$, the norm $\|g\| = \sup_{t \in \mathbf{R}} |g(t)|$ differs from the norm defined in no. 1 of I, p. 15.
- 9) Let A be a normed algebra and $b = \inf_{x \neq 0} \|x^2\| / \|x\|^2$. Then

$$b \leq \inf_{x \neq 0} \frac{\varrho(x)}{\|x\|} \leq \sqrt{b}.$$

- 10) Let A be a normed algebra and $x, y \in A$. We have $\varrho(xy) = \varrho(yx)$.
- 11) Let H be a Hilbert space of countable type and $(f_n)_{n \geq 1}$ an orthonormal basis of H . Let A be the normed algebra $\mathcal{L}(H)$. Let $x \in A$ the element defined by $x(f_i) = 2^{-i} f_{i+1}$. Show that x is quasi-nilpotent, but not nilpotent.
- ¶ 12) Let H be a separable Hilbert space and let $(f_n)_{n \geq 1}$ be an orthonormal basis of H . One defines the numbers $(\alpha_m)_{m \geq 1}$ by $\alpha_m = e^{-k}$ if m is the product of 2^k by an odd number.
- a) Let $x \in \mathcal{L}(H)$ defined by $x(f_m) = \alpha_m f_{m+1}$ for $m \geq 1$. Prove that x is not quasi-nilpotent. (Observe that

$$\|x^n\| = \sup_m (\alpha_m \alpha_{m+1} \dots \alpha_{m+n-1}),$$

and evaluate $\alpha_1 \alpha_2 \dots \alpha_{2^r-1}$.)

b) Let $x_k \in \mathcal{L}(H)$ defined by $x_k(f_m) = 0$ if m is the product of 2^k by an odd number, and $x_k(f_m) = \alpha_m f_{m+1}$ otherwise. Show that x_k is nilpotent and that (x_k) converges to x as k tends to $+\infty$.

13) Let A be a unital Banach algebra. Let $x \in A$ such that $\|x - 1\| < 1$. There exists $y \in A$ such that $y^2 = x$.

14) Let A be a unital complex normed algebra. If $\|x^{-1}\| = \|x\|^{-1}$ for every invertible element x of A , one has $A = C \cdot 1$. (Reduce to the case where A is complete. Let $x \in A$, at a distance $\alpha > 0$ from $C \cdot 1$. If $\lambda_0 \in C$ is such that $x - \lambda_0$ were invertible, then $x - \lambda$ would be invertible for $|\lambda - \lambda_0| < \alpha$. Deduce that the spectrum of x would be empty.)

15) Let A be a unital complex normed algebra in which the only left topological divisor of zero is 0 and the only right topological divisor of zero is 0. Then $A = C \cdot 1$. (If $x \in A$, and if λ belongs to the boundary of $\text{Sp}_{\widehat{A}}(x)$, then $x - \lambda$ is a topological divisor of zero in $\widehat{A}$, hence in A , therefore $x = \lambda$.)

16) Let A be a unital complex Banach algebra. Let a an element of A . The singular spectrum of A is called the set of $\lambda \in \mathbb{C}$ such that $a - \lambda e$ is a topological zero divisor on the left and on the right in A .

a) Show that the singular spectrum of A is contained in the spectrum of A , and contains the boundary of the spectrum of A .

b) Let $f \in C[X]$ a polynomial. The image under f of the singular spectrum of a is the singular spectrum of $f(a)$.

17) Let A be a normed algebra. For every $x \in A$, we set:

$$\lambda(x) = \inf_{y \neq 0} \frac{\|xy\|}{\|y\|} \quad \lambda'(x) = \inf_{y \neq 0} \frac{\|yx\|}{\|y\|}.$$

a) We have

$$|\lambda(x) - \lambda(y)| \leq \|x - y\|, \quad |\lambda'(x) - \lambda'(y)| \leq \|x - y\|,$$

$$\lambda(x)\lambda(y) \leq \lambda(xy) \leq \|x\|\lambda(y), \quad \lambda'(x)\lambda'(y) \leq \lambda'(xy) \leq \lambda'(x)\|y\|.$$

b) The set of left (resp. right) topological zero divisors in A is closed.

18) Let A be a unital Banach algebra. The set of x which are neither invertible nor topological zero divisors, is open in A .

19) Let X be a complex Banach space and $x \in A = \mathcal{L}(X)$.

a) If x is not invertible, then x is a left or right topological zero divisor. (Consider successively the following cases:)

1°) x is not injective;

2°) $\overline{x(X)} \neq X$;

3°) x is injective, $x(X)$ is dense in X and distinct from X .

b) If x maps X bicontinuously onto a closed vector subspace of X distinct from X , or if x is non-injective with image X , then x is interior to the set of non-invertible elements of A .

20) In the algebra A of Example 9 of I, p. 20, the identity function z is neither invertible nor a topological zero divisor.

21) Let A be a unital normed algebra. Let B be the normed algebra $\mathcal{L}(A)$. Then the image of A under the morphism $x \mapsto \gamma_x$ is a full subalgebra of B . Therefore, if $x \in A$, one has $\text{Sp}_A(x) = \text{Sp}_B(\gamma_x)$.

¶ 22) Let A be a Banach algebra.

a) Let S be the set of bounded sequences of elements of A , with the norm $\|(x_n)\| = \sup \|x_n\|$. Let R be the set of $(x_n) \in S$ such that $\|x_n\|$ tends to 0. Then S is a Banach algebra and R is a closed two-sided ideal of S .

Let $S' = S/R$ and $\varphi : S \rightarrow S'$ the canonical homomorphism.

b) The map θ from A into S' defined by $\theta(x) = \varphi(x, x, x, \dots)$ is an isometric isomorphism of A onto a subalgebra of S' . Every left topological divisor of zero in S' is a left divisor of zero. For $x \in A$ to be a left topological divisor of zero, it is necessary and sufficient that $\theta(x)$ be a left divisor of zero in S' .

c) Assume that A has a unit element. Prove that there exists a Banach algebra B and an isometric isomorphism of A onto a subalgebra of B such that an element of A is non-invertible if and only if its image in B is a left or right divisor of zero. (Use (a) and Exercises 19 and 21.)

23) Let A be a normed algebra and R its radical.

a) If $x \in R$, then x is quasi-nilpotent. (We have $\text{Sp}'_A(x) = \{0\}$ and a fortiori $\text{Sp}'_{\hat{A}}(x) = \{0\}$.)

b) Suppose that A is a Banach algebra. Let I be a left ideal of A all of whose elements are quasi-nilpotent. Then $I \subset R$ (We may assume that A has a unit element. If $x \in R$ and $a \in A$, one has $\text{Sp}_A(ax) = \{0\}$, hence $1 - ax$ is invertible; hence $x \in R$.)

24) Let A be a complex Banach algebra, B a complex algebra without radical, and φ a morphism from A onto B . The kernel of φ is closed.

25) Let A be a complex Banach algebra and π a nonzero irreducible representation of A in a complex vector space X . Let ξ_0 be a nonzero element of X .

a) The annihilator I of ξ_0 is a maximal regular left ideal of A ; it is closed.

b) The map $x \mapsto x\xi_0$ defines, by passage to the quotient, an isomorphism φ of the A -module A/I onto the A -module X .

c) If one transports by φ the norm of the Banach space A/I , then X becomes a Banach space, and $\|\pi(x)\| \leq \|x\|$ for all $x \in A$.

d) If A is primitive (that is, $\{0\}$ is a primitive ideal, cf. definition I, p. 11, cf. also A, VIII, p. 433, exerc. 6), then $A = \{0\}$ or $C \cdot 1$ according as A does or does not have a unit element.

(Use the preceding and cor. 4 of I, p. 26).

¶ 26) Let A be a Banach algebra and R its radical.

a) Prove that if $r \in R$, the series

$$-\frac{1}{2} \sum_{k=1}^{\infty} \binom{1/2}{k} (-4r)^k$$

converges to an element $x \in R$ such that $x^2 - x + r = 0$.

b) Let u be an element of A whose class in A/R is idempotent. There exists an idempotent of A congruent to u modulo R . (We may assume that A has a unit element. Let $q = u - u^2 \in R$ and $x \in R$ a solution, constructed with the aid of a), of $x^2 - x - q(1 - 4q)^{-1} = 0$. Then $u - (2u - 1)x$ answers the question.)

27) Let A be a complex unital Banach algebra, $x \in A$ and ζ a boundary point of $\text{Sp}_A(x)$. The resolvent of x cannot be extended to a continuous function at ζ .

28) Let A be a complex unital Banach algebra, Δ an open subset of $\mathbb{C}$ and $\lambda \mapsto R(\lambda)$ a map of Δ to A such that

$$R(\lambda) - R(\mu) = -(\lambda - \mu)R(\lambda)R(\mu) \quad (5)$$

for all $\lambda, \mu \in \Delta$.

a) Let $\lambda_0 \in \Delta$. For every $\lambda \in \Delta$ such that $|\lambda - \lambda_0| \cdot \|R(\lambda)\| < 1$, we have:

$$R(\lambda) = \sum_{n=0}^{\infty} (\lambda_0 - \lambda)^n R(\lambda_0)^{n+1}.$$

b) For every $\lambda \in \Delta$ and every integer $n \geq 0$, one has

$$\frac{\partial^n}{\partial \lambda^n} R(\lambda) = (-1)^n n! R(\lambda)^{n+1}.$$

c) If R is holomorphic at infinity, then there exists $z, j, x \in A$ such that $z^2 = 0$, $j^2 = j$

, $zj = jz = 0$, $x \in jAj$, and $R(\lambda) = z + jR(\lambda, x)$ for $|\lambda|$ sufficiently large. (Write $R(\lambda) = \sum_{n=0}^{\infty} c_n \lambda^{-n}$ for $|\lambda| > \lambda_0$, and express that R satisfies (5). One may set $c_0 = z$, $c_1 = j$, $c_2 = x$.)

d) For there to exist $a \in A$ such that R is the restriction to Δ of the resolvent $\lambda \mapsto R(a, \lambda)$ of a , it is necessary and sufficient that there exist $\lambda_0 \in \Delta$ such that $R(\lambda_0)$ be invertible.

e) For every $x \in A$, the function $\lambda \mapsto \sum_{n=0}^{\infty} (\lambda_0 - \lambda)^n x^{n+1}$ satisfies (5) for $|\lambda - \lambda_0| \cdot \|x\| < 1$.

29) a) Let E be a complex Banach space, $f : \mathbf{C} \rightarrow E$ an entire function and $\theta \mapsto P(\theta) = \sum_{\mu=k}^m c_{\mu} e^{\mu i \theta}$ a trigonometric polynomial ($c_k \neq 0$). We assume that $|P(\theta)| \cdot \|f(re^{i\theta})\| \leq Mr^{\alpha}$ for $r \geq r_0$ (M, r_0, α being constants ≥ 0). Then f is a polynomial of degree $\leq \alpha$. (Let $f(\zeta) = \sum_{\nu=0}^{\infty} a_{\nu} \zeta^{\nu}$, and

$$J(r, p) = r^{-p} \int_0^{2\pi} P(\theta) f(re^{i\theta}) e^{-(k+p)i\theta} d\theta.$$

Compute the limit of $J(r, p)$ when $r \rightarrow +\infty$ in two different ways for $p > \alpha$.

b) With the notations of a), if α is an integer and if, for every $\theta \in \mathbf{R}$, one has

$$\lim_{r \rightarrow +\infty} r^{-\alpha} |P(\theta)| \|f(re^{i\theta})\| = 0,$$

then f is a polynomial of degree $< \alpha$.

c) Let A be a unital complex Banach algebra and x a quasi-nilpotent element of A . We assume that, for some integer $n > 0$, and for every $\theta \in \mathbf{R}$, one has

$$\lim_{r \rightarrow +\infty} r^{-n} |\cos^n \theta| \cdot \|(1 - re^{i\theta}x)^{-1}\| = 0.$$

Then $x^n = 0$. (Use b).)

¶ 30) a) Let E be a complex Banach space and $x : \mathbf{C} = \{1\} \rightarrow E$ an entire function of $1/(\zeta - 1)$. One denotes by $x(\zeta) = \sum_{n=0}^{\infty} a_n \zeta^n$, $x(\zeta) = \sum_{n=0}^{\infty} b_n \zeta^{-n}$ the expansions of x for

$|\zeta| < 1$, $|\zeta| > 1$ respectively. We assume that there exists $\alpha > 0$ such that $\|a_n\| = o(n^\alpha)$, $\|b_n\| = o(n^\alpha)$. Then $x(\zeta)$ is a polynomial in $1/(\zeta - 1)$ of degree $< \alpha + 1$. (Let $\varepsilon > 0$. We have $\|x(\zeta)\| \leq \varepsilon(1 - r)^{-1-\alpha}$ for $r_\varepsilon \leq r = |\zeta| < 1$, $\|x(\zeta)\| \leq \varepsilon(1 - r^{-1})^{-1-\alpha}$ for $1 < r \leq r'_\varepsilon$. Setting $1 - \zeta = (\omega + \frac{1}{2})^{-1}$, $\omega = Re^{i\theta}$, show that, for $0 \leq r < 1$ and $R \geq 1$, one has

$$(1 - r)^{-1} \leq \frac{9}{4} \frac{R}{\cos \theta}$$

and that, for $1 < r$ and $R \geq 1$,

$$(1 - r^{-1})^{-1} \leq \frac{9}{4} \frac{R}{\cos \theta}.$$

Set $y(\omega) = x(\zeta)$ and show that $R^{-\alpha-1} |\cos \theta|^{\alpha+1} \|y(Re^{i\theta})\|$ tends to 0 as R tends to $+\infty$, uniformly in θ . Then apply exerc. 29, b).

Let A be a complex unital Banach algebra, $q \in A$ a quasi-nilpotent element and $x = 1 + q$.

b) For $q^N = 0$, it is necessary and sufficient that $\|x^{\pm n}\| = o(n^N)$ when n tends to $+\infty$. (To see that the condition is sufficient, show, by applying a), that $R(\lambda, x)$ is a polynomial in $1/(\lambda - 1)$ of degree $\leq N$. On the other hand, $R(\lambda, x) = \sum_{n=0}^{\infty} q^n (\lambda - 1)^{-n-1}$.)

c) Let R be the radical of A and G a subgroup of the group of invertible elements of A . If, for every $x \in G$, one has $\|x^n\| = o(n)$, $\|x^{-n}\| = o(n)$ for $n \rightarrow +\infty$, then the restriction to G of the canonical map $A \rightarrow A/R$ is injective.

31) a) Let A be a normed algebra, and x, y two elements of A such that $xy - yx = y$. Prove that $y^n = 0$ for $n > 2\|x\|$. (Use the formula $xy^n - y^n x = ny^n$).

b) Let L be a finite-dimensional Lie algebra over $\mathbb{C}$. Prove that, if L is not nilpotent, it contains two nonzero elements x, y such that $[x, y] = y$. (Use the fact that there exists an $x \in L$ such that $\text{ad}(x)$ is not nilpotent.)

c) Let L be a real or complex finite-dimensional Lie algebra such that the enveloping algebra of L possesses a norm compatible with its algebra structure. Prove that L is nilpotent. (Use

a) and b).

- d) Let V be a finite-dimensional real or complex vector space, $\mathfrak{L}$ a Lie subalgebra of $\mathfrak{gl}(V)$ consisting of nilpotent endomorphisms, and $x_1, \dots, x_n$ elements of $\mathfrak{L}$. Let $\alpha > 0$. Prove that there exists a Hilbert space structure on V such that $\|x_i\| \leq \alpha$ for $1 \leq i \leq n$.
- e) Let $\mathfrak{L}$ be a real or complex finite-dimensional nilpotent Lie algebra, and U its enveloping algebra. Prove that there exists on U a norm compatible with its algebra structure. (Using the method of LIE, I, §3, exerc. 5 and §7, exerc. 3, show that there exists a family (π_λ) of representations of $\mathfrak{L}$ in finite-dimensional spaces V_λ , having the following property:

for all $u \in U$, there exists an λ such that $\pi_\lambda(u) \neq 0$, and $\pi_\lambda(\mathfrak{L})$ consists, for every λ , of nilpotent endomorphisms. By endowing each V_λ with a Hilbert structure furnished by d), one obtains in the Hilbert space V , the Hilbert sum of the V_λ , a representation π of L by continuous endomorphisms, whence an injective morphism φ of U into $\mathcal{L}(V)$. Set $\|u\|_U = \|\varphi(u)\|$ for all $u \in U$).

- f) Let A be a complex unital normed algebra and x and y elements of A . If $A \neq \{0\}$, one has $xy - yx \neq 1$. (First proof: one has $\text{Sp}'_A(xy) = \text{Sp}'_A(yx)$; if $xy = yx + 1$, then $\text{Sp}_A(xy)$ is deduced from $\text{Sp}_A(yx)$ by the translation $z \mapsto z + 1$; deduce that $\text{Sp}_A(xy)$ is unbounded, which is absurd. Second proof: if $[x, y] = 1$, one has $[xy, y] = y$, whence $y^n = 0$ for n sufficiently large by a); since $[x, y^p] = py^{p-1}$ deduce that $y = 0$).

- 32) Let A be a normed algebra and $x \in A$. One says that x is topologically nilpotent if x^n tends to 0 when n tends to $+\infty$. Show that x is topologically nilpotent if and only if $\varrho(x) < 1$.

- 33) Let A be a commutative unital Banach algebra over $\mathbb{C}$. Suppose that every closed ideal I of A is finitely generated (that is, there exist $n \in \mathbb{N}$ and $x_1, \dots, x_n \in A$ such that $I = Ax_1 + \dots + Ax_n$).

- a) Every ideal I of A is closed. (Write $\bar{I} = Ax_1 + \dots + Ax_n$ with $x_1, \dots, x_n$ in A . Let $(x_{ip})_{p \geq 1}$ be a sequence of elements of I tending to x_i . For every $(y_1, \dots, y_n) \in A^n$ and every $p \geq 1$, set

$$\varphi(y_1, \dots, y_n) = x_1 y_1 + \dots + x_n y_n \in \bar{I}$$

$$\varphi_p(y_1, \dots, y_n) = x_{1p} y_1 + \dots + x_{np} y_n \in I.$$

We have $\varphi \in \mathcal{L}(A^n, \bar{I})$, $\varphi_p \in \mathcal{L}(A^n, \bar{I})$, $\|\varphi - \varphi_p\|$ tends to 0 as p tends to $+\infty$, and φ is surjective. Deduce that φ_p is surjective for p large enough, considering the transposes of φ and φ_p . Hence $I = \bar{I}$.

b) If A is an integral domain, $A = C \cdot 1$ (Use a) and Exerc. 15.)

c) If the only nilpotent element of A is 0, there exists an integer $n \geq 0$ such that A is isomorphic to C^n . (Since A is Noetherian, $\{0\}$ is the intersection of prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_m$; these are closed by a), and the algebras $A/\mathfrak{p}_i$ are isomorphic to C according to b .)

d) We have $\dim_C A < +\infty$. (Let N be the set of nilpotent elements of A , which is a (closed) ideal of A . Then $\dim(A/N) < +\infty$ according to c). Since N is a finitely generated ideal, $N^n = 0$ for n sufficiently large. Finally, each N^i/N^{i+1} is a finitely generated module over A/N , hence $\dim(N^i/N^{i+1}) < +\infty$.)

34) Let A be a unital algebra over C , equipped with a separated locally convex topology such that multiplication in A is separately continuous.

. An element $a \in A$ is said to be regular if there exists $r \geq 0$ such that $a - \lambda$ is invertible for $|\lambda| \geq r$, and such that the set of $(a - \lambda)^{-1}$, for $|\lambda| \geq r$, is bounded.

a) If a is regular, $\lambda(a - \lambda)^{-1}$ remains bounded for $|\lambda| \geq r$.

b) If the mapping $x \mapsto x^{-1}$ is defined in a neighborhood of 1 and continuous at the point 1, every element of A is regular.

c) For every $a \in A$, let U_a be the set of $\lambda \in C$ such that $(a - \mu)^{-1}$ exists and is bounded for all μ of a neighborhood of λ . Let $S_a = C - U_a$. Then S_a is closed; and, if a is regular, S_a is compact nonempty (provided that $A \neq \{0\}$).

d) Let $a \in A$. The function $\lambda \mapsto (a - \lambda)^{-1}$ is holomorphic in U_a .

e) Let $a \in A$. For $\lambda \in U_a$, it is necessary and sufficient that $a - \lambda$ has a regular inverse.

f) If every element of A is regular, and if A is a field, then we have $A = C \cdot 1$.

35) Let A be an algebra over R which is a field, equipped with a separated locally convex topology such that multiplication in A is continuous, and such that the map $x \mapsto x^{-1}$ be continuous at 1. Then A is R -isomorphic either to R , to C , or to the quaternion field H . (If A is commutative and there exists $u \in A$ with $u^2 = -1$ give A a structure of algebra over C , and apply exerc. 34, f.) If A is commutative and there exists no element $u \in A$ such that $u^2 = -1$ apply exerc. 34, f) to the field $A \otimes_R C$. If A is noncommutative, argue as in AC, VI, §6, no. 4, th. 1, third case.)

¶ 36) Let A be the algebra over C consisting of the restrictions to $[0,1]$ of rational functions in one variable with complex coefficients. This algebra is a field.

a) We endow A with the topology of convergence in measure (INT, IV, §5, no. 11). Then the mappings $(x, y) \mapsto x + y$ and $(x, y) \mapsto xy$ of $A \times A$ into A are continuous, and the map $x \mapsto x^{-1}$ from A^* to A is continuous. The topology of A is not locally convex.

b) For $n \in N^*$, we define the sequence $(w_{r,n})_{r \in \mathbb{Z}}$ by $w_{r,n} = (-r+1)^{n(-r+1)}$ if $r < 0$, $w_{0,n} = 1$ and $w_{r,n} = (r+1)^{-(r+1)/n}$ if $r > 0$. If $f \in A$, set $p_n(f) = \sum_r w_{r,n} |a_r| < +\infty$, where $\sum a_r t^r$ is the Laurent expansion of $f(t)$ at 0. Then the p_n are semi-norms that define on A a metrizable locally convex topology. The map $(x, y) \mapsto xy$ The mapping from $A \times A$ to A is continuous. The map $x \mapsto x^{-1}$ The map from A^* into A is not continuous. The algebra A is not complete for this topology.

37) a) Let A be an algebra over C equipped with a locally convex topology. The following conditions are equivalent:

(i) there exists a fundamental system (U_i) of convex balanced neighborhoods of 0 such that

$$U_i U_i \subset U_i \text{ for all } i;$$

(ii) A is isomorphic to a subalgebra of a product of normed algebras.

(iii) the topology of A can be defined by a family of semi-norms p_i satisfying $p_i(xy) \leq p_i(x)p_i(y)$ for all $x, y \in A$.

If A satisfies these conditions, we say that A is locally m -convex.

- b) Let A be a locally m -convex algebra. The map $(x, y) \mapsto xy$ The mapping from $A \times A$ into A is continuous. If A is unital, and if G denotes the set of invertible elements of A , then the map $x \mapsto x^{-1}$ of G into A is continuous.
- c) Let A be a unital locally m -convex algebra. The spectrum of every element of A is nonempty. If A is a field, then $A = C \cdot 1$.
- d) Let A be a complete locally m -convex algebra. There exists a directed set I , a family $(A_i)_{i \in I}$ of Banach algebras, continuous morphisms $\pi_{ij}: A_j \mapsto A_i$ for $i \leq j$, with $\pi_{ij} \circ \pi_{jk} = \pi_{ik}$ such that A is isomorphic to the subalgebra of $\prod_i A_i$ consisting of (x_i) such that $\pi_{ij}(x_j) = x_i$ for $i \leq j$.
- e) The algebra $\mathcal{C}_C(\mathbf{R})$, equipped with the topology of compact convergence, is locally m -convex. The set of invertible elements of $\mathcal{C}_C(\mathbf{R})$ is dense in $\mathcal{C}_C(\mathbf{R})$; it is not open.
- f) Let D be an open subset of C , and let A be the algebra of complex-valued holomorphic functions on D , equipped with the topology of compact convergence. Then A is locally m -convex. The set of invertible elements of A is closed.
- g) The algebra of infinitely differentiable complex-valued functions on $[0,1]$, equipped with the topology of uniform convergence of each derivative, is locally m -convex.
- h) Let A be the algebra of (classes of) complex functions f on $[0,1]$ such that $f \in L^p([0,1])$ for every $p > 1$. Endowed with the topology defined by the family of seminorms $f \mapsto \|f\|_p$ ($p > 1$), A is not locally m -convex. The map $(x, y) \mapsto xy$ from $A \times A$ into A is continuous.

Exercises for Section 3: Commutative Banach Algebras

In the exercises below, all algebras considered are over C , unless explicitly stated otherwise.

- 1) Let A be a commutative Banach algebra and I a maximal ideal of A . Show that there are only two possible cases: either I is the kernel of a character of A , or I is a hyperplane

containing A^2 . (Use exerc. 6 of I, p. 154). In particular, if I is not closed, I is a dense hyperplane of A containing A^2 . (To obtain an example of this latter situation, consider exerc. 3.)

- 2) Let A be the Banach algebra of sequences $x = (x_n)_{n \geq 1}$ of complex numbers tending to 0, with $\|x\| = \sup_i |x_i|$. Let I be the set of elements of A with finite support. Then I is a dense ideal of A , and is contained in no maximal ideal. (Observe that $A^2 = A$, and use exerc. 1.)
- 3) Let Λ a set, $p \in [1, +\infty[$, and $\mathbf{A} = \ell^p(\Lambda)$ the set of families $x = (x_\lambda)_{\lambda \in \Lambda}$, where $x_\lambda \in \mathbb{C}$ and

$$\|x\|_p = \left(\sum_{\lambda} |x_\lambda|^p \right)^{1/p} < +\infty.$$

- a) For componentwise addition and multiplication, and the norm $x \mapsto \|x\|_p$, the space A is a Banach algebra.
- b) We have $A^2 \neq A$ and A^2 is dense in A .
- c) The space $X(A)$ of nonzero characters of A is naturally identified with Λ endowed with the discrete topology. (Use the fact that the dual of $\ell^p(\Lambda)$ is $\ell^q(\Lambda)$, where q is the conjugate exponent of p .) The Gelfand transformation then becomes the identity map and its image in the space of continuous functions tending to 0 at infinity on Λ is dense and not closed.
- 4) Let $(\alpha_n)_{n \geq 0}$ a sequence in $\mathbf{R}_+^*$ such that $\alpha_0 = 1$, $\alpha_{m+n} \leq \alpha_m \alpha_n$ for all integers m and n , and such that $\alpha_n^{1/n} \rightarrow 0$. Let A be the set of formal series $x = \sum_{n=0}^{\infty} x_n z^n \in \mathbb{C}[[z]]$ such that

$$\|x\| = \sum_{n=0}^{\infty} |x_n| \alpha_n < +\infty.$$

Show that A is a commutative unital Banach algebra topologically generated by z , and that the unique maximal ideal of A is the set of $x \in A$ without constant term. (Observe that z is

quasi-nilpotent.)

- 5) In the algebra $C(X)$ of rational fractions over $\mathbb{C}$, the ideal $\{0\}$ is maximal, but is not of codimension 1.
- 6) Let Δ the closed unit disk in $\mathbb{C}$. Show that the group of invertible elements in $\mathcal{C}(\Delta)$ is not dense in $\mathcal{C}(\Delta)$.
- 7) Let A be a commutative unital Banach algebra and $\chi \in X(A)$. For every $x \in A$, set $\|x\|_\chi = |\chi(x)| + \|x - \chi(x)\|$. Show that $x \mapsto \|x\|_\chi$ is a norm, that $\|xy\|_\chi \leq \|x\|_\chi \|y\|_\chi$, $\|e\|_\chi = 1$, and that if $\|e\| = 1$, then $\|x\| \leq \|x\|_\chi \leq 3\|x\|$.
- 8) Let A be a commutative unital Banach algebra such that $\|1\| = 1$. Let N be the set of norms equivalent to the given norm, for which A is still a Banach algebra, and for which 1 has norm 1.
 - a) Let $x_0 \in A$ such that $\varrho(x_0) < 1$. For every $x \in A$, we set:

$$\|x\|' = \inf \sum_n \|a_n\|$$

for all representations of x in the form $a_0 + a_1 x_0 + \dots + a_n x_0^n$ (with $a_0, a_1, \dots, a_n \in A$). Show that $x \mapsto \|x\|'$ is an element of N . (Since $\varrho(x_0) < 1$, there exists k such that $\|x_0^n\| \leq k$ for all n , whence $\|x\| \leq k\|x\|'$.)

- b) Show that $\|x_0\|' \leq 1$.
- c) Deduce from a) and b) that, for every $x \in A$, one has

$$\varrho(x) = \inf_{n \in \mathbb{N}} n(x).$$

- d) If the only element of N is the given norm, we have $A = C \cdot 1$. (Use c) and exerc. 7.)
- 9) Let X be a locally compact space and A the Banach algebra $\mathcal{C}_0(X)$. Let I be an ideal of A such that, for every $x \in X$, there exists $f \in I$ such that $f(x) \neq 0$. Show that I contains the ideal of functions with compact support in X .

10) Let A be a commutative Banach algebra and D a continuous derivation of A into A , that is, a continuous linear map of A into A such that $D(ab) = (Da)b + a(Db)$ for $a, b \in A$.

a) For every $\chi \in X'(A)$ and every $\lambda \in \mathbb{C}$, the series

$$\varphi_\lambda(a) = \sum_{n=0}^{\infty} \frac{\lambda^n}{n!} \chi(D^n(a))$$

converges; $\lambda \mapsto \varphi_\lambda(a)$ is an entire function, and $a \mapsto \varphi_\lambda(a)$ is a character of A .

b) The number $\varphi_\lambda(a)$ is independent of λ . (Observe that $|\varphi_\lambda(a)| \leq \|a\|$ according to a .)
Hence $\chi(D(a)) = 0$.

c) The image of D is contained in the radical of A . Deduce from this a new proof of assertion f) of exerc. 31 of I, p. 162.

11) Let B be the algebra over $\mathbb{C}$ of infinitely differentiable functions on $[0,1]$ in $\mathbb{C}$.

a) Let A be a subalgebra of B , equipped with a norm making A a Banach algebra. Show that there exists a sequence $(m_n)_{n \geq 0}$ into $\mathbb{R}_+$ such that, for all $x \in A$, one has

$$\sup_{0 \leq t \leq 1} |x^{(n)}(t)| = O(m_n).$$

(Let B_n the Banach algebra of functions n times continuously differentiable on $[0,1]$. By prop. 9 of I, p. 40, the canonical injection $A \rightarrow B_n$ is continuous; let M_n be its norm; if $x \in A$, one has $\sup_{0 \leq t \leq 1} |x^{(n)}(t)| \leq n! M_n \|x\|$.)

b) There exists no norm on B making it a Banach algebra.

12) Let A be a unital Banach algebra and (x_n) a sequence of invertible elements of A tending to an element $x \in A$. If the sequence $(\varrho(x_n^{-1}))$ is bounded, and if $x_n x = x x_n$ for all n , then x is invertible. (By the cor. of prop. 8 of I, p. 38, $\varrho(1 - x_n^{-1}x) \leq \varrho(x_n^{-1})\varrho(x_n - x)$, hence $x_n^{-1}x$ is invertible for n sufficiently large.)

¶ 13) Let A be the Banach algebra $\ell^2(\mathbb{N})$ (exerc. 3). Let A_0 the subalgebra formed by sequences with finite support. Let B be the direct sum of A_0 and of $\mathbb{C}$, equipped with the

multiplication $(f, \alpha)(g, \beta) = (fg, 0)$ for $f, g \in A_0$, $\alpha, \beta \in \mathbb{C}$, and the norm

$$\|(f, \alpha)\| = \sup \left(\|f\|, \left| \alpha - \sum_n f(n) \right| \right).$$

- a) The completed algebra $\widehat{\mathbf{B}}$ is a commutative Banach algebra whose radical R is $\mathbb{C} \cdot (0, 1)$, and $\widehat{\mathbf{B}}/R$ is isometrically isomorphic to A .
- b) Let $\pi: \widehat{\mathbf{B}} \rightarrow \widehat{\mathbf{B}}/R$ the canonical morphism. There exists no subalgebra B_1 of $\widehat{\mathbf{B}}$ supplementary to R such that $\pi|_{B_1}: B_1 \rightarrow \widehat{\mathbf{B}}/R = A$ is a homeomorphism. (Let B_1 be such a subalgebra. Let $u_n \in A$ such that $u_n(n) = 1$, $u_n(m) = 0$ for $m \neq n$. Let $e_n \in B_1$ such that $\pi(e_n) = u_n$. Show that $e_n \in A_0$, that $\sum_{n=1}^{\infty} \frac{1}{n} u_n$ converges in A but $\sum_{n=1}^{\infty} \frac{1}{n} e_n$ does not converge in $\widehat{\mathbf{B}}$.)

14) In $\mathbb{C}^2$, let Ω the compact set defined by

$$\frac{1}{2} \leq |z_1|^2 + |z_2|^2 \leq 1,$$

and let A be the subalgebra of $\mathcal{C}(\Omega)$ consisting of functions that are holomorphic in $\mathring{\Omega}$. Let U be the open set defined by $|z_1|^2 + |z_2|^2 < 1$. By virtue of a theorem of Hartogs (cf. J.-P. DEMAILLY, Complex analytic and differential geometry, Ch. I, th. 3.28), for every $f \in A$, there exists a function $\tilde{f} \in \mathcal{C}(\overline{U})$

, unique, which is holomorphic in U and coincides with f on Ω . Show that the canonical injection from A into $\mathcal{C}(\Omega)$ is an isometric morphism h whose image is a full subalgebra of $\mathcal{C}(\Omega)$ but $X(h)$ is not surjective.

- 15) Let A and B be commutative unital Banach algebras, φ a unital morphism from A to B and $(x_\lambda)_{\lambda \in \Lambda}$ a family of elements of A such that the closed full subalgebra of A generated by the x_λ is equal to A . For $X(\varphi)$ to be surjective, it is necessary and sufficient that $\text{Sp}_A^\wedge((x_\lambda)) = \text{Sp}_B^\wedge((\varphi(x_\lambda)))$ (Use diagram (1) of no. 6 of I, p. 41, where the right-hand arrow is bijective by prop. 12 of I, p. 43.)

16) Let A be a commutative Banach algebra. The following conditions are equivalent:

- (i) A is radical-free and $\mathcal{G}(A)$ is closed in $\mathcal{C}_0(X(A))$;
 - (ii) $\mathcal{G}$ is a homeomorphism from A onto $\mathcal{G}(A)$;
 - (iii) there exists a constant $c > 0$ such that $\|x\|^2 \leq c\|x^2\|$ for all $x \in A$.
- 17) Let A be a commutative Banach algebra and I a closed ideal of A . Denote by h the canonical injection of I into A . The space $X'(I)$ is identified with the quotient space of $X'(A)$ by the equivalence relation defined by $X'(h)$. (Use part (a) of Exercise 4 of I, p. 153).
- 18) Let A be a unital commutative Banach algebra and x an element of A . Suppose that the closed full subalgebra of A generated by x is equal to A , so that the map $\chi \mapsto \chi(x)$ allows one to identify $X(A)$ and $\text{Sp}_A(x)$. Let $\mathcal{H}$ the set of functions $\mathcal{G}_y$ on $\text{Sp}_A(x)$, for y ranging over A . Then $\check{S}_{\mathcal{H}}(\text{Sp}_A(x))$ (INT, IV, §7, no. 4) is the boundary of $\text{Sp}_A(x)$ relative to $\mathbb{C}$. (To show that $\check{S}_{\mathcal{H}}(\text{Sp}_A(x))$ is contained in this boundary, use the maximum principle. Conversely, let z_0 a point of this boundary; to show that $z_0 \in \check{S}_{\mathcal{H}}(\text{Sp}_A(x))$, consider $(x - z_1)^{-1}$ for z_1 quite close to z_0 into $\mathbb{C} = \text{Sp}_A(x)$.)
- 19) Let A be a commutative unital Banach algebra, B a closed unital subalgebra of A , and T the map $\chi \mapsto \chi|_B$ from $X(A)$ into $X(B)$, and let R be the equivalence relation on $X(A)$ defined by T . Then T induces, by passing to the quotient, a homeomorphism from $X(A)/R$ onto $T(X(A))$, and for every $x \in B$, the function $f = \mathcal{G}_B(x)$ is such that $|f|$ attains its upper bound on $T(X(A))$.
- 20) Let Δ the disk $|z| \leq 1$ into $\mathbb{C}$ the set of $f \in \mathcal{C}(\Delta)$ that are holomorphic in the interior of Δ and A_1 the Banach subalgebra of $\mathcal{C}(\Delta)$ generated by A and by the function $z \mapsto |z|$. Show that $X(A_1)$ is homeomorphic to the set of $(x_1, x_2, x_3) \in \mathbb{R}^3$ such that $x_1^2 + x_2^2 \leq x_3^2$ and $0 \leq x_3 \leq 1$.
- 21) a) Let $(X_\lambda)_{\lambda \in \Lambda}$ a family of indeterminates. For any formal series $f \in \mathbb{C}[[X_\lambda]]_{\lambda \in \Lambda}$, let $\|f\|$ the sum of the absolute values of its coefficients. Let A be the set of $f \in \mathbb{C}[[X_\lambda]]_{\lambda \in \Lambda}$ such that $\|f\| < +\infty$ Equipped with the usual addition and multiplication, A is a commutative unital Banach algebra without radical.

- b) For every commutative unital Banach algebra B, there exists an algebra A of the type in a) and a continuous unital morphism from A onto B.

22) Let Ω a compact subset of $\mathbf{C}^n$. The map

$$(z_1, \dots, z_n) \mapsto (z_1, \dots, z_n, \bar{z}_1, \dots, \bar{z}_n)$$

is a homeomorphism from Ω onto a compact polynomially convex subset of $\mathbf{C}^{2n}$. (Let $A = \mathcal{C}(\Omega)$, z_i the coordinate functions on $\mathbf{C}^n$. Then $(z_i|_{\Omega}, \bar{z}_i|_{\Omega})_{1 \leq i \leq n}$ is a topological generating system of A, and the map in question is the map from $X(A) = \Omega$ into $\mathbf{C}^{2n}$ defined by this topological generating system.)

23) We denote Ω the set of $(z_1, z_2) \in \mathbf{C}^2$ such that $|z_1| \leq 1$, $|z_2| \leq 1$. Let $\alpha \in [0, 1]$ and Ω_α the set of $(z_1, z_2) \in \mathbf{C}^2$ such that

$$|z_1| \leq 1, \quad |z_2| \leq (1 - \alpha)|z_1| + \alpha.$$

- a) The complements of Ω and Ω_α into $\mathbf{C}^2$ are connected.
- b) Ω is the polynomially convex hull of Ω_α . (If $(z_1^0, z_2^0) \in \Omega$ and if $P \in \mathbf{C}[z_1, z_2]$, there exists $z \in \mathbf{C}$ such that $|z| = 1$ and $|P(z_1^0, z_2^0)| \leq |P(z, z_2^0)|$; but $|P(z, z_2^0)| \leq \sup_{(z_1, z_2) \in \Omega_\alpha} |P(z_1, z_2)|$.)

24) Let $n \geq 1$. There exists a closed convex subset of $\mathbf{C}^n$ that is not polynomially convex.

25) Let A be a complete commutative unital locally m-convex algebra (cf. I, p. 165, exerc. 37).

- a) An element x of A is invertible if and only if $\chi(x) \neq 0$ for every continuous character χ of A. The radical of A is the intersection of the kernels of the continuous characters of A, hence is closed.
- b) In example e) of exercise 37 of I, p. 165, let I be the ideal formed by the f such that $f(n) = 0$ for every integer $n > 0$ sufficiently large. Then every maximal ideal containing

I is dense and of infinite codimension.

- 26) Let X be a compact topological space and E a complex Banach space, whose norm is denoted by $\|\cdot\|_E$. We denote by $\mathcal{C}(X, E)$ the complex vector space of continuous functions from X into E .

a) Equipped with the norm

$$\|f\| = \sup_{x \in X} \|f(x)\|_E,$$

the space $\mathcal{C}(X, E)$ is a Banach space.

- b) Let B be the vector subspace of $\mathcal{C}(X, E)$ generated by the functions $x \mapsto f(x)y$, where $f \in \mathcal{C}(X)$ and $y \in E$. Show that B is dense in $\mathcal{C}(X, E)$.

In what follows, we consider a commutative Banach algebra A .

- c) Equipped with the multiplication $(fg)(x) = f(x)g(x)$ for all $x \in X$, the Banach space $\mathcal{C}(X, A)$ is a commutative Banach algebra.
- d) Every character $\chi \in X(\mathcal{C}(X, A))$ is of the form $f \mapsto \tilde{\chi}(f(x))$, where $\tilde{\chi} \in X(A)$ and $x \in X$. (First treat the case where A is unital and $\|1\| = 1$; consider the restriction of χ to $\mathcal{C}(X) \cdot 1$ and to the image of the homomorphism $A \rightarrow \mathcal{C}(X, A)$ which sends a to the constant function $x \mapsto a$).
- e) The map from $X(A) \times X$ into $X(\mathcal{C}(X, A))$ which to $(\tilde{\chi}, x)$ associates the character $f \mapsto \tilde{\chi}(f(x))$ is a homeomorphism.
- 27) Let A be a commutative unital Banach algebra over $\mathbb{C}$. A closed subset F of $X(A)$ is called a boundary of A if one has

$$\sup_{\chi \in X(A)} |\chi(x)| = \sup_{\chi \in F} |\chi(x)|$$

for all $x \in A$.

- a) Order the set of boundaries of A by inclusion. There exists a minimal boundary.

- b) Let S be a minimal boundary of A . Every boundary of A contains S , hence S is the intersection of all boundaries of A . (Let F be a boundary of A ; show that for every $x \in S$, every neighborhood U of x meets F ; for this, use the fact that $S - U$ is not a boundary of A .)

One says that S is the Shilov boundary of A , and one denotes it by ∂A .

- c) If the image of the Gelfand transform of A is dense in $\mathcal{C}(X(A))$, the Shilov boundary of A is equal to $X(A)$.
- d) Let A be the algebra of continuous functions on the unit disk Δ of $z \in \mathbb{C}$ such that $|z| \leq 1$ that are holomorphic inside of Δ (example 9 of I, p. 20). Then $X(A)$ is identified with Δ (exercise 6 of I, p. 193) and the Shilov boundary of A is identified with the unit circle U .
- 28) Let X be a compact topological space and n an integer ≥ 1 . Let I be a two-sided ideal of the Banach algebra $\mathcal{C}(X, M_n(\mathbb{C}))$. Show that there exists a unique closed subset $S \subset X$ such that I is the set of $f \in \mathcal{C}(X, M_n(\mathbb{C}))$ whose restriction to S is zero.
- 29) * Let X be a compact real manifold. We equip $\mathcal{C}^\infty(X)$ of the topology of uniform convergence of functions and all their derivatives; it is a Fréchet space. Let A be a subalgebra of $\mathcal{C}(X)$. We assume that A is equipped with a norm $f \mapsto \|f\|_A$ which makes it a Banach algebra, such that the canonical injection of A into $\mathcal{C}(X)$ is continuous. We further assume that $\mathcal{C}^\infty(X)$ is contained and dense in A .
- a) There exists a constant $C \geq 0$ and an integer $k \in \mathbb{N}$ such that for all $f \in \mathcal{C}^\infty(X)$, one has

$$\|f\|_A \leq C \sup_{x \in X} |f^{(k)}(x)|.$$

- b) The subalgebra A is full in $\mathcal{C}(X)$. (Let $f \in A$ that does not vanish on X and $\delta = \inf_{x \in X} |f(x)|$; choose $g \in \mathcal{C}^\infty(X)$ such that $\|f - g\|_A < \delta/3$, then note that $f^{-1} = g^{-1} \sum_{n \in \mathbb{N}} ((g - f)/g)^n$ into $\mathcal{C}(X)$ and verify that the series converges in A .)
- c) Deduce from this a new proof of Wiener's theorem (I, p. 38, example).

Exercises for Section 4: Holomorphic Functional Calculus

In the exercises below, all algebras considered are over $\mathbb{C}$, unless explicitly stated otherwise.

- 1) Let $n \geq 1$ be an integer and $A = \mathcal{L}(\mathbb{C}^n)$. Let $\alpha \in \mathbb{C}$ and $x \in A$ an element whose matrix relative to the canonical basis of $\mathbb{C}^n$ is a Jordan matrix of order n and eigenvalue α . The spectrum of x is reduced to α .

- a) For $f \in \mathcal{O}(\{\alpha\})$ compute the matrix relative to the canonical basis of $\mathbb{C}^n$ which represents $f(x)$.

Let $m \geq 1$ be an integer. Let B_m the unital Banach algebra of jets of functions of class C^m in a neighborhood of α endowed with the topology of C^m -uniform convergence (VAR, R2, §12).

- b) There exists a continuous morphism of unital algebras from B_{m-1} to A such that $\varphi(z) = x$, where $z \in B_{m-1}$ is the jet of order $m-1$ at α of the identity function of $\mathbb{C}$.
- c) There does not exist a continuous morphism φ of unital algebras from B_m to A such that $\varphi(z) = x$, where $z \in B_m$ is the jet of the identity function of $\mathbb{C}$.

- 2) Let A be a unital Banach algebra and G the group of invertible elements of A .

- a) For an element $x \in A$ to be of the form $\exp(y)$, where $y \in A$, it is necessary and sufficient that it be contained in a connected commutative subgroup of G .
- b) For an element $x \in A$ to be of the form $\exp(y)$, where $y \in A$, it suffices that 0 belong to the unbounded connected component of $\mathbb{C} = \text{Sp}_A(x)$.

- 3) Let A be a unital Banach algebra, G the group of invertible elements of A , and G_1 the identity component of G .

- a) Let $x \in A$ and n an integer ≥ 0 such that $x^n = e$. Show that $x \in G_1$. (The spectrum of x is finite. The set of $\lambda \in \mathbb{C}$ such that $y(\lambda) = \lambda x + e - \lambda e$ such that it is invertible has finite complement, hence is connected. Now, $y(0) = e$, $y(1) = x$.)
- b) Assume A is commutative. Every element of G/G_1 distinct from the identity element has infinite order. (If $x \in G$ and $x^n \in G_1$, one has $x^n = \exp(y)$ with a $y \in A$ by no. 10 of I,

p. 78, whence $(x \exp(-y/n))^n = e$. Apply a.)

- 4) There exists a commutative Banach algebra, non-unital and non-zero, such that $X(A)$ is compact.
- 5) Let A be a unital Banach algebra and $a \in A$. Let G be the group of invertible elements of A and G_1 the connected component of the identity in G . The exponential spectrum of A is the set of $\lambda \in \mathbb{C}$ such that $x - \lambda e$ does not belong to G_1 (cf. prop. 12 of I, p. 79).
 - a) The exponential spectrum of a is closed; it contains the spectrum of a .
 - b) The boundary of the exponential spectrum of a is contained in the singular spectrum of a (exercise 16 of I, p. 158).
 - c) The exponential spectrum of a is the union of the spectrum of a and of certain bounded connected components of $\mathbb{C} = \text{Sp}(a)$.
 - d) Let $f \in \mathbb{C}[X]$ be a polynomial. The image under f of the exponential spectrum of a contains the exponential spectrum of $f(a)$. In general, the inclusion is strict (use Fredholm theory, cf. III, to appear).
- 6) Let A be a commutative unital Banach algebra.
 - a) If j is an idempotent of A , one has $\exp(2i\pi j) = 1$ and $\exp(i\pi j) = 1 - 2j$.
 - b) Let $p \in A$ such that $\exp(p) = 1$. Then $\text{Sp}_A(p)$ consists of a finite number of points of the form $2in\pi$, where $n \in \mathbb{Z}$.
 - c) For $\lambda \notin \text{Sp}_A(p)$, we have:

$$\int_0^1 \exp(t(p - \lambda 1)) dt = (1 - \exp(-\lambda))(\lambda 1 - p)^{-1}.$$

- d) Deduce that every point of $\text{Sp}_A(p)$ is a simple pole of the resolvent $R(p, \lambda)$, then that there exists an integer $m \geq 0$, integers $n_k \in \mathbb{Z}$ and pairwise orthogonal idempotents j_k , for $1 \leq k \leq m$, such that

$$p = 2i\pi \sum_k n_k j_k.$$

¶ 7) a) Let B be a complex Banach space and j a point of B such that $\|j\| = 1$. For every $x \in B$, the limit

$$\lim_{\substack{\alpha \rightarrow 0 \\ \alpha > 0}} \frac{1}{\alpha} (\|j + \alpha x\| - 1)$$

exists. Let $\varphi(x)$ be this limit.

b) Let

$$\psi(x) = \sup_{\substack{\varepsilon \in \mathbb{C} \\ |\varepsilon|=1}} \varphi(\varepsilon x).$$

The function ψ is a semi-norm bounded above by the norm of B . One has

$$\psi(x) = \sup_f |f(x)|,$$

where f ranges over the set of elements of the dual of B such that $\|f\| = f(j) = 1$.

c) One now assumes that B is a Banach algebra and that $j = 1$ is a unit element of B . One has:

$$\varphi(x) = \lim_{\substack{\alpha \rightarrow 0 \\ \alpha > 0}} \frac{1}{\alpha} \log(\|\exp(\alpha x)\|) = \sup_{\alpha > 0} \frac{1}{\alpha} \log(\|\exp(\alpha x)\|).$$

(Observe that $\log(\|\exp((\alpha + \alpha')x)\|) \leq \log(\|\exp(\alpha x)\|) + \log(\|\exp(\alpha'x)\|)$.)

d) Using c), show that $\|\exp(\lambda x)\| \leq \exp(|\lambda| \psi(x))$. By integrating $\lambda \mapsto \lambda^{-2} \exp(\lambda x)$ over the circle determined by $|\lambda| = \varrho$, deduce that

$$\|x\| \leq \varrho^{-1} \exp(\varrho \psi(x)),$$

whence, setting $\varrho = \psi(x)^{-1}$, that $\|x\| \leq e \psi(x)$, where $e = \exp(1)$.

- e) For every x nonzero element of B , there exists a linear form $f \in B'$ such that $f(1) = \|f\| = 1$ and $f(x) \neq 0$.
- 8) Let A be a unital Banach algebra, $x \in A$, and B the subalgebra of A generated by x . For $\text{Sp}_A(x)$ to be a finite set all of whose elements are poles of the resolvent of x , it is necessary and sufficient that B be of finite dimension. (To show that the condition is necessary, if $R(x, \lambda) = \sum_{i=1}^p F_i(\lambda) a_i$, where $a_i \in A$ and where F_i is a scalar rational function, the expansion of $R(x, \lambda)$ around $\lambda = \infty$ shows that the x^n are linear combinations of the a_i .)
- 9) Let A be a unital Banach algebra, $x \in A$, U an open neighborhood of $\text{Sp}_A(x)$ having only a finite number of connected components U_i ($i \in I$), and $f \in \mathcal{O}(U)$. For $f(x) = 0$, it is necessary and sufficient that the following conditions be satisfied:
- (i) for every i such that $\text{Sp}_A(x) \cap U_i$ is infinite or contains a point which is not a pole of the resolvent, one has $f|_{U_i} = 0$;
 - (ii) every pole of order p of the resolvent is a zero of order $\geq p$ of f (For necessity of (ii), argue as in Exerc. 8.)
- 10) Let Δ the disk $|z| \leq 1$ in $\mathbb{C}$. Let A be the space of functions f that are continuous in Δ and holomorphic in the interior of Δ and such that there exists a sequence (c_n) summable of complex numbers satisfying

$$f(z) = \sum_{n \geq 0} c_n z^n$$

for all $z \in \Delta$. We then denote by $\|f\| = \sum_n |c_n|$.

- a) Equipped with the multiplication $(f, g) \mapsto fg$, the space A is a unital Banach algebra and z is a topological generator of A .
 - b) The space $X(A)$ is identified with Δ ; if $f \in A$ and if g is holomorphic in an open neighborhood of $f(\Delta)$, then $g \circ f \in A$.
- 11) Let A be a commutative unital Banach algebra and $x \in A$. Let S be a subset of $X(A)$ closed for the Jacobson topology, and let f be a complex holomorphic function in a neighborhood

of $(\mathcal{G}x)(S)$. There exists $y \in A$ such that $Gy = f \circ Gx$ on S . (Let I be the intersection of the kernels of the $\chi \in S$. Use the functional calculus in A/I .)

12) Let U be a nonempty open set in $\mathbb{C}$. Show that there exists no norm on $\mathcal{O}(U)$ for which $\mathcal{O}(U)$ is a Banach algebra. (Using th. 1 of I, p. 29, show that the functions of $\mathcal{O}(U)$ would be bounded.)

13) Let $n \in \mathbb{N}$

a) A subset V of $\mathbb{C}^n$ is polynomially convex if and only if, for every $x \in \mathbb{C}^n - V$, there exists an entire function $f : \mathbb{C}^n \rightarrow \mathbb{C}$ such that

$$|f(x)| > \sup_{y \in V} |f(y)|.$$

b) Let $K \subset \mathbb{R}^n$ a compact set. Then K is polynomially convex in $\mathbb{C}^n$.

c) Let V be a polynomially convex compact subset of $\mathbb{C}^n$ and $f \in \mathcal{O}(V)$. Then the graph of f is polynomially convex in $\mathbb{C}^{n+1}$. (Use the Oka–Weil theorem.)

14) Let K be a compact subset of $\mathbb{C}$ and let P (resp. Q) be the closure in $\mathcal{O}(K)$ of the set of germs of polynomial (resp. rational) functions holomorphic in a neighborhood of K . Show that $P = Q$ if and only if the set $\mathbb{C} - K$ is connected.

15) Let U be the set of complex numbers of modulus 1. Let K_1, K_2 be the compact subsets of $\mathbb{C}^2$ given by

$$K_1 = U \times \{0\}, \quad K_2 = \{(z, \bar{z}) \mid z \in U\}.$$

a) Show that the set of germs in a neighborhood of K_1 of polynomial functions on $\mathbb{C}$ with complex coefficients is not dense in $\mathcal{O}(K_1)$.

b) Show that the set of germs in a neighborhood of K_2 of polynomial functions on $\mathbb{C}$ with complex coefficients is dense in $\mathcal{O}(K_2)$.

16) Let A be a unital Banach algebra, $x \in A$, and k an integer ≥ 0 . Let U be the set of complex numbers of modulus 1.

- a) Suppose that $\text{Sp}_A(x) \subset U$ and that $\|R(x, \lambda)\| = O((1 - |\lambda|)^k)$ when $|\lambda|$ tends to 1. Show that $\|x^n\| = O(|n|^k)$ when n tends to $\pm\infty$. (For every $\delta > 1$, show that

$$2\pi\|x^n\| \leq M \frac{\delta^n}{(\delta - 1)^k} + M \frac{\delta^{-n}}{(1 - \delta^{-1})^k}$$

where M is independent of n and δ . Take $\delta = n/(n - k)$ when n tends to $+\infty$, $\delta = n/(n + k)$ when n tends to $-\infty$.

- b) Suppose that $\|x^n\| = O(|n|^k)$ when n tends to $\pm\infty$. Show that $\text{Sp}_A(x) \subset U$ and that $\|R(\lambda, x)\| = O((1 - |\lambda|)^{k+1})$ when $|\lambda|$ tends to 1. (We have $\varrho(x) = \varrho(x^{-1}) = 1$, hence $\text{Sp}_A(x) \subset U$. For $|\lambda| < 1$,

$$R(\lambda, x) = -x^{-1} (1 + \lambda x^{-1} + \lambda^2 x^{-2} + \lambda^3 x^{-3} + \dots).$$

Deduce an upper bound for $\|R(\lambda, x)\|$, and argue similarly if $|\lambda| > 1$.

17) Let X_1 (resp. X_2) the set of $(z_1, z_2) \in \mathbb{C}^2$ such that

$$\frac{1}{2} \leq |z_1|^2 + |z_2|^2 \leq 1, \quad \text{resp.} \quad |z_1|^2 + |z_2|^2 \leq 1.$$

Let $X = X_1 \cup \{(0, 0)\} \subset \mathbb{C}^2$. Let $A = \mathcal{C}(X)$, and a_1, a_2 the restrictions to X of the coordinate functions on $\mathbb{C}^2$.

- a) The simultaneous spectrum $\text{Sp}_A^2(a_1, a_2)$ is equal to X .
- b) Construct two distinct continuous unital morphisms from $\mathcal{O}(X)$ to A sending z_1 to a_1 and z_2 to a_2 . (Use the following theorem: for every function f holomorphic in a neighborhood of X_1 , there exists a function g holomorphic in a neighborhood of X_2 which coincides with f in a neighborhood of X_1 (cf. I, p. 168, exerc. 14).

¶ 18) Let A be a commutative unital Banach algebra.

- a) For any finite family I of elements of A , let $S(I) \subset C^I$ the simultaneous spectrum of I . If I' extends I , the canonical surjection $\text{pr}_{I,I'}$ of $C^{I'}$ on C^I is such that $\text{pr}_{I,I'}(S(I')) = S(I)$, whence an injective morphism $\varphi_{II'}$ of $\mathcal{O}(S(I))$ into $\mathcal{O}(S(I'))$. Let $\mathcal{O}(X(A))$ the inductive limit of the $\mathcal{O}(S(I))$ with respect to the $\varphi_{II'}$ (we index the finite families of elements of A by the finite subsets of $A \times \mathbb{N}$). The canonical maps

$$\varphi_I: \mathcal{O}(S(I)) \rightarrow \mathcal{O}(X(A))$$

are injective.

We shall equip $\mathcal{O}(X(A))$ of the inductive limit topology of those of $\mathcal{O}(S(I))$. For every $a \in A$ the coordinate function in a neighborhood of $S(a) \subset C$ defines an element z_a of $\mathcal{O}(X(A))$.

- b) The continuous surjection $X(A) \rightarrow S(I)$ defines a continuous morphism $\mathcal{C}(S(I)) \rightarrow \mathcal{C}(X(A))$. The morphisms $\mathcal{O}(S(I)) \rightarrow \mathcal{C}(S(I)) \rightarrow \mathcal{C}(X(A))$ define a continuous morphism:

$$\varphi: \mathcal{O}(X(A)) \rightarrow \mathcal{C}(X(A)).$$

This morphism is not injective in general.

- c) There exists a unique continuous unital morphism ψ of $\mathcal{O}(X(A))$ to A such that $\psi(z_a) = a$ for all $a \in A$ (Use th. 1 of I, p. 51.) Its composite with the Gelfand transformation is φ .
- d) Let E be a separated locally convex space, U a weakly open subset of E , and $f: U \rightarrow C$ a function. We say that f is weakly holomorphic if, for every $u \in U$ there exists a weakly open neighborhood V_u of u , a closed vector subspace F_u of finite codimension in E , and a holomorphic function g_u on $p_u(V_u)$, where p_u denotes the canonical mapping of E onto E/F_u , such that $f|V_u = g_u \circ p_u$.

Show that if f is weakly holomorphic, and if K is a weakly compact subset of U , there exists a weakly open neighborhood V_K of K in U , a closed vector subspace F_K of finite codimension

in E , and a holomorphic function g_K on $p_K(V_K)$ such that $f|_{V_K} = g_K \circ p_K$, where p_K is the canonical mapping of E onto E/F_K .

- e) Show that $\mathcal{O}(X(A))$ is the inductive limit of the $\mathcal{O}(S)$, where S ranges over the set of weakly open neighborhoods of $X(A)$ in the dual of A , and where $\mathcal{O}(S)$ denotes the algebra of weakly holomorphic functions in S , endowed with the topology of compact convergence.

Exercises for Section 5: Regular Commutative Banach Algebras

In the exercises below, all algebras considered are over $\mathbb{C}$.

- 1) Let A be a regular commutative Banach algebra and I a closed ideal of A . Then I and A/I are regular commutative Banach algebras.
- 2) Let A be a regular commutative Banach algebra without radical, I an ideal of A , χ an element of $V(I)$, and J the set of $x \in A$ such that Gx has compact support disjoint from $\{\chi\}$. Then $I + J$ is the smallest of the ideals R of A such that $R \supset I$ and $V(R) = \{\chi\}$.
- 3) Let A be a commutative Banach algebra, A' the dual Banach space of A . For every $x \in A$, let $\kappa(x)$ the linear operator in A' transposed to multiplication by x in A . For $x \in A$ and $x' \in A'$, we denote by $x \star x' = \kappa(x)(x') \in A'$. A vector subspace of A' is said to be invariant if it is stable under all the $\kappa(x)$.
 - a) For an element x' of A' to be proportional to a nonzero character of A , it is necessary and sufficient that Cx' be invariant.
 - b) The map $V \mapsto V^\circ$ is a bijection from the set of closed ideals of A onto the set of weakly closed invariant vector subspaces of A' .
 - c) If W is an invariant vector subspace of A' , we denote by $\sigma(W)$ the set of characters belonging to W . If W_1 and W_2 are weakly closed invariant subspaces, and if W is the weakly

- closed vector subspace generated by W_1 and W_2 , one has $\sigma(W) = \sigma(W_1) \cup \sigma(W_2)$ (Use b).
- d) Let W be a weakly closed invariant vector subspace of A' . Then $\sigma(W)$ is the set of $\chi \in \widehat{A}$ such that the condition $x * x' = 0$ for all $x' \in W$ implies $\chi(x) = 0$.
- e) If A admits a unit element, every weakly closed invariant vector subspace of A' contains a nonzero character.
- f) In the remainder of this exercise, we assume that A is a regular algebra without radical and with identity. If W is a weakly closed invariant vector subspace of A' and U a neighborhood of $\sigma(W)$ into $X(A)$ then W is contained in the weakly closed vector subspace of A' generated by U .
- g) Let $x' \in A'$ and let W be the weakly closed invariant vector subspace of A' generated by x' . We set $\sigma(x') = \sigma(W)$. If $x \in A$ is such that $\mathcal{G}x = 1$ on a neighborhood of $\sigma(x')$, then $x * x' = x'$. (Use f.)
- h) If $\sigma(x')$ is the union of two disjoint closed sets σ_1 and σ_2 , there exists $x'_1, x'_2 \in A'$ such that $\sigma(x'_1) = \sigma_1$, $\sigma(x'_2) = \sigma_2$ and $x' = x'_1 + x'_2$. (Set $x'_1 = u_1 * x'$, $x'_2 = u_2 * x'$, with $\mathcal{G}u_1 = 1$ (resp. 0) in a neighborhood of σ_1 (resp. σ_2), and $\mathcal{G}u_2 = 1$ (resp. 0) in a neighborhood of σ_2 (resp. σ_1).)
- 4) Let A be a commutative regular semisimple Banach algebra satisfying Ditkin's condition, and let I be a closed ideal of A . $x \in \Upsilon(V(I))$ and F the boundary of $V(Ax)$. If $F \cap V(I)$ contains no nonempty perfect set, we have $x \in I$ (Imitate the reasoning of Prop. 5 of I, p. 94.)
- 5) Let A be a regular commutative Banach algebra without radical. We assume that every closed ideal of A is an intersection of regular maximal ideals. Let $x \in A$. Let F be the set of zeros of $\mathcal{G}'(x)$. Then x is a limit of elements $u_n x$, where $u_n \in A$ and $\mathcal{G}(u_n)$ vanishes in a neighborhood of F . (Let J be the set of $y \in A$ such that $\mathcal{G}(y)$ with compact support disjoint from F . We have $x \in \overline{J}$. Use prop. 4 of I, p. 91.) In particular, A satisfies Ditkin's condition.

- 6) Let B be the algebra of continuously differentiable complex functions on $[0, 1]$, equipped with the norm $\|f\| = \sup |f| + \sup |f'|$. Let A be the closed subalgebra of B formed by the functions $f \in B$ such that $f\left(\frac{1}{2}\right) = 0$. Let I be the closed ideal of A formed by the $f \in A$ such that $f'\left(\frac{1}{2}\right) = 0$.
- a) The space $X(A)$ is identified with $[0, 1] = \left\{\frac{1}{2}\right\}$, and A is regular.
- b) The ideal I is a maximal ideal of A that is not regular.
- 7) Let R, T_1, T_2 be the subsets formed by $\mathbb{C}^4$ consisting of the elements $(z_1, z_2, z_3, z_4) \in \mathbb{C}^4$ such that:

$$R: z_1 z_2 = 2, \quad 1 \leq |z_1| \leq 2, \quad z_3 = z_4 = 0,$$

$$T_1: z_1 z_2 = 2, \quad |z_1| = 1, \quad |z_3| \leq 1, \quad z_4 = 0,$$

$$T_2: z_1 z_2 = 2, \quad |z_1| = 2, \quad |z_3| \leq 1, \quad z_4 = z_3^2.$$

- a) The set $X = R \cup T_1 \cup T_2$ is polynomially convex. (Show that X is the set of (z_1, z_2, z_3, z_4) such that

$$z_1 z_2 = 2, \quad |z_1| \leq 2, \quad |z_2| \leq 2, \quad |z_3| \leq 1, \quad z_4(z_4 - z_3^2) = 0,$$

$$|z_4 z_2^k| \leq 1 \quad \text{and} \quad |z_1^k(z_4 - z_3^2)| \leq 1 \quad \text{for} \quad k = 1, 2, \dots).$$

- b) Let A be the closed subalgebra of $\mathcal{C}(X)$ generated by the restrictions to X of polynomial functions on $\mathbb{C}^4$. Then X is identified with $X(A)$.
- c) Let $\Gamma_1, \Gamma_2 \subset \mathbb{C}$ the circles centered at 0 with radii 1, 2 oriented in the positive direction. Let μ_0, μ_1, ν the measures on $\Gamma_1, \Gamma_2, \Gamma_1$ defined by the differential forms $z^{-1}dz, -z^{-1}dz, z^{-2}dz$. Let $\mu = \mu_0 + \mu_1$, so that $\mu \otimes \nu$ is a measure on $(\Gamma_1 \cup \Gamma_2) \times \Gamma_1$.

Let B be the set of $z = (z_1, z_2, z_3, z_4) \in X$ such that $(z_1, z_3) \in (\Gamma_1 \cup \Gamma_2) \times \Gamma_1$. The map $\varphi : z \mapsto (z_1, z_3)$ from B into $(\Gamma_1 \cup \Gamma_2) \times \Gamma_1$ is a homeomorphism. Let ν the measure $\varphi^{-1}(\mu \otimes \nu)$. Show that ν is orthogonal to A .

d) Let g be the function on X that is zero on $R \cup T_1$ and equal to z_3 on T_2 . Then $g \notin A$.

e) The function g coincides with an element of A in a neighborhood of every $z \in X$.

Exercises for Section 6: Stellar Algebras

- 1) Find a unital involutive Banach algebra A and a unitary element $u \in A$ such that $\|u\| \neq 1$.
- 2) Let A be a Banach algebra and $A_1 \subset A$ a closed subalgebra. If A_1 is an involutive algebra, then every character of A_1 extends to a character of A .
- 3) Let A be a unital involutive algebra. Suppose that for every x in A , the spectrum of x does not contain -1 .
 - a) For every idempotent element x of A , show that there exists a hermitian idempotent element y of A such that $xA = yA$. (Consider the inverse z of $1 + (x - x^*)(x^* - x)$; show that z commutes with x and set $y = xx^*z$.)
 - b) For every idempotent element x of A , show that there exists a hermitian idempotent element y of A and u invertible in A such that $x = uy u^{-1}$. (If $xA = yA$, set $u = (1 - (y - x))^{-1}$.)
- 4) Let A be a normed algebra, equipped with a continuous involution. If one sets $\|x\|' = \sup(\|x\|, \|x^*\|)$ for all $x \in A$, A becomes an involutive normed algebra, and the new norm is equivalent to the old one.
- 5) Let A be a commutative Banach algebra without radical. Every involution of A is continuous. (Use prop. 9 of I, p. 40.)
- ¶ 6) Let A be a commutative unital involutive Banach algebra whose every character is hermitian. Suppose that, for every $x \in A$ non-invertible hermitian element, one has $\|(x -$

$\lambda)^{-1}\| = o(\mathcal{I}(\lambda)^{-2})$ when λ tends to 0 with $\mathcal{I}(\lambda) \neq 0$. Then every closed ideal I of A is an intersection of maximal ideals. (Let $\varphi : A \rightarrow A/I$ be the canonical morphism. Let $x \in A$ such that $\varphi(x)$ be in the radical of A/I . It must be proved that $\varphi(x) = 0$. Reduce to the case where x is hermitian. Then use exerc. 29 of I, p. 161).

- 7) Let A be a commutative unital involutive Banach algebra. For every character of A to be hermitian, it is necessary and sufficient that $1 + xx^*$ be invertible for every $x \in A$. (If every character of A is hermitian, $\mathcal{G}(1 + xx^*)$ is everywhere > 0 on $X(A)$. If a character χ of A is non-hermitian, there exists a $x \in A$ hermitian element such that $\chi(x) = i$, whence $\chi(1 + xx^*) = 0$ and $1 + xx^*$ is not invertible.)
- 8) Construct an example of a group G such that the involutive Banach algebra $\mathcal{M}^1(G)$ (I, p. 100, example 4) is not a star algebra.
- 9) Let A be a unital Banach algebra. The group of unitary elements of A is a maximal bounded subgroup of A^* .
- 10) Let A be an involutive Banach algebra and $\tilde{A}$ the algebra obtained by adjoining a unit element. Show that the norm defined in number I, p. 15 on $\tilde{A}$ does not coincide, in general, with the norm defined by prop. 3 of I, p. 105.
- 11) Let A be an involutive Banach algebra, whose involution satisfies $\|x^*x\| = \|x\| \cdot \|x^*\|$ for all $x \in A$. Then A is a star algebra. (One has $\|y^2\| = \|y\|^2$ for hermitian y , whence $\|y\| = \varrho(y)$; observing that $\text{Sp}'_A(x^*) = \overline{\text{Sp}'_A(x)}$ for all $x \in A$, we therefore have

$$\|x^*\| \cdot \|x\| = \|x^*x\| = \varrho(x^*x) \leq \varrho(x^*)\varrho(x) = \varrho(x^2) \leq \|x\|^2,$$

whence $\|x^*\| \leq \|x\|$ and $\|x^*\| = \|x\|$.

- 12) Let A be a star algebra, N the set of normal elements of A , $x \in N$ and V a neighborhood of 0 in C . There exists a neighborhood U of x in N such that, for every $y \in U$, one has $\text{Sp}_A(y) \subset \text{Sp}_A(x) + V$ and $\text{Sp}_A(x) \subset \text{Sp}_A(y) + V$. (Use the cor. of 1 of I, p. 108, prop. 3 of I, p. 23 and prop. 10 of I, p. 76.)

- 13) Let A be a unital Banach algebra over $\mathbb{C}$, $x \in A$, and $S = \text{Sp}_A(x)$. Suppose that there exists a morphism φ of $\mathcal{C}(S)$ into A which sends 1 to 1 and z to x , denoting by z the identity map of S . Then $\|\varphi(f)\| \geq \|f\|$ for every $f \in \mathcal{C}(S)$. In particular, if the morphism φ is continuous, then φ is a homeomorphism onto its image. (One may assume A commutative. Let $\lambda \in S$. There exists $\chi \in X(A)$ such that $\lambda = \chi(x) = \chi(\varphi(z)) = (X(\varphi)(\chi))(z)$. Therefore $X(\varphi)(\chi)$ be the character $f \mapsto f(\lambda)$ of $\mathcal{C}(S)$. Let $f \in \mathcal{C}(S)$. According to what precedes, there exists $\chi \in X(A)$ such that $|f|$ attains its maximum at $X(\varphi)(\chi)$. Then $|\chi(\varphi(f))| = \|f\|$, hence $\|\varphi(f)\| \geq \|f\|$.)
- 14) Let G be a locally compact group. If $\text{Stell}(G)$ has an identity element, then the group G is discrete. (Use the fact that, in the Hilbert space $L^2(G)$, the identity map is the norm limit of endomorphisms γ_f , where γ is the left regular representation and where $f \in L^1(G)$.)
- 15) a) Let A be a commutative stellar algebra. If x and y are positive elements of A such that $x \leq y$, then $x^2 \leq y^2$.
- b) There exists a stellar algebra A and elements x and y of A such that $0 \leq x \leq y$ but such that x^2 is not $\leq y^2$.
- c) There exists a stellar algebra A and elements x and y in A such that $|x + y|$ is not $\leq |x| + |y|$.
- 16) Let X be a topological space. Let $\mathcal{C}_b(X)$ be the commutative stellar algebra of bounded continuous complex-valued functions on X .
- a) Let $\widehat{X} = X(\mathcal{C}_b(X))$; the space $\widehat{X}$ is a compact space such that $\mathcal{C}(\widehat{X})$ is identified with $\mathcal{C}_b(X)$.
- b) The map $j : X \rightarrow \widehat{X}$ which sends x to the character $f \mapsto f(x)$ is continuous.
- c) If X is completely regular (TG, IX, p. 8, def. 4), the map j is a homeomorphism of X onto a dense subspace of $\widehat{X}$ (To show that the image of j is dense in $\widehat{X}$, show that a continuous complex function on $\widehat{X}$ vanishing on $j(X)$ is identically zero.)
- d) The space $\widehat{X}$ is identified with the Stone–Čech compactification of X (TG, IX, p. 9).

- 17) Let A be a C^* -algebra and $x \in A$. If $x = a - b$ with a, b positive and $ab = ba = 0$, then $a = x^+$ and $b = x^-$.
- 18) Let A be a separable stellar algebra. There exists a sequence $(e_n)_{n \in \mathbb{N}}$ into A_+ such that the associated elementary filter (TG, I, p. 42, def. 7) is an increasing approximate unit of A .
- 19) Let A be a stella algebra. An element x is said to be systematically positive if $x \in A_+$ and if xAx is dense in A .
- a) Show that a commutative stellar algebra A contains a systematically positive element if, and only if, there exists a locally compact topological space X , countable at infinity, such that A is isomorphic to $\mathcal{C}_0(X)$.
- b) If A is unital, then an element x of A is strictly positive if and only if x is positive and invertible.
- c) Let $x \in A$ an element that is systematically positive such that $\|x\| \leq 1$. Prove that the sequence $(x^{1/n})_{n \geq 1}$ is an increasing approximate unit in A .
- d) A stellate algebra A contains a systematically positive element if and only if there exists a sequence $(x_n)_{n \geq 1}$ in A , which forms an approximate identity for A .
- 20) Let A be a C^* -algebra, I a closed two-sided ideal of A , and B a closed involutive subalgebra of A . Then $B+I$ is a closed involutive subalgebra of A , and there exists an isometric isomorphism $B + I/I \rightarrow B/B \cap I$.
- 21) Let S be the closed unit disk in $\mathbb{C}$. Let A be the involutive Banach algebra of continuous functions from S to $\mathbb{C}$ that are holomorphic in the open unit disk, where the involution is given by $f^*(z) = \overline{f(\bar{z})}$. There exists a Hermitian element of A whose spectrum is not contained in $\mathbb{R}$.
- 22) Let $A = \mathcal{C}([0, 1])$. Show that the ideal I of A generated by the identity function of $[0, 1]$ is not closed.

- 23) Let A be a unital C^* -algebra. We denote by A'_+ the vector space of linear forms f on A , not necessarily continuous, such that

$$f(A_+) \subset [0, +\infty[$$

(“positive linear forms”).

- a) Let $f \in A'_+$. For every sequence $(x_n)_{n \in \mathbb{N}}$ into $A_+^{\leq 1}$, the family $(f(x_n))$ is bounded. (Show that the series $\sum_i f(x_i)y_i$ converges for every positive family (y_i) into $\ell^1(\mathbb{N})$.)
- b) The linear map f is continuous.
- c) A linear form f belongs to A'_+ if and only if f is continuous and $\|f\| = f(1)$.
- d) Let $x \in A_h$ a Hermitian element. There exists a positive linear form $f \in A'_+$ such that $|f(x)| = \|x\|$ (Use Exercise 7 of I, p. 174).
- 24) Let A be a unital Banach algebra over $\mathbb{C}$. Suppose that $x \mapsto x^*$ and $x \mapsto x^\circ$ are involutions on A such that A , equipped with each one, is a stellar algebra. We denote by A_1 and A_2 these two stellar algebras.
- a) Using the notation from Exercise 23, show that $A'_{1,+} = A'_{2,+}$. One denotes by A'_+ this set.
- b) Let $x \in A_{1,h}$ an element such that $x^* = x$. Show that for every $f \in A'_+$, one has $f(i(x - x^\circ)) = 0$. Deduce that $x \in A_{2,h}$ (Use part (b) of Exercise 23).
- c) Conclude that $A_1 = A_2$, and that a complex Banach algebra admits at most one involution making it a C^* -algebra.

- 25) Let G be a group. We denote by $C[G]$ the group algebra of G over $\mathbb{C}$. It is a subspace of the Hilbert space $\ell^2(G)$. For g in G , we denote by $[g]$ the element of $C[G]$ corresponding to g . The elements $[g]$, for $g \in G$, form an orthonormal basis of $\ell^2(G)$ and $C[G]$ is dense in $\ell^2(G)$.

- a) The map

$$\sum_g \alpha_g[g] \mapsto \sum_g \bar{\alpha}_g[g^{-1}]$$

is an involution on $\mathbf{C}[G]$.

- b) For $g \in G$, let $\pi(g)$ the map $x \mapsto gx$ of $\mathbf{C}[G]$. The map which associates $\pi(g)$ to g extends to a continuous and injective morphism of involutive algebras from $\mathbf{C}[G]$ into $\mathcal{L}(\ell^2(G))$.

We denote $\text{Stell}_r(\mathbf{G})$, and the reduced C^* -algebra of G is called the closure in $\mathcal{L}(\ell^2(\mathbf{G}))$ of the image of the map π . It is a C^* -algebra. We identify $\mathbf{C}[G]$ with its image under π into $\text{Stell}_r(\mathbf{G})$.

- c) The linear trace map $\text{Tr}: \mathbf{C}[G] \rightarrow \mathbf{C}$ such that $\text{Tr}(1) = 1$ and $\text{Tr}(g) = 0$ for $g \neq 1$ extends to a continuous linear map $\text{Stell}_r(G) \rightarrow \mathbf{C}$.
- d) We have $\text{Tr}(xy) = \text{Tr}(yx)$ for all x and y into $\text{Stell}_r(G)$.
- e) We have $\text{Tr}(xx^*) \geq 0$ for all x into $\text{Stell}_r(G)$, and $\text{Tr}(xx^*) = 0$ if and only if $x = 0$.
- f) Let e be an idempotent element of $\mathbf{C}[G]$ such that $e \neq 0$ and $e \neq 1$. Show that $0 < \text{Tr}(e) < 1$. (Use Exercise 3).
- g) Let e be an idempotent element of $\mathbf{Z}[G]$. Show that $e = 0$ or $e = 1$. ("Kaplansky's theorem").
- h) Let x be an element of $\mathbf{C}[G]$. Then x is invertible if and only if it is left-invertible, if and only if it is right-invertible.
- i) Let n be a strictly positive integer. A matrix x of type (n, n) with coefficients in $\mathbf{C}[G]$ is invertible if and only if it is left-invertible, if and only if it is right-invertible.
- 26) Let A be a unital C^* -algebra. Let a and b be normal elements of A and $x \in A^*$ such that $b = xax^{-1}$. There exists a unitary element u of A such that $b = uau^*$. (Use the preceding exercise and the polar decomposition of x , noting that $|x|$ belongs to the bicommutant of x).

- 27) Let A be a unital C^* -algebra, and let a , b , and c be elements of A . Suppose that b and c are normal. If $ac = cb$, then we have $f(a)c = cf(b)$ for every continuous function f on the union of the spectrum of a and the spectrum of b .
- 28) Let A be a C^* -algebra. Let a and b be positive elements of A such that $ab \neq 0$. Then $\text{Sp}_A(ab)$ is not reduced to 0.
- 29) a) Let G be a locally compact topological group. The Banach algebra $L^1(G)$ has an approximate identity. (Use INT, VIII, § 2, no. 7, lemma 4).
- b) Let A be a Banach algebra admitting an approximate identity. For every character $\chi \in X(A)$, one has $\|\chi\| = 1$.
- c) The Banach subalgebra of $L^1(\mathbb{Z})$ formed by the sequences (x_n) such that $x_n = 0$ if $n < 0$ has no approximate identity. (Construct a nonzero character of norm < 1 .)
- d) Let G be the free group generated by two elements a and b and μ a Haar measure on G . Let I be the closed ideal of $L^1(G)$ consisting of the $f \in L^1(G)$ such that $\mu(f) = 0$. Let $\xi = e^{2i\pi/3}$. Let $x = \varepsilon_e + \xi \varepsilon_a + \xi^{-1} \varepsilon_b \in L^1(G)$. For every $f \in L^1(G)$, one has $\|x * f - x\| \geq 1$. The Banach algebra I has no approximate identity.
- ¶ 30) Let A be a Banach algebra with an approximate identity. Denote by $\tilde{A}$ the Banach algebra obtained from A by adjoining an identity element, denoted by 1.
- a) Let $x \in A$. Then x belongs to the closed left ideal generated by x .
- b) For every $\varepsilon > 0$ and every finite family $(x_i)_{i \in I}$ of elements of A , there exists $e \in A$ such that $\|e\| \leq 1$ and $\|ex_i - x_i\| < \varepsilon$ for all $i \in I$.
- c) Let $x \in A$, $\varepsilon > 0$ and $0 < \gamma \leq 1/8$. There exists $\eta > 0$ such that for all $e \in A$ satisfying $\|e\| \leq 1$ and $\|ex - x\| < \eta$, then $\alpha = (1 - \gamma) \cdot 1 + \gamma e$ is invertible in $\tilde{A}$ and $\|x - \alpha^{-1}x\| < \varepsilon$.
- d) Let $x \in A$, $\varepsilon > 0$, $\delta > 0$ and $0 < \gamma \leq 1/8$. There exists a sequence $(e_n)_{n \geq 1}$ in A having the following properties:
- (a) We have $\|e_n\| < 2$ for all $n \geq 1$;

(b) The element

$$y_n = \sum_{k=1}^n \gamma (1 - \gamma)^{k-1} e_k + (1 - \gamma)^n \cdot 1$$

is invertible in $\widetilde{A}$ for all $n \geq 1$ and

$$\|y_n^{-1}x - y_{n-1}^{-1}x\| < \delta 2^{-n} \quad (n \geq 2), \quad \|y_1^{-1}x - x\| < \delta/2.$$

(Assuming that $(e_1, \dots, e_n)$ have been defined, consider an element e_{n+1} arbitrary of norm < 2 ; observe that

$$y_{n+1} = ((1 - \gamma) \cdot 1 + e_{n+1}) \left(\sum_{k=1}^n \gamma (1 - \gamma)^{k-1} e'_k + (1 - \gamma)^n \cdot 1 \right),$$

where $e'_k = ((1 - \gamma) \cdot 1 + e_{n+1})^{-1} e_k$, and determine e_{n+1} satisfying the required conditions by exploiting the fact that the set of invertible elements of $\widetilde{A}$ is open, and that the map $a \mapsto a^{-1}$ is continuous.)

e) For every $x \in A$, there exist y and z in A such that $x = yz$ ("Cohen's factorization theorem"). Let $\delta > 0$. One may assume that z belongs to the closed left ideal generated by x and that $\|y - z\| < \delta$. (Consider a sequence $(e_n)_{n \geq 1}$ as above and define y_n as before; the sequence $z_n = y_n^{-1}x$ converges to an element $z \in A$, and z_n belongs to the closed left ideal generated by x ; the sequence (y_n) converges to $y \in A$ and $x = yz$.)

31) Let A be the Banach space of complex integrable functions on $[0, 1]$ with respect to Lebesgue measure. Set

$$(f \cdot g)(t) = \int_0^t f(t-x)g(x)dx$$

for $t \in [0, 1]$.

a) The function $f \cdot g$ is defined almost everywhere and is an element of A .

b) Equipped with the operation $(f, g) \mapsto f \cdot g$, the space A is a commutative Banach algebra.

c) Let $x \in A$ be the constant function equal to 1. The function x^n is

$$t \mapsto \frac{t^{n-1}}{(n-1)!}.$$

d) The algebra A is topologically generated by x , the element x is quasi-nilpotent, and A is equal to its radical.

e) The algebra A admits an approximate identity.

f) The algebra A has no maximal ideal, regular or not, closed or not (use the preceding exercise and exercise 1 of I, p. 166). In particular, $X(A)$ is empty, hence compact.

32) Let G be a group. One says that G has the Powers property if, for every $n \in \mathbb{N}$ and every finite subset F of $G = \{e\}$, there exists a partition of G into two subsets A and B and a family $(g_i)_{1 \leq i \leq n}$ of elements of G such that

(i) for all $g \in F$, the sets A and gA are disjoint;

(ii) for all distinct integers j and k between 1 and n , the sets $g_j B$ and $g_k B$ are disjoint.

Let G be a group possessing the Powers property. Let I be a nonzero closed two-sided ideal of the reduced C^* -algebra $\text{Stell}_r(G)$ (exercise 25). Let $\pi : C[G] \rightarrow \text{Stell}_r(G)$ be the canonical representation.

a) There exists a family $(\lambda_g)_{g \in G - \{e\}}$ into $\mathbb{C}$ such that

$$x = \sum_{g \in G - \{e\}} \lambda_g \pi(g)$$

is a hermitian element of $\text{Stell}_r(G)$ such that $y = 1 + x$ belongs to I .

b) Let $\varepsilon > 0$ such that $\varepsilon < 1$. There exists a finite subset F of $G = \{e\}$ and an integer $n \geq 1$ such that the element

$$x' = \sum_{g \in F} \lambda_g \pi(g)$$

satisfies $\|x' - x\| < \varepsilon$ and $\|x'\| < \frac{\sqrt{n}}{2}(1 - \varepsilon)$.

- c) Let $G = A \cup B$ and $(g_i)_{1 \leq i \leq n}$ satisfying the conditions of the Powers property for the finite subset F and the integer n . Set

$$y' = \frac{1}{n} \sum_{i=1}^n \pi(g_i)x' \pi(g_i^{-1}).$$

We have $\|y'\| < 1 - \varepsilon$. (Let p_i be the orthogonal projection of $\ell^2(G)$ on $\ell^2(g_i B)$; verify that

$$(1 - p_i)\pi(g_i)x' \pi(g_i^{-1})(1 - p_i) = 0$$

for $1 \leq i \leq n$, and deduce that

$$1 + y' = 1 + \frac{1}{n} \sum_{i=1}^n p_i \pi(g_i)x' \pi(g_i^{-1}) + \left(\frac{1}{n} \sum_{i=1}^n p_i \pi(g_i)x' \pi(g_i^{-1})(1 - p_i) \right)^*,$$

then note that the endomorphisms $p_i \pi(g_i)x' \pi(g_i^{-1})$ have pairwise orthogonal images to estimate their norms.)

- d) The ideal I contains the element

$$1 + \frac{1}{n} \sum_{i=1}^n \pi(g_i)x \pi(g_i^{-1}),$$

and this element is invertible. Consequently, the stellar algebra $\text{Stell}_r(G)$ contains no closed two-sided ideal other than $\{0\}$ and of $\text{Stell}_r(G)$.

- e) Let G be a noncommutative free group. Then G has the Powers property.

Exercises for Section 7: Spectrum of Endomorphisms of Banach Spaces

In the exercises below, all Banach spaces are over $\mathbb{C}$.

- 1) Let $E = \mathcal{C}([0, 1])$ and let u be the endomorphism of E defined by

$$u(f)(t) = \int_0^t f(t)dt.$$

- a) The spectral radius of u is zero. (Estimate the norm of u^n .)
- b) The spectrum of u consists only of 0, and, for every $n \geq 1$, the kernel of u^n consists only of 0.

(In particular, 0 is an isolated point of the spectrum of u , but the closure of the union of the kernels of u^n , for $n \geq 1$, is not equal to the spectral subspace $E_0(x)$.)

- 2) There exists a Banach space E and an endomorphism u of E such that 0 is an eigenvalue of u but 0 is not isolated in $\text{Sp}(u)$.
- 3) Give an example of a Hilbert space E and an endomorphism $u \in \mathcal{L}(E)$ such that 0 is an eigenvalue of u but not of u^* .
- 4) Let $E = \ell^2(\mathbf{Z})$ and let u be the endomorphism such that $u((\lambda_n)) = (\lambda_{n+1})$.
 - a) Show that u is unitary.
 - b) Compute the spectrum of u .
 - c) The finite spectrum of u is empty.
- 5) Let s be the endomorphism of $\ell^2(\mathbf{N})$ such that

$$s(a_0, a_1, \dots, a_n, \dots) = (a_1, \dots, a_n, \dots).$$

Show that the numerical range of s is the open unit disk Δ in $\mathbf{C}$, and that the spectrum of s is the closed unit disk in $\mathbf{C}$. Determine the eigenvalues of s . Do the same for the adjoint s^* of s .

- 6) Let $u \in \mathcal{L}(\mathbf{C}^n)$ be a trace-zero endomorphism. There exists an orthonormal basis B of $\mathbf{C}^n$ such that all diagonal entries of the matrix representing u in the basis B are zero. (Prove that 0 belongs to the numerical range of u .)
- 7) Let u be an endomorphism of a complex Hilbert space E such that the spectrum of u is

contained in the open unit disk in $\mathbb{C}$. Let F be the Hilbert space $\ell_E^2(\mathbb{N})$ of square-summable sequences of elements of E (EVT, V, p. 18, example), and let s_F be the endomorphism of F such that

$$s_F(a_0, a_1, \dots, a_n, \dots) = (a_1, \dots, a_n, \dots).$$

There exists a closed subspace H of F and a continuous bijective linear map $v : E \rightarrow H$ such that $u = v^{-1}s_F v$.

- 8) Let u be an endomorphism of a complex Hilbert space E . Let S_u be the set of endomorphisms of E of the form wuw^{-1} , where w ranges over the set of invertible endomorphisms of E .

a) We have

$$\varrho(u) = \inf_{v \in S_u} \|v\|.$$

(One may assume that $\varrho(u) \leq 1$; apply the preceding exercise to the endomorphisms $u_\varepsilon = (1 + \varepsilon)^{-1}u$ for $\varepsilon > 0$.)

- b) The convex hull of the spectrum of u is equal to the intersection of the subsets $\iota(v)$, for $v \in S_u$ ("Hildebrandt's theorem").

¶ 9) Let E be a complex Banach space.

- a) Let $u \in \mathcal{L}(E)$. Consider the endomorphisms γ_u and δ_u of $\mathcal{L}(E)$ defined by $\gamma_u(v) = u \circ v$ and $\delta_u(v) = v \circ u$. We have $\text{Sp}(u) = \text{Sp}(\gamma_u) = \text{Sp}(\delta_u)$, where the last two spectra are relative to the algebra $\mathcal{L}(\mathcal{L}(E))$.
- b) For every function f holomorphic in a neighborhood of the spectrum of u , one has $\gamma_{f(u)} = f(\gamma_u)$ and $\delta_{f(u)} = f(\delta_u)$.
- c) Let u_1 and u_2 be in $\mathcal{L}(E)$. Let ϱ be the endomorphism $v \mapsto u_1 \circ v \circ u_2$ of $\mathcal{L}(E)$. Its spectrum is equal to $\text{Sp}(u_1) \text{Sp}(u_2) \subset \mathbb{C}$.

d) Let u_1 and u_2 be in $\mathcal{L}(E)$. Let ϱ be the endomorphism $v \mapsto u_1 \circ v \circ u_2$ of the space $\mathcal{L}_2(E)$ of Hilbert–Schmidt maps from E to E . Its spectrum is equal to $\text{Sp}(u_1)\text{Sp}(u_2) \subset \mathbf{C}$. (Apply a similar method.)

10) Let E be a complex Hilbert space. Let x_1 and x_2 be endomorphisms of E . Show that the spectrum of the endomorphism $x_1 \otimes x_2$ of $E \widehat{\otimes}_2 E$ is equal to $\text{Sp}(x_1)\text{Sp}(x_2)$. (Identify $\overline{E} \widehat{\otimes}_2 E$ with the space of Hilbert–Schmidt operators from E to E , cf. EVT 5, p. 52, th. 1, b), and apply the preceding exercise.)

11) Let u be an endomorphism of a complex Hilbert space E and let $(j, |u|)$ be its polar decomposition.

a) For j to commute with $|u|$, it is necessary and sufficient that u commute with u^*u .

b) Prove that one then has $\|u^*(x)\| \leq \|u(x)\|$ for all $x \in E$, and that $\varrho(u) = \|u\|$.

c) Let $E = \ell^2(\mathbf{N})$ and let u be the endomorphism of E given by

$$u((a_0, a_1, \dots)) = (0, a_0, a_1, \dots).$$

It is not normal. Compute the polar decomposition $(j, |u|)$ of u ; j commutes with $|u|$.

12) Let u be an endomorphism of a Banach space E . We denote by $\sigma_a(u)$ the set of $\lambda \in \mathbf{C}$ such that, for every $\alpha \in \mathbf{R}_+^*$, there exists $x \in E$ such that $\|\lambda x - u(x)\| < \alpha\|x\|$.

a) Prove that $\sigma_a(u)$ is a closed subset of $\sigma(u)$.

b) Prove that $\sigma_a(u)$ contains the boundary of $\sigma(u)$.

13) Let E be a complex Hilbert space. Let N be a map from $\mathcal{L}(E)$ into $\mathfrak{P}(\mathbf{C})$ such that

a) $N(u)$ is compact for every $u \in \mathcal{L}(E)$;

b) $N(\alpha u + \beta 1_E) = \alpha N(u) + \beta$ for all $u \in \mathcal{L}(E)$ and every $(\alpha, \beta) \in \mathbf{C}^2$;

c) $N(u) \subset \{z \in \mathbf{C} \mid \Re(z) \geq 0\}$ if and only if $u + u^*$ is positive.

Then $N(u) = \overline{\iota(u)}$ for all $u \in \mathcal{L}(E)$. (Show that $u \mapsto \overline{\iota(u)}$ has the indicated properties; let N_1 and N_2 be maps satisfying them; assuming that $N_1(u)$ is not contained in $N_2(u)$, find α and β in $\mathbb{C}$ such that $N_2(\alpha u + \beta)$ is contained in the half-plane $\Re(z) \geq 0$ but $N_1(\alpha u + \beta)$ is not, and conclude a contradiction using c).)

14) Give an example of a complex Hilbert space and of $u \in \mathcal{L}(E)$ such that $\iota(u)$ is not closed.

15) The set X of nonempty compact subsets of $\mathbb{C}$ is equipped with the distance defined in TG, IX, p. 91, exerc. 6. Let E be a complex Hilbert space.

a) The map $u \mapsto \overline{\iota(u)}$ from $\mathcal{L}(E)$ to X is continuous. If E is infinite-dimensional, it is not continuous when $\mathcal{L}(E)$ is equipped with the topology of simple convergence.

b) The map $u \mapsto \text{Sp}(u)$ from $\mathcal{L}(E)$ to X is not continuous.

16) Let X be a locally compact space, μ a positive measure of total mass 1 on X , and $f : X \rightarrow X$ a map such that $f(\mu) = \mu$.

a) The map $\varphi \mapsto \varphi \circ f$ induces a unitary endomorphism u_f of $L^2(X, \mu)$.

b) The real number 1 is an eigenvalue of u_f of multiplicity 1 if and only if, for every measurable subset A of X such that $f(A) = A$, one has $\mu(A) \in \{0, 1\}$. One then says that f is μ -ergodic.

c) Let $E \subset L^2(X, \mu)$ be the eigenspace of u_f for the eigenvalue 1. The sequence $(v_N)_{N \in \mathbb{N}}$ defined by

$$v_N = \frac{1}{N} \sum_{n=0}^{N-1} u_f^n$$

converges to the orthogonal projection onto E in the space $\mathcal{L}(L^2(X, \mu))$ equipped with the topology of simple convergence ("von Neumann's ergodic theorem"). (Note that the orthogonal complement of E is the closure of the image of $u_f - 1_E$; then verify that $v_N(\varphi) \rightarrow 0$ when $\varphi \in E^\circ$.)

d) One defines an endomorphism $\tilde{u}_f$ of $L^1(X, \mu)$ as in part a), and we set

$$\tilde{v}_N = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{u}_f^n.$$

For every $\varphi \in L^1(X, \mu)$ the sequence $(\tilde{v}_N(\varphi))_{N \in \mathbb{N}}$ converges in $L^1(X, \mu)$ to an element $\tilde{\varphi}$ of the eigenspace of $\tilde{u}_f$ for the eigenvalue 1. (First consider the case where φ is bounded, and apply the previous question.)

e) Let I be a finite set and $X = I^{\mathbb{Z}}$ endowed with the product measure of the measures μ_n on I such that $\mu_n(i) = 1/\text{Card}(I)$ for all $i \in I$. Set $f((i_n)) = (i_{n+1})$ for all $(i_n) \in X$. Then $f(\mu) = \mu$ and f is μ -ergodic (“Bernoulli shift”).

f) Let $X = \mathbf{R}/\mathbf{Z}$ and μ the normalized Haar measure on X . Let $a \in X$ and set $f(x) = x + a$. Then $f(\mu) = \mu$ and f is μ -ergodic if and only if a is the class of an irrational number.

Exercises for Section 8: Algebras of Continuous Functions on a Compact Space

All Banach spaces and all Banach algebras below are complex.

- 1) Let X be a topological space, let $\mathcal{C}_b(X)$ be the stellar algebra of bounded continuous complex functions on X , and let $\widehat{X} = X(\mathcal{C}_b(X))$.
 - a) Let A be a unital Banach subalgebra of $\mathcal{C}_b(X)$ and let R_A be the equivalence relation on X such that x is equivalent to x' if $f(x) = f(x')$ for every function $f \in A$. The R_A -saturation S of a compact subset K of X is closed. (Let $x \in \overline{S}$; for all $f \in A$ and every $\varepsilon > 0$, let $V_{f,\varepsilon}$ be the set of $x' \in X$ such that $|f(x) - f(x')| \leq \varepsilon$; if $f_1, \dots, f_n \in A$ and $\varepsilon_1, \dots, \varepsilon_n > 0$, one has

$$V_{f_1, \varepsilon_1} \cap \dots \cap V_{f_n, \varepsilon_n} \cap K \neq \emptyset,$$

hence

$$\bigcap_{\substack{f \in A \\ \varepsilon > 0}} V_{f, \varepsilon} \cap K \neq \emptyset$$

whence $x \in S$.)

- b) Assume X is normal. Let R be a closed equivalence relation on X and let $A = A_R$ be the subalgebra of $\mathcal{C}_b(X)$ formed by the functions constant on the classes modulo R . Show that $R = R_A$ (Use TG, IX, p. 105, exerc. 19, a).
- c) Assume X is compact. $R \mapsto A_R$ is a bijection from the set of separated equivalence relations on X onto the set of stellar subalgebras of $\mathcal{C}(X)$ (Let A be a stellar subalgebra of $\mathcal{C}(X)$. Show that R_A is separated, and apply the Weierstrass–Stone theorem to the algebra of continuous functions on X/R_A obtained from A by passing to the quotient.)
- 2) Let X be a compact topological space, R a separated equivalence relation on X , and B a closed subalgebra of $\mathcal{C}(X)$ containing A_R (notation of Exercise 1). If $f \in \mathcal{C}(X)$ coincides on each R -class c with an element g_c of B , then $f \in B$. (Let $\varepsilon > 0$. There exists a saturated open neighborhood V_c of c such that $|f - g_c| \leq \varepsilon$ on V_c . Let $V_{c_1}, \dots, V_{c_n}$ be a cover of X . Let $(u_1, \dots, u_n)$ be a partition of unity subordinate to $(V_{c_1}, \dots, V_{c_n})$ and consisting of functions of A_R . Then

$$g = u_1 g_{c_1} + \dots + u_n g_{c_n} \in B,$$

and $|f - g| \leq \varepsilon$.)

- 3) Let X be a compact topological space and, for every $x \in X$, let A_x be a commutative unital Banach algebra, with e_x denoting the identity element. We assume $\|e_x\| = 1$. Let B be the product Banach algebra of the A_x for $x \in X$. Let C be a Banach subalgebra of B having the following properties:

- (i) $e = (e_x)_{x \in X} \in C$;
- (ii) If $a = (a_x) \in C$ and $f \in \mathcal{C}(X)$, the function $fa : x \mapsto f(x)a_x$ is an element of C ;

(iii) For every $a = (a_x) \in C$, the function $x \mapsto \|a_x\|$ is upper semicontinuous.

Let $\widehat{X}$ be the disjoint union of the $X(A_x)$. For every $x_0 \in X$ and $\chi \in X(A_{x_0})$, let $\varphi(\chi)$ be the character of C defined by $\varphi(\chi)((a_x)_{x \in X}) = \chi(a_{x_0})$. Prove that φ defines a bijection from $\widehat{X}$ onto $X(C)$. (Let $\xi \in X(C)$. Since $\mathcal{C}(X)$ can be identified with a closed subalgebra of C , ξ defines a character of $\mathcal{C}(X)$, hence there exists $x_0 \in X$. Let $a = (a_x) \in C$ and $b = (b_x) \in C$, with $a_{x_0} = b_{x_0}$; let $\varepsilon > 0$; we have $\|a_x - b_x\| < \varepsilon$ on a neighborhood U of x_0 ; let $f \in \mathcal{C}(X)$ be such that $0 \leq f \leq 1$, equal to 1 at x_0 and to 0 outside U ; we have $\|fa - fb\| \leq \varepsilon$, whence $|\xi(fa - fb)| \leq \varepsilon$, whence $|\xi(a) - \xi(b)| \leq \varepsilon$, whence $\xi(a) = \xi(b)$. Deduce that there exists a $\chi \in X(A_{x_0})$ such that $\xi = \varphi(\chi)$.

4) Let X be a compact topological space such that the Banach algebra $\mathcal{C}(X)$ is generated by a sequence (j_n) of idempotents. The function

$$\omega \mapsto \sum_{n=1}^{\infty} \frac{1}{3^n} (2j_n(\omega) - 1)$$

separates the points of X . It generates the Banach algebra $\mathcal{C}(X)$.

5) Let A be the Banach algebra of functions from $[0,1]$ to C admitting continuous derivatives on $[0,1]$ up to order n (example 4 of I, p. 18). Let $\omega \in [0,1] = X(A)$.

a) The smallest closed ideal I of A such that $V(I) = \{\omega\}$ is the set of $f \in A$ such that $f, f', \dots, f^{(n)}$ vanish at ω . (Use prop. 5 of I, p. 94.)

b) If $g \in A$, the norm of its canonical image in A/I is equal to

$$\sum_{k=0}^n \frac{|g^{(k)}(\omega)|}{k!}.$$

c) The algebra A/I is isomorphic to $C[X]/(X^{n+1})$; the only maximal ideal of A containing I is the set of $f \in A$ such that $f(\omega) = 0$.

6) Let Δ be the disk $|\zeta| \leq 1$ in C and A the Banach algebra of continuous complex-valued functions on Δ that are analytic in the interior of Δ (example 9 of I, p. 20).

a) If $x \in A$, then x is the limit in A of the functions

$$\zeta \mapsto x_n(\zeta) = x\left(\frac{n\zeta}{n+1}\right).$$

The identity mapping of Δ is a generator of A .

b) The space $X(A)$ is canonically identified with Δ (Use assertion g) of prop. 1 of I, p. 142.)

c) Let S be a closed subset of Δ , identified with a closed subset of $X(A)$. If S has a non-isolated point ζ such that $|\zeta| < 1$, the closure of S for the Jacobson topology is Δ .

d) For $x \in A$, set $x^*(\zeta) = x(\overline{\zeta})$. Then $x \mapsto x^*$ is an isometric involution of A . The Hermitian characters of A are given by the real elements of Δ .

e) The map $x \mapsto x|U$ is an isomorphism from A onto $P(U)$. Every function in $\mathcal{C}_R(U)$ is the uniform limit of real parts of functions in $P(U)$.

7) Let A_1 (resp. A_2) be the unital Banach algebra considered in Exercise 5 (resp. 6). Let $A = A_1 \times A_2$. Show that A is semisimple, but that the image of the Gelfand transform of A is neither closed nor dense in $\mathcal{C}(X(A))$.

¶ 8) Let X be a compact topological space and B a unital Banach subalgebra of $\mathcal{C}(X)$ (with the induced norm), separating the points of X . One identifies X with a closed subset of $X(B) = X'$. Let B^* be the set of invertible elements of B . One says that B is logmodular if the set of functions $\log(|f|)$, for $f \in B^*$, is dense in $\mathcal{C}_R(X)$. (This is the case, in particular, if the set of functions $\mathcal{R}(f)$, for $f \in B$, is dense in $\mathcal{C}_R(X)$, because $\log(|e^f|) = \mathcal{R}(f)$.) In what follows, one assumes that B is logmodular.

a) For every $\chi \in X'$, there exists a unique measure $\mu_\chi \geq 0$ on X such that

$$\log(|\chi(f)|) = \int_X \log(|f(\omega)|) d\mu_\chi(\omega)$$

for all $f \in B^*$, hence such that $\chi(f) = \int_X f(\omega) d\mu_\chi(\omega)$ for all $f \in B$. b) One has

$$\log|\chi(f)| \leq \int_X \log|f(\omega)| d\mu_\chi(\omega)$$

for all $f \in B$ and the condition $\chi(f) = \int_X f(\omega) d\mu_\chi(\omega)$ for all $f \in B$ defines μ_χ uniquely.

- c) Let $\chi \in X'$ and μ be a positive measure on X . Write $\mu = \mu_1 + \mu_2$, where μ_1 is absolutely continuous with respect to μ_χ , and where μ_2 and μ_χ are mutually singular. Let I be the kernel of χ . Then

$$\inf_{f \in I} \int_X |1 - f|^2 d\mu = \inf_{f \in I} \int_X |1 - f|^2 d\mu_1.$$

- d) Let $\chi \in X'$ and let h be a nonnegative function in $\mathcal{L}^1(\mu_\chi)$. Let I be the kernel of χ . Then

$$\inf_{f \in I} \int_X |1 - f|^2 h d\mu_\chi = \exp\left(\int_X \log(h) d\mu_\chi\right).$$

- e) For every positive measure μ on X , and every $p \in [1, +\infty[$, let $H^p(\mu)$ be the closure of the canonical image of B in $L^p(\mu)$. Let $\chi \in X'$ and let I be the kernel of χ . Then $L^2(\mu_\chi)$ is the Hilbert sum of $H^2(\mu_\chi)$ and of the closure in $L^2(\mu_\chi)$ of the canonical image of $\bar{I}$ (the set of conjugates of functions belonging to I).
- f) The space $H^1(\mu_\chi)$ is the set of classes $\tilde{h}$ of $h \in \mathcal{L}^1(\mu_\chi)$ such that $\int_X f h d\mu_\chi = 0$ for all $f \in I$.

- g) Let h be a nonnegative function in $\mathcal{L}^1(\mu_\chi)$. Then we have $\tilde{h} = |\tilde{f}|$ with $\tilde{f} \in H^1(\mu_\chi)$ and $\int_X f d\mu_\chi \neq 0$ if and only if $\log(h) \in \mathcal{L}^1(\mu_\chi)$.

¶ 9) Let X be a compact topological space and B a unital Banach subalgebra of $\mathcal{C}(X)$ (with the induced norm). Every element x of X defines a character χ_x of B . If x and x' are in X , we write $x \sim x'$ when $\|\chi_x - \chi_{x'}\| \neq 2$. a) Let x and x' be in X . If there exists a sequence (f_n) of elements of B such that $\|f_n\| \leq 1$, $|f_n(x)|$ tends to 1, and $\liminf_{n \rightarrow \infty} |f_n(x) - f_n(x')| > 0$, then $x \not\sim x'$. (One may assume that $f_n(x)$ tends to 1. Consider $g_n = (f_n - \lambda_n)(1 - \lambda_n f_n)^{-1}$, where (λ_n) is a sequence of numbers in $[0, 1[$ tending toward 1. We have $\|g_n\| \leq 1$. If the sequence (λ_n) is well chosen, $g_n(x)$ tends to 1 and $g_n(x')$ tends to -1.)

- b) Let R be the set of real parts of the elements of B . If $x \sim x'$, there exists $c \in]0, 1[$ such that $f(x) \geq cf(x')$ and $f(x') \geq cf(x)$ for every $f \geq 0$ in R . (It suffices to obtain the

first condition; if it is not true, there exists a sequence (f_n) in R with $f_n \geq 0$, $f_n(x') = 1$, and $f_n(x)$ tending to 0; let $g_n \in B$ with $\mathcal{R}(g_n) = f_n$. Then $e^{-g_n} \in B$, $\|e^{-g_n}\| \leq 1$, $|e^{-g_n(x')}| = 1/e$, and $|e^{-g_n(x)}|$ tends to 1, which contradicts a).)

- c) There exist positive measures α and β on X such that $f(x) - cf(x') = \alpha(f)$ and $f(x') - cf(x) = \beta(f)$ for $f \in R$. Setting $\mu_x = (1 - c^2)^{-1}(c\beta + \alpha)$, $\mu_{x'} = (1 - c^2)^{-1}(c\alpha + \beta)$, one has $\mu_x(f) = f(x)$ and $\mu_{x'}(f) = f(x')$ for all $f \in B$, and $c\mu_x \leq \mu_{x'}$, $c\mu_{x'} \leq \mu_x$.
- d) We still assume $x \sim x'$. Show that if μ is a positive measure on X such that $\mu(f) = f(x)$ for all $f \in B$, there exists a positive measure μ' on X such that $\mu'(f) = f(x')$ for all $f \in B$ and $c\mu \leq \mu'$ for some $c \in]0, 1[$. (With the notations of c), set $\mu' = \mu_{x'} - c\mu_x + c\mu$.)
- e) The relation $x \sim x'$ is an equivalence relation on X . (Use d).)

¶ 10) Let X be a compact topological space and A a unital Banach subalgebra of $\mathcal{C}(X)$ for the induced norm.

- a) A subset K of X is said to be antisymmetric (relative to A) if every function $f \in A$ which is real on K is constant on K . Every antisymmetric subset is contained in a maximal antisymmetric subset. A maximal antisymmetric subset is closed. The set $\mathfrak{K}$ of maximal antisymmetric subsets is a partition of X .
- b) Let $A^\perp$ be the orthogonal of A in the Banach space of complex measures on X . Let μ be an extreme point of the unit ball of $A^\perp$. Show that the support of μ is antisymmetric.
- c) Let $f \in \mathcal{C}(X)$. If $f|_K \in A|_K$ for all $K \in \mathfrak{K}$, then $f \in A$. (Apply b) and the Krein–Milman theorem.) From this, recover the Weierstrass–Stone theorem (TG, X, p. 36, th. 3).
- d) A nonempty subset E of X is called a peak (relative to A) if there exists $f \in A$ such that $\|f\| = 1$ and E is the set of those $x \in X$ for which $f(x) = 1$. One may then assume $|f(x)| < 1$ for $x \notin E$ (replace f by $\frac{1}{2}(1 + f)$). Every nonempty countable intersection of peaks is a peak. If E_1, E_2 are peaks, then $E_1 \cup E_2$ is a peak. (Let $f_1, f_2 \in A$ with $f_i = 1$ on E_i , $|f_i| < 1$ on $X - E_i$; let $g_i = \frac{1}{4}(1 - f_i)^{1/3} \in A$; then $1 - g_1g_2 = 1$ on $E_1 \cup E_2$, $|1 - g_1g_2| < 1$ on $X - (E_1 \cup E_2)$.)
- e) Let E be an intersection of peaks. Let I be the set of $f \in A$ vanishing on E . Then $f \mapsto f|_E$

defines an isometry from A/I onto the algebra $A|E$ of restrictions to E of functions in A .

- f) Let E be a peak. Let $E' \subset E$ be a peak relative to $A|E$. Then E' is a peak relative to A .
- g) Every antisymmetric subset of X is an intersection of peaks. (Use d) and f).)
- h) For every antisymmetric subset K of X , the algebra $A|K$ is a closed subalgebra of $\mathcal{C}(K)$. (Use e and g.)
- i) Let R be the set of real parts of elements of A . Suppose that $X \in \mathfrak{R}$ and that R is stable under multiplication. Then X consists of a single point. (Let $x_0 \in X$. For $u \in R$, there exists a unique $f \in A$ such that $u = \mathcal{R}(f)$ and $f(x_0) \in \mathbf{R}$; set $N(u) = \|f\|$. Then R is a real Banach space for N ; the closed graph theorem proves that multiplication in R is separately continuous, hence continuous. Deduce from this a Banach algebra structure on $S = R + iR$. Let $p \in R$ with $p(x) > 0$ for all $x \in X$. Show that, for every character χ of S , one has $\mathcal{R}(\chi(p)) > 0$, and deduce that $\log(p) \in R$. Assuming $X \neq \{x_0\}$, one can construct an $f \in A$ such that $0 < \mathcal{R}(f) \leq 1$ on X , $f(x_0) \in \mathbf{R}$, and such that $\|f\|$ is arbitrarily large. According to what precedes, there exists $V \in A$ with $|V|^2 = \mathcal{R}(f)$. We have $\|V\| \leq 1$, hence $N(\mathcal{R}(V)) \leq 2$, but

$$N(\mathcal{R}(V)^2) \geq \frac{1}{2} (\|V\|^2 + \|f\| - |\mathcal{I}(V)(x_0)|^2)$$

is arbitrarily large.)

- j) If R is stable under multiplication, then $A = C(X)$. (Use a), c), h), i).) Consequently, if $A \neq C(X)$, there exists $u \in R$ such that $u^2 \notin R$.
- 11) Let X be a metric space, whose distance is denoted by d . A function $f : X \rightarrow \mathbf{C}$ is said to be Lipschitz if there exists a real number $c \geq 0$ such that, for all x and y in X , one has

$$|f(x) - f(y)| \leq cd(x, y)$$

(cf. FVR, IV, p. 7, def. 1). We denote by $\text{Lip}_b(X)$ the set of bounded Lipschitz functions on X .

a) The set $\text{Lip}_b(X)$ is a subalgebra of $\mathcal{C}(X)$.

b) Equipped with the norm

$$\|f\| = \sup_{x \in X} |f(x)| + \sup_{\substack{x, y \in X \\ x \neq y}} \frac{|f(x) - f(y)|}{d(x, y)},$$

and with the involution $f \mapsto \bar{f}$, the algebra $\text{Lip}_b(X)$ is a commutative involutive Banach algebra. It is not a stellar algebra in general.

c) For any commutative Banach algebra A , we denote by $X_m(A)$ the metric space $X(A)$ endowed with the distance induced by the norm of the dual A' of A . The metric space $X_m(A)$ is complete and of diameter ≤ 2 .

d) Let A be a unital commutative Banach algebra. The Gelfand transform maps A into $\text{Lip}_b(X_m(A))$. If A is semisimple, then it induces a homeomorphism from A onto its image.

In the sequel, we assume that X is a compact metric space. We denote by $A = \text{Lip}_b(X)$.

e) The map φ which sends $x \in X$ to the character $f \mapsto f(x)$ is a homeomorphism from X onto $X(A)$.

f) Let d_1 be the distance on X induced by the homeomorphism φ . Then d_1 is equivalent to d .

Chapter 2

Locally compact abelian groups

Throughout this chapter, the letter G denotes, unless otherwise stated, a locally compact abelian group equipped with a Haar measure generally denoted by dx ; for $p \in [1, +\infty]$, the space $L^p(G, dx)$ will simply be denoted $L^p(G)$, and its norm will be denoted by $f \mapsto \|f\|_p$. We identify $L^1(G)$ with a subspace of $\mathcal{M}^1(G)$ by the map $f \mapsto f \cdot dx$. Recall that the support of the Haar measure is equal to G (INT, VII, §1, no. 1, remark 3); in particular (INT, III, §2, no. 2, proposition 9), the canonical mapping from the space $\mathcal{K}(G)$ into $L^p(G)$ is injective for $p \in [1, +\infty]$. For $p \neq +\infty$, we identify $L^p(G)$ with a subspace of the space $\widetilde{\mathcal{F}}(G; \mathbb{C})$, the space of classes of complex-valued functions on G defined and finite almost everywhere (INT, IV, §3, nos. 5, 6). In particular, the notation $L^1(G) \cap L^2(G)$ denotes the intersection of $L^1(G)$ and $L^2(G)$ in this space. We denote by $f \mapsto \tilde{f}$ the involution on the involutive algebra $L^1(G)$ (example 4 of I, p. 99); we have $\tilde{f}(x) = \overline{f(x^{-1})}$ for all x in G .

Recall (INT, V, §5, no. 3, th. 1) that if μ is a complex measure on a locally compact topological space X and if f is a locally μ -integrable function on X , then the measure ν with density f with respect to μ is denoted $f \cdot \mu$ or simply $f\mu$. A function g from X to $\mathbb{C}$ is essentially integrable for the measure ν if and only if gf is essentially integrable for μ ; we then have

$$(f \cdot \mu)(g) = \int_X g d(f \cdot \mu) = \int_X gf d\mu = \mu(gf).$$

If G is a compact topological group, the normalized Haar measure on G is the unique Haar measure μ on G such that $\mu(G) = 1$.

If X is a discrete topological space, the counting measure on X is the discrete measure μ on X such that $\mu(\{x\}) = 1$ for all $x \in X$. If G is a discrete topological group, the counting measure on G is a Haar measure on G .

We will also make use of the following two lemmas.

Lemma 1. — Let X be a locally compact topological space and μ a positive measure on X . Let x be an element of the support of μ . Let U be an open neighborhood of x . There exists a function $f \in \mathcal{K}_+(X)$ whose support is contained in U such that $\int f d\mu = 1$.

By definition of the support of a measure (INT, III, §2, no. 2, def. 1), there exists a function $g \in \mathcal{K}(X)$ supported in U such that $\mu(g) \neq 0$. We then have $\mu(|g|) > 0$ since μ is positive, and the function $f = \mu(|g|)^{-1}|g|$ has the desired properties.

Lemma 2. — Let G and H be topological groups with identity elements e_G and e_H . Suppose the topological group G is Hausdorff. Let f be a morphism of topological groups from G to H . Suppose that for every neighborhood U of e_G in G , there exists a neighborhood W of e_H in H such that $f^{-1}(W) \subset U$. Then the morphism f is injective and strict (TG, III, p. 16, def. 1).

The hypotheses imply that $\text{Ker}(f)$ is contained in every neighborhood of e_G in G , hence that $\text{Ker}(f) = \{e_G\}$ since G is separated. The homomorphism from G to $f(G)$ induced by f by passing to subspaces is then bijective; denote by $g : f(G) \rightarrow G$ the inverse homomorphism. The hypotheses then imply that g is continuous at e_H , therefore continuous (TG, III, p. 15, prop. 23).

2.1 FOURIER TRANSFORMATION

2.1.1 Unitary characters of a locally compact abelian group

DEFINITION 1. — A unitary character of G is a continuous homomorphism from G into the multiplicative group U of complex numbers of modulus 1.

In other words, a unitary character is a continuous function χ on G , with complex values, such that:

$$\chi(xy) = \chi(x)\chi(y), \quad |\chi(x)| = 1 \quad (x, y \in G).$$

In this chapter, we will often simply say “character” instead of “unitary character.”

Let E be a Hilbert space of dimension 1, and let χ be a unitary character of G . The map that assigns to $x \in G$ the homothety of ratio $\chi(x)$ in E is a continuous linear isometric representation of G on E . Conversely, every bounded continuous linear representation of G on E is obtained by this process, and in particular is unitary.

It is immediate that the product of two unitary characters, the inverse of a unitary character, and the constant function equal to 1 are unitary characters. Consequently, the set $\widehat{G}$ of unitary characters of G forms a group under multiplication. This group is commutative. On the other hand, the map $(\chi_1, \chi_2) \mapsto \chi_1 \chi_2^{-1} = \chi_1 \overline{\chi_2}$ is continuous for the compact-open topology, and $\widehat{G}$, endowed with the topology of compact convergence, is a topological group (TG, X, p. 6, corollary 2 and remark 1).

DEFINITION 2. — The topological group $\widehat{G}$ is called the dual group of G .

Since G is locally compact, the map $(x, \chi) \mapsto \chi(x)$ is continuous on $G \times \widehat{G}$ (TG, X, p. 28, th. 3).

Recall that $\mathcal{M}^1(G)$ denotes the unital involutive Banach algebra of bounded complex measures on G (Example 4 of I, p. 99). For every bounded complex measure $\mu \in \mathcal{M}^1(G)$ and every $\chi \in \widehat{G}$, we denote by

$$\chi(\mu) = \int_G \chi(x) d\mu(x) \quad (1)$$

(cf. INT, VIII, §2, no. 6).

Lemma 1. — For every $\chi \in \widehat{G}$, the map $\mu \mapsto \chi(\mu)$ is a Hermitian character of the involutive Banach algebra $\mathcal{M}^1(G)$.

By INT, VIII, §3, no. 3, prop. 11, the map $\mu \mapsto \chi(\mu)$ is a character of the involutive Banach algebra $\mathcal{M}^1(G)$. Moreover, we have:

$$\chi(\mu^*) = \int_G \chi(x^{-1}) d\overline{\mu}(x) = \int_G \overline{\chi(x)} d\overline{\mu}(x) = \overline{\chi(\mu)}$$

and this character is therefore Hermitian.

Thus we have defined a mapping from $\widehat{G}$ into $X(\mathcal{M}^1(G))$; it will be called canonical.

Let $\chi \in \widehat{\mathbf{G}}$. The restriction of $\mu \mapsto \chi(\mu)$ to $L^1(\mathbf{G})$ is nonzero (cf. INT, VIII, §2, no. 7, prop. 10). By restriction to the involutive Banach subalgebra $L^1(\mathbf{G})$, we therefore obtain a map from $\widehat{\mathbf{G}}$ into $X(L^1(\mathbf{G}))$, said to be canonical. It associates to $\chi \in \widehat{\mathbf{G}}$ the Hermitian character

$$f \mapsto \chi(f) = \chi(f \cdot dx) = \int_{\mathbf{G}} f(x) \chi(x) dx \quad (f \in L^1(\mathbf{G})) \quad (2)$$

of $L^1(\mathbf{G})$.

PROPOSITION 1. — The canonical map from $\widehat{\mathbf{G}}$ into $X(L^1(\mathbf{G}))$ is a homeomorphism.

Let us denote this canonical map by ev and, for $\chi \in \widehat{\mathbf{G}}$, let ev_χ be the image of the character χ by ev , that is, the Hermitian character $f \mapsto \chi(f)$ of $L^1(\mathbf{G})$. Considered as a mapping from $\widehat{\mathbf{G}}$ into the dual of $L^1(\mathbf{G})$, the map ev is the composition of the injection of $\widehat{\mathbf{G}}$ into $L^\infty(\mathbf{G})$, equipped with the weak topology $\sigma(L^\infty(\mathbf{G}), L^1(\mathbf{G}))$, and the injection of $L^\infty(\mathbf{G})$ into the dual of $L^1(\mathbf{G})$, equipped with the topology of pointwise convergence. Since $\widehat{\mathbf{G}}$ is a bounded subset of $L^\infty(\mathbf{G})$, the first map is continuous by Lebesgue's theorem (INT, IV, §3, no. 7, th. 6). The second is also continuous, by definition. This proves that the map ev is continuous.

If $\chi \in \widehat{\mathbf{G}}$ and if $f \in L^1(\mathbf{G})$, one has $\text{ev}_\chi(\varepsilon_x * f) = \chi(x) \text{ev}_\chi(f)$. Taking f such that $\text{ev}_\chi(f) \neq 0$, we deduce that the map ev is injective.

Let $\zeta \in X(L^1(\mathbf{G}))$ and let $f \in L^1(\mathbf{G})$ such that $\zeta(f) \neq 0$. Let us define a map $\chi : \mathbf{G} \rightarrow \mathbb{C}$ by setting, for $x \in \mathbf{G}$:

$$\chi(x) = \frac{\zeta(\varepsilon_x * f)}{\zeta(f)}. \quad (3)$$

We have $\chi(e) = 1$. Since the map $x \mapsto \varepsilon_x * f = \gamma(x)(f)$ from $\mathbf{G}$ to $L^1(\mathbf{G})$ is continuous (INT, VIII, §2, n° 5, prop. 8), the map χ is continuous. It is bounded because, for all $x \in \mathbf{G}$, one has

$$|\chi(x)| \leq \frac{\|\varepsilon_x * f\|}{|\zeta(f)|} = \frac{\|f\|}{|\zeta(f)|}$$

(th. 1 of I, p. 29 and INT, VIII, loc. cit.).

Now let $\mathfrak{B}$ be a base of the filter of neighborhoods of e consisting of compact neighborhoods. For every $V \in \mathfrak{B}$, let g_V be a positive continuous function, vanishing outside V and with integral

equal to 1 (lemma 1 of II, p. 200). For every function $h \in L^1(G)$, we then have

$$\varepsilon_x * h = \lim_{V, \mathfrak{B}} (\varepsilon_x * h) * g_V = \lim_{V, \mathfrak{B}} (\varepsilon_x * g_V * h),$$

in $L^1(G)$, the limit being taken along the filter of sections of $\mathfrak{B}$ (INT, VIII, §4, n° 7, prop. 20). In particular, since $\zeta(\varepsilon_x * g_V * f) = \zeta(\varepsilon_x * g_V) \zeta(f)$, one deduces that

$$\chi(x) = \lim_{V, \mathfrak{B}} \zeta(\varepsilon_x * g_V).$$

For every $h \in L^1(G)$, we obtain

$$\zeta(\varepsilon_x * h) = \lim_{V, \mathfrak{B}} \zeta(\varepsilon_x * g_V * h) = \zeta(h) \lim_{V, \mathfrak{B}} \zeta(\varepsilon_x * g_V) = \chi(x) \zeta(h).$$

Consequently, we have, for $x, y \in G$:

$$\chi(xy) = \frac{\zeta(\varepsilon_x * \varepsilon_y * f)}{\zeta(f)} = \frac{\chi(x) \zeta(\varepsilon_y * f)}{\zeta(f)} = \chi(x) \chi(y),$$

which proves that χ is a homomorphism from G to $\mathbb{C}^*$. Since χ is bounded and continuous, it is a unitary character of G . Moreover, if $g \in L^1(G)$, one has

$$g * f = \int_G (\varepsilon_x * f) g(x) dx$$

in $L^1(G)$ (INT, VIII, §1, n° 5, prop. 7), whence

$$\begin{aligned} \zeta(g) \zeta(f) &= \zeta(g * f) = \int_G \zeta(\varepsilon_x * f) g(x) dx \\ &= \zeta(f) \int_G \chi(x) g(x) dx = \text{ev}_\chi(g) \zeta(f) \end{aligned}$$

(INT, VI, §1, n°1, prop. 1), which shows that $\zeta = \text{ev}_\chi$. Consequently, ev is surjective, hence bijective.

Finally let us show that the inverse map ev^{-1} is continuous. Let $\zeta \in X(L^1(G))$. Let $f \in L^1(G)$ be a function such that $\zeta(f) \neq 0$. The set W of $\xi \in X(L^1(G))$ such that $\xi(f) \neq 0$ is an open neighborhood of ζ in $X(L^1(G))$. For every $\xi \in W$, what precedes shows that $\text{ev}^{-1}(\xi)$ is the character

$$x \mapsto \frac{\xi(\varepsilon_x * f)}{\xi(f)}.$$

Let $\mathfrak{F}$ be a filter on $W \subset X(L^1(G))$ converging to ζ . Since the set $X(L^1(G))$ is bounded, hence equicontinuous, in $L^\infty(G)$, the uniform structure of simple convergence coincides with the uniform structure of compact convergence (TG, X, p. 16, th. 1). Let K be a compact subset of G . The set of $\varepsilon_x * f$ for $x \in K$ is compact in $L^1(G)$ (INT, VIII, §2, n° 5, prop. 8). We therefore have

$$\lim_{\xi, \mathfrak{F}} \xi(\varepsilon_x * f) = \xi(\varepsilon_x * f)$$

uniformly for $x \in K$. From this we deduce that

$$\lim_{\xi, \mathfrak{F}} \text{ev}^{-1}(\xi) = \text{ev}^{-1}(\zeta),$$

hence ev^{-1} is continuous at ζ . This completes the proof of the proposition.

From now on we identify a unitary character χ of G with the character $f \mapsto \int_G f(x) \chi(x) dx$ of $L^1(G)$.

Remarks. — 1) *The bijectivity of the map from $\widehat{G}$ onto $X(L^1(G))$ in prop. 1 is a special case of the correspondence between continuous representations of a locally compact group H (not necessarily commutative) and continuous representations of the algebra $L^1(H)$.*

2) The canonical map from $\widehat{G}$ into $X(\mathcal{M}^1(G))$ is not surjective in general (II, p. 308, exerc. 14).

COROLLARY 1. — Every character of $L^1(G)$ is Hermitian. The canonical map from $X(\text{Stell}(G))$ (def. 9 of I, p. 125) into $X(L^1(G))$ is a homeomorphism.

The first assertion follows from Proposition 1 and Lemma 1, restricted to $L^1(G)$. The second follows from the first and from the corollary of Prop. 20 of I, p. 124.

Corollary 2. — The topological group $\widehat{G}$ is locally compact.

Indeed, $X(L^1(G))$ is locally compact (corollary of Theorem 1 of I, p. 29).

We shall identify $\widehat{G}$ with $X(L^1(G))$ and $X(\text{Stell}(G))$. For $x \in G$ and $\chi \in \widehat{G}$, we shall denote

by $\langle \chi, x \rangle$ the complex number $\chi(x)$, which belongs to $\mathbb{U}$.

We say that x and χ are orthogonal if $\langle \chi, x \rangle = 1$. Let A be a subset of G (resp. of $\widehat{G}$); the set of elements of $\widehat{G}$ (resp. of G) orthogonal to A is a closed subgroup of $\widehat{G}$ (resp. of G) which is called the orthogonal of A and is denoted $A^\perp$. The orthogonal of G consists only of e .

For $x \in G$, let $\eta(x)$ be the map from $\widehat{G}$ to $\mathbb{U}$ defined by $\chi \mapsto \langle \chi, x \rangle$. By definition of multiplication in $\widehat{G}$, the map $\eta(x)$ is a group homomorphism. It is continuous since the map $(x, \chi) \mapsto \langle \chi, x \rangle$ from $G \times \widehat{G}$ to $\mathbb{U}$ is continuous (TG, X, p. 28, th. 3). We have thus defined a mapping η , called canonical, from G into the bidual group $\widehat{\widehat{G}}$; it is a group homomorphism. Moreover, the map η is continuous (TG, X, p. 28, th. 3). We shall prove later (II, p. 220, th. 2) that η is a topological group isomorphism from G onto $\widehat{\widehat{G}}$.

Let G and H be locally compact abelian groups, and $\varphi: G \rightarrow H$ a morphism of topological groups. For every $\chi \in \widehat{H}$, the map $\chi \circ \varphi$ is a character of G , denoted $\widehat{\varphi}(\chi)$. This definition translates to the formula

$$\langle \chi, \varphi(x) \rangle = \langle \widehat{\varphi}(\chi), x \rangle \quad (4)$$

for all $\chi \in \widehat{H}$ and $x \in G$. From this we deduce that $\widehat{\varphi}$ is a morphism from the topological group $\widehat{H}$ to the topological group $\widehat{G}$; we say that $\widehat{\varphi}$ is the dual of the morphism φ .

Let K be a locally compact commutative group and $\psi: H \rightarrow K$ a morphism of topological groups. The definition shows that $\widehat{\psi \circ \varphi} = \widehat{\varphi} \circ \widehat{\psi}$. If φ is the identity map of G , then $\widehat{\varphi}$ is the identity map of $\widehat{G}$. In particular, if φ is an isomorphism of topological groups, then so is $\widehat{\varphi}$, and $\widehat{\varphi}^{-1}$ is the dual of φ^{-1} .

Lemma 2. — Let G and H be locally compact abelian groups, and let $f: H \rightarrow G$ be a morphism of topological groups. The kernel of $\widehat{f}$ is the orthogonal of the image of f .

By definition, we have $\chi \in \text{Ker}(\widehat{f})$ if and only if the restriction of χ to the image of f is trivial.

PROPOSITION 2. — Let $n \geq 0$ be an integer and let $G_1, \dots, G_n$ be locally compact commutative groups. Let G be the product group of the groups G_j for $1 \leq j \leq n$. For $1 \leq j \leq n$, let λ_j be the injection of G_j into G which associates to $x \in G_j$ the element (x_k) such that $x_k = e$ if $k \neq j$ and $x_j = x$. The map

$$(\widehat{\lambda}_j)_{1 \leq j \leq n} : \widehat{G} \rightarrow \prod_{1 \leq j \leq n} \widehat{G}_j$$

is an isomorphism of topological groups.

Let m be the product map $\widehat{G}^n$ to $\widehat{G}$, and, for every j such that $1 \leq j \leq n$, let π_j be the projection of $\widehat{G}$ onto $\widehat{G}_j$. Let μ be the morphism of topological groups $m \circ (\pi_j)_j$ from $\prod \widehat{G}_j$ to $\widehat{G}$, so that

$$\langle \mu((\chi_j)), (x_j) \rangle = \prod_{j=1}^n \langle \chi_j, x_j \rangle.$$

The map μ is continuous, and one verifies that μ and $(\widehat{\lambda}_j)$ are inverse bijections of each other. The proposition follows.

Remark. — The computation of the dual group of an infinite product of compact commutative groups is the subject of Corollary 4 of II, p. 234 below. The case of a locally compact commutative group that is an arbitrary product of locally compact groups follows from these two statements, since in such a product all but finitely many factors are compact (TG, I, p. 66, Prop. 14, b)).

2.1.2 Definition of the Fourier transform

DEFINITION 3. — Let $\mu \in \mathcal{M}^1(G)$ be a bounded complex measure on G . The Fourier transform of μ is the function $\mathcal{F}_G(\mu)$ on $\widehat{G}$ defined by

$$\mathcal{F}_G(\mu)(\widehat{x}) = \int_G \overline{\langle \widehat{x}, x \rangle} d\mu(x). \quad (5)$$

We call the Fourier cotransform of μ the function $\overline{\mathcal{F}}_G(\mu)$ on $\widehat{G}$ defined by

$$\overline{\mathcal{F}}_G(\mu)(\widehat{x}) = \int_G \langle \widehat{x}, x \rangle d\mu(x). \quad (6)$$

When there is no ambiguity regarding the group G under consideration, we will also write $\mathcal{F}(\mu)$ and $\overline{\mathcal{F}}(\mu)$. We also sometimes denote by $\widehat{\mu} = \mathcal{F}_G(\mu)$.

PROPOSITION 3. — For every measure $\mu \in \mathcal{M}^1(G)$, the functions $\mathcal{F}_G(\mu)$ and $\overline{\mathcal{F}}_G(\mu)$

are continuous and bounded. The maps $\mathcal{F}_G : \mu \mapsto \mathcal{F}_G(\mu)$ and $\overline{\mathcal{F}}_G : \mu \mapsto \overline{\mathcal{F}}_G(\mu)$ are continuous morphisms of the involutive algebra $\mathcal{M}^1(G)$ into the involutive algebra of bounded continuous functions on $\widehat{G}$ (example 1 of I, p. 99).

Let $\mu \in \mathcal{M}^1(G)$. For every $\chi \in \widehat{G}$, one has

$$|\mathcal{F}(\mu)(\chi)| = \left| \int_G \overline{\langle \chi, x \rangle} d\mu(x) \right| \leq \|\mu\|_1, \quad (7)$$

so the Fourier transform of μ is bounded. Similarly, one verifies that $\overline{\mathcal{F}}(\mu)$ is bounded.

If χ tends to χ_0 in $\widehat{G}$, the function χ on G tends to χ_0 uniformly on every compact set while remaining bounded by the constant function 1, which belongs to $L^1(G, \mu)$. By Lebesgue's theorem (INT, IV, §3, n°7, th. 6), it follows that $\mathcal{F}(\mu)(\chi)$ tends to $\mathcal{F}(\mu)(\chi_0)$. Therefore $\mathcal{F}(\mu)$ is continuous. The same is true of $\overline{\mathcal{F}}(\mu)$.

For every $\chi \in \widehat{G}$, the map $\mu \mapsto \chi(\mu) = \int \langle \chi, x \rangle d\mu(x)$ is a Hermitian character of $\mathcal{M}^1(G)$ (lemma 1 of II, p. 202). This implies that $\mathcal{F}$ and $\overline{\mathcal{F}}$ are involutive algebra morphisms of $\mathcal{M}^1(G)$ into the involutive algebra of bounded continuous functions on $\widehat{G}$. The inequality (7) shows that these morphisms are continuous.

The Fourier transform of G (resp. the Fourier cotransform of G) is the mapping $\mu \mapsto \mathcal{F}_G(\mu)$ (resp. the map $\mu \mapsto \overline{\mathcal{F}}_G(\mu)$) from $\mathcal{M}^1(G)$ into $\mathcal{C}_b(\widehat{G})$.

Let us note some useful formulas for $\mu \in \mathcal{M}^1(G)$, $x \in G$, and $\chi \in \widehat{G}$:

(8)

$$\overline{\mathcal{F}}(\mu)(\chi) = \mathcal{F}(\mu)(\chi^{-1}) = \overline{\mathcal{F}(\overline{\mu})(\chi)}, \quad (9)$$

$$\|\mathcal{F}(\mu)\|_\infty = \|\overline{\mathcal{F}}(\mu)\|_\infty \leq \|\mu\|_1, \quad (10)$$

$$\begin{cases} \mathcal{F}(\varepsilon_x)(\chi) = \overline{\langle \chi, x \rangle}, \\ \overline{\mathcal{F}}(\varepsilon_x)(\chi) = \langle \chi, x \rangle, \end{cases}$$

(in particular $\mathcal{F}(\varepsilon_e) = \overline{\mathcal{F}}(\varepsilon_e) = 1$),

(11)

$$\begin{cases} \mathcal{F}(\varepsilon_x * \mu)(\chi) = \overline{\langle \chi, x \rangle} \mathcal{F}(\mu)(\chi), \\ \overline{\mathcal{F}}(\varepsilon_x * \mu)(\chi) = \langle \chi, x \rangle \overline{\mathcal{F}}(\mu)(\chi), \end{cases} \quad (12)$$

$$\begin{cases} \mathcal{F}(\chi \cdot \mu) = \varepsilon_\chi * \mathcal{F}(\mu), \\ \overline{\mathcal{F}}(\chi \cdot \mu) = \varepsilon_{\chi^{-1}} * \overline{\mathcal{F}}(\mu). \end{cases}$$

Formulas (8), (9), (10), and (11) follow from the definitions. Let us prove the first of formulas (12), the second being analogous. For all ξ in $\widehat{G}$, the equalities

$$\begin{aligned} \mathcal{F}(\chi \cdot \mu)(\xi) &= \int_G \overline{\langle \xi, x \rangle} \langle \chi, x \rangle d\mu(x) = \int_G \overline{\langle \xi \chi^{-1}, x \rangle} d\mu(x) \\ &= \mathcal{F}(\mu)(\xi \chi^{-1}) = (\varepsilon_\chi * \mathcal{F}(\mu))(\xi). \end{aligned}$$

Note moreover that for all $\chi \in \widehat{G}$ and all measures μ and ν in $\mathcal{M}^1(G)$, one has

$$(\varepsilon_\chi * \mathcal{F}(\mu))(\varepsilon_\chi * \mathcal{F}(\nu)) = \varepsilon_\chi * (\mathcal{F}(\mu)\mathcal{F}(\nu)), \quad (13)$$

since both sides of this equality are functions on $\widehat{G}$ whose value at $\xi \in \widehat{G}$ is

$$\mathcal{F}(\mu)(\xi \chi^{-1})\mathcal{F}(\nu)(\xi \chi^{-1}).$$

Let H be a locally compact abelian group, and let $\varphi: G \rightarrow H$ be a continuous morphism. Let $\mu \in \mathcal{M}^1(G)$. The image measure $\varphi(\mu)$ is defined (INT, V, §6, no. 1, remark 1), and it follows that $\mathcal{F}_H(\varphi(\mu)) = \mathcal{F}_G(\mu) \circ \widehat{\varphi}$ (cf. INT, V, §6, no. 4, prop. 7).

By restriction to the subalgebra $L^1(G)$ of $\mathcal{M}^1(G)$, we obtain the definitions of the Fourier transform and the Fourier cotransform on $L^1(G)$. We therefore have, for $f \in L^1(G)$ and $\chi \in \widehat{G}$:

$$\mathcal{F}_G(f)(\chi) = \int_G \overline{\langle \chi, x \rangle} f(x) dx, \quad \overline{\mathcal{F}}_G(f)(\chi) = \int_G \langle \chi, x \rangle f(x) dx. \quad (14)$$

In particular, $\mathcal{F}_G(f) = \overline{\overline{\mathcal{F}}_G(f)}$. We also have

$$\overline{\mathcal{F}}(f)(\chi) = \chi(f) \quad (15)$$

for all $f \in L^1(G)$ and every $\chi \in \widehat{G}$.

Let σ be an automorphism of G and Δ the modulus of σ (INT, VII, §1, no. 4, def. 4). For $f \in L^1(G)$, one has

$$\mathcal{F}(f \circ \sigma) = \Delta^{-1} \mathcal{F}(f) \circ \widehat{\sigma}^{-1} \quad (16)$$

(cf. loc. cit., formula (31)).

If one identifies $\widehat{G}$ and $X(L^1(G))$ (prop. 1 of II, p. 202), the Fourier cotransformation is none other than the Gelfand transformation of the Banach algebra $L^1(G)$ (I, p. 7, def. 5).

PROPOSITION 4. — The Fourier transform and the Fourier cotransform are injective morphisms of involutive algebras from $L^1(G)$ into the algebra $\mathcal{C}_0(\widehat{G})$ of continuous functions vanishing at infinity on $\widehat{G}$.

The Fourier cotransform is a morphism of involutive algebras from $L^1(G)$ into the algebra of bounded continuous functions on $\widehat{G}$. Since it is identified with the Gelfand transform, its image is contained in $\mathcal{C}_0(\widehat{G})$ (I, p. 37, prop. 5), and its kernel is the radical of $L^1(G)$ (prop. 8 of I, p. 38), which is zero (cor. of prop. 22 of I, p. 126).

We shall see later (corollary of Proposition 13 of II, p. 221) that the Fourier cotransformation on $\mathcal{M}^1(G)$ is also injective.

Remark. — The Fourier transform on the space $L^1(G)$ depends on the choice of the Haar measure dx , unlike the Fourier transform on $\mathcal{M}^1(G)$. If one replaces dx by the measure $a \cdot dx$ (with $a > 0$), then for any integrable function f on G , the Fourier transform of f is $a\widehat{f}$, where $\widehat{f}$ is the Fourier transform defined with respect to the measure dx .

Consider the stellar algebra $\text{Stell}(G)$ of the group G (def. 9 of I, p. 125), and identify $L^1(G)$ with a dense subalgebra of $\text{Stell}(G)$ (prop. 22 of I, p. 126).

PROPOSITION 5. — By continuity, the Fourier transform and the Fourier cotransform extend uniquely to isomorphisms of stellar algebras from $\text{Stell}(G)$ onto $\mathcal{C}_0(\widehat{G})$.

The Fourier cotransform extends by continuity to a morphism of stellar algebras from $\text{Stell}(G)$ into $\mathcal{C}_0(\widehat{G})$. If one identifies $\widehat{G}$ with $X(\text{Stell}(G))$ (cor. 1 of II, p. 204 and prop. 1 of II, p. 202), this extension is the Gelfand transform of $\text{Stell}(G)$. By th. 1 of I, p. 108, it is an isomorphism. The assertion concerning the Fourier transform follows from this.

We will always denote by $\overline{\mathcal{F}}$ and $\mathcal{F}$ the isomorphisms in Proposition 5.

COROLLARY. — L , the image of $L^1(G)$ under the Fourier transform of G , is dense in $\mathcal{C}_0(\widehat{G})$.

Since $L^1(G)$ is dense in $\text{Stell}(G)$, this follows from Proposition 5.

PROPOSITION 6. — Suppose that G is compact. The normalized Haar measure dx belongs to $\mathcal{M}^1(G)$, and its Fourier transform is φ_e , the characteristic function of $\{e\}$.

Let $\chi \in \widehat{G}$. Since $\varepsilon_y * dx = dx$ for all $y \in G$, one has

$$\mathcal{F}(dx)(\chi) = \overline{\langle \chi, y \rangle} \mathcal{F}(dx)(\chi)$$

by formula (11). If $\chi \neq 1$, there exists $y \in G$ such that $\langle \chi, y \rangle \neq 1$, hence $\mathcal{F}(dx)(\chi) = 0$. If $\chi = 1$, then $\mathcal{F}(dx)(\chi) = \int_G dx = 1$ since the measure dx is normalized.

2.1.3 Plancherel's theorem

We denote by $A(G)$ the vector subspace of $L^1(G)$ generated by the functions $f * g$ for $f, g \in L^1(G) \cap L^2(G)$.

PROPOSITION 7. — The space $A(G)$ is a self-adjoint ideal of $L^1(G)$. It is contained in $L^1(G) \cap L^2(G)$, and in the image of $C(G)$ in $L^1(G)$.

Let $f \in L^1(G)$. For every $g \in L^2(G)$, one has $f * g \in L^1(G)$. Therefore, the space $L^1(G) \cap L^2(G)$ is an ideal of $L^1(G)$, and the same is true of the space $A(G)$. The ideal $A(G)$ is self-adjoint.

Let f and g be in $L^2(G)$. The convolution product $f * g$ is then the class of the continuous function given by

$$y \mapsto \int_G f(yx^{-1})g(x)dx$$

The second assertion follows from it.

Since $\chi(f * g) = (\chi f) * (\chi g)$ for $\chi \in \widehat{G}$, $f \in L^1(G)$, and $g \in L^1(G)$ (INT, VIII, §3, no. 1, prop. 6), we have $\chi h \in A(G)$ for all $h \in A(G)$ and $\chi \in \widehat{G}$. Since $\varepsilon_x * f = \gamma(x)f$ and the linear representation γ is isometric on $L^p(G)$ for all p (INT, VIII, §2, no. 5, prop. 8), we have

$\varepsilon_x * f \in A(G)$ for all $x \in G$ and $f \in A(G)$.

We denote by $\widehat{A}(G)$ the image of $A(G)$ under the Fourier transform. It is a subspace of $\mathcal{C}_0(\widehat{G})$.

PROPOSITION 8. — There exists a filter base $\mathfrak{B}$ on $A(G) \cap \mathcal{K}_+(G)$ such that the following conditions are satisfied:

(i) For every element φ of a member of $\mathfrak{B}$, one has $\|\varphi\|_1 = 1$ and $\|\mathcal{F}(\varphi)\|_\infty \leq 1$;

(ii) We have

$$\lim_{\varphi, \mathfrak{B}} \varphi \cdot dx = \varepsilon_e$$

in the space $\mathcal{C}'(G)$ of measures with compact support on G , endowed with the topology of uniform convergence on compact subsets of $\mathcal{C}(G)$;

(iii) We have

$$\lim_{\varphi, \mathfrak{B}} \mathcal{F}(\varphi) = 1$$

for the topology of compact convergence on $\widehat{G}$;

(iv) For $p = 1$ or $p = 2$, and for every $f \in L^p(G)$, one has $\varphi * f \in A(G)$ for every φ belonging to a member of $\mathfrak{B}$, and

$$\lim_{\varphi, \mathfrak{B}} \varphi * f = f$$

in $L^p(G)$.

Let K_0 be a fixed compact neighborhood of e in G . Let B_0 be a base of the neighborhood filter of e in G consisting of compact symmetric neighborhoods contained in K_0 (cf. TG, III, p. 4). For $K \in B_0$, let X'_K be the set of functions $\psi \in \mathcal{K}_+(G)$ such that $\text{Supp}(\psi) \subset K$ and $\int \psi(x)dx = 1$; it is nonempty (Lemma 1 of II, p. 200). Let X_K be the set of functions $\psi * \psi$ for $\psi \in X'_K$. It is nonempty and contained in $A(G) \cap \mathcal{K}_+(G)$. The set $\mathfrak{B}$ whose elements are the sets X_K for K varying in B_0 is a filter base on $A(G) \cap \mathcal{K}_+(G)$. We prove that $\mathfrak{B}$ satisfies the

required properties.

If $X \in \mathfrak{B}$ and $\varphi \in X$, one has $\|\varphi\|_1 = \int_G \varphi(x) dx = 1$, hence $\|\mathcal{F}(\varphi)\|_\infty \leq 1$, which establishes property (i).

Property (ii) follows from INT, VIII, § 2, no. 7, corollary 1 of lemma 4. A compact subset of $\widehat{G}$ is a compact subset of $\mathcal{C}(G)$, hence (ii) implies $\lim_{\varphi, \mathfrak{B}} \mathcal{F}(\varphi) = 1$ for the topology of compact convergence on $\widehat{G}$, that is, (iii).

Finally, let $p = 1$ or $p = 2$. Let $f \in L^p(G)$. We have $\varphi * f \rightarrow f$ in $L^p(G)$ according to the filter $\mathfrak{B}$ (INT, VIII, §4, no. 7, prop. 20). Moreover, for every K in B_0 and $\varphi \in X_K$, there exists $\psi \in X'_K$ such that $\varphi = \psi * \psi$, whence $\varphi * f = \psi * (\psi * f)$. We have $\psi \in L^1(G) \cap L^2(G)$ and $\psi * f \in L^1(G) \cap L^2(G)$, hence $\varphi * f \in A(G)$.

COROLLARY 1. — The space $A(G)$ is dense in $L^1(G)$ and in $L^2(G)$. It is also dense in $\text{Stell}(G)$, and its image $\widehat{A}(G)$ under the Fourier transformation is dense in $\mathcal{C}_0(\widehat{G})$.

Assertion (iv) of the proposition provides the first assertion. Since $L^1(G)$ is dense in $\text{Stell}(G)$, the second follows from it, and the last then follows from prop. 5 of II, p. 209.

COROLLARY 2. — For $f \in A(G)$, let Ω_f be the set of $\chi \in \widehat{G}$ such that $\mathcal{F}(f)(\chi) \neq 0$. The sets Ω_f form an open covering of $\widehat{G}$.

This follows from the preceding corollary since, for every $\chi \in \widehat{G}$, the map $f \mapsto \mathcal{F}(f)(\chi)$ is a nonzero character of $L^1(G)$.

Recall that the left regular representation γ of $\text{Stell}(G)$ on $L^2(G)$ (cf. I, p. 125, no. 13) is denoted $\gamma(\varphi)f = \varphi * f$ for $\varphi \in \text{Stell}(G)$ and $f \in L^2(G)$.

Lemma 3. — For every $f \in A(G)$, there exists a unique bounded measure μ_f on $\widehat{G}$ such that

$$(\varphi * f)(e) = \int_{\widehat{G}} \mathcal{F}(\varphi) d\mu_f \quad (17)$$

for all $\varphi \in \text{Stell}(G)$.

Moreover, for all f and g in $A(G)$, we have the equality

$$\mathcal{F}(f) \cdot \mu_g = \mathcal{F}(g) \cdot \mu_f, \quad (18)$$

between the measure with density $\mathcal{F}(f)$ with respect to μ_g and the measure with density $\mathcal{F}(g)$ with respect to μ_f .

Let f and g be elements of $L^1(G) \cap L^2(G)$. For every $\varphi \in \text{Stell}(G)$, one has $\varphi * f \in L^2(G)$ and $\|\varphi * f\|_2 \leq \|\varphi\|_* \|f\|_2$ (I, p. 126, formula (8)). Moreover, we have $\varphi * (f * g) = (\varphi * f) * g$ (loc. cit., formula (9)). This latter function belongs to the closure $\mathcal{C}_0(G)$ of $\mathcal{H}(G)$ in $\mathcal{C}(G)$ (INT, VIII, §4, n° 5, prop. 15). Moreover, we have

$$\|\varphi * (f * g)\|_\infty \leq \|\varphi * f\|_2 \|g\|_2 \leq \|\varphi\|_* \|f\|_2 \|g\|_2.$$

Since the functions $f * g$ for f and g in $L^1(G) \cap L^2(G)$ generate $A(G)$, one deduces that $\varphi * f \in \mathcal{C}_0(G)$ for all $f \in A(G)$ and $\varphi \in \text{Stell}(G)$, and that the map $\varphi \mapsto (\varphi * f)(e)$ is a continuous linear form on $\text{Stell}(G)$. Since $\mathcal{F}$ is an isomorphism from $\text{Stell}(G)$ onto $\mathcal{C}_0(\widehat{G})$ (prop. 5 of II, p. 209), the first assertion follows from it.

Now let f and g be in $A(G)$. For $\varphi \in L^1(G)$, one has

$$\begin{aligned} (\mathcal{F}(f) \cdot \mu_g)(\mathcal{F}(\varphi)) &= \int_{\widehat{G}} \mathcal{F}(\varphi) \mathcal{F}(f) d\mu_g = \int_{\widehat{G}} \mathcal{F}(\varphi * f) d\mu_g \\ &= ((\varphi * f) * g)(e). \end{aligned} \tag{19}$$

Since $(\varphi * f) * g = (\varphi * g) * f$ and since the image of $L^1(G)$ under the Fourier transform is dense in $\mathcal{C}_0(\widehat{G})$ (cor. of II, p. 210), one deduces from formula (19) that formula (18) is satisfied for all f and g in $A(G)$.

Lemma 4. — There exists a unique measure ν on $\widehat{G}$ such that

$$\mu_f = \mathcal{F}(f) \cdot \nu$$

for all $f \in A(G)$. For $f \in A(G)$, one has $\mathcal{F}(f) \in L^1(\widehat{G}, \nu) \cap L^2(\widehat{G}, \nu)$. Let $f \in A(G)$. Denote by Ω_f the open set in $\widehat{G}$ consisting of the $\chi \in \widehat{G}$ such that $\mathcal{F}(f)(\chi) \neq 0$. Let φ be the characteristic function of $\widehat{G} - \Omega_f$. For every $g \in A(G)$, we then have

$$\int_{\widehat{G}} \mathcal{F}(g) d(\varphi \cdot \mu_f) = \int_{\widehat{G}} \varphi \mathcal{F}(f) d\mu_g = 0$$

given formula (18).

By Corollary 1 of II, p. 212, the image $\widehat{A}(G)$ of $A(G)$ under the Fourier transform is dense in $\mathcal{C}_0(\widehat{G})$. We then deduce from the preceding formula that $\varphi \cdot \mu_f = 0$, so that μ_f is concentrated on Ω_f (INT, IV, §4, no. 7, def. 4). Let ν_f be the measure on Ω_f with density $\mathcal{F}(f)^{-1}$ with respect to $\mu_f|_{\Omega_f}$.

The sets Ω_f , for $f \in A(G)$, form an open covering of $\widehat{G}$ (cor. 2). For all f and g in $A(G)$, formula (18) shows that $\nu_f|_{(\Omega_f \cap \Omega_g)} = \nu_g|_{(\Omega_f \cap \Omega_g)}$. Consequently, there exists a unique measure ν on $\widehat{G}$ such that $\nu_f = \nu|_{\Omega_f}$ for all $f \in A(G)$ (INT, III, §2, no. 1, prop. 1).

If $f \in A(G)$, the measures μ_f and $\mathcal{F}(f) \cdot \nu$ are concentrated on Ω_f , and their restrictions to Ω_f are equal to $\mathcal{F}(f) \cdot \nu_f$; these measures are therefore equal.

Since μ_f is a bounded measure, the Fourier transform $\mathcal{F}(f)$ belongs to the space $L^1(\widehat{G}, \nu)$. Moreover, since $\mathcal{F}(f)$ belongs to $\mathcal{C}_0(\widehat{G})$, we also have $\mathcal{F}(f) \in L^2(\widehat{G}, \nu)$.

Formula (17) now reads, for $\varphi \in \text{Stell}(G)$ and $f \in A(G)$:

$$(\varphi * f)(e) = \int_{\widehat{G}} \mathcal{F}(\varphi) \mathcal{F}(f) d\nu. \quad (20)$$

In particular, for f and g in $A(G)$, one has

$$\int_{\widehat{G}} \mathcal{F}(f) \mathcal{F}(g) d\nu = (f * g)(e) = \int_G f(x) g(x^{-1}) dx. \quad (21)$$

PROPOSITION 9. — The measure ν characterized by Lemma 4 is a Haar measure on $\widehat{G}$.

Let $\chi \in \widehat{G}$. For f and g in $A(G)$, let us apply formula (21) to χf and χg . The right-hand side is unchanged, hence

$$\nu(\mathcal{F}(f) \mathcal{F}(g)) = \nu(\mathcal{F}(\chi f) \mathcal{F}(\chi g)).$$

From formulas (12) of II, p. 208 and (13) of II, p. 208, it follows that

$$\nu(\mathcal{F}(f) \mathcal{F}(g)) = \nu(\varepsilon_\chi * (\mathcal{F}(f) \mathcal{F}(g))).$$

We then deduce from INT, VIII, §4, n° 3, prop. 7 that

$$\nu(\mathcal{F}(f)\mathcal{F}(g)) = (\varepsilon_{\chi^{-1}} * \nu)(\mathcal{F}(f)\mathcal{F}(g)),$$

that is,

$$(\mathcal{F}(f) \cdot \nu)(\mathcal{F}(g)) = (\mathcal{F}(f) \cdot (\varepsilon_{\chi^{-1}} * \nu))(\mathcal{F}(g)).$$

As the space $\widehat{A}(G)$ is dense in $\mathcal{C}_0(\widehat{G})$ (cor. 1), it follows that equality holds

$$\mathcal{F}(f) \cdot \nu = \mathcal{F}(f) \cdot (\varepsilon_{\chi^{-1}} * \nu).$$

The measures ν and $\varepsilon_{\chi^{-1}} * \nu$ therefore coincide on the open set Ω_f where $\mathcal{F}(f)$ is nonzero. By Corollary 2 and INT, III, §2, no. 1, cor. of Prop. 1, these measures are therefore equal.

This shows that the measure ν is proportional to a Haar measure on $\widehat{G}$. Let $f \in A(G)$. Let us take $g = \tilde{f}$ in formula (21). It is then written as

$$\int_{\widehat{G}} |\mathcal{F}(f)|^2 d\nu = \int_G |f|^2 dx, \quad (22)$$

which proves that the measure ν is not zero. The measure ν is therefore a Haar measure on $\widehat{G}$.

DEFINITION 4. — The Haar measure ν on $\widehat{G}$ from Proposition 9 is called the dual measure of the given Haar measure dx on G .

We will often denote by $d\chi$ or $d\hat{x}$ the Haar measure on $\widehat{G}$ which is dual to the Haar measure dx .

Note. — Let a be a real number > 0 . If one replaces dx by the measure $a \cdot dx$, the convolution product of the functions f and $g \in L^1(G)$ is replaced by $a(f * g)$. We have seen (II, p. 209, remark) that $\mathcal{F}(f)$ is replaced by $a\mathcal{F}(f)$. Therefore μ_f is unchanged and ν is replaced by $a^{-1} \cdot \nu$. In particular, the measure $dx \otimes d\hat{x}$ on $G \times \widehat{G}$ is independent of the choice of Haar measure on G .

Lemma 5. — The space $A(\widehat{G})$ is dense in $L^2(\widehat{G})$.

Let h be an element of $L^2(\widehat{G})$ orthogonal to $\widehat{A}(G)$. For f and g in $A(G)$, one has $\mathcal{F}(f) \cdot \mathcal{F}(g) = \mathcal{F}(f * g) \in \widehat{A}(G)$, hence $h \cdot \overline{\mathcal{F}(f)}$ is orthogonal to $\mathcal{F}(g)$. Thus, for every $f \in A(G)$,

the function $h \cdot \mathcal{F}(f)$ is orthogonal to $\widehat{A}(G)$. But $h \cdot \mathcal{F}(f) \in L^1(\widehat{G})$, and $\widehat{A}(G)$ is dense in $\mathcal{C}_0(\widehat{G})$, so the measure $h\mathcal{F}(f) \cdot \nu$ is zero, that is, $h\mathcal{F}(f)$ is ν -locally negligible (INT, V, §5, no. 3, cor. 2 of prop. 3). In particular, h is ν -locally negligible on the set Ω_f of characters χ such that $\mathcal{F}(f)(\chi) \neq 0$. By corollary 2, it follows that h is ν -locally negligible, hence zero since h belongs to $L^2(\widehat{G})$. This concludes the proof.

THEOREM 1 (Plancherel). — The restriction of the Fourier transform to the subspace $A(G)$ of $L^2(G)$ extends uniquely to an isometry Φ of $L^2(G)$ onto $L^2(\widehat{G})$.

Moreover, if $f \in L^1(G) \cap L^2(G)$, its Fourier transform belongs to $L^2(\widehat{G})$ and coincides in $L^2(\widehat{G})$ with $\Phi(f)$.

By formula (22), the restriction of $\mathcal{F}$ to $A(G)$ is an isometry of the subspace $A(G)$ of $L^2(G)$ onto the subspace $\widehat{A}(G)$ of $L^2(\widehat{G})$. Since $A(G)$ is dense in $L^2(G)$ (cor. 1 of II, p. 212), the Fourier transform extends uniquely to an isometry Φ of $L^2(G)$ onto a closed subspace of $L^2(\widehat{G})$. But since its image contains $\widehat{A}(G)$, which is dense in $L^2(\widehat{G})$ (lemma 5), the map Φ is surjective.

Now let $f \in L^1(G) \cap L^2(G)$; let us show that its Fourier transform belongs to $L^2(\widehat{G})$. By prop. 8, (iv) of II, p. 211, and the fact that $A(G)$ is an ideal of $L^1(G)$, there exists a filter base B on $A(G)$ which converges to f both in $L^1(G)$ and in $L^2(G)$. We then have

$$\Phi(f) = \lim_{g, \mathfrak{B}} \Phi(g) = \lim_{g, \mathfrak{B}} \mathcal{F}(g)$$

in $L^2(\widehat{G})$, and $\mathcal{F}(f) = \lim_{g, \mathfrak{B}} \mathcal{F}(g)$ in $\mathcal{C}_0(\widehat{G})$. There therefore exists a sequence (g_n) in $A(G)$ such that $\mathcal{F}(g_n)$ converges to $\Phi(f)$ in $L^2(\widehat{G})$ and to $\mathcal{F}(f)$ in $\mathcal{C}_0(\widehat{G})$. By INT, IV, §3, no. 4, th. 3 and cor. 1, we have $\mathcal{F}(f) = \Phi(f)$, and in particular $\mathcal{F}(f) \in L^2(\widehat{G})$. This completes the proof of the theorem.

We still denote by $\mathcal{F}$ the isometry of $L^2(G)$ onto $L^2(\widehat{G})$ defined in Theorem 1, and we call it the Fourier transformation on $L^2(G)$. Likewise, the Fourier cotransform admits a unique isometric extension to $L^2(G)$, still called the Fourier cotransform and denoted $\overline{\mathcal{F}}$.

COROLLARY. — Suppose that G is compact and that dx is the normalized Haar measure on G . Then the family of unitary characters of G is an orthonormal basis of $L^2(G)$.

Since G is compact, the characters of G belong to $L^2(G)$. For χ and ξ in $\widehat{G}$, one has

$$\int_G \overline{\langle \chi, x \rangle} \langle \xi, x \rangle dx = \mathcal{F}_G(dx)(\chi \xi^{-1}),$$

so the family of characters of G is orthonormal (prop. 6 of II, p. 210). It is moreover total because the scalar product of $\chi \in \widehat{G}$ and of $f \in L^2(G)$ is equal to $\mathcal{F}_G(f)(\chi)$, and hence f is orthogonal to $\widehat{G}$ if and only if $\mathcal{F}_G(f)$ is zero in $L^2(\widehat{G})$, if and only if f is zero (th. 1).

Remarks. — 1) Some of the formulas concerning the Fourier transformation on $L^1(G)$ extend to the Fourier transformation on $L^2(G)$. In particular, for $f \in L^2(G)$ and $\chi \in \widehat{G}$, one has

$$\overline{\mathcal{F}(f)} = (\chi \mapsto \mathcal{F}(f)(\chi^{-1})) = \overline{\mathcal{F}(\overline{f})}$$

$$\mathcal{F}(\varepsilon_x * f) = \eta(x^{-1}) \mathcal{F}(f), \quad \overline{\mathcal{F}(\varepsilon_x * f)} = \eta(x) \overline{\mathcal{F}(f)}$$

$$\mathcal{F}(\chi f) = \varepsilon_\chi * \mathcal{F}(f), \quad \overline{\mathcal{F}(\chi f)} = \varepsilon_\chi * \overline{\mathcal{F}(f)}.$$

If σ is an automorphism of G and Δ the modulus of σ (INT, VII, §1, no. 4, def. 4), then for $f \in L^2(G)$, one has

$$\mathcal{F}(f \circ \sigma) = \Delta^{-1} \mathcal{F}(f) \circ \widehat{\sigma}^{-1}$$

in $L^2(G)$.

2) The formulas

$$\|\overline{\mathcal{F}(f)}\|^2 = \|f\|^2 \tag{23}$$

for $f \in L^2(G)$, or

$$\int_G f(x) \overline{g(x)} dx = \int_{\widehat{G}} \mathcal{F}(f)(\chi) \overline{\mathcal{F}(g)(\chi)} d\chi \tag{24}$$

for f and g in $L^2(G)$, are called “Plancherel formulas”.

PROPOSITION 10. — Let $n \geq 0$ be an integer and let $G_1, \dots, G_n$ be locally compact abelian groups. Let μ_j , for $1 \leq j \leq n$, be Haar measures on G_j . Let G be the product group of the groups G_j for $1 \leq j \leq n$. Let β be the isomorphism from $\widehat{G}$ onto $\prod \widehat{G}_j$ of prop. 2 of II, p. 206. The Haar measure on $\widehat{G}$ dual to the product Haar measure $\mu = \mu_1 \otimes \dots \otimes \mu_n$ on G is identified with the product of the Haar measures $\widehat{\mu}_j$.

Indeed, for every family (f_j) of nonzero elements of $\mathcal{L}^2(G_j)$, the function f on G defined by $(x_j) \mapsto \prod f_j(x_j)$ belongs to $\mathcal{L}^2(G)$, and satisfies

$$\int_G |f|^2 d\mu = \prod_j \int_{G_j} |f_j|^2 d\mu_j = \prod_j \int_{\widehat{G}_j} |\mathcal{F}_{G_j}(f_j)|^2 d\widehat{\mu}_j,$$

by the Plancherel formula, which proves that the product Haar measure of the $\widehat{\mu}_j$ is identified with the dual Haar measure of μ .

2.1.4 The Fourier inversion formula

Recall that every element f of $A(G)$ is the class of a unique continuous function (prop. 7, b) of II, p. 210). For $x \in G$, we shall denote by $f(x)$ the value of this function at x .

PROPOSITION 11. — Let $f \in A(G)$. Then $\mathcal{F}(f) \in L^1(\widehat{G})$ and, for every $x \in G$, one has

$$f(x) = \int_{\widehat{G}} \langle \hat{x}, x \rangle \mathcal{F}(f)(\hat{x}) d\hat{x}. \quad (25)$$

In other words, for $f \in A(G)$, one has

$$f = \overline{\mathcal{F}_{\widehat{G}}(\mathcal{F}_G(f))} \circ \eta, \quad (26)$$

where η denotes the canonical mapping from G into the bidual group $\widehat{\widehat{G}}$.

By lemma 4 of II, p. 213 and proposition 9 of II, p. 214, we have $\mathcal{F}(f) \in L^1(\widehat{G})$ for every function $f \in A(G)$. By the Plancherel formula (24), for f and g in $L^2(G)$, one has

$$(f * \tilde{g})(e) = \int_{\widehat{G}} \mathcal{F}(f)(\chi) \overline{\mathcal{F}(g)(\chi)} d\hat{x}(\chi). \quad (27)$$

Let f and g be in $L^1(G) \cap L^2(G)$, and let $h = f * \tilde{g} \in A(G)$. Since the Fourier transformation

is an involutive morphism, formula (27) is assertion (25) for the function h at the point $x = e$. By linearity, we deduce that formula (25) is valid at the point $x = e$ for every function $h \in A(G)$.

Let $x \in G$ and $h \in A(G)$. Let $h_1 = \varepsilon_{x^{-1}} * h$. Then $h_1 \in A(G)$ and $h_1(e) = h(x)$. Since moreover $\mathcal{F}(h_1)(\chi) = \overline{\langle \chi, x \rangle} \mathcal{F}(f)(\chi)$ for all $\chi \in \widehat{G}$ (cf. formula (11) of II, p. 208), formula (25) for the function h_1 at the point e implies formula (25) for h at the point x .

Lemma 6. — Let $\varphi \in L^1(\widehat{G}) \cap L^2(\widehat{G})$. Then $f = \overline{\mathcal{F}_G}(\varphi) \circ \eta$ belongs to $L^2(G)$, and $\mathcal{F}_G(f) = \varphi$ in $L^2(\widehat{G})$.

The function f is continuous and bounded on G because $\varphi \in L^1(\widehat{G})$. For every function $g \in L^1(G) \cap L^2(G)$, one has

$$\begin{aligned} \int_G g(x)f(x)dx &= \int_G g(x) \left(\int_{\widehat{G}} \langle \chi, x \rangle \varphi(\chi) d\hat{x}(\chi) \right) dx \\ &= \int_{\widehat{G}} \overline{\mathcal{F}_G(g)}(\chi) \varphi(\chi) d\hat{x}(\chi), \end{aligned} \quad (28)$$

by applying the Lebesgue-Fubini theorem (INT, V, §8, no. 4, th. 1, a)) to the function $(x, \chi) \mapsto g(x)\varphi(\chi)\langle \chi, x \rangle$, which is integrable on $G \times \widehat{G}$ with respect to the product measure $dx \otimes d\hat{x}$. From this we deduce that

$$\left| \int_G g(x)f(x)dx \right| \leq \|\overline{\mathcal{F}_G(g)}\|_2 \|\varphi\|_2 = \|g\|_2 \|\varphi\|_2,$$

by the Plancherel formula. The linear form $g \mapsto \int_G fg$ is therefore continuous on $L^1(G) \cap L^2(G)$, and since $L^1(G) \cap L^2(G)$ is dense in the Hilbert space $L^2(G)$, we deduce that f belongs to $L^2(G)$.

Applying th. 1 of II, p. 215, we obtain on the other hand

$$\begin{aligned} \int_G g(x)f(x)dx &= \int_{\widehat{G}} \overline{\mathcal{F}_G(g)}(\chi) \mathcal{F}_G(f)(\chi) d\hat{x}(\chi) \\ &= \int_{\widehat{G}} \overline{\mathcal{F}_G(g)}(\chi) \mathcal{F}_G(f)(\chi) d\hat{x}(\chi) \end{aligned}$$

for all $g \in L^2(G)$. Comparing with (28), we conclude that $\varphi = \mathcal{F}_G(f)$ in $L^2(\widehat{G})$, since $A(G)$ is contained in $L^1(G) \cap L^2(G)$ and $\widehat{A}(G)$ is dense in $L^2(\widehat{G})$ (lemma 5 of II, p. 215).

PROPOSITION 12 (Fourier inversion formula)

Let $f \in L^2(G)$ be such that $\mathcal{F}_G(f) \in L^1(\widehat{G})$. Then we have $f = \overline{\mathcal{F}_G}(\mathcal{F}_G(f)) \circ \eta$ in $L^2(G)$.

In other words, for almost all $x \in G$, we have

$$f(x) = \int_{\widehat{G}} \langle \hat{x}, x \rangle \mathcal{F}_G(f)(\hat{x}) d\hat{x}.$$

The function $\varphi = \mathcal{F}_G(f)$ belongs to $L^1(\widehat{G}) \cap L^2(\widehat{G})$, and we obtain the desired formula by applying the lemma.

COROLLARY 1. — For every closed subset P of $\widehat{G}$ and every $\chi \in \widehat{G}$ not belonging to P , there exists a function $f \in L^1(G)$ such that $\mathcal{F}(f)$ is zero on P and nonzero at χ .

Since, by (12), we have $\mathcal{F}(\chi f) = \varepsilon_\chi * \mathcal{F}(f)$ for all $\chi \in \widehat{G}$, it suffices to consider the case where χ is the identity element of $\widehat{G}$.

Let U be a compact symmetric neighborhood of $e \in \widehat{G}$ such that $U^2 \cap P = \emptyset$. Let φ be a continuous positive function on $\widehat{G}$, vanishing outside U and such that $\varphi(e) = 1$. The function $\varphi_1 = \varphi * \varphi$ is then zero on P and $\varphi_1(e) > 0$. It therefore suffices to show that φ_1 belongs to the image of the Fourier transform on $L^1(G)$. Moreover, φ and φ_1 belong to $L^1(\widehat{G}) \cap L^2(\widehat{G})$. Set $f = \overline{\mathcal{F}}(\varphi) \circ \eta$ and $f_1 = \overline{\mathcal{F}}(\varphi_1) \circ \eta$. Lemma 6 implies that f and f_1 belong to $L^2(G)$ and satisfy $\varphi = \mathcal{F}(f)$ and $\varphi_1 = \mathcal{F}(f_1)$. Moreover

$$f_1 = \overline{\mathcal{F}}(\varphi * \varphi) \circ \eta = (\overline{\mathcal{F}}(\varphi) \circ \eta)^2 = f^2,$$

and therefore $f_1 \in L^1(G)$. Thus $\varphi_1 = \mathcal{F}(f_1)$ is indeed in the image of $L^1(G)$ under the Fourier transform.

COROLLARY 2. — The Banach algebra $L^1(G)$ is regular (I, p. 89, def. 1).

By prop. 1 of I, p. 88 and the identification of the Gelfand transform of $L^1(G)$ with the Fourier cotransform of G , this follows from the preceding corollary.

2.1.5 Pontryagin duality theorem

THEOREM 2 (Pontryagin). — The canonical mapping η from G into $\widehat{\widehat{G}}$ is an isomorphism of topological groups. It maps the Haar measure dx to the bi-dual Haar measure $d\hat{x}$.

Let us first prove that η is injective and strict. It suffices for this to show that for every neighborhood U of e in G , there exists a neighborhood W of e in $\widehat{G}$ such that $\overline{\eta}^{-1}(W) \subset U$ (lemma 2 of II, p. 200). Now let V be a compact symmetric neighborhood of e in G such that

$V^2 \subset U$, let f be a continuous positive function on G , with support contained in V , and such that $f(e) > 0$. Let $g = \tilde{f} * f$. Then g belongs to $A(G)$, its support is contained in U , and $g(e) > 0$. Moreover, $\mathcal{F}_G(g) \in L^1(\widehat{G})$ by prop. 11 of II, p. 217. The set W of ξ in $\widehat{G}$ such that

$$\left| \overline{\mathcal{F}_G(g)}(\xi) - \overline{\mathcal{F}_G(g)}(e) \right| < \frac{1}{2}g(e)$$

is a neighborhood of e in $\widehat{G}$ since the function $\overline{\mathcal{F}_G(g)}$ is continuous on $\widehat{G}$. Let $x \in \overline{\eta}^{-1}(W)$. By formula (26), we have

$$\overline{\mathcal{F}_G(g)}(\eta(x)) = g(x),$$

and therefore $|g(x) - g(e)| < \frac{1}{2}g(e)$. This implies $g(x) \neq 0$ and therefore $x \in U$, since the support of g is contained in U . Thus $\overline{\eta}^{-1}(W) \subset U$.

Let us prove that the map η is surjective. Since this map is a homeomorphism onto its image, the group $\eta(G)$ is a locally compact subgroup of $\widehat{G}$. It is therefore closed in $\widehat{G}$ (TG, III, p. 22, cor. 2). Let us argue by contradiction and suppose that there exists a character $\xi \in \widehat{G}$ such that $\xi \notin \eta(G)$. There then exists (corollary 1 of II, p. 219) a non-zero element f of $L^1(\widehat{G})$ such that $\mathcal{F}_G(f)$ vanishes on $\eta(G)$. Let $g \in L^1(G)$. The function $(x, \chi) \mapsto g(x)f(\chi)\overline{\langle \chi, x \rangle}$ belongs to $L^1(G \times \widehat{G})$. By the Lebesgue-Fubini theorem (INT, V, §8, no. 4, th. 1, a)), it follows that

$$\begin{aligned} \int_{\widehat{G}} f(\chi) \mathcal{F}_G(g)(\chi) d\chi &= \int_G g(x) \left(\int_{\widehat{G}} f(\chi) \overline{\langle \chi, x \rangle} d\chi \right) dx \\ &= \int_G g(x) \mathcal{F}_G(f)(\eta(x)) dx = 0. \end{aligned}$$

Since the image of the Fourier transform is dense in $\mathcal{C}_0(\widehat{G})$ (cor. of prop. 5 of II, p. 209), it follows that the measure $f \cdot d\chi$ is zero. This contradicts the fact that $f \neq 0$ in $L^1(\widehat{G})$, and proves that η is surjective.

The image measure $\eta(dx)$ and the measure ν dual to the measure $d\chi$ are Haar measures on $\widehat{G}$. Let f be a non-zero element of $A(G)$; in particular $f \in L^2(G)$. By prop. 12 of II, p. 219, $\mathcal{F}_G(f) \in L^1(\widehat{G})$ and one has

$$\int_{\widehat{G}} \left| \overline{\mathcal{F}_G(f)} \right|^2 \eta(dx) = \int_G |f|^2 dx = \int_{\widehat{G}} \left| \overline{\mathcal{F}_G(f)} \right|^2 d\nu,$$

where the second equality follows from two applications of the Plancherel formula, so the dual Haar measure of $d\chi$ is the measure $\eta(dx)$.

We shall henceforth identify G and $\widehat{G}$ by the isomorphism η . We then have:

COROLLARY. — The Fourier cotransform of $L^2(\widehat{G})$ on $L^2(G)$ and the Fourier transform of $L^2(G)$ on $L^2(\widehat{G})$ are reciprocal isometries of each other.

Remark. — Let $f \in L^2(G)$ and $g \in L^2(\widehat{G})$. Applying the Plancherel formula (24) to f and $\overline{\mathcal{F}_{\widehat{G}}(g)}$, we obtain the formula

$$\int_G f(x) \mathcal{F}_{\widehat{G}}(g)(x) dx = \int_{\widehat{G}} \mathcal{F}_G(f)(\chi) g(\chi) d\hat{\chi}(\chi) \quad (29)$$

since we have $\mathcal{F}_G(\overline{\mathcal{F}_{\widehat{G}}(g)}) = \mathcal{F}_G(\overline{\mathcal{F}_{\widehat{G}}(\bar{g})}) = \bar{g}$.

The Fourier transform and cotransform defined on $\mathcal{M}^1(\widehat{G})$ take values in the space of continuous bounded functions on G . For $\beta \in \mathcal{M}^1(\widehat{G})$ and $x \in G$, one has

$$\mathcal{F}_{\widehat{G}}(\beta)(x) = \int_{\widehat{G}} \overline{\langle \chi, x \rangle} d\beta(\chi), \quad \overline{\mathcal{F}_{\widehat{G}}(\beta)}(x) = \int_{\widehat{G}} \langle \chi, x \rangle d\beta(\chi).$$

The Fourier transforms of G and $\widehat{G}$ are also transposes of each other. More precisely:

PROPOSITION 13. — Let $\alpha \in \mathcal{M}^1(G)$ and $\beta \in \mathcal{M}^1(\widehat{G})$. We then have

$$\mathcal{F}_G(\overline{\mathcal{F}_{\widehat{G}}(\beta)} \cdot \alpha) = \beta * \mathcal{F}_G(\alpha) \quad (30)$$

and in particular

$$\int_G \mathcal{F}_{\widehat{G}}(\beta)(x) d\alpha(x) = \int_{\widehat{G}} \mathcal{F}_G(\alpha)(\chi) d\beta(\chi). \quad (31)$$

Formula (30) implies formula (31) by evaluating both sides of the identity at $\chi = 1$. Let us prove (30). Let $\chi \in \widehat{G}$. It follows that

$$\begin{aligned} (\mathcal{F}_G(\overline{\mathcal{F}_{\widehat{G}}(\beta)} \cdot \alpha))(\chi) &= \int_G \overline{\langle \chi, x \rangle} \overline{\mathcal{F}_{\widehat{G}}(\beta)}(x) d\alpha(x) \\ &= \int_G \overline{\langle \chi, x \rangle} \left(\int_{\widehat{G}} \langle \xi, x \rangle d\beta(\xi) \right) d\alpha(x). \end{aligned}$$

The function $(x, \xi) \mapsto \overline{\langle \chi, x \rangle} \langle \xi, x \rangle$ is continuous and bounded, hence integrable on $G \times \widehat{G}$ with respect to the measure $\alpha \otimes \beta$. By the Lebesgue-Fubini theorem (INT, V, §8, no. 4, th. 1,

a)), we obtain

$$\begin{aligned} (\mathcal{F}_G(\overline{\mathcal{F}_G(\beta)} \cdot \alpha))(\chi) &= \int_{\widehat{G}} \left(\int_G \overline{\langle \chi \xi^{-1}, x \rangle} d\alpha(x) \right) d\beta(\xi) \\ &= \int_{\widehat{G}} \mathcal{F}_G(\alpha)(\chi \xi^{-1}) d\beta(\xi) = (\beta * \mathcal{F}_G(\alpha))(\chi), \end{aligned}$$

as desired.

COROLLARY. — The Fourier transform $\mathcal{F}_G$ is injective on $\mathcal{M}^1(G)$.

Indeed, if $\alpha \in \mathcal{M}^1(G)$ satisfies $\mathcal{F}_G(\alpha) = 0$, we deduce from (31) that $\alpha(\mathcal{F}_{\widehat{G}}(f)) = 0$ for every $f \in L^1(\widehat{G})$. As the image of $L^1(\widehat{G})$ by the Fourier transform is dense in $\mathcal{C}_0(G)$ (corollary of Prop. 5 of II, p. 209), we therefore have $\alpha = 0$.

There exist functional spaces on G and $\widehat{G}$ other than $L^2(G)$ and $L^2(\widehat{G})$ on which $\mathcal{F}$ and $\overline{\mathcal{F}}$ are inverse isomorphisms of each other. The following theorem gives an example. We denote by $B(G)$ the vector subspace of $L^1(G)$ consisting of the elements $f \in L^1(G)$ such that $\mathcal{F}_G(f) \in L^1(\widehat{G})$. It is a subalgebra of $L^1(G)$. Indeed, let f and g be in $B(G)$. We have $f * g \in L^1(G)$ and $\mathcal{F}_G(f * g) = \mathcal{F}_G(f) \mathcal{F}_G(g) \in L^1(\widehat{G})$ since $\mathcal{F}_G(f) \in L^1(\widehat{G})$ and $\mathcal{F}_G(g) \in \mathcal{C}_0(\widehat{G})$.

THEOREM 3. — The restriction of the Fourier transform to $B(G)$ induces an isomorphism of vector spaces from $B(G)$ onto $B(\widehat{G})$, whose inverse is induced by the restriction to $B(\widehat{G})$ of the Fourier cotransform.

Let $f \in B(G)$. Denote by $g = \mathcal{F}_G(f)$. We have $g \in L^1(\widehat{G}) \cap \mathcal{C}_0(\widehat{G}) \subset L^1(\widehat{G}) \cap L^2(\widehat{G})$. Set $f_1 = \overline{\mathcal{F}_{\widehat{G}}}(g) \in L^2(G)$. For every continuous function with compact support $h \in \mathcal{H}(\widehat{G})$, one has $h \in L^1(\widehat{G}) \cap L^2(\widehat{G})$ and

$$\begin{aligned} \int_G f_1(x) \mathcal{F}_{\widehat{G}}(h)(x) dx &= \int_G \overline{\mathcal{F}_{\widehat{G}}(g)(x)} \overline{\mathcal{F}_{\widehat{G}}(h)(x)} dx \\ &= \int_{\widehat{G}} g(\chi) h(\chi) d\hat{x}(\chi) \\ &= \int_{\widehat{G}} \mathcal{F}_G(f)(\chi) h(\chi) d\hat{x}(\chi) = \int_G f(x) \mathcal{F}_{\widehat{G}}(h)(x) dx \end{aligned}$$

using the Plancherel theorem and formula (31). Since $\mathcal{H}(\widehat{G})$ is dense in $L^2(\widehat{G})$, its image under the Fourier transform is dense in $L^2(G)$. Consequently, we have $f_1 = f$ in $L^1(G)$; this shows that $g \in B(\widehat{G})$.

The formula $f_1 = f$ means that the restriction to $B(G)$ of the composition $\overline{\mathcal{F}_{\widehat{G}}} \circ \mathcal{F}_G$ is the identity map of $B(G)$. Interchanging the roles of G and $\widehat{G}$, we observe that $\mathcal{F}_G \circ \overline{\mathcal{F}_{\widehat{G}}}$ is the identity

map of $B(\widehat{G})$, which completes the proof of the theorem.

COROLLARY 1. — Let $f \in L^1(G)$. Then $f \in B(G)$ if and only if f belongs to the image of the Fourier transform $\mathcal{F}_{\widehat{G}}$ on $L^1(\widehat{G})$. In particular, we have $A(G) \subset B(G)$.

Theorem 3 proves that if $f \in B(G)$, then $f = \mathcal{F}_{\widehat{G}}(\overline{\mathcal{F}_G(f)})$, where $\overline{\mathcal{F}_G(f)}$ belongs to $L^1(\widehat{G})$. Conversely, if $f = \mathcal{F}_{\widehat{G}}(g)$, where $g \in L^1(\widehat{G})$, then we have $g \in B(\widehat{G})$ and therefore $f \in B(G)$ by the theorem. The last assertion then follows from prop. 11 of II, p. 217.

COROLLARY 2. — The vector space $B(G)$ is an algebra both for multiplication and for convolution. The Fourier transform interchanges convolution and multiplication in $B(G)$ and $B(\widehat{G})$.

We have already seen that $B(G)$ is a subalgebra of $L^1(G)$. On the other hand, if f and g belong to $B(G)$, then $fg \in L^1(G)$ since $f \in L^1(G)$ and g belongs to the image of the Fourier transform on $L^1(\widehat{G})$ (corollary 1). Since there exist f_1 and g_1 in $L^1(\widehat{G})$ such that $f = \mathcal{F}_{\widehat{G}}(f_1)$ and $g = \mathcal{F}_{\widehat{G}}(g_1)$ (loc. cit.), we have $fg = \mathcal{F}_{\widehat{G}}(f_1 * g_1)$, and hence $fg \in B(G)$ again by the preceding corollary.

PROPOSITION 14. — Let f and g be in $L^2(G)$. Then $\mathcal{F}_G(fg) = \mathcal{F}_G(f) * \mathcal{F}_G(g)$.

The equality holds if f and g belong to $B(G)$ (Corollary 2), and in particular if f and g belong to $A(G)$ since $A(G) \subset B(G)$ (Cor. 1). Since $A(G)$ is dense in $L^2(G)$ (Cor. 1 of II, p. 212), it suffices to show that the two members of the equality are continuous functions of $(f, g) \in L^2(G) \times L^2(G)$ with values in $\mathcal{C}_0(\widehat{G})$. Now the map $(f, g) \mapsto \mathcal{F}_G(fg)$ is obtained by composing the continuous map $(f, g) \mapsto fg$ of $L^2(G) \times L^2(G)$ into $L^1(G)$ and the Fourier transform $\mathcal{F}_G$ of $L^1(G)$ into $\mathcal{C}_0(\widehat{G})$, which is also continuous. Likewise, the map $(f, g) \mapsto \mathcal{F}_G(f) * \mathcal{F}_G(g)$ is obtained by composing the continuous maps $(f, g) \mapsto (\mathcal{F}_G(f), \mathcal{F}_G(g))$ of $L^2(G) \times L^2(G)$ into $L^2(\widehat{G}) \times L^2(\widehat{G})$ and $(h_1, h_2) \mapsto h_1 * h_2$ of $L^2(\widehat{G}) \times L^2(\widehat{G})$ into $\mathcal{C}_0(\widehat{G})$ (INT, VIII, §4, no. 5, prop. 15).

Remark. — See no. 9 and exercises 22 of II, p. 270 and 31 of II, p. 275 for other examples of function spaces on which the Fourier transform is an isomorphism, in the case of particular groups G .

2.1.6 Functorial properties of duality

Let G and H be locally compact commutative groups. Recall that if $\varphi: G \rightarrow H$ is a morphism of topological groups, the dual morphism $\widehat{\varphi}: \widehat{H} \rightarrow \widehat{G}$ is defined by $\langle \chi, \varphi(x) \rangle = \langle \widehat{\varphi}(\chi), x \rangle$ for all $\chi \in \widehat{H}$ and $x \in G$. This definition shows that $\widehat{\widehat{\varphi}} = \varphi$ with the identifications of G (resp. H) and $\widehat{\widehat{G}}$ (resp. $\widehat{\widehat{H}}$) of theorem 2 of II, p. 220.

Let θ be a map from $G \times H$ into U . For every $x \in G$ (respectively, every $y \in H$), let θ_x (respectively, θ^y) be the function from G into U defined by $y \mapsto \theta(x, y)$ (respectively, the function from H into U defined by $x \mapsto \theta(x, y)$). Suppose that the map $\alpha: x \mapsto \theta_x$ is an isomorphism of the topological group G onto the topological group $\widehat{H}$. For every $y \in H$ and $x \in G$, one has

$$\theta^y(x) = \theta(x, y) = \langle \alpha(x), y \rangle = \langle x, \widehat{\alpha}(y) \rangle,$$

that is, $\theta^y = \widehat{\alpha}(y)$. By th. 2 of II, p. 220, the map $\beta: y \mapsto \theta^y$ is therefore an isomorphism of the topological group H onto the topological group $\widehat{G}$. Under these conditions, we shall say that θ puts G and H in duality, or that G and H are in duality with respect to θ . We will then identify each of the groups G and H with the dual of the other. We will call the dual measure of the Haar measure dx the Haar measure on H obtained by transport of structure from the dual measure of dx .

Lemma 7. — Let $(G_i)_{i \in I}$ and $(H_i)_{i \in I}$ be finite families of locally compact topological groups. For $i \in I$, let $\theta_i: G_i \times H_i \rightarrow U$ be a map that puts the groups G_i and H_i in duality. The map θ defined by

$$\theta((g_i), (h_i)) = \prod_{i \in I} \theta_i(g_i, h_i)$$

puts the groups $\prod G_i$ and $\prod H_i$ in duality.

This follows from prop. 2 of II, p. 206 and from the definition preceding it.

DEFINITION 5. — Let G, H , and K be topological groups. Let $f: H \rightarrow G$ and $g: G \rightarrow K$ be morphisms of topological groups. The pair (f, g) is said to be an exact sequence of topological groups if it is an exact sequence of groups (A, II, p. 10, remark 5) and if f and g are strict

morphisms.

We will represent an exact sequence by the diagram

$$H \xrightarrow{f} G \xrightarrow{g} K,$$

and we will say that a diagram

$$G_1 \xrightarrow{f_1} G_2 \xrightarrow{f_2} G_3 \rightarrow \dots \rightarrow G_n \xrightarrow{f_n} G_{n+1}$$

is exact if each pair (f_i, f_{i+1}) for $1 \leq i \leq n-1$ is exact.

A sequence

$$1 \rightarrow H \xrightarrow{f} G \xrightarrow{g} K \rightarrow 1$$

is exact if and only if f is a strict injective morphism, g is a strict surjective morphism, and the kernel of g equals the image of f . If K is separated, the image of f is a closed subgroup of G .

Examples. — 1) Let $f: H \rightarrow G$ be a strict injective morphism whose image is a distinguished subgroup. The sequence

$$1 \rightarrow H \xrightarrow{f} G \xrightarrow{p} G/f(H) \rightarrow 1,$$

where p is the canonical projection, is exact. In particular, if H is a closed normal subgroup of G , the sequence of topological groups

$$1 \rightarrow H \xrightarrow{j} G \xrightarrow{p} G/H \rightarrow 1,$$

where j is the inclusion and p the canonical projection, is exact.

2) Let $g: G \rightarrow K$ be a strict surjective morphism. The sequence

$$1 \rightarrow \text{Ker}(g) \xrightarrow{j} G \xrightarrow{g} K \rightarrow 1,$$

where j is the inclusion, is exact.

Theorem 4. — A sequence

$$H \xrightarrow{f} G \xrightarrow{g} K$$

of locally compact commutative topological groups is exact if and only if the dual sequence

$$\widehat{K} \xrightarrow{\widehat{g}} \widehat{G} \xrightarrow{\widehat{f}} \widehat{H}$$

is exact.

We begin by proving a few lemmas. Note that each of them is, moreover, an easy consequence of the assertion of Thm. 4.

Lemma 8. — Let $g: G \rightarrow K$ be a morphism of locally compact commutative topological groups. If the morphism g is surjective and strict, then $\widehat{g}$ is injective and strict.

Since g is surjective, the morphism $\widehat{g}$ is injective (lemma 2 of II, p. 205). To prove that $\widehat{g}$ is a strict morphism, it suffices to show that for every neighborhood U of e in $\widehat{K}$, there exists a neighborhood V of e in $\widehat{G}$ such that $\widehat{g}^{-1}(U) \subset V$ (lemma 2 of II, p. 200). Let U be such a neighborhood of e in $\widehat{K}$. By definition of the topology of $\widehat{K}$, there exists a compact subset X of K and a number $\varepsilon > 0$ such that U contains the set of $\widehat{z} \in \widehat{K}$ which, for every $z \in X$, satisfy $|\langle \widehat{z}, z \rangle - 1| < \varepsilon$. Since g is strict and surjective, there exists, by TG, I, p. 80, prop. 10, a compact subset X_0 of G such that $g(X_0) = X$. Let V be the neighborhood of e in $\widehat{G}$ consisting of the elements $\widehat{\chi} \in \widehat{G}$ such that, for all $x \in X_0$, one has $|\langle \widehat{\chi}, x \rangle - 1| < \varepsilon$. We then have $\widehat{g}^{-1}(U) \subset V$. This proves the assertion.

Lemma 9. — Let $f: H \rightarrow G$ be a morphism of locally compact topological groups. If the morphism f is injective and strict, then $\widehat{f}$ is surjective and strict.

Suppose that f is injective and strict. The morphism $\widehat{f}$ induces, by passing to the quotient, a morphism $q: \widehat{G}/\text{Ker}(\widehat{f}) \rightarrow \widehat{H}$. It remains to show that this is an isomorphism of topological groups; by duality, it suffices for this to show that its dual $\widehat{q}$ is an isomorphism.

Let L denote the dual group of $\widehat{G}/\text{Ker}(\widehat{f})$ and $p: \widehat{G} \rightarrow \widehat{G}/\text{Ker}(\widehat{f})$ the canonical projection. We shall first prove that $\widehat{p}$ induces, by passing to subspaces, an isomorphism from L onto $f(H)$.

We have $q \circ p = \widehat{f}$, whence $\widehat{p} \circ \widehat{q} = f$. The image of $\widehat{p}$ therefore contains $f(H)$.

Since f is strict, its image $f(H)$ is a locally compact subgroup of G , and is therefore closed (TG, III, p. 22, cor. 2). Let $K = G/f(H)$ and consider the exact sequence

$$1 \rightarrow H \xrightarrow{f} G \xrightarrow{g} K \rightarrow 1$$

associated with example 1. By duality, the morphism $\hat{f} \circ \hat{g}$ is trivial, and hence the image of $\hat{g}$ is contained in $\text{Ker}(\hat{f}) = \text{Ker}(p)$. Thus $p \circ \hat{g}$ is the trivial morphism and, again by duality, $g \circ \hat{p}$ is also trivial. It follows that the image of $\hat{p}$ is contained in the kernel of g , which is equal to $f(H)$. We conclude that the image of $\hat{p}$ is equal to $f(H)$.

Moreover, since p is a surjective and strict morphism, the dual morphism $\hat{p}$ is a strict injective morphism from L to G (Lemma 8). It follows that $\hat{p}$ induces an isomorphism of topological groups from L onto $f(H)$. Since $\hat{p} \circ \hat{q} = \hat{f}$ and since $\hat{f}$ induces an isomorphism from H onto its image $f(H)$, the morphism $\hat{q}$ is an isomorphism.

Lemma 10. — Let

$$H \xrightarrow{f} G \xrightarrow{g} K$$

an exact sequence of locally compact abelian groups. The kernel of $\hat{f}$ is equal to the image of $\hat{g}$.

The homomorphism $\hat{f} \circ \hat{g}$ is trivial by duality, so the image of $\hat{g}$ is contained in the kernel of $\hat{f}$. Conversely, let χ be in the kernel of $\hat{f}$. This means that $\text{Im}(f) = \text{Ker}(g)$ is contained in the kernel of χ , hence that there exists a character η of $\text{Im}(g)$ such that $\eta \circ g = \chi$. Since the inclusion of $\text{Im}(g)$ in K is strict, the dual map of restriction of characters from K to $\text{Im}(g)$ is surjective (Lemma 9). There therefore exists a character β of K such that η is the restriction of β , and it follows that $\chi = \beta \circ g = \hat{g}(\beta)$. From this we conclude that the kernel of $\hat{f}$ is contained in the image of $\hat{g}$.

Now let us prove Theorem 4. By duality, it suffices to show that the sequence $\hat{K} \xrightarrow{\hat{g}} \hat{G} \xrightarrow{\hat{f}} \hat{H}$ is exact when the sequence $H \xrightarrow{f} G \xrightarrow{g} K$ is exact. Now, by Lemmas 8 and 9, the morphisms $\hat{f}$ and $\hat{g}$ are strict and, by Lemma 10, the kernel of $\hat{f}$ is equal to the image of $\hat{g}$.

COROLLARY 1. — Let

$$1 \rightarrow H \xrightarrow{f} G \xrightarrow{g} K \rightarrow 1$$

an exact sequence of locally compact commutative topological groups. The morphism $\hat{g}$ induces an isomorphism between $\hat{K}$ and $f(H)^\perp$, and $\hat{f}$ induces, by passing to the quotient, an isomorphism between $\hat{G}/f(H)^\perp$ and $\hat{H}$.

By Theorem 4, the sequence

$$1 \rightarrow \hat{K} \xrightarrow{\hat{g}} \hat{G} \xrightarrow{\hat{f}} \hat{H} \rightarrow 1 \quad (32)$$

is exact. The morphism $\hat{g}$ therefore induces an isomorphism from $\hat{K}$ onto $\text{Ker}(\hat{f}) = f(H)^\perp$ (lemma 2 of II, p. 205), and $\hat{f}$ induces, by passing to the quotient, an isomorphism from $\hat{G}/\text{Ker}(\hat{f}) = \hat{G}/f(H)^\perp$ onto $\hat{H}$ (loc. cit.).

COROLLARY 2. — Let $f: G \rightarrow H$ be a morphism of locally compact commutative topological groups. The morphism f is strict if and only if $\hat{f}$ is strict.

By the canonical decomposition (E, II, p. 44) of a strict morphism, this follows from Lemmas 8 and 9.

COROLLARY 3. — Let H be a subgroup of G . Then $(H^\perp)^\perp = \overline{H}$.

Since $H^\perp = \overline{H}^\perp$, we may assume that H is closed. Let $f: H \rightarrow G$ be the canonical injection and $p: G \rightarrow G/H$ the canonical projection. We have $H^\perp = \text{Ker}(\hat{f})$ (Lemma 10). Let k be the canonical injection of $H^\perp$ into $\hat{G}$. By Theorem 4, the morphism $\hat{p}$ induces an isomorphism $\hat{p}_H: \widehat{G/H} \rightarrow H^\perp$ of topological groups and we have $k \circ \hat{p}_H = \hat{p}$. Consequently (Corollary 1), we obtain

$$(H^\perp)^\perp = \text{Ker}(\hat{k}) = \text{Ker}(\hat{p}_H \circ \hat{k}) = \text{Ker}(p) = H.$$

COROLLARY 4. — Let I be a set and let $(H_i)_{i \in I}$ be a family of closed subgroups of G . The orthogonal of the closed subgroup generated by the H_i is $\bigcap_{i \in I} H_i^\perp$. The orthogonal of $\bigcap_i H_i$ is the closed subgroup generated by the subgroups $H_i^\perp$.

The first assertion follows from the definition of the orthogonal. Applying this result and Cor. 3 to the family of closed subgroups $(H_i^\perp)_{i \in I}$ of $\hat{G}$ we see that $\bigcap_i H_i$ is the orthogonal of

the closed subgroup generated by the subgroups $H_i^\perp$ and the second assertion is then obtained by duality.

COROLLARY 5. — Let $\varphi : G \rightarrow H$ be a morphism of locally compact commutative groups. Then the subgroup $\overline{\text{Im}(\varphi)}$ of H and the subgroup $\text{Ker}(\widehat{\varphi})$ of $\widehat{H}$ are orthogonal to each other. In particular, for $\widehat{\varphi}$ to be injective, it is necessary and sufficient that the image of φ be dense in H .

We have $\text{Ker}(\widehat{\varphi}) = \varphi(G)^\perp$ (lemma 2 of II, p. 205), whence the result by cor. 3.

COROLLARY 6. — Let $k \in \mathbf{Z}$. Then the kernel of the homomorphism $x \mapsto x^k$ from G to G and the closure of the image of the morphism $\chi \mapsto \chi^k$ from $\widehat{G}$ to $\widehat{G}$ are orthogonal to each other.

This follows from the preceding corollary, since the morphisms $x \mapsto x^k$ from G to G and $\chi \mapsto \chi^k$ from $\widehat{G}$ to $\widehat{G}$ are dual to each other.

Recall (A, X, p. 17) that a commutative group A is divisible if, for every nonzero $n \in \mathbf{Z}$, the map $a \mapsto a^n$ from A to A is surjective.

COROLLARY 7. — Let G be a locally compact abelian group.

- a) If G is divisible, then the dual group $\widehat{G}$ is torsion-free;
- b) If the dual group $\widehat{G}$ is torsion-free, and if $k \in \mathbf{Z}$ is nonzero, then the image of the homomorphism $x \mapsto x^k$ from G to G is dense in G ;
- c) Suppose G is discrete or compact. For G to be divisible it is necessary and sufficient that $\widehat{G}$ be torsion-free.

The assertions *a*) and *b*) follow from cor. 6. If G is discrete or compact, the image of the homomorphism $x \mapsto x^k$ from G to G is closed, and *c*) follows from *a*) and *b*).

Remark. — There exist locally compact abelian groups G that are not divisible and such that $\widehat{G}$ is torsion-free (exercise 63 of II, p. 299).

2.1.7 The Poisson formula

In this section, we consider a closed subgroup H of G . We denote $\beta = dx$ the Haar measure on G and $\widehat{\beta}$ the dual Haar measure on $\widehat{G}$. One denotes by α a Haar measure on H and $\widehat{\alpha}$ the dual Haar measure on the dual group $\widehat{H}$, which we identify with $\widehat{G}/H^\perp$ (Theorem 4 of

II, p. 226). We also identify $\widehat{G}/\widehat{H}$ with $H^\perp$ via the dual mapping of the canonical projection $G \rightarrow G/H$ (loc. cit.).

We shall denote by $\dot{x}$ the canonical image of an element x of G in G/H and by $\dot{\chi}$ the canonical image of an element χ of $\widehat{G}$ in $\widehat{G}/H^\perp$.

We denote by γ the Haar measure β/α on G/H (INT, VII, §2, no. 2, def. 1 and no. 7, prop. 10), and $\widehat{\gamma}$ the dual Haar measure on $H^\perp$. Recall (INT, VII, §2, n° 3, prop. 5, c)) that the measure γ is characterized by the following property: for every $f \in \mathcal{L}^1(G)$, the function $y \mapsto f(xy)$ on H is α -integrable for β -almost all $x \in G$; its integral depends only on $\dot{x}$ and the function defined γ -almost everywhere on G/H by

$$f^b : \dot{x} \mapsto \int_H f(xh) d\alpha(h)$$

belongs to $L^1(G/H, \gamma)$ and satisfies

$$\int_{G/H} f^b d\gamma = \int_G f d\beta. \quad (33)$$

PROPOSITION 15. — Let $f \in L^1(G)$ such that the restriction to $H^\perp$ of the continuous function $\mathcal{F}_G(f)$ is integrable with respect to $\widehat{\gamma}$. Then, for almost all $x \in G$, the function $y \mapsto f(xy)$ on H is α -integrable, and one has:

$$\int_H f(xy) d\alpha(y) = \int_{H^\perp} \langle \chi, x \rangle \mathcal{F}_G(f)(\chi) d\widehat{\gamma}(\chi).$$

Based on the foregoing, the function f^b defined almost everywhere on G/H by

$$f^b(\dot{x}) = \int_H f(xy) d\alpha(y)$$

belongs to $L^1(G/H)$. The Fourier transform of f^b is identified with the function on $H^\perp = \widehat{G}/\widehat{H}$ given for $\chi \in H^\perp$ by

$$\begin{aligned} \mathcal{F}_{G/H}(f^b)(\chi) &= \int_{G/H} \overline{\langle \chi, \dot{x} \rangle} f^b(\dot{x}) d\gamma(\dot{x}) \\ &= \int_G \overline{\langle \chi, x \rangle} f(x) d\beta(x) = \mathcal{F}_G(f)(\chi) \end{aligned}$$

by formula (33), applied to the integrable function $x \mapsto \overline{\langle \chi, x \rangle} f(x)$. By hypothesis, the

function $\mathcal{F}(f)|_{H^\perp} = \mathcal{F}_{G/H}(f^b)$ belongs to $L^1(H^\perp)$, and hence the function f^b belongs to the space $B(G/H)$. It follows (th. 3 of II, p. 222) that f^b coincides almost everywhere with $\overline{\mathcal{F}_{G/H}}(\mathcal{F}_{G/H}(f^b))$. For almost every $\dot{x} \in G/H$, we therefore have

$$f^b(\dot{x}) = \int_{H^\perp} \langle \chi, x \rangle \mathcal{F}_{G/H}(f^b)(\chi) d\widehat{\gamma}(\chi) = \int_{H^\perp} \langle \chi, x \rangle \mathcal{F}_G(f)(\chi) d\widehat{\gamma}(\chi).$$

This concludes the proof.

COROLLARY (Poisson formula). — Let $f \in \mathcal{L}^1(G)$. It is assumed that the following conditions are satisfied:

- (i) The restriction of $\mathcal{F}_G(f)$ to $H^\perp$ is integrable;
- (ii) For every $x \in G$, the function $y \mapsto f(xy)$ on H is integrable;
- (iii) The map $x \mapsto \int_H f(xy) d\alpha(y)$ is continuous on G . Then one has

$$\int_H f(y) d\alpha(y) = \int_{H^\perp} \mathcal{F}_G(f)(\chi) d\widehat{\gamma}(\chi). \quad (34)$$

Indeed, taking up the notation of the proof of the preceding proposition, the functions f^b and $\overline{\mathcal{F}_{G/H}}(\mathcal{F}_{G/H}(f^b))$ on G/H are continuous and equal almost everywhere. They are therefore equal everywhere and in particular at e , which gives formula (34).

PROPOSITION 16. — The measure $\widehat{\alpha}$ on $\widehat{H} = \widehat{G}/H^\perp$ is equal to $\widehat{\beta}/\widehat{\gamma}$.

Fix $f \in \mathcal{K}(G)$ nonzero. For $x \in G$ and $\chi \in \widehat{G}$, set

$$\varphi(x, \chi) = \int_H f(xy) \langle \chi, y \rangle d\alpha(y).$$

The function φ is continuous on $G \times \widehat{G}$ (INT, IV, §4, no. 3, cor. 1 of th. 2). For x fixed, $\varphi(x, \chi)$ depends only on the class of χ in $\widehat{G}/H^\perp = \widehat{H}$. For χ fixed, $\langle \chi, x \rangle \varphi(x, \chi)$ depends only on the class of x in G/H , and the function $\dot{x} \mapsto \langle \chi, x \rangle \varphi(x, \chi)$ on G/H has compact support.

Let $x \in G$. The function $\dot{\chi} \mapsto \varphi(x, \chi)$ on $\widehat{H}$ is the Fourier cotransform of the function $y \mapsto f(xy)$ on H . The latter is square-integrable, so by the Plancherel formula (23) of II, p. 217, we have

$$\int_{\widehat{G}/H^\perp} |\varphi(x, \chi)|^2 d\widehat{\alpha}(\dot{\chi}) = \int_H |f(xy)|^2 d\alpha(y). \quad (35)$$

Let $\chi \in \widehat{G}$. The function $\dot{x} \mapsto \langle \chi, x \rangle \varphi(x, \chi)$ belongs to $\mathcal{H}(G/H)$, hence to $L^1(G/H)$. Its Fourier cotransform is the function on $H^\perp$

whose value at $\xi \in H^\perp$ is

$$\begin{aligned} \int_{G/H} \langle \xi, \dot{x} \rangle \langle \chi, x \rangle \varphi(x, \chi) d\gamma(\dot{x}) &= \int_{G/H} \left(\int_H \langle \chi \xi, xy \rangle f(xy) d\alpha(y) \right) d\gamma(\dot{x}) \\ &= \int_G \langle \chi \xi, x \rangle f(x) d\beta(x) = \overline{\mathcal{F}_G(f)}(\chi \xi) \end{aligned}$$

by formula (33). Therefore

$$\int_{G/H} |\varphi(x, \chi)|^2 d\gamma(\dot{x}) = \int_{H^\perp} |\overline{\mathcal{F}_G(f)}(\chi \xi)|^2 d\widehat{\gamma}(\xi) \quad (36)$$

by the Plancherel formula again.

One then finally computes

$$\begin{aligned} \int_{\widehat{G}} |\overline{\mathcal{F}_G(f)}|^2 d\widehat{\beta} &= \int_G |f|^2 d\beta && \text{(by (23))} \\ &= \int_{G/H} d\gamma(\dot{x}) \int_H |f(xy)|^2 d\alpha(y) && \text{(by (33))} \\ &= \int_{G/H} d\gamma(\dot{x}) \int_{\widehat{G}/H^\perp} |\varphi(x, \chi)|^2 d\widehat{\alpha}(\dot{\chi}) && \text{(by (35))} \\ &= \int_{\widehat{G}/H^\perp} d\widehat{\alpha}(\dot{\chi}) \int_{G/H} |\varphi(x, \chi)|^2 d\gamma(\dot{x}) \\ &= \int_{\widehat{G}/H^\perp} d\widehat{\alpha}(\dot{\chi}) \int_{H^\perp} |\overline{\mathcal{F}_G(f)}(\chi \xi)|^2 d\widehat{\gamma}(\xi) && \text{(by (36))} \end{aligned}$$

where INT, V, §8, no. 3, prop. 5 has been applied to the continuous positive function $(\dot{x}, \dot{\chi}) \mapsto |\varphi(x, \chi)|^2$ on $G/H \times \widehat{G}/H^\perp$.

Comparing this equality with the integration formula (33) for the group $\widehat{G}$, one concludes that the Haar measures $\widehat{\alpha}$ and $\widehat{\beta}/\widehat{\gamma}$ coincide.

2.1.8 Examples of duality

PROPOSITION 17. — Let $n \geq 1$ be an integer. Denote by μ_n the group of n -th roots of unity in $\mathbf{C}$. The groups $\mathbf{Z}/n\mathbf{Z}$ and μ_n are in duality with respect to the map induced by passage to the quotient of the map $\mathbf{Z} \times \mu_n \rightarrow \mathbf{U}$ defined by $(m, z) \mapsto z^m$.

The group $\mathbf{Z}/n\mathbf{Z}$ coincides with the set of homomorphisms χ from $\mathbf{Z}/n\mathbf{Z}$ to $\mathbf{U}$. These are of the form $m \mapsto \chi(1)^m$ where $\chi(1)$ is an arbitrary element of $\mathbf{U}$ such that $\chi(1)^n = 1$, whence the result.

COROLLARY 1. — Let G be a finite commutative group. The dual group $\widehat{G}$ is isomorphic to G .

The group G is isomorphic to a finite product of cyclic groups (A, VII, p. 22, th. 3), and its dual group is isomorphic to the product of the dual groups of these (prop. 2 of II, p. 206). One is therefore reduced to the case where G is cyclic, which follows from proposition 17 since the group μ_n is cyclic of order n (A, V, p. 75, th. 1).

COROLLARY 2. — Let G be a locally compact commutative group. The group G is finite if and only if $\widehat{G}$ is finite. A closed subgroup H of G has finite index if and only if its orthogonal is finite.

By duality, the first assertion follows from the fact that the dual of a finite group is finite (corollary 1). The second follows from it, since $\widehat{G/H}$ is identified with $H^\perp$ (th. 4 of II, p. 226).

PROPOSITION 18. — For $\widehat{G}$ to be compact, it is necessary and sufficient that G be discrete. If G is discrete, the dual measure of the counting measure on G is the normalized Haar measure on $\widehat{G}$. If G is compact, the dual measure of the normalized Haar measure is the counting measure on $\widehat{G}$.

Suppose G is discrete, and let α be the counting measure on G . Let φ be the characteristic function of $e \in G$. We have $\mathcal{F}_G(\varphi) = 1$ on $\widehat{G}$. Since $\mathcal{F}_G(\varphi)$ tends to 0 at infinity, the group $\widehat{G}$ is compact.

Furthermore, for the dual measure $\widehat{\alpha}$ of α , the function $\mathcal{F}_G(\varphi)$ must have integral $\varphi(e) = 1$ (prop. 12 of II, p. 219). Therefore $\widehat{\alpha}(\widehat{G}) = 1$.

Suppose G is compact. Then the Haar measure dx belongs to $\mathcal{M}^1(G)$. Its Fourier transform is strictly positive at $\chi = e$ and zero for $\chi \neq 0$ (prop. 6 of II, p. 210). Since it is continuous on $\widehat{G}$, the group $\widehat{G}$ is discrete. If the measure of G is 1, we deduce by duality from the previous case that the dual measure of the measure dx is the counting measure on $\widehat{G}$.

COROLLARY 1 (Orthogonality relations). — Suppose G is discrete and equipped with the counting measure. For x and y in G , we have

$$\int_{\widehat{G}} \chi(x) \overline{\chi(y)} d\chi = \begin{cases} 0 & \text{if } x \neq y \\ 1 & \text{if } x = y. \end{cases}$$

This follows from the cor. of theo. 1 of II, p. 215 and from duality.

COROLLARY 2. — Let H be a closed subgroup of G .

- a) For H to be compact, it is necessary and sufficient that $H^\perp$ be open in $\widehat{G}$;
- b) For H to be open, it is necessary and sufficient that $H^\perp$ be compact in $\widehat{G}$.
- c) To say that $H^\perp$ is open amounts to saying that $\widehat{G}/H^\perp$ is discrete, but $\widehat{G}/H^\perp$ is isomorphic to $\widehat{H}$ (th. 4 of II, p. 226); the assertion therefore follows from prop. 18. Assertion b) follows by duality from assertion a) applied to $H^\perp$.

COROLLARY 3. — Let $(H_i)_{i \in I}$ be a decreasing filtered family of compact subgroups of G . For G to be identified with the projective limit of the groups G/H_i , it is necessary and sufficient that $\widehat{G}$ be the union of the open subgroups $H_i^\perp$.

To say that G is identified with the projective limit of the G/H_i amounts to saying that $\bigcap_i H_i = \{e\}$ (TG, III, p. 60, prop. 2), that is, that $\bigcup_i H_i^\perp$ is dense in $\widehat{G}$ (cor. 4 of th. 4 of II, p. 226). Now $\bigcup_i H_i^\perp$ is an open, hence closed, subgroup of $\widehat{G}$.

COROLLARY 4. — Let I be a set and let $(H_i)_{i \in I}$ be a family of compact groups. The dual of the product group of H_i is the discrete group direct sum of the groups $\widehat{H}_i$.

This is a special case of Cor. 3.

PROPOSITION 19. — Let K be a locally compact, non-discrete field, not necessarily commutative, and let G be the additive group of K , whose group law is written additively. Let χ be a unitary character of G distinct from 1. For $x, y \in G$, set $\theta(x, y) = \chi(xy)$. Then G is in duality with itself with respect to θ .

For $y \in G$, let χ_y be the map from G into $\mathbb{C}$ such that $\chi_y(x) = \chi(xy)$. We have $\chi_y \in \widehat{G}$, and it must be shown that $\beta : y \mapsto \chi_y$ is a topological group isomorphism from G into $\widehat{G}$.

The map β is an injective homomorphism from G into $\widehat{G}$; it is continuous (TG, X, p. 28, th. 3 applied to the continuous map θ of $G \times G$ into $\mathbb{C}$). Let us prove that θ is a homeomorphism onto its image. It suffices (Lemma 2 of II, p. 200) to show that for every neighborhood U of 0

in K , there exists a neighborhood V of e in $\widehat{G}$ such that $\overline{\beta}^{-1}(V) \subset U$. Let $x \mapsto |x|$ be an absolute value on K defining the topology of K (AC, VI, §9, no. 1, prop. 1), and let $x_0 \in K$ be such that $\chi(x_0) \neq 1$; let us denote $\eta = |\chi(x_0) - 1| > 0$. Let U be a neighborhood of 0 in K . There exists $\delta > 0$ such that U contains the set of $y \in K$ such that $|y| < \delta$. Let V be the set of characters $\xi \in \widehat{G}$ such that $|\langle \xi, x \rangle - 1| < \eta$ for every element $x \in K$ satisfying $|x| \leq |x_0|/\delta$. It is a neighborhood of e in $\widehat{G}$. If $y \neq 0$ is such that $\beta(y)$ belongs to V , we therefore have $|\chi(xy) - 1| < |\chi(x_0) - 1|$ for all x such that $|x| \leq |x_0|/\delta$. Consequently, we have $|x_0 y^{-1}| > |x_0|/\delta$, and hence $|y| < \delta$, so that $y \in U$.

Since β is a homeomorphism onto its image, its image is closed in $\widehat{G}$ (TG, III, p. 22, cor. 2). But on the other hand, the orthogonal of the image of β is the set of elements x of G such that $\chi(xy) = 1$ for all $y \in G$, and is therefore reduced to $\{0\}$. The image of β is therefore dense in $\widehat{G}$ (corollary 3 of II, p. 228). We conclude that β is surjective.

COROLLARY 1. — Let K be a locally compact, non-discrete, not necessarily commutative field, and χ a nontrivial unitary character of the additive group of K . Let E be a finite-dimensional topological vector space over K . The map θ from $E \times E'$ into U defined by $\theta(x, \lambda) = \chi(\langle \lambda, x \rangle)$ for $(\lambda, x) \in E' \times E$ puts the topological groups E and E' in duality.

Let n be the dimension of E and $(e_1, \dots, e_n)$ a basis of E . It allows one to identify E and K^n (EVT, I, p. 14, th. 2). The result then follows from prop. 19 and prop. 2 of II, p. 206.

We denote by T the group R/Z .

COROLLARY 2. — a) The group R is in duality with itself with respect to the mapping $(x, y) \mapsto \exp(2i\pi xy)$, and the dual measure of Lebesgue measure is Lebesgue measure;

b) The groups Z and T are in duality with respect to the map obtained by passing to the quotient from the map from $Z \times R$ to U such that $(n, x) \mapsto \exp(2i\pi nx)$. The dual Haar measure of the counting measure on Z is the normalized Haar measure on R/Z .

The group $\mathbf{R}$ is in duality with itself relative to the mapping $(x, y) \mapsto \exp(2i\pi xy)$ by prop. 19. Let us identify $\widehat{\mathbf{R}}$ with $\mathbf{R}$. The orthogonal of Z in $\widehat{\mathbf{R}} = \mathbf{R}$ is then Z , and b) follows from th. 4 of II, p. 226.

Let α be the counting measure on Z and γ the normalized Haar measure on T . If β denotes Lebesgue measure on $\mathbf{R}$, we have $\gamma = \beta/\alpha$ since these two Haar measures on R/Z have mass 1.

The Haar measure $\widehat{\alpha}$ on $\widehat{Z} = T$ is the normalized Haar measure (prop. 18), and the Haar measure $\widehat{\gamma}$ is the counting measure on $\mathbf{Z}$ (loc. cit.). By Prop. 16 of II, p. 231, the dual measure of β is therefore the measure β .

Remark. — In particular, one recovers the determination of $X(L^1(\mathbf{Z}))$ as done in Example 4 of I, p. 36.

For every integer $n \geq 0$ and $(x, y) \in \mathbf{R}^n \times \mathbf{R}^n$, we denote by

$$x \cdot y = \sum_{j=1}^n x_j y_j.$$

COROLLARY 3. — Let $n \geq 1$ an integer. The group $\mathbf{R}^n$ is in duality with itself relative to the mapping $(x, y) \mapsto \exp(2i\pi x \cdot y)$ and the dual measure of the Lebesgue measure on $\mathbf{R}^n$ is the Lebesgue measure. The groups $\mathbf{Z}^n$ and $\mathbf{T}^n = \mathbf{R}^n/\mathbf{Z}^n$ are in duality with respect to the map obtained by passing to the quotient from the map $(n, x) \mapsto \exp(2i\pi x \cdot y)$, and the dual Haar measure of the counting measure on $\mathbf{Z}^n$ is the normalized Haar measure on $(\mathbf{R}/\mathbf{Z})^n$.

This follows from Lemma 7 of II, p. 225, from Proposition 10 of II, p. 217, and from Corollary 2.

Remark. — Given a subgroup H of $\mathbf{R}^n$, there corresponds its orthogonal $H^\perp$, a subgroup of $\widehat{\mathbf{R}^n} = \mathbf{R}^n$, which is none other than the subgroup associated with H defined in TG, VII, p. 6, no. 3.

In what follows, we shall identify the dual of $\mathbf{R}^n$ (resp. of $\mathbf{T}^n$) with $\mathbf{R}^n$ (resp. with $\mathbf{Z}^n$) by means of the duality of the corollary. In particular, for $f \in L^1(\mathbf{R}^n)$, its Fourier transform is identified with the function on $\mathbf{R}^n$ with values in $\mathbf{C}$ which, to $y \in \mathbf{R}^n$, associates

$$\mathcal{F}(f)(y) = \int_{\mathbf{R}^n} f(x) \exp(-2i\pi x \cdot y) dx.$$

COROLLARY 4. — The group $\mathbf{R}^*$ is in duality with the group $\{-1, 1\} \times \mathbf{R}$ by the map $(x, (\sigma, t)) \mapsto \sigma(x/|x|)|x|^{it}$. The group $\mathbf{R}_+^*$ is in duality with $\mathbf{R}$ by the map $(x, t) \mapsto x^{it}$.

Indeed, the map $x \mapsto (x/|x|, \log(|x|))$ is a topological group isomorphism from $\mathbf{R}^*$ onto $\{-1, 1\} \times \mathbf{R}$. The assertion then follows from lemma 7 of II, p. 225, from corollary 2, and from the fact that the unitary characters of $\{-1, 1\}$ are 1 and $x \mapsto x$.

Let p be a prime number. The field $\mathbf{Q}_p$ of p -adic numbers is the completion of $\mathbf{Q}$ with respect to the p -adic valuation (INT, VII, § 1, no. 6, example, and AC, VI, § 3, no. 4, example 4). For every $x \in \mathbf{Q}_p$, there exists a unique integer $\nu \geq 0$ and a unique integer q satisfying $0 \leq q < p^\nu$ such that $qp^{-\nu} - x \in Z_p$ (A, VII, p. 10, th. 2, applied to the principal ideal domain Z_p and to the set R_p of integers j such that $0 \leq j < p$). We denote by $\lambda(x) = qp^{-\nu}$.

PROPOSITION 20. — The mapping $x \mapsto \exp(2i\pi \lambda(x))$ is a unitary character of $\mathbf{Q}_p$ whose kernel is Z_p .

For x_1 and x_2 in $\mathbf{Q}_p$, by definition $\lambda(x_1 + x_2) = \lambda(x_1) + \lambda(x_2) \in Z_p \cap \mathbf{Q} = \mathbf{Z}$. The map λ is moreover locally constant since $\lambda(x + y) = \lambda(x)$ if $y \in Z_p$. It then follows that $x \mapsto \exp(2i\pi \lambda(x))$ is a unitary character of $\mathbf{Q}_p$. Since $\lambda(x) \in \mathbf{Z}$ if and only if $x \in Z_p$, the kernel of this character is Z_p .

Recall that one calls normalized Haar measure on the additive group of $\mathbf{Q}_p$ the unique Haar measure μ such that $\mu(Z_p) = 1$ (INT, VII, §1, no. 6, example).

COROLLARY. — a) The group $\mathbf{Q}_p$ is in duality with itself relative to the mapping $(x, y) \mapsto \exp(2i\pi \lambda(xy))$. The normalized Haar measure on $\mathbf{Q}_p$ is then its own dual;

b) The groups Z_p and $\mathbf{Q}_p/Z_p$ are in duality with respect to the map obtained by passing to the quotient from the map defined by $(z, x) \mapsto \exp(2i\pi \lambda(zx))$, and the dual measure of the normalized Haar measure on Z_p is the counting measure on $\mathbf{Q}_p/Z_p$.

The proof follows step by step that of Cor. 2 of Prop. 19.

2.1.9 Euclidean Fourier transform and Fourier series

* Let $n \in \mathbf{N}$. We identify $\mathbf{R}^n$ and its dual as in Cor. 3 of II, p. 236. The dual measure of the Lebesgue measure is then the Lebesgue measure. We equip $\mathbf{R}^n$ with the Euclidean norm. For every multi-index $\alpha \in \mathbf{N}^n$, and every $x = (x_1, \dots, x_n) \in \mathbf{R}^n$, we shall denote by $x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}$, and we denote by X^α the function $x \mapsto x^\alpha$ on $\mathbf{R}^n$.

Let $m \in \mathbf{R}^m$. Every continuous homomorphism of commutative groups from $\mathbf{R}^m$ into $\mathbf{R}^n$ is a linear map $\sigma \in \mathcal{L}(\mathbf{R}^n, \mathbf{R}^m)$ (TG, VII, p. 11, prop. 1). The dual homomorphism $\widehat{\sigma}$ is identified with the linear map ${}^t\sigma$.

The Fourier transform in $\mathbf{R}^n$ takes on a particularly convenient form in the context of the Schwartz function space and its dual (IV, to appear). We summarize here the main results.

Let $\mathcal{S}(\mathbf{R}^n)$ be the space of infinitely differentiable functions φ on $\mathbf{R}^n$ with complex values, such that, for every multi-index $\alpha \in \mathbf{N}^n$ and every integer $k \in \mathbf{N}$, the function

$$x \mapsto \|x\|^k \partial^\alpha \varphi(x)$$

is bounded on $\mathbf{R}^n$. We endow $\mathcal{S}(\mathbf{R}^n)$ with the locally convex topology defined by the seminorms

$$p_{k,\alpha} : \varphi \mapsto \sup_{x \in \mathbf{R}^n} \|x\|^k |\partial^\alpha \varphi(x)|.$$

We say that $\mathcal{S}(\mathbf{R}^n)$ is the space of Schwartz functions on $\mathbf{R}^n$.

For every $\alpha \in \mathbf{N}^n$, the maps $\varphi \mapsto \partial^\alpha \varphi$ and $\varphi \mapsto X^\alpha \varphi$ are continuous from $\mathcal{S}(\mathbf{R}^n)$ into itself. The space $\mathcal{S}(\mathbf{R}^n)$ is a topological algebra; it is a Fréchet space and a Montel space (EVT IV, p. 18, def. 4). For every $p \in [1, +\infty]$, the space $\mathcal{S}(\mathbf{R}^n)$ is contained in $\mathcal{L}^p(\mathbf{R}^n)$ and the canonical injection of $\mathcal{S}(\mathbf{R}^n)$ into $\mathcal{L}^p(\mathbf{R}^n)$ is continuous. The image of $\mathcal{S}(\mathbf{R}^n)$ into $L^p(\mathbf{R}^n)$ is dense if $p \neq +\infty$.

Since every Schwartz function φ is integrable on $\mathbf{R}^n$ it admits a Fourier transform denoted $\widehat{\varphi}$ which is identified with the continuous function on $\mathbf{R}^n$ defined by

$$y \mapsto \int_{\mathbf{R}^n} \varphi(x) \exp(-2i\pi x \cdot y) dx.$$

The Fourier cotransform of φ is identified, in turn, with the continuous function defined by

$$y \mapsto \int_{\mathbf{R}^n} \varphi(x) \exp(2i\pi x \cdot y) dx.$$

Let $\varphi \in \mathcal{S}(\mathbf{R}^n)$. Let $\alpha \in \mathbf{N}^n$ be a multi-index. We have

$$\mathcal{F}(\partial^\alpha \varphi) = (2i\pi)^{|\alpha|} X^\alpha \mathcal{F}(\varphi),$$

$$\mathcal{F}(X^\alpha \varphi) = (-2i\pi)^{-|\alpha|} \partial^\alpha (\mathcal{F}(\varphi)).$$

Proposition 21. — The restriction to $\mathcal{S}(\mathbf{R}^n)$ of the Fourier transform is an automorphism of topological vector spaces whose inverse is the restriction of the Fourier cotransform.

Let $\Lambda \subset \mathbf{R}^n$ be a lattice (TG, VII, p. 4), and let $\Lambda^* \subset \mathbf{R}^n$ be the associated lattice (TG, VII, p. 6), also sometimes called the dual lattice.

We call the measure of $\mathbf{R}^n / \Lambda$ for the Haar measure induced by the Lebesgue measure on $\mathbf{R}^n$ the covolume of the lattice Λ , and denote it by $V(\Lambda)$ (cf. INT, VIII, §5, no. 5, example). For every function $f \in \mathcal{S}(\mathbf{R}^n)$ and every $y \in \mathbf{R}^n$ we have the Poisson formula

$$\sum_{x \in \Lambda} f(x + y) = \frac{1}{V(\Lambda)} \sum_{z \in \Lambda^*} \hat{f}(z) \exp(2i\pi y \cdot z).$$

Remarks. — 1) More generally, by the corollary of Proposition 15 of II, p. 230, this formula holds for any complex function integrable on $\mathbf{R}^n$ such that

$$\sum_{x \in \Lambda} |f(x + y)| < +\infty,$$

for all $y \in \mathbf{R}^n$ and such that the function on T^n defined by

$$y \mapsto \sum_{x \in \Lambda} f(x + y)$$

is continuous and admits an absolutely convergent Fourier series (cf. below).

2) There exist functions $f \in B(\mathbf{R})$ such that the series $\sum_{n \in \mathbf{Z}} f(n)$ diverge (exercise 4 of II, p. 263).

Example. — Let Q be a positive definite quadratic form on $\mathbf{R}^n$. The function defined by $\varphi(x) = \exp(-\pi Q(x))$ belongs to $\mathcal{S}(\mathbf{R}^n)$. There exists $\sigma \in GL(n, \mathbf{R})$ such that $Q(x) = \|\sigma(x)\|^2$ for all $x \in \mathbf{R}^n$. The Fourier transform of φ is given for all $y \in \mathbf{R}^n$ by

$$\hat{\varphi}(y) = \frac{1}{|\det(\sigma)|} \exp(-\pi Q^*(y))$$

where $Q^*(y) = \|\sigma^{-1}(y)\|^2$ (cf. INT, IX, §6, nos. 4–5 and exercise 1, c) of II, p. 262).

DEFINITION 6. — We call the space of tempered distributions on $\mathbf{R}^n$ the dual space of $\mathcal{S}(\mathbf{R}^n)$ endowed with the topology of bounded convergence. It is denoted $\mathcal{S}'(\mathbf{R}^n)$.

Since $\mathcal{S}(\mathbf{R}^n)$ is bornological, the space $\mathcal{S}'(\mathbf{R}^n)$ is complete and bornological (EVT, III, p. 24, cor. 1 and 2). Since $\mathcal{S}(\mathbf{R}^n)$ is a Montel space, so is $\mathcal{S}'(\mathbf{R}^n)$ (EVT, IV, p. 19, prop. 9).

Let $\alpha \in \mathbf{N}^n$. We still denote by $f \mapsto X^\alpha f$ the transpose of the endomorphism $\varphi \mapsto X^\alpha \varphi$ of $\mathcal{S}(\mathbf{R}^n)$, and we denote by $f \mapsto \partial^\alpha f$ the endomorphism of $\mathcal{S}'(\mathbf{R}^n)$ defined by

$$\langle \partial^\alpha f, \varphi \rangle = (-1)^{|\alpha|} \langle f, \partial^\alpha \varphi \rangle$$

for $f \in \mathcal{S}'(\mathbf{R}^n)$ and $\varphi \in \mathcal{S}(\mathbf{R}^n)$.

Let f be a linear map from $\mathcal{S}(\mathbf{R}^n)$ into $\mathbf{C}$. Then f is a tempered distribution if, and only if, for every family $(M_{k,\alpha})_{(k,\alpha) \in \mathbf{N} \times \mathbf{N}^n}$ in $\mathbf{R}_+$, the linear form f is bounded on the set of functions $\varphi \in \mathcal{S}(\mathbf{R}^n)$ such that for all $(k, \alpha) \in \mathbf{N} \times \mathbf{N}^n$, one has $p_{k,\alpha}(\varphi) \leq M_{k,\alpha}$.

A sequence $(f_m)_{m \in \mathbf{N}}$ of tempered distributions converges to a tempered distribution f if, and only if, one has $\langle f_m, \varphi \rangle \rightarrow \langle f, \varphi \rangle$ for all $\varphi \in \mathcal{S}(\mathbf{R}^n)$.

Example. — A measure ν on $\mathbf{R}^n$ is said to be tempered if there exists a positive integer r such that the continuous map $x \mapsto (1 + \|x\|)^{-r}$ is ν -integrable on $\mathbf{R}^n$. The restriction of ν to $\mathcal{S}(\mathbf{R}^n)$ is a tempered distribution. It is zero if and only if the measure ν is zero.

Let $p \in [1, +\infty]$ and $f \in \mathcal{L}^p(\mathbf{R}^n)$. Then the measure $f \cdot dx$ of density f with respect to Lebesgue measure is tempered. In particular, Lebesgue measure μ on $\mathbf{R}^n$ is tempered, and every bounded measure on $\mathbf{R}^n$ is tempered.

For every $p \in [1, +\infty]$ one can identify $L^p(\mathbf{R}^n)$ with a subspace of $\mathcal{S}'(\mathbf{R}^n)$ by the linear map $f \mapsto f \cdot dx$; this map is continuous.

DEFINITION 7. — We call the transpose of the Fourier transform on $\mathcal{S}(\mathbf{R}^n)$ (respectively, of the Fourier cotransform) the Fourier transform (respectively, Fourier cotransform) on $\mathcal{S}'(\mathbf{R}^n)$, and denote them by $\mathcal{F}$ and $\overline{\mathcal{F}}$, respectively.

For $f \in \mathcal{S}'(\mathbf{R}^n)$, the tempered distribution $\mathcal{F}(f)$ (resp. $\overline{\mathcal{F}}(f)$) is defined by $\varphi \mapsto \langle f, \mathcal{F}(\varphi) \rangle$ for $\varphi \in \mathcal{S}(\mathbf{R}^n)$ (resp. by $\varphi \mapsto \langle f, \overline{\mathcal{F}}(\varphi) \rangle$).

The Fourier transform on $\mathcal{S}'(\mathbf{R}^n)$ is an automorphism of topological vector spaces whose

inverse is the Fourier cotransform $\overline{\mathcal{F}}$.

PROPOSITION 22. — Let f be a tempered distribution belonging to $\mathcal{M}^1(\mathbf{R}^n)$ (respectively to $L^2(\mathbf{R}^n)$). The Fourier transform of f in $\mathcal{S}'(\mathbf{R}^n)$ is the tempered distribution associated with the Fourier transform of f in $\mathcal{C}_0(\mathbf{R}^n)$ (respectively in $L^2(\mathbf{R}^n)$). The same holds for the Fourier cotransform.

Remark. — The elementary formulas concerning the Fourier transform of measures remain valid for the Fourier transform of tempered distributions.

Thus, if $\alpha \in \mathbf{N}^n$ and $f \in \mathcal{S}'(\mathbf{R}^n)$, one has

$$\mathcal{F}(\partial^\alpha f) = (2i\pi)^{|\alpha|} X^\alpha \mathcal{F}(f),$$

$$\mathcal{F}(X^\alpha f) = (-2i\pi)^{-|\alpha|} \partial^\alpha (\mathcal{F}(f)).$$

Let $y \in \mathbf{R}^n$. Denote by $\gamma(y)$ the endomorphism of $\mathcal{S}'(\mathbf{R}^n)$ defined by

$$\langle \gamma(y)f, \varphi \rangle = \langle f, \gamma(-y)\varphi \rangle$$

for $f \in \mathcal{S}'(\mathbf{R}^n)$ and $\varphi \in \mathcal{S}(\mathbf{R}^n)$. Denote by e_y the character of $\mathbf{R}^n$ such that $e_y(x) = \exp(2i\pi x \cdot y)$. Then $e_y \in \mathcal{S}'(\mathbf{R}^n)$. We have $\mathcal{F}(e_y) = \varepsilon_y$ and more generally

$$\mathcal{F}(e_y f) = \gamma(y) \mathcal{F}(f)$$

for all $f \in \mathcal{S}'(\mathbf{R}^n)$. *

Let $n \geq 1$ be an integer and $G = T^n$, equipped with the normalized Haar measure. The dual group of G is identified with Z^n by the map $h \mapsto \chi_h$, where χ_h is the unitary character of T^n obtained by passing to the quotient from the character $x \mapsto \exp(2i\pi h \cdot x)$ of $\mathbf{R}^n$ (corollary 3 of II, p. 236). The Fourier transform of a measure μ on T^n is identified with the family $(\widehat{\mu}(h))_{h \in Z^n}$ where

$$\widehat{\mu}(h) = \int_{T^n} e^{-2i\pi h \cdot x} d\mu(x).$$

The series

$$\sum_{h \in \mathbf{Z}^n} \widehat{\mu}(h) \chi_h$$

is called the Fourier series of μ .

If $f \in L^1(\mathbf{T}^n)$ is such that its Fourier series converges absolutely in $L^1(\mathbf{Z}^n)$, we then have $f \in \mathcal{C}(\mathbf{T}^n)$ and

$$f(x) = \sum_{h \in \mathbf{Z}^n} \widehat{f}(h) e^{2i\pi h \cdot x}$$

for all $x \in \mathbf{T}^n$ (Theorem 3 of II, p. 222), where

$$\widehat{f}(h) = \int_{\mathbf{T}^n} f(x) e^{-2i\pi h \cdot x} dx, \quad h \in \mathbf{Z}^n.$$

For $f \in L^2(\mathbf{T}^n)$, the Fourier inversion formula (prop. 12 of II, p. 219) states that, if the series with general term $\widehat{f}(h)$ converges absolutely, we have

$$f(x) = \sum_{h \in \mathbf{Z}^n} \widehat{f}(h) e^{2i\pi h \cdot x}$$

for almost every x in $\mathbf{T}^n$.

However, even if f is continuous, the Fourier series of f does not in general converge to $f(x)$ for all x (exerc. 30 of II, p. 274). The following result is all the more useful.

PROPOSITION 23 (Fejér's Theorem). — Let $n \geq 1$ be an integer. For every $h = (h_i) \in \mathbf{Z}^n$, we denote by $|h| = \sup_i |h_i|$. Let $f \in \mathcal{C}(\mathbf{T}^n)$. For every integer $N \geq 1$, let f_N be the function on $\mathbf{T}^n$ such that

$$f_N(x) = \sum_{\substack{h \in \mathbf{Z}^n \\ |h| \leq N}} \widehat{f}(h) e^{2i\pi h \cdot x} \prod_{j=1}^n \left(1 - \frac{|h_j|}{N}\right)$$

for $x \in \mathbf{T}^n$. Then f_N converges to f in $\mathcal{C}(\mathbf{T}^n)$.

Lemma 11. — For every $N \geq 1$, let μ_N be the measure on $\mathbf{T}^n$ whose density is the continuous map

$$F_N: x \mapsto \sum_{\substack{h \in \mathbf{Z} \\ |h| \leq N}} e^{2i\pi h \cdot x} \prod_{j=1}^n \left(1 - \frac{|h_j|}{N}\right).$$

The sequence of measures $(\mu_N)_{N \geq 1}$ converges to ε_0 in the space $\mathcal{M}^1(\mathbf{T}^n)$ endowed with the topology of compact convergence in $\mathcal{C}(\mathbf{T}^n)$.

We reduce to the case $n = 1$ by noting that μ_N is the product of measures of the same type for $n = 1$. It then suffices to verify that the sequence (μ_N) satisfies the hypotheses of Lemma 4 of INT, VIII, §2, n°7 with $a = 0$.

To this end, we first note that F_N is the Fourier cotransform of the map $\varphi_N: h \mapsto (1 - |h|/N)$ on $\mathbf{Z}$. This is written as $\varphi_N = N^{-1} \psi_N * \widetilde{\psi}_N$, where ψ_N is the characteristic function of the set defined by $-N/2 < |h| \leq N/2$. Consequently, $F_N = N^{-1} |\mathcal{F}(\psi_N)|^2 \geq 0$. Thus, μ_N is a positive measure; we have $\mu_N(\mathbf{T}) = 1$, which proves (i) and (iii) in loc. cit.

Let us prove condition (ii) of loc. cit. Let U be an open neighborhood of 0 in $\mathbf{T}$. It suffices to show that $\mu_N(U) \rightarrow 1$ when $N \rightarrow +\infty$. Let K be a compact symmetric neighborhood of 0 such that $K^2 \subset U$ and ψ the characteristic function of K . Set $\varphi = \psi * \psi$. This is an element of $A(\mathbf{T})$ with support contained in U . The real number $m = \varphi(0)$ is the measure of the set K , and hence $m > 0$. Moreover, we have $0 \leq \varphi \leq m$ since $\varphi(x)$ is the measure of the set $K \cap xK$. We have

$$\mu_N(U) \geq \frac{1}{m} \int_{\mathbf{T}} \varphi(x) \mu_N(x) = \frac{1}{m} \sum_{h \in \mathbf{Z}} \mathcal{F}(\varphi)(h) \varphi_N(h)$$

by the transposition properties of the Fourier transform (prop. 13 of II, p. 221). Since $\varphi \in A(\mathbf{T})$, its Fourier transform belongs to $L^1(\mathbf{Z})$ and φ satisfies the Fourier inversion formula (prop. 11 of II, p. 217). Since $\varphi_N(h) \rightarrow 1$ for all $h \in \mathbf{Z}$ and $|\varphi_N(h)| \leq 1$, Lebesgue's theorem (INT, IV, §3, no. 7, th. 6) and the Fourier inversion formula imply that

$$\begin{aligned} \liminf_{N \rightarrow +\infty} \mu_N(U) &\geq \frac{1}{m} \lim_{N \rightarrow +\infty} \sum_{h \in \mathbf{Z}} \mathcal{F}(\varphi)(h) \varphi_N(h) = \\ &\frac{1}{m} \sum_{h \in \mathbf{Z}} \mathcal{F}(\varphi)(h) = \frac{1}{m} \varphi(0) = 1. \end{aligned}$$

Let us prove the proposition. We have $f * F_N = f_N$ for $N \geq 1$. The regular representation γ of $\mathbf{T}^n$ into $\mathcal{C}(\mathbf{T}^n)$ (INT, VIII, §2, no. 3) is continuous and satisfies $f * F_N = \gamma(\mu_N)f$ (INT, VIII, §4, no. 5, prop. 5 (iv)). The map $\mu \mapsto \gamma(\mu)f$ is continuous from $\mathcal{M}^1(\mathbf{T}^n)$ into $\mathcal{C}(\mathbf{T}^n)$ (INT,

VI, §1, no. 6, prop. 14). By the lemma, we therefore have

$$\lim_{N \rightarrow +\infty} f_N = \lim_{N \rightarrow +\infty} f * F_N = \lim_{N \rightarrow +\infty} \gamma(\mu_N)f = \gamma(\varepsilon_0)(f) = f$$

in $\mathcal{C}(\mathbf{T}^n)$.

Note. — There exist functions $f \in L^1(\mathbf{T})$ whose Fourier series diverges at every point $x \in T$ (Kolmogorov's theorem, cf. Exercise 51 of II, p. 289).

A theorem of Carleson ⁽¹⁾ proves that the symmetric partial sums of the Fourier series of f converge to $f(x)$ for almost all $x \in T$ if $f \in \mathcal{L}^2(T)$.

2.2 CLASSIFICATION

2.2.1 Groups generated by a compact subset

Lemma 1. — Let H be a locally compact group, and let R be one of the groups $\mathbf{R}$ or $\mathbf{Z}$. Let φ be a continuous morphism from R to H . If φ is not a topological isomorphism of R onto a subgroup of H , then the image of R in H is relatively compact.

Let I be the image of φ . Replacing H by $\bar{I}$, we may assume that I is dense in H . We must then show that H is compact if φ is not a topological isomorphism of R onto I .

Suppose there exists a neighborhood V of e in H and an integer $M > 0$ such that, for all $t > M$ in R , we have $\varphi(t) \notin V$. Then φ is injective: if $\varphi(u) = e$, one has $\varphi(nu) = e$ for every integer $n \geq 1$, hence the set $\mathbf{N}u$ is bounded in R , which means that $u = 0$. The restriction of φ to $[-M, M] \cap R$ is therefore a homeomorphism onto its image, which contains $V \cap I$. The restriction of φ^{-1} to $V \cap I$ being continuous, it follows that φ is a topological isomorphism of R onto I .

Suppose now that φ is not a topological isomorphism from R onto I . Let W be a relatively compact open neighborhood of e in H , and V a symmetric neighborhood of e such that $V^2 \subset W$. For every $x \in H = \bar{I}$, there exists an element $s \in R$ such that $x \in \varphi(s)V$. From the preceding paragraph and the hypothesis on φ , there exists $t \in R$ such that $t > |s|$ and $\varphi(t) \in V$. We then have $x \in \varphi(t+s)\varphi(t)^{-1}V \subset \varphi(t+s)W$, and $t+s > 0$. Consequently, the open sets $\varphi(u)W$

for $u > 0$ form an open covering of H . Since W is relatively compact, there exists an integer $n \geq 1$ and elements $u_1, \dots, u_n$ of $\mathbb{R}$, strictly positive, such that $\overline{W} \subset \bigcup_{1 \leq i \leq n} \varphi(u_i)W$. Let U be the largest of the u_i .

Let $x \in H$ and let $s = \inf\{t \in \mathbb{R} | t \geq 0, \varphi(t)x^{-1} \in \overline{W}\}$. Since $\overline{W}$ is compact, we then have $\varphi(s)x^{-1} \in \overline{W}$. There exists an integer i such that $\varphi(s)x^{-1} \in \varphi(u_i)W$, whence $\varphi(s-u_i)x^{-1} \in \overline{W}$. The definition of s implies $s - u_i < 0$, whence $s \leq U$. It follows that $H = \varphi([0, U] \cap \mathbb{R})\overline{W}$ is compact.

Lemma 2. — If G is generated by a compact subset V , then there exists an integer $n \geq 0$ and a discrete subgroup D of G isomorphic to $\mathbb{Z}^n$ such that G/D is compact.

Up to replacing V by $V \cup V^{-1}$, we may assume that V is symmetric; the hypothesis then means that G is the union of the sets V^n where $n \in \mathbb{N}$.

Since V^2 is compact, there exists an integer $k \geq 1$ and elements $x_1, \dots, x_k \in G$ such that $V^2 \subset \bigcup_{1 \leq i \leq k} x_i V$. Let D_0 be the subgroup of G generated by the family $(x_i)_{1 \leq i \leq k}$. We have $V^2 \subset D_0 V$, whence by induction $V^n \subset D_0 V$ for every integer $n \geq 1$, and hence $G = D_0 V$ since V generates G . Let then J be a subset of $\{1, 2, \dots, k\}$ such that the subgroup D generated by the family $(x_i)_{i \in J}$ is topologically isomorphic to $\mathbb{Z}^{\text{Card}(J)}$ and maximal for this property. Let us show that G/D is compact.

Let p be the canonical surjection from G onto G/D . Let $i \in \{1, 2, \dots, k\} - J$. If the subgroup H_i of G/D generated by $p(x_i)$ is topologically isomorphic to $\mathbb{Z}$, the subgroup of G generated by D and x_i is discrete and the map $(d, n) \mapsto dx_i^n$ is an isomorphism from $D \times \mathbb{Z}$ onto this subgroup, contrary to the maximality of J . Lemma 1 therefore implies that $\overline{H_i}$ is compact. Hence $G/D = (\prod_{i \notin J} \overline{H_i})p(V)$ is compact.

Lemma 3. — Let A and B be commutative groups such that A is divisible. Let C be a subgroup of B and φ a morphism from C to A . There exists a morphism from B to A extending φ .

Let $\mathcal{O}$ be the set of pairs (X, f) , where X is a subgroup of B containing C and f a morphism from X to A extending φ . Order $\mathcal{O}$ by the relation " $X \subset X'$ and f' extends f ". One verifies that $\mathcal{O}$ is inductive. Let (X, f) be a maximal element of $\mathcal{O}$ (E, III, p. 20, th. 2). If $X \neq B$, take an element b of $B-X$ and let X' be the subgroup generated by X and b . The commutativity of

B shows that X' is the set of elements $b^n x$ for $n \in \mathbf{Z}$ and $x \in X$. Suppose first that $b^n \notin X$ for every integer $n \neq 0$ and define f' as a morphism from X' to A by taking an arbitrary element $y \in A$ and setting $f'(b^n x) = y^n f(x)$ for all $n \in \mathbf{Z}$ and every $x \in X$. Since A is commutative, f' is a morphism, and it extends f . Suppose now that there exists $n \neq 0$ such that $b^n \in X$ and let $m > 0$ such that $m\mathbf{Z} = \{n \in \mathbf{Z} \mid b^n \in X\}$. Since A is divisible, there exists an element $y \in A$ such that $y^m = f(b^m)$. One then extends f to a morphism from X' to A by $f'(b^n x) = y^n f(x)$ for $n \in \{0, 1, \dots, m-1\}$ and $x \in X$. In both cases, (X, f) would not be maximal. Hence we have $X = B$ and the lemma is proved.

Remark. — In the language of categories, the lemma says that divisible groups are injective objects in the category of commutative groups; cf. A, VII, p. 53, exerc. 3.

PROPOSITION 1. — The following conditions are equivalent:

- (i) G is generated by a compact subset;
- (ii) there exist positive integers p and q and a compact group K such that G is isomorphic to $R^p \times Z^q \times K$;
- (iii) there exists an integer $n \geq 0$ such that $\widehat{G}$ is locally isomorphic to R^n ;
- (iv) there exist positive integers p and q and a discrete group D such that $\widehat{G}$ is isomorphic to $R^p \times T^q \times D$.
- (v) $\Rightarrow$ (iii): if G has property (i), there exists an integer $n \geq 0$ and a subgroup D of G isomorphic to Z^n such that G/D is compact (lemma 2). Then $D^\perp$, which is identified with the dual of G/D , is discrete (th. 4 of II, p. 226 and prop. 18 of II, p. 233). Hence $\widehat{G}$ is locally isomorphic to $\widehat{G}/D^\perp$, that is to say to $\widehat{D}$, which is isomorphic to T^n (th. 4 of II, p. 226 and prop. 18 of II, p. 233). Now T^n is locally isomorphic to R^n .
- (vi) $\Rightarrow$ (iv): if $\widehat{G}$ is locally isomorphic to $\mathbf{R}^n$, there exists an integer p such that $0 \leq p \leq n$ so that the neutral component $\widehat{G}_0$ of $\widehat{G}$ is an open subgroup isomorphic to $\mathbf{R}^p \times \mathbf{T}^{n-p}$ (TG, VII, p. 13, th. 1). In particular, $\widehat{G}_0$ is a divisible group. Applying lemma 3 to the identity map of the subgroup $\widehat{G}_0$ of the group $\widehat{G}$ into the divisible group $\widehat{G}_0$, there exists a morphism π of $\widehat{G}$ into $\widehat{G}_0$ which is the identity map on $\widehat{G}_0$. Consequently, we have $\pi \circ \pi = \pi$.

, and π is a projector. It is continuous, since its restriction to the open subgroup $\widehat{G}_0$ is. Hence, $\widehat{G}$ is the direct product of $\widehat{G}_0$ and the subgroup $\overline{\pi}^{-1}(e)$, which is discrete since it is isomorphic to $\widehat{G}/\widehat{G}_0$ (TG, III, p. 47, cor.) (iv) $\Rightarrow$ (ii): follows from prop. 2 of II, p. 206, from prop. 18 of II, p. 233 and from corollary 3 of II, p. 236.

(vii) $\Rightarrow$ (i): for every compact group K , the group $R^p \times Z^q \times K$ is generated by the compact set $[0, 1]^p \times \{0, 1\}^q \times K$.

COROLLARY 1. — Suppose that G is generated by a compact neighborhood of e .

- a) There exists a compact subgroup K of G and positive integers p and q such that G is isomorphic to $R^p \times Z^q \times K$.
- b) Conversely, let K be a compact group, let p and q be positive integers, and let G be a group isomorphic to $R^p \times Z^q \times K$. Then K is the unique maximal compact subgroup of G , and the integers (p, q) are uniquely determined by G .

Assertion a) follows from prop. 1. Let then K be a compact group, and p, q positive integers. Suppose that G is isomorphic to the group $R^p \times Z^q \times K$, and we identify G with this group. Under the canonical projection of G onto $R^p \times Z^q$, the image of every compact subgroup of G is a compact subgroup of $R^p \times Z^q$; therefore it is reduced to the identity element. Therefore $K' \subset K$ and K is the largest compact subgroup of G . The subgroup $R^p \times K$ is also unique because R^p is the neutral component of G/K . Taking into account TG, VII, p. 13, cor. 3, the integer p is uniquely determined by G . Since $G/(R^p \times K)$ is isomorphic to Z^q , the integer q is also uniquely determined by G .

Note. — By duality, $\widehat{G}$ is isomorphic to $R^p \times T^q \times D$ (prop. 3 (iv)), where the subgroups $R^p \times T^q$ and T^q , and the integers p and q , are uniquely determined.

COROLLARY 2. — The following conditions are equivalent:

- (i) The groups G and $\widehat{G}$ are generated by compact subsets;
- (ii) There exist positive integers n and m such that G is locally isomorphic to R^m and $\widehat{G}$ to R^n ;
;

(iii) There exist positive integers p, q , and r and a finite group A such that G is isomorphic to

$$R^p \times T^q \times Z^r \times A ;$$

(iv) There exist positive integers p, q , and r and a finite group A such that $\widehat{G}$ is isomorphic to a product $R^p \times Z^q \times T^r \times A$.

We have (i) $\Leftrightarrow$ (ii) by Prop. 1, and (iii) $\Leftrightarrow$ (iv) by duality, hence (iii) $\Rightarrow$ (i). Finally, if (i) is true, then $\widehat{G}$ is isomorphic to $\mathbf{R}^p \times \mathbf{T}^r \times \mathbf{D}$, where $\mathbf{D}$ is discrete (prop. 1). The group $\mathbf{D}$ is generated by a compact subset and is therefore finite. Consequently, there exists $q \geq 0$ such that $\mathbf{D}$ is isomorphic to $\mathbf{Z}^q \times \mathbf{A}$, where $\mathbf{A}$ is a finite group (A, VII, p. 22, th. 3).

Note. — With the notations of Cor. 2, if one identifies G with $R^p \times T^q \times Z^r \times A$ the subgroup $R^p \times T^q$ is the identity component of G , the subgroup $T^q \times A$ is its largest compact subgroup and T^q is the identity component of it; the integers p, q, r are uniquely determined by G according to the preceding remark, and the group A is determined by G up to isomorphism.

PROPOSITION 2. — Suppose G is compact. There exists a decreasing filtered family $(H_i)_{i \in I}$ of closed subgroups of G such that

- a) the group G is identified with the projective limit of the G/H_i ;
- b) for every i , there exists an integer $q \geq 0$ and a finite group A such that G/H_i is isomorphic to $T^q \times A$.

Indeed, $\widehat{G}$ is discrete (prop. 18 of II, p. 233), hence is the union of an increasing filtered family $(D_i)_{i \in I}$ of finitely generated subgroups. Set $H_i = D_i^\perp$; the group G is identified with the projective limit of the G/H_i (II, p. 234, cor. 3), and (H_i) is a decreasing filtered family.

Let $i \in I$. There exists an integer $q \geq 0$ and a finite group A such that the group D_i is isomorphic to $Z^q \times A$ (A, VII, p. 22, th. 3), hence G/H_i is isomorphic to $T^q \times \widehat{A}$ (cf. corollary 1 of II, p. 232).

COROLLARY. — If the group G is generated by a compact subset, then it is a projective limit of groups isomorphic to groups of the form $R^p \times T^q \times Z^r \times A$, where A is a finite group and p, q, r are positive integers.

By (ii) of prop. 1 of II, p. 246, the corollary follows from prop. 2.

2.2.2 General case

In this section, G denotes a locally compact commutative group.

PROPOSITION 3. — a) There exists an integer $n \geq 0$ and a subgroup L such that G is the direct product of L and a subgroup isomorphic to $\mathbf{R}^n$, and moreover L admits a compact open subgroup K such that L/K is discrete;

b) The group G is the union of an increasing filtered family of open subgroups, each being a projective limit of groups isomorphic to groups of the form $R^p \times T^q \times Z^r \times A$, where A is a finite group and p, q, r are positive integers.

We prove *b*). For every compact neighborhood V of e , let G_V denote the subgroup of G generated by V . It is open, and by the corollary of prop. 2, the group G_V is a projective limit of groups of the form $\mathbf{R}^p \times \mathbf{T}^q \times \mathbf{Z}^r \times A$, where A is a compact group and p, q and r in $\mathbf{N}$. When V ranges over the compact neighborhoods of e , these subgroups G_V form a filtered family (since G_V and G_W are contained in $G_{V \cup W}$ for all compact neighborhoods V and W of e). Finally, the group G is the union of the subgroups G_V .

Let H be an open subgroup of G generated by a compact neighborhood of e . There exist a compact group K and positive integers p and q such that H is isomorphic to $R^p \times Z^q \times K$ (prop. 1 of II, p. 246); identify H with this product. The canonical surjection of H onto the divisible group R^p extends to a morphism π of G onto R^p (lemma 3 of II, p. 245). It is a projector π of G onto R^p , which is continuous since its restriction to the open subgroup H is continuous. Hence G is the direct product of R^p and of the kernel L of π (TG, III, p. 47, cor.). We have $Z^q \times K = H \cap L$, hence $Z^q \times K$ is an open subgroup of L . Thus K is a compact open subgroup of L , and consequently L/K is discrete.

PROPOSITION 4. — Let B_G be the set of elements of G that generate a relatively compact subgroup of G . Then B_G is a closed subgroup of G and $B_G^\perp$ is the identity component of $\widehat{G}$.

The set B_G is a subgroup of G , since the product of two compact subsets of G is a compact subset of G .

Let H be an open subgroup of G generated by a compact neighborhood of e . There exist positive integers p and q and a compact group K such that the subgroup H is isomorphic to

$R^p \times Z^q \times K$ (prop. 1 of II, p. 246). If one identifies these groups, one sees that $B_G \cap H = K$ is closed in H . Since the family of open subgroups H generated by compact neighborhoods is an open cover of G (for example, x belongs to the subgroup generated by $U \cap \{x\}$ for every fixed compact neighborhood U of e), it follows that B_G is closed.

Now let us calculate $B_G^\perp$. By prop. 3, a), there exists a positive integer $n \geq 0$ and a group L admitting an open compact subgroup such that G can be identified with $R^n \times L$. Then B_G is identified with $\{0\} \times B_L$, and $B_G^\perp$ with $R^n \times B_L^\perp$. We are thus reduced to the case where $G = L$ admits a compact open subgroup K .

We then have $K \subset B_G$. If one identifies $\widehat{G/K}$ with $K^\perp$ (Theorem 4 of II, p. 226), the orthogonal $(B_G/K)^\perp$ of B_G/K in $\widehat{G/K}$ is identified with $B_G^\perp$. But on the other hand $B_G/K = B_{G/K}$, and since $K^\perp$ is an open subgroup of $\widehat{G}$ (cor. 2 of II, p. 233), the identity component $\widehat{G}_0$ of $\widehat{G}$ is also the identity component of $K^\perp$. Thus the assertion for the discrete group G/K is equivalent to that for G .

Finally, suppose that G is discrete. The group $\widehat{G}$ is then compact (prop. 18 of II, p. 233). The neutral component $(\widehat{G})_0$ is the intersection of the open subgroups of $\widehat{G}$ (TG, III, p. 35, prop. 14) and a subgroup of $\widehat{G}$ is open if and only if it is closed and of finite index, or equivalently if its orthogonal is finite (Corollary 2 of II, p. 233); Corollary 4 of II, p. 228 shows that $(\widehat{G})_0^\perp$ is the union of the finite subgroups of G , which is none other than B_G since G is discrete. We conclude by duality.

COROLLARY 1. — Suppose G is compact. Then the following conditions are equivalent:

- (i) The group G is connected;
- (ii) The group $\widehat{G}$ is torsion-free;
- (iii) The group G is divisible.

This follows from prop. 4 and cor. 7 of II, p. 229, since $B_G = G$.

COROLLARY 2. — Suppose G is compact. Then G is totally disconnected if and only if $\widehat{G}$ is a torsion group.

The group G is totally disconnected if and only if its identity component is reduced to $\{e\}$; prop. 4 shows that this condition is equivalent to $B_{\widehat{G}} = \widehat{G}$. Since the group $\widehat{G}$ is discrete (prop.

18 of II, p. 233), this amounts to saying that each element of $\widehat{G}$ generates a finite group, hence that $\widehat{G}$ is torsion.

COROLLARY 3. — If G is connected, then G is divisible.

Indeed, there exist a compact connected group K and an integer $n \geq 0$ such that G is isomorphic to $R^n \times K$ (II, p. 248, prop. 3, where one must have $L = K$). Corollary 1 shows that K is divisible, and therefore G is divisible.

2.3 INVARIANT SUBSPACES

The goal of this section is the study of certain subspaces invariant under translation in the spaces $L^1(G)$, $L^2(G)$ and $L^\infty(G)$.

2.3.1 The case of Hilbert space $L^2(G)$

For every measurable subset M of $\widehat{G}$, we denote by E_M the set of $f \in L^2(\mathbf{G})$ such that the Fourier transform $\mathcal{F}_G(f)$ is zero almost everywhere on $\widehat{G}$. Let φ_M be the characteristic function of M . The space E_M is the kernel of the continuous linear map $f \mapsto \varphi_M \mathcal{F}_G(f)$ of $L^2(\mathbf{G})$ into $L^2(\widehat{\mathbf{G}})$, and it is therefore a closed subspace of $L^2(\mathbf{G})$.

PROPOSITION 1. — a) Let M be a measurable subset of $\widehat{G}$. For every $x \in G$, the space E_M is stable under the map $f \mapsto \varepsilon_x * f$;

b) Let M and N be measurable subsets of G . One has $E_M = E_N$ if and only if M and N are equal up to a locally negligible set;

c) Every subspace of $L^2(G)$ stable under the maps $f \mapsto \varepsilon_x * f$ for all $x \in G$ is of the form E_M for some measurable subset M of $\widehat{G}$.

This result will be proved later (cf. V, to appear).

2.3.2 Closed ideals of $L^1(G)$

The Fourier cotransform on the Banach algebra $L^1(G)$ is identified with the Gelfand transform of $L^1(G)$ (II, p. 209). With this identification, recall that if I is an ideal of $L^1(G)$ we denote by

$V(I)$ the closed set in $\widehat{G}$ of characters $\chi \in \widehat{G}$ such that, for every function $f \in I$, the Fourier cotransform of f vanishes at χ (cf. I, p. 30). For every subset M of $\widehat{G}$, we denote by $\Upsilon(M)$ the closed ideal consisting of the $f \in L^1(G)$ such that $\overline{\mathcal{F}_G(f)}$ vanishes on M (I, p. 30).

By prop. 2 of II, p. 219, the Banach algebra $L^1(G)$ is regular. By § 5 of I, p. 88, we therefore deduce the following properties of the Fourier transform and cotransform:

- 1) If F is a closed subset of $\widehat{G}$ and K is a compact subset of $\widehat{G}$ such that $F \cap K = \emptyset$, there exists a function $f \in L^1(G)$ such that $\mathcal{F}_G(f)$ is equal to 0 on F and to 1 on K (I, p. 88, prop. 1; for this fact and the following ones, one passes from the Fourier cotransform to the Fourier transform by means of formula (8) of II, p. 207).
- 2) Let M be a closed subset of $\widehat{G}$. The set of ideals I of $L^1(G)$ such that $V(I) = M$ has $\Upsilon(M)$ as its greatest element and, as its smallest element, the set of $f \in L^1(G)$ whose Fourier cotransform has compact support disjoint from M (I, p. 91, prop. 4).
- 3) Let I be an ideal of $L^1(G)$, and $g: \widehat{G} \rightarrow C$ a continuous function. We assume that for every $\chi \in \widehat{G}$, there exists a function $f_\chi \in I$ such that g is equal to $\mathcal{F}_G(f_\chi)$ in a neighborhood of χ . We assume in addition that there exists a function $f_\infty \in I$ such that g is equal to $\mathcal{F}_G(f_\infty)$ in the complement of a compact subset of $\widehat{G}$, this last condition being always satisfied if G is discrete. Then there exists a function $f \in I$ such that $g = \mathcal{F}_G(f)$ (I, p. 91, cor. 2).

Lemma 1. — The space of functions in $L^1(G)$ whose Fourier transform has compact support is dense in $L^1(G)$.

Since $\mathcal{K}(\widehat{G})$ is dense in $L^2(\widehat{G})$ and since the Fourier transform of $L^2(G)$ is an isometry on $L^2(\widehat{G})$ (Theorem 1 of II, p. 215), the subspace V of $L^2(G)$ consisting of the $f \in L^2(G)$ such that $\mathcal{F}_G(f) \in \mathcal{K}(\widehat{G})$ is dense in $L^2(G)$.

Let $g \in L^1(G)$. There exists $g_1, g_2 \in L^2(G)$ such that $g = g_1 g_2$ (one can, for example, take $g_1 = |g|^{1/2}$, and $g_2(x) = 0$ if $g(x) = 0$, $g_2(x) = g(x)/g_1(x)$ otherwise). From the above, we therefore deduce that g is the limit of a sequence of functions of the form $h_1 h_2$, where h_1 and h_2 belong to V . However, $\mathcal{F}_G(h_1 h_2) = \mathcal{F}_G(h_1) * \mathcal{F}_G(h_2)$ (II, p. 223, prop. 14), and $\mathcal{F}_G(h_1) * \mathcal{F}_G(h_2)$ belongs to $\mathcal{K}(G)$. The lemma follows.

PROPOSITION 2. — Let I be a closed ideal of $L^1(G)$, and let $f \in L^1(G)$. If $\overline{\mathcal{F}_G}(f)$ vanishes on a neighborhood of $V(I)$, then f belongs to I .

Let $\varepsilon > 0$. There exists a function $g \in L^1(G)$ such that $\|f - f * g\|_1 < \varepsilon$ (prop. 8 of II, p. 211 (iv)). Let $h \in L^1(G)$ such that the support of $\overline{\mathcal{F}_G}(h)$ is compact and $\|f\|_1 \|g - h\|_1 < \varepsilon$ (lemma 1). We have

$$\|f - f * h\|_1 \leq \|f - f * g\|_1 + \|f * (g - h)\|_1 < 2\varepsilon.$$

By the hypothesis on f , the function $\overline{\mathcal{F}_G}(f * h) = \overline{\mathcal{F}_G}(f) \overline{\mathcal{F}_G}(h)$ is compactly supported and disjoint from $V(I)$, which implies that $f * h \in I$ (remark 2 above). Since ε is arbitrarily small, we have $f \in \bar{I} = I$.

THEOREM 1. — Let I be a closed ideal of $L^1(G)$ distinct from $L^1(G)$. There exists a character $\hat{x} \in \widehat{G}$ such that $\mathcal{F}_G(f)(\hat{x}) = 0$ for every $f \in I$.

Since the algebra $L^1(G)$ is regular (prop. 2 of II, p. 219) and semisimple (cor. of prop. 22 of I, p. 126), and since the set of functions whose Fourier cotransform has compact support is dense in

$L^1(G)$ (lemma 1), cor. 1 of I, p. 92 shows that the ideal I is contained in a regular maximal ideal of $L^1(G)$, that is, in the kernel of a character $\hat{y}$ of $L^1(G)$ (th. 2 of I, p. 30); one can then take $\hat{x} = \hat{y}^{-1}$ (cf. formula (8) of II, p. 207).

COROLLARY 1 (Wiener's tauberian theorem)

Let $f \in L^1(G)$. If the Fourier transform of f does not vanish, the functions $f * \varepsilon_x : g \mapsto f(gx^{-1})$, where x ranges over G , form a total set in $L^1(G)$ (EVT, I, p. 12, def. 1).

Let V be the closed vector subspace of $L^1(G)$ generated by the $f * \varepsilon_x$. By INT, VIII, §4, cor. of prop. 20, the space V is a closed ideal of $L^1(G)$. By th. 1 we have $V = L^1(G)$.

DEFINITION 1. — Let g be a complex function on G and let Φ be a filter on G . We say that g is slowly oscillating along Φ if, for every $\varepsilon > 0$, there exists a set $M \in \Phi$ and a neighborhood V of e in G such that

$$x \in M \quad \text{and} \quad y \in V \quad \Rightarrow \quad |g(xy) - g(x)| \leq \varepsilon.$$

COROLLARY 2. — Let Φ be a filter on G invariant under translation. Let $f \in L^1(G)$ such that the Fourier transform of f does not vanish and such that $\int_G f(x)dx = 1$. Let $g \in L^\infty(G)$. We assume that $f * g$ has a finite limit α along Φ .

- a) For every function $h \in L^1(G)$ such that $\int_G h(x)dx = 1$, the limit of $h * g$ along Φ is equal to α ;
- b) Suppose moreover that g is slowly oscillating along Φ . Then g tends to α along Φ .

By replacing g by $g - \alpha$, one reduces to the case where $\alpha = 0$. Let I be the set of functions $h \in L^1(G)$ such that $h * g$ tends to 0 along Φ . The set I is a translation-invariant vector subspace of $L^1(G)$. It is a closed vector subspace. Indeed, let $h \in \bar{I}$. For every function $h_0 \in L^1(G)$ and every $x \in G$, one has

$$\begin{aligned} |(h * g)(x)| &\leq |((h - h_0) * g)(x)| + |(h_0 * g)(x)| \\ &\leq \|h - h_0\|_1 \|g\|_\infty + |(h_0 * g)(x)|. \end{aligned}$$

For every $\varepsilon > 0$, there exists $h_0 \in I$ such that $\|h - h_0\|_1 \|g\|_\infty < \varepsilon$. Let $M \in \Phi$ such that $|(h_0 * g)(x)| < \varepsilon$ for all $x \in M$. We then have $|(h * g)(x)| < 2\varepsilon$ for all $x \in M$, hence $h * g$ converges to 0 along Φ . This shows that $h \in I$.

The space I is therefore a closed ideal of $L^1(G)$. We have $f \in I$ by hypothesis, therefore $I = L^1(G)$ by th. 1, since the Fourier transform of f does not vanish. This implies a).

Assume now the hypotheses of b). Let $\varepsilon > 0$. Since g is slowly oscillating along Φ , there exists $M \in \Phi$ and a compact neighborhood V of e such that

$$x \in M \quad \text{and} \quad y \in V \implies |g(y^{-1}x) - g(x)| \leq \varepsilon.$$

Let φ be the characteristic function of V and $\mu = \int \varphi(x)dx$. For every $x \in G$, one has

$$\frac{1}{\mu}(\varphi * g)(x) = \frac{1}{\mu} \int_V g(y^{-1}x)dy = g(x) + \frac{1}{\mu} \int_V (g(y^{-1}x) - g(x))dy.$$

Therefore for all $x \in M$, one has

$$\left| \frac{1}{\mu} (\varphi * g)(x) - g(x) \right| \leq \varepsilon.$$

Since, by a), the limit of $\varphi * g$ along Φ is zero, we have $\limsup_{\Phi} |g| \leq \varepsilon$. Since ε is arbitrary, we conclude that the limit of g along Φ is zero, which proves b).

Lemma 2. — Let K be a compact subset of G . For every $\eta > 0$, there exists a function $j \in L^1(G)$ such that:

- a) $\|j\|_1 \leq \sqrt{2}$;
- b) the function $\mathcal{F}_G(j)$ is equal to 1 in a neighborhood of the identity element of $\widehat{G}$;
- c) for all $x \in K$, one has $\|j - j * \varepsilon_x\|_1 \leq \eta$.

The set U_1 of elements $\hat{x} \in \widehat{G}$ such that

$$|\langle \hat{x}, x \rangle - 1| \leq \frac{\eta}{4}$$

for all $x \in K$ is a neighborhood of e in $\widehat{G}$. Let $U \subset U_1$ be an open, symmetric neighborhood integrable with respect to Haar measure $m = d\hat{x}$ of $\widehat{G}$ dual to the measure dx . Let $V \subset U$ be a compact symmetric neighborhood of e such that $m(V) \geq \frac{1}{2}m(U)$. Denote by φ_U (resp. φ_V) the characteristic function of U (resp. of V). Since φ_U belongs to $L^2(G)$, there exists $u \in L^2(G)$ such that $\varphi_U = \mathcal{F}_G(u)$ (th. 1 of II, p. 215). Similarly, there exists a function $v \in L^2(G)$ such that $\varphi_V = \mathcal{F}_G(v)$. We shall show that the function $j = \frac{1}{m(V)}uv$ satisfies the required properties. We have $j \in L^1(G)$.

- a) By the Plancherel theorem and the condition $m(V) \geq \frac{1}{2}m(U)$, one has

$$\|j\|_1 \leq \frac{\|u\|_2 \|v\|_2}{m(V)} = \frac{\|\mathcal{F}_G(u)\|_2 \|\mathcal{F}_G(v)\|_2}{m(V)} = \frac{\sqrt{m(U)m(V)}}{m(V)} \leq \sqrt{2}.$$

- b) There exists a neighborhood W of e in $\widehat{G}$ such that $WV \subset U$ (TG, II, p. 31, prop. 4). For every $\hat{x} \in W$, one has $\hat{x}V \subset U$, and prop. 14 of II, p. 223 implies

$$\begin{aligned}
\mathcal{F}_G(j)(\hat{x}) &= \frac{1}{m(V)} (\mathcal{F}_G(u) * \mathcal{F}_G(v))(\hat{x}) \\
&= \frac{1}{m(V)} \int_{\widehat{G}} \varphi_U(\hat{y}) \varphi_V(\hat{y}^{-1}\hat{x}) dm(\hat{y}) \\
&= \frac{m(U \cap \hat{x}V^{-1})}{m(V)} = \frac{m(\hat{x}V)}{m(V)} = 1
\end{aligned}$$

since V is symmetric.

c) If $x \in K$, one has

$$\|u - u * \varepsilon_x\|_2^2 = \int_{\widehat{G}} |\mathcal{F}_G(u)(\hat{x})(1 - \overline{\langle x, \hat{x} \rangle})|^2 dm(\hat{x}) \leq m(U) \left(\frac{\eta}{4}\right)^2$$

since $U \subset U_1$, and likewise $\|v - v * \varepsilon_x\|_2^2 \leq m(V) \left(\frac{\eta}{4}\right)^2$. Therefore

$$\begin{aligned}
\|j - j * \varepsilon_x\|_1 &= \frac{1}{m(V)} \|u(v - v * \varepsilon_x) + (v * \varepsilon_x)(u - u * \varepsilon_x)\|_1 \\
&\leq \frac{\eta}{4m(V)} (\|u\|_2 \sqrt{m(V)} + \|v\|_2 \sqrt{m(U)}) \\
&= \frac{\eta \sqrt{m(U)m(V)}}{2m(V)} < \eta.
\end{aligned}$$

PROPOSITION 3. — The algebra $L^1(G)$ satisfies the Ditkin condition (I, p. 92, def. 2).

Let χ be a character of $L^1(G)$. We distinguish two cases according as χ is zero or not. If χ is zero, it must be verified that for every function $f \in L^1(G)$ there exists a sequence $(f_n)_{n \geq 1}$ in $L^1(G)$ such that $\mathcal{F}(f_n)$ vanishes outside a compact subset of $\widehat{G}$ and such that $f_n * f$ tends to f in $L^1(G)$. The existence of such a sequence follows from Lemma 1 above and from Prop. 8 of II, p. 211.

Suppose now that χ is nonzero, hence $\chi \in X(L^1(G)) = \widehat{G}$ (prop. 1 of II, p. 202). Let $f \in L^1(G)$ such that $\mathcal{G}_{L^1(G)}(f)(\chi) = \overline{\mathcal{F}(f)}(\chi) = 0$. It remains to prove the existence of a sequence $(f_n)_{n \geq 1}$ in

$L^1(G)$ such that $f * f_n$ converges to f in $L^1(G)$ and such that $\overline{\mathcal{F}(f_n)}$ vanishes in a neighborhood of χ . One may assume that $\|f\|_1 = 1$. By translation in $\widehat{G}$, one reduces to the case where $\chi = e$.

Let K_n be a compact subset of G such that

$$\int_{G-K_n} |f(x)| dx \leq \frac{1}{n}.$$

Let $u_n \in L^1(G)$ be a function ≥ 0 such that $\|u_n\|_1 = 1$ and

$$\|f - f * u_n\|_1 \leq \frac{1}{n}$$

(cf. prop. 8 of II, p. 211, (iii)). By Lemma 2, there exists a function j_n in $L^1(G)$ such that $\|j_n\|_1 \leq \sqrt{2}$ whose Fourier cotransform equals 1 in a neighborhood of e and moreover such that $\|j_n - j_n * \varepsilon_x\|_1 \leq n^{-1}$ for all $x \in K_n$. We set

$$f_n = u_n - j_n * u_n.$$

We shall show that the sequence $(f_n)_{n \geq 1}$ has the required properties. First of all, we have

$$\mathcal{F}(f_n) = \mathcal{F}(u_n) - \mathcal{F}(j_n)\mathcal{F}(u_n) = (1 - \mathcal{F}(j_n))\mathcal{F}(u_n)$$

so the Fourier transform of f_n vanishes in a neighborhood of $\chi = e$. On the other hand,

$$\|f * f_n - f\|_1 \leq \|f * u_n - f\|_1 + \|f * j_n\|_1 \|u_n\|_1 \leq \frac{1}{n} + \|f * j_n\|_1.$$

Now, for almost every $y \in G$, one has

$$(f * j_n)(y) = \int_G f(x) j_n(x^{-1}y) dx = \int_G f(x) (j_n(x^{-1}y) - j_n(y)) dx$$

since, by hypothesis, we have $\mathcal{F}(f)(e) = \int_G f(x) dx = 0$. Hence

$$\begin{aligned} \|f * j_n\|_1 &\leq \int_G |f(x)| \|j_n * \varepsilon_x - j_n\|_1 dx \\ &= \int_{K_n} |f(x)| \|j_n * \varepsilon_x - j_n\|_1 dx \\ &\quad + \int_{G-K_n} |f(x)| \|j_n * \varepsilon_x - j_n\|_1 dx \\ &\leq \frac{1}{n} \int_{K_n} |f(x)| dx + 4 \int_{G-K_n} |f(x)| dx \leq \frac{5}{n}. \end{aligned}$$

Finally, $\|f * f_n - f\|_1 \leq 6n^{-1}$ and therefore $f * f_n$ converges to f in $L^1(G)$, as desired.

Applying prop. 5, we then obtain the following result:

THEOREM 2. — Let I be a closed ideal of $L^1(G)$ such that the boundary of $V(I)$ contains no nonempty perfect set. Then I is the set of functions $f \in L^1(G)$ such that $\mathcal{F}(f)$ vanishes on

$V(I)$.

For any closed ideal of $L^1(G)$, the conclusion of Thm. 2 is in general false (cf. Exerc. 12 of II, p. 314). More precisely, one can show that if G is noncompact, there exists a closed ideal of $L^1(G)$ which is not self-adjoint (see, for example, W. RUDIN, Fourier analysis on groups, Interscience tracts in pure and applied mathematics, theorem 7.7.1.)

COROLLARY. — If a closed ideal I of $L^1(G)$ is contained in a single regular maximal ideal, then I is itself regular maximal.

2.3.3 Weakly closed invariant subspaces of $L^\infty(G)$

In this issue, we identify $L^\infty(G)$ with the dual of $L^1(G)$, and we endow it with the weak topology $\sigma(L^\infty(G), L^1(G))$. One denotes by $(f, g) \mapsto \langle f, g \rangle$ the bilinear map defining this duality for $f \in L^1(G)$ and $g \in L^\infty(G)$.

The map $W \mapsto W^\circ$ is a bijection from the set of weakly closed vector subspaces of $L^\infty(G)$ onto the set of closed vector subspaces of $L^1(G)$ (EVT, II, p. 55, prop. 10).

On the other hand, if $f \in L^1(G)$ and $x \in G$, the endomorphism $g \mapsto f * g$ (resp. $g \mapsto \varepsilon_x * g$) of the Banach space $L^1(G)$ has as its transpose the endomorphism $h \mapsto \check{f} * h$ (resp. $h \mapsto \varepsilon_{x^{-1}} * h$) of the Banach space $L^\infty(G)$ (INT, VIII, §4, no. 3, example 6). For a closed vector subspace of $L^1(G)$ to be an ideal of $L^1(G)$, it is necessary and sufficient that it be invariant under translations of G . Therefore, for a weakly closed vector subspace of $L^\infty(G)$ to be stable under convolution with elements of $L^1(G)$, it is necessary and sufficient that it be invariant under translations of G .

Let W be a weakly closed vector subspace of $L^\infty(G)$. Suppose W (and hence also W°) is invariant under translations of G . Let $f \in L^1(G)$. For every $g \in L^\infty(G)$, one has $(\check{f} * g)(x) = \langle \varepsilon_x * f, g \rangle = \langle f, \varepsilon_{x^{-1}} * g \rangle$. Therefore, for f to belong to W° , it is necessary and sufficient that $\check{f} * g = 0$ for all $g \in W$.

If W is a weakly closed vector subspace of $L^\infty(G)$ invariant under translations of G , we denote by $A(W)$ the set of characters $\chi \in \widehat{G}$ that belong to W . This is a closed subset of $\widehat{G}$. If F is a closed subset of $\widehat{G}$, we denote by $Y(F)$ the weakly closed vector subspace of $L^\infty(G)$ generated by the elements of F ; since every translation of G transforms each character into a

function proportional to that character, the space $Y(F)$ is translation-invariant.

Let W be a weakly closed subspace of $L^\infty(G)$ invariant under translations of G . By the bipolar theorem (EVT, II, p. 48, th. 1), a character χ belongs to W if and only if it belongs to $(W^\circ)^\circ$; this latter space is the set of functions $g \in L^\infty(G)$ such that $\langle f, g \rangle = 0$ for every $f \in W^\circ$. We have $\langle f, \chi \rangle = \overline{\mathcal{F}(f)}(\chi)$, and hence

$$A(W) = V(W^\circ).$$

Similarly, a function $f \in L^1(G)$ belongs to $\Upsilon(F)^\circ$ if and only if $\langle f, \chi \rangle = 0$ for all $\chi \in F$, which is equivalent to $\overline{\mathcal{F}(f)}(\chi) = 0$ for $\chi \in F$, that is to say to $f \in \Upsilon(F)$. Therefore (loc. cit.) we have

$$Y(F) = \Upsilon(F)^\circ.$$

The relations $V(\Upsilon(F)) = F$ (I, p. 13 and I, p. 30) and $\Upsilon(V(I)) \supset I$, combined with the bipolar theorem (EVT, II, p. 48, th. 1), then imply

$$A(Y(F)) = F, \quad Y(A(W)) \subset W.$$

PROPOSITION 4. — Let W be a nonzero weakly closed translation-invariant vector subspace of $L^\infty(G)$. Then W contains at least one character of G .

We have seen that $A(W) = V(W^\circ)$. Since $W \neq 0$, one has $W^\circ \neq L^1(G)$, and then $V(W^\circ)$ is nonempty by th. 1 of II, p. 252.

PROPOSITION 5. — Let W be a weakly closed translation-invariant vector subspace of $L^\infty(G)$.

- a) For any neighborhood U of $A(W)$ in $\widehat{G}$, every function in W is a weak limit of linear combinations of characters belonging to U ;
- b) If the boundary of $A(W)$ contains no nonempty perfect set, every function of W is a weak limit of linear combinations of characters belonging to W .

To prove *a*), it suffices by the bipolar theorem to show that if f is a function of $L^1(G)$

orthogonal to the elements of U , then f is orthogonal to W . Now, the Fourier cotransform $\overline{\mathcal{F}}(f)$ then vanishes on the neighborhood U of $A(W) = V(W^\circ)$, so that prop. 2 of II, p. 252 indeed shows that $f \in W^\circ$. Assertion b) is established in an analogous manner, using th. 2 of II, p. 257 instead of prop. 2 of II, p. 252.

Exercises for Section 1: Fourier Transformation

In all exercises of Chapter II, we identify the dual of R^n (resp. of $(\mathbf{R}/\mathbf{Z})^n$, of Z^n) with R^n (resp. with Z^n , with $(\mathbf{R}/\mathbf{Z})^n$) according to corollary 3 of II, p. 236. For x and y in R^n , we denote by $x \cdot y = \sum_i x_i y_i$. We set $T = \mathbf{R}/\mathbf{Z}$, and we endow T with its structure of real Lie group (LIE, III, p. 105, prop. 11). We shall often denote by $\hat{f}$ the Fourier transform of a function f .

For every nonzero real number t , we denote $s(t) = t/|t|$, and we set $s(0) = 0$ (sign function).

If E is a topological vector space and $(x_h)_{h \in \mathbf{Z}}$ a family of elements of E , we say that the series with general term x_h converges symmetrically in E to $x \in E$ if the sequence $(s_n)_{n \geq 1}$ defined by

$$s_n = \sum_{-n \leq h \leq n} x_h$$

converges to x in E .

A probability measure on a locally compact topological space X is a positive measure of total mass 1. We denote by $\mathcal{P}(X)$ the set of probability measures on X .

Unless stated otherwise, G denotes a locally compact commutative topological group.

- 1) a) Let $\omega, p, q \in \mathbf{R}$ with $p \leq q$. We set $f(x) = e^{i\omega x}$ for $p \leq x \leq q$, $f(x) = 0$ for $x > q$ or $x < p$. Then for $y \in \mathbf{R}$, one has

$$\mathcal{F}(f)(y) = i \frac{e^{ip(\omega - 2\pi y)} - e^{iq(\omega - 2\pi y)}}{\omega - 2\pi y}.$$

- b) Let $\omega \in \mathbf{R}$, $\beta > 0$. We set $f(x) = e^{(-\beta + i\omega)x}$ for $x \geq 0$, $f(x) = 0$ for $x < 0$. Then for $y \in \mathbf{R}$, one has

$$\mathcal{F}(f)(y) = \frac{i}{\omega - 2\pi y + i\beta}.$$

c) Let $a > 0$. We set $f(x) = \frac{1}{a}e^{-a|x|}$ for $x \in \mathbf{R}$. Then for $y \in \mathbf{R}$, one has

$$\mathcal{F}(f)(y) = \frac{2}{4\pi^2 y^2 + a^2}.$$

d) We set $f(x) = e^{-\pi x^2}$ for $x \in \mathbf{R}$. Then for $y \in \mathbf{R}$, one has $\mathcal{F}(f)(y) = e^{-\pi y^2}$. (Find a first-order linear differential equation satisfied by the Fourier transform of f , and use the formula $\int_0^{+\infty} e^{-t^2} dt = \frac{1}{2}\sqrt{\pi}$ of FVR, VII, §1, no. 3.)

e) Let $n \in \mathbf{N}$ and let Q be a positive definite quadratic form on $\mathbf{R}^n$. Let σ a linear map on $\mathbf{R}^n$ such that $Q(x) = \|\sigma(x)\|^2$. Let Q^* the positive definite quadratic form such that $Q^*(y) = \|^t \sigma^{-1}(y)\|^2$ for all $y \in \mathbf{R}^n$. Let $z \in \mathbf{C}$ a complex number such that $\mathcal{I}(z) > 0$. The function f from $\mathbf{R}^n$ to $\mathbf{C}$ defined by $f(x) = \exp(i\pi z Q(x))$ belongs to $\mathcal{S}(\mathbf{R}^n)$ and we have

$$\mathcal{F}(f)(y) = |\det(\sigma)|^{-1} \left(\frac{i}{z}\right)^{n/2} \exp(i\pi z^{-1} Q^*(y))$$

for all $y \in \mathbf{R}^n$.

f) Let $f \in \mathcal{C}(\mathbf{R})$ the function defined by $f(x) = (\sin(\pi x)/(\pi x))^2$ for $x \neq 0$ and $f(0) = 1$. We have $\mathcal{F}(f)(y) = 0$ if $|y| > 1$ and $\mathcal{F}(f)(y) = 1 - |y|$ otherwise. Let $g(x) = xf(x)$. We have $\mathcal{F}(g)(y) = 0$ if $|y| > 1$ and $\mathcal{F}(g)(y) = s(y)/(2i\pi)$ otherwise.

2) We identify the interval $[0,1[$ with $\mathbf{T} = \mathbf{R}/\mathbf{Z}$ by the canonical projection. Let $0 \leq a < b < 1$. Let f the continuous function from $\mathbf{R}/\mathbf{Z}$ into $\mathbf{R}$ such that $f(x) = 0$ if $x \notin [a, b]$, $f((a+b)/2) = 1$, and such that f is affine on the intervals $[a, (a+b)/2]$ and $[(a+b)/2, b]$.

Compute the Fourier transform of f and prove that there exists a constant $C \geq 0$ such that

$$|\hat{f}(n)| \leq \inf\left(b - a, \frac{C}{n^2}\right)$$

for all $n \in \mathbf{Z} - \{0\}$.

3) a) The functions $t \mapsto t^n e^{-\pi t^2}$ for n integer ≥ 0 form a total set in $L^2(\mathbf{R})$.

(Let $f \in L^2(\mathbf{R})$ an element orthogonal to the $t^n e^{-\pi t^2}$. For $z \in \mathbf{C}$, set $F(z) = \int_{-\infty}^{+\infty} f(t) e^{-\pi t^2} e^{2i\pi t z} dt$, and show that F is an entire function whose derivatives at 0 are all zero. Deduce that the Fourier transform of the function $t \mapsto f(t) e^{-\pi t^2}$ on $\mathbf{R}$ is zero.)

b) For $k \in \mathbf{N}$, one has $\partial_t^k (e^{-2\pi t^2}) = P_k(t) e^{-2\pi t^2}$, where P_k is a polynomial of degree k , with leading coefficient $(-4\pi)^k$.

c) Let $k \in \mathbf{N}$, $P \in C[t]$ nonzero and α the leading coefficient of P . Then

$$\int_{\mathbf{R}} P_k(t) e^{-2\pi t^2} P(t) dt = \begin{cases} 0 & \text{if } \deg(P) < k \\ (-1)^k k! 2^{-1/2} \alpha & \text{if } \deg(P) = k. \end{cases}$$

d) For $k \in \mathbf{N}$ and $t \in \mathbf{R}$, we set:

$$\mathcal{H}_k(t) = (-1)^k \frac{2^{1/4-k}}{\sqrt{\pi^k k!}} e^{\pi t^2} \partial_t^k (e^{-2\pi t^2})$$

(“Hermite functions”). The family $(\mathcal{H}_k)$ is an orthonormal basis of the space $L^2(\mathbf{R})$. (Use a), b), c)).

e) We have $\mathcal{F}(\mathcal{H}_k) = (-i)^k \mathcal{H}_k$

4) a) Let $\alpha > 0$, $h > 0$, and $f : \mathbf{R} \rightarrow \mathbf{R}$ the continuous function vanishing outside the interval $[-\alpha, \alpha[$, linear in $[-\alpha, 0]$ and $[0, \alpha]$, equal to h at 0. We have:

$$\mathcal{F}(f)(t) = \frac{h}{\alpha \pi^2} \frac{\sin^2(\pi \alpha t)}{t^2}$$

for $t \in \mathbf{R}^*$.

b) Let $N \in \mathbf{N}$. Set $g(x) = \sum_{n=-N}^N f(x+n)$. Then:

$$\mathcal{F}(g)(t) = \frac{h}{\alpha \pi^2} \frac{\sin^2(\pi \alpha t)}{t^2} \frac{\sin((2N+1)\pi t)}{\sin(\pi t)}$$

for $t \in \mathbf{R}^*$

c) We choose sequences $(\alpha_i)_{i \geq 1}$, $(h_i)_{i \geq 1}$, $(N_i)_{i \geq 1}$, whence, for each integer i , a function g_i constructed by the procedure of b). If $\sum_i h_i < +\infty$, the series $\sum_i g_i$ converges uniformly to a continuous function G on $\mathbf{R}$.

d) If

$$\sum_i \alpha_i h_i N_i < +\infty, \quad \sum_i \frac{h_i}{\alpha_i} \log N_i < +\infty, \quad \sum_i h_i N_i = +\infty,$$

then G and $\mathcal{F}(G)$ are integrable on $\mathbf{R}$ but

$$\sum_{n=-\infty}^{+\infty} G(n) = +\infty.$$

Show that there exist examples of such sequences.

5) Let $\mu \in \mathcal{M}^1(G)$. We assume that $\mathcal{F}(\mu) \in L^1(\widehat{G})$. Then μ is a measure with base the Haar measure on G (INT, V, § 5, no. 2, def. 2), and admits a continuous density with respect to it. (Use the filter base $\mathfrak{B}$ of prop. 8 of II, p. 211. Show that $\lim_{\varphi, \mathfrak{B}} \varphi * \mu = \mu$ for the vague topology, and for φ in an element of $\mathfrak{B}$, one has $\mathcal{F}(\varphi * \mu) = \mathcal{F}(\varphi) \mathcal{F}(\mu)$; but $\mathcal{F}(\mu) \in L^1(\widehat{G})$, hence $\varphi * \mu = \overline{\mathcal{F}(\mathcal{F}(\varphi * \mu))}$ tends to $\overline{\mathcal{F}(\mathcal{F}(\mu))}$ into $\mathcal{C}_0(G)$. Thus $d\mu(x) = f(x)dx$ where $f = \overline{\mathcal{F}(\mathcal{F}(\mu))}$.)

6) Let H be a subgroup of G , and $\mu \in \mathcal{M}^1(\widehat{G})$. For $\mathcal{F}(\mu)$ to be invariant under translations by elements of H , it is necessary and sufficient that the support of μ be contained in $H^\perp$.

7) Let $f \in L^1(G)$. Let F a holomorphic function defined in an open set U of $\mathbf{C}$. We assume that U contains the set of values of $\mathcal{F}(f)$, and if G is not discrete, we assume that $0 \in U$ and $F(0) = 0$.

Then there exists $g \in L^1(G)$ such that $F \circ \mathcal{F}(f) = \mathcal{F}(g)$ (Use the holomorphic functional calculus.)

8) Let $p \in [1, 2]$ and $q \in [2, +\infty]$ such that $1/p + 1/q = 1$.

a) Let $f \in \mathcal{K}(G)$. Show that $\|\mathcal{F}(f)\|_q \leq \|f\|_p$ (This inequality is valid for $p = 1$ and $p = 2$. In the general case, use the M. Riesz inequality (INT, IV, §6, exerc. 18).)

- b) Deduce that the restriction of $\mathcal{F}$ to $\mathcal{H}(G)$ extends to a continuous linear map from $L^p(G)$ into $L^q(\widehat{G})$ which coincides with the Fourier transform on $L^p(G) \cap L^1(G)$ and on $L^p(G) \cap L^2(G)$.

9) Let G be a locally compact group (not necessarily commutative) equipped with a left Haar measure.

- a) If $f \in L^1(G)$, there exists f_1 and f_2 into $L^1(G)$ such that $f_1 * f_2 = f$ (Use exercise 30 of I, p. 185.)
- b) Let $f_1, f_2 \in L^1(G)$ with $f_1 \geq 0, f_2 \geq 0$. Show that $f_1 * f_2$ is almost everywhere equal to a lower semicontinuous function. (Consider f_1 as the simple limit of an increasing sequence of functions ≥ 0 of $L^\infty(G)$.)
- c) Take $G = \mathbb{R}$. For every open interval $I =]a, b[$, set

$$A(I) =]a, a + (b - a)/6[, \quad A'(I) =]a + 5(b - a)/6, b[.$$

Define intervals I_n for $n \geq 1$ and I'_n for $n \geq 0$ by $I'_0 =]0, 1[$ and

$$I_n = A(I'_{n-1}), \quad I'_n = A'(I'_{n-1})$$

for $n \geq 1$. Let F be the union of the intervals I_n for $n \geq 1$. There exists no pair (F_1, F_2) of subsets of $\mathbb{R}$ such that $F_1 + F_2 = F$.

- d) If f is a continuous function ≥ 0 on $\mathbb{R}$ such that $F = \{x \in \mathbb{R} | f(x) > 0\}$, then there exists no pair of functions (f_1, f_2) into $L^1(\mathbb{R})$ such that $f_1 \geq 0, f_2 \geq 0$ and $f = f_1 * f_2$. (Suppose such an equality. Let E_i the set of $x \in \mathbb{R}$ such that $f_i(x) > 0$. Let F_i be the set of points of density of E_i (INT, V, §6, exerc. 15). Prove that $F = F_1 + F_2$.)

- 10) Let $k \in \mathbb{Z}$. The map $x \mapsto x^k$ the map from G to G is injective if and only if the map $\chi \mapsto \chi^k$ of $\widehat{G}$ into $\widehat{G}$ is surjective.

- 11) Let G be a finite abelian group and d an integer. ≥ 1 Prove that for all $x \in G$ and every $\chi \in \widehat{G}$, one has

$$\sum_{\substack{y \in G \\ y^d = x}} \chi(y) = \sum_{\substack{\eta \in \widehat{G} \\ \eta^d = \chi}} \eta(x).$$

12) Let $(G_i)_{i \in I}$ a family of locally compact abelian groups. For every $i \in I$, let H_i an open compact subgroup of G_i . Let G be the local direct product of the G_i with respect to the H_i . For every $\chi \in \widehat{G}$, let χ_i the restriction of χ to G_i . Show that $\chi \mapsto (\chi_i)_{i \in I}$ is an isomorphism of the topological group $\widehat{G}$ onto the local direct product of the $\widehat{G_i}$ with respect to the $H_i^\perp$.

13) Let G_d the group G equipped with the discrete topology and χ a unitary character of G_d . For every integer $n \geq 1$, all elements $x_1, \dots, x_n \in G$ and every $\varepsilon > 0$, there exists $\hat{x} \in \widehat{G}$ such that $|\langle \hat{x}, x_i \rangle - \chi(x_i)| \leq \varepsilon$ for $1 \leq i \leq n$. (The canonical injection of G_d into G has as its dual a morphism from $\widehat{G}$ into $\widehat{G_d}$ whose image is dense by Cor. 5 of II, p. 229.)

14) a) The space $L^1(G)$ is a closed ideal in $\mathcal{M}^1(G)$, hence $\widehat{G}$ identifies with an open subset of $X(\mathcal{M}^1(G))$ (cf. exercise 17 of I, p. 169).

b) Let $\mathcal{M}_d(G)$ (resp. $\mathcal{M}_a(G)$) the set of $\mu \in \mathcal{M}^1(G)$ which are diffuse (resp. atomic). Then $\mathcal{M}_d(G)$ is a closed ideal of $\mathcal{M}^1(G)$ and $\mathcal{M}_a(G)$ is a closed subalgebra of $\mathcal{M}^1(G)$.

15) For every integer $k \geq 1$ for all elements $x_1, \dots, x_k$ pairwise distinct in E and for all integers $n_1, \dots, n_k$, the relation $x_1^{n_1} \dots x_k^{n_k} = e$ implies $x_1^{n_1} = \dots = x_k^{n_k} = e$. We say that E is a Kronecker set if every continuous complex-valued function on E with absolute value 1 is a uniform limit on E of unitary characters of G . If q is an integer ≥ 1 and $G = (\mathbb{Z}/q\mathbb{Z})^{\mathbb{N}}$ we say that E is of type K_q if every continuous map from E into the set of roots q characters of the unit coincide on E with a unitary character of G .

a) Let P be a compact Kronecker set. If $\mu \in \mathcal{M}^1(G)$ is concentrated on P , we have $\|\mathcal{F}(\mu)\|_\infty = \|\mu\|$. Hence deduce that every complex function continuous on P is the restriction of the Fourier transform of an element $f \in L^1(\widehat{G})$.

b) Let $E \subset G$ a finite independent set. Let f be a complex function of absolute value 1 on E .

Suppose that $f(x)^q = 1$ when $x \in E$ is of order q . The function f is the uniform limit on E of unitary characters of G . (Use exerc. 13.)

- c) Every Kronecker set is independent and contains only elements of infinite order.
- d) Every set of type K_q is independent.
- e) Let $k \geq 1$ an integer and let $V_1, \dots, V_k$ of nonempty disjoint open subsets of G . If every neighborhood of e in G contains an element of infinite order, there exists $x_i \in V_i$ for $1 \leq i \leq k$ such that $\{x_1, \dots, x_k\}$ let be a Kronecker set. If $G = (\mathbf{Z}/q\mathbf{Z})^N$, there exists $x_i \in V_i$ such that $\{x_1, \dots, x_k\}$ let be of type K_q .
- f) Let $E \subset G$ a compact independent set, $F = E \cup E^{-1}$. Let μ a bounded diffuse measure on G , concentrated on F . Then the measures μ^n , for $n \geq 0$ are pairwise disjoint, where one sets $\mu^0 = \varepsilon_e$ and $\mu^n = \mu * \mu^{n-1}$ for $n \geq 1$. (Show that, if $m < n$, the set of $(x_1, \dots, x_n) \in G^n$ such that $x_1 \dots x_n \in F^m$ is negligible for $\mu \otimes \dots \otimes \mu$). Deduce that, if $\mu \geq 0$, one has $\|\sum_{k=0}^n \alpha_k \mu^k\| = \sum_{k=0}^n |\alpha_k| \|\mu\|^k$ whatever the complex numbers $\alpha_0, \dots, \alpha_n$.
- g) Let $r \geq 1$ an integer. Let $E \subset G$ a compact set independent, $\mu_1, \dots, \mu_r$ of measures ≥ 0 of mass 1, concentrated on disjoint parts $E_1, \dots, E_r$ of E . Let $z_1, \dots, z_r \in \mathbf{C}$ with $|z_i| \leq 1$ for all i . There exists a character χ of $\mathcal{M}^1(G)$ such that $\chi(\mu_i) = z_i$ for $1 \leq i \leq r$ (First show, by reasoning as in f), that if $(n_1, \dots, n_r) \neq (m_1, \dots, m_r)$ the measures $\mu_1^{n_1} * \dots * \mu_r^{n_r}$ and $\mu_1^{m_1} * \dots * \mu_r^{m_r}$ are disjoint. Next, assuming $|z_i| = 1$ for all i , show that the spectral radius of $\varepsilon_e + \bar{z}_1 \mu_1 + \dots + \bar{z}_r \mu_r$ is $r + 1$, whence a character χ of $\mathcal{M}^1(G)$ such that $\chi(\varepsilon_e + \bar{z}_1 \mu_1 + \dots + \bar{z}_r \mu_r) = r + 1$; this character answers the question. In the general case, write $z_i = \frac{1}{2}(z'_i + z''_i)$ with $|z'_i| = |z''_i| = 1$, and $\mu_i = \frac{1}{2}(\mu'_i + \mu''_i)$ where $\mu'_1, \mu''_1, \dots, \mu'_r, \mu''_r$ satisfy the same hypotheses as $\mu_1, \dots, \mu_r$.)
- h) Let $\mathcal{M}_d(E)$ the Banach space of diffuse measures on G concentrated on E . Every continuous linear form of norm ≤ 1 on $\mathcal{M}_d(E)$ extends to a character of $\mathcal{M}^1(G)$ (Use g).

16) For every complex number s such that $\mathcal{R}(s) > 1$, we denote by

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s}$$

(“Riemann zeta function”).

- a) The function ζ is holomorphic on the open set of $\mathbf{C}$ consisting of complex numbers s such that $\Re(s) > 1$.
- b) For $y \in \mathbf{R}_+^*$, let

$$\theta(y) = \sum_{n \in \mathbf{Z}} e^{-\pi n^2 y};$$

This series converges absolutely and uniformly on compact subsets of $\mathbf{R}_+^*$. For every s such that $\Re(s) > 1$, one has

$$\pi^{-s/2} \Gamma(s/2) \zeta(s) = \frac{1}{2} \int_0^{+\infty} \theta(y) y^{s/2} \frac{dy}{y}$$

(see FVR, VII, p. 10 for the gamma function in the complex plane).

- c) For every $y \in \mathbf{R}_+^*$, one has $\theta(y^{-1}) = \sqrt{y} \theta(y)$. (Use the Poisson formula and exercise 1.)
- d) For every s such that $\Re(s) > 1$, one has

$$\pi^{-s/2} \Gamma(s/2) \zeta(s) = \frac{1}{s(s-1)} + \frac{1}{2} \int_1^{+\infty} (\theta(y) - 1) (y^{s/2} + y^{(1-s)/2}) \frac{dy}{y}.$$

- e) There exists a unique holomorphic function on $\mathbf{C} \setminus \{1\}$ whose restriction to the open set of complex numbers with real part > 1 coincides with ζ . This function, still denoted ζ , satisfies $(s-1)\zeta(s) \rightarrow 1$ when $s \rightarrow 1$ (*in other words, it has a simple pole with residue 1 at $s = 1_*$).
- f) The function Λ defined by

$$\Lambda(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

is holomorphic on $\mathbf{C} = \{0, 1\}$; it has simple poles with residue 1 at $s = 0$ and $s = 1$. We have $\Lambda(1 - s) = \Lambda(s)$ for all $s \in \mathbf{C} = \{0, 1\}$.

g) Show that for all $s \in \mathbf{C}$ such that $\Re(s) > 1$, one has

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1},$$

where the product is taken over the set of prime numbers and is absolutely convergent (“Euler product”).

h) When $\sigma > 1$ converges to 1, we have

$$\sum_p p^{-\sigma} = \log\left(\frac{1}{\sigma - 1}\right) + O(1),$$

where the sum is over the prime numbers.

i) Everything $s \in \mathbf{C} - \{1\}$ such that $\zeta(s) = 0$ is either of the form $s = -2k$, where $k \geq 1$ is an integer, or satisfies $0 \leq \Re(s) \leq 1$.

j) We have $\zeta(s) \neq 0$ if $\Re(s) = 1$ and $s \neq 1$. (Show using the Euler product that

$$\zeta(\sigma)^3 \zeta(\sigma + it)^4 \zeta(\sigma + 2it) \geq 1$$

for all $\sigma > 1$ and $t \in \mathbf{R}$, then consider t such that $\zeta(1 + it) = 0$.

17) Let $q \geq 1$ an integer, and let $a \geq 1$ be an integer coprime to q . Let $\widetilde{\chi}$ a character of the group $(\mathbf{Z}/q\mathbf{Z})^*$. One denotes by χ the map from $\mathbf{Z}$ into $\mathbf{C}$ such that $\chi(n) = 0$ if n is not coprime to q and $\chi(n) = \widetilde{\chi}(n \pmod{q})$ in the opposite case (“Dirichlet characters”).

a) Suppose $\widetilde{\chi}$ nontrivial. The function defined by

$$L(s, \chi) = \sum_{n \geq 1} \chi(n) n^{-s}$$

is holomorphic on the open set U of $\mathbf{C}$ consisting of the complex numbers such that $\Re(s) > 1$ and there exists a unique holomorphic function on $\Re(s) > 0$ which coincides with it on U . We

have

$$L(s, \chi) = \prod_p (1 - \chi(p)p^{-s})^{-1}$$

for $s \in U$, where the product is over the prime numbers and converges absolutely and uniformly on compact subsets of U (“Dirichlet L-function”).

b) Suppose that $\tilde{\chi}^2$ is not trivial. We have $L(1, \chi) \neq 0$. (Adapt the method of question j) of the previous exercise.)

c) Suppose that $\tilde{\chi}$ has order 2. For every integer $n \geq 1$, we denote by

$$r_\chi(n) = \sum_{d|n} \chi(d)$$

where the sum is over the divisors $d \geq 1$ of n . There exists a real number $c > 0$ such that, for every $x \geq 2$, one has

$$\sum_{n \leq x} \frac{r_\chi(n)}{\sqrt{n}} \geq c \log(x)$$

(note that the map r_χ is ≥ 0 is multiplicative, that is, $r_\chi(nm) = r_\chi(n)r_\chi(m)$ if n and m are coprime, and note that $r_\chi(n) \geq 1$ if n is the square of an integer.)

d) One still assumes that $\tilde{\chi}$ has order 2. One has

$$\sum_{n \leq x} \frac{r_\chi(n)}{\sqrt{n}} = \frac{1}{2} L(1, \chi) x^{1/2} + O(1)$$

when $x \rightarrow +\infty$. Deduce that $L(1, \chi) \neq 0$.

e) When $\sigma > 1$ converges to 1, one has

$$\sum_{p \equiv a \pmod{q}} p^{-\sigma} = \frac{1}{\varphi(q)} \sum_p p^{-\sigma} + O(1),$$

where $\varphi(q)$ is the Euler indicator of q (A, V, p. 76).

f) There exist infinitely many primes congruent to a modulo q (“Dirichlet’s arithmetic pro-

gression theorem”).

- 18) a) Let $n \in \mathbb{N}$ and let $f \in \mathcal{S}(\mathbf{R}^n)$. The function of $\mathbf{R} \times \mathbf{R}^n$ to $\mathbb{C}$ defined by

$$u(t, x) = \int_{\mathbf{R}^n} \hat{f}(y) \exp(2\pi i x \cdot y - 4\pi^2 t \|y\|^2) dy$$

for $(t, x) \in \mathbf{R} \times \mathbf{R}^n$ satisfies

$$\begin{cases} u(0, x) = f(0, x) \\ \partial_t u = i\Delta_2 u, \end{cases} \quad (1)$$

(“Schrödinger equation”) where

$$\Delta_2 u = \sum_{i=1}^n \partial_{x_i}^2 u.$$

- b) The function u is the unique solution of (1) such that $t \mapsto (x \mapsto u(t, x))$ belongs to $\mathcal{C}^\infty(\mathbf{R}, \mathcal{S}(\mathbf{R}^n))$.
- 19) Let $n \in \mathbb{N}$ and let $f \in L^1(\mathbf{R}^n)$ (resp. $L^2(\mathbf{R}^n)$) such that $f \circ \sigma = f$ for every element σ of the special orthogonal group $\text{SO}(n, \mathbf{R})$. Then the Fourier transform $\hat{f}$ of f satisfies $\hat{f}(\sigma(y)) = \hat{f}(y)$ for all $\sigma \in \text{SO}(n, \mathbf{R})$ and every $y \in \mathbf{R}^n$ (resp. almost every $y \in \mathbf{R}^n$).
- 20) Let $n \geq 1$ an integer. One identifies $R_+^* \times S_n$, and $\mathbf{R}^{n+1} - \{0\}$ by the map $(t, z) \mapsto tz$. Let ω the unique measure on the sphere $S_n \subset \mathbf{R}^{n+1}$ invariant under $\mathbf{O}(n+1, \mathbf{R})$ such that the measure $t^n dt \otimes \mu$ is identified with the Lebesgue measure on $\mathbf{R}^{n+1} - \{0\}$ (cf. INT, VII, p. 116, exercise 8). Let μ be the measure on $\mathbf{R}^{n+1}$ image of the measure ω by the injection of S_n into $\mathbf{R}^{n+1}$.
- a) We have $\mu(S_n) = 2\pi^{(n+1)/2} \Gamma(\frac{n+1}{2})^{-1}$ (cf. INT, V, §8, no. 7).
- b) For every real number $k \geq 0$, one defines the function $J_k: \mathbf{R}_+ \rightarrow \mathbb{C}$ by

$$J_k(t) = \frac{(t/2)^k}{\Gamma(k + \frac{1}{2})\Gamma(\frac{1}{2})} \int_{-1}^1 e^{iut} (1 - u^2)^{k-1/2} du$$

for $t \in \mathbf{R}_+$ (“Bessel function of the first kind”). For every $x \in \mathbf{R}^{n+1}$, one has

$$\mathcal{F}(\mu)(x) = \frac{2\pi}{\|x\|^{(n-1)/2}} J_{(n-1)/2}(2\pi\|x\|)$$

where $\|x\|$ is the Euclidean norm.

c) If $k \in \mathbf{N}$, one has

$$J_k(t) = \frac{1}{2\pi} \int_0^{2\pi} \cos(t \sin(u) - ku) du$$

for all $t \in \mathbf{R}_+$.

d) If n is even, show that $\mathcal{F}(\mu)(x)$ is a polynomial in $\|x\|^{\pm 1/2}$, $\cos(2\pi\|x\|)$ and $\sin(2\pi\|x\|)$; for $n = 2$, one has

$$\mathcal{F}(\mu)(x) = \frac{\sin(2\pi\|x\|)}{2\pi\|x\|}.$$

e) Let g be a continuous function with compact support in $\mathbf{R}_+^*$. Set $f(x) = g(\|x\|^2)\|x\|^{1-n/2}$ for all $x \in \mathbf{R}^n$. Then we have $\hat{f}(y) = h(\|y\|^2)\|y\|^{1-n/2}$ for $y \in \mathbf{R}^n$, where

$$h(s) = \pi \int_0^{+\infty} g(t) J_{n/2-1}(2\pi\sqrt{ts}) dt$$

for all $s \in \mathbf{R}_+$ (“Hankel transform”).

21) Let H be a closed subgroup of G and f an element of $L^1(G)$ such that $\mathcal{F}(f)$ vanishes on $H^\perp$. Let $\varepsilon > 0$. There exists a measure $\mu \in \mathcal{M}^1(G)$ concentrated on H such that $\|\mu\| \leq 2$, $\|f * \mu\| \leq \varepsilon$ and such that $\mathcal{F}(\mu) = 1$ in a neighbourhood of $H^\perp$.

22) Let p a prime number. Let K a finite extension of the field $\mathbf{Q}_p$ of p -adic numbers; it is a locally compact non-discrete field. Let $t : K \rightarrow \mathbf{Q}_p$ the trace map. Denote by $\lambda : \mathbf{Q}_p \rightarrow \mathbf{Q}$ the map defined in prop. 20 of II, p. 237.

a) The additive group of K is in duality with itself by means of the map $(x, y) \mapsto \exp(2i\pi \lambda(t(xy)))$. One identifies K and its dual by this map.

For every integer $n \geq 1$, we denote by $\mathcal{S}(K^n)$ the complex vector space of functions from K^n to $\mathbf{C}$ that are locally constant and have compact support.

b) The restriction of the Fourier transform to $\mathcal{S}(K^n)$ is an automorphism whose inverse is the restriction of the Fourier cotransform.

c) For every real number $p \in [1, +\infty[$, the space $\mathcal{S}(K^n)$ is dense in $L^p(K^n)$.

23) Let $p \geq 3$ a prime number and $k \geq 1$ an integer. We denote $G = F_p^k$. One equips G with the Haar measure of total mass 1, and one identifies $\widehat{G}$ and G via the map that to $n = (n_1, \dots, n_k) \in G$ associates the character

$$e_n: x \mapsto \exp\left(\frac{2i\pi}{p} \sum_{j=1}^k n_j x_j\right).$$

For functions f_1, f_2, f_3 from G to $\mathbb{C}$, we denote by

$$\Lambda(f_1, f_2, f_3) = \frac{1}{p^{2k}} \sum_{x \in G} \sum_{h \in G} f_1(x) f_2(x+h) f_3(x+2h).$$

We denote $\|f\|$ the norm of f into $L^2(G)$.

a) We have

$$\Lambda(1, f_2, f_3) = \left(\frac{1}{p^k} \sum_{x \in G} f_2(x)\right) \left(\frac{1}{p^k} \sum_{x \in G} f_3(x)\right)$$

and

$$\Lambda(1, 1, f_3) = \frac{1}{p^k} \sum_{x \in G} f_3(x).$$

b) We have

$$\Lambda(f_1, f_2, f_3) = \sum_{t \in G} \widehat{f_1}(t) \widehat{f_2}(-2t) \widehat{f_3}(t),$$

and

$$|\Lambda(f_1, f_2, f_3)| \leq \|f_1\| \|f_2\| \|\widehat{f_3}\|_{\infty}.$$

c) Let $f: G \rightarrow \mathbb{R}$ a function such that $\sum f(x) = 0$. There exists an affine hyperplane $H \subset G$

such that

$$\frac{1}{p^{k-1}} \sum_{x \in H} f(x) \geq \frac{1}{2} \|\hat{f}\|_{\infty}$$

(show that there exist elements $t \in G$ and $\theta \in \mathbf{R}$ such that the real part of $p^{-k} \sum_x f(x) (e_t(x) \exp(i\theta) + 1)$ is equal to $\|\hat{f}\|_{\infty}$).

- d) Let $A \subset G$ contain no arithmetic progression of length 3, that is, there do not exist $x \in A$ and $h \in G = \{0\}$ such that $x + h$ and $x + 2h$ also belong to A . Show that there exists an affine hyperplane $H \subset G$ such that

$$\frac{\text{Card}(A \cap H)}{\text{Card}(H)} \geq \frac{\text{Card}(A)}{\text{Card}(G)} + \frac{1}{2} \left(\left(\frac{\text{Card}(A)}{\text{Card}(G)} \right)^2 - \frac{1}{p^k} \right)$$

(apply the previous questions to $f = \varphi_A - |A|/p^k$, where φ_A is the characteristic function of A).

- e) Let $A \subset G$ contain no arithmetic progression of length 3. Show that $\text{Card}(A) \leq 3p^k/k$ ("Roth's theorem for F_p^k ").

24) a) Let $\delta > 0$. If p is a sufficiently large prime number and if A is a subset of $\mathbf{Z}/p\mathbf{Z}$ such that $\text{Card}(A) \geq \delta p$, then A contains an arithmetic progression of length 3. (Adapt the method of the preceding exercise.)

- b) Let $\delta > 0$. If N is a sufficiently large integer and if $A \subset \{1, \dots, N\}$ satisfies $\text{Card}(A) \geq \delta N$ then A contains an arithmetic progression of length 3. (Reduce to the case of the previous question by choosing p appropriately as a function of N .)

25) We denote μ_G the given Haar measure on G . Let μ a complex measure on G .

- a) The image of the Fourier transform of μ is finite if and only if μ is a linear combination of idempotent measures, that is, of measures ν such that $\nu * \nu = \nu$.
- b) The support of $\mathcal{F}_G(\mu)$ is finite if and only if μ is a finite linear combination of measures $\hat{x}\mu_G$, where $\hat{x} \in \widehat{G}$.

- c) We denote A_μ the set of measures of the form $\hat{x}\mu$ for $\hat{x} \in \widehat{G}$. The set of measures $\nu \in A_\mu$ such that $\nu(G) \neq 0$ is finite if and only if there exists a closed subgroup H of G , a finite set K , characters $\hat{x}_k$ for $k \in K$ and complex numbers t_k for $k \in K$ such that

$$\nu = \left(\sum_{k \in K} t_k \hat{x}_k \right) i(\mu_H)$$

where i denotes the canonical inclusion of H into G .

- 26) For every compact abelian group H , we denote by μ_H the normalized Haar measure on H .

Assume G is compact. We denote E_G the set of complex measures μ on G such that the Fourier transform of μ takes integer values. This is a commutative subgroup of $\mathcal{M}^1(G)$ which contains the Haar measure μ_G . One denotes by F_G the subgroup of E_G generated by the measures $\hat{x}i_H(\mu_H)$, where H ranges over the closed subgroups of G i_H denoting the canonical inclusion of H into G), and where $\hat{x}$ ranges over $\widehat{G}$.

The goal of this exercise is to prove that $E_G = F_G$ (Cohen's idempotent theorem).

- a) Let $\varphi: G_1 \rightarrow G_2$ a morphism of compact commutative groups. The map $\mu \mapsto \varphi(\mu)$ is a group homomorphism from E_{G_1} into E_{G_2} .
- b) For every $\mu \in E_G$ and every $\hat{x} \in \widehat{G}$, one has $\hat{x}\mu_G \in E_G$.

Let μ a complex measure on G . We denote θ the function on G such that $\mu = \theta|\mu|$. One denotes by $A_\mu \subset \mathcal{M}^1(G)$ the set of measures of the form $\hat{x}\mu$. Let ν a point adherent to A_μ into $\mathcal{M}^1(G)$.

- c) If ν does not belong to A_μ , then ν is singular with respect to μ_G .
- d) Let $\varepsilon > 0$. Let $f \in \mathcal{C}(G)$ of norm ≤ 1 such that $\nu(f) > (1 - \varepsilon)\|\mu\|$ and $\hat{x} \in \widehat{G}$ such that $\mathcal{R}(\mu(\hat{x}f)) > (1 - \varepsilon)\|\mu\|$. Show that

$$\int_G |1 - f\hat{x}\theta|d|\mu| \leq 3\varepsilon^{1/2}\|\mu\|.$$

- e) Suppose that $\mu \in E_G$. Show that $\|\nu\| \leq \|\mu\| - 1/(36\|\mu\|)$. (Apply the preceding question to two characters $\hat{x}_1$ and $\hat{x}_2$ such that $\hat{x}_1\mu \neq \hat{x}_2\mu$, and note that $\|\hat{x}_1\mu - \hat{x}_2\mu\| \geq 1$.)
- f) Conclude that $E_G = F_G$ and more precisely that every measure μ into E_G can be written as

$$\mu = \sum_{k \in K} n_k \hat{x}_k i(\mu_{H_k})$$

where the set K is finite, $n_k \in \mathbb{Z}$, $\hat{x}_k \in \widehat{G}$, and the measures $i(\mu_{H_k})$ are mutually disjoint. (Let $\mu \in E_G$; show that there exists a measure $\nu \neq 0$ in the closure of $\{\nu \in A_\mu \mid \nu(G) \neq 0\}$ whose norm is minimal; then show, using e), that $\{\hat{x} \in \widehat{G} \mid \nu(\hat{x}) \neq 0\}$ is finite, and apply the preceding exercise.)

- 27) Let f a function in $\mathcal{S}(\mathbf{R})$ of norm L^2 equal to 1. We have (uncertainty principle) the inequality

$$\left(\int_{\mathbf{R}} x^2 |f(x)|^2 dx \right) \left(\int_{\mathbf{R}} t^2 |\hat{f}(t)|^2 dt \right) \geq \frac{1}{(4\pi)^2}$$

(observe that

$$-\frac{1}{2} = \Re \left(\int_{\mathbf{R}} x f(x) \overline{f'(x)} dx \right),$$

and apply the Cauchy-Schwarz inequality). Determine the cases of equality.

- 28) a) Let $\delta > 0$ a real number. There does not exist a function $f : \mathbf{R} \rightarrow \mathbf{C}$ integrable, nonzero, and compactly supported, such that $\hat{f}(y) = O(\exp(-\delta|y|))$ when $|y| \rightarrow +\infty$ (In particular, there does not exist such a function such that $\hat{f}$ also compactly supported.)
- b) Let $(\varrho_n)_{n \geq 1}$ a summable sequence of strictly positive real numbers. The product

$$g(x) = \prod_{n \geq 1} \frac{\sin(\varrho_n x)}{\varrho_n x},$$

defined as equal to 1 for $x = 0$, converges absolutely and uniformly on every compact subset of $\mathbf{R}$. It defines a continuous, nonzero, even function such that $g \in L^1(\mathbf{R})$.

c) Let ϱ the sum of the series $\sum \varrho_n$. The support of the Fourier transform of g is contained in $[-\varrho, \varrho]$.

d) Let $r > 0$ a real number. Let $\delta : \mathbf{R}_+ \rightarrow \mathbf{R}_+^*$ a decreasing function such that $\delta(y) \rightarrow 0$ when $y \rightarrow +\infty$, such that $y \mapsto \delta(y)/y$ is integrable on $[1, +\infty[$ with respect to Lebesgue measure, and such that $y\delta(y) \rightarrow +\infty$ when $y \rightarrow +\infty$. There exists a continuous function $f \in L^1(\mathbf{R})$ with compact support contained in $[-r, r]$ such that

$$\hat{f}(y) = O(\exp(-\delta(|y|)|y|))$$

when $|y| \rightarrow +\infty$ (First show that there exists a summable decreasing sequence (ϱ_n) of strictly positive real numbers such that $\varrho \leq r$ and $\varrho_n \geq e\delta(n)n^{-1}$ for all n large enough.)

29) Let $f \in L^2(\mathbf{R})$ such that

$$\int_{\mathbf{R}} |f(x)|^2 e^{x^2} dx \leq 1, \quad \int_{\mathbf{R}} |\hat{f}(y)|^2 e^{y^2} dy \leq 1.$$

a) For every integer $n \in \mathbf{N}$, the function g_n defined by

$$g_n(z) = \left(\int_{\mathbf{R}} f(x) x^n \exp(-zx^2/2) dx \right)^2$$

is holomorphic in the open set $U = \{z \in \mathbf{C} \mid \Re(z) > -1\}$ of $\mathbf{C}$.

b) For every integer $n \in \mathbf{N}$, the function h_n defined by

$$h_n(z) = \frac{(-1)^n}{z^{2n+1}} \left(\int_{\mathbf{R}} \hat{f}(y) y^n \exp(-y^2/(2z)) dy \right)^2$$

is holomorphic in the open set $V = \{z \in \mathbf{C} \mid \Re(z^{-1}) > -1\} = \{z \in \mathbf{C} \mid |z + 1/2| > 1/2\}$ of $\mathbf{C}$.

c) There exists a holomorphic function k_0 into $U \cup V = \mathbf{C} - \{-1\}$ such that $k_0|_U = g_0$ and

$$k_0|V = h_0 .$$

- d) The function k_0 is zero. (Show that there exists $c \in \mathbb{C}$ such that $|k_0(z)| \leq c/|z+1|$ for all $z \in \mathbb{C} - \{-1\}$ and apply Liouville's theorem to $z \mapsto (z+1)k_0(z)$.)
- e) Prove by induction on $n \in \mathbb{N}$ that there exists a holomorphic function k_n into $U \cup V = \mathbb{C} - \{-1\}$ such that $k_n|U = g_n$ and $k_n|V = h_n$, then that $k_n = 0$ f) We have $f = 0$.
- f) Let $a, b > 0$ such that $ab \geq 1$. If $f \in L^2(\mathbb{R})$ satisfies

$$\int_{\mathbb{R}} |f(x)|^2 e^{ax^2} dx < +\infty, \quad \int_{\mathbb{R}} |\hat{f}(y)|^2 e^{by^2} dy < +\infty,$$

then $f = 0$ ("Hardy's uncertainty principle").

- 30) a) For every $x_0 \in \mathbb{T}$, there exists $f \in \mathcal{C}(\mathbb{T})$ such that the Fourier series of f to x_0 does not converge symmetrically to $f(x_0)$. (Let $s_n(f)$ the symmetric partial sums of the Fourier series of f ; consider the linear maps $f \mapsto s_n(f)(x_0)$ and apply the Banach–Steinhaus theorem.)
- b) There exists $f \in L^1(\mathbb{T})$ such that the Fourier series of f does not converge symmetrically to f in $L^1(\mathbb{T})$.
- c) The Fourier transform is not surjective from $L^1(\mathbb{T})$ in the Banach space of sequences $(c_n)_{n \in \mathbb{Z}}$ tending to 0 at infinity. (Apply Cor. 1 of EVT, I, p. 19.)
- d) Suppose that $f \in \mathcal{C}^1(\mathbb{T})$. The sequence $s_n(f)$ converges to f into $\mathcal{C}(\mathbb{T})$.
- 31) Let $r \geq 1$ an integer. We denote $\mathcal{S}(\mathbb{T}^r)$ the complex vector space of infinitely differentiable functions on $\mathbb{T}^r$, and $\mathcal{S}(\mathbb{Z}^r)$ the complex vector space of functions $f: \mathbb{Z}^r \rightarrow \mathbb{C}$ such that, for every real $A > 0$, one has $f(n) = O(\|n\|^{-A})$ when $\|n\| \rightarrow +\infty$.
- a) Equipped with the seminorms

$$p_A(f) = \sup_{n \in \mathbb{Z}^r} \|n\|^A |f(n)|,$$

the space $\mathcal{S}(\mathbb{Z}^r)$ is a Fréchet space.

- b) For every real number $p \in [1, +\infty[$, the space $\mathcal{S}(\mathbf{Z}^r)$ (resp. the space $\mathcal{S}(\mathbf{T}^r)$) is dense in $L^p(\mathbf{Z}^r)$ (respectively in $L^p(\mathbf{T}^r)$).
- c) The restriction of the Fourier transform of T^r to $\mathcal{S}(\mathbf{T}^r)$ is an isomorphism from $\mathcal{S}(\mathbf{T}^r)$ into $\mathcal{S}(\mathbf{Z}^r)$, whose inverse is the restriction of the Fourier cotransform.
- d) The topological vector space $\mathcal{S}(\mathbf{Z}^r)$ is isomorphic to $\mathcal{S}(\mathbf{R}^n)$, in particular c is a Montel space. (Use Exercise 3.)

32) Let G be a finite abelian group. We endow G with the Haar measure of total mass 1.

- a) Show that if f is nonzero, then

$$\text{Card}(\text{Supp}(f)) \text{Card}(\text{Supp}(\hat{f})) \geq \text{Card}(G)$$

Uncertainty principle; note that

$$\|\hat{f}\|_{\infty} \leq \frac{1}{\text{Card}(G)} \sum_{x \in G} |f(x)|$$

and apply the Cauchy-Schwarz inequality and the Plancherel formula.)

- b) Determine the cases of equality.

33) For every integer $r \geq 1$, we denote by $V_r = \det((x_i^j)_{\substack{1 \leq i \leq r \\ 0 \leq j < r}}) \in \mathbf{Z}[x_1, \dots, x_r]$ the Vandermonde determinant (A, III, p. 99, example 1).

- a) Let $f_1, \dots, f_r$ infinitely differentiable functions on $\mathbf{R}$. For all real numbers $x_1 < \dots < x_r$, there exist real numbers $(t_0, \dots, t_{r-1})$ into $[x_1, x_r]$ such that

$$\det(f_i(x_j)) = \frac{1}{1!2! \dots (r-1)!} V_r(x_1, \dots, x_r) \det((f_j^{(i)}(t_i))_{\substack{1 \leq i \leq r \\ 0 \leq j < r}})$$

(cf. FVR, I, p. 48, exerc. 11).

- b) Let r a positive integer and $m_1, \dots, m_r$ of positive integers. Let

$$\Delta_r = \det((x_i^{m_j})_{1 \leq i, j \leq r}) \in \mathbf{Z}[x_1, \dots, x_r].$$

The polynomial Δ_r is divisible by V_r into $\mathbf{Z}[x_1, \dots, x_r]$. One denotes by $F_r = \Delta_r/V_r$.

c) Let p be a prime number. Suppose that

$$0 \leq m_1 < m_2 < \dots < m_r \leq p-1.$$

Let $\xi_1, \dots, \xi_r$ of the roots p -th roots of unity distinct. We have

$$F_r(\xi_1, \dots, \xi_r) \neq 0.$$

(“Chebotarev’s Theorem”; let ξ a primitive root p th root of unity; we have $F_r(\xi_1, \dots, \xi_r) = P(\xi)$ where $P \in \mathbf{Z}[x]$ has degree $\leq p-1$ by the irreducibility of cyclotomic polynomials, the condition $F_r(\xi_1, \dots, \xi_r) = 0$ would imply that $F_r(1, \dots, 1)$ is divisible by p ; using a) with $x_i \rightarrow 1$ to prove that this is not the case.)

d) Let $G = \mathbf{Z}/p\mathbf{Z}$. For every function $f \neq 0$ from G to $\mathbf{C}$, we have

$$\text{Card}(\text{Supp}(f)) + \text{Card}(\text{Supp}(\hat{f})) \geq p+1.$$

e) For all nonempty subsets A and B of G such that

$$\text{Card}(A) + \text{Card}(B) \geq p+1,$$

there exists $f: G \rightarrow \mathbf{C}$ whose support is A and such that the support of $\hat{f}$ is B . (First consider the case where $\text{Card}(A) + \text{Card}(B) = p+1$.)

f) Let A and B be arbitrary subsets of G . We have

$$\text{Card}(A+B) \geq \inf(\text{Card}(A) + \text{Card}(B) - 1, p)$$

(by the Cauchy-Davenport theorem; apply the previous question to (A, C) and to (B, D) , where C and D are subsets of cardinality respectively $p+1 - \text{Card}(A)$ and $p+1 - \text{Card}(B)$, such that $C \cap D$ a cardinal $\sup(\text{Card}(A) + \text{Card}(B) - p, 1)$.)

g) Let $(a_1, \dots, a_{2p-1})$ be a family of elements of $\mathbb{F}_p$. There exists a subset $I \subset \{1, \dots, 2p-1\}$ of cardinality p such that

$$\sum_{i \in I} a_i = 0.$$

This property is false in general for a family in $\mathbb{F}_p^{2p-2}$. (Let $0 \leq b_1 \leq \dots \leq b_{2p-1} < p$ integers whose reductions modulo p form a permutation of $(a_1, \dots, a_{2p-1})$; directly handle the case where there exists i such that $a_i = a_{i+p-1}$, and apply the previous question in the contrary case.)

h) Let $k \geq 1$ be an integer. Let $v \geq 0$ be the integer such that $k = p^v r$ where p does not divide r . Denote by γ the integer

$$\gamma = \begin{cases} v + 1 & \text{if } p \geq 3 \\ v + 2 & \text{otherwise.} \end{cases}$$

If $\ell \geq 4k$ is an integer, then there exist integers $(n_1, \dots, n_\ell)$ such that p does not divide all the n_i and such that

$$n_1^k + \dots + n_\ell^k \equiv 0 \pmod{p^\gamma}.$$

34) For every compact abelian group G , we denote by μ_G the normalized Haar measure on G .

a) Let G be a compact commutative group. Let $(\nu_n)_{n \in \mathbb{N}}$ be a sequence of measures on G such that $\nu_n(G) = 1$ for all n . Then (ν_n) converges vaguely to μ_G if and only if $\nu_n(\hat{x})$ converges to 0 for every character $\hat{x} \neq 1$ of G ("Weyl's equidistribution criterion").

b) Let $r \geq 1$ be an integer and $x = (x_1, \dots, x_r) \in \mathbb{T}^r$. For $n \geq 0$, we denote by ν_n be the measure on T^r such that

$$\nu_n(f) = \frac{1}{n} \sum_{k=0}^{n-1} f(kx)$$

for $f \in \mathcal{C}(\mathbf{T}^r)$. The sequence (ν_n) converges vaguely to the normalized Haar measure μ_H , where H is the closure in $\mathbf{T}^r$ of the set $\{kx \mid k \in \mathbf{Z}\}$.

c) Let y_i of real numbers such that x_i is the class of y_i . If the family $(1, y_1, \dots, y_n)$ is $\mathbf{Q}$ -linearly independent, then we have $H = \mathbf{T}^r$ ("Kronecker's theorem.")

d) Let P be the set of prime numbers and $G = \mathbf{U}^P$. For every real number $T > 0$, let λ_T the measure $T^{-1} \lambda$ on $[0, T]$, where λ is the Lebesgue measure on $\mathbf{R}$. Let ν_T the image measure of λ_T under the continuous map $t \mapsto (p^{it})_{p \in P}$ of $[0, T]$ in G . When $T \rightarrow +\infty$, the family (ν_T) converges vaguely to μ_G .

35) a) For every integer $N \geq 1$ for every family $(a_n)_{1 \leq n \leq N}$ of complex numbers and every integer $H \geq 1$, one has

$$\left| \sum_{n=1}^N a_n \right|^2 \leq \left(1 + \frac{N+1}{H} \right) \sum_{|h| < H} \left(1 - \frac{|h|}{H} \right) \sum_{\substack{1 \leq n \leq N \\ 1 \leq n+h \leq N}} a_{n+h} \bar{a}_n$$

("van der Corput inequality"). (Set $a_n = 0$ if n is not between 1 and N ; write $\sum_n a_n = \sum_n a_{n+h}$ for $0 \leq h < H$, sum over h , then apply the Cauchy-Schwarz inequality.)

We denote $\pi: \mathbf{R} \rightarrow \mathbf{R}/\mathbf{Z}$ the canonical projection. A sequence (x_n) of real numbers is said to be equidistributed modulo 1 if the sequence (ν_n) of measures on $\mathbf{R}/\mathbf{Z}$ defined by

$$\nu_n(f) = \frac{1}{n} \sum_{k=1}^n f(\pi(x_k)), \quad f \in \mathcal{C}(\mathbf{R}/\mathbf{Z})$$

converges vaguely to the Haar measure of total mass 1 on $\mathbf{R}/\mathbf{Z}$.

b) If, for every integer $h \geq 1$ the sequence $(x_{n+h} - x_n)_n$ is equidistributed modulo 1, then (x_n) is equidistributed modulo 1. (Use the preceding exercise and the van der Corput inequality.)

c) Let $d \geq 1$ be an integer and $P = \alpha_d X^d + \dots + \alpha_0 \in R[X]$ a polynomial such that $\alpha_d \notin \mathbf{Q}$. Then the sequence $(P(n))$ is equidistributed modulo 1. (Reason by induction on $d \geq 1$.)

36) Let σ a positive real number. We denote by $\mathcal{E}_\sigma$ the complex vector space of entire functions f on $\mathbf{C}$ such that, for every real number $\varepsilon > 0$, one has

$$|f(z)| = O(e^{2\pi(\sigma+\varepsilon)|z|}), \quad \text{as } |z| \rightarrow +\infty.$$

a) Suppose $\sigma > 0$. Let $g \in L^2([-\sigma, \sigma])$. The function

$$f(z) = \int_{-\sigma}^{\sigma} g(x) e^{-2i\pi xz} dx$$

belongs to $\mathcal{E}_\sigma$. The restriction of f to $\mathbf{R}$ belongs to $L^2(\mathbf{R})$.

Let $f \in \mathcal{E}_\sigma$ such that the restriction of f to $\mathbf{R}$ belongs to $L^2(\mathbf{R})$. Let $g \in L^2(\mathbf{R})$ the Fourier transform of the restriction of f to $\mathbf{R}$. The goal of this exercise is to prove that the support of g is contained in $[-\sigma, \sigma]$; by question a), this provides a characterization of the space $\mathcal{E}_\sigma$ (“Paley-Wiener theorem”).

b) For every real number θ , the function

$$h_\theta(z) = e^{-i\theta} \int_0^{+\infty} f(re^{-i\theta}) \exp(-2\pi z r e^{-i\theta}) dr$$

is holomorphic in the half-plane H_θ of $\mathbf{C}$ determined by the inequality

$$\Re(z) \cos(\theta) + \Im(z) \sin(\theta) > \sigma.$$

c) If θ_1 and θ_2 are such that $|\theta_1 - \theta_2| < \pi$, then $h_{\theta_1}|_{H_{\theta_1} \cap H_{\theta_2}} = h_{\theta_2}|_{H_{\theta_2} \cap H_{\theta_1}}$. (Apply the Cauchy formula.)

d) The function h_0 (resp. h_π) is holomorphic in the open set of $z \in \mathbf{C}$ such that $\Re(z) > 0$ (resp. $\Re(z) < 0$).

e) Let $U = \mathbf{C} - i[-\sigma, \sigma]$. There exists a unique holomorphic function h on U such that $h(z) = h_0(z)$ if $\Re(z) > 0$ and $h(z) = h_\pi(z)$ if $\Re(z) < 0$.

f) When $x \rightarrow 0$ through values > 0 , the functions

$$y \mapsto h_0(x + iy) - h_\pi(-x + iy)$$

converge to g into $L^2(\mathbf{R})$.

g) Conclude that g is zero almost everywhere outside $[-\sigma, \sigma]$.

37) We keep the notations of the preceding exercise.

a) Let $f: \mathbf{C} \rightarrow \mathbf{C}$ be an element of $\mathcal{E}_{1/2}$ such that the restriction of f to $\mathbf{R}$ belongs to $L^2(\mathbf{R})$.

For every $z \in \mathbf{C} - \mathbf{Z}$, one has

$$f(z) = \frac{\sin(\pi z)}{\pi} \sum_{n \in \mathbf{Z}} (-1)^n \frac{f(n)}{z - n}.$$

Use the Paley-Wiener theorem and the formula a) of exercise 1.)

b) Let $f: \mathbf{C} \rightarrow \mathbf{C}$ be an element of $\mathcal{E}_{1/2}$ such that f is bounded on $\mathbf{R}$. For every $z \in \mathbf{C} - \mathbf{Z}$, one has

$$f(z) = f(0) + \frac{\sin(\pi z)}{\pi} \left(f'(0) + \sum_{n \in \mathbf{Z} - \{0\}} (-1)^n z \frac{f(n) - f(0)}{n(z - n)} \right)$$

(apply the previous question to the function defined by $z \mapsto (f(z) - f(0))/z$ for $z \neq 0$ and which maps 0 to $f'(0)$).

c) Let f be an element of $\mathcal{E}_{1/2}$ bounded on $\mathbf{R}$. Then f' is bounded on $\mathbf{R}$ and we have

$$\sup_{x \in \mathbf{R}} |f'(x)| \leq 2 \sup_{x \in \mathbf{R}} |f(x)|$$

("Bernstein's inequality"; reduce to bounding $|f'(1/2)|$ and apply the preceding formula).

d) Let f be an element of $E_{1/2}$ such that the restriction of f to $\mathbf{R}$ belongs to $L^2(\mathbf{R})$. Let u and v be the 1-periodic functions on $\mathbf{R}$ defined by

$$u(t) = \hat{f}(t) + \hat{f}(t-1), \quad v(t) = 2i\pi(t\hat{f}(t) + (t-1)\hat{f}(t-1))$$

for $0 \leq t < 1$. For almost every $t \in [-1, 1]$, one has

$$\hat{f}(t) = (1 - |t|)u(t) + (2i\pi)^{-1}s(t)v(t),$$

and the coefficients of the Fourier series of u and v are $f(-n)$ and $f'(-n)$, respectively, for $n \in \mathbb{Z}$.

e) Let $f \in \mathcal{E}_{1/2}$ as in the previous question. We have for all $z \in \mathbb{C}$ the formula

$$f(z) = \left(\frac{\sin(\pi z)}{\pi} \right)^2 \left\{ \sum_{n \in \mathbb{Z}} \frac{f(n)}{(z-n)^2} + \sum_{n \in \mathbb{Z}} \frac{f'(n)}{z-n} \right\},$$

where the series converges symmetrically in $\mathcal{O}(\mathbb{C})$, that is, uniformly on compact subsets of $\mathbb{C}$. (Use the formulas from the question f) of exercise 1.)

38) For every $n \in \mathbb{Z}$, we denote by e_n the character $x \mapsto e^{2i\pi nx}$ of $\mathbb{T}$. We denote by P the subspace of $\mathcal{C}(\mathbb{T})$ generated by the e_n .

Let $a = (a_h)_{h \in \mathbb{Z}}$ a family of complex numbers. The conjugate-harmonic family is the family $\tilde{a} = (-is(h)a_h)_{h \in \mathbb{Z}}$, where $s(h)$ denotes the sign function.

For $f \in P$, we denote by $\tilde{f} \in P$ the function on $\mathbb{T}$ such that $\mathcal{F}(\tilde{f}) = \widetilde{\mathcal{F}(f)}$ (conjugate-harmonic function of f).

a) Let $f \in P$ such that $\mathcal{F}(f)(0) = 0$ and such that f is real-valued. For every integer $k \geq 1$, there exists a constant C_k , independent of f , such that

$$\|\tilde{f}\|_{2k} \leq C_k \|f\|_{2k}.$$

(Observe that the Fourier coefficients of $g = f + i\tilde{f}$ are supported on $\mathbb{N}^*$, write

$$\mathcal{R} \left(\int_{\mathbb{T}} g(x)^{2k} dx \right) = 0$$

and expand using the binomial formula.)

b) Let $1 < p < +\infty$. The map $f \mapsto \tilde{f}$ of P into P admits a unique continuous linear extension from $L^p(\mathbb{T})$ into $L^p(\mathbb{T})$ such that $\mathcal{F}(\tilde{f}) = \widetilde{\mathcal{F}(f)}$ for all $f \in L^p(\mathbb{T})$ (cf. exercise 8). (For $p \geq 2$, using the previous question and the Riesz inequality (INT, IV, §6, exerc. 18); for $1 < p < 2$ (using duality.)

c) The result of the previous question does not extend to the case $p = 1$. (Use part (b) of Exercise 30.)

39) Let $\nu \in \mathcal{P}(\mathbf{R})$ a probability measure on $\mathbf{R}$ The characteristic function of ν is the function φ_ν of $\mathbf{R}$ into $\mathbf{C}$ such that

$$\varphi_\nu(x) = \int_{\mathbf{R}} e^{itx} d\nu(t)$$

for all $x \in \mathbf{R}$.

- a) For ν_1, ν_2 into $\mathcal{P}(\mathbf{R})$, one has $\nu_1 = \nu_2$ if and only if $\varphi_{\nu_1} = \varphi_{\nu_2}$
- b) For ν_1, ν_2 into $\mathcal{P}(\mathbf{R})$, one has $\nu_1 * \nu_2 \in \mathcal{P}(\mathbf{R})$ and $\varphi_{\nu_1 * \nu_2} = \varphi_{\nu_1} \varphi_{\nu_2}$.
- c) Let $\nu \in \mathcal{P}(\mathbf{R})$. The function φ_ν is bounded by 1 and uniformly continuous on $\mathbf{R}$.
- d) Let $\nu \in \mathcal{P}(\mathbf{R})$ For all real numbers $a < b$, one has

$$\nu([a, b]) + \frac{1}{2}\nu(\{a, b\}) = \lim_{T \rightarrow +\infty} \frac{1}{2T} \int_{-T}^T \varphi_\nu(x) \frac{e^{-ixa} - e^{-ixb}}{ix} dx,$$

and in particular the limit always exists. Moreover, for every $T > 0$, we have

$$\nu([-2T^{-1}, 2T^{-1}]) \leq \frac{1}{T} \int_{-T}^T (1 - \Re(\varphi_\nu(x))) dx.$$

- e) Let $(\nu_n)_{n \in \mathbf{N}}$ a sequence of probability measures on $\mathbf{R}$ converging narrowly (INT, IX, p. 59, def. 1) to a measure ν ; then $\nu \in \mathcal{P}(\mathbf{R})$ and $\varphi_{\nu_n}(x)$ converges to $\varphi_\nu(x)$ for all $x \in \mathbf{R}$.

Let $(\nu_n)_{n \in \mathbf{N}}$ a sequence of probability measures on $\mathbf{R}$. Suppose that for every $x \in \mathbf{R}$ the sequence $(\varphi_{\nu_n}(x))_n$ converges to a complex number $\varphi(x)$, and that the function $x \mapsto \varphi(x)$ is continuous at 0.

- f) Let $\varepsilon > 0$. There exists $a > 0$ such that

$$\nu_n([-a, a]) \geq 1 - \varepsilon$$

for every integer $n \in \mathbf{N}$.

- g) The sequence (ν_n) is relatively compact for the narrow topology.
- h) The sequence (ν_n) converges narrowly to a probability measure ν whose φ is the characteristic function (“P. Lévy’s continuity theorem”).
- i) There exist sequences (ν_n) of probability measures on $\mathbf{R}$, not converging tightly to a probability measure, such that $\varphi_{\nu_n}(x)$ converges for all $x \in \mathbf{R}$.

40) Let ν a probability measure on $\mathbf{R}$ such that the identity function x of $\mathbf{R}$ belongs to $L^2(\nu)$.

Denote by $m = \nu(x) \in \mathbf{R}$ and let $\sigma \geq 0$ such that $\sigma^2 = \nu((x - m)^2)$. We suppose $\sigma > 0$.

- a) Let $f(t) = (t - m)/\sigma$. The image measure $\mu = f(\nu)$ belongs to $\mathcal{P}(\mathbf{R})$, it satisfies $\mu(x) = 0$ and $\mu(x^2) = 1$.
- b) For $n \geq 1$, let $\mu^n = \mu * \dots * \mu$ the n -th convolution power of μ . The sequence $(\mu^n)_{n \geq 1}$ converges narrowly to the probability measure with density

$$\frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

with respect to Lebesgue measure on $\mathbf{R}$, called the “standard Gaussian law” (“Fundamental limit theorem of probability theory”; apply the preceding exercise).

- c) Let $d \geq 2$ an integer. For every integer $n \geq 1$, we denote by s_n the map from $[0, 1]$ into $\{0, \dots, d - 1\}$ which assigns to x the n -th term of the expansion of x in base d , if it is unique, or the n -th term of the limited expansion of x in base d , otherwise (TG, IV, p. 43, no. 5).

Let λ the Lebesgue measure on $[0, 1]$. Denote by $m_d = (d - 1)/2$ and $\sigma_d = \sqrt{(d^2 - 1)/12}$.

For all real numbers $a < b$, one has

$$\lim_{n \rightarrow +\infty} \lambda \left(\left\{ x \in [0, 1] \mid a < \frac{s_1(x) + \dots + s_n(x) - nm_d}{\sigma_d \sqrt{n}} < b \right\} \right) = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-x^2/2} dx.$$

- 41) Let $\lambda \in \mathbf{R}_+^*$. One denotes by ϖ_λ the discrete measure on $\mathbf{R}$ supported on $\mathbf{N}$ such that

$$\varpi_{\lambda}(k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

(“Poisson law with parameter λ ”).

- a) The measure ϖ_{λ} is a probability measure. Compute the characteristic function of ϖ_{λ} ; show that the identity function x of $\mathbf{R}$ belongs to $L^2(\mathbf{R}, \varpi_{\lambda})$ and compute $\varpi(x)$, $\varpi(x^2)$.
- b) The family of measures

$$\frac{\varpi_{\lambda} - \lambda}{\sqrt{\lambda}}$$

converges narrowly as $\lambda \rightarrow +\infty$ to the standard Gaussian measure (cf. preceding exercise, *b*).

- 42) Let $n \geq 2$ an integer. We denote by G the group S_n endowed with its Haar measure μ_n of total mass 1. Let $\varpi : S_n \rightarrow \mathbf{R}$ be the map which to σ , associates $a + b$, where a is the number of cycles in the decomposition of σ into a product of disjoint cycles (A, I, p. 60, prop. 7) and b is the number of fixed points of σ . One denotes by ν_n the image measure $\nu_n = \varpi(\mu_n)$ on $\mathbf{R}$. It is a probability measure.

- a) The characteristic function φ_n of ν_n satisfies

$$\varphi_n(t) = \prod_{j=1}^n \left(1 - \frac{1}{j} + \frac{e^{it}}{j} \right)$$

for all $t \in \mathbf{R}$.

- b) Let ϖ_n a Poisson law with parameter $\log(n)$ and ψ_n its characteristic function. When $n \rightarrow +\infty$, one has

$$\varphi_n(t) = \frac{1}{\Gamma(it)} \psi_n(t) (1 + o(1))$$

uniformly for $t \in \mathbf{R}$. (Use Gauss’s formula for the gamma function, cf. FVR, VII, p. 11, formula (3).)

- c) Let $f_n(x) = (x - \log(n)) / \sqrt{\log(n)}$ for $x \in \mathbb{R}$. The sequence of measures $f_n(\mu_n)$ converges narrowly to the standard Gaussian law when $n \rightarrow +\infty$.
- 43) Let $k \geq 1$ an integer. We denote λ the Lebesgue measure on $\mathbb{R}^k$. Let $(\mu_n)_{n \in \mathbb{N}}$ a sequence of probability measures on $\mathbb{R}^k$ and let μ a probability measure on $\mathbb{R}^k$. Suppose that the following conditions are satisfied:
- (i) The Fourier transform of μ belongs to $L^1(\mathbb{R}^k)$;
 - (ii) There exists a sequence (σ_n) into $GL(k, \mathbb{R})$ such that the sequence of image measures $(\sigma_n(\mu_n))$ converges narrowly to μ ;
 - (iii) For every positive real number τ , let $\varphi_{n,\tau}$ the characteristic function of the set of $t \in \mathbb{R}^k$ such that $\|\sigma_n(t)\| \leq \tau$. We have

$$\lim_{a \rightarrow +\infty} \sup_{n \in \mathbb{N}} \int_{\|t\| \geq a} |\widehat{\mu}_n(\sigma_n(t))| \varphi_{n,\tau}(t) dt = 0.$$

By Exercise 5, condition (i) implies that the measure μ has a continuous density ψ with respect to Lebesgue measure. We denote by $\alpha = \psi(0)$. a) Let $f \in \mathcal{C}_b(\mathbb{R}^k)$ an integrable function such that $\widehat{f}$ has compact support. We have

$$\lim_{n \rightarrow +\infty} \frac{1}{|\det(\sigma_n)|} \int_{\mathbb{R}^k} f \mu_n = \alpha \int_{\mathbb{R}^k} f.$$

(Apply the transposition formula for the Fourier transform.)

- b) The formula of the preceding question is valid when $f \in \mathcal{H}(\mathbb{R}^k)$. For every measurable bounded subset $B \subset \mathbb{R}^k$ such that the boundary of B is negligible for Lebesgue measure, we have

$$\lim_{n \rightarrow +\infty} \frac{1}{|\det(\sigma_n)|} \mu_n(B) = \alpha \lambda(B).$$

(For every function $f \in \mathcal{H}_{\mathbb{R}}(\mathbb{R}^k)$, and every $\varepsilon > 0$, there exist integrable functions f_1 and f_2 whose Fourier transforms have compact support and such that $f_1 \leq f \leq f_2$ and $\int (f_2 - f_1) \leq \varepsilon$.)

- c) Suppose that $\sigma_n^{-1} \rightarrow 0$ in the space of real matrices of type (k, k) and that there exists a continuous function $\Phi: \mathbf{R}^k \rightarrow \mathbf{C}^*$ such that

$$\widehat{\mu}_n \cdot \Phi^{-1} \cdot (\widehat{\mu} \circ {}^t\sigma_n^{-1})^{-1} \rightarrow 1$$

uniformly on compact subsets of $\mathbf{R}^k$. Then the above conditions are satisfied.

- d) Take $k = 1$. Let μ be a probability measure on $\mathbf{R}$ with base Lebesgue measure such that the identity function x of $\mathbf{R}$ belongs to $L^2(\nu)$ and such that $\mu(x) = 0$. Let $\sigma \geq 0$ such that $\sigma^2 = \mu(x^2)$. We suppose $\sigma > 0$. For every integer $n \geq 1$, let μ^n the n -th convolution power of μ . For every bounded measurable subset $B \subset \mathbf{R}$ whose boundary has measure zero with respect to Lebesgue measure, we have

$$\lim_{n \rightarrow +\infty} \sigma \sqrt{n} \mu^n(B) = \frac{1}{2\sqrt{\pi}} \lambda(B)$$

("fundamental local limit theorem").

- 44) Let $I = [a, b]$ be a compact interval of $\mathbf{R}$. Let f be a continuous function from I into $\mathbf{R}^2$ which is injective on the interior of I , and which is C^1 piecewise (that is, there exists a finite partition of I into intervals J such that $f|_J$ is differentiable, and $f'(f|_J)$ is continuous). We say that f is a parametrized curve in the plane. We say that

$$\ell(f) = \int_I |f'(t)| dt$$

(where $f'(t)$ is identified with a complex number) is the length of the parametrized curve f . If moreover $f(a) = f(b)$, we say that f defines a Jordan curve. There then exists a unique bounded connected component C of $\mathbf{R}^2 - f(I)$ (TA, to appear). The Lebesgue measure of C is called the area enclosed by the Jordan curve f , and is denoted $A(f)$.

- a) Suppose that f is a Jordan curve. For every $t \in I$, the restriction f_t of f to $[a, t]$ is a parametrized curve and the function $s: t \mapsto \ell(f_t)$ is a strictly increasing bijection from I into $J = [0, \ell(f)]$. The function $g = f \circ s^{-1}: J \rightarrow \mathbf{R}^2$ is a Jordan curve, and for every

$t \in J$, one has $\ell(g_t) = t$. b) Let $f = (f_1, f_2)$ be a Jordan curve. We have

$$A(f) = \frac{1}{2} \int_I (f_1 f_2' - f_2 f_1'),$$

where the integral is with respect to Lebesgue measure on I . (Apply Stokes' formula.) Moreover, we have $A(g) = A(f)$, where g is the function of question a).

c) We have $4\pi A(f) \leq \ell(f)^2$ ("isoperimetric inequality in the plane"). (Reduce to the case where $\ell(f) = 1$; consider the Fourier series of the real and imaginary parts of g and show that $|g'(t)|^2 = 1$ for $t \in J$.)

d) If $4\pi A(f) = \ell(f)^2$, then the image of f is a circle in $\mathbf{R}^2$.

45) For every real number $h > 0$, let Δ_h the continuous function on $\mathbf{R}$ such that $\Delta_h(t) = 0$ if $|t| > h$ and $\Delta_h(t) = 1 - |t|/h$ for $t \in [-h, h]$.

a) The family in $\mathcal{C}([0, 1])$ consisting of the functions $f_0 = 1$, $f_1 : t \mapsto t$, and the functions

$$f_{k,l} : t \mapsto \Delta_{2^{-k}} \left(t - \frac{2l+1}{2^k} \right)$$

for $k \geq 1$ and $0 \leq l < 2^{k-1}$ is a Banach basis of the Banach space $\mathcal{C}([0, 1])$ (cf. EVT, IV, p. 70, exerc. 14). More precisely, for every $f \in \mathcal{C}([0, 1])$, one has

$$f = f(0)f_0 + (f(1) - f(0))f_1 + \sum_{k \geq 1} \sum_{0 \leq l < 2^{k-1}} \left(f \left(\frac{2l+1}{2^k} \right) - \frac{1}{2} \left(f \left(\frac{l}{2^{k-1}} \right) + f \left(\frac{l+1}{2^{k-1}} \right) \right) \right) f_{k,l}$$

into $\mathcal{C}([0, 1])$.

b) The family of functions $t \mapsto e^{2i\pi ht}$ for $h \in \mathbf{Z}$ is not a Schauder basis of the space of functions $f \in \mathcal{C}([0, 1])$ such that $f(0) = f(1)$.

46) For every integer $k \geq 0$, let $D_k \subset [0, 1]$ the set of numbers $j/2^k$ for $0 \leq j \leq 2^k$ integer. Let D be the union of the sets D_k for $k \geq 0$. For every $k \geq 1$, and every $t \in D_{k+1} - D_k$, let t_+ (resp. t_-) the smallest element of D_{k+1} such that $t < t_+$ (resp. the greatest element of D_{k+1} such that $t_- < t$); t_+ and t_- belong to D_k .

- a) Let $\sigma : D \rightarrow [0, 1]$ a strictly increasing function such that $\sigma(0) = 0$ and $\sigma(1) = 1$ whose image is dense in $[0, 1]$. Then there exists a unique strictly increasing homeomorphism from $[0, 1]$ into itself that extends σ .

Let $f : [0, 1] \rightarrow \mathbf{R}$ a continuous function. We call a level sketch k of f a strictly increasing mapping $\sigma : D_k \rightarrow [0, 1]$ such that $\sigma(0) = 0$, $\sigma(1) = 1$ and such that for every integer j such that $0 \leq j < 2^k$, one of the following conditions holds:

- (i) $f(\sigma(j/2^k)) \neq f(\sigma((j+1)/2^k))$;
- (ii) there exists an open interval I contained in $\sigma(j/2^k), \sigma(j+1)/2^k$ such that, for every $t \in I$, one has

$$f(t) = f(\sigma(j/2^k)).$$

We denote t_σ the smallest strictly positive element of D_k such that

$$\inf_{t' < t_\sigma} |\sigma(t_\sigma) - \sigma(t')| = \sup_{t \in D_k} \inf_{t' < t} |\sigma(t) - \sigma(t')|.$$

One says that a sketch τ of level $k+1$ is a good extension of a draft σ of level k if τ extends σ and if, for all $t \in D_{k+1} - D_k$, one has

$$f(\tau(t)) = \frac{1}{2}(f(\sigma(t_-)) + f(\sigma(t_+)))$$

if $t_- \neq t_\sigma$

- b) There exists an increasing homeomorphism σ of $[0, 1]$ of itself into itself such that
 - (i) For every integer $k \geq 1$, the restriction σ_k of σ to D_k is a draft of level k ;
 - (ii) For every integer $k \geq 1$, the map σ_{k+1} is a good extension of σ_k .
- c) Suppose that $f(0) = f(1)$. Denote by σ the increasing homeomorphism of $[0, 1]$ defined by a mapping $\sigma : D \rightarrow [0, 1]$ satisfying the conditions of the previous question. There exists a constant $C \geq 0$ such that the continuous function $f \circ \sigma$ identified with a continuous

function on T , satisfies

$$|\widehat{(f \circ \sigma)}(n)| \leq \frac{C}{|n|}$$

for all $n \in \mathbb{Z} - \{0\}$ ("Sahakyan's Theorem"; use Exercise 2 of II, p. 262 and the preceding exercise.)

d) There exists an increasing homeomorphism σ of $[0, 1]$ into itself such that the Fourier series of $f \circ \sigma$ converges symmetrically in $\mathcal{C}(T)$ ("Bohr-Pál Theorem").

e) Does Sahakyan's theorem hold for complex-valued continuous functions?

f) Is there a function $f : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{C}$ whose image has nonempty interior and which satisfies the condition $\sup_{h \in \mathbb{Z} - \{0\}} |h| |\hat{f}(h)| < +\infty$?

47) Let X be a locally compact topological space and μ a probability measure on X . Let $(f_n)_{n \in \mathbb{N}^*}$ an orthonormal family in $L^2(X, \mu)$. Let $(a_n)_{n \in \mathbb{N}^*}$ a sequence of complex numbers such that the family $(a_n \log(n))_{n \in \mathbb{N}^*}$ belongs to $\ell^2(\mathbb{N}^*)$.

For every finite set of integers I , we denote by

$$s_I = \sum_{n \in I} a_n f_n,$$

and we also write $s_n = s_{\{1, \dots, n\}}$.

a) Let $N \geq 1$ There exists a family I of nonempty intervals of $\{1, \dots, N\}$ and a constant $C \geq 0$ independent of N , satisfying the following conditions:

(i) Every interval of $\{1, \dots, N\}$ of the form $I = \{1, \dots, M\}$ is a disjoint union of at most $C \log(M + 1)$ intervals in $\mathcal{I}$;

(ii) Every integer $n \in \{1, \dots, N\}$ belongs to at most $C \log(N + 1)$ intervals in I .

b) There exists a constant $C > 0$ such that for all $N \in \mathbb{N}^*$, one has

$$\int_X \left(\sup_{1 \leq n \leq N} |s_n|^2 \right) d\mu \leq C \sum_{n=1}^N |a_n|^2.$$

(Let $\tau(x)$ the smallest integer $\leq N$ such that $\sup_{1 \leq n \leq N} |s_n(x)| = |s_{\tau(x)}(x)|$, and let $S(x) = s_{\tau(x)}(x)$. Let $\mathcal{J}$ a family of intervals as in the previous question; write

$$S = \sum_{I \in \mathcal{J}} s_I$$

where J is a disjoint family of intervals in I , and then apply the Cauchy–Schwarz inequality.)

c) For $j \in \mathbb{N}^*$, set

$$S_j = \sum_{2^j < n \leq 2^{j+1}} a_n f_n.$$

The series with general term $(j+1)|S_j|^2$ converges μ -almost everywhere in X .

d) The sequence $(s_{2^j}(x))_{j \in \mathbb{N}^*}$ is a Cauchy sequence for μ -almost all $x \in X$. Denote by $f(x)$ its limit, defined μ -almost everywhere.

e) The sequence $(s_n(x))_{n \in \mathbb{N}^*}$ converges to $f(x)$ for μ -almost all $x \in X$.

f) Let $f \in L^1(\mathbb{T})$ such that

$$\sum_{h \in \mathbb{Z}} \log(|h|+1)^2 |\hat{f}(h)|^2 < +\infty.$$

For almost all x , the Fourier series of f to x converge symmetrically to $f(x)$ ("Rademacher-Menshov theorem").

48) a) Let $p \in [1, +\infty[$. A subset X of $L^p(\mathbf{R})$ is relatively compact if and only if it is bounded and satisfies the conditions

$$\lim_{y \rightarrow 0} \sup_{f \in X} \|\gamma(y)f - f\|_p = 0$$

$$\lim_{R \rightarrow +\infty} \sup_{f \in X} \|(1 - \varphi_R)f\|_p = 0,$$

where φ_R is the characteristic function of the interval $[-R, R]$ (cf. INT, VIII, p. 205, exercise 26).

b) Let X be a bounded subset of $L^2(\mathbf{R})$. Prove that X is precompact if and only if

$$\lim_{R \rightarrow +\infty} \sup_{f \in X} \int_{\mathbf{R}} (1 - \varphi_R)(|f|^2 + |\hat{f}|^2) dx = 0.$$

c) Let (f_n) a sequence in $L^2(\mathbf{R})$ orthonormal in $L^2(\mathbf{R})$ such that $\sup_n |f_n|$ belongs to $L^2(\mathbf{R})$.

Then $\sup_n |\widehat{f_n}|$ does not belong to $L^2(\mathbf{R})$.

49) Let k a finite field of characteristic p and cardinality q . We endow k of counting measure and we denote $\hat{f}$ the inverse Fourier transform of a function $f : k \rightarrow \mathbf{C}$.

a) Let χ a non-trivial character of k^* . We extend χ to k by setting $\chi(0) = 0$. Prove that

$$|\widehat{\chi}(\psi)| = \sqrt{q} \text{ for all } \psi \in \hat{k} \text{ such that } \psi \neq 1. \text{ ("Gauss sums.")}$$

b) Let $d \mid q - 1$ an integer. Prove that for every character $\psi \neq 1$ of k , one has

$$\left| \sum_{x \in k} \psi(x^d) \right| \leq (d-1)\sqrt{q},$$

with equality if $d = 2$.

c) Let χ_1 and χ_2 of the non-trivial characters of k^* , extended by 0 to k as in question a).

We set

$$J(\chi_1, \chi_2) = \sum_{x \in k} \chi_1(x) \chi_2(1-x)$$

("Jacobi sums"). If $\chi_1 \chi_2 \neq 1$, then for every character $\psi \neq 1$ of k , one has

$$J(\chi_1, \chi_2) = \frac{\widehat{\chi_1}(\psi) \widehat{\chi_2}(\psi)}{\widehat{\chi_1 \chi_2}(\psi)}, \quad |J(\chi_1, \chi_2)| = \sqrt{q}.$$

d) Suppose that p is a prime congruent to 1 modulo 4. There exist integers a and b such that $a^2 + b^2 = p$ ("Fermat's two-square theorem"). (Consider a Jacobi sum for characters of suitably chosen order of F_p^* .)

- e) Suppose that p is odd. Let $\lambda \in \hat{k}^*$ be the unique character of order 2 (“Legendre character”). Let χ be a non-trivial character of k^* and set $\eta = \chi^2$. Prove that for every character $\psi \neq 1$ of k , we have

$$\hat{\eta}(\psi) \hat{\lambda}(\psi) = \chi(4) \widehat{\chi}(\psi) \widehat{\chi \lambda}(\psi)$$

(“Hasse–Davenport formula”).

- f) Suppose that p is odd. Let N_p the number of $(x, y) \in \mathbf{F}_p \times \mathbf{F}_p$ such that $y^2 = x^4 + 1$. We have

$$|N_p - p| \leq 2\sqrt{p}.$$

(When p is congruent to 3 modulo 4, show that $N_p = p$; otherwise, verify that $N_p - p = -\sum_x \lambda(x^4 + 1)$, where λ is the Legendre character modulo p , and express this sum in terms of Jacobi sums.)

- g) Compare these formulas with those satisfied by the gamma function (FVR, VII).

- 50) Let k a finite field of characteristic $p \neq 2$ and cardinality q . We endow k of counting measure and we denote $\hat{f}$ the inverse Fourier transform of a function $f : k \rightarrow \mathbf{C}$. We endow k^* of the counting measure and fix a nontrivial character ψ of k . One denotes by λ the unique character of order 2 of k^* , extended to k by setting $\lambda(0) = 0$.

For every $a \in k^*$, set

$$S(a) = \sum_{x \in k^*} \lambda(x) \psi(ax + x^{-1})$$

(“Salié sums”). We denote by $\mathcal{F}(S)$ the Fourier transform of the function S on k^* .

- a) Let $\chi \in \hat{k}^*$ and set $\eta = \chi^2$. We have

$$\mathcal{F}(S)(\chi) = \widehat{\chi}(\psi) \widehat{\chi \lambda}(\psi) = \chi(4) \widehat{\lambda}(\psi) \widehat{\eta}(\psi)$$

(use the Hasse-Davenport formula).

b) Deduce that

$$S(a) = \lambda(a) \widehat{\lambda}(\psi) \sum_{\substack{y \in k^* \\ y^2 = 4a}} \psi(y), \quad |S(a)| \leq 2\sqrt{q}.$$

c) Analogy with Bessel functions $J_{k+1/2}$ (exercise 20) for $k \in \mathbf{N}$.

51) We identify $\mathbf{T}$ and the interval $[0,1[$. For $n \in \mathbf{Z}$, let e_n the unitary character of $\mathbf{T}$ defined by $t \mapsto \exp(2i\pi nt)$. For every measure $\mu \in \mathcal{M}^1(\mathbf{T})$ and every integer $m \geq 1$, we denote by

$$S_m(\mu) = \sum_{h=-m}^m \widehat{\mu}(h) e_h \in \mathcal{C}(\mathbf{T})$$

the m -th symmetric partial sum of the Fourier series of μ .

A finite family $(y_1, \dots, y_k)$ of real numbers is said to be $\mathbf{T}$ -independent if $(1, y_1, \dots, y_k)$ is $\mathbf{Q}$ -linearly independent. This amounts to saying that the equation

$$\sum_{j=1}^k m_j y_j = 0$$

into $\mathbf{T}$, with $m_j \in \mathbf{Z}$, has as its unique solution the zero family. For $t \in \mathbf{R}$, we denote by $t - (y_1, \dots, y_k)$ the family $(t - y_j)_j$.

a) Let $n \geq 10^6$ an integer. There exist real numbers $x_1, \dots, x_n$ into $[0,1]$ such that

$$|x_j - j/n| \leq 10^{-4} n^{-1}$$

and such that the family $E_n = (x_1, \dots, x_n)$ is $\mathbf{T}$ -independent.

We denote

$$\mu_n = \frac{1}{n} \sum_{j=1}^n \varepsilon_{x_j}.$$

This is a probability measure on $\mathbf{T}$.

b) Let $t \in \mathbf{R}$. We have

$$\frac{1}{n} \sum_{j=1}^n \frac{1}{|\sin(\pi(t - x_j))|} \geq \frac{1}{5} \log n.$$

c) Let $t \in \mathbf{R}$ such that the family $t - E_n$ is T-independent. There exists $m \in N^*$ such that

$$|S_m(\mu_n)(t)| \geq \frac{2}{15} \log n.$$

(apply Kronecker's theorem, cf. exercise 34).

d) Let $t \in \mathbf{R}$ such that the family $t - E_n$ is not T-independent and let $m_1, \dots, m_n$ be integers, not all zero, such that

$$\sum_{j=1}^n m_j(t - x_j) = 0$$

in T. If two of the integers m_j are nonzero, there exists $m \geq 0$ such that

$$|S_m(\mu_n)(t)| \geq \frac{1}{10} \log n$$

(let j be such that $m_j \neq 0$ and $|x_j - t|$ is maximal; note that the family $t - (x_i)_{i \neq j}$ is T-independent and apply Kronecker's theorem).

e) We keep the hypothesis from the previous question, but we assume that only one of the coefficients m_j is nonzero, and that $t \notin E_n$. There exists $m \geq 0$ such that

$$|S_m(\mu_n)(t)| \geq \frac{1}{10} \log n$$

(observe that the family $m_j(t - x_i)_{i \neq j}$ is T-independent and apply Kronecker's theorem).

f) Suppose that $t \in E_n$. There exists $m \geq 0$ such that

$$|S_m(\mu_n)(t)| \geq \frac{1}{10} \log n$$

(the family $(x_i)_{x_i \neq t}$ is T-independent).

g) There exists an integer $M \geq 0$ such that, for every $t \in \mathbf{T}$, one has

$$\sup_{0 \leq m \leq M} |S_m(\mu_n)(t)| \geq \frac{1}{20} \log n.$$

h) There exists a function $f_n \in \mathcal{C}^\infty(\mathbf{T})$ and an integer $M \geq 0$ such that

$$\begin{aligned} \text{Supp}(f_n) &\subset \bigcup_{j=1}^n \left[\frac{j}{n} - \frac{2}{10^4 n}, \frac{j}{n} + \frac{2}{10^4 n} \right] \\ \int_{j/n-2/(10^4 n)}^{j/n+2/(10^4 n)} f_n &= \frac{1}{n}, \quad 1 \leq j \leq n \\ \sup_{0 \leq m \leq M} |S_m(f_n)(t)| &\geq \frac{1}{40} \log n, \quad \text{for all } t \in \mathbf{T}. \end{aligned}$$

i) There exists $M \geq 1$ and a finite family $(a_n)_{|n| \leq M}$ of complex numbers such that the function

$$g_M(t) = \sum_{k=-M}^M a_k e_k n$$

satisfies

$$\int_{\mathbf{T}} |g_M(t)| dt \leq 2, \quad \inf_{t \in \mathbf{T}} \sup_{0 \leq m \leq M} |S_m(g_M)(t)| \geq \frac{1}{40} \log n.$$

j) There exists $\varphi \in L^1(\mathbf{T})$ such that $\widehat{\varphi}(h) = 0$ if $h < 0$ and such that the series

$$\sum_{h \geq 1} \widehat{\varphi}(h) e_h(t)$$

diverges for every $t \in \mathbf{T}$ ("Kolmogorov's theorem").

$$g_k = \sum_{n=-N_k}^{N_k} a_{k,n} e_n$$

(There exists a sequence N_k of integers and a sequence of functions

such that $\|g_k\|_1 \leq 2$ and

$$\inf_{t \in \mathbf{T}} \sup_{0 \leq m \leq N_k} |S_m(g_k)(t)| \geq 2^{2k}.$$

Let $\tilde{g}_k = e_{M_k} g_k$, where M_k is a sufficiently large integer; set $\varphi = \sum_{k \geq 1} 2^{-k} \tilde{g}_k$.

- 52) Let H be a countable discrete group, not necessarily commutative. a) The group H is amenable (EVT, IV, p. 73, exercise 4) if and only if there exists an increasing sequence $(F_N)_N$ of finite subsets of H such that, for every $x \in H$, one has

$$\lim_{N \rightarrow +\infty} \frac{1}{\text{Card}(F_N)} \text{Card}((F_N - xF_N) \cup (xF_N - F_N)) = 0$$

(“Følner sequence”). If H is commutative, then it is amenable.

- b) Let G be a compact commutative group. Fix a sequence $(F_N)_{N \geq 1}$ of finite subsets of the discrete group $\widehat{G}$ as in a) and denote φ_N be the characteristic function of F_N . Set

$$\psi_N = \frac{1}{\text{Card}(F_N)} \varphi_N * \widetilde{\varphi}_N.$$

The function ψ_N is finitely supported, it satisfies $0 \leq \psi_N \leq 1$, and

$$\lim_{N \rightarrow +\infty} \psi_N(\chi) = 1$$

for all $\chi \in \widehat{G}$.

- c) Let μ_N the measure on G with density the continuous map

$$\sum_{\chi \in \widehat{G}} \psi_N(\chi) \chi.$$

The sequence of measures $(\mu_N)_{N \geq 1}$ converges to ε_e in the space $\mathcal{M}^1(G)$ endowed with the topology of compact convergence in $\mathcal{C}(G)$ (Apply Lemma 4 of INT, VIII, §2, no. 7.)

- d) Let $f \in \mathcal{C}(G)$. We then have

$$f = \lim_{N \rightarrow \infty} \sum_{\chi \in \widehat{G}} \psi_N(\chi) \mathcal{F}(f)(\chi) \chi$$

into $\mathcal{C}(G)$.

- e) Retrieve Fejér’s theorem (prop. 23 of II, p. 242) as a special case of this statement.

53) Let $f \in \mathcal{L}^1(\mathbf{T})$

a) Let $k \geq 1$ an integer and suppose that $f \in \mathcal{C}^k(\mathbf{T})$. The Fourier series of f converges to f in $\mathcal{C}^{k-1}(\mathbf{T})$. If $k \geq 2$, the Fourier series of f converges absolutely to f in $\mathcal{C}(\mathbf{T})$.

b) For every integer $N \geq 1$, let f_N the function on $\mathbf{T}$ such that

$$f_N(t) = \sum_{\substack{h \in \mathbf{Z} \\ |h| \leq N}} \hat{f}(h) e^{2i\pi ht} \left(1 - \frac{|h|}{N}\right)$$

for $t \in \mathbf{T}$. Let $X \subset \mathbf{T}$ a set such that the restriction of f to X is continuous. Prove that f_N converges to f uniformly on X .

c) Let $X \subset \mathbf{T}$ an open set such that the restriction of f to X is of class C^1 . Prove that the Fourier series of f converges symmetrically to f uniformly on X .

54) Suppose that G is not compact. Let $\widehat{G}_d$ the dual group $\widehat{G}$ endowed with the discrete topology and $\widetilde{G}$ the dual group of $\widehat{G}_d$; this is a compact topological group.

a) The map which associates to $x \in G$ the character $\chi \mapsto \chi(x)$ is a continuous injective homomorphism of G into $\widetilde{G}$; its image is dense in $\widetilde{G}$.

From now on we identify G with a subset of $\widetilde{G}$.

b) Let K be a compact topological group and $\varphi: G \rightarrow K$ a continuous homomorphism. There exists a unique homomorphism $\widetilde{\varphi}$ of $\widetilde{G}$ into K which extends φ .

c) Let $f \in \mathcal{C}_b(G)$. The following conditions are equivalent:

- (i) There exists a function $\widetilde{f} \in \mathcal{C}(\widetilde{G})$ which extends f ;
- (ii) The function f belongs to the closure in $\mathcal{C}(G)$ of the subspace of $\mathcal{C}(G)$ generated by the characters $\chi \in \widehat{G}$;
- (iii) The set of $\gamma(x)f$ for $x \in G$ is precompact in $\mathcal{C}(G)$.

(To verify that (iii) implies (i), show that f is uniformly continuous. Then consider the closure of the $\gamma(x)f$; this is a compact metric space K . Prove that $\varrho: x \mapsto \gamma(x)$ is a continuous

homomorphism of G into the group of isometries of K ; then apply question b) to extend ϱ to $\tilde{\varrho}$ and set $\tilde{f}(x) = (\tilde{\varrho}(x)f)(e)$.

One says that a function $f \in \mathcal{C}_b(G)$ satisfying these equivalent conditions is almost periodic.

d) Let f be an almost periodic function on G . For every $\varepsilon > 0$ and for every compact subset F of G , there exists $x \in G - F$ such that $\|\gamma(x)f - f\| \leq \varepsilon$.

e) The set of almost periodic functions on G is a star subalgebra of $\mathcal{C}_b(G)$.

55) We keep the notation from the previous exercise. We endow $\tilde{G}$ of the normalized Haar measure ν . Let f be an almost periodic function on G and $\tilde{f}$

its extension to $\tilde{G}$. We set

$$A(f) = \int_{\tilde{G}} \tilde{f}(x) \nu(x).$$

a) Let $\chi \in \widehat{\tilde{G}}$. The function $\overline{\chi}f$ is almost periodic. We then define $\mathcal{P}(f)(\chi) = A(\overline{\chi}f)$.

We have

$$A(|f|^2) = \sum_{\chi \in \widehat{\tilde{G}}} |\mathcal{P}(f)(\chi)|^2$$

and in particular $A(\overline{\chi}f) = 0$ except for a countable set of χ . b) Let f be an almost periodic function on $\mathbb{R}$. For every $y \in \mathbb{R}$, one has

$$\mathcal{P}(f)(y) = \lim_{T \rightarrow +\infty} \frac{1}{2T} \int_{-T}^T f(x) e^{-2i\pi xy} dx.$$

c) Let Λ a finite subset of $\mathbf{R}$. The function defined on $\mathbf{R}$ by

$$f(x) = \sum_{\lambda \in \Lambda} e^{i\lambda x}$$

is almost periodic on $\mathbf{R}$ to determine as a function of Λ the set of $y \in \mathbf{R}$ such that $\mathcal{P}(f)(y) \neq 0$.

.

56) We set

$$k(z) = \left(\frac{\sin(\pi z)}{\pi z} \right)^2$$

if $z \in \mathbf{C} - \mathbf{Z}$ and $k(z) = 1$ if $z \in \mathbf{Z}$. The function k belongs to the space $\mathcal{E}_{1/2}$ (Exercise 36).

a) For every complex number $z \in \mathbf{C} - \mathbf{Z}$, one has

$$\sum_{n \in \mathbf{Z}} \frac{1}{(z - n)^2} = \left(\frac{\pi}{\sin(\pi z)} \right)^2$$

(apply the Poisson formula).

b) For $z \in \mathbf{C} - \mathbf{Z}$, we define

$$h(z) = \left(\frac{\sin(\pi z)}{\pi} \right)^2 \left\{ \sum_{n \in \mathbf{Z}} \frac{s(n)}{(z - n)^2} + \frac{2}{z} \right\},$$

where $s(n)$ denotes the sign function. The function h extends to an entire function belonging to the space E_2 . We set $b = h + k$.

c) For every $x \in \mathbf{R}$, one has $1 - k(x) \leq h(x) \leq 1$.

d) For every $x \in \mathbf{R}$, one has $|s(x) - h(x)| \leq k(x)$.

e) We have $b \geq s$; the function $b - s$ is integrable on $\mathbf{R}$ and satisfies $\int (b - s) = 1$. f) Let $\alpha < \beta$ be real numbers and φ the characteristic function of the interval $[\alpha, \beta]$. The function $f_{\alpha, \beta}(z) = \frac{1}{2}(b(\beta - z) + b(z - \alpha))$ satisfies $\varphi \leq f_{\alpha, \beta}$ and $\int_{\mathbf{R}} (f_{\alpha, \beta} - \varphi) = 1$.

g) *There exists an entire function σ such that $f_{\alpha, \beta}(x) = |\sigma(x)|^2$ for all $x \in \mathbf{R}$. The Fourier transform of the restriction of σ to $\mathbf{R}$ is supported in $[-1/2, 1/2]$.*

57) We keep the notations of the previous exercise. Let $M, N \geq 1$ be integers and let $(a_n)_{M < n \leq M+N}$ be a family of complex numbers. Let $\delta > 0$ be a real number, $R \geq 1$ be an integer and $(\xi_r)_{1 \leq r \leq R}$ a family of real numbers such that $\inf_{h \in \mathbf{Z}} |\xi_r - \xi_{r'} - h| \geq \delta$ for all $r \neq r'$.

For every $x \in \mathbf{R}$, set

$$S(x) = \sum_{n=M+1}^{M+N} a_n e^{2i\pi nx}.$$

a) We have

$$\sum_{r=1}^R |S(\xi_r)|^2 \leq (N + \delta^{-1} - 1) \sum_{n=M+1}^{M+N} |a_n|^2$$

("large sieve inequality").

Consider the function $\tau : z \mapsto \sigma(\delta z)$, where σ is the function of the questions f) and g) of the preceding exercise for the interval $[\alpha, \beta] = [\delta(M+1), \delta(M+N)]$; set

$$S^*(x) = \sum_{n=M+1}^{M+N} a(n) \tau(n)^{-1} e^{2i\pi nx}$$

and observe that

$$S(x) = \int_{-\delta/2}^{\delta/2} \widehat{\tau}(t) S^*(t+x) dt.$$

Set $x = \xi_r$, apply the Cauchy-Schwarz inequality and sum over r .)

b) The constant $N + \delta^{-1} - 1$ is optimal.

58) For every nonzero real number λ , we denote by μ_λ the discrete probability measure on $\mathbf{R}$ such that $\mu_\lambda(\lambda) = \mu_\lambda(-\lambda) = \frac{1}{2}$.

We fix $\lambda \in]0, 1[$.

a) Let us set $\nu_0 = \mu_1$ and $\nu_{n+1} = \mu_{\lambda^{n+1}} * \nu_n$ for every integer $n \geq 0$. The sequence $(\nu_n)_{n \in \mathbf{N}}$ converges to a probability measure ν_λ on $\mathbf{R}$. The characteristic function of ν_λ is the function on $\mathbf{R}$ defined by

$$\varphi_\lambda(x) = \prod_{n=0}^{+\infty} \cos(\lambda^n x).$$

b) The measure ν_λ is the unique positive measure ν of total mass 1 on $\mathbf{R}$ such that

$$\nu = \frac{1}{2}\ell_1(\nu) + \frac{1}{2}\ell_2(\nu),$$

where ℓ_1 (resp. ℓ_2) is the map $x \mapsto \lambda x - 1$ (resp. $x \mapsto \lambda x + 1$) from $\mathbb{R}$ to $\mathbb{R}$.

- c) If the measure ν_λ is not a base measure, the Lebesgue measure, then ν_λ is singular with respect to the Lebesgue measure. (Consider the Lebesgue decomposition of ν_λ with respect to the Lebesgue measure, cf. INT, V, §5, no. 7, th. 3.)
- d) If $0 < \lambda < \frac{1}{2}$, then the measure ν_λ is singular with respect to the Lebesgue measure.
- e) If $\lambda = \frac{1}{2}$, then the measure ν_λ is equal to the uniform measure on $[-2, 2]$. If $\lambda > 1/2$, the support of ν_λ is the interval $[-(1 - \lambda)^{-1}, (1 - \lambda)^{-1}]$.
- f) A Pisot number is a real algebraic integer x such that all the conjugates $y \in \mathbb{C}$ of x , other than x , satisfy $|y| < 1$. If λ^{-1} is a Pisot number, then the function φ_λ does not tend to 0 at infinity. In particular, the measure ν_λ is singular with respect to the Lebesgue measure.
- g) There exists $\lambda > 1/2$ such that ν_λ is singular with respect to the Lebesgue measure.

We now assume that φ_λ does not tend to 0 at infinity, and we denote by $\theta = 1/\lambda$. For every real number x , we denote by $\langle x \rangle$ the unique real number in $[0, 1[$ such that $x - \langle x \rangle \in \mathbb{Z}$.

- h) There exists a real number $u \neq 0$ such that the series with general term $\sin(u\theta^n)^2$ converges; the series with general term $\langle u\theta^n \rangle$ converges.
- i) For every $n \in \mathbb{N}$, let c_n be an integer such that $|u\theta^n - c_n| \leq \frac{1}{2}$. The formal series $g(z) = \sum c_n z^n$ is a rational function. (Use the criterion from A, IV, p. 85, Exercise 1, and Hadamard's inequality, EVT, V, p. 37, Cor. 3, to show that the Hankel determinants that arise tend to 0; conclude by observing that these determinants are integers.)
- j) There exist polynomials with integer coefficients f_1 and f_2 such that $g = f_1/f_2$ and $f_2(0) = 1$.
- k) The number θ is a Pisot number ("Salem's theorem"). (Note that $f_2(\lambda) = 0$ and that the other roots of f_2 have modulus > 1 .)

59) Let $n \geq 1$ be an integer and equip $\mathbf{R}^n$ with the Euclidean norm. We denote by μ the Lebesgue measure on $\mathbf{R}^n$.

Let $r > 0$. A packing of spheres of radius r in $\mathbf{R}^n$ is a set P of disjoint subsets of $\mathbf{R}^n$, such that every $S \in P$ is a closed ball of radius r . We say that P is periodic if there exists a lattice $\Lambda \subset \mathbf{R}^n$ such that $S + h \in P$ for all $S \in P$ and every $h \in \Lambda$.

Let A_P be the union of the elements of P . It is a measurable set. We call the density of P the real number

$$\Delta(\mathcal{P}) = \limsup_{R \rightarrow +\infty} \frac{\mu(A_{\mathcal{P}} \cap B_R)}{\mu(B_R)}$$

where B_R is the closed ball of radius R in $\mathbf{R}^n$. We denote by $\Delta_n = \sup_{\mathcal{P}} \Delta(\mathcal{P})$. a) We have

$$\Delta_n = \sup_{\mathcal{P} \text{ periodic}} \Delta(\mathcal{P}).$$

b) Let P be a periodic sphere packing, invariant under the lattice Λ . Let $C \subset \mathbf{R}^n$ be a set of representatives of the centers of the balls $S \in P$ modulo Λ . We have

$$\Delta(\mathcal{P}) = \mu(B_1) \frac{r^n \text{Card}(C)}{V(\Lambda)}$$

where $V(\Lambda)$ is the covolume of Λ .

c) Let $a > 0$ be a real number. Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ be a Schwartz function such that

(i) We have $f(x) \leq 0$ if $\|x\| \geq 2a$;

(ii) The Fourier transform of f is real-valued and positive, and satisfies $\hat{f}(0) > 0$.

Then we have

$$\Delta_n \leq \mu(B_1) a^n \frac{f(0)}{\hat{f}(0)}.$$

(It suffices to consider periodic sphere packings of radius a ; let $\mathcal{P}$ be one of them, and let Λ and $C \subset \mathbf{R}^n$ be as in the previous question; using the Poisson formula, prove that

$$\sum_{(c,d) \in \mathbb{C}^2} \sum_{n \in \Lambda} f(d - c + n) = \frac{1}{V(\Lambda)} \sum_{h \in \Lambda^*} \left| \sum_{c \in \mathbb{C}} e^{2i\pi c \cdot h} \right|^2 \hat{f}(h),$$

where Λ^* is the associated lattice, and conclude.)

60) a) Let $\mathbf{H} \subset \mathbf{C}$ be the open set of complex numbers z such that $\mathcal{I}(z) > 0$. The function

$$\Theta(z) = \sum_{n \in \mathbb{Z}} e^{2i\pi n^2 z}$$

is holomorphic on $\mathbf{H}$, and it satisfies $\Theta(z) = (z/i)^{-1/2} \Theta(-1/z)$ for $z \in \mathbf{H}$, where $z \mapsto (z/i)^{1/2}$ is the unique holomorphic function on $\mathbf{H}$ that extends the function $iy \mapsto \sqrt{y}$. (Use Exercise 16, b).

b) Let m and n be strictly positive integers. We have

$$\frac{1}{\sqrt{m}} \sum_{k=0}^{m-1} e^{2i\pi k^2 n/m} = \frac{e^{i\pi/4}}{\sqrt{2n}} \sum_{k=0}^{2n-1} e^{-2i\pi k^2 m/(2n)}.$$

(Apply the relation from the previous question with $z = -2in/m + iy$ and let $y \rightarrow 0$.)

c) For every integer $m \geq 1$ and every $n \in \mathbb{Z}$, set

$$\tau(n, m) = \sum_{k=0}^{m-1} e^{2i\pi k^2 n/m}.$$

For all strictly positive integers m_1 and m_2 we have

$$\tau(1, m_1 m_2) = \tau(m_2, m_1) \tau(m_1, m_2).$$

d) If m is odd, we have $\tau(1, m) = i^{((m-1)/2)^2} \sqrt{m}$. (Use a).

e) If p is an odd prime number, then $\tau(n, p) = \ell_p(n) \tau(1, p)$, where ℓ_p is the unique character of order 2 of $(\mathbb{Z}/p\mathbb{Z})^*$, extended to $\mathbb{Z}/p\mathbb{Z}$ by setting $\ell_p(0) = 0$.

f) Let $p \neq q$ be two odd prime numbers. We have

$$\ell_p(q)\ell_q(p) = \frac{\tau(1,pq)}{\tau(1,p)\tau(1,q)}.$$

- g) Deduce from this a new proof of the law of quadratic reciprocity (cf. A, V, p. 156, Exercise 23): for all odd prime numbers $p \neq q$, one has

$$\ell_p(q)\ell_q(p) = (-1)^{(p-1)(q-1)/4}.$$

- 61) Let $r \geq 1$ be an integer. Let $A = (a_{i,j})$ be a symmetric positive definite matrix of type (r, r) with integer coefficients, and let $N \geq 1$ be an integer such that the matrix $A^* = NA^{-1}$ also has integer coefficients. We denote by Q (resp. $\tilde{Q}$) the integral quadratic form associated with A (resp. with A^{-1}), that is, one has $Q(x) = {}^t x A x$ for all $x \in \mathbf{Z}^r$. We denote by $(a, b) \mapsto \langle a|b \rangle_A$ the bilinear form associated with A , so that $\langle a|b \rangle_A = {}^t a A b$.

We denote by H the set of complex numbers z such that $\mathcal{I}(z) > 0$. This is an open subset of $\mathbf{C}$. We denote by $\log: H \rightarrow \mathbf{C}$ the restriction to H of the principal determination of the logarithm (FVR, III, p. 10). Let $k \in \mathbf{N}$ be an odd integer. For $z \in H$, we denote by $z^k = \exp(k \log(z))$.

- a) The group $SL_2(\mathbf{Z})$ acts transitively on $\mathbf{H}$ by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az + b}{cz + d}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbf{Z})$$

(cf. TA, I, p. 150, Exercise 5).

- b) For every complex number $z \in \mathbf{H}$ and every $x \in \mathbf{Z}^r$, the series

$$\sum_{\substack{m \in \mathbf{Z}^r \\ m \equiv x \pmod{N}}} \exp(i\pi Q(m)z)$$

converges absolutely and uniformly on compact sets.

We denote by $\theta_A(z; x)$ or $\theta_Q(z; x)$ the sum of this series (“theta function associated with Q ”). It is a holomorphic function on $\mathbf{H}$.

- c) For every $y \in \mathbf{Z}^r$, one has

$$\sum_{m \in \mathbf{Z}^r} \exp(i\pi Q(m+y)z) = \frac{1}{\sqrt{\det(A)}} \left(\frac{i}{z}\right)^{r/2} \sum_{m \in \mathbf{Z}^r} \exp\left(-i\pi \frac{\tilde{Q}(m)}{z} + m \cdot y\right),$$

(Apply Poisson's formula and Exercise 1.)

- d) We denote by H the subgroup of $(\mathbf{Z}/N\mathbf{Z})^r$ consisting of the elements $\alpha \in (\mathbf{Z}/N\mathbf{Z})^r$ such that $A\alpha = 0$ in $\mathbf{Z}/N\mathbf{Z}$. The group H has cardinality $\det(A)$. Every character of H is of the form

$$\psi_\beta(\alpha) = \exp\left(2i\pi \frac{\langle \alpha | \beta \rangle_A}{N^2}\right),$$

where $\beta \in H$.

- e) For every $\alpha \in H$ and $z \in \mathbf{H}$, one has $\theta_Q(z+2; \alpha) = \exp\left(\frac{2i\pi Q(\alpha)}{N^2}\right) \theta_Q(z; \alpha)$.
- f) For every $\alpha \in H$ and $z \in \mathbf{H}$, one has

$$\theta_Q\left(-\frac{1}{z}; \alpha\right) = \frac{1}{\sqrt{\det(A)}} (-iz)^{r/2} \sum_{\beta \in H} \psi_\beta(\alpha) \theta_Q(z; \beta).$$

- g) Let $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbf{Z})$ such that $c \neq 0$, $d > 0$ and $b \equiv c \equiv 0 \pmod{2}$. For $\alpha \in H$ and $z \in \mathbf{H}$, one has

$$\theta_Q(g \cdot z; \alpha) = \frac{1}{d^{r/2} \det(A)} (cz + d)^{r/2} \sum_{\beta \in H} \Phi(\alpha, \beta) \theta_Q(z; \beta),$$

where

$$\Phi(\alpha, \beta) = \sum_{\gamma \in H} \psi_\gamma(\alpha) \left(\sum_{\substack{\delta \bmod dN \\ \delta \equiv \alpha \bmod N}} \exp\left(\frac{i\pi}{dN^2} (bQ(\delta) + 2\langle \gamma | \delta \rangle_A - cQ(\gamma))\right) \right)$$

- (First consider $\theta_Q(\tilde{g} \cdot z; \alpha)$ where $\tilde{g} = g \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Note the formula $d\tilde{g} \cdot z = b - (dz - c)^{-1}$ and apply the preceding question; note that $\Phi(\alpha, \beta)$ depends only on β modulo N).

- h) Suppose moreover that $c \equiv 0 \pmod{2N}$. Then

$$\theta_Q(g \cdot z; x) = \frac{\tau}{d^{r/2}} (cz + d)^k \theta_Q(z; a\alpha),$$

where

$$\tau = \sum_{\substack{\delta \bmod dN \\ \delta \equiv x \bmod N}} \exp\left(\frac{i\pi bQ(\delta)}{dN^2}\right).$$

i) Suppose moreover that d is odd. We have $\tau = \exp\left(\frac{i\pi abQ(\alpha)}{N^2}\right) \tau_Q(c; d)$, where

$$\tau_Q(c, d) = \sum_{x \in (\mathbf{Z}/d\mathbf{Z})^r} e\left(-\frac{i\pi cQ(x)}{d}\right)$$

(“Gauss sum.”)

j) Let us write $d = p_1 \cdots p_k$, where $p_1, \dots, p_k$ are prime numbers. We denote

$$\left(\frac{n}{d}\right) = \prod_{i=1}^k \ell_{p_i}(n)$$

where, for every odd prime number p , we denote by ℓ_p the unique character of order 2 of $(\mathbf{Z}/p\mathbf{Z})^*$, extended to $\mathbf{Z}/p\mathbf{Z}$ by setting $\ell_p(0) = 0$ (“Jacobi symbol”). If d and $2c \det(A)$ are coprime, then

$$\tau_Q(c, d) = \left(\frac{\det(A)}{d}\right) \left(\bar{\varepsilon}_d\left(\frac{2c}{d}\right) \sqrt{d}\right)^r,$$

where

$$\varepsilon_d = \begin{cases} 1 & \text{if } d \equiv 1 \pmod{4} \\ i & \text{if } d \equiv -1 \pmod{4}. \end{cases}$$

(Diagonalize the quadratic form and use the preceding exercise.)

k) If g is congruent to the identity modulo $4N$, then

$$\theta_Q(g \cdot z; \alpha) = \left(\frac{2c}{d}\right)^r (cz + d)^k \theta_Q(z; \alpha).$$

* This means that the map $z \mapsto \theta_q(z; \alpha)$ is a holomorphic modular form of weight $r/2$ and level $4N$. (FM, in preparation.) *

62) Let us identify $\mathbf{T}$ with the interval $[-1/2, 1/2]$. Let f be the function on $\mathbf{R}/\mathbf{Z}$ defined by $f(x) = s(x)$ (sign function). Let s_n be the symmetric partial sum

$$s_n(x) = \sum_{h=-n}^n \hat{f}(h) e^{2i\pi hx}$$

of the Fourier series of f .

a) For every $x \neq 0$ in $\mathbf{T}$, one has $s_n(x) \rightarrow f(x)$ when $n \rightarrow +\infty$.

b) Let $n \geq 1$ and $0 < x \leq 1/2$. We have

$$s_{2n}(x) = \frac{\sin(2\pi nx)}{2 \sin(\pi x)}.$$

c) We have

$$\lim_{n \rightarrow +\infty} s_{2n}\left(\frac{1}{4n}\right) = 2 \int_0^{1/2} \frac{\sin(2\pi x)}{\pi x} dx,$$

and this limit belongs to the interval $]1, 2[$ ("Gibbs phenomenon").

63) Let G be the subgroup of $T^{\mathbf{N}}$ consisting of the elements $x = (x_n)$ such that $2x_n = 0$ for all $n \in \mathbf{N}$ except for at most finitely many.

a) For every finite set $\Lambda \subset \mathbf{N}$, let $U_\Lambda \subset G$ be the set of (x_n) such that $x_n = 1$ for $n \in \Lambda$ and $2x_n = 0$ for $n \notin \Lambda$. There exists a topological group structure on G such that the family of sets U_Λ is a base of open neighborhoods of e .

b) The topological group G is locally compact.

c) The group G is not divisible.

d) The dual group $\widehat{G}$ is torsion-free.

64) Let μ be a measure on $\mathbf{T}$.

a) For every $x \in \mathbf{T}$, one has

$$\mu(\{x\}) = \lim_{H \rightarrow +\infty} \frac{1}{2H+1} \sum_{|h| \leq H} \widehat{\mu}(h) e^{2i\pi hx}.$$

b) Let us write $\mu = \mu_d + \mu_a$, where μ_a is an atomic measure and μ_d is a diffuse measure (INT, V, §5, no. 10, Prop. 15). Let S be the support of μ_a . We have

$$\lim_{H \rightarrow +\infty} \frac{1}{2H+1} \sum_{|h| \leq H} |\widehat{\mu}(h)|^2 = \sum_{x \in S} |\mu(\{x\})|^2$$

(consider $(\mu * \widetilde{\mu})(\{0\})$).

c) Suppose that μ is a probability measure on $\mathbf{T}$. The following conditions are equivalent:

- (i) There exists $x \in \mathbf{T}$ such that $\mu = \varepsilon_x$;
- (ii) We have $\lim_{|h| \rightarrow +\infty} |\widehat{\mu}(h)|^2 = 1$;
- (iii) We have

$$\lim_{H \rightarrow +\infty} \frac{1}{2H+1} \sum_{|h| \leq H} |\widehat{\mu}(h)|^2 = 1.$$

65) Assume that G is compact and equipped with its normalized Haar measure μ . Let $f : G \rightarrow G$ be a continuous and surjective homomorphism.

- a) We have $f(\mu) = \mu$.
- b) The map f is μ -ergodic (cf. exercise 16 of I, p. 190) if and only if there exists no character $\chi \neq e_{\widehat{G}}$ of G and integer $n \geq 1$ such that $\chi \circ f^n = \chi$.
- c) Let $d \geq 1$ be an integer and let a be a matrix of type (d, d) with integer coefficients and determinant 1. The map $f_a : T^d \rightarrow T^d$ defined by $f_a(x) = ax$ for all $x \in T^d$ is ergodic with respect to the normalized Haar measure on T^d if and only if the spectrum of the matrix a in $\mathcal{L}(\mathbf{C}^n)$ contains no root of unity.

¶ 66) We denote $e: \mathbf{T} \rightarrow \mathbf{C}^*$ the character defined by passing to the quotient by the character $x \mapsto e^{2i\pi x}$ of $\mathbf{R}$.

Let $d \geq 2$ be an integer. Let $\alpha \in \mathbf{R}$ and let $g \in R[X]$ be a polynomial whose degree is $< d$. Let a and q be coprime integers, with $q \neq 0$, such that

$$\left| \alpha - \frac{a}{q} \right| \leq \frac{1}{q^2}.$$

a) Suppose that $\alpha \notin \mathbf{Z}$. Let N and M be integers with $M \geq 0$. We have

$$\left| \sum_{n=N}^{N+M} e^{2i\pi \alpha n} \right| \leq \frac{1}{|\sin(\pi \alpha)|}.$$

b) We denote $D = 2^{d-1}$. For every $\varepsilon > 0$, there exists a real number $C(\varepsilon, d) \geq 0$ such that for every integer $P \geq 1$, one has

$$\left| \sum_{n=1}^P e^{2i\pi (an^d + g(n))} \right| \leq C(\varepsilon, d) P^{1+\varepsilon} \left(P^{-1/D} + q^{-1/D} + \left(\frac{P^d}{q} \right)^{-1/D} \right).$$

(Proceed by induction on d , reducing the square of the left-hand side of the inequality to a family of polynomials of degree $\leq d-1$; when $d=2$, apply part a). One may prove and use the bound

$$d(n) = O(n^\varepsilon)$$

when $n \rightarrow +\infty$, where $n \geq 1$ and $d(n)$ is the number of divisors of n in $\mathbf{N}$, valid for every real number $\varepsilon > 0$.

c) Deduce a new proof of the assertion of part c) of Exercise 35.

¶ 67) Let $k \geq 2$ and $\ell > k$ be natural numbers. For every integer $N \geq 1$, we denote by $r_{k,\ell}(N)$ the number of families $(n_1, \dots, n_\ell) \in \mathbf{N}^\ell$ such that

$$n_1^k + \dots + n_\ell^k = N.$$

The goal of this exercise is to show that, if ℓ is sufficiently large as a function of k , then

$r_{k,\ell}(N) \geq 1$ (“Waring’s problem”) for every $N \geq 1$.

We denote by $e: \mathbf{T} \rightarrow \mathbf{C}^*$ the character defined by passing to the quotient by the character $x \mapsto e^{2i\pi x}$ of $\mathbf{R}$.

Let $P \geq 1$ be an integer. We denote by S_P the function on $\mathbf{T}$ such that

$$S_P(x) = \sum_{n=1}^P e(xn^k).$$

We denote by dx the normalized Haar measure on $\mathbf{T}$.

a) If $P > N^{1/k}$, then we have

$$r_{k,\ell}(N) = \int_{\mathbf{T}} S_P(x)^\ell e(-Nx) dx.$$

b) Let δ be a real number such that $0 < \delta < 1/3$ and let $P = [N^{1/k}]$. We assume that N is sufficiently large, so that $P \geq 1$. Let a and q be pairwise coprime integers such that $1 \leq q \leq P^\delta$. We denote by $M_{a,q}$ the image in $\mathbf{T}$ of the open interval

$$\left] \frac{a}{q} - \frac{1}{P^{k-\delta}}, \frac{a}{q} + \frac{1}{P^{k-\delta}} \right[.$$

For $x \in M_{a,q}$, one has

$$\sum_{n=1}^P e^{2i\pi x n^k} = \frac{1}{q} S(a; q) I\left(x - \frac{a}{q}\right) + O(P^{2\delta})$$

as $N \rightarrow +\infty$, where

$$S(a; q) = \sum_{x=1}^q e\left(\frac{ax^k}{q}\right), \quad I(y) = \int_0^P e(yt^k) dt.$$

c) We denote by M the union of the sets $M_{a,q}$, where q ranges over the integers such that $1 \leq q \leq P^\delta$ and a ranges over the integers coprime to q such that $1 \leq a < q$. If $\delta < 1/5$, then there exists a real number $\delta' > 0$ such that

$$\int_M S_P(x)^\ell e(-Nx) dx = P^{\ell-k} \mathfrak{S}(P^\delta, N) J(P^\delta) + O(P^{\ell-k-\delta'})$$

as $N \rightarrow +\infty$, where we set, for $Q \geq 1$

$$\mathfrak{S}(Q, N) = \sum_{1 \leq q \leq Q} \sum_{\substack{a=1 \\ (a, q)=1}}^q \left(\frac{1}{q} S(a; q) \right)^\ell e\left(-\frac{aN}{q}\right)$$

$$J(Q) = \int_{-Q}^Q \left(\int_0^1 e(yt^k) dt \right)^\ell e(-y) dy.$$

d) We have

$$J(Q) = \frac{\Gamma(1 + 1/k)^\ell}{\Gamma(\ell/k)} + O(Q^{-(\ell/k-1)})$$

when $Q \rightarrow +\infty$. (First show that

$$J(Q) = \frac{1}{k^\ell} \int_{X_\ell} (y_1 \cdots y_{\ell-1} (1 - y_1 - \cdots - y_{\ell-1}))^{-1+1/k} dy + O(Q^{-(\ell/k-1)})$$

where

$$X_\ell = \left\{ (y_1, \dots, y_{\ell-1}) \in (\mathbf{R}_+^*)^{\ell-1} \mid y_1 + \cdots + y_{\ell-1} < 1 \right\},$$

then evaluate this integral; cf. FVR VII, p. 8, prop. 4.)

e) Suppose that $\ell > \delta^{-1} k 2^k$. There exists $\delta' > 0$ such that

$$\int_{\mathbf{T}-\mathbf{M}} |S_P(x)|^\ell e(-Nx) dx = O(P^{\ell-k-\delta'})$$

when $N \rightarrow +\infty$. (If N is sufficiently large and if $x \in R$ has image in $\mathbf{T}-\mathbf{M}$, there exist coprime integers a and q such that

$$P^\delta \leq q \leq P^{k-\delta}, \quad \left| x - \frac{a}{q} \right| \leq \frac{1}{q^2}.$$

Then use Exercise 66.)

f) For every prime number p and every integer $m \geq 1$, set

$$A(p^m) = \sum_{\substack{a=1 \\ (a,q)=1}}^{p^m} \left(\frac{1}{p^m} S(a; p^m) \right)^\ell e\left(-\frac{aN}{p^m} \right)$$

and

$$N(p^m) = \text{Card} \left(\{ (x_1, \dots, x_\ell) \in (\mathbf{Z}/p^m\mathbf{Z})^\ell \mid x_1^k + \dots + x_\ell^k = N \bmod p^m \} \right).$$

When $\ell > 2^k$, one has

$$1 + \sum_{m=1}^{+\infty} A(p^m) = \lim_{m \rightarrow +\infty} \frac{N(p^m)}{p^{m(\ell-1)}}.$$

g) When $\ell > 2^k$, one has $\mathfrak{S}(Q, N) = \mathfrak{S}(N) + o(1)$ when $Q \rightarrow +\infty$, where

$$\mathfrak{S}(N) = \prod_p \left(1 + \sum_{m \geq 1} A(p^m) \right),$$

the product over the prime numbers being absolutely convergent.

h) Let p be a prime number. Let $\nu \geq 0$ be the integer such that $k = p^\nu r$ where p does not divide r . Denote by γ the integer

$$\gamma = \begin{cases} \nu + 1 & \text{if } p \geq 3 \\ \nu + 2 & \text{otherwise.} \end{cases}$$

If there exist integers $(n_1, \dots, n_\ell)$ such that p does not divide all the n_i and such that

$$x_1^k + \dots + x_\ell^k \equiv 0 \bmod p^\gamma,$$

then

$$\lim_{m \rightarrow +\infty} \frac{N(p^m)}{p^{m(\ell-1)}} > 0.$$

i) There exists $\ell > k$ such that, if N is sufficiently large, then $\mathfrak{S}(N) > 0$. (Use the previous question and part h) of Exercise 33.)

j) There exists an integer $\ell > k$ such that $r_{k,\ell}(N) \geq 1$ for every integer $N \geq 1$.

68) We write the group law of G additively. Let Γ be a discrete subgroup of G such that G/Γ is compact. Let $\Lambda = \Gamma^\perp$ be the orthogonal complement of Γ in $\widehat{G}$; since $\widehat{G/\Gamma}$ is identified with Λ , which is a discrete subgroup of $\widehat{G}$, and $\widehat{G}/\Lambda$ is compact (since its dual identifies with Γ).

We assume that the Haar measure on G is chosen so that the quotient of the dual measure $d\chi$ on $\widehat{G}$ by the counting measure on Λ is the normalized Haar measure on $\widehat{G}/\Lambda$ (INT, VII, § 2, no. 2, def. 1).

a) There exists a measurable subset Ω of $\widehat{G}$ which meets each class of $\mathbf{G}$ modulo Λ at one point and only one; the measure of Ω is equal to 1.

b) For $x \in \mathbf{G}$, set

$$\varphi(x) = \int_{\Omega} \chi(x) d\chi.$$

The function φ is continuous; it satisfies $\varphi(0) = 1$ and $\varphi(\gamma) = 0$ for all $\gamma \in \Gamma - \{0\}$.

c) Prove that φ belongs to $L^2(G)$ and that $\|\varphi\|_2 = 1$. For every $\gamma \in \Gamma - \{e\}$, one has

$$\int_G \varphi(x) \overline{\varphi(\gamma - x)} dx = 0.$$

d) Let $f \in L^2(G)$ such that $\mathcal{F}(f)$ is zero almost everywhere outside of Ω . Prove that

$$f = \sum_{\gamma \in \Gamma} f(\gamma) \delta_\gamma(\varphi)$$

in $L^2(G)$, where $\delta_\gamma(\varphi)(x) = \varphi(x - \gamma)$. Moreover, we have

$$\int_G |f(x)|^2 dx = \sum_{\gamma \in \Gamma} |f(\gamma)|^2.$$

e) There exists a unique continuous function on G which coincides almost everywhere with

f. If f is continuous, then

$$f = \sum_{\gamma \in \Gamma} f(\gamma) \delta_{\gamma}(\varphi)$$

in $\mathcal{C}(G)$.

f) Specialize to the case where $G = \mathbb{R}$ and $\Lambda = h\mathbb{Z}$ for some $h > 0$ by taking $\Omega = [-h, h[$.

Exercises for Section 2: Classification

- 1) a) For G to have a countable base, it is necessary and sufficient that $\widehat{G}$ have a countable base. (If G has a countable base, then $L^1(G)$ has a countable base, hence $X(L^1(G))$ has a countable base.)
- b) For G to be countable at infinity, it is necessary and sufficient that $\widehat{G}$ admit an open subgroup with a countable base, or equivalently that $\widehat{G}$ be metrizable. (Use Prop. 3 of II, p. 248.)
- c) Suppose G is compact. For G to be metrizable, it is necessary and sufficient that $\widehat{G}$ be countable, or equivalently that G be isomorphic to a closed subgroup of $T^{\mathbb{N}}$. (A countable discrete commutative group is a quotient of $\mathbb{Z}^{(\mathbb{N})}$.)
- 2) a) Every infinite commutative group Γ admits an infinite countable quotient group. (Embed an infinite countable subgroup Δ of Γ in a countable divisible group Δ' by applying A, II, p. 185, Exercise 14; then extend to Γ the identity morphism of Δ into Δ' .)
- b) Every infinite compact commutative group contains an infinite metrizable closed subgroup. (Use a.)
- 3) a) Let K be a compact group and $K_{(n)}$ the set of elements of order n of K . If $K \neq E_n$ for every n , then the set of elements of infinite order of K is dense in K . (If $K_{(n)}$ has nonempty interior, we have $K_{(m)} = K$ for some integer $m \geq n$.)
- b) Let Γ be a discrete commutative group whose elements are not of bounded order. Then Γ admits a countable quotient with the same property. (Reason as in part a) of Exercise 2.)

- c) If every neighborhood of e in G contains an element of infinite order, then G possesses a metrizable closed subgroup with the same property. (Reduce to the case where G is compact, and use a) and b).) Otherwise, if G is not discrete, there exists an integer $q > 1$ such that G contains a closed subgroup isomorphic to $(\mathbf{Z}/q\mathbf{Z})^{\mathbf{N}}$. (Use A, VII, p. 54, exerc. 4, c).)
- 4) If G is compact and if $\widehat{G}$ has an element of infinite order, then there exists a nontrivial continuous homomorphism from $\mathbf{R}$ to G . (Since $\mathbf{R}$ is divisible, there exists a nontrivial homomorphism from $\widehat{G}$ into $\mathbf{R}$.)
- 5) Let p be a prime number. The group $\mathcal{Q}_p$ is not the product of a compact group and a discrete group. (Every compact subgroup of $\mathcal{Q}_p$ is of the form $p^n \mathbf{Z}_p$ for some integer $n \in \mathbf{Z}$.)
- 6) If G is a torsion group, then G and $\widehat{G}$ are totally disconnected. ($\widehat{G}$ is totally disconnected by Cor. 2 of II, p. 250; to prove that G is totally disconnected, reduce to the case where G is compact using Prop. 3 of II, p. 248, and then prove that $\widehat{G}$ is a torsion group.)
- 7) An element x of G is said to be of infinite height if, for every $n \in \mathbf{Z}$, there exists $y \in G$ such that $y^n = x$. Prove that, if G is compact, the set of elements of infinite height is the neutral component of G .

¶ 8) The group G is locally connected if and only if G is isomorphic to a group $\mathbf{R}^n \times E \times \widehat{D}$ where E and D are discrete and every finite-rank subgroup of D is free.

- 9) a) Let $a = (a_n)_{n \geq 0}$ be a sequence of integers > 1 . We equip $\mathbf{Z}$ with the topological ring structure for which the subsets $\mathbf{Z}a_0a_1 \cdots a_n$, where $n \geq 0$, form a fundamental system of neighborhoods of 0. We denote by Δ_a the completion of this ring. It is compact, metrizable, and totally disconnected. If p is a prime number, and if $a_i = p$ for all i , we have $\Delta_a = \mathbf{Z}_p$.
- b) We denote by $\mathbf{Z}(a^\infty)$ the multiplicative group of complex numbers of the form $\exp(2i\pi l / (a_0a_1 \cdots a_r))$ where $l \in \mathbf{Z}$ and $r \in \mathbf{N}$, equipped with the discrete topology. Then $\chi \mapsto \chi(1)$ is an isomorphism from $\widehat{\Delta_a}$ onto $\mathbf{Z}(a^\infty)$.

c) Let B be the subgroup of $\mathbf{R} \times \Delta_a$ consisting of the (n, n) for $n \in \mathbf{Z}$; it is discrete. We set $\Sigma_a = (\mathbf{R} \times \Delta_a)/B$. It is a compact, metrizable, connected group. (Note that the canonical image of $\mathbf{R}$ into Σ_a is dense, and use Lemma 1 of II, p. 244.)

d) Let Γ_a be the additive subgroup of the discrete group $\mathbf{Q}$ consisting of rational numbers of the form $l/(a_0 a_1 \cdots a_r)$ where $l \in \mathbf{Z}$ and $r \in \mathbf{N}$. If $\chi \in \widehat{\Sigma}_a$, then χ defines a unitary character of $\mathbf{R} \times \Delta_a$, hence a unitary character of $\mathbf{R}$ of the form $x \mapsto \exp(2i\pi \alpha_\chi x)$ where $\alpha_\chi \in \mathbf{R}$. Then $\chi \mapsto \alpha_\chi$ is an isomorphism from $\widehat{\Sigma}_a$ onto Γ_a .

e) Show that $\widehat{\mathbf{Q}}$ is canonically isomorphic to Σ_a , where the sequence a is defined by $a_n = n + 2$ for all $n \geq 0$.

f) Let R_d be the group $\mathbf{R}$ equipped with the discrete topology. Show that $\widehat{R}_d$ is isomorphic to $(\widehat{\Sigma}_a)^\mathfrak{c}$ where $a_n = n + 2$ and where $\mathfrak{c}$ is the cardinality of the continuum. (Consider a basis of $\mathbf{R}$ as a $\mathbf{Q}$ -vector space.)

¶ 10) A topological group is said to be monothetic if there exists an element of the group whose powers are dense in the group, and solenoidal if there exists a continuous homomorphism from $\mathbf{R}$ into the group whose image is dense.

a) With the notation of Exercise 9, the group Δ_a is monothetic.

b) Every compact monothetic totally disconnected group is topologically isomorphic to a group of the form Δ_a .

c) Suppose G is compact. For G to be monothetic, it is necessary and sufficient that $\widehat{G}$ is isomorphic to a subgroup of T equipped with the discrete topology.

d) Suppose G is compact. The following conditions are equivalent:

(i) G is solenoidal;

(ii) $\widehat{G}$ is isomorphic to a subgroup of $\mathbf{R}$ endowed with the discrete topology;

(iii) $\widehat{G}$ is torsion-free, and $\text{Card}(\widehat{G}) \leq \mathfrak{c}$, the cardinality of the continuum;

(iv) G is a quotient of $(\Sigma_a)^\mathfrak{c}$ with $a = (n + 2)_{n \geq 0}$.

- e) If G is a compact solenoidal group, then G is monothetic.
- 11) We denote by $L(G)$ the set of continuous representations of G into $\mathbf{R}$. A one-parameter subgroup of G is the image of a continuous homomorphism from $\mathbf{R}$ into G .
- a) The following conditions are equivalent:
- (i) G is the union of one-parameter subgroups;
 - (ii) every homomorphism from $\mathbf{Z}$ into G extends to a continuous homomorphism from $\mathbf{R}$ into G ;
 - (iii) every continuous homomorphism from $\widehat{G}$ into $\mathbf{R}/\mathbf{Z}$ arises by passing to the quotient from an element of $L(\widehat{G})$;
 - (iv) every unitary character of $\widehat{G}$ is of the form $e^{i\lambda}$ where $\lambda \in L(\widehat{G})$.
- b) If G has a countable base, the conditions of a) are equivalent to: G is isomorphic to $R^n \times T^I$ where I is a countable set.
- c) The union of the one-parameter subgroups of G is a dense subgroup of the neutral component of G . (Use Exerc. 4.)
- d) Let $t \mapsto \chi_t$ be a continuous map from $[0, 1]$ into $\widehat{G}$ such that χ_0 is the trivial character. There exists a map $t \mapsto \lambda_t$ from $[0, 1]$ into $L(G)$ with $\lambda_0 = 0$ such that, for all t , we have $\chi_t = \exp(2i\pi \lambda_t)$, and, for every $x \in G$, the map $t \mapsto \lambda_t(x)$ is continuous.
- e) The union of the one-parameter subgroups of G is also the union of the arcs of G that contain e , an arc of G being the image of a continuous map from $[0, 1]$ into G . (Use d).
- 12) We use the notation $L(G)$ from Exercise 11.
- a) Let H be a closed subgroup of G . Every element of $L(H)$ extends to an element of $L(G)$. (First reduce to the case where $G = R^n \times D$ with D discrete, then in the case where the given element of $L(H)$ is trivial on R^n and then to the case where G is discrete. Then use the fact that $\mathbf{R}$ is divisible.)

- b) Every continuous morphism from H into $R^n \times T^I$, where I is an arbitrary set, extends to a continuous morphism from G into $R^n \times T^I$ (Use a) and Th. 4 of II, p. 226.)
- c) Conversely, let A be a locally compact commutative group. Suppose that, for every locally compact commutative group G , for every closed subgroup H of G , and every continuous morphism φ of H into A , φ extends to a continuous morphism from G into A . Then there exist an integer n and a set I such that A is isomorphic to $R^n \times T^I$ (Taking $G = R$, $H = Z$, one sees that A is connected, hence of the form $R^n \times \widehat{D}$ with D discrete. Show that D is a projective Z -module, hence free.)

13) Let C be a compact subset of G and $\varepsilon > 0$.

- a) Let α be the Haar measure of G . There exists a compact subset D of G such that $\alpha(CD) \leq (1 + \varepsilon)\alpha(D)$. (The case where $G = R^n$ being immediate, we are reduced to the case where G admits a compact open subgroup H , and we may assume C is saturated with respect to H ; we are then reduced to the case where G is discrete; we may next assume G is finitely generated, and finally $G = Z^p$.)
- b) There exists $k \in L^1(\widehat{G})$ such that $\mathcal{F}(k) \in \mathcal{H}(G)$ and $\mathcal{F}(k)(x) = 1$ for $x \in C$, with $\|k\|_1 \leq 1 + \varepsilon$. (Take k of the form λfg , where λ is a constant and where $f, g \in L^2(\widehat{G})$ have as Fourier transforms the characteristic functions of suitable sets, using a).)

¶ 14) a) Suppose that every neighborhood of e in G contains an element of infinite order. There exists a compact metrizable perfect totally disconnected subset P of G which is a Kronecker set (Exercise 15 of II, p. 265), and a nonzero diffuse measure concentrated on P . (Reduce to the case where G is metrizable using Exercise 3, c). Then imitate the construction of TG, IX, p. 64, Lemma 3, and at the same time use part e) of Exercise 15 of II, p. 265.)

- b) If $G = (Z/qZ)^N$, there exists a compact perfect totally disconnected subset P of G which is of type K_q , and a nonzero diffuse measure concentrated on P .
- c) Assume G is non-discrete. There exists an element τ of $\mathcal{M}^1(G)$ such that $\|\mathcal{F}(\tau)\|_\infty < \varrho(\tau)$, where $\varrho(\tau)$ is the spectral radius in $\mathcal{M}^1(G)$. There exists a non-invertible element τ' of $\mathcal{M}^1(G)$ such that $\mathcal{F}(\tau') \geq 1$ everywhere on $\widehat{G}$. There exists a non-Hermitian

character of $\mathcal{M}^1(G)$. (Use a), b), exerc. 3, c), and part f) of exerc. 15 of II, p. 265.)

d) Deduce from c) that $\widehat{G}$, which is open in $X(\mathcal{M}^1(G))$, is not dense in $X(\mathcal{M}^1(G))$.

Exercises for Section 3: Invariant Subspaces

1) Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of complex numbers such that there exists a real number $A \geq 0$ satisfying $n|a_n| \leq A$ for all $n \in \mathbb{N}$. For $z \in \mathbb{C}$ of modulus < 1 , set $\varphi(z) = \sum_{n \geq 0} a_n z^n$.

One assumes that there exists $\ell \in \mathbb{C}$ such that

$$\lim_{\substack{x \in \mathbf{R}_+ \\ x < 1}} \varphi(x) = \ell.$$

For $t \in \mathbf{R}_+^*$, set

$$g(t) = \sum_{0 \leq n \leq t^{-1}} a_n, \quad F(t) = \varphi(e^{-t}).$$

We denote by dx the Lebesgue measure on $\mathbf{R}$ as well as its restriction to $\mathbf{R}_+^*$.

a) Let $s \leq t$ be elements of $\mathbf{R}_+^*$. We have

$$|g(s) - g(t)| \leq |a_{[t^{-1}+1]}| + A \log\left(\frac{t}{s}\right).$$

b) The function g is slowly oscillating for $t \rightarrow 0$.

c) For every $t \in \mathbf{R}_+^*$, the function $x \mapsto g(t/x)xe^{-x}$ is integrable on $\mathbf{R}_+^*$ with respect to the Haar measure $x^{-1}dx$.

d) For $0 < x < 1$, one has

$$\varphi(x) = \sum_{n \geq 0} g\left(\frac{1}{n}\right) (x^n - x^{n+1}).$$

e) Let $f(x) = xe^{-x}$ for $x \in \mathbf{R}_+^*$. We have $f \in L^1(\mathbf{R}_+^*, dx/x)$; its integral is equal to 1, and we have $F = g * f$.

f) The function g belongs to $L^\infty(\mathbf{R}_+^*)$. (Show that $F - g$ is bounded using a) as well as the hypothesis.)

g) The Fourier transform of the function f on the group $\mathbf{R}_+^*$ does not vanish. (Identify $\mathcal{F}(f)$ to the function $y \mapsto \Gamma(1 + iy)$ and use FVR, VII.)

h) We have

$$\sum_{n=0}^{+\infty} a_n = \ell.$$

This statement is Littlewood's "Tauberian" theorem. The hypothesis concerning the sequence (a_n) , that is, $a_n = O(1/n)$, can moreover be weakened: it suffices that there exist a constant $A \geq 0$ such that $na_n \geq -A$ for every integer $n \geq 1$ ("Hardy–Littlewood Tauberian theorem," see for example J. KOREVAAR, Tauberian theory, Grundlehren math. wiss. 329, Springer (2004), Th. 7.4, p. 16; the reader should take care not to confuse this theorem with that of FVR, I, p. 50, Exercise 18).

2) Let F be a closed subset of $\widehat{G}$. Let K be a compact subset of $\widehat{G}$ such that $F \cap K = \emptyset$.

Show that there exists a continuous function g integrable on G such that $\mathcal{F}(g)$ is equal to 0 on F and to 1 on K . (Let $f \in L^1(G)$ be such that $\mathcal{F}(f)$ is zero on F and equal to 1 on K ; consider $f * \varphi$, where φ is the Fourier transform of a function of the form $h * h'$ with $h \in \mathcal{K}(\widehat{G})$, $h' \in \mathcal{K}(\widehat{G})$, and where $h * h' = 1$ on K .)

3) Let g be a complex function defined on $]0, +\infty[$, integrable with respect to Lebesgue measure. We assume that $\int_0^{+\infty} g(t)dt = 1$ and that $\int_0^{+\infty} g(t)t^{ix}dt \neq 0$ for all $x \in \mathbf{R}$. Let f be a complex measurable and bounded function in $]0, +\infty[$, such that $\frac{1}{x} \int_0^{+\infty} g(t/x)f(t)dt$ tends to a finite limit ℓ when x tends to $+\infty$.

a) For any complex-valued function h defined on $]0, +\infty[$, integrable with respect to Lebesgue measure and with integral 1, we have

$$\lim_{x \rightarrow +\infty} \frac{1}{x} \int_0^{+\infty} h\left(\frac{t}{x}\right)f(t)dt = \ell.$$

In particular,

$$\lim_{x \rightarrow +\infty} \frac{1}{x} \int_0^x f(t) dt = \ell.$$

(Setting $g_1(t) = tg(t)$, $h_1(t) = th(t)$, we have $g_1 \in L^1(\mathbf{R}_+^*)$; the Fourier transform $\mathcal{F}(g_1)$ does not vanish at ℓ , hence $h_1 * f$ tends to ℓ .) b) If $f(t)$ is slowly oscillating on $\mathbf{R}_+^*$ when t tends to $+\infty$, then f tends to ℓ when t tends to $+\infty$.

4) Let $G = \mathbf{R}$. For every real number $\alpha > 1$, we define $f_\alpha \in L^1(\mathbf{R})$ by

$$f_\alpha(x) = \begin{cases} 2 & \text{if } 0 < x < 1 \\ 1 & \text{if } 1 \leq x < \alpha \\ 0 & \text{otherwise.} \end{cases}$$

a) The functions $f * \varepsilon_x$ for $x \in \mathbf{R}$ form a total set in $L^1(\mathbf{R})$ if and only if α is irrational.

b) The functions $f * \varepsilon_x$ for $x \in \mathbf{R}$ form a total set in $L^2(\mathbf{R})$.

5) Let $f \in L^\infty([0, +\infty[)$ and $k \in \mathbf{R}_+^*$. If

$$\lim_{x \rightarrow +\infty} \frac{1}{x} \int_0^{+\infty} e^{-t/x} f(t) dt = \ell$$

then

$$\lim_{x \rightarrow +\infty} \frac{k}{x^k} \int_0^x (x-t)^{k-1} f(t) dt = \ell$$

(Use the functions $g(t) = e^{-t}$, and $h(t) = k(1-t)^{k-1}$ for $0 \leq t \leq 1$, $h(t) = 0$ for $t > 1$.)

6) Let $f: \mathbf{R} \rightarrow \mathbf{R}$ and $g: \mathbf{R} \rightarrow \mathbf{C}$ be two functions. We assume that $f \in L^\infty(\mathbf{R})$ and that $g \in L^1(\mathbf{R})$, $\int_{\mathbf{R}} g(x) dx = 1$ and that $\mathcal{F}(g)$ does not vanish. We assume that f is slowly decreasing, that is, for $y > x$ such that $y - x$ tends to 0 and $x \rightarrow +\infty$, one has

$$\liminf (f(y) - f(x)) \geq 0.$$

If $(g * f)(x)$ tends to ℓ as x tends to $+\infty$, then $f(x)$ tends to ℓ as x tends to $+\infty$.

7) Let $g : \mathbf{R} \rightarrow \mathbf{C}$ be a continuous function such that:

$$\sum_{n=-\infty}^{+\infty} \sup_{n \leq t \leq n+1} |g(t)| < +\infty.$$

a) We have $g \in L^1(\mathbf{R})$.

b) Suppose that g has integral 1 and that $\mathcal{F}(g)$ does not vanish. Let μ be a complex measure on $\mathbf{R}$ such that $|\mu|([x, x+1])$ is bounded as x ranges over $\mathbf{R}$. If $(g * \mu)(x)$ tends to ℓ as x tends to $+\infty$ then, for every continuous function $h : \mathbf{R} \rightarrow \mathbf{C}$ such that

$$\sum_{n=-\infty}^{+\infty} \sup_{n \leq t \leq n+1} |h(t)| < +\infty,$$

of integral 1, the limit of $(h * \mu)(x)$ when x tends to $+\infty$ is equal to ℓ . (Show that $h * \mu$ is slowly oscillating and that $(g * (h * \mu))(x)$ tends to ℓ when x tends to $+\infty$.)

8) Let g be a function satisfying the hypotheses of Exerc. 3. Moreover, suppose $g \geq 0$ and $\liminf_{x \rightarrow 0} g(x) > 0$. Let f be a real-valued measurable function bounded below on $]0, +\infty[$. If

$$\lim_{x \rightarrow +\infty} \frac{1}{x} \int_0^{+\infty} g\left(\frac{t}{x}\right) f(t) dt = \ell$$

then

$$\sigma(x) = \frac{1}{x} \int_0^x f(t) dt$$

tends to ℓ when x tends to $+\infty$. (Show that σ is bounded for $x \geq 1$, then that σ is slowly decreasing, then that

$$\frac{1}{x} \int_0^{+\infty} g\left(\frac{t}{x}\right) \sigma(t) dt$$

tends to ℓ when x tends to $+\infty$.)

9) Let $\varphi : [0, +\infty[\rightarrow \mathbf{R}$ be a nonnegative increasing function, such that the function $t \mapsto$

$e^{-\sigma t} \varphi(t)$ is integrable with respect to Lebesgue measure when $\sigma > 1$. For every $s \in \mathbb{C}$ such that $\Re(s) > 1$, set

$$f(s) = \int_0^{+\infty} e^{-st} \varphi(t) dt$$

(“Laplace transform of φ ”).

Suppose there exists $A \in \mathbb{C}$ with the following property: as σ tends to 1 through values > 1 , the function

$$t \mapsto f(\sigma + i\tau) - \frac{A}{\sigma + i\tau - 1} \quad \tau \in \mathbb{R}$$

converges uniformly to a function g on every compact subset of $\mathbb{R}$. Then

$$\lim_{t \rightarrow +\infty} \varphi(t) e^{-t} = A$$

(“Ikehara tauberian theorem”).

(We set $a(t) = e^{-t} \varphi(t)$ for $t > 0$, $a(t) = 0$ for $t \leq 0$, $A(t) = A$ for $t > 0$, $A(t) = 0$ for $t \leq 0$

. Let $\lambda > 0$. We set:

$$k_\lambda(t) = \lambda \sqrt{\frac{2}{\pi}} \left(\frac{\sin 2\lambda t}{2\lambda t} \right)^2$$

$$K_\lambda(t) = \begin{cases} 1 - \left| \frac{t}{2\lambda} \right| & \text{if } |t| \leq 2\lambda \\ 0 & \text{if } |t| > 2\lambda. \end{cases}$$

Using exercise 1 of II, p. 262, the Lebesgue–Fubini theorem, and the Plancherel theorem, show that:

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} k_\lambda(x-t) (a(t) - A(t)) e^{-\varepsilon t} dt = \frac{1}{\sqrt{2\pi}} \int_{-2\lambda}^{2\lambda} K_\lambda(y) e^{-ixy} g(y) dy$$

and deduce that

$$\lim_{x \rightarrow +\infty} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} k_\lambda(x-t) a(t) dt = A.$$

Prove then that a is bounded and slowly decreasing. One can finally apply exerc. 6.)

10) We define the von Mangoldt function on $\mathbf{N}^*$ by $\Lambda(n) = 0$ if n is not of the form p^k where p is prime and $k \geq 1$, and $\Lambda(p^k) = \log(p)$.

We denote by ζ the holomorphic function on $\mathbf{C} - \{1\}$ whose restriction to the set of $s \in \mathbf{C}$ such that $\Re(s) > 1$ coincides with the Riemann zeta function (Exercise 16 of II, p. 266).

For $x > 0$, we set

$$\psi(x) = \sum_{1 \leq n \leq x} \Lambda(n).$$

a) Show that for all $s \in \mathbf{C}$ such that $\Re(s) > 1$, one has

$$\sum_{n \geq 1} \Lambda(n) n^{-s} = -\frac{\zeta'(s)}{\zeta(s)}.$$

b) We have

$$\lim_{x \rightarrow +\infty} \frac{\psi(x)}{x} = 1.$$

(Use Exercise 16 of II, p. 266 and Exercise 9).

c) Let $\pi(x)$ be the number of prime numbers $\leq x$. Show that

$$\pi(x) \sim \frac{x}{\log(x)}$$

when $x \rightarrow +\infty$ ("prime number theorem," due to Hadamard and de la Vallée Poussin).

11) One defines the Möbius function μ on $\mathbf{N}^*$ by $\mu(n) = (-1)^k$ if n is the product of k prime factors pairwise distinct and $\mu(n) = 0$ otherwise.

For $x > 0$, we set

$$M(x) = \sum_{n \leq x} \mu(n), \quad f(x) = \frac{M(x)}{x}.$$

a) Show that for $\Re(s) > 1$, one has

$$\sum_{n \geq 1} \mu(n) n^{-s} = \frac{1}{\zeta(s)}.$$

- b) Show that f is bounded, and that $f(y) - f(x) = O((y - x)/x)$ for $y \geq x \geq 1$, so that f is slowly oscillating on the group R_+^* as x tends to $+\infty$.
- c) Let $a, b \in R_+^*$. For $x > 0$, let $[x]$ be the integer part of x . We set $g_0(t) = [t^{-1}]$ for $t \in R_+^*$ and

$$g(t) = 2g_0(t) - ag_0(at) - bg_0(bt).$$

The function g is integrable on R_+^* with respect to Lebesgue measure. If $s \in C$ and $\Re(s) > 0$, one has

$$\int_0^{+\infty} g(t) t^s dt = (2 - a^{-s} - b^{-s}) \frac{\zeta(1+s)}{1+s}.$$

- d) We have $\int_0^{+\infty} g(t) t^{ix} dt \neq 0$ for all $x \in \mathbf{R}$. (Use the part j) of Exercise 16 of II, p. 266.)
- e) Show that $\int_0^{+\infty} g(t/x) f(t) dt = o(x)$ when $x \rightarrow +\infty$. Deduce that $M(x) = o(x)$ when $x \rightarrow +\infty$.

(One can show in an elementary way that this result is equivalent to the prime number theorem, cf. Exercise 10; see for example H. IWANIEC and E. KOWALSKI, Analytic number theory, A.M.S Coll. Publ. 53 (2004), pp. 31–32.)

- 12) Let J be the set of $f \in \mathcal{S}(\mathbf{R}^3)$ such that $\mathcal{F}(f)$ vanishes on the sphere $\mathbf{S}_2$. Let I be the set of $f \in J$ such that $(\partial/\partial y_1)(\mathcal{F}(f))$ vanishes on $\mathbf{S}_2$. Let $\bar{I}$ and $\bar{J}$ be the closures of I and J in $L^1(\mathbf{R}^3)$. Then we have $V(\bar{I}) = V(\bar{J}) = \mathbf{S}_2$ but $\bar{I} \neq \bar{J}$. (Let μ be the positive measure of mass 1 on $\mathbf{S}_2$ invariant under the orthogonal group of $\mathbf{R}^3$; show, using Exercise 20 of II, p. 269, c), that $f(t) = t_1 \mathcal{F}(\mu)(t)$ is an element of $L^\infty(\mathbf{R}^3)$ orthogonal to I but not to J .)
- 13) One calls a C-set in $\widehat{\mathbf{G}}$ a closed subset E of $\widehat{\mathbf{G}}$ having the following property: if $f \in L^1(\mathbf{G})$, if $\mathcal{F}(f)$ vanishes on E , and if $\varepsilon > 0$, there exists a function $g \in L^1(\mathbf{G})$ such that $\|f - f * g\| \leq \varepsilon$ and such that $\text{Supp}(\mathcal{F}(g))$ be compact and disjoint from E .

- a) If E is a C-set in $\widehat{G}$, there exists only one closed ideal I of $L^1(G)$ such that $V(I) = E$.
- b) Every singleton subset of $\widehat{G}$ is a C-set.
- c) The union of two C-sets is a C-set.
- d) Every set obtained by translation from a C-set is a C-set.
- e) Let H be a closed subgroup of $\widehat{G}$, let E be a closed subset of H , and let E' be the boundary of E relative to H . If E' is a C-set relative to $\widehat{G}$, then E is a C-set with respect to $\widehat{G}$. (Use exerc. 21 of II, p. 270.)
- f) In $\mathbb{R}^n$, every intersection E of a finite number of closed half-spaces is a C-set. (Reason by induction on the dimension of the affine subspace generated by E , using b), c), d), e).
- 14) Let E be a closed subset of $\mathbb{R}^n$. Suppose that there exists an interior point p of E such that every line passing through p meets the boundary of E in at most two points. Then there exists only one closed ideal I of $L^1(\mathbb{R}^n)$ such that $\mathcal{F}(f)$ vanishes on E . (Consider f as a limit in $L^1(\mathbb{R}^n)$ of functions deduced from f by homotheties with center p).
- ¶ 15) We denote by μ a Haar measure of G and by B the filter of neighborhoods of e in $\widehat{G}$.
- a) Let U be an integrable open neighborhood of e in G . There exists a function $a \in L^1(G)$, $a \geq 0$, of integral 1, vanishing outside U , such that $\mathcal{F}(a) \in L^1(\widehat{G})$, and

$$\int_G a(x)^2 dx \leq \frac{2}{\mu(U)}.$$

(Let V be a compact neighborhood of e such that $V \subset U$, $\mu(U) < \sqrt{2}\mu(V)$; let W be a compact symmetric neighborhood of e such that $VW^2 \subset U$; take $a = \lambda f * g$, where λ is a constant > 0 , and where f, g are the characteristic functions of the sets VW and W .)

- b) If $f \in L^\infty(G)$, we denote by $A(f)$ the set of elements of $\widehat{G}$ which belong to the weakly closed vector subspace of $L^\infty(G)$ invariant under translation generated by f . If $f, g \in L^\infty(G)$, one has

$$A(fg) \subset \overline{A(f)A(g)}.$$

(Let U be a neighborhood of e . Show that fg is the weak limit of linear combinations of elements of $\widehat{G}$ belonging to $A(f)UA(g)U$.)

c) Let $g \in L^\infty(G)$, K a compact subset of $\widehat{G}$, and f a function in $L^1(G)$ such that $\mathcal{F}(f)$ vanishes on $A = A(g)$ and outside K . For every $V \in \mathcal{B}$, let $\omega(V)$ be the supremum of $|\mathcal{F}(f)|$ on AV . If

$$\liminf_{\mathfrak{B}} \frac{\omega(V)^2 \mu((AV - A) \cap K)}{\mu(V)} = 0,$$

then $\langle f, g \rangle = 0$. (Let $\varepsilon > 0$. There exists a compact subset H of G such that $\int_{G-H} |f| dx \leq \varepsilon$. Then there exists an open neighborhood U of e in $\widehat{G}$ such that $|\langle x, \hat{x} \rangle - 1| \leq \varepsilon$ for $x \in H$, $\hat{x} \in U$. Applying a) to $\widehat{G}$ and U , one obtains a function $a \in L^1(\widehat{G})$. Let $b = \mathcal{F}(a) \in L^1(G)$. Using b), show that $\mathcal{F}(bg)$ vanishes outside AU . On the other hand,

$$\left| \int_G fg dx - \int_G fgb dx \right| \leq \varepsilon (\|f\|_1 + 2) \|g\|_\infty$$

and, since $f, gb \in L^2(G)$,

$$\int_G fgb dx = \int_{\widehat{G}} \mathcal{F}(f) \mathcal{F}(gb) d\hat{x} = \int_{(AU-A) \cap K} \mathcal{F}(f) \mathcal{F}(gb) d\hat{x}.$$

Bound the absolute value of this last expression by:

$$2\|g\|_\infty \frac{\omega(U) \sqrt{\mu((AU - A) \cap K)}}{\sqrt{\mu(U)}}$$

to conclude.)

d) Let I be a closed ideal of $L^1(G)$, $A = V(I)$, f a function of $L^1(G)$ such that $\mathcal{F}(f)$ vanishes on A , S the set of points where $\mathcal{F}(f)$ does not vanish, S' the boundary of S , and B the largest perfect set contained in $S \cap A$. For $V \in \mathfrak{B}$, let $\omega(V)$ be the supremum of $|\mathcal{F}(f)|$ on AV . We assume that every point of B has a neighborhood K in $\widehat{G}$ such that

$$\liminf_{\mathfrak{B}} \frac{\omega(V)^2 \mu((AV - A) \cap K)}{\mu(V)} = 0.$$

Then $f \in I$. (Use the fact that $L^1(G)$ satisfies Ditkin's condition, and argue as in Prop. 5 of I, p. 94; let $G' = \widehat{G} \cup \{\infty\}$ be the Alexandroff compactification of $\widehat{G}$ and let N be the set of $\chi \in G'$ such that $\mathcal{F}(f)$ does not belong to $\mathcal{F}(I)$ in a neighborhood of χ . First show that $N \subset B \cup \{\infty\}$. Then, using c), show that $N \subset \{\infty\}$. Finally, show that $N = \emptyset$.)

e) Let I be a closed ideal of $L^1(\mathbf{R}^n)$, $A = V(I)$, and f a function in $L^1(\mathbf{R}^n)$ such that $\mathcal{F}(f)$ vanishes on A . For $h > 0$, let A_h be the set of points of $\mathbf{R}^n$ outside A and located at a distance from A less than h , and let $\omega(h) = \sup_{x \in A_h} |\mathcal{F}(f)(x)|$. If

$$\liminf_{h \rightarrow 0} \frac{\omega(h)^2 \mu(A_h)}{h^n} = 0$$

then $f \in I$. (Use d .)

f) Let $f \in L^1(\mathbf{R})$, $\alpha \in]0, 1[$, and suppose $\int_{\mathbf{R}} |y|^\alpha |f(y)| dy < +\infty$. Then there exists a constant k such that $|\mathcal{F}(f)(x+h) - \mathcal{F}(f)(x)| \leq kh^\alpha$ for all $x \in \mathbf{R}$ and every $h \in \mathbf{R}$.

g) Let I be a closed ideal of $L^1(\mathbf{R})$, f a function of $L^1(\mathbf{R})$ such that $\mathcal{F}(f)$ vanishes on $V(I)$. If

$$\int_{\mathbf{R}} |y|^{1/2} |f(y)| dy < +\infty,$$

then we have $f \in I$. (Use parts e) and f).

16) Let $f \in L^\infty(G)$. We say that f satisfies the spectral synthesis condition if it belongs to $Y(A(W_f))$, where W_f is the intersection of all translation-invariant weakly closed subspaces of $L^\infty(G)$ which contain f .

a) Let $\mu \in \mathcal{M}^1(\widehat{G})$ and $f = \overline{\mathcal{F}}_{\widehat{G}}(\mu) \in L^\infty(G)$. Then $A(W_f)$ is the support of μ . Moreover, f satisfies the spectral synthesis condition.

b) If G is discrete, every function $f \in L^2(G)$ satisfies the spectral synthesis condition.

17) For every $p \geq 1$, we identify $L^p(\mathbf{R}_+)$ with a closed subspace of $L^p(\mathbf{R})$. Let $f \in L^1(\mathbf{R}_+)$

be the function defined by $f(x) = 2e^{-x}$ for $x \in \mathbf{R}_+$, and let $g \in L^1(\mathbf{R})$ be the function such that $g(x) = f(-x)$.

- a) Show that $f + g = f * g$.
- b) Let A (resp. B) be the subalgebra of $L^1(\mathbf{R})$ generated by f (resp. by f and g). Show that A is dense in $L^1(\mathbf{R}_+)$. (Let $h \in L^\infty(\mathbf{R}_+)$ such that $\langle h, f^{*n} \rangle = 0$ for every integer $n \geq 1$, where f^{*n} is the n-fold iterated convolution of f; show that the Laplace transform

$$F(z) = \int_0^{+\infty} e^{-xz} h(x) dx,$$

defined for all $z \in C$ such that $\Re(z) > 0$ is zero.) Deduce that B is dense in $L^1(\mathbf{R})$.

- c) Let C be a proper closed subalgebra of $L^1(\mathbf{R})$ containing A. We have $1 \in \text{Sp}_C(f)$. (Show that otherwise, part (a) and the holomorphic functional calculus in C would imply that $g \in C$.)
- d) There exists a character χ of C such that $\chi(f) = 1$; we have

$$\chi(\varphi) = \int_0^{+\infty} e^{-x} \varphi(x) dx$$

for every function $\varphi \in L^1(\mathbf{R}_+)$.

- e) There exists a measure $\mu \in \mathcal{M}^1(\mathbf{R})$ of norm 1 such that $\chi(\varphi) = \int \widehat{\varphi} d\mu$ for every function $\varphi \in C$.
- f) The Fourier transform of μ is the function $x \mapsto e^{-|x|}$.
- g) Let $\varphi \in C$. For every $y > 0$, one has

$$\int_{\mathbf{R}} \varphi(x) (e^{-|x+y|} - e^{-|x|-y}) dx = 0.$$

(Using the relation $\chi(\varphi * \psi) = \chi(\varphi)\chi(\psi)$ for $\psi \in L^1(\mathbf{R}_+)$.)

h) We have $C = L^1(\mathbf{R}_+)$. (Thus, the subalgebra $L^1(\mathbf{R}_+)$ is a maximal closed subalgebra of $L^1(\mathbf{R})$.)

18) Let G be a discrete ordered commutative topological group. We write the group law of G additively, and denote its identity by 0 . We denote by G^+ the set of elements $x \geq 0$ of G .

a) The space L_+^1 of functions $f \in L^1(G)$ vanishing outside of G^+ is a closed subalgebra of $L^1(G)$.

b) We assume that G is not Archimedean (TG, V, p. 16, exercise 1 of §3). The algebra L_+^1 is not maximal in $L^1(G)$.

Assume henceforth that G is Archimedean. Let A be a closed subalgebra of $L^1(G)$ containing L_+^1 . For every $x \in G$, let $\varphi_x \in L^1(G)$ be the characteristic function of $\{x\}$. It is invertible in $L^1(G)$ and its inverse is φ_{-x} .

c) There exists $y \in G_+$ such that $\varphi_y \in A$ and $\varphi_{-y} = \varphi_y^{-1} \notin A$ and there exists a character $\chi \neq 0$ of A such that $\chi(y) = 0$.

d) For every $x > 0$ in G , we have $\chi(x) = 0$. (There exists an integer $k \geq 1$ such that $-y + kx > 0$.)

e) There exists a measure $\mu \in \mathcal{M}^1(G)$ of norm 1 such that χ is the restriction of μ to the algebra A viewed as a subspace of $\mathcal{C}(G)$. (For $f \in A$, one has $|\chi(f)| \leq \varrho(f)$ and moreover $\varrho(f) = \|f\|_\infty$.)

f) Let $\widehat{\mu}: G \rightarrow \mathbf{C}$ be the inverse Fourier transform of μ . We have $\widehat{\mu}(x) = 0$ if $x > 0$.

g) The measure μ is the counting measure on G .

h) The subalgebra L_+^1 is maximal in $L^1(G)$.

19) For $f \in L^1(\mathbf{R})$, we denote by A_f the closed subalgebra of $L^1(\mathbf{R})$ generated by f .

a) There exists $f \in L^1(\mathbf{R})$ such that A_f is a proper maximal subalgebra of $L^1(\mathbf{R})$.

In what follows, we assume that $f \in L^1(\mathbf{R})$ is a function such that A_f is a proper maximal subalgebra of $L^1(\mathbf{R})$.

- b) We denote $S = \{0\} \cup \hat{f}(\mathbf{R}) \subset \mathbf{C}$. Show that the function $z \mapsto \bar{z}$ is not a uniform limit on S of polynomial functions. (First show that C-S is not connected.)
- c) The subalgebra of $L^1(\mathbf{R})$ generated by f and $\tilde{f}$ is dense in $L^1(\mathbf{R})$.
- d) The Fourier transform of f is injective and does not vanish on $\mathbf{R}$. (Show that S cannot have two bounded connected components.)

FORMULA SHEET FOR FOURIER THEORY

Function spaces

Let G be a locally compact commutative group. We recall the principal function spaces on G and their relation to the Fourier transformation.

The left column defines a function space E on G , and indicates whether it is a subspace of $L^1(G)$, of $L^2(G)$, or of both. The right column indicates analogous inclusions for the image of E under the transformation and cotransformation of Fourier (the latter being either that defined on $L^1(G)$, or that on $L^2(G)$, as the case may be). The notation F denotes either the Fourier transformation or the Fourier cotransformation.

$$\mathcal{M}^1(G)$$

$$\mathcal{F}(\mathcal{M}^1(G)) \subset \mathcal{C}_b(\widehat{G})$$

(space of bounded measures)

(prop. 3, p. 207)

$$L^1(G)$$

$$\mathcal{F}(L^1(G)) \subset \mathcal{C}_0(\widehat{G})$$

(integrable functions)

$$L^2(G)$$

$$\mathcal{F}(L^2(G)) = L^2(\widehat{G})$$

(square-integrable functions) (th. 1, p. 215)

$$B(G) \subset L^1(G) \quad \mathcal{F}(B(G)) = B(\widehat{G})$$

space of $f \in L^1(G)$ such that (th. 3, p. 222)

$$\mathcal{F}_G(f) \in L^1(\widehat{G})$$

$A(G) \subset L^1(G) \cap L^2(G) \cap B(G) \quad \mathcal{F}(A(G)) \subset \mathcal{C}_0(\widehat{G}) \cap L^1(\widehat{G}) \cap L^2(\widehat{G})$ (space spanned by $f * g$ where $f, g \in L^1(G) \cap L^2(G)$) (cor. of prop. 8, p. 211, lemma 5, p. 215, prop. 11, p. 217)

Formulary in $\mathbf{R}^n$

We recall here the most important formulas concerning the Fourier transform in $\mathbf{R}^n$, equipped with the Euclidean norm. We identify $\mathbf{R}^n$ with its dual via the map $(x, y) \mapsto \exp(2i\pi x \cdot y)$, where $x \cdot y = \sum_j x_j y_j$ for all $x = (x_j)$ and $y = (y_j)$ in $\mathbf{R}^n$.

$f \in L^1(\mathbf{R}^n)$	$\hat{f}(y) = \int_{\mathbf{R}^n} f(x) \exp(-2i\pi x \cdot y) dx$ ($y \in \mathbf{R}^n$; Def. 3, p. 206)
$f \in L^1(\mathbf{R}^n) \text{ or } L^2(\mathbf{R}^n)$	$\hat{f}_h(y) = \exp(2i\pi h \cdot y) \hat{f}(y)$
$y \in \mathbf{R}^n$	$f_h(x) = f(x + h)$ (Eq. (11), p. 208)
$f \in L^1(\mathbf{R}^n) \text{ or } L^2(\mathbf{R}^n)$	$\hat{g}_h(y) = \hat{f}(y - h)$
$h \in \mathbf{R}^n$	$g_h(x) = f(x) \exp(2i\pi h \cdot x)$ (Eq. (12), p. 208)
$f \in L^1(\mathbf{R}^n) \text{ or } L^2(\mathbf{R}^n)$	$\hat{f} \circ \sigma = \frac{1}{ \det(\sigma) } \hat{f} \circ {}^t\sigma^{-1}$
$\sigma \in \text{GL}(n, \mathbf{R})$	(Eq. (16), p. 208)
$f(x) = \exp(-Q(x)),$	$\hat{f}(y) = \frac{1}{\det(\sigma)} \exp(-Q^*(y)),$
$Q(x) = \ \sigma(x)\ ^2$	$Q^*(y) = \ {}^t\sigma^{-1}(y)\ ^2$
Q positive definite quadratic form	(example, p. 239)
$f \in L^2(\mathbf{R}^n)$	$\int_{\mathbf{R}^n} f(x) ^2 dx = \int_{\mathbf{R}^n} \hat{f}(y) ^2 dy$ (Th. 1, p. 215)
$f \in L^2(\mathbf{R}^n)$ such that	$f(x) = \int_{\mathbf{R}^n} \hat{f}(y) \exp(2i\pi x \cdot y) dx$
$\hat{f} \in L^1(\mathbf{R}^n)$	(almost everywhere ; Prop. 12, p. 219)
$f \in A(\mathbf{R}^n)$	$f(x) = \int_{\mathbf{R}^n} \hat{f}(y) \exp(2i\pi x \cdot y) dx$ ($x \in \mathbf{R}^n$; Prop. 11 p. 217)

Formulary in $(\mathbf{R}/a\mathbf{Z})^n$

For the convenience of the reader, we state the main formulas of Fourier theory for the group $(\mathbf{R}/a\mathbf{Z})^n$, where $a > 0$, equipped with the Haar measure of total mass 1, denoted $a^{-n}dx$. When $n = 1$, this corresponds to periodic functions of period a . The dual group of $(\mathbf{R}/a\mathbf{Z})^n$ is identified with the group $\mathbf{Z}^n$ via the map $(\mathbf{R}/a\mathbf{Z})^n \times \mathbf{Z}^n \rightarrow \mathbf{U}$ induced by passing to the quotient from the map defined by

$$(x, y) \mapsto \exp\left(\frac{2i\pi}{a}x \cdot y\right)$$

for all $x \in \mathbf{R}^n$ and every $y \in \mathbf{Z}^n$. The dual measure on $\mathbf{Z}^n$ is then the counting measure.

$$\begin{array}{ll}
f \in L^1((\mathbf{R}/a\mathbf{Z})^n) & \hat{f}(y) = \frac{1}{a^n} \int_{(\mathbf{R}/a\mathbf{Z})^n} f(x) \exp\left(-\frac{2i\pi}{a} y \cdot x\right) dx \\
& (y \in \mathbf{Z}^n; \text{Def. 3, p. 206}) \\
f \in L^1((\mathbf{R}/a\mathbf{Z})^n) \text{ or } & \hat{f}_h(y) = \exp\left(\frac{2i\pi}{a} h \cdot y\right) \hat{f}(y) \\
L^2((\mathbf{R}/a\mathbf{Z})^n) & f_h(x) = f(x + h) \\
h \in (\mathbf{R}/a\mathbf{Z})^n & (\text{Eq. (11), p. 208}) \\
f \in L^1((\mathbf{R}/a\mathbf{Z})^n) \text{ or } & \hat{g}_h(y) = \hat{f}(y - h) \\
L^2((\mathbf{R}/a\mathbf{Z})^n) & g_h(x) = f(x) \exp\left(\frac{2i\pi}{a} h \cdot x\right) \\
h \in (\mathbf{R}/a\mathbf{Z})^n & (\text{Eq. (12), p. 208}) \\
f \in L^2((\mathbf{R}/a\mathbf{Z})^n) & \frac{1}{a^n} \int_{(\mathbf{R}/a\mathbf{Z})^n} |f(x)|^2 dx = \sum_{y \in \mathbf{Z}^n} |\hat{f}(y)|^2 \\
& (\text{Th. 1, p. 215}) \\
f \in L^2((\mathbf{R}/a\mathbf{Z})^n) \text{ such that } & f(x) = \sum_{y \in \mathbf{Z}^n} \hat{f}(y) \exp\left(\frac{2i\pi}{a} y \cdot x\right) \\
\hat{f} \in L^1(\mathbf{Z}^n) & (\text{almost everywhere; Prop. 12, p. 219}) \\
f \in A((\mathbf{R}/a\mathbf{Z})^n) & f(x) = \sum_{y \in \mathbf{Z}^n} \hat{f}(y) \exp\left(\frac{2i\pi}{a} y \cdot x\right) \\
& (x \in (\mathbf{R}/a\mathbf{Z})^n; \text{Prop. 11 p. 217})
\end{array}$$

Poisson's formula in $\mathbf{R}^n$

We state here the Poisson formula in the particularly important case where $n = 1$ (corollary of Proposition 15 of II, p. 230).

Let $f \in L^1(\mathbf{R})$ and $a > 0$. Suppose that

$$\sum_{h \in \mathbf{Z}} |\hat{f}(a^{-1}h)| < +\infty,$$

$$\sum_{h \in \mathbf{Z}} |f(x + ah)| < +\infty \text{ for all } x \in \mathbf{R},$$

$$x \mapsto \sum_{h \in \mathbf{Z}} f(x + ah) \text{ continuous on } \mathbf{R}.$$

Then, for all $x \in \mathbf{R}$, one has

$$\sum_{h \in \mathbf{Z}} f(x + ah) = \frac{1}{a} \sum_{k \in \mathbf{Z}} \hat{f}\left(\frac{k}{a}\right) \exp\left(\frac{2i\pi}{a} kx\right).$$

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